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Quantitative
Positive Energy District
Definition, Contextualization and Assessment

Doctoral Thesis in Electrical and Computer Engineering

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RESUMO

Os Bairros de Energia Positiva (PEDs) são promovidos como um instrumento-chave para o desenvolvimento urbano climaticamente neutro. No entanto, a sua implementação é condicionada por definições vagas, qualitativas e pouco sensíveis ao contexto. Os conceitos de PED existentes assentam frequentemente em reivindicações de "energia positiva" formuladas de modo pouco preciso, com pressupostos implícitos quanto às fronteiras do sistema, aos métodos de ponderação e às convenções de contabilização. Tal limita a comparabilidade entre casos, fragiliza a sua utilidade para a governação e a certificação, e pode deturpar o contributo dos bairros para os objetivos climáticos nacionais, sobretudo em áreas urbanas densas e em contextos de reabilitação. Neste contexto, a presente tese procura responder à seguinte questão: como podem os PEDs ser definidos e avaliados de forma quantitativa, sensível ao contexto e relevante para a governação, mantendo-se simultaneamente coerentes com metas de descarbonização de nível superior.

A tese desenvolve uma definição quantitativa de PED que trata os balanços energéticos e de emissões de gases com efeito de estufa à escala do bairro como instrumentos de avaliação concebidos intencionalmente, e não como indicadores meramente descritivos. O desempenho dos PEDs é operacionalizado através de critérios de suficiência orientados para a energia primária e para as emissões de gases com efeito de estufa ao longo do ciclo de vida, incorporando explicitamente fatores de contexto estrutural, em particular a densidade edificada, bem como a mobilidade quotidiana e as emissões incorporadas. Estes fatores de contexto ajustam as metas, e não os fluxos físicos de energia, permitindo comparações mais significativas entre tipologias urbanas heterogéneas, ao mesmo tempo que asseguram a coerência com trajetórias nacionais de descarbonização. Para apoiar a aplicação prática, a tese introduz um quadro de simulação e avaliação horária e multienergética, destinado a contextos de planeamento preliminar e de governação. Aplicado a um vasto conjunto de estudos de caso de bairros europeus e integrado num sistema austríaco de declaração PED, este quadro demonstra que a viabilidade dos PEDs depende fortemente do contexto e que sistemas multienergéticos energeticamente flexíveis, bem como uma elevada penetração de veículos elétricos, constituem fatores facilitadores importantes.

A tese conclui que as definições de PED devem ser entendidas como instrumentos normativos e explicitamente concebidos. Só se as fronteiras do sistema, as opções de ponderação e os ajustamentos contextuais forem tornados transparentes poderão os PEDs servir como instrumentos robustos de planeamento, governação e certificação. Ao articular a avaliação à escala do bairro com os objetivos

nacionais em matéria de clima e energia, a abordagem proposta oferece uma base mais coerente para avaliar os PEDs e apoiar a sua implementação em contextos reais de desenvolvimento urbano.

Palavras-chave: Distritos de Energia Positiva, sistemas energéticos urbanos, densidade edificada, emissões de gases com efeito de estufa ao longo do ciclo de vida, balanço de energia primária, modelação energética urbana, avaliação sensível ao contexto, instrumentos de planeamento e governação

ABSTRACT

Positive Energy Districts (PEDs) are promoted as a key instrument for climate-neutral urban development. However, their implementation is constrained by vague, qualitative, and context-insensitive definitions. Existing PED concepts often rely on loosely specified "positive energy" claims with implicit assumptions regarding system boundaries, weighting methods, and accounting conventions. This limits comparability between cases, weakens their usefulness for governance and certification, and can misrepresent district-level contributions to national climate objectives, especially in dense urban areas and renovation contexts. Against this background, this thesis asks how PEDs can be defined and assessed in a quantitative, context-aware, and governance-relevant way that remains consistent with higher-level decarbonization targets.

The thesis develops a quantitative PED definition that treats district energy and greenhouse gas balances as designed assessment instruments rather than descriptive indicators. It operationalizes PED performance through sufficiency-oriented primary energy and life-cycle greenhouse gas criteria and explicitly incorporates structural context factors, most notably building density, as well as everyday mobility and embodied emissions. These context factors adjust targets rather than physical energy flows, enabling more meaningful comparison across heterogeneous urban typologies while maintaining consistency with national decarbonization pathways. To support practical application, the thesis introduces an hourly, multi-energy simulation and assessment framework for early planning and governance contexts. Applied to a broad set of European district case studies and embedded in an Austrian PED declaration system, the framework shows that PED feasibility is strongly context-dependent, and that energy-flexible multi-energy systems and high electric vehicle penetration are important enabling factors.

The thesis concludes that PED definitions must be understood as normative and explicitly designed instruments. Only if system boundaries, weighting choices, and contextual adjustments are made transparent can PEDs serve as robust tools for planning, governance, and certification. By linking district-scale assessment to national climate and energy objectives, the proposed approach provides a more consistent basis for evaluating PEDs and for supporting their implementation in real urban development contexts.

Keywords: Positive Energy Districts, urban energy systems, building density, life-cycle greenhouse gas emissions, primary energy balance, urban building energy modelling, context-aware assessment, planning and governance tools

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The research presented in this thesis was developed in connection with the following research projects. The projects provided funding, case studies, data, methodological development contexts, and opportunities for practical implementation.

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kolPEQ	2019–2022	City of Vienna, Municipal Department 23, Call 24: *Stiftungsprofessuren und Kompetenzteams für Lehre*	24-14
Zukunftsquartier **Österreich**	2021–2022	Austrian Research Promotion Agency (FFG)	
Plus-Energie-Campus	2020–2022	Austrian Research Promotion Agency (FFG), *Stadt der Zukunft*, 7th call, 2019	3900269
Cities4PEDs	2021–2023	Austrian Research Promotion Agency (FFG), JPI Urban Europe, Pilot Call *Positive Energy Districts*	4126826
SimplyPositive	2023–2024	Austrian Research Promotion Agency (FFG), JPI Urban Europe, Urban Europe Call 2021	4459627
ZQ3Demo	2021–2026	Austrian Research Promotion Agency (FFG), *Stadt der Zukunft*, 8th call, 2021	4123865
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ACRONYMS

Acronym	Meaning
BAU	Business as Usual
BEM	Building Energy Modelling
BESS	Battery Energy Storage System
CAPEX	Capital Expenditure
CEA	City Energy Analyst
CFD	Density Context Factor
CFR	Renovation Context Factor
CO_2	Carbon Dioxide
COP	Coefficient of Performance
DH	District Heating
DHW	Domestic Hot Water
DSS	Decision Support System
EPBD	Energy Performance of Buildings Directive
EUI	Energy Use Intensity
EV	Electric Vehicle
FAR	Floor Area Ratio
GFA	Gross Floor Area
GHG	Greenhouse Gas
GIS	Geographic Information System
GSHP	Ground-Source Heat Pump
HP	Heat Pump
HVAC	Heating, Ventilation and Air Conditioning
KPI	Key Performance Indicator
LCA	Life Cycle Assessment
MES	Multi-Energy System
MIM	Motorized Individual Mobility

Acronym	Meaning
NEB	New European Bauhaus
NFA	Net Floor Area
nZEB	Nearly Zero Energy Building
NZEB	Net Zero Energy Building
OPEX	Operational Expenditure
PE	Primary Energy
PED	Positive Energy District
PV	Photovoltaics
RES	Renewable Energy Sources
SET Plan	Strategic Energy Technology Plan
SoC	State of Charge
TABS	Thermally Activated Building System
UBEM	Urban Building Energy Modelling
V2B	Vehicle-to-Building
V2G	Vehicle-to-Grid



Introduction

What do buildings and districts look like in a decarbonized future? Are they still net energy consumers or are they self-sufficient? Or even net-suppliers of electricity? Which role do they play in the greater energy system in terms of storage and demand-side management? How do they interact with the mobility sector? How much embodied emissions can they cause? And what can we infer for our planning practice today? This chapter introduces the motivation, relevance and aims of the thesis and provides the reader with an overview of the pursued methodology, main contributions and structure of the thesis.

1.1. Motivation and Relevance

The decarbonization of the building sector is a central prerequisite for achieving national, European, and global climate targets. Buildings and districts account for a substantial share of final energy demand and greenhouse gas (GHG) emissions, not only through their operation but increasingly through construction, refurbishment, and associated mobility. While decades of policy and research have focused on improving energy efficiency at the building scale, it has become evident that efficiency gains alone are insufficient to deliver climate neutrality within the required timeframes. Rebound effects, absolute growth in floor area, and the slow turnover of the building stock limit the effectiveness of incremental improvements. As a result, the focus of both research and policy has shifted from relative efficiency gains toward absolute performance criteria aligned with long-term climate targets.

Within this context, the concept of **Positive Energy Districts** (PEDs) has emerged as a prominent paradigm. PEDs are promoted as urban-scale systems that combine high energy efficiency, local renewable energy generation, and energy flexibility to achieve net-zero or net-positive energy and

high quality of life. At European level, PEDs have been positioned as a key implementation vehicle for climate-neutral cities, most notably through the Strategic Energy Technology (SET) Plan and related research and innovation programs (Europe & Action 3.2, 2020; SET Plan, 2018): The ambition to initiate 100 PEDs by the mid-2020s across Europe illustrates the political relevance attributed to the concept. PEDs are expected to act not only as technical demonstrations, but as scalable instruments capable of guiding mainstream urban development toward climate neutrality.

Despite this prominence, a fundamental gap persists between the political ambitions associated with PEDs and their practical operationalization. Existing PED definitions are predominantly qualitative, descriptive, or project specific. They articulate broad aspirations, such as energy efficiency, renewable integration, flexibility, and quality of life, but remain insufficiently specified to support consistent assessment, comparison, or governance. As a result, districts labelled as "Positive Energy Districts" often differ substantially in system boundaries, performance metrics, and implicit assumptions. This heterogeneity undermines the credibility of the PED concept as a policy instrument and limits its usefulness for planners, municipalities, developers, and funding bodies.

A central challenge lies in the interpretation of the "positive energy" criterion itself. While the term suggests a clear quantitative condition, in practice it is implemented through widely varying energy balance formulations, temporal scopes, weighting schemes, and boundary choices. Identical districts can appear compliant or non-compliant depending on methodological decisions that are often implicit, undocumented, or tailored to specific project contexts. Without a shared and transparent quantitative basis, PED assessments risk becoming arbitrary and susceptible to greenwashing, particularly when positive energy claims are used to justify public funding or regulatory privileges.

These challenges are compounded by strong contextual dependencies. The feasibility of achieving positive or net-zero energy balances at district scale is heavily influenced by factors such as building density, urban form, climate, available renewable resources, and the composition of the existing building stock and underlying energy system. Uniform PED targets implicitly favour low-density or greenfield developments with abundant surface area for renewable generation, while dense urban or historic districts – where transformation is most urgently needed – are structurally disadvantaged. If unaddressed, these biases risk reinforcing unsustainable development patterns and contradicting broader objectives related to land-use efficiency, mobility reduction, and urban compactness.

At the same time, the transformation of the energy system toward high shares of renewable energy introduces new performance dimensions that are not captured by static annual balances. Temporal alignment between demand and generation, the provision of energy flexibility, and interaction with surrounding energy systems become decisive for system compatibility. Similarly, everyday mobility

increasingly overlaps with district energy systems through electrification, while embodied emissions from construction and infrastructure represent a growing share of total climate impact. Yet these dimensions are often excluded from PED assessments due to methodological complexity or unclear allocation of responsibility.

From a governance perspective, the absence of a robust quantitative PED definition poses a significant obstacle. Policymakers require instruments that translate abstract climate targets into operational criteria that are nationally fair, comparable across contexts, and stable over time. Planners and developers require early-stage assessment methods that provide reliable orientation without presupposing detailed design decisions. Certification bodies and funding agencies require transparent and auditable criteria that support binary compliance decisions. Existing PED frameworks struggle to satisfy these requirements simultaneously.

This motivates this thesis: While the PED concept is widely recognized as a promising approach for urban decarbonization, its current formulations lack the quantitative rigor, contextual sensitivity, and governance compatibility required to function as a reliable climate instrument. Addressing this gap is not merely a technical exercise. Any quantitative PED definition implicitly encodes normative decisions about effort sharing, resource allocation, and feasibility across different urban contexts. Making these decisions explicit, transparent, and anchored in national climate pathways is therefore essential for both scientific credibility and political legitimacy.

This thesis responds to this challenge by treating the PED definition itself as a design problem. Rather than asking whether a given district qualifies as a PED under a predefined label, the work asks what a PED definition must achieve to be useful, fair, and climate-compatible across diverse contexts. By anchoring district-level targets in national decarbonization pathways, explicitly accounting for contextual constraints such as density, and integrating flexibility, mobility, and life-cycle emissions into a coherent quantitative framework, the thesis seeks to transform the PED concept from a loosely defined aspiration into an operational planning and governance instrument that provides a transparent answer to the often voiced “How much is enough?” in planning practice.

In doing so, the thesis positions PEDs not as isolated showcase projects, but as repeatable and comparable planning objects that can act as gauge of the decarbonization of the building sector. The central motivation is therefore to reconcile ambition with feasibility, and innovation with governance, by providing a quantitative foundation that allows PEDs to fulfil their intended role in climate-neutral urban development.

1.2. Objectives

This thesis is organized around the following central hypothesis:

The PED concept can be operationalized as a quantitative sufficiency criterion for district-scale energy and greenhouse gas performance that is nationally fair, attainable across heterogeneous urban contexts and densities, and aligned with sectoral decarbonization pathways. And that this definition can be implemented in a policy-relevant national declaration system.

Building on this hypothesis, the thesis pursues three tightly coupled objectives that correspond to theory development, methodological operationalization, and institutional application. A detailed overview of the individual contributions is given in chapter 1.4.

1.2.1. Objective 1: Development of a quantitative, context-aware PED definition

The first objective is to develop a theoretical framework for a **quantitative PED definition** centred on energy and greenhouse gas balancing. The definition is constructed around a small number of system-critical performance metrics, most prominently the primary energy balance and the life cycle GHG balance, which are interpreted as **sufficiency criteria** rather than relative efficiency indicators.

A core aspect of this objective is the explicit treatment of **contextual constraints** that structurally influence PED feasibility, such as building vintage and density, and characteristics of the surrounding energy and transport systems. These influences are formalized through quantitative context factors that modify PED target reference values without altering physical energy flows. By anchoring these targets in national decarbonization pathways, the definition aims to ensure comparability, fairness, and climate compatibility across diverse implementation contexts.

1.2.2. Objective 2: Development of a definition-consistent PED simulation and assessment framework

The second objective is to develop a PED simulation and assessment framework that operationalizes the quantitative definition and enables its application under real-world planning constraints. The framework is designed to evaluate district energy systems at hourly resolution while remaining suitable for early planning phases characterized by limited and uncertain data.

Rather than optimizing individual system configurations, the model prioritizes definition consistency, transparency, and robustness, allowing compliance with the PED definition to be tested across a wide range of contexts and scenarios. The framework aims to integrate multiple energy carriers, flexibility effects, and – where applicable – everyday mobility and life-cycle emissions,

while complying with established national and European accounting standards. The framework aims to serve as a bridge between abstract definition logic and practical assessment.

1.2.3. Objective 3: Empirical and institutional application of the PED framework

The third objective is to apply and validate the PED framework across heterogeneous European and national contexts and to demonstrate its suitability for institutional use. This objective is pursued through two complementary application modes.

First, the framework is applied analytically to a range of European district case studies and scenario-based investigations. These applications are used to explore feasibility limits, boundary effects, and context sensitivity, and to validate that the same underlying definition remains coherent when used for exploratory analysis rather than formal declaration.

Second, the framework is embedded in a **national declaration system**, where it is translated into fixed assessment scopes, compliance rules, and auditable outputs suitable for certification and governance. This tests whether the definition and model can function under institutional constraints such as stability, reproducibility, and parameter governance.

Together, these applications aim to demonstrate that the PED definition can support different responsibilities and decision contexts such as policy, planning, and research.

1.3. Methodology

This thesis follows a definition-driven and iterative methodological approach**.** Conceptual development, quantitative formalization, modelling, and application were not conducted as strictly sequential steps, but evolved in parallel, with continuous feedback between theory, implementation, and empirical use. This reflects both the complexity of the PED concept and the applied planning and policy contexts in which PEDs are developed.

The methodological foundation of this work is a structured analysis of existing PED definitions, assessment frameworks, and district energy modelling approaches. At the institutional and planning level, recurring questions emerged concerning the objectives of PED implementation, appropriate levels of ambition, and the meaning of “positive” performance under evolving energy system conditions. Addressing these questions required moving beyond descriptive comparison toward a prescriptive perspective, motivating a comparative analysis focused on uncovering structural inconsistencies, implicit assumptions, and limitations in assessment logic, comparability, and governance applicability.

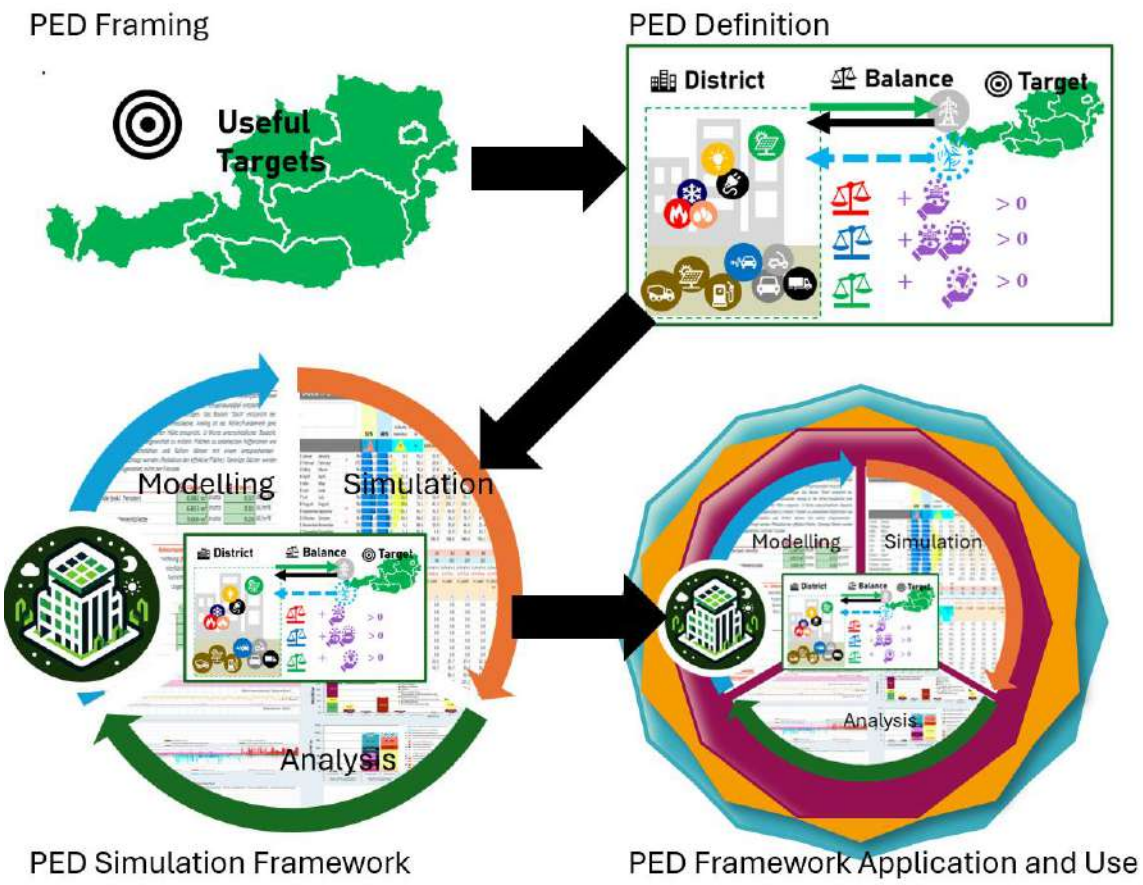


Figure 1. Methodology of thesis development

This analytical synthesis provided the basis for reframing the PED definition as a design problem rather than a descriptive label. Quantitative methods were used to construct a context-aware PED definition anchored in national decarbonization pathways. District-level performance requirements were formulated as sufficiency criteria expressed through primary energy and life-cycle greenhouse gas balances, with contextual influences such as density and mobility incorporated via explicit target-modifying mechanisms rooted in national energy and emission stock-taking and decarbonization pathways.

To operationalize the definition, an hourly, multi-energy simulation and assessment framework was developed. Districts are represented as aggregated systems at high temporal and reduced spatial resolution, enabling the assessment of energy balances, flexibility effects, and, where applicable, mobility and embodied emissions under uncertainty.

Empirical and institutional application forms a final part of the methodology. The framework was applied in several national and European research projects and to multiple European district cases for analytical validation and was embedded in a national Austrian PED declaration context for institutional validation.

1.4. Contributions

This thesis makes **conceptual, methodological, and applied contributions** to the definition, assessment, and governance of PEDs, which form the three pillars of the work. Together they aim to establish a continuous methodological chain from conceptual clarification (Chapter 3.1), through quantitative definition (Chapter 3.2), modelling and simulation (Chapter 3.3), to institutional and empirical application (Chapter 3.4):

Table 1. Structure of the thesis contributions

Pillar	**Core Contribution**
Theory	Quantitative, context-aware definition of PEDs
Simulation Model	Definition-consistent, hourly, multi-energy PED assessment framework
Application	Translation into planning practice, declaration, and validation through case studies

Chapter 3 presents the author's own scientific contributions as documented in peer-reviewed journal articles (J), conference papers (C) and author-led research outputs (R), as listed in the following subsection.

Table 2. Publication Overview

	Publications:	[Schneider-2023 a]	[Schneider-2023 b]	[Schneider-2024 a]	[Schneider-2024 b]	[Schneider-2024 c]	[Schneider-2024 d]	[Bruckner-2025]	[Schneider-2025 a]	[Schneider-2025 b]	[Schneider-2025 c]	[Schneider-2025 d]
		1J	**2R**	**3J**	**4C**	**5C**	**6C**	**7J**	**8C**	**9R**	**10C**	**11J**
Topic												
1 Framing PEDs as a Design Problem		✓	✓									
2 Construction of a quantitative PED Definition	Goals	✓	✓	✓						✓		✓
	National targets	✓	✓	✓						✓		✓
	Density	✓	✓				✓	✓		✓		
	Flexibility	✓			✓	✓	✓		✓		✓	
	Mobility	✓							✓			✓
	Embodied Emissions											✓
3 Modelling and Simulation Framework	Requirements					✓					✓	✓
	Approach					✓			✓		✓	✓
	Testing	✓	✓			✓			✓	✓	✓	✓
4 Application and Use of PED Framework	Declaration	✓			✓					✓		✓
	Comparative cases	✓	✓			✓	✓	✓		✓		✓

Publication list

1 (J) Simon Schneider, Thomas Zelger, David Sengl, and José Baptista. „A Quantitative Positive Energy District Definition with Contextual Targets“. *Buildings* 13, Nr. 5 (2023): 5. https://doi.org/10.3390/buildings13051210.

Innovation: PED balance target Contextualization method for density and mobility contexts and Context Calibration to national decarbonization pathways.

2 (R) Simon Schneider. Cities4PEDs – Work Package 2 Deliverable: Definition of Positive Energy Districts. JPI UE PED Project Cities4PEDs, 2023. Available at: https://energy-cities.eu/wp-content/uploads/2023/07/Cities4PEDs-WP2-Definition-.pdf

Innovation: Conceptual framing of PED definitions as instruments linking European climate objectives to local implementation. Identification of PED definition trilemma and of definition regionality levels. Assigning Method of normative target setting to the national level while preserving project-level flexibility.

3 (J) Simon Schneider, Edit Paráda, David Sengl, José Baptista, and Paulo Moura Oliveira. „Allocation of National Renewable Expansion and Sectoral Demand Reduction Targets to Municipal Level“. *Frontiers in Sustainable Cities* 5 (January 2024). https://doi.org/10.3389/frsc.2023.1294361.

Innovation: Top-down allocation methodology translating national renewable expansion and sectoral demand reduction targets to the municipal level, preserving consistency with national decarbonization pathways.

4 (C) Simon Schneider, Albert Treytl, Stefan Wilker, Michael Schmidthaler, Thomas Zelger, and Elisabeth Kerschbaum. „Digital Positive Energy Districts: A Scalable Strategy for Urban Heat and Power Transition?“. Conference paper presented at *18. Symposium Energieinnovation*, 14.-16.02.2024, Graz, Austria. 16. February 2024. https://www.tugraz.at/fileadmin/user_upload/tugrazExternal/f560810f-089d-42d8-ae6d-8e82a8454ca9/files/lf/Session_F6/661_LF_Schneider.pdf.

Innovation: Integration of PED criteria into urban development competitions and early-stage project evaluation.

5 (C) Simon Schneider, Raphael Drexel, Thomas Zelger, and José Baptista. „PEExcel: A Fast One-Stop-Shop Assessment and Simulation Framework for Positive Energy Districts“. *BauSim Conference*, V. 10 (2024): 80–88. https://doi.org/10.26868/29761662.2024.11.

Innovation: Lightweight, spreadsheet-based PED simulation framework enabling rapid, hourly PED assessment consistent with a quantitative definition.

6 (C) Simon Schneider, Svitlana Alyokhina, Helmut Bruckner, und José Baptista. „Can Urban Retrofitting Achieve a Positive Energy Balance? a Case Study of Four European Positive Energy Districts“. *2024 International Conference on Sustainable Technology and Engineering (i-COSTE)*, Dezember 2024, 01–06. https://doi.org/10.1109/i-COSTE63786.2024.11024809.

Innovation: Systematic, multi-context application of the density-contextualized PED energy balance framework to real European retrofitting districts.

7 (J) Simon Schneider, Svitlana Alyokhina, et al. „Lessons Learned from Four Real-Life Case Studies: Energy Balance Calculations for Implementing Positive Energy Districts “. *Energies* 18, Nr. 3 (2025): 560. https://doi.org/10.3390/en18030560.

Innovation: Sensitivity analysis on density context operationalization

8 (C) Simon Schneider, and José Baptista. „Annual Hourly E-Mobility Modelling and Assessment in Climate Neutral Positive Energy Districts“. *2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe* (EEEIC / I&CPS Europe), Juli 2025, 1–7. https://doi.org/10.1109/EEEIC/ICPSEurope64998.2025.11169136.

Innovation: Annual, hourly e-mobility modelling approach integrated into district-scale energy balance assessment framework. Demonstrates the impact of charging behaviour, flexibility assumptions, and electrified mobility on PED feasibility.

9 (R) Simon Schneider, Raphael Drexel, and Constanze Frech. Klimaneutrales Alliiertenviertel. Wien, 2025. https://www.data.gv.at/katalog/dataset/a6bb0ae7-6059-47c1-8ab3-bdfb2dbddb77.

Innovation: Full-scale, real-world application of the PED assessment framework in a dense urban retrofit context under municipal planning constraints.

10 (C) Simon Schneider, Raphael Drexel, and José Baptista. „From Fossil Fuel Dependency to Renewable Sufficiency: Predictive Energy Modelling for District Decarbonization Scenarios“. Conference paper presented at the CESB25 – *Central Europe towards Sustainable Building 2025*, Prague. 17. September 2025. https://cesb.cz/cesb25-proceedings/.

Innovation: Comparative modelling approach combining building-level simulations using a secondary engine (City Energy Analyst, CEA) with district-scale assessment using PEExcel. Cross-Validation of aggregation logic and boundary assumptions across modelling scales, highlighting consistency limits between bottom-up building simulations and district energy balance frameworks.

11(J) Simon Schneider, Thomas Zelger, Raphael Drexel, Manfred Schindler, Paul Krainer, and José Baptista. „Declaration-Ready Climate-Neutral PEDs: Budget-Based, Hourly LCA Including Mobility and Flexibility“. **Designs** 9, Nr. 6 (2025): 123. https://doi.org/10.3390/designs9060123.

Innovation: Declaration-ready assessment framework, method of emission balancing and budget derivation.

1.4.1. Conceptual Contributions (Chapter 3.1)

The first contribution concerns the conceptual interpretation of PED definitions. Chapter 3.1 establishes this thesis' understanding of PED definitions as instruments that translate system-level climate and energy objectives into district-scale planning requirements. This framing clarifies the roles PED definitions are expected to fulfil and provides a conceptual basis for quantitative operationalization. Concrete conceptual novelties include:

- The framing of PED definitions as designed instruments with explicit functional requirements, rather than arbitrary boundary-setting.
- The formulation of a multi-level responsibility model that distinguishes European principles, national quantitative target setting, and local project implementation.
- The explicit treatment of contextual diversity as a design input, providing a conceptual justification for context-dependent target formulation in later chapters

These conceptual contributions define the design space within which the quantitative PED definition is constructed in Chapter 3.2.

1.4.2. Methodological Contributions (Chapters 3.2 and 3.3)

Building on the conceptual framing, the thesis delivers its central methodological contributions through the construction and operationalization of a quantitative PED definition. Chapters 3.2 and 3.3 translate abstract PED ambitions into assessable and comparable performance criteria aligned with national decarbonization pathways. Concrete methodological novelties include:

- The formulation of a quantitative PED definition as a sufficiency criterion, expressed through primary energy and life-cycle greenhouse gas balances anchored in national climate targets (Chapter 3.2).
- The integration of energy flexibility into primary energy accounting, allowing temporal system effects to be reflected within a single balance metric (Section 3.2.4).
- The introduction of density-dependent target references that correct for structural differences in renewable energy potential across urban forms without modifying physical energy flows (Section 3.2.6).
- The inclusion of everyday motorized mobility via bounded system extensions based on reference mobility budgets, enabling transparent attribution of responsibility (Section 3.2.7).
- The derivation and application of floor-area-specific, Paris-compatible cumulative greenhouse gas budgets that combine operational and embodied emissions (Section 3.2.8).
- The development of a definition-consistent PED simulation and assessment framework designed for early planning phases, combining hourly temporal resolution with reduced spatial detail (Chapter 3.3).

1.4.3. Applied Contributions (Chapter 3.4)

The third contribution concerns the translation of the quantitative PED framework into institutional and empirical contexts**.** Chapter 3.4 demonstrates how the same definition and assessment logic can support governance, certification, and analytical exploration. Concrete institutional and applied novelties include:

- The analytical application of the PED framework to heterogeneous European district case studies**,** providing empirical insight into feasibility limits, boundary effects, and context sensitivity.
- The implementation of the quantitative PED framework in a national declaration system, demonstrating its suitability for policy and certification contexts. The explicit alignment of modelling logic with declaration requirements such as auditability, reproducibility, and transparency.

- The development of a scope-based declaration structure that distinguishes operational PEDs, mobility-inclusive PEDs, and climate-neutral PEDs while maintaining a single underlying definition.

These contributions aim to show that the PED framework functions not only as an analytical tool, but as a stable and transferable instrument for planning and governance.

1.5. Thesis Organization

This thesis is organized in four chapters. Following the introduction, chapter 2 details the most important background information. Chapter 3 presents the key contributions of this thesis. Chapter 4 presents conclusions and examines perspectives of future work.

Chapter 2 establishes the problem space addressed by the thesis. It reviews the current state of research, policy documents, assessment frameworks, and applied PED projects, with the explicit aim of identifying inconsistencies, gaps, and internal contradictions in existing PED concepts.

Chapter 3.1 provides the conceptual reframing of PED definitions that underpins the thesis.

Chapter 3.2 constitutes the normative core of the thesis. Building on the conceptual insights of Chapter 3.1, it constructs a quantitative, context-aware PED definition that is anchored in national climate neutrality pathways. The chapter formalizes the requirements a PED must satisfy to be nationally fair, comparable across contexts, and compatible with absolute climate constraints.

Chapter 3.3 addresses the operationalization of the quantitative PED definition. It develops and justifies the modelling and simulation framework.

Chapter 3.4 demonstrates the empirical and institutional use of the PED framework. It distinguishes between two complementary modes of application: a normative mode, in which the framework is embedded in a declaration or certification context with fixed rules and responsibilities, and an analytical mode, in which the same framework is applied to empirical case studies for exploratory analysis and stress-testing.

2 Background and State of the Art

2.1. Climate-neutral urban development

Urban climate mitigation strategies increasingly converge on climate neutrality as a long-term objective, driven by European and national policy frameworks (Bundesministerium für Klimaschutz, 2023; EU Performance of Buildings Directive (EPBD) 2024/1275, 2024). While early approaches focused predominantly on individual buildings or city-wide balances (Aelenei & Gonçalves, 2014; Atanasiu et al., 2011; Becchio et al., 2016), recent research and policy practice have identified the district scale as a critical intermediary level for implementation (Becchio et al., 2018; Esfandi et al., 2024). Districts enable the aggregation of buildings, infrastructures, and users while remaining small enough to reflect concrete spatial, social, and governance contexts.

2.1.1. Increasing focus on the district level

At the district scale, interactions between buildings, energy systems, mobility, and public space become visible and can be deliberately designed. This scale enables shared energy infrastructures, local renewable generation, coordinated renovation strategies, and pooled mobility and storage solutions that are rarely feasible at the level of individual buildings, while avoiding the abstraction of city-wide modelling. Through aggregation, district-scale systems achieve higher efficiency and flexibility, making districts a preferred testing ground for climate-neutral and climate-positive solutions, particularly in dense urban areas and large renovation projects.

Beyond technical advantages, the growing relevance of districts reflects institutional and planning realities. Districts align well with existing planning instruments, development processes, and neighbourhood-based governance structures, allowing energy and sustainability considerations to be

integrated earlier and more consistently into the planning process. This early integration supports coherent long-term strategies across multiple buildings and projects rather than fragmented, case-by-case optimization.

The district perspective also enables a more comprehensive treatment of qualitative development goals. Issues such as the design and continuity of green and open spaces, social inclusion, quality of life, and local organization are more effectively addressed when multiple buildings and functions are planned in a coordinated manner (Šimkovicová et al., 2024). In addition, the deliberate mixing of residential, commercial, and service uses creates complementary demand profiles, improving both economic performance and the utilization of locally generated renewable energy.

District-scale development allows for more ambitious sustainability targets through holistic optimization. Performance differences between buildings or subsystems can be balanced at the district level, enabling trade-offs and synergies that are inaccessible in isolated building-based approaches. This flexibility supports integrated environmental, social, and economic objectives and establishes the district as a key operational unit for sustainable urban development.

2.1.2. Energy and GHG targets of the built environment are insufficient or insufficiently specified

Energy and greenhouse gas (GHG) reduction targets for the built environment are predominantly defined at national and municipal levels and embedded in long-term climate neutrality or decarbonization strategies. However, a consistent finding across the literature is that these targets are unlikely to be achieved under current business-as-usual trajectories (IEA, 2021, 2024b, 2024a). While headline objectives such as climate neutrality by mid-century or fossil-free heating are clearly articulated, they are only partially translated into binding, operational requirements at the level of building codes, refurbishment obligations, and district-scale implementation mechanisms (Steininger et al., 2020).

Quantitatively, national decarbonization pathways typically assume annual renovation rates of 2–3% combined with deep retrofit levels (Amann et al., 2020, 2021), alongside a rapid phase-out of fossil heating systems and large-scale electrification of heat supply. In practice, observed renovation rates in most European countries remain closer to 1% or below (Directorate-General for Energy (European Commission) et al., 2019), often dominated by shallow or component-level refurbishments that deliver only limited long-term emission reductions. As a result, compliance with existing building regulations does not necessarily imply consistency with national climate targets, revealing a structural gap between policy ambition and regulatory implementation.

This gap is evident at the municipal level as well: The City of Vienna for example, has set the strategic objective of phasing out gas-based heating systems by 2040 (*Wiener Klima-Fahrplan - Unser Weg zur klimagerechten Stadt*, 2022). While this goal provides a clear political direction, it is not uniformly enforced through binding building codes, and current refurbishment and system replacement rates (1.6% by 2025 according to (IIBW & Umweltbundesamt, 2026)) are insufficient to transition the entire building stock within the remaining timeframe. Similar patterns can be observed in other European cities, where long-term targets coexist with regulatory frameworks that still permit fossil-based systems or do not mandate renovation depth aligned with decarbonization pathways, a problem now addressed by the latest EU Performance of Buildings Directive (EPBD) which mandates Member States to submit National Building Renovation Plans (NBRPs) to decarbonize the building stock by 2050 (EU Performance of Buildings Directive (EPBD) 2024/1275, 2024).

At the national scale, decarbonization plans implicitly rely on substantial demand reductions, high shares of renewable energy, and coordinated system transformation across buildings, energy networks, and mobility (Krutzler et al., 2023; Zelger et al., 2025). Yet these assumptions are rarely decomposed into **explicit, enforceable performance requirements** at the building or district level, leaving open questions of responsibility, sufficiency, and sequencing. This lack of specification makes it difficult to assess whether individual projects or developments sufficiently contribute to collective targets.

Against this backdrop, district-scale approaches such as PEDs are increasingly discussed as potential intermediaries between abstract climate targets and building-level regulation. By aggregating buildings, infrastructures, and energy services, districts offer a scale at which national decarbonization assumptions – regarding renovation depth, system integration, and renewable supply – can be operationalized and assessed more explicitly. However, this potential can only be realized if district concepts are underpinned by **quantitative, sufficiency-oriented requirements** that translate high-level targets into concrete performance criteria, rather than remaining aspirational labels.

2.2. Positive Energy Districts

Positive Energy Districts can be characterized by the following framework definition of the Joint Programme Initiative Urban Europe (JPI Urban Europe / SET Plan Action 3.2, 2020): “Positive Energy Districts are energy-efficient and energy-flexible urban areas or groups of connected buildings which produce net zero greenhouse gas emissions and actively manage an annual local or regional surplus production of renewable energy. They require integration of different systems and

infrastructures and interaction between buildings, the users and the regional energy, mobility and ICT systems, while securing the energy supply and a good life for all in line with social, economic and environmental sustainability."

Although often associated with the notion of a positive annual energy balance, most sources emphasize that this balance is not in itself the defining characteristic of a Positive Energy District. Instead, PEDs are framed as integrated urban energy systems in which efficiency, flexibility, system interaction, and long-term climate objectives are at least equally central (Albert-Seifried et al., 2022; Bossi et al., 2020; Derkenbaeva et al., 2022; Kozlowska et al., 2024; Lindholm et al., 2021)

One of the strengths of the PED concept is its clearly defined main components: PEDs are based on three conceptual pillars: Energy Efficiency, Local renewable energy supply and Energy flexibility (Kozlowska et al., 2024; Shnapp et al., 2020) as shown in Figure 2, which should be optimized with the target of climate neutrality and energy surplus, taking also into account quality of life, inclusiveness and sustainability (Šimkovicová et al., 2024).

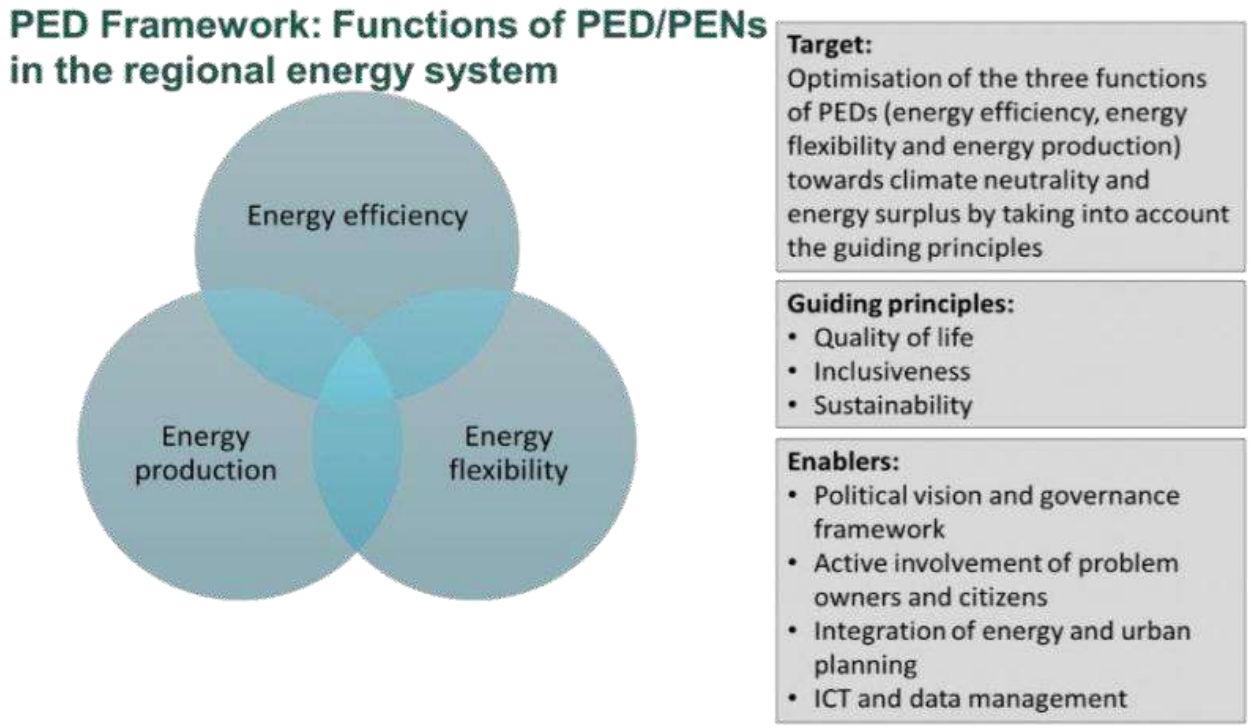


Figure 2. PED framework definition (JPI Urban Europe / SET Plan Action 3.2, 2020)

The emergence of the PED concept is the consequent continuation of building energy research of the past decades: The predominant paradigm was energy efficiency (Aelenei & Gonçalves, 2014; Atanasiu et al., 2011; Attia et al., 2013; Camarasa et al., 2019; Schneider et al., 2019a; Shnapp et al., 2020). The trend however changed around 2015, with research focus shifting away from building energy efficiency exclusively (Cristino et al., 2022). Instead, it widened to other aspects of the

building energy system, to decrease all negative impacts of the built environment and make it a "net positive" (Attia, 2016).

In particular, the concept of Net Zero Energy Buildings (NZEBs) has been coined and thoroughly studied in the past decade (Aelenei & Gonçalves, 2014; Ala Hasan & Francesco Reda, 2022; Berggren et al., 2013; Bourrelle et al., 2013; Federico Noris et al., 2013; Garde et al., 2014; Jensen et al., 2017a; Sartori et al., 2012). Already back then, the adjacency and distinction between concepts was grounds for extensive discussion, e.g., to that of "Nearly Zero Energy Buildings" (Atanasiu et al., 2011; Becchio et al., 2016; BPIE, 2014; D'Agostino & Mazzarella, 2019; Ferrari & Beccali, 2017; Ferreira et al., 2016; Iturriaga et al., 2018; Köhler et al., 2019; Kolokotsa et al., 2022; Saheb et al., 2019; Sleiman et al., 2017). In parallel, more and more researchers called to question the notion of "net-zero" itself in favour of a more encompassing and appealing "net-positive", reinterpreting the building as a potentially positive contributor rather than just having close to zero negative impacts (Attia, 2016; Attia et al., 2013; Cole, 2015; Cole & Fedoruk, 2015).

Earlier conceptual predecessors of PEDs can also be found in approaches such as the Passive House standard, which established internationally influential models of holistic design and highly efficient building operation (Lackner et al., 2025).

Positive Energy Districts are increasingly framed as a promising paradigm for decarbonizing cities, supporting electricity system balancing, and improving urban quality of life (Gabaldón Moreno et al., 2021). The European Union states that the ambition of PEDs is to **"**go well beyond what is already requested in the Energy Performance of Buildings Directive" (European Commission, 2019).

At the same time, substantial uncertainty remains regarding how this paradigm can be operationalized in concrete European and national contexts, where regulatory frameworks, energy systems, and planning responsibilities differ markedly.

As PED discourse has evolved, additional dimensions have gained prominence beyond local renewable generation alone, including energy flexibility, demand-side management, and the consideration of embodied and scope-3 emissions. This shift reflects a growing recognition of systemic effects such as rebound phenomena (Wei & Liu, 2017) and the limits of purely supply-oriented strategies. Consequently, while local renewable energy remains a central pillar of PEDs, its role and sufficiency cannot be assessed in isolation, raising open questions about scale, system boundaries, and the interaction between local generation, demand reduction, and broader energy system dynamics.

2.2.1. Local renewable energy supply

Local renewable energy supply gains particular relevance at the district scale due to the emphasis on geographically bounded energy generation. In most definitions, this implies that renewable energy is produced primarily within the physical boundaries of the district. Alternative approaches adopt less spatially constrained definitions, such as virtual power plants or the use of green certificates (Albert-Seifried et al., 2022; Semprini et al., 2025). However, the reduced traceability and accountability associated with these approaches increase the risk of greenwashing and can undermine confidence in their contribution to systemic sufficiency.

In practice, strictly on-site renewable energy generation is largely limited to photovoltaic and solar thermal systems. While certain districts benefit from exceptional conditions that enable the use of unconventional energy sources or storage solutions, such as thermal energy storage in repurposed oil caverns or the availability of large open areas suitable for wind power, these cases are atypical. In most urban environments, and even in rural settlement areas, wind energy plays only a marginal role, as there is a gap in methodology as to how to include "offsite" renewable energy supply (Cirrincione et al., 2023; Leone et al., 2022).

2.2.2. Energy flexibility

Energy flexibility is the potential of an energy system to dynamically change its energy demand in terms of time, duration, and/or magnitude. It is typically categorized into load curtailment, load shifting, valley filling, peak shaving (Jensen et al., 2017b).

Energy flexibility plays an important role in the decarbonization of the energy system in the upcoming decades as it has the potential to significantly reduce the amount of energy storage required (Koolen, Derck et al., 2023), promising a substantial decrease in system costs as well as costs for flexibility providers (Semmelmann et al., 2026).

The assessment of energy flexibility in PEDs is recognized as a crucial yet unresolved issue in the field (Gondeck et al., 2024). Industry practitioners agree on the necessity of calculating dynamic matching across all energy uses, encompassing even household appliances. This highlights the importance of comprehensive energy management strategies in PEDs (Ala-Juusela et al., 2021). Furthermore, there is an identified need for in-depth research exploring the synergistic potential of enhanced demand flexibility, achieved through building hull improvements and advanced ventilation heat recovery, in conjunction with predictive controls. These controls would optimally leverage flexibility, informed by forecasts encompassing weather, usage patterns, and renewable energy generation, potentially extending up to a week (Erba & Pagliano, 2021).

Although established metrics such as the Grid Support Coefficient (Klein, Killinger, et al., 2016a; Klein, Langner, et al., 2016) or the Smart Readiness Indicator (SRI) (Knotzer et al., 2020; Märzinger & Österreicher, 2019; Verbeke S et al., 2018) allow for detailed assessments of grid serviceability, they significantly increase modelling and verification complexity.

As summarized in (Lackner et al., 2025), one way of dealing with Energy Flexibility is through demand response, which is thoroughly studied within Multi-Energy Systems (Albadi & El-Saadany, 2008; Erba & Pagliano, 2021; F. Lopes & Algarvio, 2018; R. A. Lopes et al., 2016; Paterakis et al., 2017). Strategies and methods differ considerably in their temporal scope, ranging from the long-term planning of renewable energy systems (RES) and storage in smart energy hubs (Honarmand & Rashid, 2022), to day-ahead and hourly scheduling approaches (Wu et al., 2013; Zhao et al., 2019). They also address a wide range of operational questions, including the maintenance of thermal comfort (Da Fonseca et al., 2021) and the practical integration of demand response strategies (Bahrami & Sheikhi, 2015; Vahid-Ghavidel et al., 2021). Moreover, the widespread deployment of demand-side response could substantially support the integration of volatile renewable energy (Kiviluoma et al., 2022; Rasku & Kiviluoma, 2019).

Current flexibility studies on PEDs mostly use cost savings as indicators (Marotta, Guarino, Cellura, et al., 2021), and to a lesser extent carbon emissions (Marotta, Guarino, Cellura, et al., 2021).

In this context, thermally activated building core as storage (TABS) is a particularly promising option to efficiently use increasingly volatile renewables such as wind and solar power with the use of heat pumps, providing energy flexibility by the district (Klein, Killinger, et al., 2016b). A key enabler is the user's willingness to participate or at least tolerate such flexible control and potential indoor temperature fluctuations (Bell et al., 2021), as illustrated in Figure 3.

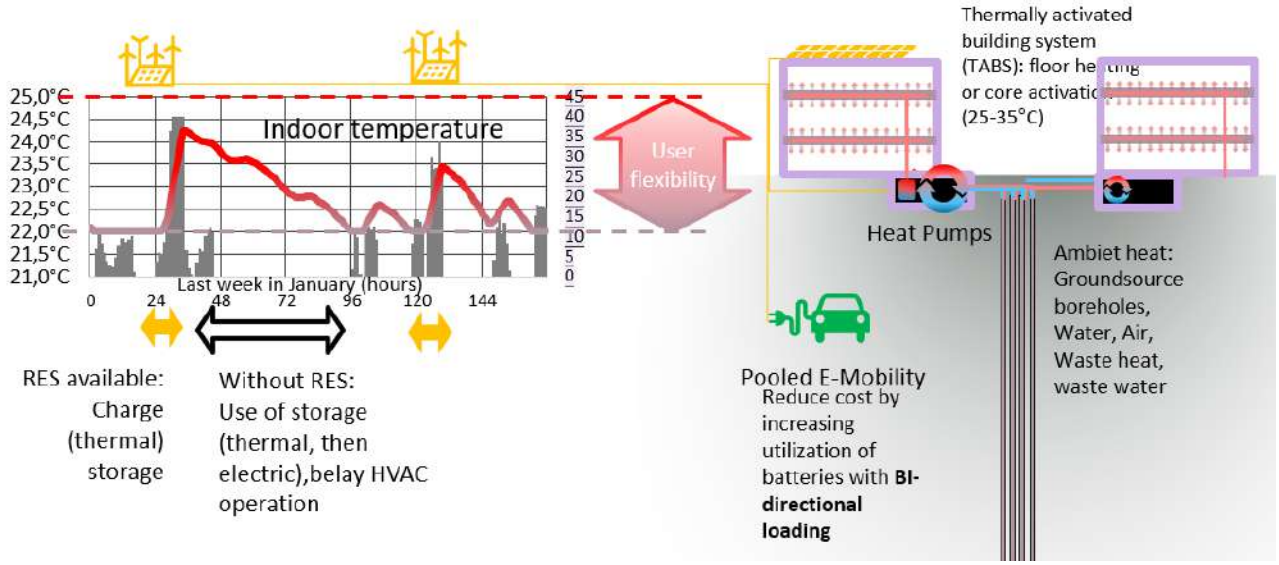


Figure 3. Schematic visualization of flexibility through thermally activated building mass and user flexibility

2.2.3. Green infrastructure

Urban green infrastructure addresses both the quantity and the quality of green spaces within districts (Reinwald et al., 2021). These spaces provide multiple benefits for residents as well as for flora and fauna, including improvements in outdoor comfort and quality of stay, air quality, biodiversity, rainwater retention, and humidity regulation (Kardan et al., 2015). In this sense, green infrastructure contributes not only to environmental performance but also to the multifunctionality and liveability of dense urban districts. This can be measured by a "green and open space factor", as shown in (Ring et al., 2021). The design and placement of greenery also matter, as tree location affects solar exposure and shading and can thereby reduce cooling loads, mitigate urban heat island effects, and improve climate resilience (Stojakovic et al., 2020).

A particularly relevant synergy arises in combination with photovoltaic systems, where evaporative cooling from vegetation can increase PV efficiency (Moren & Korjenic, 2017; Penaranda Moren & Korjenic, 2017).

2.2.4. Co-Benefits

Another important aspect in PED conceptualization and assessment is that of so-called "Co-benefits", such as increased climate resilience, user comfort and many more that should be actively screened and at least qualitatively assessed (Bisello, 2020). A similar perspective is adopted in the PEDRERA project, whose primary objective is to develop a district simulation model capable of generating and analysing reliable predictions of potential business scenarios for large-scale retrofit actions, while also assessing the overall co-benefits arising from the renovation of a cluster of buildings (Civiero et al., 2021).

Due to their heterogeneous, often qualitative nature, assessing co-benefits poses operational challenges, for which frameworks exist that area near maturity (Vandevyvere & Wyckmans, 2023).

2.2.5. Sufficiency

The long-term increase in residential floor area per inhabitant in Austria and Europe underscores the relevance of sufficiency in the built environment: in Austria, average floor area per flat increased from 87.6 m² in 1995 to 101.7 m² in 2024 (Statistik Austria, 2025), while across the EU floor space per person grew by more than 20% between 1990 and 2007 (Lavagna et al., 2018). Since growing floor area per capita is closely linked to higher operational and embodied emissions, it also contributes to rising individual carbon footprints and can offset gains achieved through technical efficiency improvements. This is why sufficiency is an important complement to efficiency: Technological measures need to be complemented by lifestyle and behavioural change (Abadi et al., 2025; Gifford

et al., 2011). In practice, this perspective can be translated into concrete interventions in building design and operation. For instance, higher cooling setpoints can maintain nearly equivalent indoor comfort while significantly reducing energy demand (Alaux et al., 2024; Morley, 2026). Likewise, simple enabling measures such as installing watertaps on balconies can facilitate the use of sprinklers for evaporative cooling on extreme days (Erba & Pagliano, 2021).

2.2.6. PEDs Today

Positive Energy Districts (PEDs) have moved from a largely conceptual policy target into a diversified European project landscape spanning (i) large multi-city demonstration programmes, (ii) method and tool development projects, and (iii) institutionalization efforts that aim to make PED planning reproducible, comparable, and scalable (Turci et al., 2022). Early consolidation occurred through JPI Urban Europe / SET-Plan Action 3.2 and Horizon 2020 Smart Cities and Communities "lighthouse" projects, which established PED-like demonstration sites and created much of today's practice-based evidence base. Typical examples repeatedly referenced in the PED literature and project mapping work include multi-site programs such as *MAKING-CITY*, *+CityxChange*, *POCITYF*, *ATELIER*, *SPARCS*, *Syn.ikia*, *RESPONSE* and related initiatives (as compiled in the PED-focused project landscape).

A practical indicator of maturity is the emergence of shared registries and harmonized descriptors. The PED-EU-NET mapping work provides the methodological foundation for an interoperable PED database and distinguishes PED-related implementations by case status and maturity. The PED-EU-NET database mapping has been used to structure the European portfolio into PED case studies, PED Labs, and PED-relevant cases, and to characterize development stages (planning, implementation, completed, operation), highlighting that the European PED landscape is dominated by projects still transitioning from planning to implementation rather than long-term operation (Turci et al., 2022).

Since 2022, the project landscape has further expanded under the Driving Urban Transitions (DUT) Partnership, which frames PED work as one of three transition pathways and organizes activity around key action areas such as integrated energy systems, energy markets/commercial viability, urban regeneration, and mission-oriented urban planning. The current DUT PED project portfolio (calls 2022–2024 visible in the DUT repository) shows a marked shift from "single demonstration districts" toward replication-oriented enablers, notably:

- Governance and justice (e.g., *JUST PEPP, PED-JUST, CO2PED, Coop4PEDs, PERSIST*),
- Digital twins and decision-support (e.g., *DigiTwins4PEDs*, *Making PEDs*, *Scale-UP*),

- Flexibility, grid interaction, and sector coupling (e.g., *FlexPED, FLEdge, V2G-Quests, LIGHT, LTDE–repBC*),
- Policy/market sandboxes and community platforms (e.g., *Pol4PED, OPEN4CEC, DECODE*).

Overall, this portfolio composition indicates that PEDs today are increasingly treated not only as place-based demonstrations, but as transferable planning, governance, and interoperability problems addressed by reusable methods, digital infrastructures, and institutional processes.

Recent review literature broadly supports this interpretation: systematic and comparative reviews report that a significant share of PED publications still rely on project-funded case studies, while pointing to persistent gaps in holistic integration (social/non-technical dimensions), comparability of assessment approaches, and the link between design/operation and governance requirements (Kozlowska et al., 2024; Sassenou et al., 2024; Yang et al., 2025). Across these reviews and Annex 83 synthesis outputs, recurring needs include clearer and more consistent assessment conventions, stronger treatment of system boundaries and responsibility allocation, and better integration of multi-vector operation and flexibility into evaluative practice.

In summary, the European PED landscape can be read as a progression from (1) demonstration-heavy lighthouse projects, to (2) structured registries and harmonization efforts (PED databases and taxonomies), and now to (3) scaling-oriented DUT projects that prioritize decision support, governance models, and replicable methods. This progression also explains why current PED practice remains heterogeneous: many "PED" projects are still in mid-maturity stages and are optimized for learning and replication rather than for long-term audited operation at scale (Semprini et al., 2025).

2.3. Related concepts

Several studies highlight strong overlaps among various district-scale urban sustainability concepts that are described using different terminology, as summarized by (Lackner et al., 2025): These concepts are often used interchangeably and share substantial similarities, to the extent that clear distinctions between them are frequently difficult to establish.

- "Zero Emission Neighbourhoods" (ZEN): (Lausselet et al., 2019; Pinel, 2020; Wiberg et al., 2019; Wiik et al., 2018; Wiik, Fufa, et al., 2019; Wiik, Lolli, et al., 2019) and for buildings (Fufa et al., 2016; Kristjansdottir Torhildur, 2019)
- "(Net and Nearly) Zero Energy Buildings" (Net ZEB, NZEB) (D'Agostino & Mazzarella, 2019; Hawila et al., 2022; Jensen et al., 2017a; Khan et al., 2017; Péan et al., 2017; Salom et al., 2014; Shiraishi & Ishibashi, 2019; Stein et al., 2014)
- "Net Zero-Energy District" (NZED) Turin (Becchio et al., 2018)

- "Zero Energy Communities" (Mittal et al., 2019)
- "Sustainable Positive Energy Neighbourhoods" (SPEN): (Salom et al., 2021a; Salom & Tamm, 2020)
- "Climate-friendly neighbourhoods" (Brozovsky et al., 2021).

Apart from these congruent concepts, there are other concepts (with great synergy with the PED concept, as argued in (Clerici Maestosi et al., 2021): Climate Neutral Cities, Circular Economies and 15-minute Cities.

2.3.1. Multi-Energy Systems

PEDs can be seen as a form of Multi-Energy Systems (MES). These systems integrate various energy sectors like electricity generation, building operation and mobility to improve synergies and performance within districts, cities, or regions. They outperform traditional, separate energy systems both operationally and in planning. MES, especially those with distributed generation of multiple energy sources, are essential for decarbonization, but they require complex modelling and tools. Additionally, defining performance metrics that consider costs and benefits across different criteria is challenging. (Mancarella, 2014) provides a comprehensive overview of models and assessment techniques for analysing MES, including concepts like energy hubs (Aien et al., 2016; Geidl et al., 2007; Lasemi et al., 2022; T. Liu et al., 2019; Mohammadi et al., 2017, 2018; Sadeghi et al., 2019), microgrids (Bessler, 2018; Diab et al., 2022; Riou et al., 2021), and virtual power plants (Hashemipour et al., 2021), across energy, environmental, and techno-economic criteria. This includes a number of new methods and algorithms to use AI and machine learning (C.-R. Chen et al., 2021; J. Chen et al., 2022; Sarhan et al., 2022; Zhong et al., 2022) and information theory (Javadi et al., 2017; Jordehi et al., 2021; Y. Liu et al., 2022; Mohammadi-Ivatloo et al., 2013; Najafi et al., 2019; Najafi-Ghalelou et al., 2018), novel dispatch and operation optimization models for combined heat and power plants (Ginidi et al., 2021; Lund, 2018; Najafi-Ghalelou et al., 2018), greenhouse operation (Lee et al., 2021), day-ahead scheduling of integrated energy systems (C.-R. Chen et al., 2021; J. Chen et al., 2022; Najafi et al., 2019; Wu et al., 2013; Zhao et al., 2019), hydropower (Song & Wei, 2022), propulsion maintenance and efficiency (Fitri et al., 2022; Kadarohman et al., 2021; Łosiewicz et al., 2022), hydro-connected floating PV RES and onshore wind power (Nyoni et al., 2021), hydrogen sourcing, MES dispatch and long-term storage (Farahani et al., 2020; Frischmuth, 2022; Kamal & Adouane, 2022; Löhr et al., 2022; O'Dwyer et al., 2022), power filters (Nikkhah Kashani et al., 2022), power-to-gas (Brennenstuhl et al., 2021; Feng et al., 2022) and other cogeneration methods of heat and power (García-Domínguez & Marcos, 2021; R. Liu et al., 2021; Z. Liu et al.,

2023). (Greiml et al., 2022) even analysed Austria as an MES with high spatial and temporal resolution. A number of other publications focused on various aspects of the Austrian energy system as an energy hub (Abart-Heriszt et al., 2020; Kranzl et al., 2019; Monsberger et al., 2020).

2.3.2. New European Bauhaus

The New European Bauhaus (NEB) is a European initiative that links the goals of the European Green Deal to the transformation of everyday living environments (European Union, 2020). It frames sustainability as a cultural and societal project based on three inseparable values: sustainability, aesthetics, and inclusion. In doing so, it positions the built environment as a central arena for climate action while also addressing spatial quality, social cohesion, and governance. Rather than prescribing technical targets, the NEB provides a normative and integrative framework for how transformation should be conceived and evaluated, an approach that has also been translated into national quality criteria catalogues (B. Feller, R. Halbartschlager, R. Hammer, S. Hofer, C. Ohrhallinger, R. Temel, 2024).

This distinguishes it from PEDs, which are techno-systemic constructs focused on district-scale energy and emission performance through quantitative indicators and defined system boundaries. While PEDs operationalize climate-neutral and climate-positive urban development through energy, flexibility, mobility, and infrastructure design, the NEB offers a broader value-based scope that addresses dimensions often left implicit in PED frameworks, such as architectural quality, cultural identity, affordability, inclusion, and participatory governance. Frameworks such as the NEB Impact Model further translate these values into flexible assessment structures that complement existing indicator systems (Vandevyvere & Wyckmans, 2023).

In this thesis, the NEB is therefore treated not as an alternative to PEDs, but as a complementary framing: PEDs provide the analytical and technical backbone, while the NEB situates them within broader societal goals of quality of life, inclusiveness, and spatial coherence.

2.4. Problems with the PED Definition

Despite its promise, the PED concept has well-documented weaknesses related to its formulation, and particularly its definition.

2.4.1. What is a “positive energy balance”?

One would assume that the defining characteristic of a PED, the “positive energy balance”, is relatively straightforward. However, the last years have shown that the gap between the *idea* of a

"positive energy balance" and how it can be interpreted in practice is so wide that it is hardly bridged systematically, but rather only for individual PED projects independently and very heterogeneously.

Why is that? Early works on NZEBs of Koutra and Sartori (Sartori et al., 2012) provided a useful framework for identifying the components of a balance: The system boundaries, the type of balance and the weighting system, all offering a great many choices:

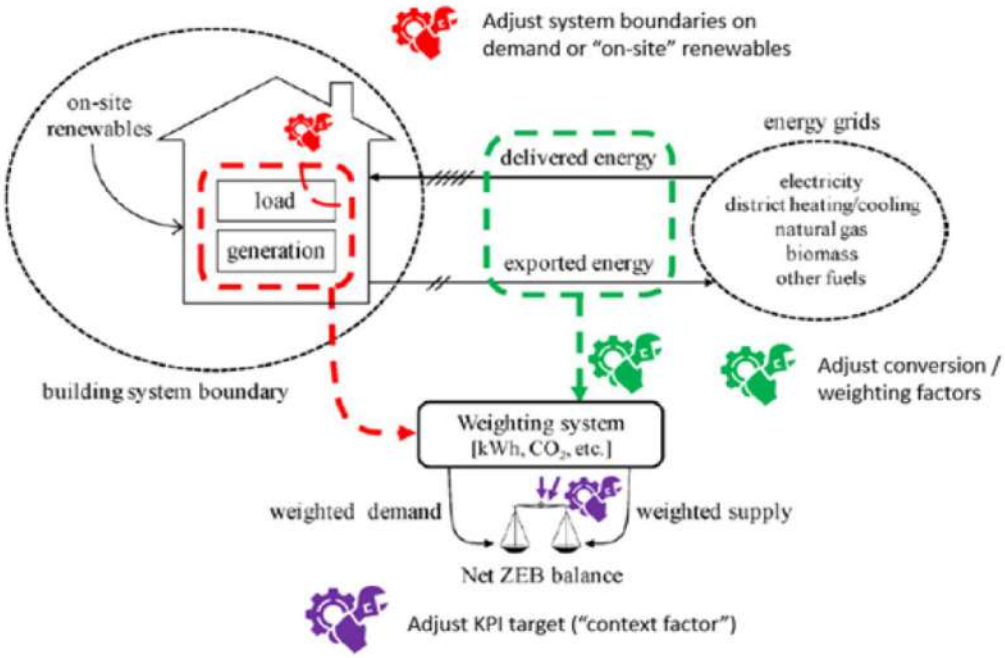


Figure 4. PED assessment balance components adapted from (Sartori et al., 2012)

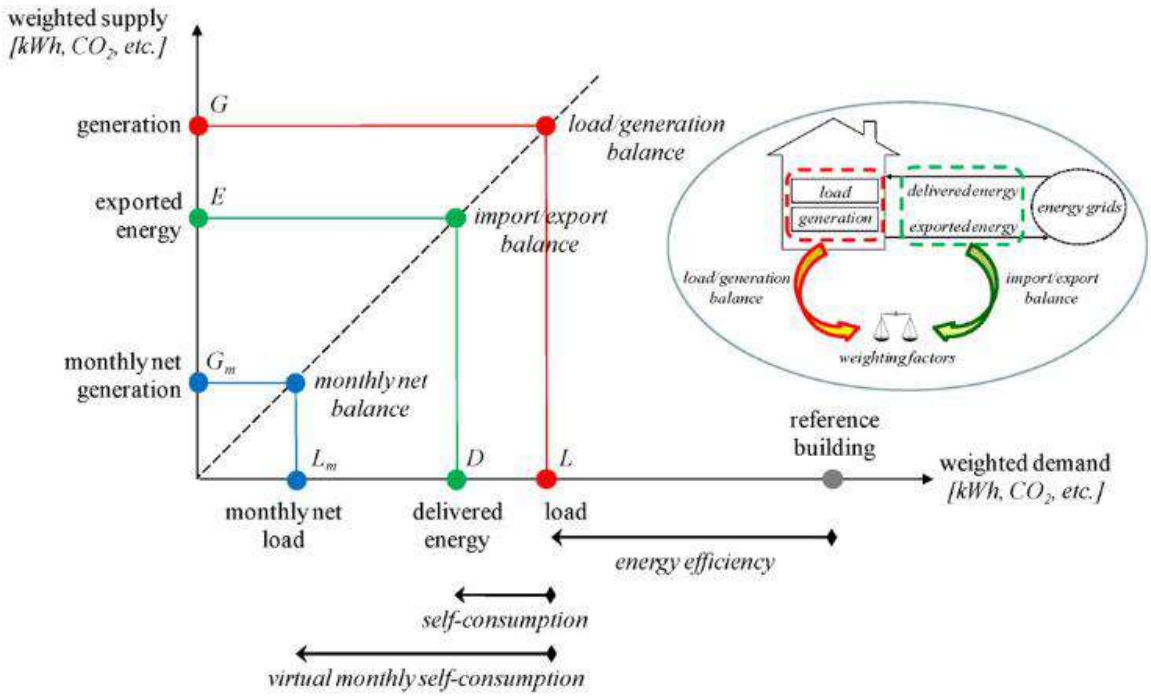


Figure 5. Illustration of different kinds of energy balances (Sartori et al., 2012)

In many contemporary works however, the balance target is implied by the 45°-line delineating net positive from net negative regime, as is the case in the figure above. This will be an important topic for definition construction in Chapter 3.2.

2.4.2. What even is balanced? And why?

Even though system boundaries are routinely drawn spatially around buildings and districts, this does not automatically imply that all energy demands occurring within are also covered in the balance: Historically, energy balancing focused on building HVAC operation, often omitting user plug-loads as "out-of-our-hands". Today, many actors agree that building plug loads alongside operation must be part of the energy demand of a district balance (Ala-Juusela et al., 2021).

But it does not stop there: What about streetlights, mobility demand and embodied energy of construction materials and even food consumption? Here lies a fundamental dilemma of energy and emission balancing: the more that is included, the greater the fidelity, but at the same time: the lower the achievability of balancing targets in general. This is illustrated in the following Figure 6.

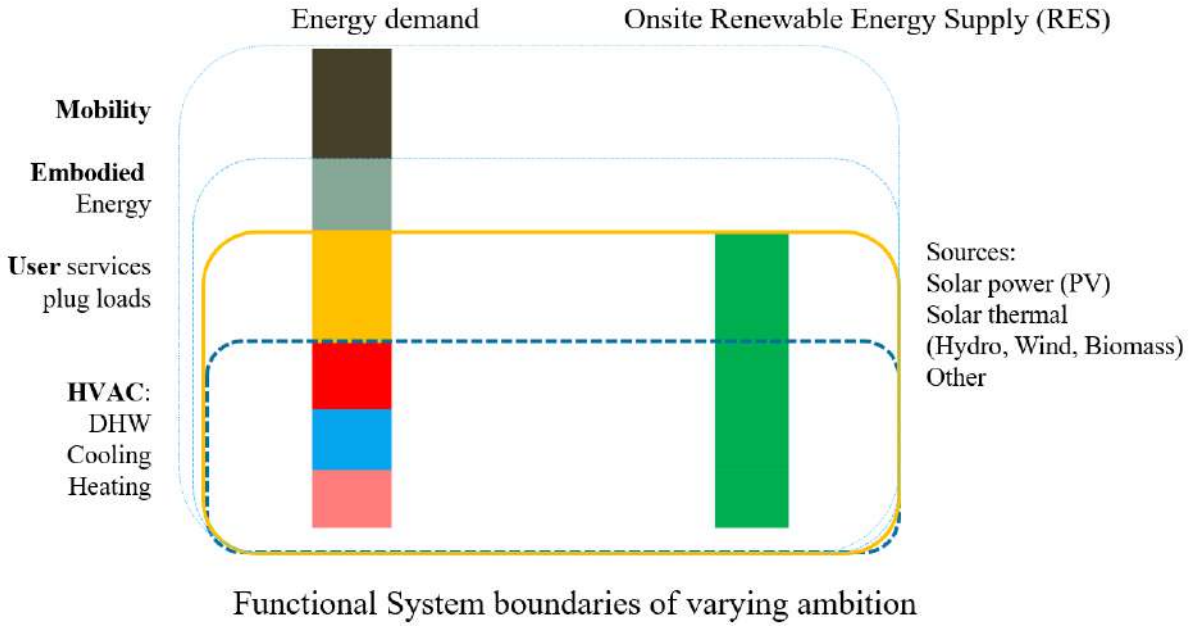


Figure 6. Energy balance illustration for different functional system boundaries

Mobility

Mobility to and within a district can represent a major part of district energy demands and may even exceed that of building operation, particularly where travel is still dominated by individual motorized modes such as fossil-fuelled cars, mopeds, and similar vehicles (Castillo-Calzadilla et al., 2021). Despite its relevance, mobility has often played a limited role in district-scale energy concepts, and even less so in balancing. This is likely due to the absence of methods or standards for their

integration, with notable exceptions remaining highly localized (Dahlen et al., 2020). Open questions include whether public transport energy use should be accounted for, which trips can reasonably be attributed to a district, how such trips can be calculated or estimated, and whether private long-distance travel, such as air transport, or logistics and goods transport should be considered. Given the lack of standardized answers and the fact that including mobility typically increases the difficulty of achieving ambitious energy targets, it is frequently excluded from assessments altogether.

Even when certain mobility aspects are included, district developers face limited direct influence over their decarbonization. Nevertheless, a range of measures is known to positively affect modal split and overall travel demand. These include high-quality public transport, reduced parking provision, sufficient bicycle parking and storage, shared mobility services, additional local public services, and a strong local supply of everyday goods and services. While the impacts of such measures are difficult to quantify precisely in terms of energy balancing and accounting, several approaches have been proposed to address this challenge.

Embodied energy and emissions: With increased energy efficiency, GHG emissions caused by building operation are more and more overshadowed by so-called "embodied" GHG emissions associated with building material production and processing (Röck et al., 2023). Life cycle assessments therefore consider all relevant energy and material flows, including construction and assembly, refurbishment and maintenance, and ultimately deconstruction, recycling, and disposal. (Berggren et al., 2013) demonstrated that as the operational energy demand of nearly zero-energy buildings is reduced, the relative contribution of embodied energy increases. For nearly zero-energy buildings, (Giordano et al., 2015) reported that embodied energy accounts for approximately 25–30% of the total annual energy demand over a 50-year assessment period. Early life cycle assessment results for zero-energy buildings further indicate that their ecological footprints are not necessarily lower than those of reference buildings, once the additional energy and material requirements associated with their construction are taken into account (Yi et al., 2017).

2.4.3. Lack of contextualization

It's quite obvious that the base PED concept favours certain locations, conditions and contexts, such as districts with low density (Neumann et al., 2021) or favourable climate conditions (Bruck et al., 2021; Bruck, Casamassima, et al., 2022) over others.

Climate

Ambient climate determines the balance and magnitude of heating and cooling loads. Climate change is driving heating demand down and cooling demand up in many European regions. Heating degree-

days decrease and cooling degree-days increase under warming scenarios, reducing heating requirements but increasing cooling loads over time. This alters the temporal structure of energy demand that PEDs must meet (Chidiac et al., 2022; Spinoni et al., 2018).

Technologies like borehole heat exchangers benefit from seasonally balanced heating and cooling loads because excess heat from cooling can regenerate the ground thermal resource; in predominantly heating climates without significant cooling demand, seasonal imbalance can limit performance and feasibility. By contrast, in climates where cooling loads are significant or growing, multi-energy systems that integrate cooling provision (e.g., heat pumps sized for both heating and cooling) offer co-benefits for comfort and marketability that were previously uncommon in heating-dominant regions (Bayer et al., 2014).

Cold, high-latitude climates typically have lower solar potential due to reduced irradiance, which can constrain the renewable energy supply side of PEDs. In warmer climates with higher irradiation, PV generation can better match increased cooling loads and support multi-energy solutions, though demand for dehumidification and its impact on electrical loads also becomes important.

Without explicit consideration of climate and its influence on heating vs. cooling demand, assessments of PED achievability may misrepresent technology suitability or performance across regions, since design strategies that work in one climate may be infeasible or suboptimal in another.

Building Density

Building density is a key determinant of PED performance because it directly links demand-side scaling with supply-side constraints. Urban density is commonly described through a set of metrics: floor area ratio (FAR), site Cover Ratio, and number of floors, which together characterize different urban form types, as illustrated in Figure 7.

From an energy perspective, final energy demand scales primarily with gross floor area and is therefore positively correlated with FAR. At the same time, demand per unit floor area decreases with increasing building compactness, as more compact forms reduce envelope losses (Natanian et al., 2019; Quan, 2016).

Conversely, on-site renewable generation potential scales with available plot or roof area, which is constrained as density increases. This creates a structural tension: higher-density developments concentrate demand while limiting surface area for local renewable supply. As a result, the achievable energy balance per unit floor area decreases with increasing density, making PED targets progressively harder to achieve in dense urban forms without access to external renewable sources or flexibility mechanisms.

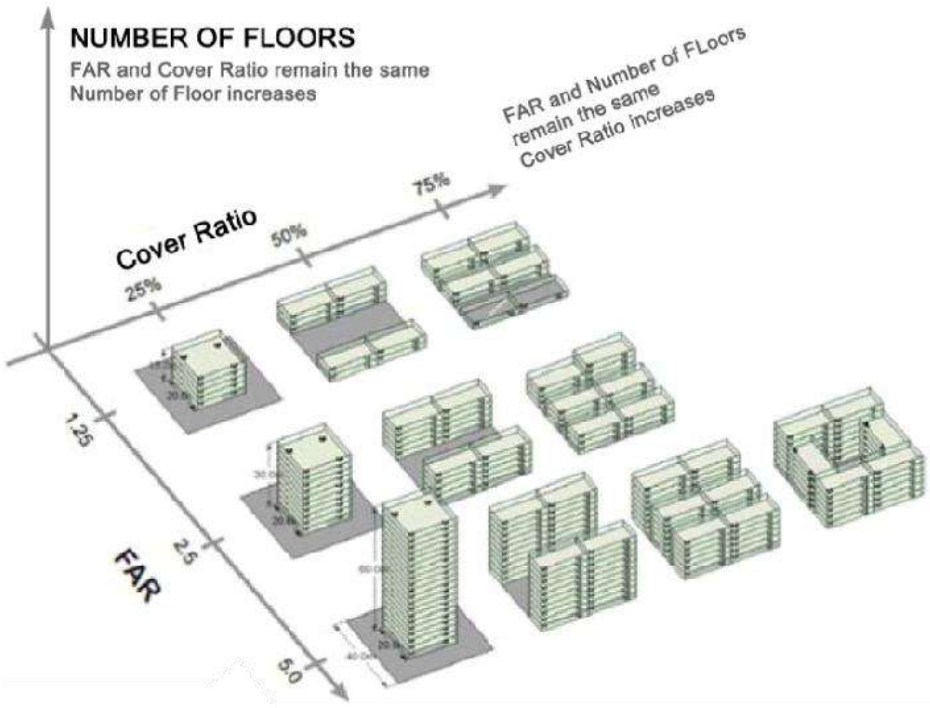


Figure 7. Illustration of the relationship between FAR, cover ratio and number of floors (MIT Density Atlas 2011) according to (Quan, 2016)

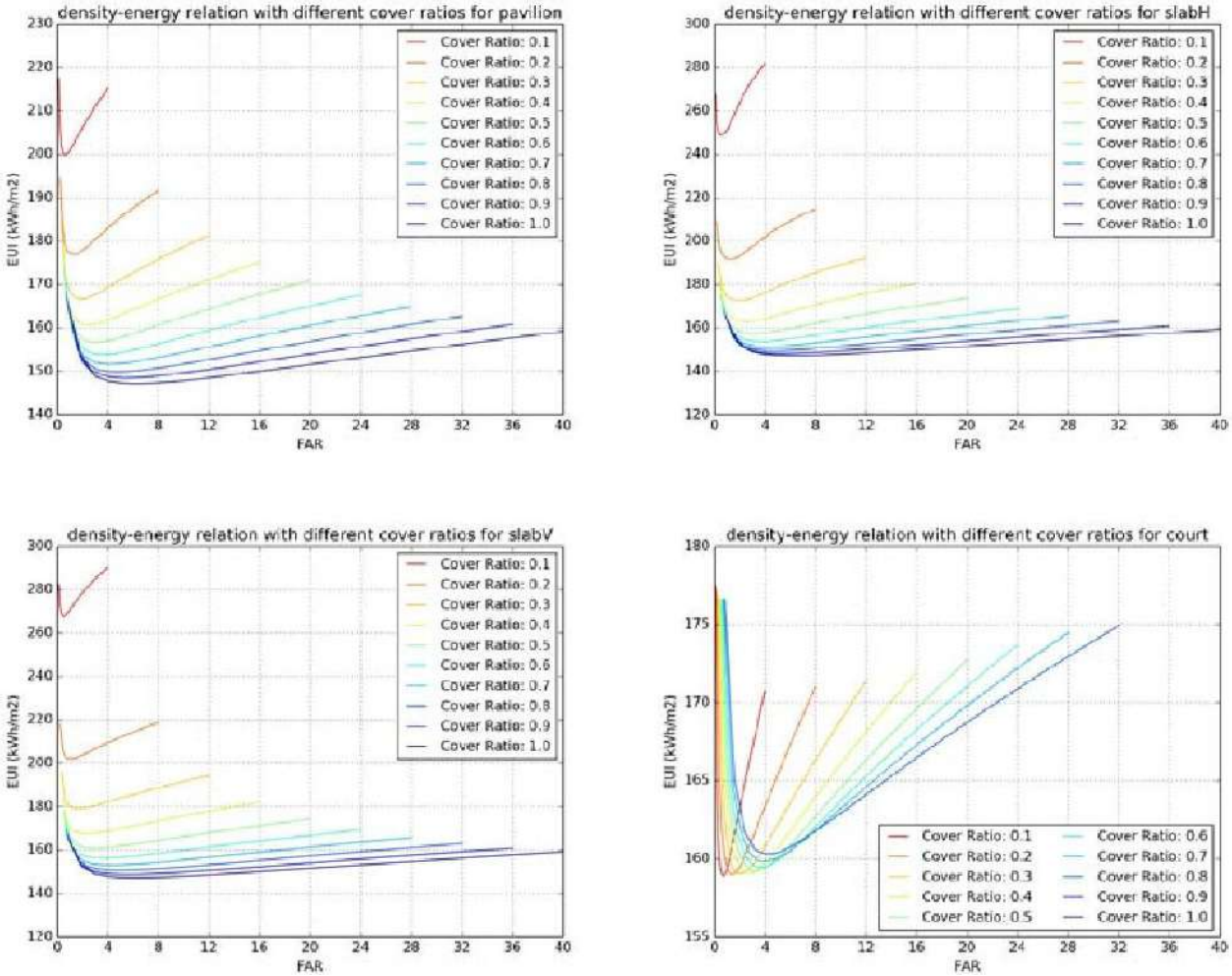


Figure 8. Energy use intensity as a function of density (FAR) according to (Quan, 2016)

As can be seen in Figure 8, the relationship between density (expressed as FAR) and energy use intensity (EUI) across urban form types exhibits a characteristic hockey-stick shape governed by compactness effects. At very low FAR, buildings are weakly compact, with a high envelope-to-floor-area ratio. In this regime, EUI decreases sharply as FAR increases: Small increases in density substantially improve compactness and reduce transmission losses. This steep 1/FAR-dominated segment explains the left-hand side of the curve. As FAR increases further, buildings become progressively more compact. Beyond a threshold (≈ FAR 3–4), additional densification yields diminishing compactness gains. EUI reaches a shallow minimum and then increases only gradually for very high FAR values, when compactness is again reduced by height, also reflecting secondary effects such as increased structural constraints, and reduced daylight access in very tall buildings.

In the European context, this distinction is critical: most buildings, including those in urban areas, operate at FAR < 4 (Mahdavi, 2022). Consequently, PED-relevant assessments are dominated by the steep, compactness-driven part of the curve, while the gradual EUI increase at very high densities is largely negligible in practice.

Compared to the corresponding solar energy production intensity curves, the EUI variation with FAR is relatively modest. While EUI changes are measurable, they are far less pronounced than for on-site solar potential, where density induces an almost hyperbolic decline. This confirms that density constrains PEDs primarily through renewable supply limitations rather than through disproportionate increases in operational energy demand:

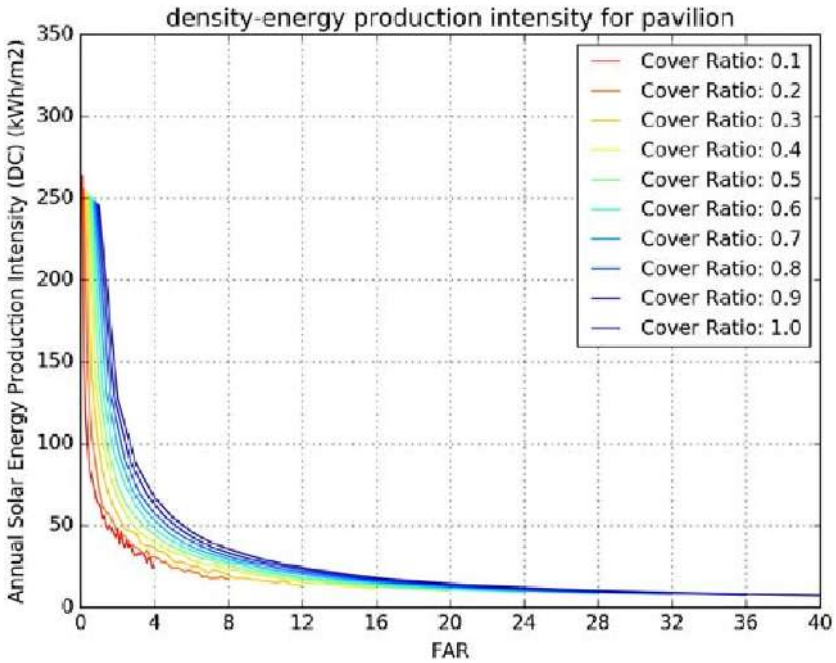


Figure 9. Annual on-site solar energy production intensity for different FAR and CR regimes according to (Quan, 2016)

(Resch et al., 2016) report that, for typical urban building heights, operational heating energy intensity decreases approximately inversely with the number of above-ground storeys (~1/x), due to the declining envelope-to-floor-area ratio as buildings become taller. This effect is strongest at low to medium heights and gradually levels off as additional storeys provide diminishing marginal reductions in energy intensity.

An analysis of the energy performance of 141 best-practice buildings and districts reported by (Fellner et al., 2018) identifies a consistent correlation that does not depend on project location, geometry, or use type. The primary energy balance (PEB) is primarily associated with (i) building age, (ii) the share of renewable energy in the energy system, and (iii) floor space index. The highest PEB values occur in projects combining state-of-the-art building envelopes with renewable energy supply systems (Schneider et al., 2019b).

These findings highlight a central problem for quantitative PED definitions: Uniform PED targets implicitly reward low-density development and penalize compact urban forms, contradicting broader sustainability objectives related to land use, transport demand, and infrastructure efficiency. Without explicit density consideration, PED definitions risk

- discouraging urban densification,
- favouring greenfield development,
- and misrepresenting the quality of district-level decarbonization efforts.

Building Vintage

Several studies emphasize that achieving energy-positive performance is particularly challenging in historic districts (Gouveia et al., 2021). This highlights the need for strategies tailored to existing urban structures, including the combination of revitalization and greening measures as well as stakeholder involvement, especially of landowners and citizens (Bossi et al., 2020).

Another approach to contextualization has been proposed through GIS-based comparability methods that combine heat demand, floor space index, and residential share within raster-based spatial analyses (Bruck, Casamassima, et al., 2022).

Without consideration of these contextual factors, concepts tend to be too simplistic and fail to consider the inherent real-life PED environments (Derkenbaeva et al., 2022). Even though PED concepts target dense urban areas and the existing building stock in particular, energy balance assessment results of these projects indicate that in these environments, districts can only partially cover their energy demand for HVAC and user electricity (Blumberga et al., 2020). Several studies

have concluded that in dense urban environments, a net positive energy balance is mostly infeasible (Neumann et al., 2022; Schneider et al., 2019a).

Some studies even suggest locating the 100 PEDs promised by the EU to be underway by 2025 simply in these conditions (Bruck, Díaz Ruano, et al., 2022). (Lindholm et al., 2021) found in their survey, that "it can be concluded that the techno-economic potential of different renewable energy and energy storage technologies varies between different EU countries and cities. […] The economic viability of wind power is, for instance, greater in regions close to the Northern Atlantic than in the heart of Central Europe".

This ties back into the question of PED definition, and when a district can claim to be a "PED": Many studies have highlighted the need for further clarification of the definition of Positive Energy Districts, especially taking into account the specific context of the district (Albert-Seifried et al., 2022; Guarino et al., 2023; Steemers et al., 2022). Further work on the PED definition is needed to take this into account (Williams et al., 2022).

(Gondeck et al., 2024) concludes that for these reasons, many researchers argue for the development of a more holistic PED definition and assessment methodology that enables different types of districts to qualify as PEDs, regardless of their specific boundary conditions (Brozovsky et al., 2021; Sareen et al., 2022; Vandevyvere et al., 2022).

2.4.4. How to assess energy flexibility?

Similarly, many actors in the field consider the assessment of energy flexibility in the context of a PED an important, yet unresolved question: Assessing positive energy performance solely on an annual balance risks obscuring temporal mismatches between local generation and demand, potentially imposing additional strain on upstream grid levels that only becomes visible when hourly (or finer) dynamics are explicitly considered (Casamassima et al., 2022; Gondeck et al., 2024). (Ala-Juusela et al., 2021) finds that when asking practitioners in the field "Almost all [of 40] respondents thought that dynamic matching should be calculated for all uses, including appliances".

According to (Lackner et al., 2025), one way of dealing with Energy Flexibility is with the well-established field of demand response, in particular in context of MES, which are well reviewed and summarized (Albadi & El-Saadany, 2008; F. Lopes & Algarvio, 2018; R. A. Lopes et al., 2016; Paterakis et al., 2017; Wang et al., 2017). Demand response strategies and methods vary in their timeframe from long-term planning of RES and storage in smart energy hubs (Honarmand & Rashid, 2022), to day-ahead and hourly scheduling (Najafi et al., 2019; Wu et al., 2013; Zhao et al., 2019),

and cover topics such as maintaining thermal comfort (Da Fonseca et al., 2021) and how to integrate the demand response operation in practice (Bahrami & Sheikhi, 2015; Vahid-Ghavidel et al., 2021).

Further, the increasing integration of volatile renewable energy (VRE) could also significantly benefit from widespread demand-side response (Kiviluoma et al., 2022; Rasku & Kiviluoma, 2019).

The outcome of the PED flexibility optimization depends greatly on the minimization objective (Backe et al., 2022), which is why a standardized assessment of PED flexibility would be of utility.

2.4.5. Tension with district stakeholder needs

Stakeholders with diverging goals and needs for a definition lead to very different definitions for individual exploration and implementation projects. Most of them focus on feasibility of a certain district. The resulting definition interpretations and operationalizations reflect more than anything the focus and context of the attached project site. This might not constitute a problem – each project utilizes the PED concept to suit their needs. However, this might hinder the deployment of the PED concept in a more standardized way of a mainstream development. Individual project definitions are mutually incompatible, meaning a PED will most likely not comply with the definition of a different project and vice versa. If the PED concept is to be used as a mainstream vehicle towards sustainable, climate-neutral buildings and cities, individual district efforts need to be comparable along a standardized definition. This is also the prerequisite for a PED certification, which is of great interest to developers and municipalities.

2.4.6. A lack of standardization

All the above lead to the fact that PEDs are a nice concept, but do not upscale, which would require standardization. When it comes to standardizing PEDs in the form of a label or certification to be used in city and district planning and development, there are several challenges addressed in literature:

- Which Key performance indicators (KPI) should be included in the assessment of the concept, and with which targets and why (Angelakoglou et al., 2020)?
- Quantifying and comparing key aspects of PEDs remains infeasible: “There is a lack of consistency between PED and similar definitions and concepts that often causes misinterpretations.“ (Derkenbaeva et al., 2022; Guarino et al., 2022)
- „Many studies […] address technical, economic, social aspects of PED, but very limited studies are found in characterizing PED.“ (Zhang et al., 2021)
- “A shared positive energy building definition is needed in order to enable the design, assessment, and documentation of positive energy buildings” (Ala-Juusela et al., 2021)

- "The PED definition is still in a conceptualization phase, and further research is therefore needed in order to initiate a discussion on a societal level" (Lindholm et al., 2021)

A review of existing assessment schemes and certifications shows a heavy focus on quantification, but relatively low targets (Koutra et al., 2018). Further harmonization of PED assessment methodologies is needed with reference to modelling assumptions and methodological choices (Marotta, Guarino, Longo, et al., 2021).

As of 2026, there are 114 PED-related projects mapped on the PED database (COST & PED-EU-NET, 2026). The lack of comparability is evident when analysing the only energy-related metric – energy demand: It is compiled only in absolute terms, which is either not reported or not comparable across districts of different sizes.

2.4.7. Lack of connection to overarching goals

Ultimately, the PED concept is normatively aligned with overarching climate-neutrality objectives, yet it remains weakly connected to these goals at an operational and system level. This disconnect is reflected in a frequently raised argument against further refining PED definitions: If electricity supply is expected to become largely or fully renewable by 2040, the achievement of a local positive energy balance at district scale is often portrayed as secondary or even irrelevant.

Such reasoning, however, neglects the central role of energy demand reduction as a prerequisite for the cost-optimality of national and European decarbonization pathways (IEA, 2021; Terés-Zubiaga et al., 2020). Even under fully renewable supply assumptions, reducing final energy demand remains critical for limiting infrastructure expansion, system costs, and peak capacity requirements (van Zuijlen et al., 2019; Victoria et al., 2020; Weidner & Guillén-Gosálbez, 2023). A PED definition that focuses primarily on surplus generation therefore risks obscuring the systemic value of efficiency and sufficiency measures.

More fundamentally, the PED concept lacks a clear methodological integration of the surrounding energy system as contextual reference. While a positive annual energy balance may serve as an effective benchmark for lighthouse and demonstration projects, its necessity and relevance for most districts, particularly those constrained by density, heritage, or limited on-site renewable potential, remain unresolved. This raises open questions regarding which aspects of PED performance should be mandatory, which should be contextualized against external system conditions, and how district-level assessments can be meaningfully aligned with broader decarbonization strategies rather than evaluated in isolation.

2.5. Approaches to PED definition

2.5.1. Positive energy balance as core criterion

Despite the aforementioned definition problems, a large share of recent PED literature continues to operationalize PEDs primarily as technical assessment problems, focusing on district-scale energy balance modelling, system feasibility, and scenario-based design (Binda et al., 2022; Cellura et al., 2025; Nair et al., 2024; Nikolic et al., 2025; Siakas et al., 2025; Yang et al., 2025). In these studies, the "positive" character of PEDs is typically inferred from modelled surplus energy at annual resolution, while questions of sufficiency, boundary responsibility, and alignment with national decarbonization pathways remain largely unaddressed. As a result, PED performance is treated as a project-specific outcome rather than as a declaration-ready, governance-relevant criterion.

2.5.2. Define geographic system boundaries

A popular approach to PED definition (due to its lack of alternatives) was developed within IEA EBC Annex 83, COST Action focusing on system boundaries mainly in a geographic sense, as shown in Figure 10.

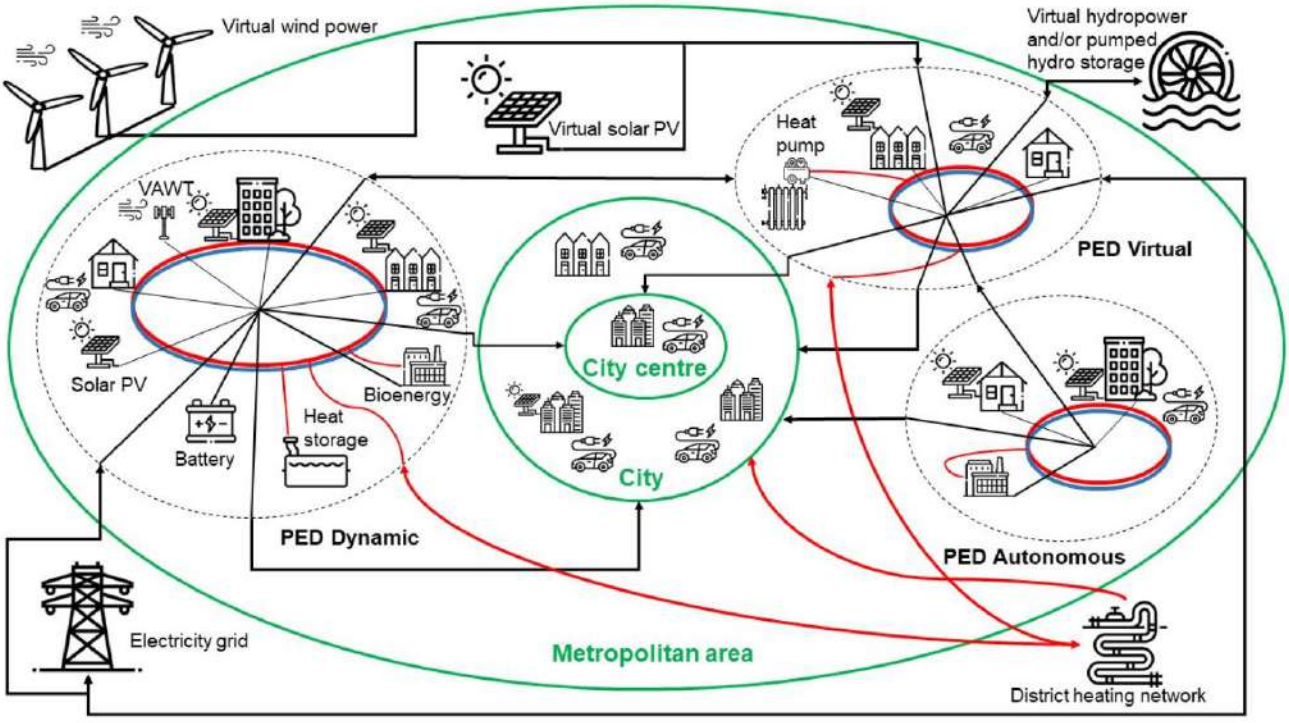


Figure 10. Geographic PED types: Autonomous, Dynamic and Virtual (Lindholm et al., 2021)

This focus is well suited for characterizing district and project approaches but does not provide definitions related to the PED energy target. It emphasizes the difference in energy autonomy by defining different levels of "hinterland" interaction but does not embed them in a normative frame.

The challenge of ambitious energy balance targets is obfuscated and open for design by not explicitly addressing functional system boundaries, i.e. which energy services are included (operation, plug loads, mobility, embodied emissions).

The most interactive type is that of the "V*irtual PED*", which employs virtual system boundaries to include offsite generation. However, the virtual inclusion of external RES such as wind farms and hydropower is rarely defined from a wider system perspective addressing the questions how and why these virtual RES are distributed and available for a certain district: As (Thellufsen et al., 2020) have argued, "the allocation and balancing of these resources must be decided on municipal, county or national level: A [district,] region, city and municipality need to situate themselves within national and international contexts in order to make a coherent strategy to transition to for instance 100% renewable energy. It is therefore necessary to develop methods that enable cities to assess the compatibility of the local [i.e. district] renewable energy strategy to the surrounding national and global energy systems. […] Cities and municipalities should act locally with regard to local demands but acknowledge the national and global context when addressing resources, industry and transport".

2.5.3. Omit the energy positive balance target

Another approach in the literature is to deliberately omit the requirement of a positive energy balance and to frame the PED primarily as a conceptual or guiding idea rather than as a quantitatively verifiable performance target. In this interpretation, PEDs function as reference models that articulate ambitions related to sustainability, innovation, and urban transformation, without requiring compliance with a specific energy balance criterion. Several policy and research-oriented contributions emphasize PEDs as process-oriented or vision-driven constructs, focusing on integrated urban planning, experimentation, and transition management rather than on strict performance accounting (European Commission. Directorate General for Research and Innovation., 2020; Lindholm et al., 2021).

This conceptual framing is also reflected in reviews that highlight the lack of a universally accepted PED definition and the wide variation in how "positive" performance is interpreted across projects and contexts (Albert-Seifried et al., 2022, p. 22). In such approaches, qualitative objectives – such as stakeholder engagement, governance innovation, and system integration – take precedence over quantitative thresholds, with energy positivity treated as an aspiration rather than a defining requirement (Bossi et al., 2020; Derkenbaeva et al., 2022). While this increases conceptual flexibility and inclusiveness, it also limits operational comparability and constrains the use of PEDs as auditable instruments for policy implementation and target tracking.

2.5.4. Broaden the assessment

Across European and international PED initiatives, including the IEA EBC Annex 83, and related flagship projects, the PED concept has increasingly been positioned as a vehicle for a wide range of objectives, including governance innovation, social inclusion, quality of life, circularity, participation processes, digitalization, and economic development. While these dimensions are undoubtedly relevant for sustainable urban development, their aggregation within a single PED definition has led to two recurring methodological issues:

First, PED frameworks have increasingly gravitated toward large indicator sets ("KPI Zoos") and points-based scoring systems**,** aiming to capture incremental improvement across many dimensions simultaneously (Fatima et al., 2023). Such approaches are particularly prevalent in contexts where comprehensive national sustainability labelling systems are not yet established. While useful for awareness-raising and early-stage benchmarking, they tend to dilute the analytical depth of core energy and climate metrics and obscure trade-offs between ambition, feasibility, and system impact. Holistic assessments need to cover a large number of dimensions and concerns, leading to a dense forest of KPIs, e.g.: „*POCITYF* has adopted all eight dimensions towards setting a holistic performance framework, corresponding with the type of solutions and actions to be implemented" (Angelakoglou et al., 2020). The *Syn.ikia* Methodology Framework for assessing "Sustainable Positive Energy Neighborhoods (SPEN)" covers 44 KPIs in five main categories (Salom et al., 2021a; Salom & Tamm, 2020).

Second, points-based and multi-KPI approaches often shift the focus from absolute performance thresholds to relative or incremental improvement. In the context of climate neutrality, this can be problematic: incremental gains do not guarantee alignment with finite energy and GHG budgets, and high scores can coexist with trajectories that remain incompatible with long-term climate targets.

2.5.5. Certification and standardization efforts

As PEDs move from exploratory concepts to broader implementation, questions of certification, standardization, and comparability have gained increasing importance. Existing sustainability certification schemes at the district scale, most prominently *LEED* for Neighbourhood Development and *BREEAM* Communities, provide structured, indicator-based assessment frameworks, but were not designed to capture the specific characteristics of PEDs, such as net-positive energy balances, multi-energy system integration, or operational flexibility (*BREEAM - BRE Group*, n.d.; *LEED Rating System | U.S. Green Building Council*, n.d.). As a result, PED performance is only partially and indirectly reflected in current certification outcomes.

(Volpatti et al., 2024) therefore proposes a methodological approach to adapt existing certification protocols by identifying PED-relevant criteria, separating PED and non-PED score components, and introducing a dedicated PED credit through score reallocation rather than redefining the entire protocol structure. This approach highlights the potential of leveraging established certification systems as institutional entry points for PED evaluation, while also revealing structural limitations, such as opaque weighting schemes, aggregation effects, and insufficient representation of operational energy performance.

Beyond individual certification schemes, broader standardization efforts for PEDs are still in an active and formative phase. Research-driven initiatives such as PED-EU-NET and IEA EBC Annex 83 have focused on harmonizing terminology, system boundaries, and performance indicators, while explicitly acknowledging that no consolidated PED standard yet exists. In parallel, formal standardization bodies are beginning to engage with PEDs more directly: the ongoing work coordinated by CEN-CENELEC, including dedicated workshops on PEDs in 2025, indicates that PEDs are now being discussed as a potential subject of future European standardization. At present, however, these activities remain exploratory and consensus-building in nature, without resulting normative documents or certification standards (*Launch of the CEN Workshop "Guidelines for the Integrated and Context-Sensitive Development of Positive Energy Districts (PEDs),"* n.d.).

At the national level, Austria illustrates how PED-relevant assessment practices can emerge outside formal PED standards. A well-established landscape of qualitative and semi-quantitative certification and assessment systems exists, including *klimaaktiv Gebäude*, *klimaaktiv Siedlungen und Quartiere*, *Total Quality Building*, and *aspern klimafit 2.0*. These frameworks address energy efficiency, renewable energy use, comfort, and aspects of sustainability and governance, but rely predominantly on self-reporting, qualitative criteria and aggregated scoring approaches rather than explicit, time-resolved energy balances. They provide institutionalized entry points for district-scale sustainability assessment, yet stop short of delivering declaration-ready PED assessments, highlighting the gap between existing national certification practice and the emerging requirements of quantitative, auditable PED definitions (*Der Klimaaktiv Gebäudestandard 2025 | ÖGUT*, n.d.; O. Mair am Tinkhof, 2017; Thomas Zelger, et al., 2024).

2.6. Building, district, and urban building energy modelling (UBEM) for PEDs

(Natanian et al., 2024) synthesize the state of tools and methods for PED design. The review shows that while a wide range of modelling tools exists, the PED tool landscape remains highly fragmented, with limited interoperability across design phases, performance domains, and stakeholders. Urban

Building Energy Modelling (UBEM) is identified as the analytical backbone of most PED tools but is insufficient on its own to address system integration, operational flexibility, life-cycle impacts, and governance needs. Four modelling approaches can be distinguished: physics-based, reduced-order, data-driven, and metadata-driven, each trading off accuracy, computational cost, and suitability for optimization. Across tools, energy demand and supply dominate, while mobility, social aspects, life-cycle emissions, and post-occupancy operation are systematically underrepresented. The authors conclude that scaling PEDs depends less on developing new standalone tools than on integrated digital workflows, standardized metrics, and assessment frameworks that link modelling outputs to decision-making and governance processes (Becchio et al., 2017; Cellura et al., 2019; Congedo & Baglivo, 2021; García-Fuentes et al., 2018; Kılkış, 2014; Koutra et al., 2018; Salom et al., 2021b; Sibilla & Abanda, 2022).

UBEM methods estimate energy demand (and sometimes HVAC operation and on-site generation potential) for many buildings simultaneously, typically at neighbourhood to city scale. UBEM is used to generate comparable demand profiles, identify retrofit and electrification hotspots, support district energy system sizing, and test policy scenarios under data scarcity. The field has converged on a small number of modelling "families" that trade off input requirements, such as geometry fidelity (De Jaeger et al., 2017, 2018), interpretability, and computational cost (Ali et al., 2021):

- **Top-down (statistical/econometric):** Relates aggregated energy use to macro drivers (e.g., climate, floor area, socio-economic factors); suitable for rapid city-wide screening and policy analysis, but limited for design, sizing, and operational questions.
- **Bottom-up physics-based:** Simulates building-level thermal and energy behaviour (often archetype-based) to generate end-use and hourly profiles; supports retrofit and system design but is constrained by data availability, calibration effort, and computational cost.
- **Bottom-up reduced-order / surrogate:** Uses simplified RC models or metamodels calibrated on detailed simulations or measurements; enables fast district-scale iteration, optimization, and uncertainty analysis with reduced accuracy.
- **Bottom-up data-driven:** Applies regression or ML to predict loads from measured data and contextual features; effective for large-scale prediction and calibration where data exist, but weak for evaluating novel technologies or control strategies without physics coupling.

Most UBEM implementations follow a pipeline: (i) urban/building stock representation (GIS, CityGML, footprints + heights), (ii) semantic enrichment (use type, vintage, envelope/HVAC archetypes, occupancy schedules), (iii) weather and boundary conditions, (iv) simulation/prediction, (v) calibration/validation (billing, metered subsets), and (vi) aggregation and post-processing (district peaks, coincidence factors, temporal profiles). Key recurring challenges are incomplete building

attributes, privacy restrictions around measured data, occupant behaviour assumptions, and uncertainty propagation.

UBEM methods operationalize the spatial distribution and temporal structure of building energy demand at district to city scale; PED assessment builds on these demand profiles but additionally requires explicit multi-energy system modelling, import/export accounting, and flexibility/control representation to support auditable district-level "positive" claims. PED assessment requires more than annual demand estimation:

- **Temporal resolution:** PED flexibility and grid interaction depend on hourly (or sub-hourly) coincidence between demand, local generation, and storage/control.
- **Sector coupling:** PEDs typically require integrated electricity + heat (and often mobility) modelling, not only building thermal/electric loads.
- **System boundary consistency:** PED declaration logic needs explicit treatment of imports/exports, conversion efficiencies, and responsibility boundaries (what the district can influence vs exogenous factors).
- **Operational strategies:** load shifting, storage dispatch, heat pump control, EV charging schedules, and curtailment affect both PED balance and "grid friendliness".

Apart from the fact that applied PED assessment frameworks are largely incomparable, another issue lies in the lack of dynamic energy system models as part of the assessment (Gabaldón Moreno et al., 2021). However, this is necessary to assess the gains of delivering HVAC, e-mobility and plug loads with onsite RES and minimal grid interaction *flexibly*. Several studies have thus stated the importance of suitable timesteps in assessing energy flexibility, e.g.: „Future outlooks should be directed towards the development of dynamic environmental analyses taking into account long-term uncertainties and energy flexible control" (Marotta, Guarino, Longo, et al., 2021).

For resolution in time, most sources agree that a stationary calculation of energy balances is insufficient, mainly due to being unable to resolve flexible control measures and interplay between district demand and onsite RES. Having high temporal resolutions on the other hand, typically comes with the disadvantage of long compute times and resulting unwieldy models. Testament to computational complexity: „The housing stock was depicted using two archetype houses modelled using a simple lumped capacitance approach, integrally optimized as a part of a stochastic linear programming unit commitment model of the power system" (Rasku & Kiviluoma, 2019).The sweet spot seems to be at an hourly resolution on the scale of a full operational year (Ala-Juusela et al., 2021; Laitinen et al., 2021).

Recent work on Virtual Positive Energy District Laboratories (VPED-Labs) further clarifies the role of UBEM within PED-oriented modelling environments. VPED-Labs are conceived as digital, district-scale environments that integrate simulation, data management, and stakeholder interaction to support PED design, testing, and governance, explicitly positioning UBEM as a core but non-sufficient component of such systems (Semprini et al., 2025). While UBEM provides the spatially and temporally resolved demand backbone required for district-scale analysis, VPED-Lab research highlights three recurring limitations when UBEM is applied in isolation to PED assessment: the lack of explicit multi-energy system integration (electricity, heat, storage, mobility), insufficient representation of operational and control strategies that determine temporal coincidence and flexibility, and the absence of clear responsibility and system boundary definitions required for auditable import/export accounting. Accordingly, VPED-Labs conceptualize UBEM as an input layer feeding higher-level simulation and decision-support functions such as energy management, scenario testing, and regulatory evaluation, which reinforces the distinction made in this thesis between energy modelling and declaration-ready PED assessment frameworks.



Contributions

3.1. Framing the problem

This chapter introduces foundational observations and deductions that serve as a platform for the subsequent development of the thesis.

3.1.1. Problem diagnosis: Are PED definitions meaningless?

Existing PED definitions provide a broad articulation of ambitions, including energy efficiency, flexibility, renewable energy production, and system integration. While valuable as a shared vision, such definitions remain insufficiently specified to support assessment, comparison, or governance in practice. Comparative analysis and stakeholder exchange in [Schneider-2023 b] yielded three recurring and competing definition requirements:

1. **Simplicity**: the definition should be easy to communicate and understand.
2. **Determinism**: the definition should enable unambiguous assessment.
3. **Achievability**: the definition should be achievable across different district types.

In practice, these requirements are difficult to satisfy simultaneously. Definitions that prioritize simplicity and determinism tend to render energy targets unachievable for dense or otherwise constrained districts. Definitions that prioritize simplicity and achievability often lose quantitative rigor. Definitions that aim to remain both deterministic and broadly achievable require greater methodological complexity.

This tension constitutes a definition trilemma, as shown in Figure 11: Any PED definition involves an explicit position on ambition, effort sharing, methodological complexity, and feasibility.

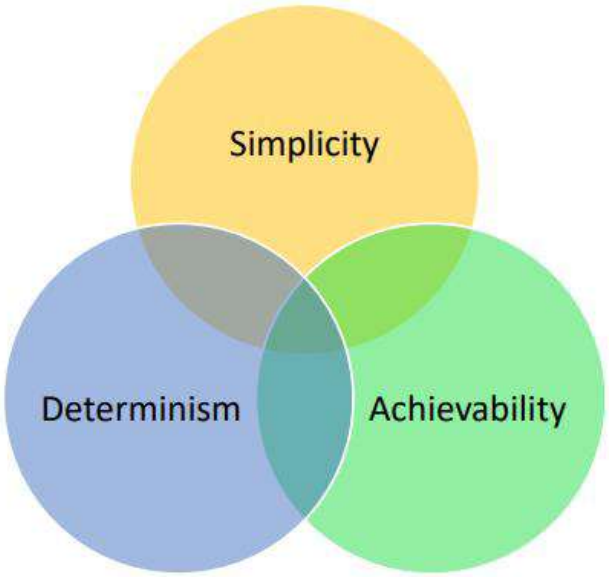


Figure 11. Definition trilemma [Schneider-2023 b]

3.1.2. The solution: PED definitions are design problems

PED definitions are expected to translate abstract climate and energy objectives into district-scale requirements, enable comparison between projects in different spatial, climatic, and regulatory contexts, and remain understandable and actionable for practitioners and decision-makers. These expectations are partly contradictory and cannot be resolved through indicator selection alone.

This thesis therefore treats the development of a PED definition as a multidimensional design problem, following the conceptual framing introduced in [Schneider-2023 b]. The guiding question shifts from:

"What is a PED?"

to:

"*What must a PED definition do to be useful, comparable, and operational across different contexts?"*

This shift does not eliminate the need for descriptive indicators, system boundaries, or declarative thresholds. Instead, these elements are derived from explicitly defined functional goals. The selected goals act as design requirements that guide the choice and operationalization of suitable definition components, as illustrated in Figure 12:



Figure 12. Schematic approach for the development of a PED definition [Schneider-2023 b]

3.1.3. PED Definitions Across Levels of Regionality

[Schneider-2023 a] and [Schneider-2023 b], later reinforced by [Schneider-2024 d], [Schneider-2025 a] and [Schneider-2025 d], show that contextual differences between districts are not marginal deviations, but dominant drivers of positive energy balance feasibility. These differences include, among others: building density and urban form, climate and renewable energy availability, green or brownfield developments (renovations). The analysis demonstrates that a strictly uniform and fully operational PED definition applicable across all contexts is neither feasible nor desirable. Attempts to enforce such uniformity result either in definitions that are too vague to be meaningful or too restrictive to be applicable. Comparative analysis across international PED projects shows that energy systems, building practices, climatic conditions, data availability, and legislative frameworks differ fundamentally between countries. As a result, identical PED criteria can imply vastly different levels of effort, feasibility, and system impact depending on context.

To resolve the tension between uniformity and locality, the author introduces a multi-level interpretation of PED definitions, distinguishing between different regimes of regionality, as shown in Figure 13:

- a **European level**, defining minimum common principles and intent,
- a **national (or regional) level**, responsible for quantitative operationalization and target setting,
- a **local or project level**, where implementation-specific design choices are made.

In this framing, a PED is not subject to a single monolithic definition, but to multiple, simultaneous definitions operating at different levels of abstraction and responsibility. This approach allows common goals to be preserved while enabling nationally and locally appropriate operationalization. Crucially, this **is** not equivalent to letting each project define its own PED criteria freely. Instead, responsibility for defining ambition, effort sharing, and feasibility is explicitly assigned to higher governance levels, reducing arbitrariness at the project level.

This suggests that the national level represents the highest level of stability and homogeneity at which PEDs can be operationalized in a concrete and enforceable manner. National contexts provide consistent climatic datasets, harmonized legislation and standards, coherent decarbonization pathways, and clearly defined governance responsibilities.

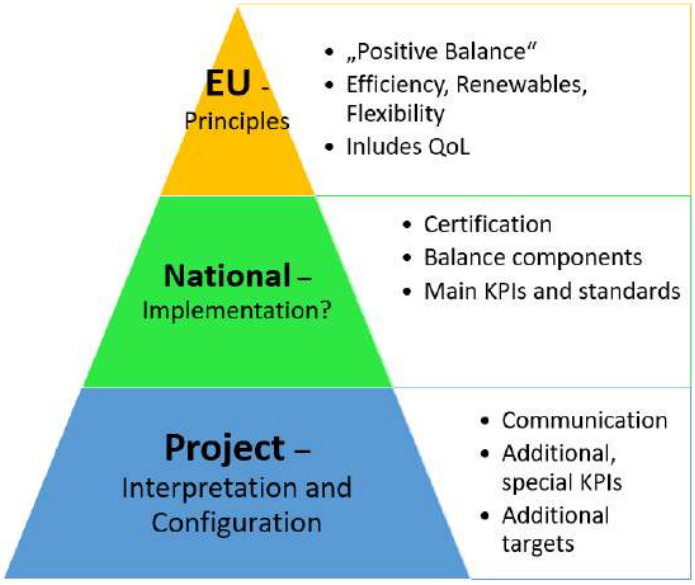


Figure 13. Three layers of regionality with different responsibilities of definition [Schneider-2023 b]

3.1.4. Requirements for a Quantitative PED Definition

Framing the PED definition as a design problem has direct implications for the construction of quantitative assessment methods. It clarifies that:

- a positive **energy balance alone is insufficient** without explicit definition of system boundaries, weighting schemes, and balance targets,
- feasibility across contexts requires **explicit mechanisms** rather than implicit boundary manipulation,
- quantitative PED definitions inevitably encode normative decisions about effort sharing.

These implications form the conceptual foundation for the quantitative PED definition developed in the following sections. Concretely, [Schneider-2023 a] and [Schneider-2023 b] demonstrate that a

PED definition must answer a limited set of structurally unavoidable questions in a transparent and reproducible way. In extension to (Sartori et al., 2012), they establish three core design dimensions that any operational PED definition must explicitly address: system boundaries, weighting of energy and emission flows, and allocation of external resources expressed as quantitative targets as shown in Figure 14. Together, these dimensions allow the quantification of energy and emission balances against their defined sufficiency targets. A quantitative PED definition must make these three boundary choices explicit, standardized, and auditable**,** even where these choices remain normative.

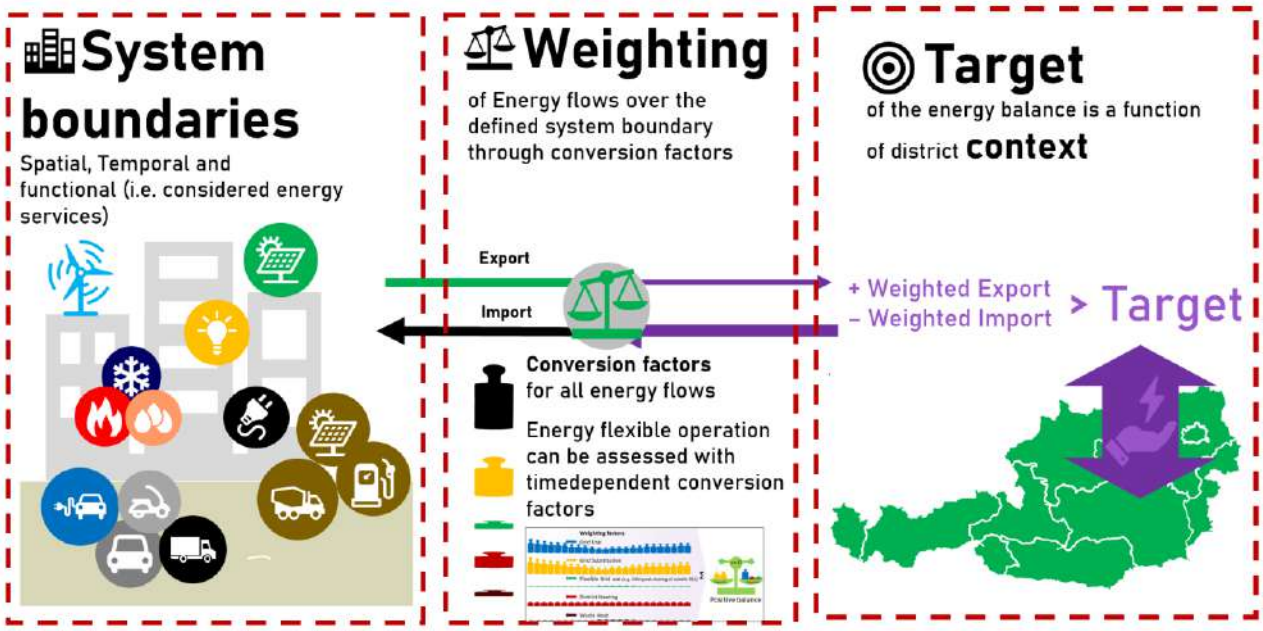


Figure 14. The three main design elements of a quantitative positive energy balance assessment [Schneider-2023 b]

3.1.5. "Context factors": Quantitative contextualizations of PED balance targets

[Schneider-2023 a] and [Schneider-2023 b] show that "context factors" can take on both the role of explicit targets (relative to an absolute zero target) and also the role of other budgets and credits (like offsite RES credits proposed in virtual schemes), as illustrated in Figure 15. Using this paradigm, a quantitative PED definition can be designed with a positive balance target using the context factors for the appropriate system boundaries. The inclusion of these virtual factors in the balance can be designed in any number of ways. Their use must be rooted in its comprehensibility and link to the definition goals, which are examined in the following chapter.

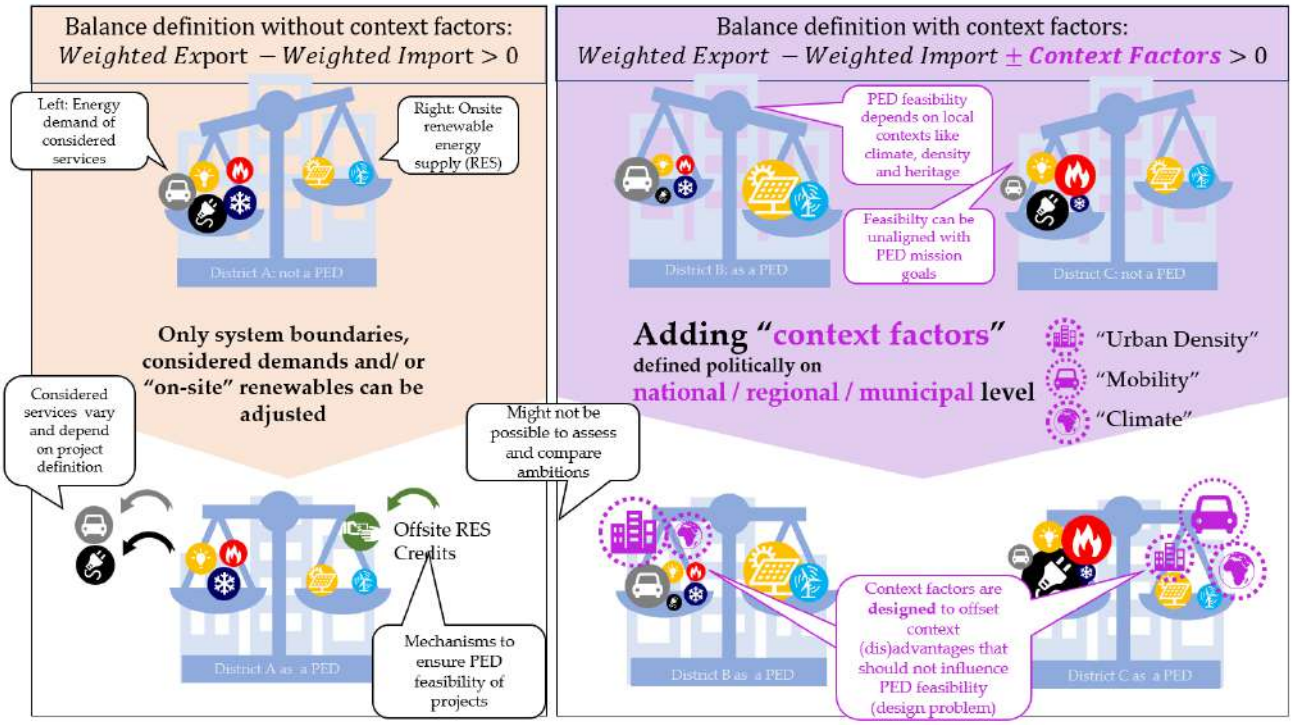


Figure 15. Conceptual differences without (left) and with (right) explicit context factors [Schneider-2023 a]

3.2. Quantitative PED definition

This chapter presents the main contributions of this thesis on the developed PED definition framework.

3.2.1. Strategic Goals of the Quantitative PED Definition

Based on the identified gap in quantitative, declarative PED definitions, a stakeholder participation process identified the following **strategic goals** for the PED definition design in [Schneider-2023 b]:

1. Translate higher-level climate and energy goals to the district scale: European, national, and regional decarbonization targets must be transformed into quantifiable district-level objectives that are feasible within planning and development processes.
2. Ensure comparability across contexts without enforcing uniformity**:** The definition must allow PEDs in different climatic, spatial, and regulatory contexts to be compared meaningfully, without assuming a one-size-fits-all solution.
3. Support governance, labelling, and planning processes**:** The PED definition should provide a mobilizing reference for cities and stakeholders, enabling coordination, recognition, and if possible financial or regulatory support mechanisms.
4. Enable high-quality transformations: The definition should not merely certify minimum compliance, but actively *push* districts toward ambitious, system-relevant solutions. PED criteria thus act as a quality framework that makes ambition measurable and comparable.

Figure 16 shows all design goals of a common PED definition in [Schneider-2023 b]. For this thesis, not all identified project goals are recognized: Transnational comparability, Coverage of all relevant Phases (Monitoring is future work), Collaboration support (is only implied but not enforced). The above goals are augmented by [Schneider-2025 d] with three more from a Climate-neutrality Perspective:

- **Integrated and adaptable system boundaries**: Assessment boundaries extend beyond individual buildings to the full PED system, covering on-site generation, storage, flexibility, and interconnections with surrounding grids. The framework retains adaptability in data sources, emission factors, and system configurations, allowing regional customization while ensuring methodological comparability.
- **Temporal differentiation**: Emissions must be assessed over a 50-year timeframe (2025–2075), with the assumption that climate neutrality is achieved for all services by 2050, and resulting decarbonization pathways

- Science-based, budget-oriented compliance thresholds: The method is anchored in downscaled CO_2 budgets consistent with Austria's 2040 climate neutrality target and the 1.5 °C global carbon budget.

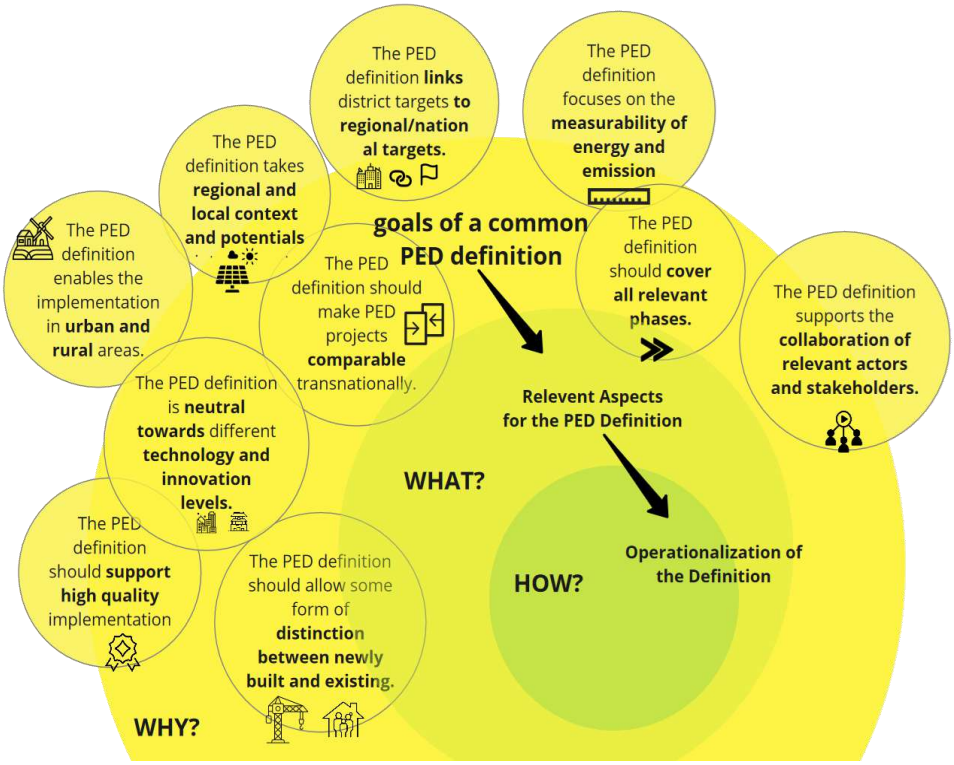


Figure 16. Resulting design goals of a common PED definition [Schneider-2023 b]

Non-Goals and Exclusions

The quantitative PED definition developed in this thesis deliberately excludes several extensions of the PED concept that are common in international research and practice, in particular the inclusion of extensive sets of qualitative indicators or broad collections of key performance indicators. This exclusion is deliberately tailored to the Austrian context: It has a well-established landscape of qualitative and semi-quantitative assessment and certification systems, including *klimaaktiv Gebäude*, *klimaaktiv Siedlungen und Quartiere*, *Total Quality Building*, and *aspern klimafit* (*Der Klimaaktiv Gebäudestandard 2025 | ÖGUT*, n.d.; O. Mair am Tinkhof, 2017; Thomas Zelger, et al., 2024). These frameworks comprehensively address aspects such as urban quality, building standards, planning and governance processes, comfort, materials, and stakeholder involvement. Replicating these dimensions within a PED definition would introduce unnecessary redundancy, increase complexity without improving decision relevance, and risk producing inconsistent or even conflicting assessment outcomes. The quantitative PED definition therefore does not seek to replace or duplicate existing sustainability assessment systems.

Another explicit non-goal is the formulation of a globally uniform, fully operational PED definition. The definition is designed to be anchored at the national level, while remaining compatible with higher-level European principles and adaptable to local implementation.

3.2.2. System Boundary Definition

[Schneider-2023 a] and [Schneider-2023 b] derive three concrete system boundary definition aspects as the first functional definition requirement:

- **Functional scope:** which energy services and emission sources are considered part of the PED system (e.g., building operation, local energy generation, mobility-related energy use, embodied emissions).
- **Spatial scope:** which renewable energy sources are considered "local" and therefore attributable to the district.
- **Temporal scope:** Aggregation timeframe.

The **functional system boundary** specifies which energy services and emission sources are included in the PED balance and how they are treated. Following the framework proposed by [Schneider-2023a, b], three levels of functional scope can be distinguished, back then referred to as *Alpha*, *Beta*, and *Omega* boundaries, and since then renamed to "PED", "PED+M", and "CN PED".

Table 3. Considered energy services within defined PED boundaries [Schneider-2023 a]

Energy Services		Alpha	Beta	Omega	Implicit *
Building operation	Heating	✓	✓	✓	-
	Cooling	✓	✓	✓	-
	Humidification and dehumidification	✓	✓	✓	-
	Ventilation	✓	✓	✓	-
	Auxiliary power of the building services system	✓	✓	✓	-
	General power and lift	✓	✓	✓	-
	Lighting	✓	✓	✓	-
District operation Industry, agriculture	Power requirements of users (plug loads)	✓	✓	✓	-
	Operating power (office, retail, school)	✓	✓	✓	-
	Process heat	-	-	-	✓
	Process cooling	-	-	-	✓
	Electricity demand for industrial production processes	-	-	-	✓
	Electricity demand for general use (incl. services)	✓	✓	✓	-
Mobility	Motorized private transport	-	✓	✓	-
	Public transport	-	-	-	✓
	Other mobility	-	-	-	-
Embodied Energy	Components of the Austrian energy certificate	-	-	✓	-
	Accessory components (cellars, underground parking, garages, carports, bicycle storage areas, balconies and terraces, other outbuildings)	-	-	✓	-
	Building and energy equipment	-	-	✓	-
	Vehicles and infrastructure for mobility	-	-	✓	-
	Public transport	-	-	-	✓

(✓) Included in system boundary, (-) not included in system boundary. * **Implicit**: Not included in any system boundary but instead part of the district context of the surrounding energy scenario that in turn influences the balance target as context factors.

At a minimum (*Alpha* boundary or "PED in Operation"), the functional scope includes all operational energy demands associated with building operation. This comprises space heating and cooling, ventilation, humidification and dehumidification, domestic hot water, lighting, auxiliary electricity for building services, general electricity use, and lift operation. These services are directly linked to the core function of buildings and can be reliably quantified using established building energy assessment methods.

An extended functional scope (*Beta* boundary or "PED with mobility") additionally includes district-level operational energy demands that are not attributable to individual buildings alone. This includes electricity demand for common district functions such as offices, retail, schools, and general services, as well as energy use associated with district infrastructure. In some definitions, selected mobility-related energy uses, particularly motorized private transport within or directly linked to the district, may also be included at this level.

The most comprehensive scope (*Omega* boundary or "climate-neutral PED") further expands the functional system boundary to include additional emission sources that influence the overall climate impact of the district. This includes embodied energy and emissions associated with buildings, infrastructure, and technical equipment, as well as emissions related to vehicles construction and everyday use. Certain elements, such as public transport energy use or goods and services produced or consumed in the district, are treated implicitly on the target side: They are not included directly in the PED balance, but they influence the boundary conditions of the surrounding energy system and thus the district's emission factors, as shown in Chapter 3.2.8. Figure 17 shows a schematic overview of emissions caused by operational energy flows in the climate-neutral PED framework, illustrating the treatment of local PV direct use and grid feed in, waste heat imports, district cooling emissions, and other energy carriers.

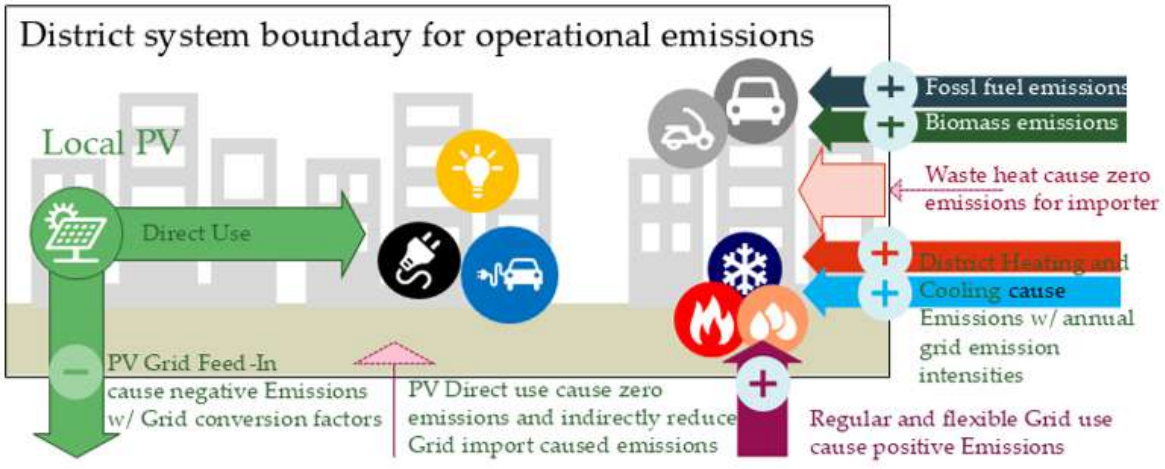


Figure 17. Climate-neutral PED system boundary for operational emissions [Schneider-2025 d]

Spatial system boundary: The spatial system boundary corresponds to the physical boundary of the district and covers the area required for its full functional use. Boundary manipulation for the selective inclusion or exclusion of energy sources or emission sinks must be avoided; boundaries should be as convex as possible and only as concave as necessary. All local renewable energy sources within this area may be included in the energy balance, regardless of conversion technology, provided their use does not disadvantage surrounding areas. This is particularly relevant for shared or flowing resources such as water used for cooling or small-scale hydropower. Spatial boundaries should therefore be defined to ensure fairness and avoid negative impacts on adjacent districts.

Temporal system boundary: Operational energy balances (Alpha and Beta) are aggregated over one operational year. Life-cycle greenhouse gas emissions (Omega) are assessed over a 50-year period (2025–2075), reflecting the service life of buildings and infrastructure.

3.2.3. Balance Weighting of Energy and Emission Flows

The quantitative PED definition is intentionally focused on only two system-critical key performance indicators: the primary energy balance, and the life-cycle greenhouse gas balance. Following [Schneider-2023a, b], energy and emission flows are weighted using primary energy and life-cycle greenhouse gas emission factors, aligned with existing planning practice, building codes, and district heating regulations. To reflect seasonal system impacts, in the absence of hourly factors, monthly conversion factors must be applied, distinguishing between summer renewable feed-in and winter grid imports.

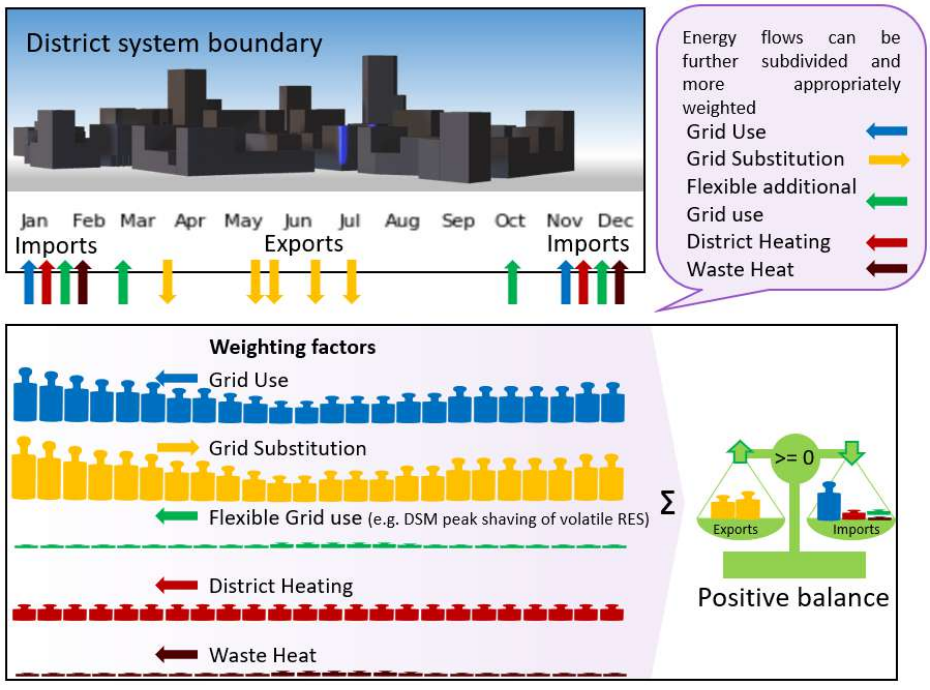


Figure 18. Illustration of time-dependent energy flow weighting [Schneider-2023 b]

Grid-supportive and time-sensitive energy use and feed-in are explicitly valued: energy that is otherwise unavailable in the surrounding system is treated with an explicitly different conversion factor, as illustrated in Figure 18 and described in the following chapter.

3.2.4. Energy flexibility assessment

Instead of introducing an additional indicator, energy flexibility assessment is incorporated by applying time-dependent weighting factors to energy imports and exports, in particular grid electricity.

This captures system-relevant flexibility and allows energy efficiency, local renewable generation, and flexibility measures to be reflected within a single, declaration-ready balance without prescribing specific technologies [Schneider-2023 a, 2025 d]. Flexibility may be provided through a range of measures, including storage, load shifting, thermal building mass, or sector coupling, but compliance is determined solely by system behaviour as reflected in the weighted balance. Concrete weights must be chosen to fit the use case:

- Hourly primary energy and emission intensities of grid electricity allow direct assessment, scientific analysis and optimization, but are not widely available and unpractical for declarative purposes
- Proxies for flexibility demand signal times of high opportunity for renewable uptake from the grid: Wind power, residual loads or forecasting errors from wind parks.

The latter represent situational renewable energy surpluses that can increase renewable uptake, but only by sufficiently energy-flexible districts. Time-dependent use of the latter can be credited favourably in the balance, by using conversion factors of zero to these situational renewable imports, as signalled by proxy.

3.2.5. Balance Target definition

Based on the previously defined system boundaries, balance metrics, and weighting schemes, the final step of the balance definition concerns the formulation of targets. In line with the PED paradigm, both KPI PED balances – primary energy and GHG emissions – are required to be positive. However, this requirement is not applied in isolation. Instead, the balance targets are evaluated in combination with three explicitly defined “context factors” that are operationalized at the national level. These context factors adjust the balance requirements to reflect structural conditions that cannot be fully influenced at district scale and are introduced and specified in the following chapters:

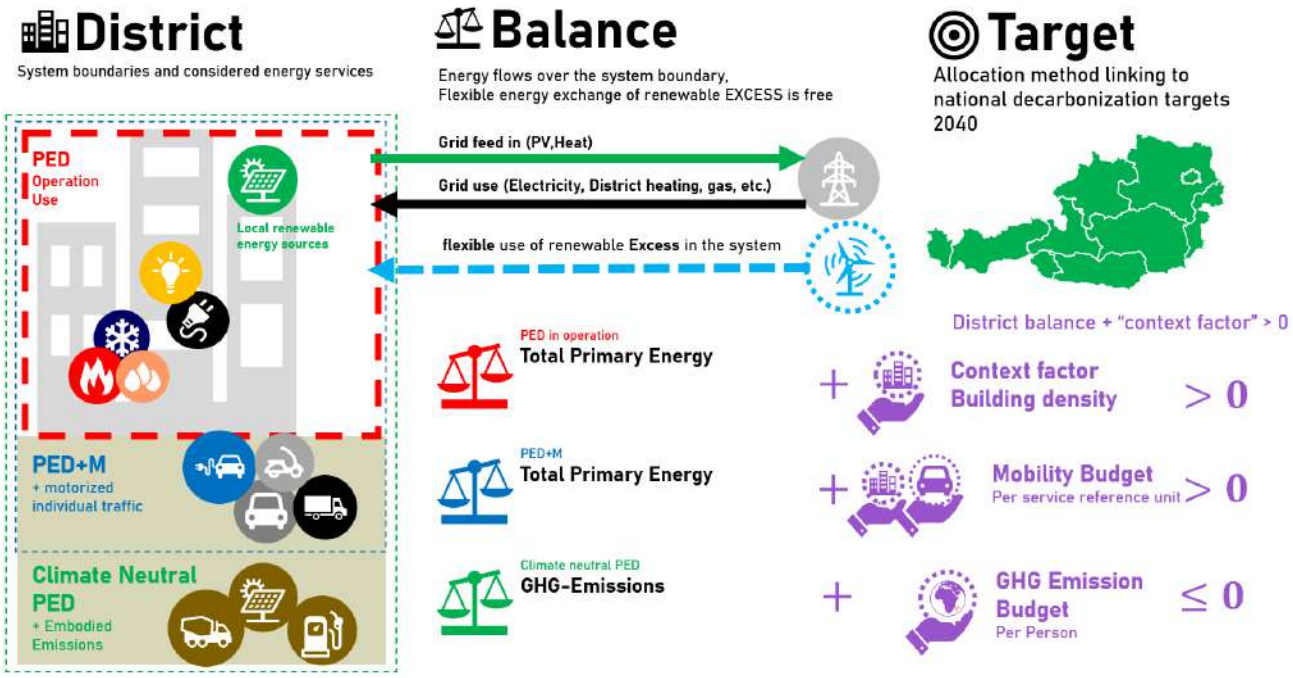


Figure 19. PED balance definition overview

3.2.6. PED Target Context factor 1: Density

The density context factor links the PED target value to building density, expressed by the floor area ratio (FAR), to account for the structural mismatch between energy demand and local renewable potential [Schneider-2023a]. It can either be seen as an explicit target function, or – if the balance target should remain as > 0 for narrative reasons, can be formulated as virtual balance component of opposite sign. For example, a low-density district with FAR of 0.25 receives a negative density context, incurring a +125 kWh_{PE}/m^2_{NFA} virtual demand to its energy balance, which is equivalent to a balance target of +125 kWh_{PE}/m^2_{NFA} without context factor. Conversely, a high-density district of FAR = 4 receives a positive context factor of approximately 40 kWh_{PE}/m^2_{NFA} as a virtual supply, equating to an effective target of −40 kWh_{PE}/m^2_{NFA} without context factor.

Figure 20 shows the derived target function, i.e. the negative context factor in comparison to achievable energy balances of investigated case studies. The function includes offsets and cutoffs to limit extreme values. The function shape is derived from physical building and plot properties, but its parameters are not intended to represent technical maxima. They are operationalized to map politically defined effort-sharing across district typologies within the Austrian building sector.

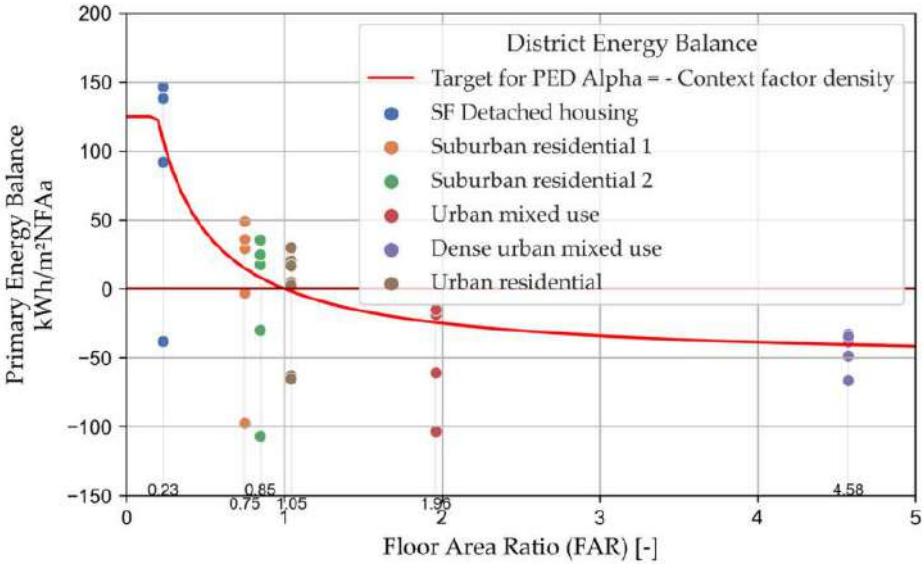


Figure 20. Primary energy balances of example districts as a function of density (FAR) [Schneider-2023 a]

The parameterization is anchored in national decarbonization targets for 2040 and calibrated such that the aggregated performance of all district typologies matches the building sector's allocated primary energy balance target. As a result, low-density districts are not structurally favoured, dense districts remain achievable within the PED framework, and individual projects may still exceed the adjusted target where technically and economically feasible.

3.2.7. PED Target Context factor 2: Mobility

Integrating everyday mobility into the PED balance generally reduces the likelihood of achieving a positive balance unless part of the mobility-related energy demand can be offset externally. To address this, a virtual mobility context factor is introduced. This factor is derived top-down from the Austrian national energy system and represents the share of renewable energy that can be allocated to districts for covering private everyday mobility without requiring on-site generation.

Assuming Austria's policy target of a fully renewable energy system by 2040, projected national scenarios indicate a surplus from large-scale renewable power plants, primarily wind, hydropower, and biomass. This surplus is interpreted as a nationally available mobility budget and is distributed equally across the population. For Austria, this results in a total primary energy surplus of approximately 10.7 TWh per year available to cover motorized individual mobility in the built environment.

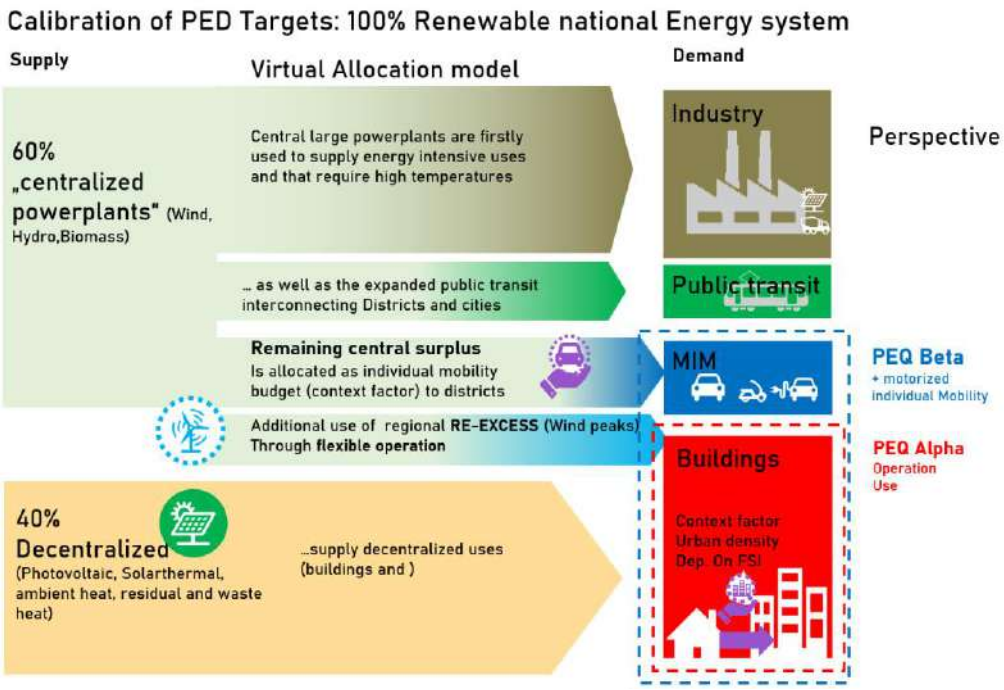


Figure 21. Sectoral allocation model for a 100% renewable Austrian energy system [Schneider-2023 a]

This national surplus is allocated to districts according to the average Austrian distribution of destination traffic. Residential use accounts for roughly 50% of all trips, with the remainder attributed to office and commercial, education, and retail uses. When distributed over the corresponding national usable floor area, this allocation yields use-specific mobility context factors expressed in kWh of primary energy per square meter and year.

The resulting context factor is then compared with the statistically derived energy demand of private everyday mobility, based on average travel distances, vehicle technologies, and spatial characteristics. On average, the allocated budget is insufficient to fully cover mobility energy demand, making PED Beta targets more demanding than PED Alpha. Moreover, mobility energy demand varies significantly with location: districts with high public transport accessibility can fall well below the national average, while rural areas with limited alternatives to private car use may exceed it by up to 50%. Consequently, the mobility context factor introduces a location-sensitive constraint that systematically differentiates the feasibility of PED Beta across urban and rural contexts.

3.2.8. PED Target Context factor 3: Climate-Neutral Emission budget

Operational energy positivity alone is insufficient to ensure climate compatibility at the district scale. As demonstrated in [Schneider-2023 a] and formalized in the declaration-ready framework [Schneider-2025 d]), embodied greenhouse gas (GHG) emissions associated with construction, refurbishment, and infrastructure represent a significant and temporally front-loaded contribution to

total climate impact. Their exclusion leads to systematic underestimation of district-level emissions, particularly in new developments.

In [Schneider-2025 d], climate-neutrality is operationalized through Paris-aligned carbon budgets derived top-down from national emission pathways consistent with a 1.5 °C target. These budgets are (i) allocated to the building sector, (ii) translated into cumulative allowable emissions over a defined assessment horizon, and (iii) normalized by gross floor area to yield floor-area–specific GHG reference budgets. These budgets act as a hard constraint on the *sum* of operational and embodied emissions.

The framework adopts a life-cycle perspective covering the period 2025–2075, including construction stages (A1–A3), major maintenance and replacement (B2–B5), and operational energy use. Operational emissions account for electricity imports and exports as well as other carriers (e.g., gas, biomass, district heating and cooling), with time-dependent decarbonization of conversion factors towards zero by 2050. Construction emissions include material-specific assumptions and explicitly consider biogenic carbon storage where applicable, ensuring consistency with LCA standards while remaining compatible with policy-level carbon budgeting. Figure 22 shows the system boundaries and scope of the life-cycle assessment (LCA), covering 2025–2075 and including both operational and construction emissions. Operational emissions comprise grid electricity use and substitution as well as other carriers (natural gas, biomass, fossil, district heating and cooling), with annual decarbonization of conversion factors to zero by 2050 where applicable. Table 4 shows the considered emissions based on EN:15978 as presented in [Schneider-2025 d].

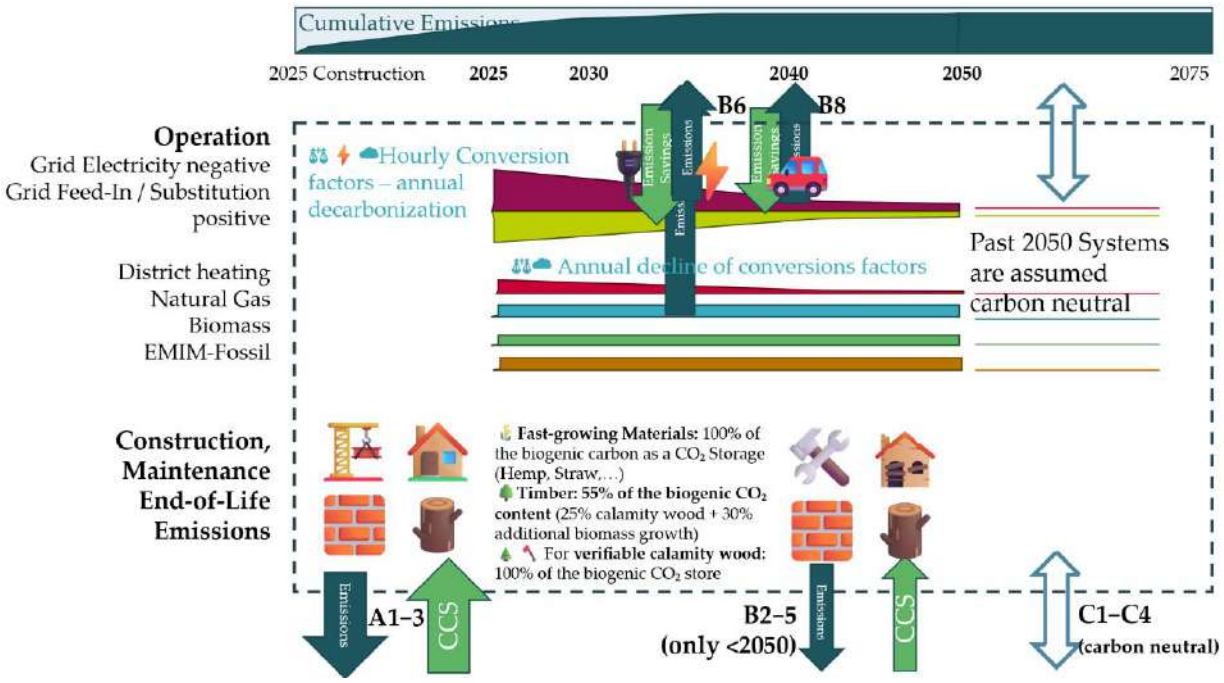


Figure 22. LCA system boundaries of the climate-neutral PED [Schneider-2025 d]

Table 4. Life Cycle Stage Inclusion in the assessment framework.

Life Cycle Stage	Sub-Stage	Module	Considered	Notes/Comments
A—pre-construction	Pre-construction	A0	No	Outside EN 15978 scope here
A—Product stage	Product stage	A1	Yes	Raw material supply (embodied)
		A2	Yes	Transport to manufacturing
		A3	Yes	Manufacturing emissions
A—Construction process stage	Construction	A4	Yes	Assessed if practical / data available
		A5	Yes	Assessed if practical / data available
B—Use stage	Use (non-energy)	B1	No	Not considered
	Maintenance	B2	Yes	Included; post-2050 assumed carbon-neutral
	Repair	B3	Yes	Same as B2
	Replacement	B4	Yes	Same as B2
	Refurbishment	B5	Yes	Same as B2
	Operational energy use	B6	Yes	Dynamic/Monthly factors (to 0 by 2050)
	Operational water use	B7	No	Water-related GHG excluded
	Other building-related activities	B8	Partly	EMIM B8.1 and B8.2 included (Section 2.4.2) Operational mileage + one-time vehicle embodied (≤2050)
C—End-of-life stage	End-of-life	C1	Yes	Included; post-2050 carbon-neutral
		C2	Yes	Same as C1
		C3	Yes	Same as C1
		C4	Yes	Same as C1
D—Beyond system boundary	Reuse/recycling recovery	D1	Yes	
	Exported utilities	D2	Yes	PV grid feed in credited as substitution with negative hourly/monthly factors
Contextual	Off-plot public infrastructure	-	No	Optional reporting only (not for compliance)

The budget-based approach introduces temporal decarbonization pathways shown in Figure 23: Early-phase embodied emissions consume a disproportionate share of the allowable budget, thereby tightening the feasible operational emissions envelope over time. Within the PED framework, the climate-neutral emission budget functions as a binding context factor that constrains achievable energy positivity claims and prevents shifting emissions across time or system boundaries.

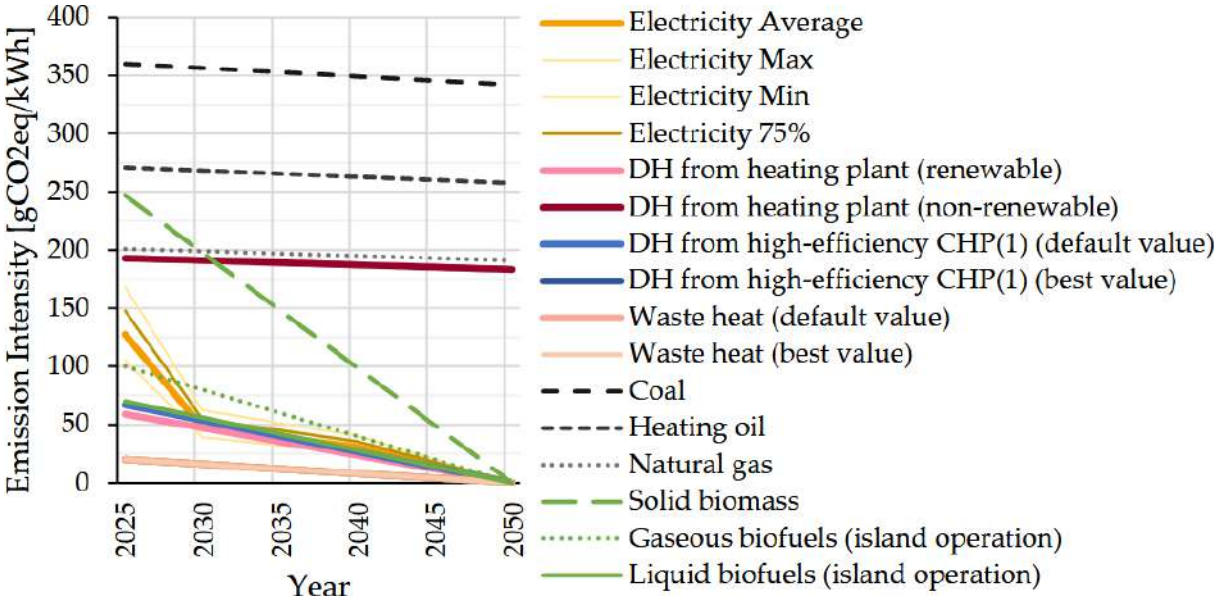


Figure 23. Decarbonization pathways [Schneider-2025 d]

3.3. Simulation and Modelling Framework

This chapter presents the author's contribution to the methodological design and implementation of a simulation framework capable of assessing the PED definition developed in chapter 3.2.

3.3.1. Intended Use, Validity Domain, and Limits of the Modelling Framework

The modelling framework is intended as a definition-consistent assessment instrument for early-phase PED planning and governance, where design information is incomplete, uncertain, and subject to change, and is intended for continuous use and refinement throughout the planning process until completion. It is however **not** designed for detailed thermal zoning, geometry-dependent or shading analysis, techno-economic optimization, adaptive or learning-based control strategies, or real-time or predictive operational performance modelling. These aspects are intentionally excluded, as they require levels of design specification and operational certainty that are unavailable in the intended application contexts.

The framework is positioned as a planning-oriented decision support system (DSS). Its primary purpose is to enable rapid comparison of alternative district concepts, exploration of feasibility under contextual constraints, and translation of quantitative PED targets into standardized and auditable planning outcomes compatible with institutional declaration and certification processes. Accordingly, its validity is assessed in terms of robustness of trends, internal consistency with the PED definition, stability under scenario variation, and suitability for decision-making, rather than numerical accuracy at component level. Validation follows a use-oriented approach based on transparent assumptions, reproducible accounting logic, consistency with established energy and GHG accounting standards, and internal checks confirming plausible and explainable system behaviour across different district typologies and dominant drivers such as density, renewable availability, and mobility demand.

3.3.2. Model Requirements Derived from the PED Definition

The established PED definition in Section 3.2 entails a set of binding modelling requirements that follow directly from the scope, system boundaries, and target logic embedded in the definition:

- **Integrated system coverage:** Joint modelling of building operation, district energy systems, everyday mobility, and embodied emissions within one internally consistent framework.
- **High temporal resolution:** Explicit representation of hourly system dynamics to capture flexibility effects and mobility–energy interactions.
- **Scenario-based application:** Support for rapid generation and comparison of alternative planning scenarios under high uncertainty in early design phases.

- **Transparency and auditability:** Fully traceable, explainable, and reproducible open-source calculation logic suitable for governance, certification and declaration use.
- **Practical deployability:** Tool maturity, robustness, and usability compatible with established professional workflows; direct applicability by certifiers and energy engineers, avoiding bespoke research code environments.

3.3.3. Simulation Model Structure and Functional Scope

The simulation model translates project-specific district data into declaration-ready indicators for the assessment of climate-neutral plus-energy districts. It combines an hourly operational energy model with annual and life-cycle accounting procedures. The model represents energy demand, local renewable generation, storage and flexibility options, electricity import and export, mobility-related energy demand, and life-cycle greenhouse gas emissions. Its primary purpose is transparent and reproducible assessment of district performance against the declaration logic of the *klimaaktiv* framework. An overview of the model structure is shown in Figure 24.

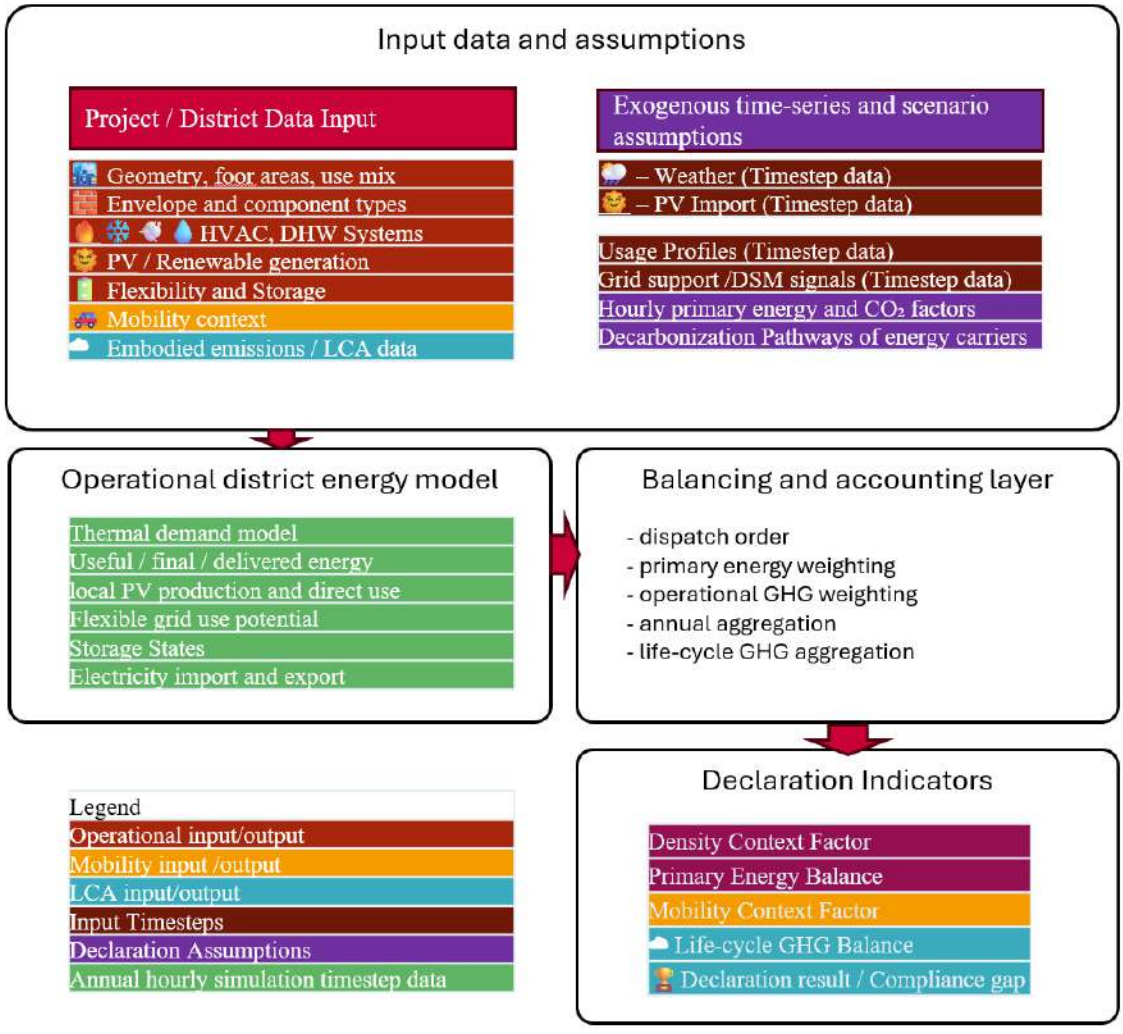


Figure 24. Simulation model overview

The simulation framework presented in [Schneider-2024 c] is designed to support declaration-ready PED assessment by prioritizing temporal system behaviour over detailed system and spatial representation. This choice reflects the specific requirements of PED assessment, where system-level interactions over time are decisive for performance, whereas fine-grained spatial effects typically refine results without altering their qualitative interpretation. Figure 25 shows an overview of the modelled energy system components and system boundary for balancing:

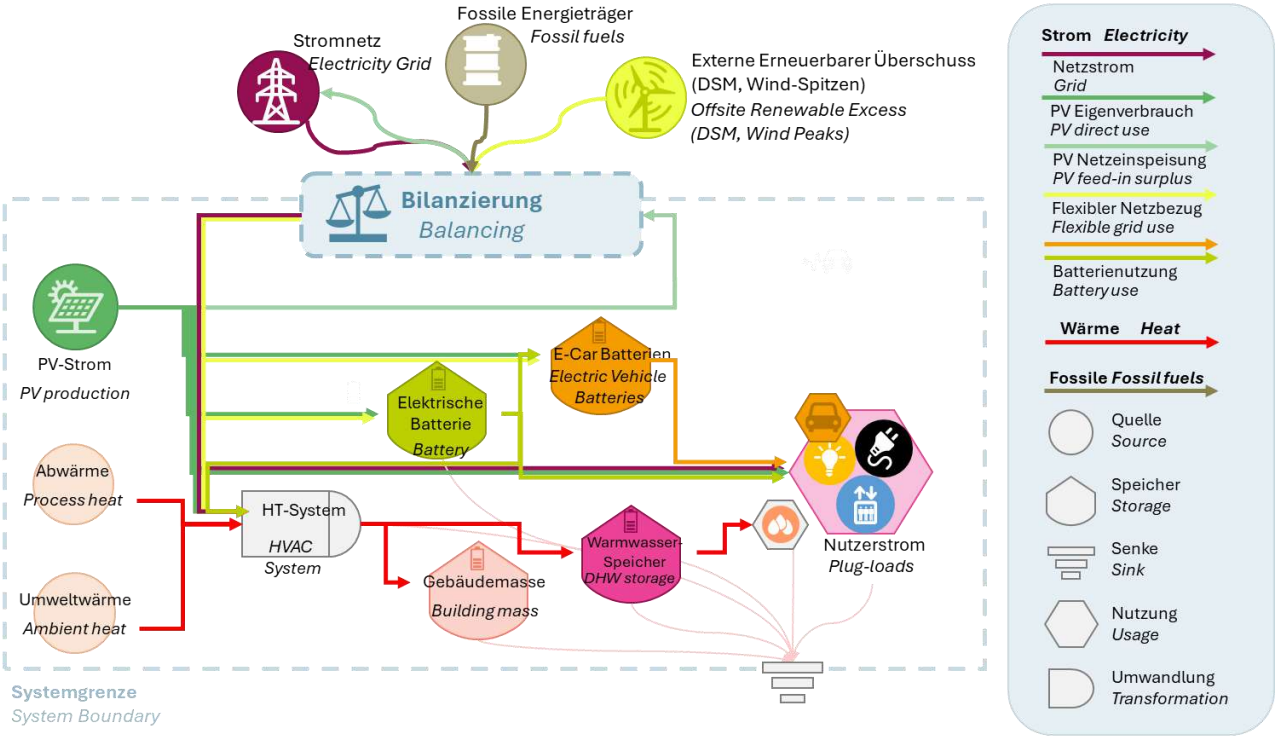


Figure 25. Components of the implemented PED simulation and modelling framework

Spatially, the district is represented as an aggregated system. Buildings, infrastructure, and storage are condensed into representative elements described by abstracted geometric and typological parameters such as gross floor area, floor area ratio, usage categories, and typology shares sufficient to derive space heating and cooling demand, domestic hot water demand, electricity demand profiles, and density-dependent context factors. Explicit building geometry and fine-grained spatial effects, including orientation-dependent solar gains, inter-building shading, or room-level thermal behaviour, are omitted. Instead, two thermally homogeneous zones – with and without active cooling – are modelled with energy flows shown in Figure 26.

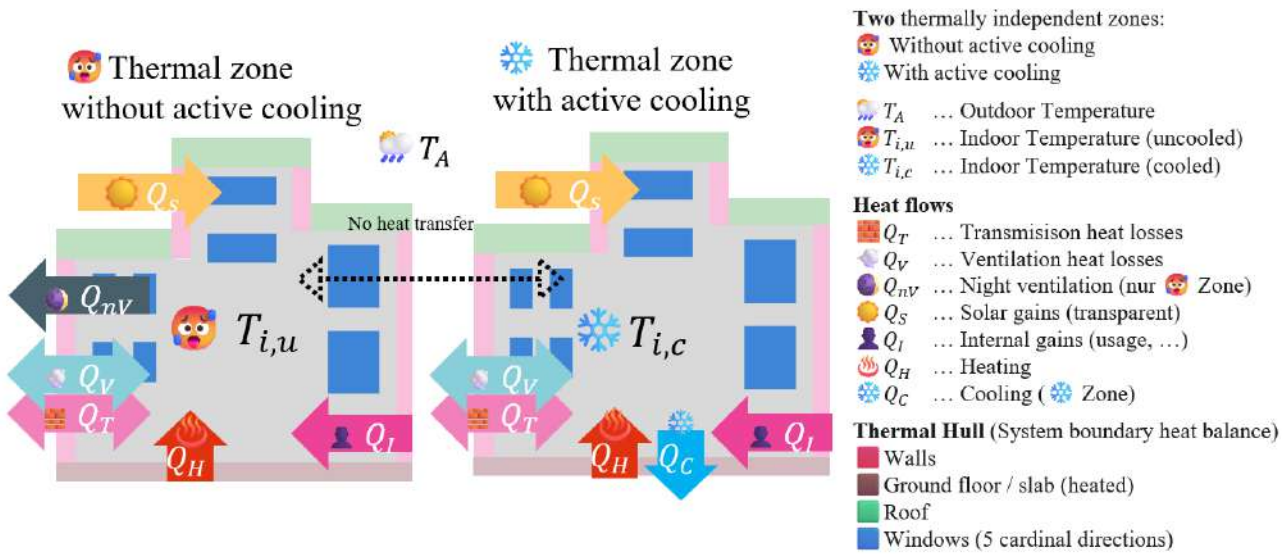


Figure 26. Energy flows of the two-zone thermal model

Temporally, the model operates on an hourly time step over a full annual cycle. This resolution is required to capture renewable generation variability, storage charging and discharging, interactions between building demand and mobility charging, and the application of time-dependent primary energy and emission factors. For PED assessment, these temporal effects dominate outcomes, whereas additional spatial detail primarily affects absolute values without altering qualitative conclusions. The resulting modelling approach enables full-year dynamic assessment while maintaining short runtimes suitable for iterative scenario analysis, as demonstrated in Schneider (2024b). The effectiveness of this trade-off is demonstrated in the modelling framework documented in [Schneider-2024 b], where simulation runtimes remain below 10 seconds per scenario while retaining the ability to represent flexibility and sector coupling effects that stationary methods cannot capture.

Within the aggregated spatial representation, the district energy system is modelled as an integrated thermoelectric system. It includes space heating, cooling, domestic hot water, and electricity demand; local renewable electricity and heat generation; electrical and thermal storage; and residual interfaces for external energy imports and exports. The level of system detail is deliberately reduced to what is necessary to represent sector coupling and flexibility effects relevant to PED assessment.

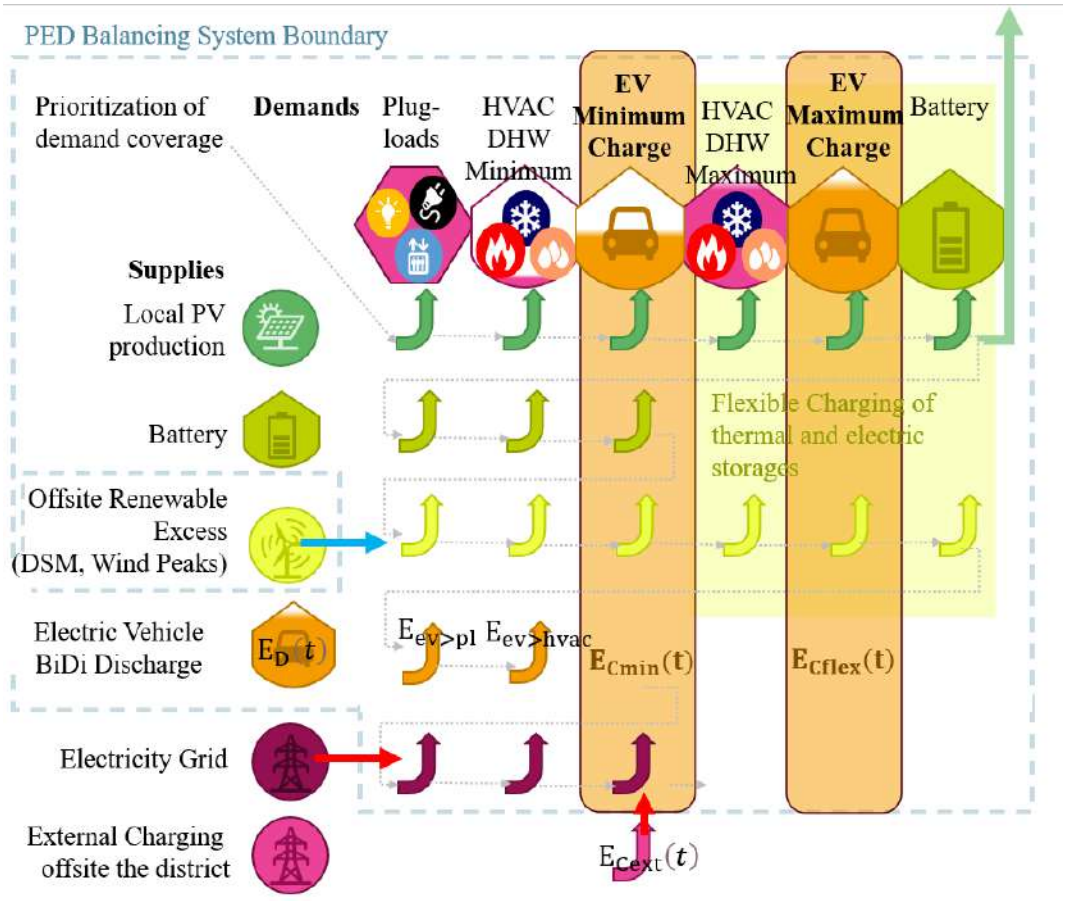


Figure 27. Energy dispatch sequence including provisioning of flexibility

As shown in Figure 27, energy flows are resolved using a fixed and transparent dispatch sequence rather than optimization-based control. Local renewable generation is prioritized for direct use, followed by storage operation, the use of external flexibility options, and finally residual imports or exports. This rule-based dispatch ensures computational efficiency for hourly full-year simulations and avoids reliance on optimization assumptions – such as perfect foresight, price-based control, or rational actor behaviour – that are typically unjustified in early planning and declaration contexts. Transparency and interpretability are prioritized at this stage, leaving detailed system optimization to subsequent planning stages.

External energy flexibility opportunities are represented by using discrete, time-explicit signals that are sourced from data such as historical wind power availability, residual loads or price signals.

3.3.4. Mobility modelling implementation in the PED framework

[Schneider-2025 a] presents how mobility is implemented in the PED assessment framework as an aggregated, hourly, full-year energy model for everyday motorized individual mobility (MIM) that is tightly coupled to the district energy simulation. The implementation is designed to be computationally lightweight, declaration-compatible, and consistent with limited data availability in

early planning stages, while still capturing the dominant temporal and system-level effects of electric mobility on district energy balances.

The model does not simulate individual vehicles as agents. Instead, it represents electric vehicles (EVs) in aggregate by usage type (residential, work, education, retail), which allows the integration of mobility without exploding model size or computation time. For each usage, the expected number of "averagely mobile" persons is derived from floor-area–based census indicators, combined with region-specific annual mileage data. These annual mileages are adjusted for EV penetration rates, simultaneity factors, and optional district-level mobility measures that reduce motorized travel. This yields an annual EV mileage demand per usage, which is then distributed over the year using standardized hourly mobility and presence profiles.

A central implementation feature is the explicit hourly tracking of EV presence at the district. For each usage type, the model determines the probability that vehicles are onsite, offsite, leaving, or returning in each hour of the year, based on empirically derived weekly profiles, as shown in Figure 28. Only vehicles that are present at the district are eligible for charging or discharging, while driving energy consumption is attributed to offsite vehicles. This separation allows the model to realistically constrain charging availability without simulating individual trips or routes.

Hourly SoC changes account for six processes: self-discharge, driving energy consumption, charging to a minimum SoC, optional charging to maximum SoC when surplus energy is available, bidirectional discharging to the district (for residential use), and SoC exchange due to vehicles leaving or returning. This structure ensures internal energy consistency while avoiding agent-based complexity.

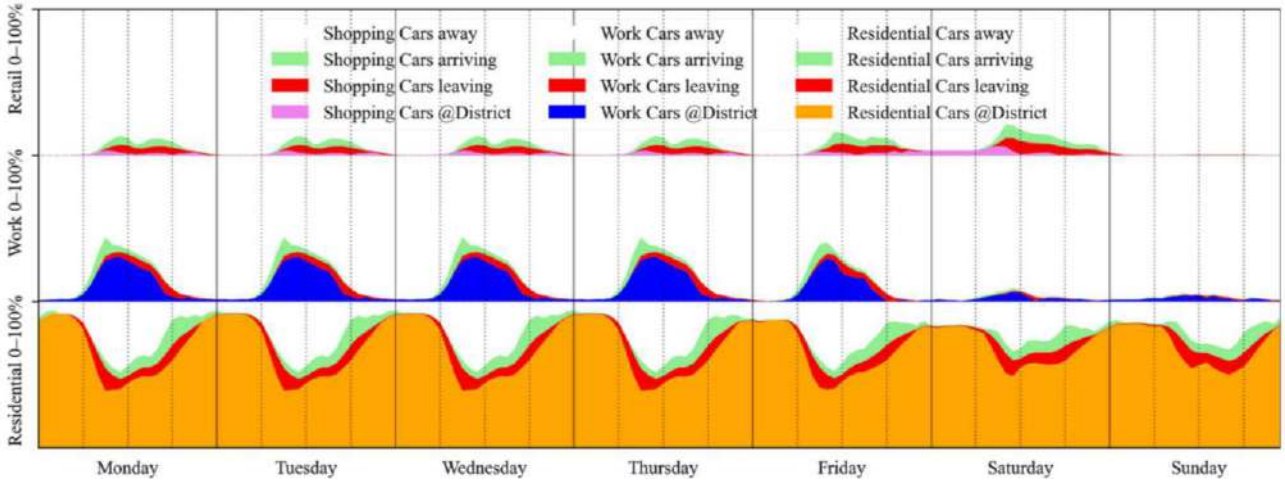


Figure 28. Average EV onsite availability for operationalized usages [Schneider-2025 a]

Charging and discharging follow a rule-based, fixed dispatch logic aligned with the overall PED energy dispatch sequence. Charging to minimum SoC is prioritized to ensure mobility functionality,

while additional charging to full capacity occurs only when local or external renewable surpluses are available. Bidirectional discharging is restricted to residential vehicles and only activated when district electricity demand is uncovered and vehicle SoC remains above a defined minimum threshold. Offsite charging is triggered if the average SoC of offsite vehicles drops below a minimum value, representing charging at external infrastructure.

3.3.5. Life-Cycle Assessment and GHG Accounting

Life-cycle GHG emissions are integrated into the simulation framework primarily as a model input following the methodology introduced in [Schneider-2025 d], which includes:

- embodied emissions associated with buildings (categorized in Figure 29) and energy infrastructure,
- operational emissions from energy import and export,
- cumulative emissions over the defined assessment horizon.

Results are evaluated against **floor-area-specific GHG budgets** derived from national carbon budgets, enabling direct assessment of Paris compatibility. Life-cycle impacts are calculated alongside energy balances, ensuring that trade-offs between operational efficiency and construction-related emissions are made explicit (IBO, 2026; Thomas Zelger, et al., 2024).

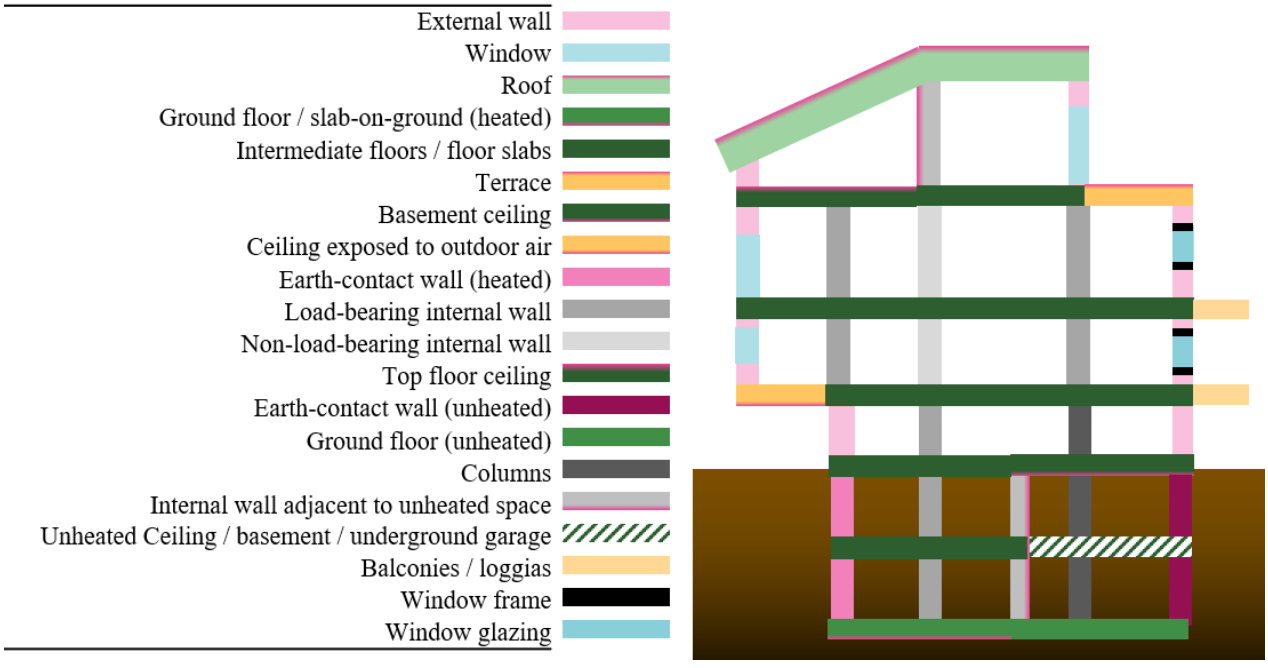


Figure 29. Categorization of construction components

3.3.6. Outputs, Reporting, and Role as a Planning Decision Support Tool

The simulation framework is designed to produce ready-made outputs tailored to PED assessment and declaration contexts shown in Figure 30, including:

- standardized energy and GHG balances,
- context-corrected PED target evaluations,
- scenario comparisons across design alternatives,
- visual diagrams and tables suitable for communication with non-expert stakeholders.
- Presets, defaults, and parameter libraries enable rapid model setup and iteration, supporting exploratory planning.

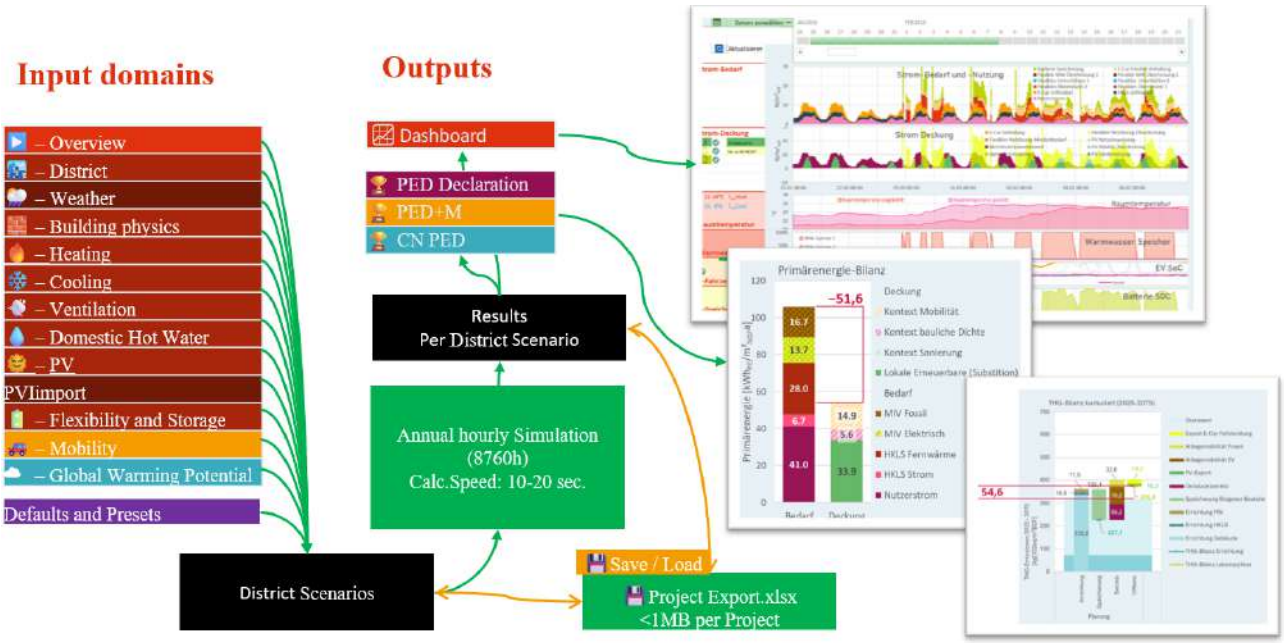


Figure 30. Overview of the assessment tool structure

In this sense, the framework functions as a planning decision support tool, bridging the gap between strategic target setting and later-stage detailed modelling. It provides structured, comparable evidence to inform design direction, while leaving optimization and control refinement to subsequent project phases. While the framework represents multiple energy carriers, conversion processes, and storage elements within a unified system boundary, its primary contribution lies not in advancing multi-energy system theory, but in making such systems assessable, comparable, and usable in early PED planning.

3.4. Application

The quantitative PED framework developed in Sections 3.2 and 3.3 is conceived as an **actionable assessment instrument** rather than a purely analytical construct. It is designed to support concrete planning, policy, and research tasks, but its application does not follow a single, uniform mode. In practice, the same definition and modelling logic is applied under different objectives, constraints, and degrees of institutional formalization. This chapter documents two complementary application contexts.

Analytical applications use the framework to explore feasibility, boundary effects, and context sensitivity in concrete districts, primarily to advance methodological understanding.

Institutional applications operationalize the framework as a declaration or certification instrument with explicit rules, governance procedures, and compliance statements. Institutional declaration constitutes the primary application, as it represents the most stringent test of operationalizability under legal and administrative constraints. Analytical applications play a supporting role by exercising the framework in contexts where declaration is not intended or methodologically appropriate, such as comparative case studies or exploratory scenario analysis.

Beyond demonstrating usability, the documented applications collectively serve as **validation of the PED framework**. The evidence base comprises declaration-aligned implementations, comparative European case studies, scenario-based district analyses, and project demonstrators with heterogeneous data quality.

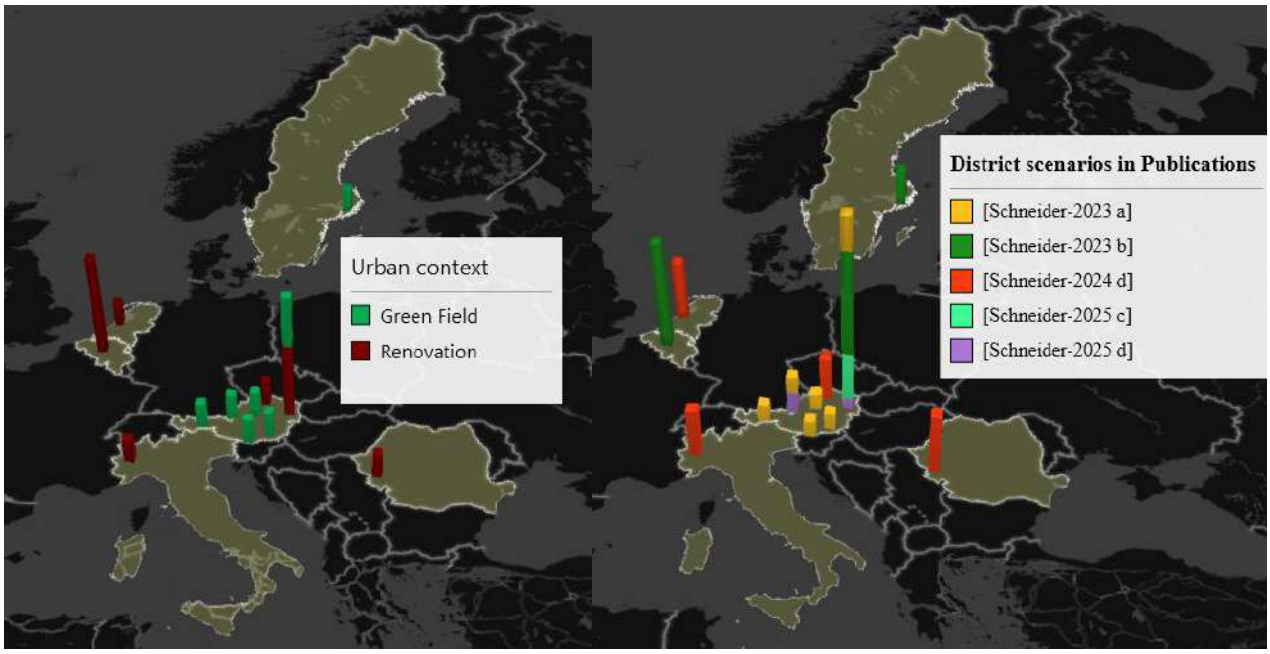


Figure 31. Investigated districts and scenarios

Figure 31 provides a geographic overview of all PEDs and district-scale case studies investigated in this thesis. The map distinguishes between dominant urban context (greenfield, brownfield, renovation). The spatial distribution illustrates the diversity of climatic, regulatory, and urban contexts in which the PED framework has been applied, ranging from dense inner-city renovation areas to low-density peripheral developments across multiple European countries.

Table 5 summarises the set of investigated PEDs and analytical case studies used throughout this chapter. For each case, the table reports location, urban context, dominant use mix, scale, application type (institutional declaration or analytical study), data availability, and the specific role of the case in validating the PED framework. The table highlights that, despite heterogeneity in context and data quality, all cases are assessed using the same PED definition and modelling logic, enabling systematic comparison across applications and supporting the consistency claims underlying the framework.

Table 5. Overview of investigated district case-studies

District	City	Country	Source	Research Project	Urban context	Scale	FAR	Scenarios	Dominant usage
PE Campus	Vienna	Austria	[Schneider-2023 a]	PE-Campus	Renovation	District	5.80	3	Secondary Edu.
Pilzgasse	Vienna	Austria	[Schneider-2023 a]	ZQ AT	Greenfield	Block	4.58	3	Mixed-use
Graz SC Mitte	Graz	Austria	[Schneider-2023 a]	ZQ AT	Greenfield	District	1.96	3	Mixed-use
Gewin Gneis	Salzburg	Austria	[Schneider-2023 a]	ZQ AT	Greenfield	District	1.05	3	Residential
Am Bichl III	Innsbruck	Austria	[Schneider-2023 a]	ZQ AT	Greenfield	Block	0.85	3	Residential
An der Glan	Klagenfurt	Austria	[Schneider-2023 a]	ZQ AT	Greenfield	District	0.75	3	Residential
Aichinger	Steyr	Austria	[Schneider-2023 a]	ZQ AT	Greenfield	Building	0.25	3	Residential
Royal Seaport	Stockholm	Sweden	[Schneider-2023 b]	Cities4PEDs	Greenfield	District	2.74	7	Mixed-use
Seestadt Aspern Nord	Vienna	Austria	[Schneider-2023 b]	Cities4PEDs	Greenfield	District	2.70	19	Mixed-use
Foyer Laekenois	Brussels	Belgium	[Schneider-2023 b]	Cities4PEDs	Renovation	District	1.3	5	Residential
Masui	Brussels	Belgium	[Schneider-2023 b]	Cities4PEDs	Renovation	District	2.9	5	Residential
Maximilien	Brussels	Belgium	[Schneider-2023 b]	Cities4PEDs	Renovation	District	4	5	Mixed-use
Vergotedok	Brussels	Belgium	[Schneider-2023 b]	Cities4PEDs	Renovation	District	1	5	Mixed-use
Alliiertenviertel	Vienna	Austria	[Schneider-2025 c]	WieNeu+	Renovation	District	3.75	9	Residential
Großschönau	Großschönau	Austria	[Schneider-2024 d]	SimplyPositive	Renovation	District	0.15	8	Residential
Lunca Pomostului	Reşiţa	Romania	[Schneider-2024 d]	SimplyPositive	Renovation	District	0.9	11	Residential
Fiat Village	Sett. Torinese	Italy	[Schneider-2024 d]	SimplyPositive	Renovation	District	1.49	9	Mixed-use
Beursplein	Amsterdam	Netherlands	[Schneider-2024 d]	SimplyPositive	Renovation	District	3.29	11	Mixed-use
RugInnovation	Vienna	Austria	[Schneider-2025 d]	-	Renovation	Building	1.14	2	Mixed-use
Südtiroler District	Salzburg	Austria	[Schneider-2025 d]	-	Renovation	District	3.75	4	Mixed-use
Elementary School	Vienna	Austria	[Schneider-2025 d]	-	Greenfield	Building	0.8	1	Primary Edu.

3.4.1. District case studies

[Schneider-2023 a] showed, using seven Austrian district typologies spanning very low to very high density (FAR ≈0.23 to 5.80; from detached housing to dense urban education and mixed-use), that the ability to achieve a positive annual district energy balance is strongly density-dependent: As can be seen in Figure 32, high-FAR districts could not reach positivity even under "maximum ambition" packages of on-site renewables, flexible operation, and partial electrification/reduction of private motorized mobility ("With additional Measures" points below 45° break-even of supply and demand), while low-density districts could reach or even exceed positivity with comparatively modest measures. The case studies further showed how successive accounting extensions (i.e. adding a density context factor, then including motorized individual mobility energy, and finally adding a mobility-related context compensation) systematically shift balance points in demand-supply space and can invert apparent PED outcomes, illustrating that comparability across typologies requires explicit treatment of density and mobility context in the balance logic.

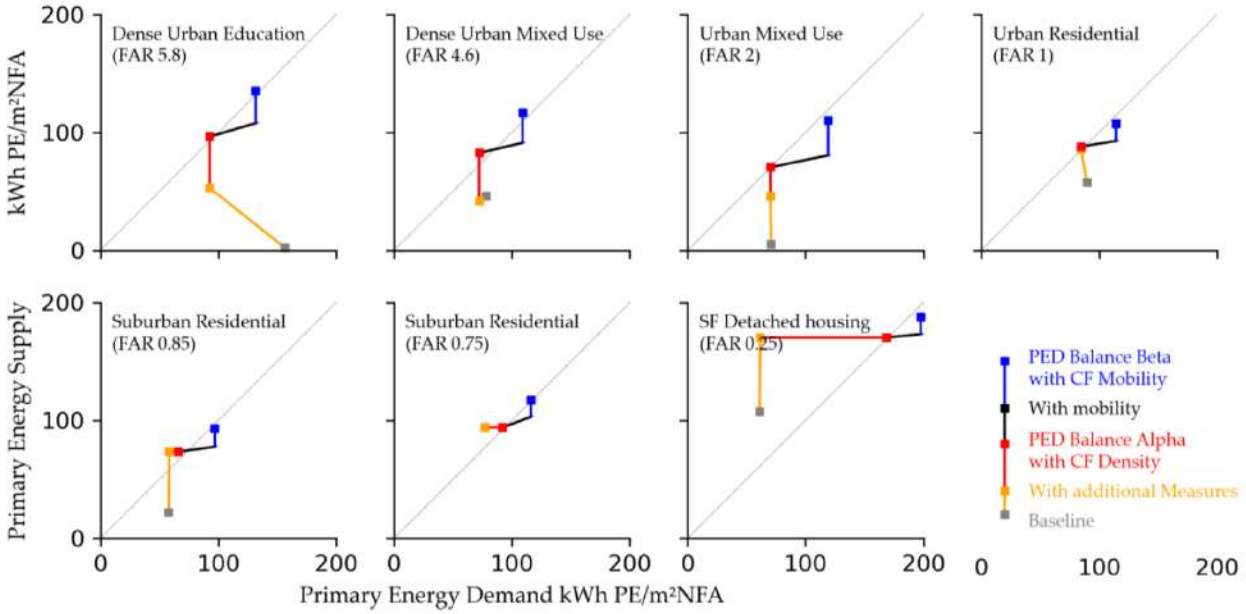


Figure 32. Annul operational energy balance for Austrian districts of varying density

[Schneider-2023 b] concluded based on three Cities4PEDs example districts, that dense urban projects can only be assessed as "positive" under a primary-energy PED definition if roof-PV limits are made explicit and compensating context factors (density and, when included, mobility) are applied consistently:

Operational PE-balances for the *Stockholm Royal Seaport* ("SRS", FAR 2.74, mixed-use) obtained in [Schneider-2023 b] shown in Figure 33, showed that even substantial PV deployment (25–90% roof utilization) is insufficient to make the district positive without the density context factor (red

hatched area), because roof-mounted PV cannot cover plug-load and lighting demand. Moderate thermal flexibility (≈0.5 K) improves PV self-consumption and reduces feed-in but does not change the fundamental roof-area constraint. Extending the boundary to include motorized individual mobility shifts demand strongly upward; achieving a positive balance then depends on high EV shares and the mobility context credit derived from sectoral renewable allocation.

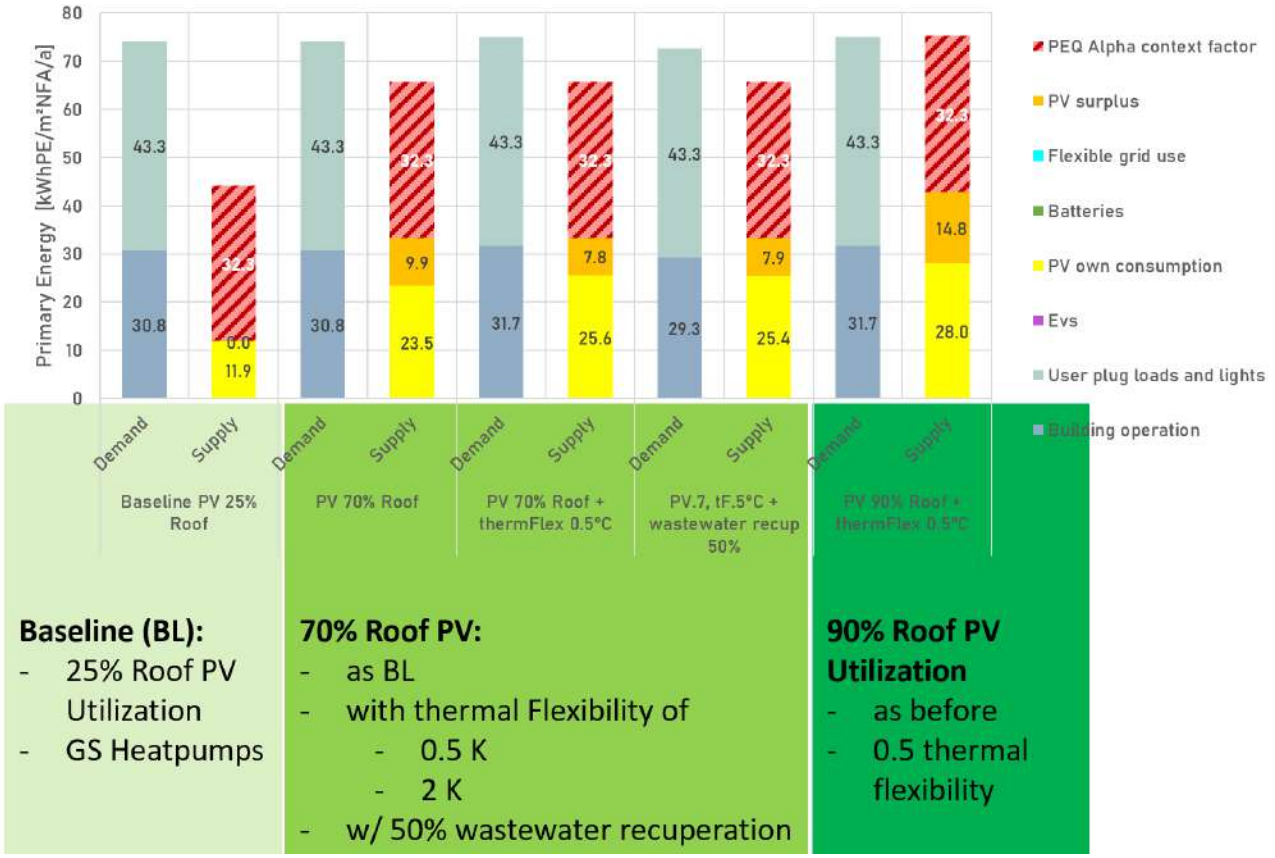


Figure 33. Stockholm Royal Seaport PE-balance for PED Alpha (Operation) [Schneider 2023 b]

For the Brussels *North District* brownfield renovation context [Schneider-2023 b],

Figure 34 showed that PE import/export performance is strongly correlated with density across representative blocks, and that the gap to an "operation-only" PED balance varies systematically by typology. The simplified block-based test was used to demonstrate that brownfield/heritage constraints can be interpreted as a structural limitation of on-site supply potential, motivating a dedicated contextual treatment (i.e., an indicative "heritage" effort-sharing factor) when transferring the definition to refurbishment districts.

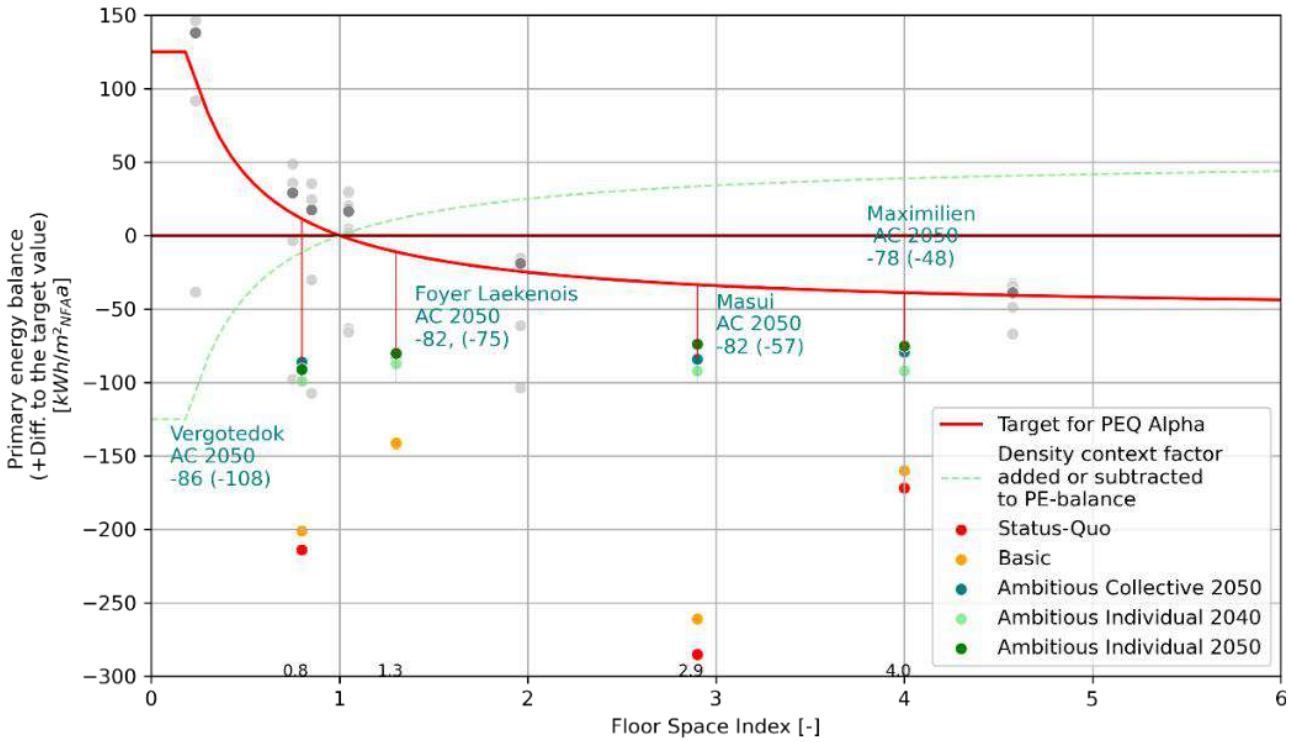


Figure 34. Brussels North District PE-balance [Schneider 2023 b]

Figure 35 in [Schneider-2023 b] showed for Vienna *Seestadt Aspern North* (FAR 2.70; mixed-use, modelled via representative block F12) that operational positivity (PED Alpha) becomes feasible either by PV expansion beyond roof-only limits (≥105% roof-equivalent via façades) or by combining ~70% roof PV with flexible use of offsite renewable surpluses ("WPS"), while thermal flexibility mainly increases self-consumption and reduces summer surplus. When mobility is included (PED Beta), the model indicates that positivity requires high EV shares (≈60%+ in the shown configurations) and/or additional demand-side measures; EV charging is also implemented as a flexibility resource that can shift demand into renewable oversupply periods, improving the effective balance under the same installed PV capacity.

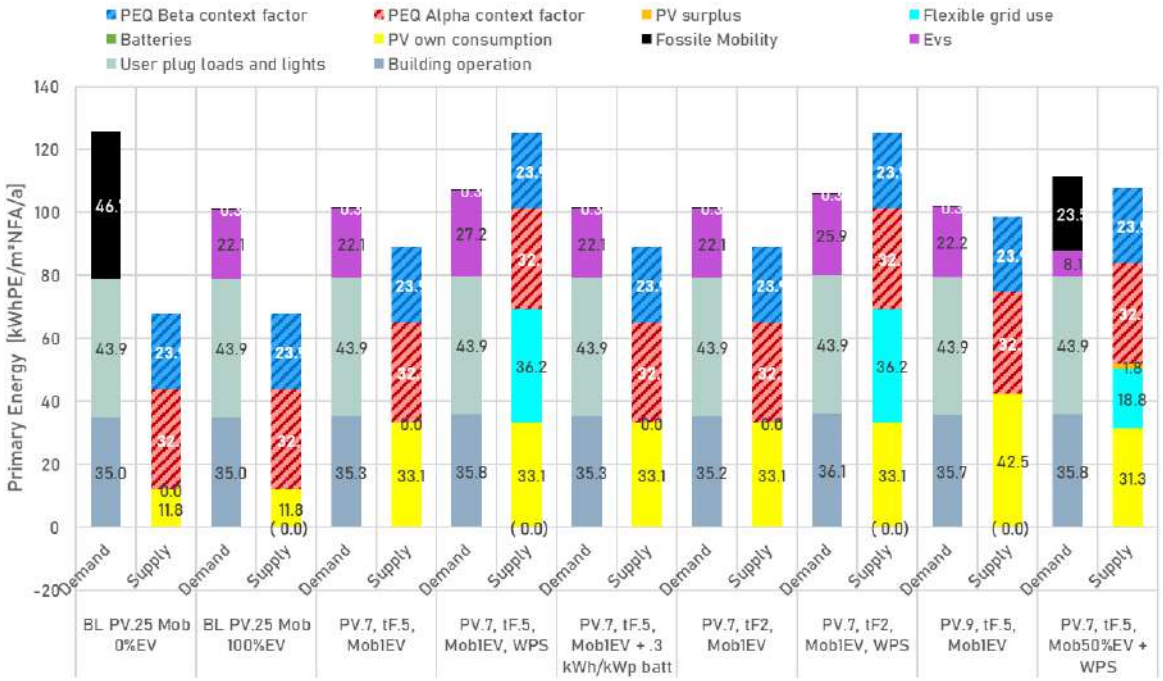


Figure 35. Vienna Seestadt Aspern North PE-balance Beta (Operation and Mobility) [Schneider-2023 b]

[Schneider-2024 c] showed, using the historic Viennese "Alliiertenviertel" (68 buildings, ~90% residential, mixed construction periods, ~20% renovated), that the PED-oriented Excel-based district energy model produces heating-demand results that are well correlated with a state-of-the-art urban building energy model (City Energy Analyst, CEA) when identical geometry and envelope assumptions are applied, as shown in Figure 36. For the existing building stock, both models align closely with statistically typical age- and typology-specific benchmarks, whereas for the renovation cases, the Excel results were actually closer to the statistical renovation estimates for the building types than the more detailed CEA model results, confirming that the aggregation and simplification approach preserves first-order demand characteristics at district scale.

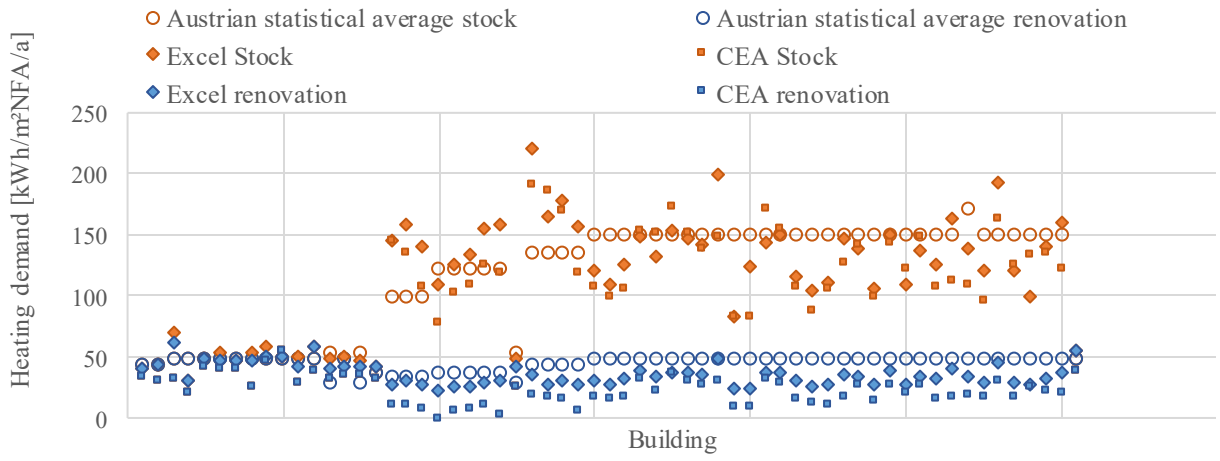


Figure 36. Assessment tool heat demand comparison of 68 buildings with CEA and statistical data for "stock" and "renovation" scenarios [Schneider-2024 c]

[Schneider-2024 d] showed, using four European retrofit case studies, that large reductions in district heating and cooling demand (≈40–80%) are achievable across climates, but that attaining a positive primary energy balance depends primarily on district density and flexibility, rather than climate severity alone. Across all districts, roof-mounted PV contributes up to ~90 kWhPE·m^{-2}NFA, while flexible HVAC operation enables an additional ~20 kWhPE·m^{-2}NFA of renewable use by shifting demand away from carbon-intensive grid periods (see Figure 37). Additionally, [Schneider-2024 d] showed that PED feasibility in retrofit contexts requires density-sensitive context factors and their extension (e.g., modified CFD formulations and renovation context factors), as well as flexibility measures, while heritage-protected districts represent a structural boundary where PED targets may need alternative governance treatment rather than stricter technical requirements.

[Schneider-2024 d] showed, through a comparative methodological analysis of four European retrofit districts, that primary energy balance outcomes are dominated by density effects rather than climate, making the treatment of density a decisive modelling choice. Across all cases, renovation and PV deployment alone shift districts closer to balance (see Figure 37), but only the inclusion of explicit context factors systematically repositions scenarios into the PED-compliant domain, as can be seen in Figure 38. Methodologically, this demonstrated that the density context factor (CFD) is essential to ensure effort sharing between low- and high-density districts: without CFD, dense urban districts systematically fail the positive energy criterion despite high absolute renewable deployment, while low-density districts appear unrealistically over-compliant. However, the analysis also showed that a single FAR-based CFD formulation over-penalizes very low-density retrofit contexts, where PV potential scales poorly with plot area.

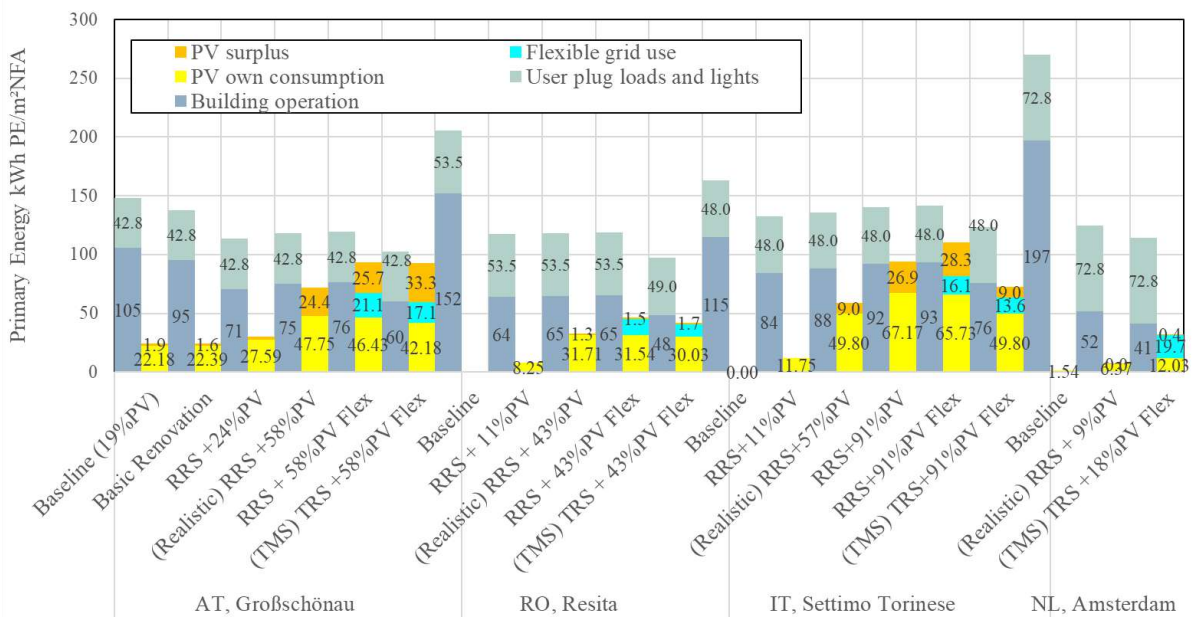


Figure 37. Annual primary energy balance by district [Schneider-2024 d]

To address this, [Schneider-2024 d] refined the density treatment by introducing combined context formulations that decouple PV potential from plot area alone and partially cap virtual compensation effects, as shown in Figure 38. The results showed that extended formulations (CFD* and CFD combined with a renovation context factor, CFR) preserve the intended density-offsetting effect for compact urban districts while preventing infeasible PV requirements for rural or heritage-constrained areas. This refinement enables PED feasibility across the full density spectrum without collapsing the normative ambition of dense urban districts.

Overall, the case studies established the density context factor not as a corrective add-on but as a structural modelling component, whose formulation directly determines whether PED pathways are interpretable, comparable, and institutionally usable across heterogeneous urban contexts.

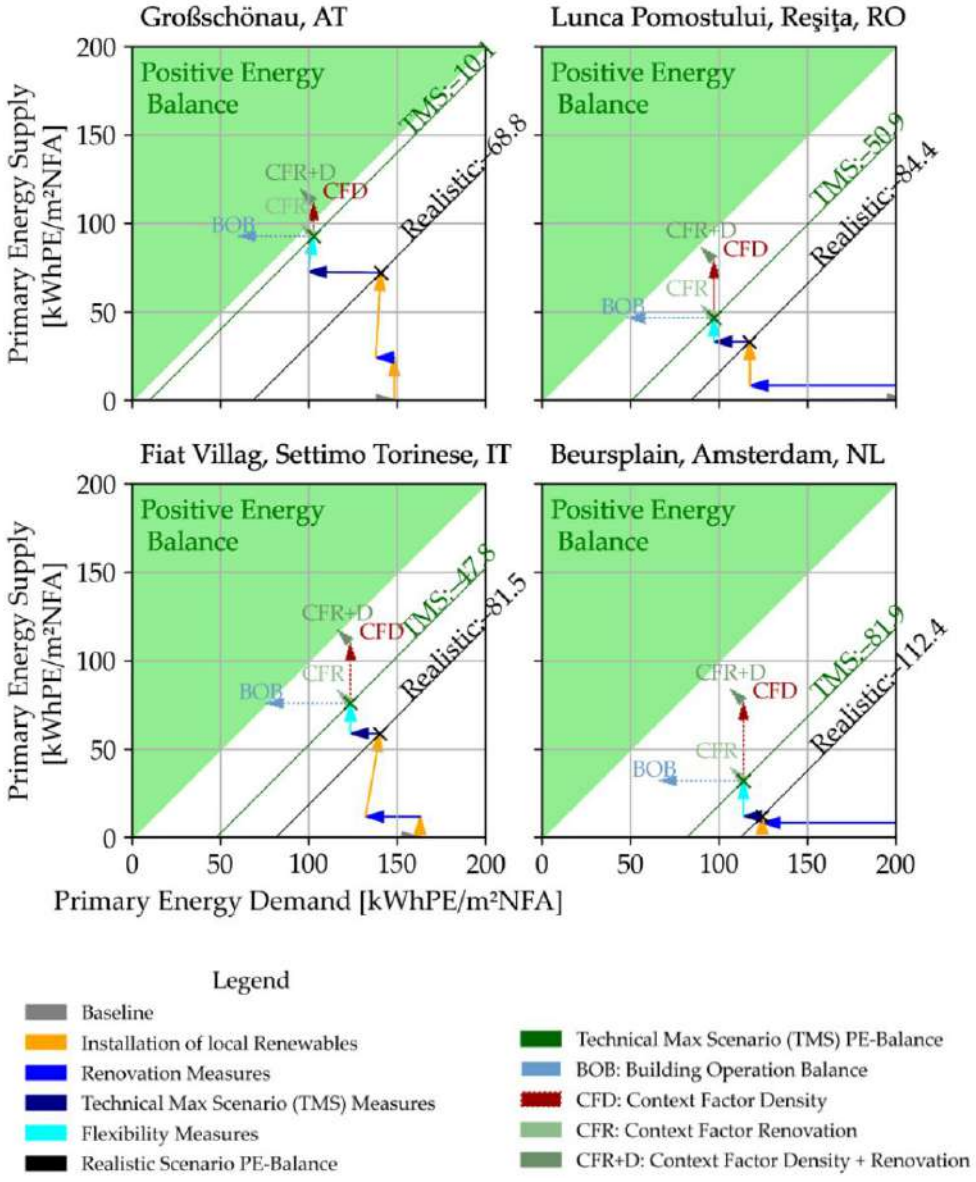


Figure 38. Primary energy balance of focus districts with revised context factors

[Schneider-2025 a] showed that explicitly modelling electric-vehicle-based motorized individual mobility (EV-MIM) at district level fundamentally changes feasible PV sizing and utilization across different building uses. Using four representative usage types (residential, office, retail, education, illustrated in Figure 39), the study demonstrated that EV mobility demand (≈53–205 kWh/m²$_{NFA}$·a) acts as a strong additional sink for onsite renewables, raising PV self-utilization far beyond levels achievable with building operation alone: up to ~80% for office and ~90% for retail uses with full EV integration and PV systems 5−10 times the required size.

The case studies further showed that usage mix and temporal availability of vehicles are decisive: non-residential buildings benefit strongly from daytime EV presence and from winter wind-driven flexibility signals, while residential districts gain most from bidirectional charging, which effectively substitutes stationary batteries. Allowing V2B discharge to 60% SoC reduced annual grid electricity demand by ~28% and increased PV self-utilization to ~95%, albeit at higher battery cycling.

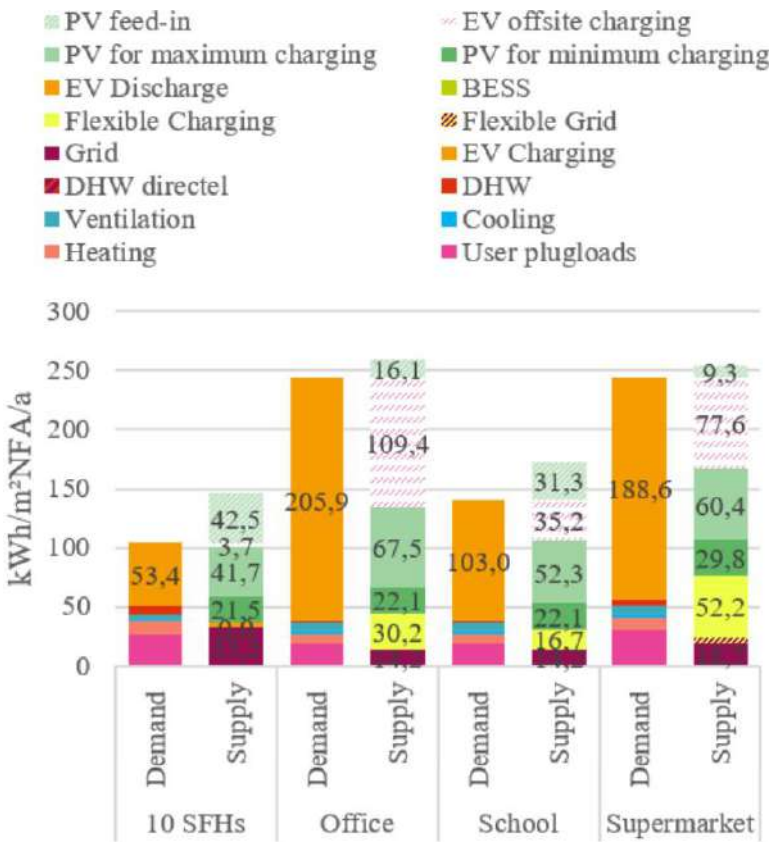


Figure 39. Annual electricity demand (left) and supply (right) for usages residential, work, education and retail

[Schneider-2025 c] showed that building heat demand correlates strongly with compactness across construction periods, and that the marginal effect of renovation decreases for newer buildings, allowing typology-based ranges of expected performance to be derived for rapid stakeholder reference, as can be seen in Figure 40.

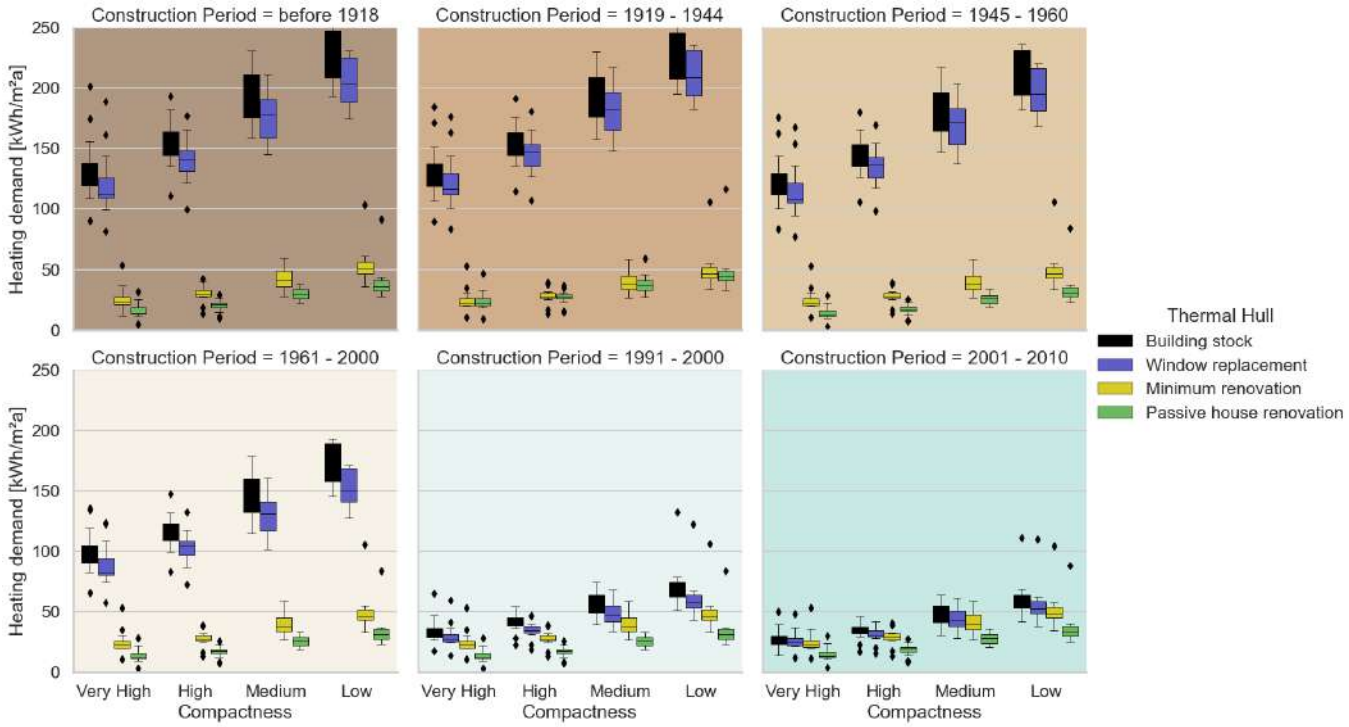


Figure 40. Heating demand per compactness, construction period and thermal quality of building.

Aggregated scenario results demonstrated that Vienna's municipal target of a 30 % reduction in heating energy by 2040 is met even under business-as-usual retrofit rates, while full climate neutrality is not achieved due to externally fixed grid and district-heating emission factors. By presenting results in a GIS-based format, the study illustrated the suitability of CEA as a municipal decision-support tool for exploring spatially explicit renovation impacts without requiring building-specific engineering models.

[Schneider-2025 d] showed that applying a declaration-ready, budget-based, hourly life-cycle assessment to six baseline districts yields strongly heterogeneous compliance outcomes, as shown in Figure 41 and Figure 42: Only D2 (school campus) and D4 (ecologic renovation with GSHP/TABS and mobility measures) meet the L3 GHG budget and maintain a positive PE-balance, while several districts pass one criterion but fail the other (e.g., D3 meets L3 but fails PE due to district heating constraints), and some overshoot the GHG budget despite positive PE (notably D6, dominated by embodied and mobility-related loads). Across the comparison, failures are typically driven by front-loaded embodied emissions and mobility, whereas PE failures in otherwise compliant districts are mainly linked to limited supply-side flexibility and unfavourable conversion factors for district heating systems.

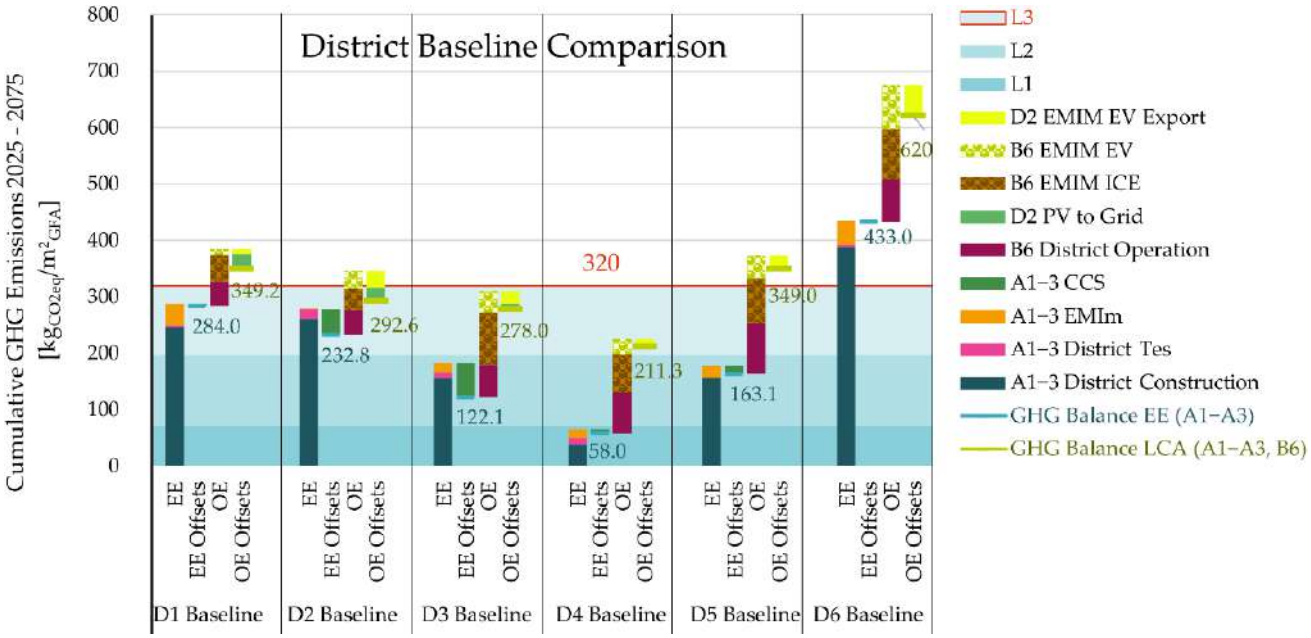


Figure 41. Baseline life-cycle GHG balance by district [Schneider-2025 d]

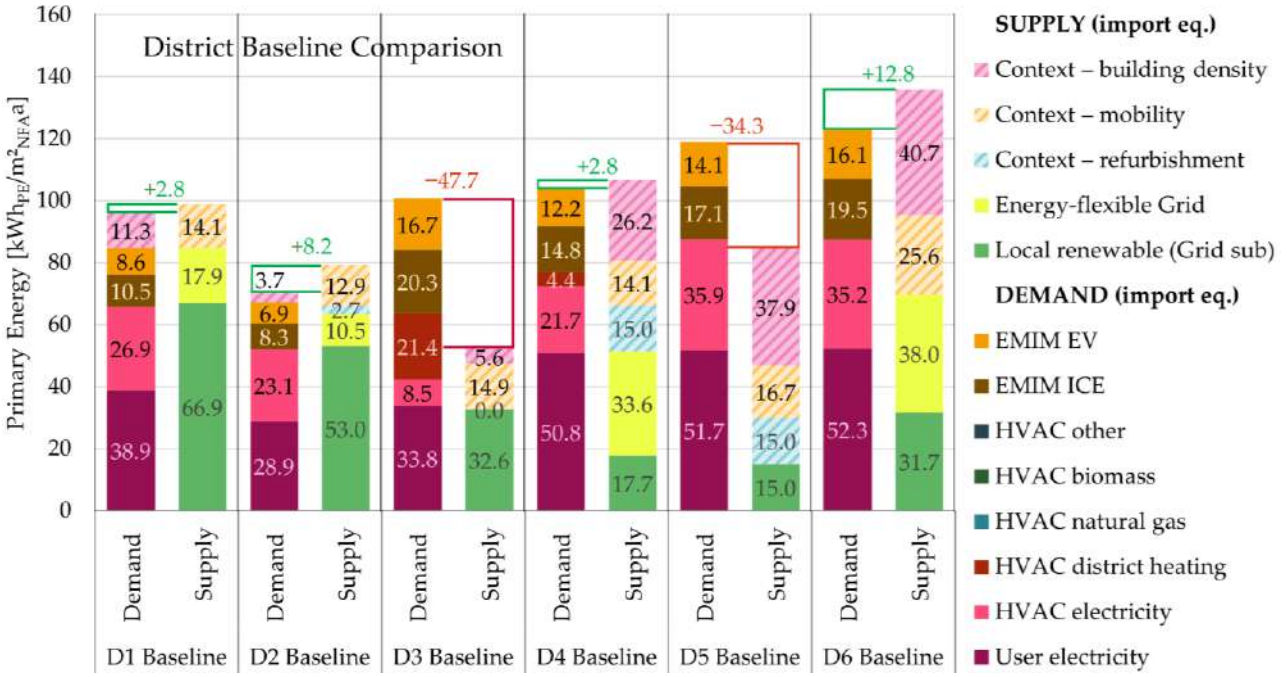


Figure 42 Annual PE-balance by district [Schneider-2025 d]

District 3 showed the redevelopment case study (urban medium-density residential) that ecological construction is a necessary condition for GHG budget compliance (see Figure 43), but not sufficient for climate-neutral PED status (Figure 44): fossil-intensive construction exceeds the GHG emission budget (L3) regardless of operation, while ecological variants stay below L3 and (only when combined with substantial PV expansion and demand-side mobility measures) can reach precautionary compliance levels and achieve a positive PE balance. The case also showed that achieving a net-positive PE balance remains difficult under district heating even with favourable emissions assumptions, whereas on-site GSHP supply combined with PV most consistently enables simultaneous compliance with both the cumulative GHG budget and PE criterion.

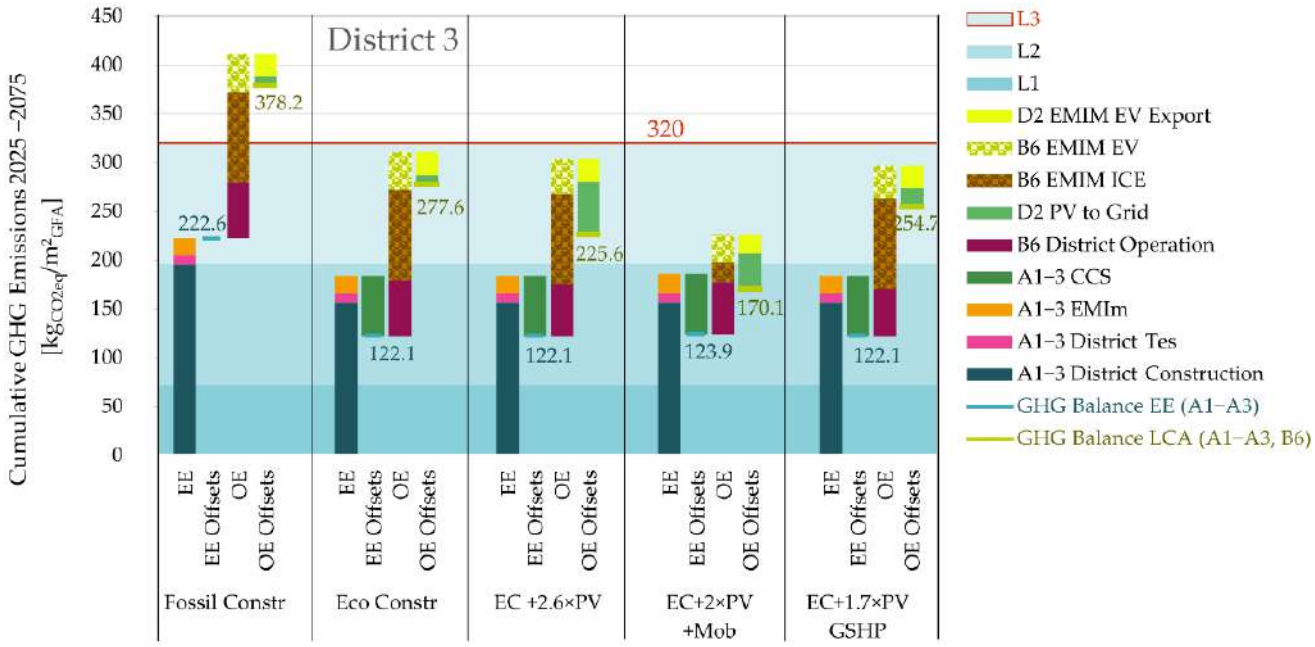


Figure 43. Cumulative GHG LCA emissions of District 3 [Schneider-2025 d]

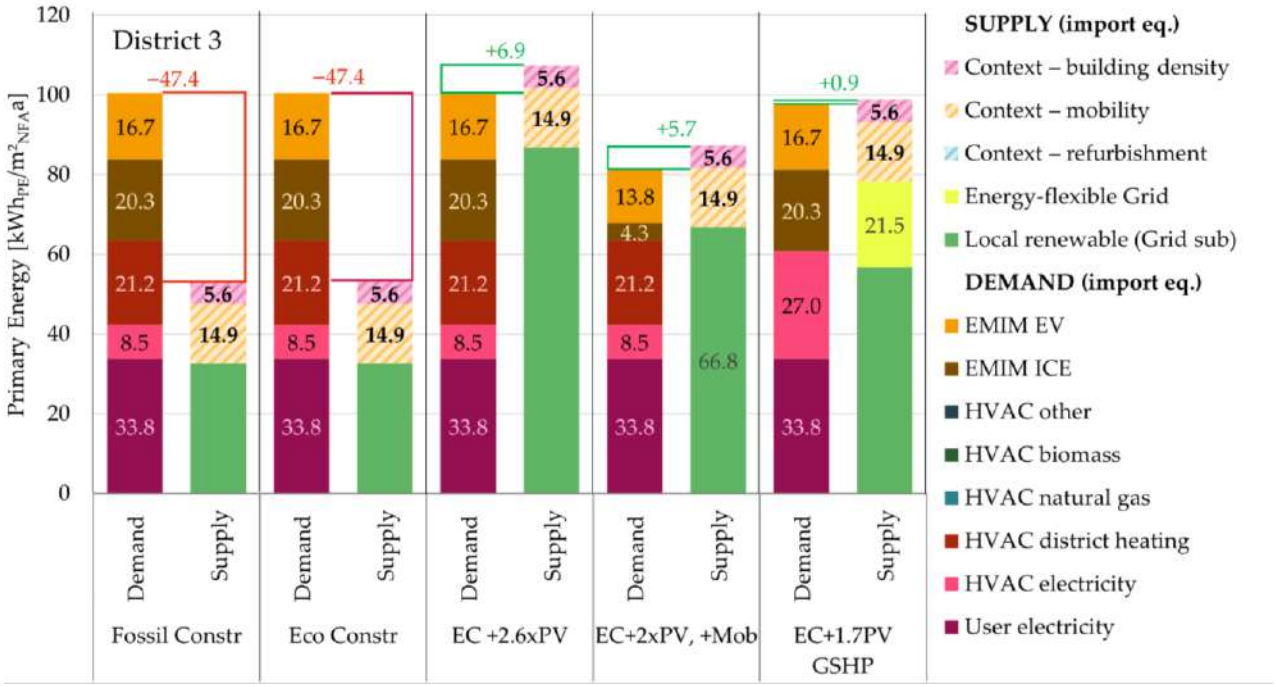


Figure 44. Annual PE-balance of District 3 [Schneider-2025 d]

District 4 showed that the choice of temporal emission factors materially changes operational accounting and makes flexibility effects measurable (as seen in Figure 45): monthly or annual factors systematically under-represent the winter import penalty because districts typically import electricity in high-intensity periods and export PV in lower-intensity periods. Switching to hourly factors and applying rolling-average GHG signals enables flexible loads (TABS), storage (BESS), and bidirectional EV charging (V2B/V2G) to target low-intensity hours, reducing effective

import/charging emission intensity substantially and lowering net operational emissions despite slightly higher electricity throughput from storage and charging losses.

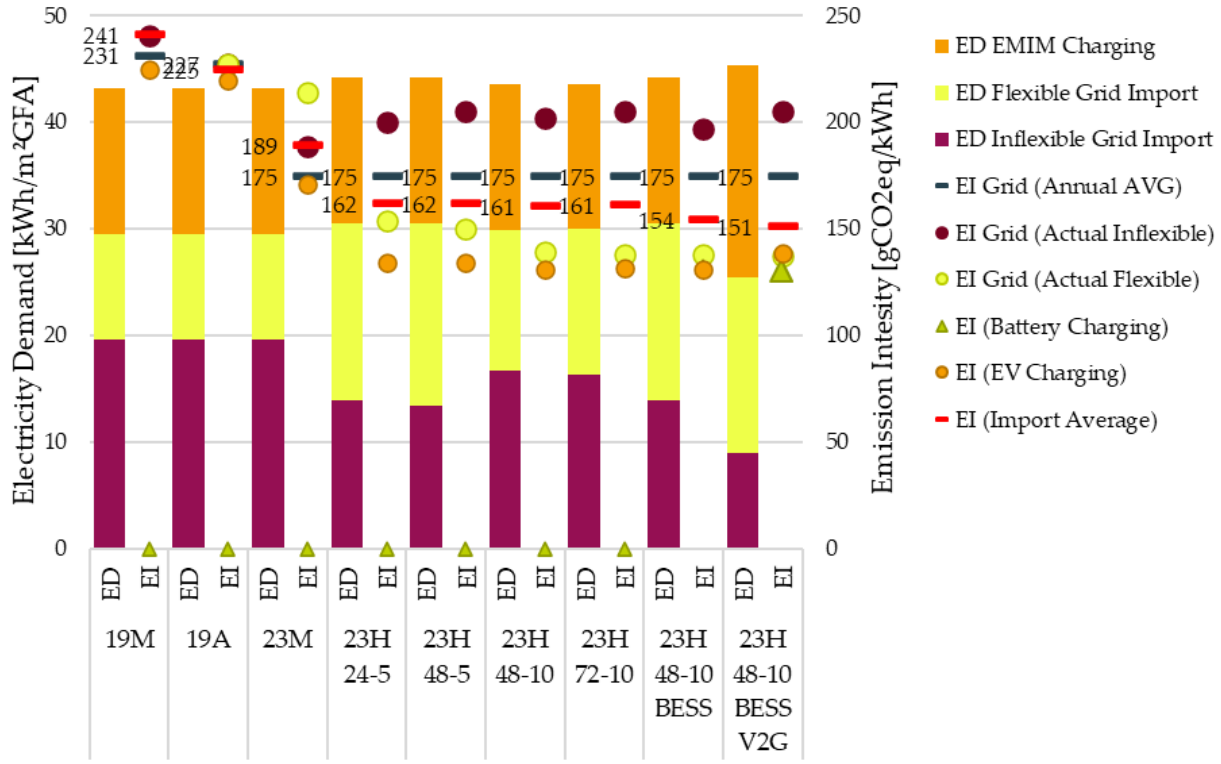


Figure 45. District 4 electricity demands (left axis) and emission intensities (right axis) under GHG emission factor scenarios [Schneider-2025 d]

District 5 showed that mobility context can dominate life-cycle results and that electrification alone is insufficient without strong accompanying measures (compare Figure and Figure 47): location-driven mileage differences shift cumulative mobility emissions by orders of magnitude, and at intermediate EV shares (30–70%) mobility impacts remain largely fossil-dominated over the horizon. In the refurbishment sensitivity set, “supply-only” measures such as switching from gas to district heating can achieve GHG compliance in some cases but still fail the PE/PED criterion without efficiency and flexibility; the first configuration achieving both GHG budget compliance and a positive PE balance combines non-fossil supply (GSHP), conventional renovation plus flexibility, and high EV shares, illustrating that climate-neutral PED status in dense urban renovation is difficult but feasible only through bundled measures.

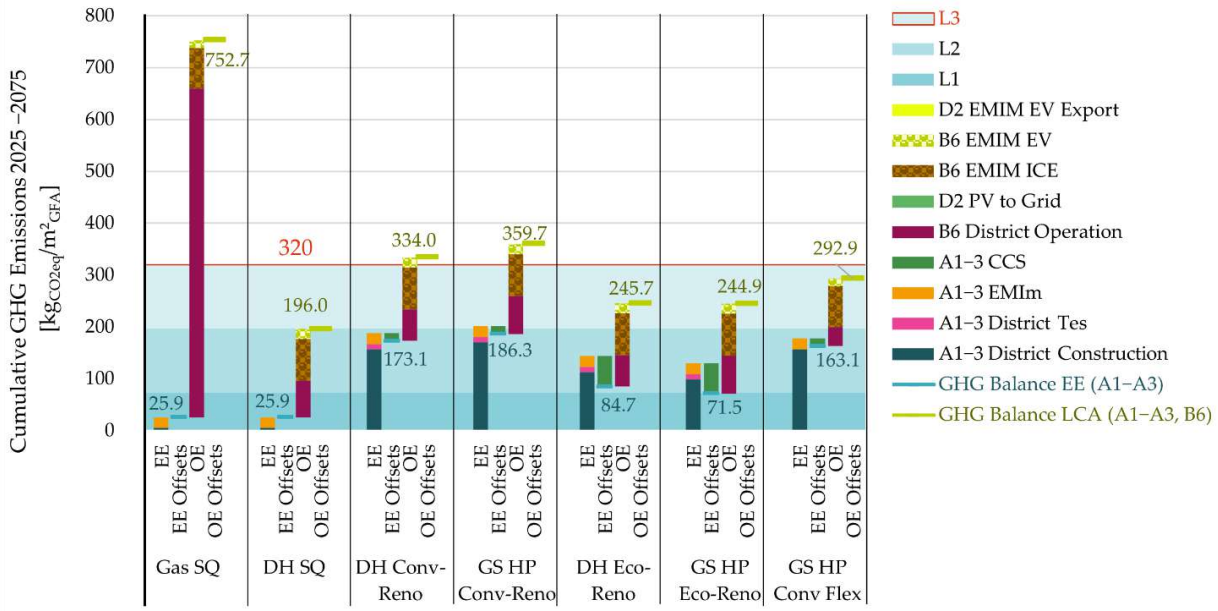


Figure 46. Cumulative GHG LCA emissions of District 5 [Schneider-2025 d]

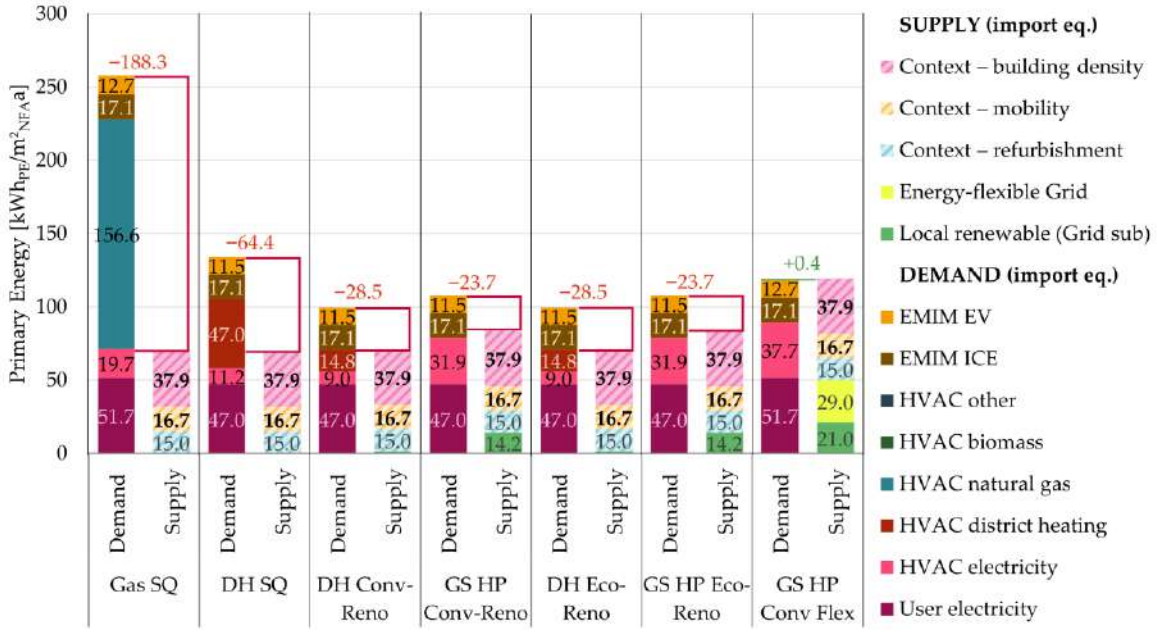


Figure 47. Annual PE- balance of District 5 [Schneider-2025 d]

District 6 (greenfield, mixed-use) showed that construction practice and materials can be the dominant determinant of compliance, even when operational emissions are comparatively low (see Figure 48): a conventional concrete-based concept overshoots the budget strongly, incremental "eco" adjustments reduce but do not resolve the overshoot, and near-compliance requires a mixed concrete-timber structural approach. Full compliance is only achieved in an extreme "Max-Eco" variant with very low concrete use and high biogenic storage credits under the simplified method, highlighting that embodied emissions (EE) can overwhelm operational improvements and that compliance

pathways in such cases are primarily material- and structure-driven, with secondary levers in PV sizing and mobility electrification.

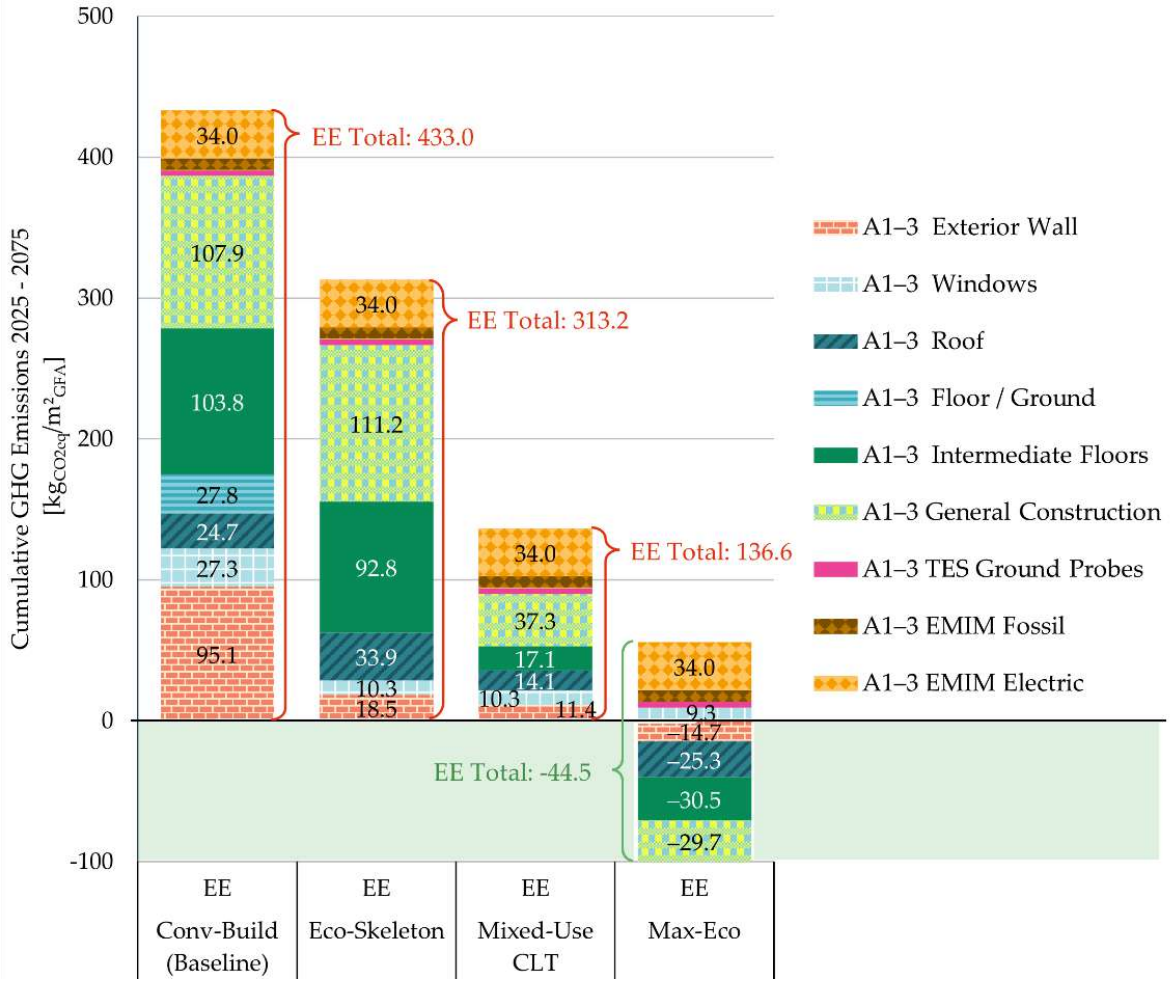


Figure 48. Cumulative A1–A3 EE by component for four District 6 concepts [Schneider-2025 d]

3.4.2. Austrian PED Declaration

Declarational use constitutes the primary institutional application of the PED framework. In this mode, the framework is employed as a rule-based assessment instrument that produces auditable compliance statements based on predefined system boundaries, calculation rules, and parameter sets. The declaration does not modify the PED definition or modelling logic; it constrains their use to ensure stability, comparability, and institutional defensibility.

Declaration-Specific Modelling Constraints:

- **Assessment logic stability:** System boundaries, balance formulations, context interpretation, and compliance logic are fixed. Adjustable elements are limited to parameter sets such as reference scenarios, conversion factors, emission coefficients, and national budgets.
- **Auditability and reproducibility:** All declaration results must be reproducible by third parties. The framework therefore employs deterministic calculation procedures, documented defaults,

explicit handling of missing data, and excludes stochastic, adaptive, or optimization-based routines.

- **Input standardization:** Inputs are divided into mandatory project-specific inputs and standardized defaults. Defaults are externally defined and documented in declaration outputs.
- **Discrete compliance evaluation:** All calculations are continuous. Compliance is evaluated only at the final step using threshold- or budget-based criteria. Weighted scoring systems are not used.
- **Explicit responsibility delimitation:** Each included system component is paired with a reference budget or zero-sum context factor. Responsibility is assessed relative to reference conditions, and exclusions are stated explicitly.
- **Energy flexibility** opportunities are derived from aggregated wind power outputs exceeding a predefined threshold expressed as a share of rated capacity. Hours above this cut-off are interpreted as periods of high renewable availability, during which flexible electricity consumption is considered system-supportive. This provides a computationally lightweight approximation of grid-side flexibility requirements, grounded in historically observed wind generation patterns, while remaining compatible with the hourly resolution and balance-based logic of the PED assessment framework.

Declarational application is organized through three scope levels, which define assessment boundaries, required inputs, and evaluation criteria. The scope levels are not hierarchical performance stages and do not imply incremental scoring. Each scope represents a distinct, self-contained assessment configuration anchored to the same PED definition and modelling framework, as illustrated in Figure 49.

Operational energy configuration **(Positive Energy District):** This configuration assesses district operational energy performance. It includes aggregated building energy demand, local renewable generation, energy-flexible operation, and evaluation of the primary energy balance against density-adjusted targets. Required model inputs include aggregated building demand profiles, representation of on-site renewable supply, hourly balancing logic, and the Floor Area Ratio (FAR). FAR is calculated as gross floor area divided by buildable plot area, excluding non-buildable land uses that prohibit renewable deployment.

Operational energy and mobility configuration **(Positive Energy District with Mobility):** This configuration extends the system boundary to include everyday motorized individual mobility associated with the district. Mobility is represented as an energy demand and flexibility component and evaluated relative to reference mobility budgets. Declaration-level specification requires municipality, district usage mix, and electric vehicle penetration. For declaration purposes, EV penetration is fixed at 70% over the 50-year assessment horizon.

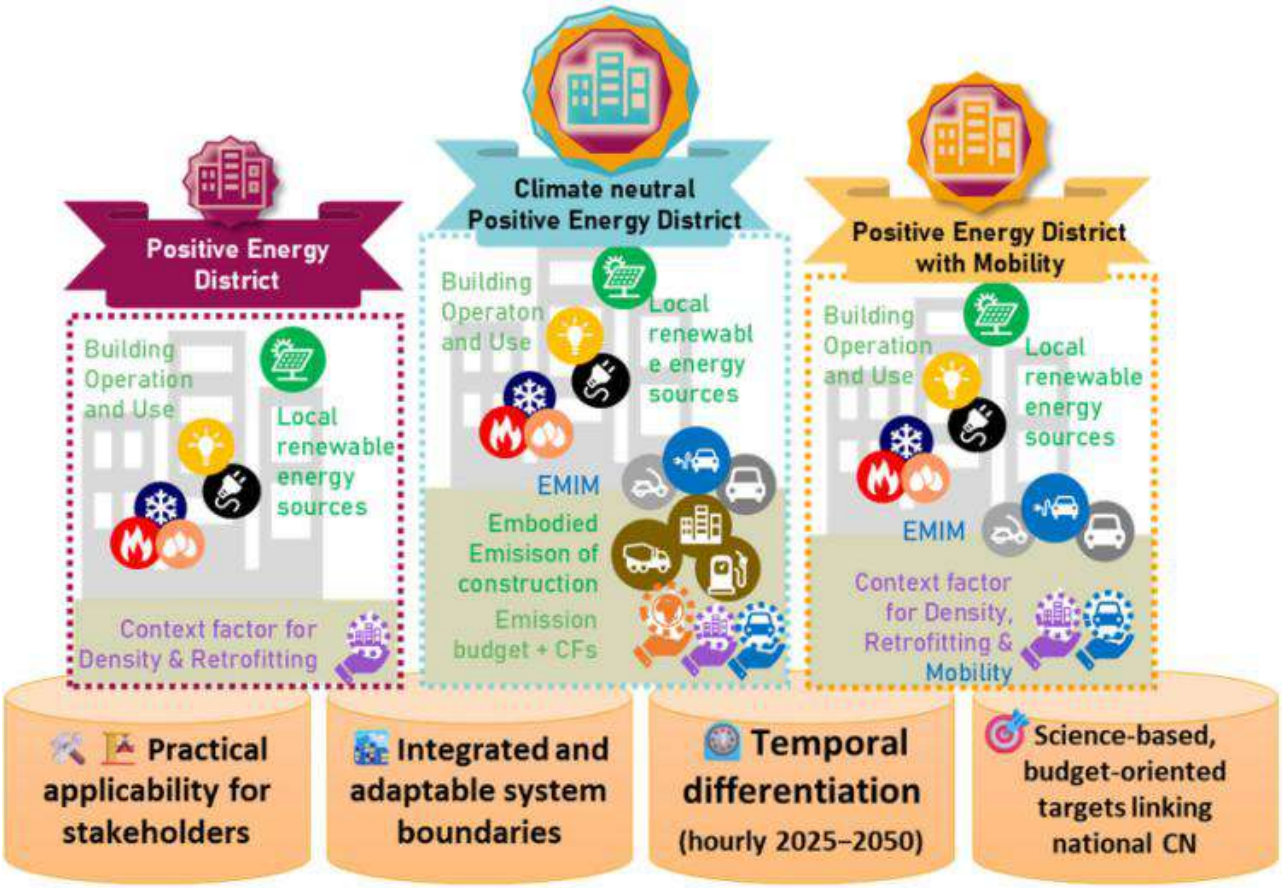


Figure 49. Methodological PED definition design approach and components [Schneider-2025 d]

Climate-neutral configuration **(Climate Neutral Positive Energy District):** This configuration integrates life-cycle greenhouse gas emissions. Assessment includes embodied emissions from buildings and energy infrastructure and operational emissions over the assessment horizon. Compliance is evaluated against floor-area-specific GHG budgets derived from national carbon budgets.

Declaration requirements and auditability

PED declaration is subject to an audit process documented through a comprehensive audit report. This report must transparently outline all relevant project parameters and modelling assumptions, including but not limited to district areas and surfaces, building physics and archetypes, energy systems, and on-site photovoltaic installations. All input data used in the assessment must be fully documented and traceable, and the report must include supporting materials such as plans, schematics, renderings, and technical descriptions sufficient to substantiate the provisioning and correctness of the underlying modelling approach.

In addition, results generated by the assessment tool must be included in the audit documentation and presented in a reproducible manner to demonstrate compliance with the targeted declaration level.

Where favourable flexibility weighting is claimed, the audit report must further document the existence of appropriate technical equipment and operational control strategies, ensuring that declared performance is supported by implementable and verifiable district operation rather than by purely theoretical potential.

Declaration Examples

The declaration framework was applied to a set of declaration-aligned example districts. These examples are not presented as representative or exhaustive, nor are they intended to benchmark PED performance. Their purpose is to demonstrate that the declaration logic can be applied consistently across different spatial contexts, development types, and data situations, while preserving traceability and auditability.

[Schneider-2025 b] showed, using the Alliiertenviertel in Vienna, that a typology-based district model can translate municipal climate goals into operationally comparable renovation pathways even when building-level input uncertainty prevents address-specific statements. Instead of evaluating single addresses, the study simulated all buildings across a typology matrix defined by compactness, age, and condition, demonstrating that heating demand reductions scale strongly with surface-to-volume ratio and that the marginal benefit of deep retrofit diminishes for newer, already efficient buildings. Aggregated scenarios can be assessed against the PED declaration target goals, as shown in Figure 50.

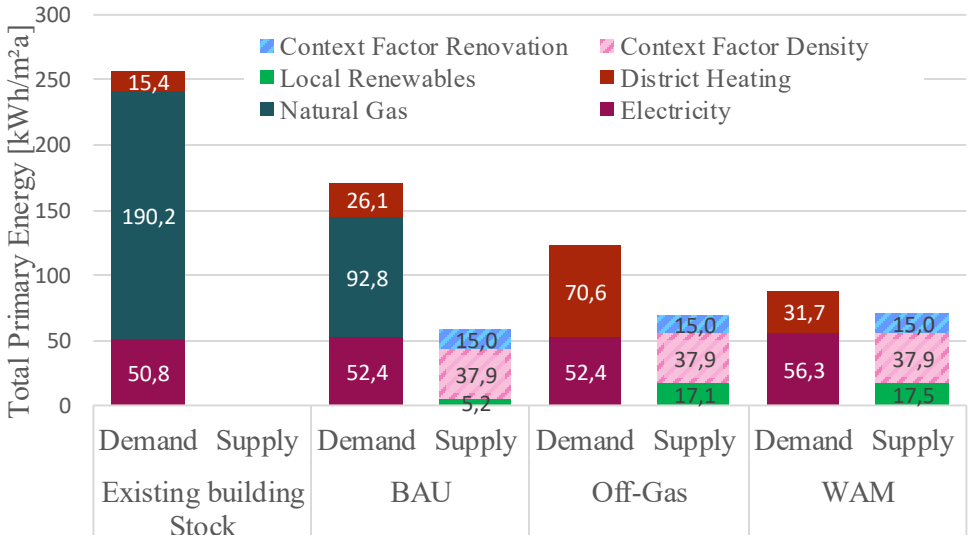


Figure 50. PE-balance of investigated district scenarios, translated from German in [Schneider-2025 b]

[Schneider-2025 b] showed that scenario portfolios aligned with municipal policy instruments, specifically the "Smart Klima City" targets (end-energy reduction, operational CO_2 reduction, and "Off-Gas" by 2040) enable a transparent effort-outcome comparison: technical maximum scenarios

define an upper bound (WAM), while business as usual (BAU) and "Wiener Scenario" (Off-Gas) variants expose how renovation rates and boiler replacement rates control both the magnitude and timing of emission reductions. The comparison highlighted that "Off-Gas" is achievable in the modelled pathways, but that lower renovation rates significantly delay the emissions decline, making target achievement a question of implementation speed rather than only technical feasibility (see Figure 51).

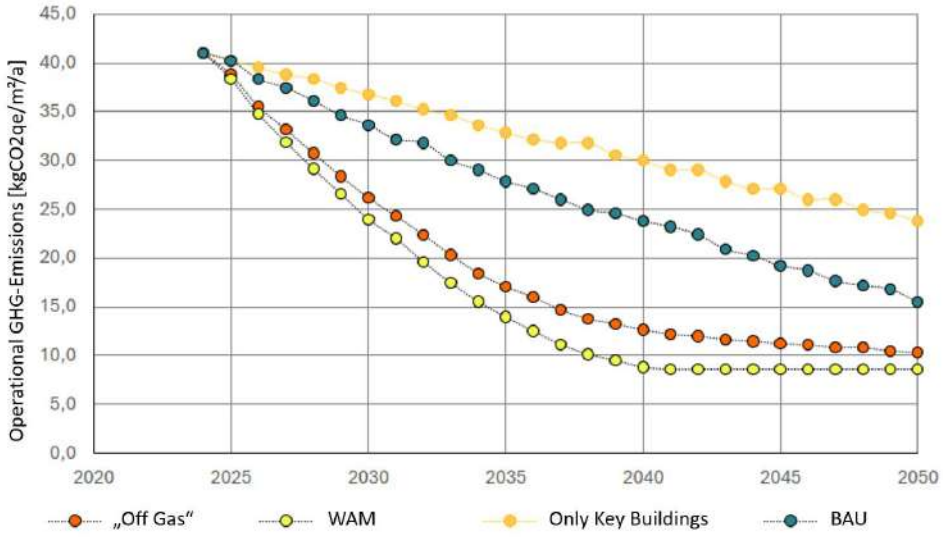


Figure 51. Decarbonization pathways of investigated district scenarios

This work also showed the practical value of the method as a policy-support tool by benchmarking the same scenario set against different governance frames: while several scenarios can meet or approach municipal decarbonisation targets, as can be seen in Figure 50, none reaches the requirement of the *klimaaktiv* declaration "Positive Energy District" under the restricted system boundary (operation only; roof-PV only; no flexibility, storage, façade PV, or heat pumps). This demonstrated how the framework can be used to identify whether a given municipal strategy is compatible with a specific label or target logic, and which additional levers would need to be opened to close remaining gaps.

3.5. Further Contributions

The following contributions are not directly connected to the main topic of PED definition, but offer supporting methods and evidence in adjacent topics:

3.5.1. Municipal Effort sharing as a missing link between national and district targets

The allocation methodology presented in [Schneider-2024 a] introduces a municipal perspective between national decarbonization pathways and district-level assessments. While the quantitative

PED definition developed in this thesis is anchored at the national level, municipal allocation represents a potential refinement layer for contextualizing district targets.

The approach distributes national renewable expansion potentials and sectoral demand reduction targets to municipalities using transparent, structure-based indicators such as population, settlement density, building stock characteristics, and sectoral energy demand profiles. The resulting municipal reference values remain fully consistent with national climate pathways while reflecting structural differences in local contribution potential.

For PED assessment, such municipal reference values enable differentiated district targets and more meaningful comparisons across heterogeneous urban contexts. District performance can be interpreted relative to municipal expectations rather than uniform national benchmarks, reducing the risk of conflating project-level performance with structural advantages or constraints.

Within this thesis, municipal allocation is not implemented as a formal context factor nor integrated into the declaration framework, which intentionally relies on stable national reference values. Instead, it is presented as a complementary option for future applications, offering a potential bridge between national target setting and district-level assessment in coordinated municipal decarbonization strategies.

By situating district performance within a municipal reference frame, this approach illustrates how PED assessment could evolve from a purely national benchmarking instrument toward a multi-level effort-sharing framework, without compromising consistency with national climate objectives.

3.5.2. Using PED criteria in urban development competitions

[Schneider-2024 b] demonstrates how PED criteria can be integrated into early-stage urban development competitions, where key design decisions are made before detailed energy modelling is available. To this end, a points-based evaluation catalogue is introduced that translates PED ambitions into assessable criteria under conditions of limited information.

Shown in Figure 52, the catalogue structures qualitative competition submissions along core PED objectives, enabling consistent comparison without implying formal compliance. Criteria span strategic district-level aspects, plausibility of implementation at micro-district scale, and the contribution of competition outcomes to institutional learning and future procurement.

Task	Criterion	Weight
Task A: District and Neighborhood	Achievement of quantitative PED Targets (Primary Energy Balance)	10
	Design of the Development Process	5
	Climate Resilience, Avoidance of Heat Islands, Rainwater Management	5
	Energy Flexibility and Storage	5
	Suitability for Integration of Digital Components	5
	Sector Coupling and Grid Services	5
	Fair Effort Sharing for the Transformation of the Entire District	5
	Integration of Stakeholders and Users	10
Task B: Micro-Districts	Proof of PED Targets	20
	Greening Concept	10
	Economic Efficiency over the Life Cycle	5
	Probability of Implementation	5
Task C	Improvement of Tender Documents	10

Figure 52. Example of an early planning process assessment catalogue for PEDs [Schneider-2024 b]

The approach shows how PEDs can inform governance and selection processes prior to formal assessment or declaration, providing a bridge between early strategic ambition and later quantitative verification, and positioning PEDs as a guiding framework rather than a purely ex-post compliance instrument.

4 Conclusions and future work

Section 4.1 present the most important conclusions that resulted from the work developed in the scope of this thesis. Perspectives of future work are presented in section 4.2.

4.1. Main conclusions

This thesis set out to address a central limitation in the current discourse and practice surrounding Positive Energy Districts (PEDs): the absence of a quantitative, context-aware, and institutionally robust definition capable of supporting assessment, comparison, and governance across heterogeneous urban contexts. While PEDs have been widely adopted as a strategic concept in European climate and urban policy, their operationalization has remained inconsistent, largely qualitative, and insufficiently anchored in national decarbonization pathways. The work presented in this thesis demonstrates that this limitation is not incidental, but structural, and that it can be resolved only by explicitly treating PED definitions as designed instruments rather than descriptive labels.

The first key conclusion is that **uniform, context-blind PED definitions are neither feasible nor desirable**. Empirical application across a wide range of European district typologies shows that structural factors (most prominently building density, urban form, and mobility demand) dominate the feasibility of achieving positive energy or climate-neutral balances. Without explicit contextualization, PED criteria systematically disadvantage dense urban and renovation districts, thereby contradicting broader sustainability objectives related to land-use efficiency, compact urban development, and infrastructure utilization. The introduction of quantitative context factors is therefore not a corrective add-on but a necessary design element for any PED definition intended for fair and comparable application.

Second, the thesis demonstrates that **PED performance must be interpreted as a sufficiency problem rather than an efficiency problem**. By formulating PED requirements in terms of primary energy and life-cycle greenhouse gas (GHG) balances anchored in national climate targets, the work shifts the focus from relative improvement to absolute compatibility with finite energy and emission budgets. This perspective reveals that operational or physical energy positivity is insufficient to ensure climate compatibility. Life-cycle GHG emissions, particularly embodied emissions associated with construction and refurbishment represent the real constraint that can dominate long-term district assessment and have important implications for design and planning decisions.

Third, the results show that **energy flexibility and mobility are decisive system-level drivers** that cannot be treated as peripheral extensions to PED assessment. Hourly, time-resolved modelling demonstrates that flexibility affects not only grid interaction but also the effective primary energy and emission performance of districts. Similarly, everyday motorized mobility can exceed building operational energy demand and fundamentally alter the feasibility of positive balances. The thesis confirms that both dimensions can be integrated coherently into a single quantitative framework when responsibility boundaries and reference budgets are explicitly defined at the national level.

A further conclusion concerns methodology. The development and application of a **definition-consistent, hourly, multi-energy simulation framework** demonstrates that high temporal resolution and institutional usability are not mutually exclusive. By prioritizing transparency, robustness, and auditability over optimization and fine-grained spatial detail, the framework proves suitable for early planning, scenario comparison, and declaration contexts characterized by limited and uncertain data. This confirms that PED assessment does not require maximal modelling complexity, but rather methodological consistency with the underlying definition and governance requirements.

Finally, the institutional application of the framework in a national PED declaration context confirms that **quantitative PED definitions can be made declaration-ready without sacrificing analytical rigor**. Fixed system boundaries, standardized defaults, explicit parameter governance, and discrete compliance logic allow PEDs to function as stable and auditable planning instruments. At the same time, analytical applications show that the same definition can support exploratory research and feasibility analysis without loss of coherence. This dual usability validates the central thesis claim that PEDs can bridge policy, planning, and research domains when their definition, modelling, and application are aligned.

In summary, this thesis establishes a continuous methodological chain from conceptual clarification, through quantitative definition and modelling, to institutional application. By explicitly addressing contextual constraints, anchoring district performance in national decarbonization

pathways, and integrating energy, mobility, flexibility, and life-cycle emissions within a coherent framework, the work transforms Positive Energy Districts from aspirational concepts into **operational instruments for climate-neutral urban development**.

4.2. Perspectives on Future Work

While this thesis establishes a quantitative, context-aware framework for defining and assessing PEDs, several avenues for further research and methodological refinement remain. These perspectives primarily concern the extension of contextualization logic, the deepening of life cycle and flexibility treatment, and the strengthening of institutional and operational applicability.

The first area for future work lies in **extending contextualization beyond density and mobility**. While these factors capture major structural influences on district energy performance, additional quantitative context factors warrant investigation. These include constraints related to heritage protection, plot fragmentation, limited access to renewable resources, and other site-specific conditions that affect feasible intervention depth. In this context, renovation-specific structural limitations could be formalized as a distinct class of constraints, rather than being implicitly subsumed under density. Further research is also needed to explore climate-zone-specific contextualization under harmonized national targets, enabling consistent PED ambition levels across heterogeneous climatic conditions.

Second, the **life-cycle greenhouse gas (GHG) budgeting approach** could be refined. Future work should address the temporal allocation of embodied emissions beyond simplified front-loading assumptions, particularly for long-lived structures and phased renovation strategies. This includes integrating dynamic material substitution, reuse, and circularity scenarios, as well as aligning the framework with evolving European standards and regulations on whole-life carbon accounting. Such extensions would improve consistency between PED assessment and emerging regulatory practice.

A third research direction concerns the **explicit operationalization of uncertainty and robustness**. Early-stage PED assessment is inherently affected by uncertainties related to building characteristics, mobility behaviour, technology uptake, and future energy system conditions. Rather than treating compliance as a binary outcome, future developments could introduce uncertainty-aware assessment outputs, such as scenario envelopes, robustness indicators, and sensitivity-informed performance ranges. This would enhance the usefulness of PED declarations as decision-support instruments under incomplete information.

Further refinement is also required in the **treatment of energy flexibility**. While the current framework captures flexibility implicitly through balance-based representations, future work could

integrate explicit flexibility metrics, such as residual load smoothing, curtailment avoidance, or peak coincidence reduction. In addition, the investigation of seasonal and multi-day flexibility potentials, beyond hourly dispatch, remains an open topic. These methodological advances should be complemented by an assessment of regulatory and market conditions under which flexibility can be credibly valued and credited in PED assessment.

The framework could also benefit from **closer integration with higher-resolution urban and building energy models**. Systematic coupling with UBEM and building-scale simulation tools would enable a smoother transition from early-stage, declaration-oriented assessment to later planning and design phases. Automated upscaling and downscaling between building, block, district, and municipal levels would further support consistency across scales. As PEDs increasingly enter operational phases, validation against measured district performance data will become essential for refining assumptions and calibration.

Beyond methodological extensions, future work should explore **multi-level effort-sharing frameworks**. This includes integrating municipal and regional allocation mechanisms into PED target-setting logic, developing nested frameworks that link national, municipal, and district-level contributions, and comparatively evaluating different anchoring strategies. Such work would further clarify the role of PEDs within broader climate governance architectures.

Another important perspective concerns **long-term monitoring and post-declaration evaluation**. The framework is currently designed for ex-ante assessment; future research should address the transition toward ex-post monitoring and verification. This includes developing PED performance indicators suitable for operational tracking and establishing feedback mechanisms through which monitored performance informs future declaration parameters and benchmarks.

Finally, broader **institutional and policy embedding** represents a key area for future work. This includes testing the transferability of the framework to national contexts beyond Austria, adapting it to different regulatory cultures and energy system configurations, and clarifying the role of PEDs within complementary policy instruments such as renovation strategies, urban development competitions, or funding schemes.

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Annexes

Annex A. PED Simulation and Modelling Framework

The PED Simulation and Modelling Framework developed in this thesis is publicly available and documented at https://github.com/simonschaluppe/peexcel (Tool Shema Version 1.11.4)

A1. Input Variables

Tool Shema Version 1.11.4

Column	Meaning
var_name	internal model variable
domain	model domain
measure	calculation group
spatial_scope	Reference area for the specified quantity
temporal scope	Timeframe such as winter, summer, annual, life cycle
entity_group	If variable is applicable only to a certain entity of a group
entity_key	Which entity the variable belongs to
unit	physical unit
formula	implemented formula for calculation or lookup

A1. Input Variables

Simulation Tool Sheet “IN” Table “IN”

var_name	domain	measure	unit	entity_group	entity_key	Formula
scenario_version	tool	version	text			
input_version_number	tool	input version	text			
cfd_fPE	declaration	fPE	kWhPE/kWhEE			
cfd_A	declaration	A	kWhEE/m²GF			
cfd_dx	declaration	dx	-			
cfd_EUI	declaration	EUI	kWhEE/m²NGFa			
cfd_cutoff	declaration	cutoff	kWhPE/m²NGFa			
cfm_budget_residential	declaration	context factor	kWhPE/m²NGFa	mobility		
cfm_budget_office	declaration	context factor	kWhPE/m²NGFa	mobility		
cfm_budget_school	declaration	context factor	kWhPE/m²NGFa	mobility		
cfm_budget_retail	declaration	context factor	kWhPE/m²NGFa	mobility		
cfr_budget	declaration	context factor	kWhPE/m²NGFa			
project_name	project	name	text			!$D$5
project_url	project	url	text			!D6
location_address	project	address	text			!$D$12
municipality_name	project	municipality	text			!D14
project_creation_date	project	created	date			!$D$7
project_description	project	description	text			!D8
project_scenario_name	project	scenario_name	text			!D9
location_postcode	project	postcode	text			!$D$15
mobility_mode	declaration	mobility_mode	text			
weather_name	weather	name	text			!$D$19
weather_index	weather	index	int			VERGLEICH(N28;list_weathers;0)

var_name	domain	measure	unit	entity_group	entity_key	Formula
GFA_residential	usage	area	m²	usage	residential	!D28
GFA_office	usage	area	m²	usage	office	!D29
GFA_schoolsec	usage	area	m²	usage	schoolsec	!D30
GFA_schoolprim	usage	area	m²	usage	schoolprim	!D31
GFA_retailfood	usage	area	m²	usage	retailfood	!D32
GFA_retailother	usage	area	m²	usage	retailother	!D33
GFA_other	usage	area	m²	usage	other	!D34
GFA_total	usage	area	m²			SUMME(N30:N36)
NFA_to_GFA_ratio_residential	usage	NFA_to_GFA	ratio	usage	residential	!F42
NFA_to_GFA_ratio_office	usage	NFA_to_GFA	ratio	usage	office	!F43
NFA_to_GFA_ratio_schoolsec	usage	NFA_to_GFA	ratio	usage	schoolsec	!F44
NFA_to_GFA_ratio_schoolprim	usage	NFA_to_GFA	ratio	usage	schoolprim	!F45
NFA_to_GFA_ratio_retailfood	usage	NFA_to_GFA	ratio	usage	retailfood	!F46
NFA_to_GFA_ratio_retailother	usage	NFA_to_GFA	ratio	usage	retailother	!F47
NFA_to_GFA_ratio_other	usage	NFA_to_GFA	ratio	usage	other	!F48
NFA_to_GFA_ratio	usage	NFA_to_GFA	ratio			NFA_total/GFA_total
NFA_residential	usage	area	m²	usage	residential	N30*N38
NFA_office	usage	area	m²	usage	office	N31*N39
NFA_schoolsec	usage	area	m²	usage	schoolsec	N32*N40
NFA_schoolprim	usage	area	m²	usage	schoolprim	N33*N41
NFA_retailfood	usage	area	m²	usage	retailfood	N34*N42
NFA_retailother	usage	area	m²	usage	retailother	N35*N43
NFA_other	usage	area	m²	usage	other	N36*N44
NFA_total	usage	area	m²			SUMME(N46:N52)
per_NFA	usage	area_inverse	1/m²NGF			1/NFA_total
plot_area	district	area	m²			!D55
FSI	district	floor_space_index	-			GFA_total/N55

var_name		domain	measure	unit	entity_group	entity_key	Formula
renovation_ratio_residential		district	renovation	ratio	usage	residential	!D61
renovation_ratio_office		district	renovation	ratio	usage	office	!D62
renovation_ratio_schoolsec		district	renovation	ratio	usage	schoolsec	!D63
renovation_ratio_schoolprim		district	renovation	ratio	usage	schoolprim	!D64
renovation_ratio_retailfood		district	renovation	ratio	usage	retailfood	!D65
renovation_ratio_retailother		district	renovation	ratio	usage	retailother	!D66
renovation_ratio_other		district	renovation	ratio	usage	other	!D67
renovation_share		district	renovation	ratio			SUMMENPRODUKT(Renovation_shares;NFA_shares)
building_count		district	buildings	-			
storey_count_avg		district	storeys	-			
rh_residential		geometry	height	m	usage	residential	!D42
rh_office		geometry	height	m	usage	office	!D43
rh_schoolsec		geometry	height	m	usage	schoolsec	!D44
rh_schoolprim		geometry	height	m	usage	schoolprim	!D45
rh_retailfood		geometry	height	m	usage	retailfood	!D46
rh_retailother		geometry	height	m	usage	retailother	!D47
rh_other		geometry	height	m	usage	other	!D48
rh_avg		geometry	height	m			SUMMENPRODUKT(N67:N73;N46:N52)/NFA_total
NFV_total		geometry	volume	m³net			NFA_total*rh_avg
NFV_u		geometry	volume	m³net			SUMMENPRODUKT(N46:N52;N67:N73;cool_passive_shares)
NFV_c		geometry	volume	m³net			SUMMENPRODUKT(N46:N52;N67:N73;cool_active_shares)
rh_ceiling		geometry	height	m			
GFV		geometry	volume	m³gross			NFA_total*(rh_avg+N78)
hull_ext_wall_wo_window_m2		geometry	area	m²	hull	wall	!D11
hull_roof_m2		geometry	area	m²	hull	roof	!D12
hull_fundament_m2		geometry	area	m²	hull	fundament	!D13

var_name	domain	measure	unit	entity_group	entity_key	Formula
hull_windows_north_m2	geometry	area	m²	hull	window_north	!D18
hull_windows_east_m2	geometry	area	m²	hull	window_east	!D19
hull_windows_south_m2	geometry	area	m²	hull	window_south	!D20
hull_windows_west_m2	geometry	area	m²	hull	window_west	!D21
hull_windows_horizontal_m2	geometry	area	m²	hull	window_horizontal	!D22
hull_window_m2	geometry	area	m²	hull	windows	SUMME(N83:N86)
hull_fenestration_rate	geometry	area	m²	hull		hull_window_m2/(hull_window_m2+hull_ext_wall_wo_window_m2)
hull_m2	geometry	area	m²	hull		hull_ext_wall_wo_window_m2+hull_roof_m2+hull_fundament_m2+hull_window_m2
lc	geometry	lc	m	hull		GFV/(hull_ext_wall_wo_window_m2+hull_roof_m2+hull_fundament_m2+hull_window_m2)
hull_transmittance_walls	Thermal Hull	transmittance	W/m²K	hull	wall	!F11
hull_transmittance_roof	Thermal Hull	transmittance	W/m²K	hull	roof	!F12
hull_transmittance_fundament	Thermal Hull	transmittance	W/m²K	hull	fundament	!F13
hull_transmittance_windows_north	Thermal Hull	transmittance	W/m²K	hull	windows	!F28
hull_transmittance_windows_east	Thermal Hull	transmittance	W/m²K	hull	windows	!F29
hull_transmittance_windows_south	Thermal Hull	transmittance	W/m²K	hull	windows	!F30
hull_transmittance_windows_west	Thermal Hull	transmittance	W/m²K	hull	windows	!F31
hull_transmittance_windows_horizontal	Thermal Hull	transmittance	W/m²K	hull	windows	!F32
hull_transmittance_heatbridge	Thermal Hull	transmittance_heatbridge_factor	ratio			!D36
transmittance_walls_Wm2NFA	Thermal Hull	transmittance	W/m²(NFA)K	hull	wall	hull_ext_wall_wo_window_m2*hull_transmittance_walls*per_NFA

var_name	domain	measure	unit	entity_group	entity_key	Formula
transmittance_roof_Wm2NFA	Thermal Hull	transmittance	W/m²(NFA)K	hull	roof	hull_roof_m2*hull_transmittance_roof*per_NFA
transmittance_fundament_Wm2NFA	Thermal Hull	transmittance	W/m²(NFA)K	hull	fundament	N82*N103*per_NFA
transmittance_windows_Wm2NFA	Thermal Hull	transmittance	W/m²(NFA)K	hull	windows	SUMMENPRODUKT(N83:N87;N104:N108)*per_NFA
transmittance_heatbridge_Wm2NFA	Thermal Hull	transmittance_heatbridge_factor	W/m²(NFA)K			SUMME(N110:N113)*N109
transmittance_MW	Thermal Hull	transmittance	MW/K			transmittance_Wm2*NFA_total/1000000
transmittance_Wm2	Thermal Hull	transmittance	W/m²K			SUMME(N110:N114)
heat_capacity_effective_m²	Thermal Hull	heat_capacity	Wh/m²K			!D77
heat_cap_eff_uncooled_m2	Thermal Hull	heat_capacity	Wh/m²K			heat_capacity_effective_m2
heat_cap_eff_cooled_m2	Thermal Hull	heat_capacity	Wh/m²K			heat_capacity_effective_m2
window_total_transmittance_north	Solar Gains	solar_gains_coefficient	ratio	hull	window_north	!D28
window_total_transmittance_east	Solar Gains	solar_gains_coefficient	ratio	hull	window_east	!D29
window_total_transmittance_south	Solar Gains	solar_gains_coefficient	ratio	hull	window_south	!D30
window_total_transmittance_west	Solar Gains	solar_gains_coefficient	ratio	hull	window_west	!D31
window_total_transmittance_horizontal	Solar Gains	solar_gains_coefficient	ratio	hull	window_horizontal	!D32
window_irradiation_factor_winter_north	Solar Gains	sgc_factor	ratio	hull	window_north	!D55
window_irradiation_factor_winter_east	Solar Gains	sgc_factor	ratio	hull	window_east	!D56
window_irradiation_factor_winter_south	Solar Gains	sgc_factor	ratio	hull	window_south	!D57
window_irradiation_factor_winter_west	Solar Gains	sgc_factor	ratio	hull	window_west	!D58
window_irradiation_factor_winter_horizontal	Solar Gains	sgc_factor	ratio	hull	window_horizontal	!D59

var_name	domain	measure	unit	entity_group	entity_key	Formula
window_irradiation_factor_summer_north	Solar Gains	sgc_factor	ratio	hull	window_north	!F55
window_irradiation_factor_summer_east	Solar Gains	sgc_factor	ratio	hull	window_east	!F56
window_irradiation_factor_summer_south	Solar Gains	sgc_factor	ratio	hull	window_south	!F57
window_irradiation_factor_summer_west	Solar Gains	sgc_factor	ratio	hull	window_west	!F58
window_irradiation_factor_summer_horizontal	Solar Gains	sgc_factor	ratio	hull	window_horizontal	!F59
irradiation_opaque_surcharge	Solar Gains	sgc_factor	-			
fc_residential	Solar Gains	shading_coefficient	ratio	usage	residential	!D64
fc_office	Solar Gains	shading_coefficient	ratio	usage	office	!D65
fc_schoolsec	Solar Gains	shading_coefficient	ratio	usage	schoolsec	!D66
fc_schoolprim	Solar Gains	shading_coefficient	ratio	usage	schoolprim	!D67
fc_retailfood	Solar Gains	shading_coefficient	ratio	usage	retailfood	!D68
fc_retailother	Solar Gains	shading_coefficient	ratio	usage	retailother	!D69
fc_other	Solar Gains	shading_coefficient	ratio	usage	other	!D70
fc_u	Solar Gains	shading_coefficient	ratio			SUMMENPRODUKT(N152:N158;N46:N52;cool_passive_shares)*per_NFA_uncooled
fc_c	Solar Gains	shading_coefficient	ratio			SUMMENPRODUKT(N152:N158;N46:N52;cool_active_shares)*per_NFA_cooled
illuminance_min_residential	Solar Gains	illuminance_min	lux	usage	residential	!F64
illuminance_min_office	Solar Gains	illuminance_min	lux	usage	office	!F65
illuminance_min_schoolsec	Solar Gains	illuminance_min	lux	usage	schoolsec	!F66
illuminance_min_schoolprim	Solar Gains	illuminance_min	lux	usage	schoolprim	!F67
illuminance_min_retailfood	Solar Gains	illuminance_min	lux	usage	retailfood	!F68
illuminance_min_retailother	Solar Gains	illuminance_min	lux	usage	retailother	!F69
illuminance_min_other	Solar Gains	illuminance_min	lux	usage	other	!F70

var_name	domain	measure	unit	entity_group	entity_key	Formula
QS_max_shading_u	Solar Gains	max_irradiation	W/m²			SUMMENPRODUKT(NFAs;col_passive_shares;Lux_minimums)*0,01*per_NFA_uncooled
QS_max_shading_c	Solar Gains	max_irradiation	W/m²			SUMMENPRODUKT(NFAs;col_active_shares;Lux_minimums)*0,01*per_NFA_cooled
illuminance_efficiency_dirt	Solar Gains	sgc_factor				
illuminance_efficiency_glazing_ratio	Solar Gains	sgc_factor				!D25
mob_shading_factor_u	Solar Gains					fc_u
mob_shading_factor_c	Solar Gains					fc_c
vent_air_tightness	Ventilation	air_tightness	1/h			!D41
vent_tightness_ratio_blowerdoor_to_real	Ventilation	blowerdoor_to_real	ratio			
vent_infiltration_ACH	Ventilation	ach	1/h			N175/N176
vent_share_mechanical_residential	Ventilation	mechanical_share	ratio	usage	residential	L!D19
vent_share_mechanical_office	Ventilation	mechanical_share	ratio	usage	office	L!D20
vent_share_mechanical_schoolsec	Ventilation	mechanical_share	ratio	usage	schoolsec	L!D21
vent_share_mechanical_schoolprim	Ventilation	mechanical_share	ratio	usage	schoolprim	L!D22
vent_share_mechanical_retailfood	Ventilation	mechanical_share	ratio	usage	retailfood	L!D23
vent_share_mechanical_retailother	Ventilation	mechanical_share	ratio	usage	retailother	L!D24
vent_share_mechanical_other	Ventilation	mechanical_share	ratio	usage	other	L!D25
NFA_windowvent	Ventilation		m²NGF			NFA_total-NFA_mechvent
NFV_windowvent	Ventilation		m³netto			NFV_total-N195
NFA_mechvent	Ventilation		m²NGF			SUMMENPRODUKT(N$178:N$184;N46:N52)
NFV_mechvent	Ventilation		m³netto			SUMMENPRODUKT(N$178:N$184;N46:N52;N67:N73)

var_name	domain	measure	unit	entity_group	entity_key	Formula
vent_night_residential	Ventilation			usage	residential	L!D54
vent_night_office	Ventilation			usage	office	L!D55
vent_night_schoolsec	Ventilation			usage	schoolsec	L!D57
vent_night_schoolprim	Ventilation			usage	schoolprim	L!D56
vent_night_retailfood	Ventilation			usage	retailfood	L!D58
vent_night_retailother	Ventilation			usage	retailother	L!D59
vent_night_other	Ventilation			usage	other	L!D60
ACH_night_m³	Ventilation		m³/h			SUMMENPRODUKT(N185:N191;N67:N73;N46:N52;SVERWEIS(N196:N202;night_ventilation_types;2;FALSCH))
vent_ach_max_residential	Ventilation	air_change	1/h	usage	residential	L!D8
vent_ach_max_office	Ventilation	air_change	1/h	usage	office	L!D9
vent_ach_max_schoolsec	Ventilation	air_change	1/h	usage	schoolsec	L!D10
vent_ach_max_schoolprim	Ventilation	air_change	1/h	usage	schoolprim	L!D11
vent_ach_max_retailfood	Ventilation	air_change	1/h	usage	retailfood	L!D12
vent_ach_max_retailother	Ventilation	air_change	1/h	usage	retailother	L!D13
vent_ach_max_other	Ventilation	air_change	1/h	usage	other	L!D14
vent_scale_residential	Ventilation	air_change_scale	raio	usage	residential	N204/MAX(Nutzung_Wohnen[Bedarf Luftwechsel])
vent_scale_office	Ventilation	air_change_scale	raio	usage	office	N205/MAX(Nutzung_Büro[Bedarf Luftwechsel])
vent_scale_school_sec	Ventilation	air_change_scale	raio	usage	schoolsec	N206/MAX(Nutzung_Schule[Bedarf Luftwechsel])
vent_scale_school_prim	Ventilation	air_change_scale	raio	usage	schoolprim	N207/MAX(Nutzung_KIGA[Bedarf Luftwechsel])
vent_scale_supermarket	Ventilation	air_change_scale	raio	usage	retailfood	N208/MAX(Nutzung_Handel_Food[Bedarf Luftwechsel])
vent_scale_retail	Ventilation	air_change_scale	raio	usage	retailother	N209/MAX(Nutzung_Handel_NonFood[Bedarf Luftwechsel])
vent_scale_other	Ventilation	air_change_scale	raio	usage	other	
Ev_scale_residential	Ventilation	el_scale	Anteil	usage	residential	L!D41
Ev_scale_office	Ventilation	el_scale	Anteil	usage	office	L!D42

var_name	domain	measure	unit	entity_group	entity_key	Formula
Ev_scale_school_sec	Ventilation	el_scale	Anteil	usage	schoolsec	L!D43
Ev_scale_school_prim	Ventilation	el_scale	Anteil	usage	schoolprim	L!D44
Ev_scale_retail_food	Ventilation	el_scale	Anteil	usage	retailfood	L!D45
Ev_scale_retail_other	Ventilation	el_scale	Anteil	usage	retailother	L!D46
Ev_scale_other	Ventilation	el_scale	Anteil	usage	other	L!D47
vent_heat_recovery_winter_residential	Ventilation	heat_recovery_winter	ratio	usage	residential	L!D30
vent_heat_recovery_winter_office	Ventilation	heat_recovery_winter	ratio	usage	office	L!D31
vent_heat_recovery_winter_schoolsec	Ventilation	heat_recovery_winter	ratio	usage	schoolsec	L!D32
vent_heat_recovery_winter_schoolprim	Ventilation	heat_recovery_winter	ratio	usage	schoolprim	L!D33
vent_heat_recovery_winter_retailfood	Ventilation	heat_recovery_winter	ratio	usage	retailfood	L!D34
vent_heat_recovery_winter_retailother	Ventilation	heat_recovery_winter	ratio	usage	retailother	L!D35
vent_heat_recovery_winter_other	Ventilation	heat_recovery_winter	ratio	usage	other	L!D36
vent_heat_recovery_summer_residential	Ventilation	heat_recovery_summer	ratio	usage	residential	L!F30
vent_heat_recovery_summer_office	Ventilation	heat_recovery_summer	ratio	usage	office	L!F31
vent_heat_recovery_summer_schoolsec	Ventilation	heat_recovery_summer	ratio	usage	schoolsec	L!F32
vent_heat_recovery_summer_schoolprim	Ventilation	heat_recovery_summer	ratio	usage	schoolprim	L!F33
vent_heat_recovery_summer_retailfood	Ventilation	heat_recovery_summer	ratio	usage	retailfood	L!F34
vent_heat_recovery_summer_retailother	Ventilation	heat_recovery_summer	ratio	usage	retailother	L!F35
vent_heat_recovery_summer_other	Ventilation	heat_recovery_summer	ratio	usage	other	L!F36
Vent_share_window_uncooled	Ventilation					WENNFEHLER(SUMMENPRODUKT(N46:N52;cool_passive

var_name	domain	measure	unit	entity_group	entity_key	Formula
						_shares;MTRANS(vent_window_shares);N67:N73)/NFV_total;0)
Vent_share_window_cooled	Ventilation					WENNFEHLER(SUMMENPRODUKT(N46:N52;cool_active_shares;MTRANS(vent_window_shares);N67:N73)/NFV_total;0)
Vent_share_mech_uncooled	Ventilation					WENNFEHLER(SUMMENPRODUKT(N46:N52;cool_passive_shares;MTRANS(vent_mech_shares);N67:N73)/NFV_total;0)
Vent_share_mech_cooled	Ventilation					WENNFEHLER(SUMMENPRODUKT(N46:N52;cool_active_shares;MTRANS(vent_mech_shares);N67:N73)/NFV_total;0)
test_NFV_shares	Ventilation					100%=SUMME(Vent_share_window_uncooled;Vent_share_window_cooled;Vent_share_mech_uncooled;Vent_share_mech_cooled)
usage_concurrency_winter_residential	Space Use	concurrency	ratio	usage	residential	!F21
usage_concurrency_winter_office	Space Use	concurrency	ratio	usage	office	!F22
usage_concurrency_winter_schoolsec	Space Use	concurrency	ratio	usage	schoolsec	!F23
usage_concurrency_winter_schoolprim	Space Use	concurrency	ratio	usage	schoolprim	!F24
usage_concurrency_winter_retailfood	Space Use	concurrency	ratio	usage	retailfood	!F25
usage_concurrency_winter_retailother	Space Use	concurrency	ratio	usage	retailother	!F26
usage_concurrency_winter_other	Space Use	concurrency	ratio	usage	other	!F27

var_name	domain	measure	unit	entity_group	entity_key	Formula
usage_concurrency_summer_residential	Space Use	concurrency	ratio	usage	residential	!I21
usage_concurrency_summer_office	Space Use	concurrency	ratio	usage	office	!I22
usage_concurrency_summer_schoolsec	Space Use	concurrency	ratio	usage	schoolsec	!I23
usage_concurrency_summer_schoolprim	Space Use	concurrency	ratio	usage	schoolprim	!I24
usage_concurrency_summer_retailfood	Space Use	concurrency	ratio	usage	retailfood	!I25
usage_concurrency_summer_retailother	Space Use	concurrency	ratio	usage	retailother	!I26
usage_concurrency_summer_other	Space Use	concurrency	ratio	usage	other	!I27
DHW_demand_residential_kWhm2	Space Use	dhw_demand	kWh/m²NGFa	usage	residential	WW!D33
DHW_demand_office_kWhm2	Space Use	dhw_demand	kWh/m²NGFa	usage	office	WW!D34
DHW_demand_schoolsec_kWhm2	Space Use	dhw_demand	kWh/m²NGFa	usage	schoolsec	WW!D35
DHW_demand_schoolprim_kWhm2	Space Use	dhw_demand	kWh/m²NGFa	usage	schoolprim	WW!D36
DHW_demand_retailfood_kWhm2	Space Use	dhw_demand	kWh/m²NGFa	usage	retailfood	WW!D37
DHW_demand_retailother_kWhm2	Space Use	dhw_demand	kWh/m²NGFa	usage	retailother	WW!D38
DHW_demand_other_kWhm2	Space Use	dhw_demand	kWh/m²NGFa	usage	other	WW!D39
aux_el_demand_residential_kWhm2	Space Use	aux_el_demand	kWh/m²NGFa	usage	residential	!L9
aux_el_demand_office_kWhm2	Space Use	aux_el_demand	kWh/m²NGFa	usage	office	!L10
aux_el_demand_schoolsec_kWhm2	Space Use	aux_el_demand	kWh/m²NGFa	usage	schoolsec	!L11
aux_el_demand_schoolprim_kWhm2	Space Use	aux_el_demand	kWh/m²NGFa	usage	schoolprim	!L12

var_name		domain	measure	unit	entity_group	entity_key	Formula
aux_el_demand_retailfood_kWhm2	👤	Space Use	aux_el_demand	kWh/m²NGFa	usage	retailfood	👤 !L13
aux_el_demand_retailother_kWhm2	👤	Space Use	aux_el_demand	kWh/m²NGFa	usage	retailother	👤 !L14
aux_el_demand_other_kWhm2	👤	Space Use	aux_el_demand	kWh/m²NGFa	usage	other	👤 !L15
Plugloads_scale_residential	👤	Space Use	plugload_scale	ratio	usage	residential	N702/O702
Plugloads_scale_office	👤	Space Use	plugload_scale	ratio	usage	office	N703/O703
Plugloads_scale_schoolsec	👤	Space Use	plugload_scale	ratio	usage	schoolsec	N704/O704
Plugloads_scale_schoolprim	👤	Space Use	plugload_scale	ratio	usage	schoolprim	N705/O705
Plugloads_scale_retailfood	👤	Space Use	plugload_scale	ratio	usage	retailfood	N706/O706
Plugloads_scale_retailother	👤	Space Use	plugload_scale	ratio	usage	retailother	N707/O707
Plugloads_scale_other	👤	Space Use	plugload_scale	ratio	usage	other	N708/O708
density_NFApPers_residential	🚗	Space Use	occupancy	m²NF/Pers	usage	residential	
density_NFApPers_office	🚗	Space Use	occupancy	m²NF/Pers	usage	office	
density_NFApPers_schoolsec	🚗	Space Use	occupancy	m²NF/Pers	usage	schoolsec	
density_NFApPers_retail	🚗	Space Use	occupancy	m²NF/Pers	usage	retailfood	
density_NFApPers_schoolprim	🚗	Space Use	occupancy	m²NF/Pers	usage	schoolprim	
density_NFApPers_retail_other	🚗	Space Use	occupancy	m²NF/Pers	usage	retailother	
mob_simultaneity_edu	🚗	Space Use	concurrency_mobility	mobility	education	🚗 !D74	
mob_simultaneity_retail	🚗	Space Use	concurrency_mobility	mobility	retail	🚗 !D75	
mob_simultaneity_office	🚗	Space Use	concurrency_mobility	mobility	work		
mob_motorization_perNFA_residential	🚗	Space Use		1/100m²	mobility	education	
mob_motorization_perNFA_work	🚗	Space Use		1/100m²	mobility	retail	
mob_motorization_perNFA_retail	🚗	Space Use		1/100m²	mobility	work	🚗 !D33
Plight_max_office	💡	Space Use	lighting_power_max	W/m²	usage	office	👤 !U10

var_name	domain	measure	unit	entity_group	entity_key	Formula
Plight_max_schoolsec	Space Use	lighting_power_max	W/m²	usage	schoolsec	!U11
Plight_max_schoolprim	Space Use	lighting_power_max	W/m²	usage	schoolprim	!U12
Plight_min_office	Space Use	lighting_power_max	W/m²	usage	office	!T10
Plight_min_schoolsec	Space Use	lighting_power_max	W/m²	usage	schoolsec	!T11
Plight_min_schoolprim	Space Use	lighting_power_max	W/m²	usage	schoolprim	!T12
lighting_factor_office	Space Use	lighting_scale	ratio	usage	office	!V10
lighting_factor_schoolsec	Space Use	lighting_scale	ratio	usage	schoolsec	!V11
lighting_factor_schoolprim	Space Use	lighting_scale	ratio	usage	schoolprim	!V12
lighting_factor_retailfood	Space Use	lighting_scale	ratio	usage	retailfood	!V13
lighting_factor_retailother	Space Use	lighting_scale	ratio	usage	retailother	!V14
lighting_factor_other	Space Use	lighting_scale	ratio	usage	other	!V15
Daylightcoefficient_office	Space Use	daylight_coefficient	-	usage	office	!R10
Daylightcoefficient_schoolsec	Space Use	daylight_coefficient	-	usage	schoolsec	!R11
Daylightcoefficient_schoolprim	Space Use	daylight_coefficient	-	usage	schoolprim	!R12
daylightcontr_office	Space Use	daylight_control_share	ratio	usage	office	
daylightcontr_schoolsec	Space Use	daylight_control_share	ratio	usage	schoolsec	
daylightcontr_schoolprim	Space Use	daylight_control_share	ratio	usage	schoolprim	
heating_month1	Heating	is_heating_period	bool			
heating_month2	Heating	is_heating_period	bool			
heating_month3	Heating	is_heating_period	bool			
heating_month4	Heating	is_heating_period	bool			
heating_month5	Heating	is_heating_period	bool			
heating_month6	Heating	is_heating_period	bool			
heating_month7	Heating	is_heating_period	bool			
heating_month8	Heating	is_heating_period	bool			
heating_month9	Heating	is_heating_period	bool			
heating_month10	Heating	is_heating_period	bool			
heating_month11	Heating	is_heating_period	bool			
heating_month12	Heating	is_heating_period	bool			

var_name	domain	measure	unit	entity_group	entity_key	Formula
QHmax_room_m²	Heating	emission_power	W/m²			H!D11
QHmax_room_MW	Heating	emission_power	MW			N323*N53/1000000
QHmax_1el_m2	Heating	generation_power	W/m²	heating	system_1	H!D25
QHmax_2th_m2	Heating	generation_power	W/m²	heating	system_2	H!D32
QHmax_3el_m2	Heating	generation_power	W/m²	heating	system_3	H!D39
QHmax_4th_m2	Heating	generation_power	W/m²	heating	system_4	H!D46
QH_distr_loss_1el	Heating	distribution_losses	ratio	heating	system_1	H!D27
QH_distr_loss_2th	Heating	distribution_losses	ratio	heating	system_2	H!D34
QH_distr_loss_3el	Heating	distribution_losses	ratio	heating	system_3	H!D41
QH_distr_loss_4th	Heating	distribution_losses	ratio	heating	system_4	H!D48
heat_cop_1el	Heating	generation_efficiency	ratio	heating	system_1	H!D26
heat_cop_2th	Heating	generation_efficiency	ratio	heating	system_2	H!D33
heat_cop_3el	Heating	generation_efficiency	ratio	heating	system_3	H!D40
heat_cop_4th	Heating	generation_efficiency	ratio	heating	system_4	H!D47
QH_generation_eff_1el	Heating	generation_efficiency_inverse	ratio	heating	system_1	WENNFEHLER(1/heat_cop_1el;0)
QH_generation_eff_2th	Heating	generation_efficiency_inverse	ratio	heating	system_2	WENNFEHLER(1/heat_cop_2th;0)
QH_generation_eff_3el	Heating	generation_efficiency_inverse	ratio	heating	system_3	WENNFEHLER(1/heat_cop_3el;0)
QH_generation_eff_4th	Heating	generation_efficiency_inverse	ratio	heating	system_4	WENNFEHLER(1/heat_cop_4th;0)
QH_aux_wasteheat	Heating	auxiliary_power_ratio	ratio			
QH_aux_el_to_th_1el	Heating	auxiliary_power_ratio	ratio	heating	system_1	H!D28
QH_aux_el_to_th_2th	Heating	auxiliary_power_ratio	ratio	heating	system_2	H!D35
QH_aux_el_to_th_3el	Heating	auxiliary_power_ratio	ratio	heating	system_3	H!D42
QH_aux_el_to_th_4th	Heating	auxiliary_power_ratio	ratio	heating	system_4	H!D49
heat_th2_carrier_type	Heating	carrier_type		heating	system_2	H!D31
heat_th4_carrier_type	Heating	carrier_type		heating	system_4	H!D45
Tsetheat_min	Heating	minimum_setpoint	°C			H!D7
cooling_month1	Cooling	is_cooling_period	Bool			

var_name	domain	measure	unit	entity_group	entity_key	Formula
cooling_month2	Cooling	is_cooling_period	Bool			
cooling_month3	Cooling	is_cooling_period	Bool			
cooling_month4	Cooling	is_cooling_period	Bool			
cooling_month5	Cooling	is_cooling_period	Bool			
cooling_month6	Cooling	is_cooling_period	Bool			
cooling_month7	Cooling	is_cooling_period	Bool			
cooling_month8	Cooling	is_cooling_period	Bool			
cooling_month9	Cooling	is_cooling_period	Bool			
cooling_month10	Cooling	is_cooling_period	Bool			
cooling_month11	Cooling	is_cooling_period	Bool			
cooling_month12	Cooling	is_cooling_period	Bool			
cool_share_residential	Cooling	active_cooling_share	ratio	usage	residential	K'!D10
cool_share_office	Cooling	active_cooling_share	ratio	usage	office	K'!D11
cool_share_schoolsec	Cooling	active_cooling_share	ratio	usage	schoolsec	K'!D12
cool_share_schoolprim	Cooling	active_cooling_share	ratio	usage	schoolprim	K'!D13
cool_share_retailfood	Cooling	active_cooling_share	ratio	usage	retailfood	K'!D14
cool_share_retailother	Cooling	active_cooling_share	ratio	usage	retailother	K'!D15
cool_share_other	Cooling	active_cooling_share	ratio	usage	other	K'!D16
NFA_cooled	Cooling	area	m²			SUMMENPRODUKT(N361:N367;N46:N52)
NFAfrac_c	Cooling	active_cooling_share	ratio			NFA_cooled/NFA_total
NFAfrac_u	Cooling	passive_cooling_shar	ratio			1-N376
QCmax_room_m2	Cooling	emission_power	W/m²			K'!D23
QCmax_room_MW	Cooling	emission_power	MW			QCmax_room_m2*NFA_cooled/1000000
QCmax_freecooling	Cooling	generation_power	W/m²	cooling		K'!D29
QCmax_1el	Cooling	generation_power	W/m²	cooling	system_1	K'!D35
QCmax_2th	Cooling	generation_power	W/m²	cooling	system_2	K'!D43
QCmax_3el	Cooling	generation_power	W/m²	cooling	system_3	K'!D50
QC_distr_losses_1el	Cooling	distribution_losses	ratio	cooling	system_1	K'!D37
QC_distr_losses_2th	Cooling	distribution_losses	ratio	cooling	system_2	K'!D45

var_name		domain	measure	unit	entity_group	entity_key	Formula
QC_distr_losses_3el	❄	Cooling	distribution_losses	ratio	cooling	system_3	❄K'!D52
cool_cop_1el	❄	Cooling	generation_efficiency	ratio	cooling	system_1	❄K'!D36
cool_cop_2th	❄	Cooling	generation_efficiency	ratio	cooling	system_2	❄K'!D44
cool_cop_3el	❄	Cooling	generation_efficiency	ratio	cooling	system_3	❄K'!D51
QC_to_EC_1	❄	Cooling	generation_efficiency	ratio	cooling	system_1	WENNFEHLER(1/N387;0)
QC_generation_eff_2th	❄	Cooling	generation_efficiency_inverse	ratio	cooling	system_2	WENNFEHLER(1/N388;0)
QC_to_EC_3	❄	Cooling	generation_efficiency_inverse	ratio	cooling	system_3	WENNFEHLER(1/N389;0)
QC_aux_fc	❄	Cooling	auxiliary_power_ratio	ratio	cooling		❄K'!D30
QC_aux_1el	❄	Cooling	auxiliary_power_ratio	ratio	cooling	system_1	❄K'!D38
QC_aux_2th	❄	Cooling	auxiliary_power_ratio	ratio	cooling	system_2	❄K'!D46
QC_aux_3el	❄	Cooling	auxiliary_power_ratio	ratio	cooling	system_3	❄K'!D53
cool_th2_carrier_type	❄	Cooling	carrier_type		cooling	system_2	❄K'!D42
Tsetcool_max	❄	Cooling	minimum_setpoint	°C			❄K'!D7
DHW_Tmin	💧	DHW	minimum_setpoint	°C			💧WW!D48
DHW_Tmax_input	💧	DHW		°C			🔋!D43
DHW_Tmax	💧	DHW	minimum_setpoint	°C			MIN(N401;DHW_Tmax_constant)
DHW_1_share_residential	💧	DHW	system_1_share	ratio	usage	residential	💧WW!D22
DHW_1_share_office	💧	DHW	system_1_share	ratio	usage	office	💧WW!D23
DHW_1_share_schoolsec	💧	DHW	system_1_share	ratio	usage	schoolsec	💧WW!D24
DHW_1_share_schoolprim	💧	DHW	system_1_share	ratio	usage	schoolprim	💧WW!D25
DHW_1_share_retailfood	💧	DHW	system_1_share	ratio	usage	retailfood	💧WW!D26
DHW_1_share_retailother	💧	DHW	system_1_share	ratio	usage	retailother	💧WW!D27
DHW_1_share_other	💧	DHW	system_1_share	ratio	usage	other	💧WW!D28
DHW_storage_1_liter	💧	DHW	capacity	liter	dhw	system_1	SUMMENPRODUKT(N403:N409;N410:N416)
DHW_storage_2_liter	💧	DHW	capacity	liter	dhw	system_2	SUMME(N410:N416)-DHW_storage_1_liter
DHW_1_is_used	💧	DHW	system_used	bool	dhw	system_1	WENN(DHW_storage_1_liter;WAHR;FALSCH)

var_name	domain	measure	unit	entity_group	entity_key	Formula
DHW_2_is_used	DHW	system_used	bool	dhw	system_2	WENN(DHW_storage_2_liter;WAHR;FALSCH)
DHW_thermal_power_1_kW	DHW	generation_power	kW			DHW_thermal_power_Wpl*DHW_storage_1_liter/1000
DHW_losses_1	DHW	storage_losses	1/h	dhw	system_1	WW!D10
DHW_losses_2	DHW	storage_losses	1/h	dhw	system_2	WW!D17
DHW_efficiency_distribution_1	DHW	distribution_efficiency	dhw	system_1	WW!D9	
DHW_efficiency_distribution_2	DHW	distribution_efficiency	dhw	system_2	WW!D16	
DHW_1_incl_distribution_factor	DHW	distribution_factor	dhw	system_1	WENNFEHLER(1/DHW_efficiency_distribution_1;0)	
DHW_2_incl_distribution_factor	DHW	distribution_factor	dhw	system_2	WENNFEHLER(1/DHW_efficiency_distribution_2;0)	
DHW_carriertype_1	DHW	carrier_type		dhw	system_1	WW!D7
DHW_1_is_electric	DHW	is_electric		dhw	system_1	WENN(DHW_carriertype_1="Strom";WAHR;FALSCH)
DHW_carriertype_2	DHW	carrier_type		dhw	system_2	WW!D14
DHW_2_is_electric	DHW	is_electric		dhw	system_2	WENN(DHW_carriertype_2="Strom";WAHR;FALSCH)
DHW_1_efficiency	DHW	generation_efficiency	dhw	system_1	WW!D8	
DHW_2_efficiency	DHW	generation_efficiency	dhw	system_2	WW!D15	
DHW_conversion_1	DHW	generation_efficiency_inverse	dhw	system_1	WENNFEHLER(1/DHW_1_efficiency;0)	
DHW_conversion_2	DHW	generation_efficiency_inverse	dhw	system_2	WENNFEHLER(1/DHW_2_efficiency;0)	
DHW_1_el_aux	DHW	auxiliary_power_ratio	dhw	system_1		
DHW_2_el_aux	DHW	auxiliary_power_ratio	dhw	system_2	WW!D18	
DHW_occupancy_residential	DHW	occupancy	m²NF/Pers	usage	residential	WW!D60
DHW_occupancy_office	DHW	occupancy	m²NF/Pers	usage	office	WW!D61
DHW_occupancy_schoolsec	DHW	occupancy	m²NF/Pers	usage	schoolsec	WW!D62
DHW_occupancy_schoolprim	DHW	occupancy	m²NF/Pers	usage	schoolprim	WW!D63
DHW_occupancy_retailfood	DHW	occupancy	m²NF/Pers	usage	retailfood	WW!D64

var_name		domain	measure	unit	entity_group	entity_key	Formula
DHW_occupancy_retailother	💧	DHW	occupancy	m²NF/Pers	usage	retailother	💧 WW!D65
DHW_occupancy_other	💧	DHW	occupancy	m²NF/Pers	usage	other	💧 WW!D66
DHW_concurrency_residential	💧	DHW	concurrency	usage	residential	💧 WW!D51	
DHW_concurrency_office	💧	DHW	concurrency	usage	office	💧 WW!D52	
DHW_concurrency_schoolsec	💧	DHW	concurrency	usage	schoolsec	💧 WW!D53	
DHW_concurrency_schoolprim	💧	DHW	concurrency	usage	schoolprim	💧 WW!D54	
DHW_concurrency_retailfood	💧	DHW	concurrency	usage	retailfood	💧 WW!D55	
DHW_concurrency_retailother	💧	DHW	concurrency	usage	retailother	💧 WW!D56	
DHW_concurrency_other	💧	DHW	concurrency	usage	other	💧 WW!D57	
DHW_storage_liter_pPerson	💧	DHW	storage_capacity	l/Pers/Tag			💧 WW!D46
DHW_thermal_power_pPerson	💧	DHW	generation_power	W/Pers			
DHW_thermal_power_Wpl	💧	DHW	generation_power	W/l			DHW_thermal_power_pPerson/ DHW_storage_liter_pPerson
DHW_storage_env_temp_default	💧	DHW	ambient_temperature	°C			💧 WW!D47
PV_is_used	🌞	PV	is_used	bool			
PV_profile_name	🌞	PV	name	text			🌞 PV!D8
PV_id	🌞	PV	index	int			VERGLEICH(PV_profile_name;pv_profile_names;0)
PV_scale	🌞	PV	pv_scale	ratio			🌞 PV!D9
PV_efficiency	🌞	PV	efficiency				
PV_m2_per_kWp	🌞	PV	area	m²/kWp			🌞 PV!D10
PV_kWp	🌞	PV	installed_power	kWp			PV_scale*INDEX('PV-Import'!$B$19:$AV$19;1;VERGLEICH(PV_profile_name;pv_profile_names;0))
PV_module_area	🌞	PV	area	m²			PV_kWp*PV_m2_per_kWp
FLEX_PV_is_used	🔋	Flexibility	flexible_PV_use	bool			

var_name	domain	measure	unit	entity_group	entity_key	Formula
FLEX_is_used	Flexibility	flexible_grid_use	bool			
FLEX_signal_name	Flexibility	signal_name	text			Signals!D5
FLEX_signal_ID	Flexibility	signal_index	int			VERGLEICH(FLEX_signal_name;flex_signal_names;0)
FLEX_grid_maxpower_Wm2	Flexibility	max_grid_power	W/m²			
FLEX_is_used_for_plugloads	Flexibility	plugloads_use	bool			
FLEX_is_used_for_HVAC_min	Flexibility	HVAC_min_use	bool			
FLEX_is_used_for_ev_min	Flexibility	EV_min_use	bool			
flex_heat_dT	Flexibility	max_temperature_increase	K			!D22
Tsetheat_flex	Flexibility	maximum_setpoint_temperature	°C			Tsetheat_min+N473
FLEX_choice_heat_system	Flexibility	heating_system_selection	text			!D23
FLEX_heat1_use	Flexibility	is_flexible	bool	heating	system_1	WENN(N475="System 1";WAHR;FALSCH)
FLEX_heat3_use	Flexibility	is_flexible	bool	heating	system_3	WENN(N475="System 3";WAHR;FALSCH)
flex_cool_dT	Flexibility	max_temperature_decrease	K			!D25
Tsetcool_flex	Flexibility	maximum_setpoint_temperature	°C			Tsetcool_max-N478
FLEX_choice_cool_system	Flexibility	cooling_system_selection	text			!D26
FLEX_cool1_use	Flexibility	is_flexible	bool	cooling	system_1	WENN(N480="System 1";WAHR;FALSCH)
FLEX_cool3_use	Flexibility	is_flexible	bool	cooling	system_3	WENN(N480="System 3";WAHR;FALSCH)
Batt_is_used	Battery	is_used	bool			
Batt_cap_kWh	Battery	capacity	kWh			!D34
Batt_cap_Wh_per_NFA	Battery	capacity	W/m²			Batt_cap_kWh/NFA_total*1000
Batt_c_rate	Battery	c_rate	ratio			!D35

var_name		domain	measure	unit	entity_group	entity_key	Formula
Batt_max_power_specific	🔋	Battery	power	W/m²			Batt_cap_Wh_per_NFA*Batt_c_rate
Batt_eff_factor_charge	🔋	Battery	efficiency	ratio			🔋 !D36
Batt_eff_factor_discharge	🔋	Battery	efficiency	ratio			🔋 !D37
Batt_self_discharge_rate	🔋	Battery	storage_losses	1/week			🔋 !D38
Batt_auto_discharge_factor	🔋	Battery	storage_losses	1/h			Batt_self_discharge_rate/7/24
Batt_SOC_init	🔋	Battery	SOC_init	ratio			
Batt_is_used_for_plugloads	🔋	Battery	plugloads_use	bool			
Batt_is_used_for_HVACminimum	🔋	Battery	HVAC_min_use	bool			
Batt_is_used_for_EV	🔋	Battery	EV_min_use	bool			
Batt_is_gridcharged	🔋	Battery	flexible_grid_use	bool			
flex_dhw_use	🔋	Flexibility	dhw_use	bool			
flex_Signals_selected_column	🔋	Flexibility	signal_index	int			VERGLEICH(FLEX_signal_name;Signals[#Kopfzeilen];0)
Batt_is_not_used_during_signals	🔋	Flexibility	is_not_discharged_during_signals				
mobility_is_included	🚗	Mobility	is_used	bool			
mobility_region	🚗	Mobility	selection	str			WENNFEHLER(SVERWEIS(N22;gemeinden;3;FALSCH)&WENN(mobility_mode="Vereinfacht";" optimiert";""); "Keine Region ausgewählt.")
mobshare_residential	🚗	Mobility	target_share	ratio	mobility	residential	SVERWEIS(mobility_region;Regionstypen; 2;FALSCH)
mobshare_office	🚗	Mobility	target_share	ratio	mobility	work	SVERWEIS(mobility_region;Regionstypen; 3;FALSCH)
mobshare_school	🚗	Mobility	target_share	ratio	mobility	education	SVERWEIS(mobility_region;Regionstypen; 4;FALSCH)
mobshare_retail	🚗	Mobility	target_share	ratio	mobility	retail	SVERWEIS(mobility_region;Regionstypen; 5;FALSCH)
pkm_pedestrian	🚗	Mobility	mileage	km			SVERWEIS(mobility_region;Regionstypen; 6;FALSCH)

var_name	domain	measure	unit	entity_group	entity_key	Formula
pkm_bike	Mobility	mileage	km			SVERWEIS(mobility_region; Regionstypen; 7;FALSCH)
pkm_mofa	Mobility	mileage	km			SVERWEIS(mobility_region; Regionstypen; 8;FALSCH)
pkm_car_driver	Mobility	mileage	km			SVERWEIS(mobility_region; Regionstypen; 9;FALSCH)
pkm_car_passenger	Mobility	mileage	km			SVERWEIS(mobility_region; Regionstypen; 10;FALSCH)
pkm_bus	Mobility	mileage	km			SVERWEIS(mobility_region; Regionstypen; 11;FALSCH)
pkm_tram_metro	Mobility	mileage	km			SVERWEIS(mobility_region; Regionstypen; 12;FALSCH)
pkm_train	Mobility	mileage	km			SVERWEIS(mobility_region; Regionstypen; 13;FALSCH)
pkm_distancebus	Mobility	mileage	km			SVERWEIS(mobility_region; Regionstypen; 14;FALSCH)
mobility_vehicle_count	Mobility	vehicles	-			!D77
EV_share	Mobility	ev_share	ratio			!D38
EV_demand_kWhpkm	Mobility	demand	kWh/km			!D59
EV_battsize_kWh	Mobility	capacity	kWh/EV			!D55
EV_storage_total_kWh	Mobility	capacity	kWh			N518*N515
EV_self_discharge_per_week	Mobility	storage_losses	1/week			!D56
EV_selfdischarge_per_h	Mobility	storage_losses	1/h			N520/24/7
EV_soc_minimum	Mobility	minimum_setpoint	ratio	mobility	residential	
EV_charging_efficiency	Mobility	charging_efficiency	ratio			!D57
EV_charging_losses_surcharge_factor	Mobility	charging_efficiency_inverse	ratio			WENNFEHLER(1/N523;0)
EV_max_charging_power_ratio	Mobility	c_rate	ratio			
EV_soc_min_work	Mobility	minimum_setpoint	ratio	mobility	work	
EV_soc_min_retail	Mobility	minimum_setpoint	ratio	mobility	retail	
moped_factor	Mobility	moped_to_car	ratio			

var_name		domain	measure	unit	entity_group	entity_key	Formula
EV_share_constant_charging	🚗	Mobility	nonflexible_charging_share	ratio			
mob_pkm_factor	🚗	Mobility	mileage_reduction	ratio			🚗 !D29
EV_experimental_calculation	🚗	Mobility					
EV_mileage_residential	🚗	Mobility	mileage	Pkm/a	mobility	residential	mobility_is_included*(EV_share*StatPAX_residential*pkm_car*mob_pkm_factor)
EV_mileage_work	🚗	Mobility	mileage	Pkm/a	mobility	work	mobility_is_included*(mob_pkm_factor*EV_share*pkm_car*(mob_simultaneity_office*StatPAX_office+mob_simultaneity_edu*(StatPAX_schoolprim+StatPAX_schoolsec)))
EV_mileage_retail	🚗	Mobility	mileage	Pkm/a	mobility	retail	mobility_is_included*(mob_pkm_factor*EV_share*pkm_car*mob_simultaneity_retail*(StatPAX_retail+StatPAX_retailother))
EV_soc_min_ext	🚗	Mobility	minimum_setpoint	ratio			🚗 !D66
EV_soc_min_discharge	🚗	Mobility	minimum_setpoint	ratio			🚗 !D71
fossile_demand_kWhpkm	🚗	Mobility	demand	kWh/km			
EV_count_residential	🚗	Mobility	ev_count	-	mobility	residential	EV_share*NFA_residential/100*mob_motorization_perNFA_residential*mobility_is_included
EV_count_work	🚗	Mobility	ev_count	-	mobility	work	EV_share*(NFA_office*mob_simultaneity_office+mob_simultaneity_edu*(NFA_schoolprim+NFA_schoolsec))/100*mob_motorization_perNFA_work*mobility_is_included
EV_count_retail	🚗	Mobility	ev_count	-	mobility	retail	EV_share*mob_simultaneity_retail*(NFA_retailfood+NFA_retailother)/100*mob_motorization_perNFA_retail*mobility_is_included

var_name	domain	measure	unit	entity_group	entity_key	Formula
ev_bidirectional_use 🚗	Mobility	is_bidirectional	bool			
GWP_walls_name ☁	Construction	component_type	component		THG!F15	
GWP_windows_name ☁	Construction	component_type	component		THG!F16	
GWP_roof_name ☁	Construction	component_type	component		THG!F17	
GWP_ground_name ☁	Construction	component_type	component		THG!F18	
GPW_ceilings_name ☁	Construction	component_type	component		THG!F19	
COMP_name_terrace	Construction	component_type	component		THG!F20	
COMP_name_basement_ceiling	Construction	component_type		component		THG!F21
COMP_name_fundament	Construction	component_type	component		THG!F22	
COMP_name_ceil_to_air	Construction	component_type	component		THG!F23	
COMP_name_wall_earth_contacted	Construction	component_type		component		THG!F24
COMP_name_internal_wall_load	Construction	component_type		component		THG!F25
COMP_name_internal_wall_nonload	Construction	component_type		component		THG!F26
COMP_name_ceiling_topfloor	Construction	component_type		component		THG!F27
COMP_name_wall_ec_unheated	Construction	component_type		component		THG!F28
COMP_name_basement_floor_unheated	Construction	component_type		component		THG!F29
COMP_name_columns	Construction	component_type	component		THG!F30	
COMP_name_internal_wall_unheated	Construction	component_type		component		THG!F31
COMP_name_unheated_horizontal	Construction	component_type		component		THG!F32
COMP_name_balconies	Construction	component_type	component		THG!F33	
COMP_name_windowframe	Construction	component_type		component		THG!F34
COMP_name_windowglazing	Construction	component_type		component		THG!F35
COMP_name_other	Construction	component_type	component		THG!F36	

var_name		domain	measure	unit	entity_group	entity_key	Formula
GWP_general_name	☁						☁THG'!D26
GWP_other_name	☁						☁THG'!D28
GWP_ref_area_walls	☁	Construction	component_fu	m² FU	component		THG!C15
GWP_ref_area_windows	☁	Construction	component_fu	m² FU	component		THG!C16
GWP_ref_area_roof	☁	Construction	component_fu	m² FU	component		THG!C17
GWP_ref_area_fundament	☁	Construction	component_fu	m² FU	component		THG!C18
GWP_ref_area_ceilings	☁	Construction	component_fu	m² FU	component		THG!C19
COMP_area_terrace		Construction	component_fu	m² FU	component		THG!C20
COMP_area_basement_ceiling		Construction	component_fu	m² FU	component		THG!C21
COMP_area_fundament		Construction	component_fu	m² FU	component		THG!C22
COMP_area_ceil_to_air		Construction	component_fu	m² FU	component		THG!C23
COMP_area_wall_earth_contacted		Construction	component_fu	m² FU	component		THG!C24
COMP_area_internal_wall_load		Construction	component_fu	m² FU	component		THG!C25
COMP_area_internal_wall_nonload		Construction	component_fu	m² FU	component		THG!C26
COMP_area_ceiling_topfloor		Construction	component_fu	m² FU	component		THG!C27
COMP_area_wall_ec_unheated		Construction	component_fu	m² FU	component		THG!C28
COMP_area_basement_floor_unheated		Construction	component_fu	m² FU	component		THG!C29
COMP_area_columns		Construction	component_fu	m² FU	component		THG!C30
COMP_area_internal_wall_unheated		Construction	component_fu	m² FU	component		THG!C31
COMP_area_unheated_horizontal		Construction	component_fu	m² FU	component		THG!C32
COMP_area_balconies		Construction	component_fu	m² FU	component		THG!C33
COMP_area_windowframe		Construction	component_fu	m² FU	component		THG!C34
COMP_area_windowglazing		Construction	component_fu	m² FU	component		THG!C35
GWP_refratio_walls	☁			m²AW/m²BGF			N568/GFA_total

var_name		domain	measure	unit	entity_group	entity_key	Formula
GWP_refratio_windows	☁			m²Fe/m²BGF			N569/GFA_total
GWP_refratio_roof	☁			m²Dach/m²BGF			N570/GFA_total
GWP_refratio_fundament	☁			m²KB/m²BGF			N571/GFA_total
GWP_refratio_ceilings	☁			m²Zwd/m²BGF			N572/GFA_total
GWP_pv_name	☁	lca	component_type			☁THG'!D42	
GWP_refratio_pv	☁	lca		m²PV/m²BGF			PV_module_area/GFA_total
GWP_boreholes_name	☁	lca	component_type			☁THG'!D43	
borehole_count	☁	lca		Stk Erdsonden			☁THG'!E43
GWP_ventilation_name	☁	lca	component_type			☁THG'!D44	
GWP_solarthermal_name	☁	lca	component_type			☁THG'!D45	
GWP_battery_name	☁	lca	component_type			☁THG'!D47	
GWP_storage_name	☁	lca	component_type			☁THG'!D48	
GWP_tga_general_name	☁						☁THG'!D49
GWP_tga_other_name	☁						☁THG'!D51
GWP_refratio_boreholes	☁			StkErdsonden/m²BGF			borehole_count/GFA_total
GWP_direct_fossile	☁						☁THG'!D30
GWP_direct_biogenic	☁						☁THG'!D31
GWP_direct_biogenic_share	☁						
GWP_direct_life	☁						
GWP_miv_count_ev	☁	lca	vehicle_count	Anzahl Fahrzeuge			N615*EV_share
GWP_miv_count_fossile	☁	lca	vehicle_count	Anzahl Fahrzeuge			N615-N616
GWP_mobility_construction_fossil	☁	lca		t CO2/Fahrzeug			☁THG'!D56
GWP_mobility_construction_ev	☁	lca		t CO2/Fahrzeug			
GHG_LCA_timeframe_years	☁	lca					
GWP_mobility_moped_gpPKm	☁	lca		GWP 100S (gCO2equiv/Pkm)			
GWP_mobility_car_gpPKm	☁	lca		GWP 100S (gCO2equiv/Pkm)			
year_rcp0	☁	lca	pathway_reduction_year				

var_name		domain	measure	unit	entity_group	entity_key	Formula
year_rcp1	☁	lca	pathway_reduction_year				
year_rcp2	☁	lca	pathway_reduction_year				
year_rcp3	☁	lca	pathway_reduction_year				
grid_rcp1	☁	lca	pathway_reduction_factor	energy_carrier	grid		
grid_rcp2	☁	lca	pathway_reduction_factor	energy_carrier	grid		
grid_rcp3	☁	lca	pathway_reduction_factor	energy_carrier	grid		
rcp1_renewable	☁	lca	pathway_reduction_factor	energy_carrier	renewables		
rcp2_renewable	☁	lca	pathway_reduction_factor	energy_carrier	renewables		
rcp3_renewable	☁	lca	pathway_reduction_factor	energy_carrier	renewables		
rcp1_dh	☁	lca	pathway_reduction_factor	energy_carrier	district_heating		WENNFEHLER(WENN(FINDEN("rneuerbar";SVERWEIS(heat_th2_carrier_type;cf_constant;7;FALSCH));rcp1_renewable;#NV);rcp1_fossil)
rcp2_dh	☁	lca	pathway_reduction_factor	energy_carrier	district_heating		WENNFEHLER(WENN(FINDEN("rneuerbar";SVERWEIS(heat_th2_carrier_type;cf_constant;7;FALSCH));rcp2_renewable;#NV);rcp2_fossil)
rcp3_dh	☁	lca	pathway_reduction_factor	energy_carrier	district_heating		WENNFEHLER(WENN(FINDEN("rneuerbar";SVERWEIS(heat_th2_carrier_type;cf_constant;7;FALSCH));rcp3_renewable;#NV);rcp3_fossil)
rcp1_fossil	☁	lca	pathway_reduction_factor	energy_carrier	fossile		
rcp2_fossil	☁	lca	pathway_reduction_factor	energy_carrier	fossile		
rcp3_fossil	☁	lca	pathway_reduction_factor	energy_carrier	fossile		
GWP_rcpi_grid	☁	lca	pathway_integral	energy_carrier	grid		(year_rcp1-year_rcp0+1)*(1+grid_rcp1)/2+(year_rcp2-year_rcp1+1)*(grid_rcp1+grid_rcp2)/2+(year_rcp

var_name		domain	measure	unit	entity_group	entity_key	Formula
							3-year_rcp2+1)*(grid_rcp3+grid_rcp2)/2-grid_rcp1-grid_rcp2
GWP_rcpi_grid_substition	☁	lca	pathway_integral	energy_carrier	renewables		(year_rcp1-year_rcp0+1)*(1+grid_rcp1)/2+(year_rcp2-year_rcp1+1)*(grid_rcp1+grid_rcp2)/2+(year_rcp3-year_rcp2+1)*(grid_rcp3+grid_rcp2)/2-grid_rcp1-grid_rcp2
GWP_rcpi_district_heating	☁	lca	pathway_integral	energy_carrier	district_heating		(year_rcp1-year_rcp0+1)*(1+rcp1_dh)/2+(year_rcp2-year_rcp1+1)*(rcp1_dh+rcp2_dh)/2+(year_rcp3-year_rcp2+1)*(rcp2_dh+rcp3_dh)/2-rcp1_dh-rcp2_dh
GWP_rcpi_fossil	☁	lca	pathway_integral	energy_carrier	fossile		(year_rcp1-year_rcp0+1)*(1+rcp1_fossil)/2+(year_rcp2-year_rcp1+1)*(rcp1_fossil+rcp2_fossil)/2+(year_rcp3-year_rcp2+1)*(rcp2_fossil+rcp3_fossil)/2-rcp1_fossil-rcp2_fossil
GWP_rcpi_natural_gas	☁	lca	pathway_integral	energy_carrier	natural_gas		GWP_rcpi_fossil
GWP_rcpi_biomass	☁	lca	pathway_integral	energy_carrier	biomass		GWP_rcpi_natural_gas
GWP_rcpi_mob_fossile	☁	lca	pathway_integral	energy_carrier	fossile		GWP_rcpi_biomass
GWP_rcpi_renewable	☁	lca	pathway_integral	energy_carrier	renewables		(year_rcp1-year_rcp0+1)*(1+rcp1_renewable)/2+(year_rcp2-year_rcp1+1)*(rcp1_renewable+rcp2_renewable)/2+(year_rcp3-year_rcp2+1)*(rcp2_renewable+rcp3_renewable)/2-rcp1_renewable-rcp2_renewable
fPE_grid_profile	⚖						
district_heating_conversion_profile	⚖	assessment	district_heating_index				
fGHG_grid_profile	⚖	assessment	grid_name				🚦 Signals!D3
fPE_grid_column	⚖	primary_energy	conversion_factor_index	int			VERGLEICH(fPE_grid_profile;fPE[#Kopfzeilen];0)

var_name		domain	measure	unit	entity_group	entity_key	Formula
fGHG_grid_column	⚖	GHG	conversion_factor_index	int			VERGLEICH(fGHG_grid_profile;fGHG[#Kopfzeilen];0)
heat_th2_is_dh	⚖	primary_energy	is_district_heating	bool	heating	system_2	-- ISTZAHL(FINDEN("Fernwärme";heat_th2_carrier_type))
heat_th4_is_dh	⚖	primary_energy	is_district_heating	bool	heating	system_4	-- ISTZAHL(FINDEN("Fernwärme";heat_th4_carrier_type))
cool_th2_is_dh	⚖	primary_energy	is_district_heating	bool	cooling	system_2	-- ISTZAHL(FINDEN("Fernwärme";cool_th2_carrier_type))
dhw1_is_dh	⚖	primary_energy	is_district_heating	bool	dhw	system_1	-- ISTZAHL(FINDEN("Fernwärme";DHW_carriertype_1))
dhw2_is_dh	⚖	primary_energy	is_district_heating	bool	dhw	system_2	-- ISTZAHL(FINDEN("Fernwärme";DHW_carriertype_2))
heat_th2_is_ng	⚖	primary_energy	is_natural_gas	bool	heating	system_2	-- ISTZAHL(FINDEN("Erdgas";heat_th2_carrier_type))
heat_th4_is_ng	⚖	primary_energy	is_natural_gas	bool	heating	system_4	-- ISTZAHL(FINDEN("Erdgas";heat_th4_carrier_type))
cool_th2_is_ng	⚖	primary_energy	is_natural_gas	bool	cooling	system_2	-- ISTZAHL(FINDEN("Erdgas";cool_th2_carrier_type))
dhw1_is_ng	⚖	primary_energy	is_natural_gas	bool	dhw	system_1	-- ISTZAHL(FINDEN("Erdgas";DHW_carriertype_1))
dhw2_is_ng	⚖	primary_energy	is_natural_gas	bool	dhw	system_2	-- ISTZAHL(FINDEN("Erdgas";DHW_carriertype_2))
heat_th2_is_bio	⚖	primary_energy	is_biomass	bool	heating	system_2	-- ISTZAHL(FINDEN("Biomasse";heat_th2_carrier_type))

var_name		domain	measure	unit	entity_group	entity_key	Formula
heat_th4_is_bio	⚖	primary_energy	is_biomass	bool	heating	system_4	-- ISTZAHL(FINDEN("Biomasse";heat_th4_carrier_type))
cool_th2_is_bio	⚖	primary_energy	is_biomass	bool	cooling	system_2	-- ISTZAHL(FINDEN("Biomasse";cool_th2_carrier_type))
dhw1_is_bio	⚖	primary_energy	is_biomass	bool	dhw	system_1	-- ISTZAHL(FINDEN("Biomasse";DHW_carriertype_1))
dhw2_is_bio	⚖	primary_energy	is_biomass	bool	dhw	system_2	-- ISTZAHL(FINDEN("Biomasse";DHW_carriertype_2))
heat_th2_is_other	⚖	primary_energy	is_other	bool	heating	system_2	-- NICHT(ODER(N659;N664;N669))
heat_th4_is_other	⚖	primary_energy	is_other	bool	heating	system_4	-- NICHT(ODER(N660;N665;N670))
cool_th2_is_other	⚖	primary_energy	is_other	bool	cooling	system_2	-- NICHT(ODER(N661;N666;N671))
dhw1_is_other	⚖	primary_energy	is_other	bool	dhw	system_1	-- UND(NICHT(ODER(N662;N667;N672)); DHW_1_is_used)
dhw2_is_other	⚖	primary_energy	is_other	bool	dhw	system_2	-- UND(NICHT(ODER(N663;N668;N673));DHW_2_is_used)
fPE_flex_factor	⚖	primary_energy	flex_share				
pe_conversion_factor_gasoline	⚖	primary_energy	conversion_factor	kWhPE/kWhEE			
StatPAX_residential	🚗	mobility	statistical_persons	-	usage	residential	NFA_residential/density_NFApPers_residential
StatPAX_office	🚗	mobility	statistical_persons	-	usage	office	NFA_office/density_NFApPers_office
StatPAX_schoolsec	🚗	mobility	statistical_persons	-	usage	schoolsec	NFA_schoolsec/density_NFApPers_schoolsec

var_name		domain	measure	unit	entity_group	entity_key	Formula
StatPAX_schoolprim	🚗	mobility	statistical_persons	-	usage	schoolprim	NFA_schoolprim/density_NFApPers_schoolprim
StatPAX_retail	🚗	mobility	statistical_persons	-	usage	retailfood	NFA_retailfood/density_NFApPers_retail
StatPAX_retailother	🚗	mobility	statistical_persons	-	usage	retailother	NFA_retailother/density_NFApPers_retail_other
e_kWhm2a	☑	tool	precision	kWh/m²NFA			
EV_charging_stations	🚗	mobility	charging_stations	-			🚗 !D48
EV_charging_station_power	🚗	mobility	charging_station_power	kW			🚗 !D49
seasonal_interpolation	🔧	tool	seasonal_interpolation	text			
plugloads_residential	👤	Space Use	plugload	kWh/m²NFA	usage	residential	👤 !D9
plugloads_office	👤	Space Use	plugload	kWh/m²NFA	usage	office	👤 !D10
plugloads_schoolsec	👤	Space Use	plugload	kWh/m²NFA	usage	schoolsec	👤 !D11
plugloads_schoolprim	👤	Space Use	plugload	kWh/m²NFA	usage	schoolprim	👤 !D12
plugloads_retailfood	👤	Space Use	plugload	kWh/m²NFA	usage	retailfood	👤 !D13
plugloads_retailother	👤	Space Use	plugload	kWh/m²NFA	usage	retailother	👤 !D14
plugloads_other	👤	Space Use	plugload	kWh/m²NFA	usage	other	👤 !D15
seasonal_phase_h	🔧	tool	hour_phase	h			

A2. Simulation energy flow and state timestep variables

Tool Shema Version 1.11.4

Column	Meaning
var_name	internal model variable
label	readable name
unit	physical unit
domain	model domain
measure	calculation group
source	input / output / intermediate
formula	implemented formula
comment	methodological note

A2. Simulation model and indicator calculation

Simulation Tool Sheet “SIM2” Table “Sim”: Energy and state time step data variables

var_name	Domain	Measure		Unit	Formula
h					1
Monat_nr	Datetime	Datetime		int	1
Monat	Datetime	Datetime		monat	Januar
date	Datetime	Datetime		date	43101
hour_of_the_day	Datetime	Datetime		hotd	0
day_of_the_year	Datetime	Datetime		doty	1
Ti0uncooled			21	°C	=IF(1-NFA_cooled/NFA_total>0,Tsetheat_min,#N/A)
Ti0cooled			21	°C	=IF(NFA_cooled/NFA_total>0,Tsetheat_min,#N/A)
Ta	Weather	Temperature		°C	=OFFSET(Weather!M25,0,(weather_index-1)*7)
rel_humidity	Weather	Luftfeuchte		%	=OFFSET(Weather!N25,0,(weather_index-1)*7)
Irr_nord	Weather	Irradiation		W/m²Nord	=OFFSET(Weather!H25,0,(weather_index-1)*7+4)
Irr_east	Weather	Irradiation		W/m²Ost	=OFFSET(Weather!I25,0,(weather_index-1)*7)
Irr_south	Weather	Irradiation		W/m²Süd	=OFFSET(Weather!J25,0,(weather_index-1)*7)
Irr_west	Weather	Irradiation		W/m²West	=OFFSET(Weather!K25,0,(weather_index-1)*7)
Irr_horizontal	Weather	Irradiation		W/m²Hori	=OFFSET(Weather!L25,0,(weather_index-1)*7-4)
is_heating_period	Control	Control		bool	=--INDEX(heating_months, [@[Monat_nr]])
is_cooling_period	Control	Control		bool	=--INDEX(cooling_months, [@[Monat_nr]])
season_step	Control	Control		%	=0.5*([@[is_heating_period]]+NOT([@[is_cooling_period]]))
Übergangszeit	Control	Control		bool	=IF([@[season_step]]=0.5,1,0)
season_wave	Control	Control		%	=COS(([@h]+seasonal_phase_h)/8760*2*PI())/2+0.5
percent_winter	Control	Control		%	=IF(seasonal_interpolation="step", [@[season_step]],[@[season_wave]])
dT_uncooled	Control	Control			=IFNA([@Ta]-[@Ti0uncooled],0)
dT_cooled	Control	Control			=IFNA([@Ta]-[@Ti0cooled],0)
mobile_shading	Control	Control		%	=fc_residential
Qvinf_u	Thermal	Infiltration		Wh/m²uncooled	=[@[dT_uncooled]]*NFV_u*vent_infiltration_ACH*cp_air*per_NFA_uncooled
Qvinf_c	Thermal	Infiltration		Wh/m²cooled	=[@[dT_cooled]]*NFV_c*vent_infiltration_ACH*cp_air*per_NFA_cooled

var_name	Domain	Measure	Unit	Formula
ACH_residential	Thermal	Air Change per Hour	m³/h	=NFA_residential*rh_residential*vent_scale_residential*Usage_Wohnen[@[Demand Luftwechsel]]
ACH_office	Thermal	Air Change per Hour	m³/h	=NFA_office*rh_office*vent_scale_office*Usage_Büro[@[Demand Luftwechsel]]
ACH_edusec	Thermal	Air Change per Hour	m³/h	=NFA_schoolsec*rh_schoolsec*vent_scale_school_sec*Usage_Schule[@[Demand Luftwechsel]]
ACH_eduprim	Thermal	Air Change per Hour	m³/h	=NFA_schoolprim*rh_schoolprim*vent_scale_school_prim*Usage_KIGA[@[Demand Luftwechsel]]
ACH_retfood	Thermal	Air Change per Hour	m³/h	=NFA_retailfood*rh_retailfood*vent_scale_supermarket*Usage_Handel_Food[@[Demand Luftwechsel]]
ACH_retail	Thermal	Air Change per Hour	m³/h	=NFA_retailother*rh_retailother*vent_scale_retail*Usage_Handel_NonFood[@[Demand Luftwechsel]]
ACH_otherusage	Thermal	Air Change per Hour	m³/h	0
ACH_mechvent_therm_u	Thermal	Air Change per Hour		=SUMPRODUCT(sim[@[ACH_residential]:[ACH_otherusage]],vent_mech_shares,cool_passive_shares_T,1-((heat_recovery_rates_summer)+[@[percent_winter]]*(heat_recovery_rates_winter-heat_recovery_rates_summer)))
ACH_mechvent_therm_c	Thermal	Air Change per Hour	m³/h	=SUMPRODUCT(sim[@[ACH_residential]:[ACH_otherusage]],vent_mech_shares,cool_active_shares_T,1-((heat_recovery_rates_summer)+[@[percent_winter]]*(heat_recovery_rates_winter-heat_recovery_rates_summer)))
ACH_mechvent_u	Thermal	Air Change per Hour	m³/h	=SUMPRODUCT(sim[@[ACH_residential]:[ACH_otherusage]],vent_mech_shares,cool_passive_shares_T)
ACH_mechvent_c	Thermal	Air Change per Hour	m³/h	=SUMPRODUCT(sim[@[ACH_residential]:[ACH_otherusage]],vent_mech_shares,cool_active_shares_T)
Qvmechvent_u	Thermal	Ventilation	Wh/m²uncooled	=[@[dT_uncooled]]*[@[ACH_mechvent_therm_u]]*cp_air*per_NFA_uncooled
Qvmechvent_c	Thermal	Ventilation	Wh/m²cooled	=[@[dT_cooled]]*[@[ACH_mechvent_therm_c]]*cp_air*per_NFA_cooled
Qvwindow_u	Thermal	Ventilation	Wh/m²uncooled	=[@[dT_uncooled]]*cp_air*SUMPRODUCT(sim[@[ACH_residential]:[ACH_otherusage]],vent_window_shares,cool_passive_shares_T)*per_NFA_uncooled
Qvwindow_c	Thermal	Ventilation	Wh/m²NFA	=[@[dT_cooled]]*cp_air*SUMPRODUCT(sim[@[ACH_residential]:[ACH_otherusage]],vent_window_shares,cool_active_shares_T)*per_NFA_cooled
QT_u	Thermal	Transmission	Wh/m²uncooled	=[@[dT_uncooled]]*transmittance_Wm2

var_name	Domain	Measure	Unit	Formula
QT_c	Thermal	Transmission	Wh/m²cooled	=[@[dT_cooled]]*transmittance_Wm2
QSwinter	Thermal	Solar gains	Wh	=SUMPRODUCT(sim[@[Irr_nord]:[Irr_horizontal]],apertures_winter)
QSsummer	Thermal	Solar gains	Wh	=SUMPRODUCT(sim[@[Irr_nord]:[Irr_horizontal]],apertures_summer)
QS_u_unshaded	Thermal	Solar gains	Wh/m²uncooled	=([@[percent_winter]]*[@QSwinter]+(1-[@[percent_winter]])*[@QSsummer])*per_NFA*(NFAfrac_u>0)
QS_c_unshaded	Thermal	Solar gains	Wh/m²cooled	=([@[percent_winter]]*[@QSwinter]+(1-[@[percent_winter]])*[@QSsummer])*per_NFA*(NFAfrac_c>0)
QS_u	Thermal	Solar gains	Wh/m²uncooled	=[@[QS_u_unshaded]]*(1-(1-mob_shading_factor_u)*AND([@[is_cooling_period]],QS_max_shading_u<[@[QS_u_unshaded]]))
QS_c	Thermal	Solar gains	Wh/m²cooled	=[@[QS_c_unshaded]]*(1-(1-mob_shading_factor_c)*AND([@[is_cooling_period]],QS_max_shading_c<[@[QS_c_unshaded]]))
QS	Thermal	Solar gains	Wh/m²NFA	=([@[percent_winter]]*[@QSwinter]+(1-[@[percent_winter]])*[@QSsummer])*per_NFA
QI_residential	Thermal	Internal gains	Wh/m²	=(usage_concurrency_winter_residential*[@[percent_winter]]*Usage_Wohnen[@[Innere Wärmen Min]]+usage_concurrency_summer_residential*(1-[@[percent_winter]])*Usage_Wohnen[@[Innere Wärmen Max]])
QI_office	Thermal	Internal gains	Wh/m²	=(usage_concurrency_winter_office*[@[percent_winter]]*Usage_Büro[@[Innere Wärmen Min]]+usage_concurrency_summer_office*(1-[@[percent_winter]])*Usage_Büro[@[Innere Wärmen Max]])
QI_edusec	Thermal	Internal gains	Wh/m²	=(usage_concurrency_winter_schoolsec*[@[percent_winter]]*Usage_Schule[@[Innere Wärmen Min]]+usage_concurrency_summer_schoolsec*(1-[@[percent_winter]])*Usage_Schule[@[Innere Wärmen Max]])
QI_eduprim	Thermal	Internal gains	Wh/m²	=(usage_concurrency_winter_schoolprim*[@[percent_winter]]*Usage_KIGA[@[Innere Wärmen Min]]+usage_concurrency_summer_schoolprim*(1-[@[percent_winter]])*Usage_KIGA[@[Innere Wärmen Max]])
QI_retfood	Thermal	Internal gains	Wh/m²	=(usage_concurrency_winter_retailfood*[@[percent_winter]]*Usage_Handel_Food[@[Innere Wärmen Winter]]+usage_concurrency_summer_retailfood*(1-[@[percent_winter]])*Usage_Handel_Food[@[Innere Wärmen Sommer]])
QI_retail	Thermal	Internal gains	Wh/m²	=(usage_concurrency_winter_retailother*[@[percent_winter]]*Usage_Handel_NonFood[@[Innere Wärmen Winter]]+usage_concurrency_summer_retailother*(1-[@[percent_winter]])*Usage_Handel_NonFood[@[Innere Wärmen Sommer]])
QI_otherusage	Thermal	Internal gains	Wh/m²	=(usage_concurrency_winter_other*[@[percent_winter]]*Usage_Handel_NonFood[@[Innere Wärmen Winter]]+usage_concurrency_summer_other*(1-[@[percent_winter]])*Usage_Handel_NonFood[@[Innere Wärmen Sommer]])

var_name	Domain	Measure	Unit	Formula
QI_u	Thermal	Internal gains	Wh/m²uncooled	=SUMPRODUCT(sim[@[QI_residential]:[QI_otherusage]],cool_passive_shares_T,NFA_shares_T)*NFA_total*per_NFA_uncooled
QI_c	Thermal	Internal gains	Wh/m²cooled	=SUMPRODUCT(sim[@[QI_residential]:[QI_otherusage]],cool_active_shares_T,NFA_shares_T)*NFA_total*per_NFA_cooled
Ti_passive_uncooled	Thermal		°C	=[@Ti0uncooled]+([@[Qvinf_u]]+[@[Qvmechvent_u]]+[@[Qvwindow_u]]+[@[QT_u]]+[@[QS_u]]+[@[QI_u]])/heat_capacity_effective_m2
Ti_passive_cooled	Thermal		°C	=[@Ti0cooled]+([@[Qvinf_c]]+[@[Qvmechvent_c]]+[@[Qvwindow_c]]+[@[QT_c]]+[@[QS_c]]+[@[QI_c]])/heat_capacity_effective_m2
dT_heating_uncooled_min	Nutzenergie Bedarfe	Temperature-Differenz		=MAX(0,IFNA(Tsetheat_min-[@[Ti_passive_uncooled]],0))
dT_heating_cooled_min		Temperature-Differenz		=MAX(0,IFNA(Tsetheat_min-[@[Ti_passive_cooled]],0))
dT_heating_uncooled_max		Temperature-Differenz		=MAX(0,IFNA(Tsetheat_flex-[@[Ti_passive_uncooled]],0))
dT_heating_cooled_max		Temperature-Differenz		=MAX(0,IFNA(Tsetheat_flex-[@[Ti_passive_cooled]],0))
Qh_to_room_uncooled_min	Heat demand	Heat provision	Wh/m²uncooled	=MIN(QHmax_room_m2,[@[is_heating_period]]*([@[dT_heating_uncooled_min]]*heat_cap_eff_uncooled_m2))
Qh_to_room_cooled_min	Heat demand	Heat provision	Wh/m²cooled	=MIN(QHmax_room_m2,[@[is_heating_period]]*([@[dT_heating_cooled_min]]*heat_cap_eff_cooled_m2))
Qh_to_room_uncooled_max	Heat demand	Heat provision	Wh/m²uncooled	=MIN(QHmax_room_m2,AND([@[is_heating_period]],NOT([@[is_cooling_period]]))*([@[dT_heating_uncooled_max]]*heat_cap_eff_uncooled_m2))
Qh_to_room_cooled_max	Heat demand	Heat provision	Wh/m²cooled	=MIN(QHmax_room_m2,AND([@[is_heating_period]],NOT([@[is_cooling_period]]))*([@[dT_heating_cooled_max]]*heat_cap_eff_cooled_m2))
Qh_to_room_min	Heat demand	HWB	Wh/m²NFA	=SUMPRODUCT(sim[@[Qh_to_room_uncooled_min]:[Qh_to_room_cooled_min]],$H$4:$I$4)
Qh_to_room_flex_u	Heat demand	HWB	Wh/m²uncooled	=[@[Qh_to_room_uncooled_max]]-[@[Qh_to_room_uncooled_min]]
Qh_to_room_flex_c	Heat demand	HWB	Wh/m²cooled	=[@[Qh_to_room_cooled_max]]-[@[Qh_to_room_cooled_min]]
Qh_to_room_flex_val	Heat demand	HWB		=[@[Qh_to_room_flex_u]]*NFAfrac_u+[@[Qh_to_room_flex_c]]*+NFAfrac_c
Qh_to_room_flex	Heat demand	HWB	Wh/m²NFA	=SUMPRODUCT(sim[@[Qh_to_room_uncooled_max]:[Qh_to_room_cooled_max]],$H$4:$I$4)-[@[Qh_to_room_min]]
waste_heat_potential	Heat demand		Wh/m²NFA	0
Qh_min_wasteheat	Heat demand		Wh/m²	=MAX(0,MIN([@[waste_heat_potential]],[@[Qh_to_room_min]]))

var_name	Domain	Measure	Unit	Formula
Qh_min_1el	Heat demand	Minimale Deckung raumseitig	Wh/m²NFA	=MIN([@[Qh_to_room_min]]-[@[Qh_min_wasteheat]],QHmax_1el_m2)
Qh_min_2th	Heat demand	Minimale Deckung raumseitig	Wh/m²NFA	=MIN([@[Qh_to_room_min]]-SUM(sim[@[Qh_min_wasteheat]:[Qh_min_1el]]),QHmax_2th_m2)
Qh_min_3el	Heat demand	Minimale Deckung raumseitig	Wh/m²NFA	=MIN([@[Qh_to_room_min]]-SUM(sim[@[Qh_min_wasteheat]:[Qh_min_2th]]),QHmax_3el_m2)
Qh_min_4th	Heat demand	Minimale Deckung raumseitig	Wh/m²NFA	=MIN([@[Qh_to_room_min]]-SUM(sim[@[Qh_min_wasteheat]:[Qh_min_3el]]),QHmax_4th_m2)
Qh_wasteheat_flex	Heat demand	🔥 + Flexibles Potential	Wh/m²NFA	=MIN([@[Qh_to_room_flex]],[@[waste_heat_potential]]-[@[Qh_min_wasteheat]])
Qh_flex_1el_potential	Heat demand	🔥 + Flexibles Potential	Wh/m²NFA	=FLEX_heat1_use*MIN(QHmax_1el_m2-[@[Qh_min_1el]],MAX([@[Qh_to_room_flex]]-[@[Qh_wasteheat_flex]],0))
Qh_flex_3el_potential	Heat demand	🔥 + Flexibles Potential	Wh/m²NFA	=FLEX_heat3_use*MIN(QHmax_3el_m2-[@[Qh_min_3el]],[@[Qh_to_room_flex]]-[@[Qh_wasteheat_flex]])
Qc_from_room_min	Cooling demand	room	Wh/m²cooled	=MIN(QCmax_room_m2,[@[is_cooling_period]]*MAX(IFNA([@[Ti_passive_cooled]]-Tsetcool_max,0),0)*heat_cap_eff_cooled_m2)
Qc_from_room_flex	Cooling demand	room	Wh/m²cooled	=MIN(QCmax_room_m2-[@[Qc_from_room_min]],AND([@[is_cooling_period]],NOT([@[is_heating_period]]))*MAX(IFNA([@[Ti_passive_cooled]]-Tsetcool_flex,0),0)*heat_cap_eff_cooled_m2+[@[Qc_from_room_min]])
Qc_min_0fc	Cooling demand	room	Wh/m²cooled	=MIN([@[Qc_from_room_min]],QCmax_freecooling)
Qc_min_1el	Cooling demand	room	Wh/m²cooled	=MIN([@[Qc_from_room_min]]-[@[Qc_min_0fc]],QCmax_1el)
Qc_min_2th	Cooling demand	room	Wh/m²cooled	=MIN([@[Qc_from_room_min]]-SUM(sim[@[Qc_min_0fc]:[Qc_min_1el]]),QCmax_2th)
Qc_min_3el	Cooling demand	room	Wh/m²cooled	=MIN([@[Qc_from_room_min]]-SUM(sim[@[Qc_min_0fc]:[Qc_min_2th]]),QCmax_3el)
Qc_flex_1el	Cooling demand	room	Wh/m²cooled	=FLEX_cool1_use*MIN(QCmax_1el-[@[Qc_min_1el]],[@[Qc_from_room_flex]])
Qc_flex_3el	Cooling demand	room	Wh/m²cooled	=FLEX_cool3_use*MIN(QCmax_3el-[@[Qc_min_3el]],[@[Qc_from_room_flex]])

var_name	Domain	Measure	Unit	Formula
Tdhw1_0	DHW	DHW Temperature	°C	=DHW_Tmin*DHW_1_is_used
Tdhw2_0	DHW	DHW Temperature	°C	=DHW_Tmin*DHW_2_is_used
DHW_residential_kW	DHW	HEAT DEMAND per usage	kW	=NFA_residential*Usage_Wohnen[@[Warmwasserbedarf_W_m2]]*DHW_demand_residential_kWhm2/DHW_default_residential_kWhm2/1000
DHW_office_kW	DHW	HEAT DEMAND per usage	kW	=NFA_office*Usage_Büro[@[Warmwasserbedarf_W_m2]]*DHW_demand_office_kWhm2/DHW_default_office_kWhm2/1000
DHW_schoolsec_kW	DHW	HEAT DEMAND per usage	kW	=NFA_schoolsec*Usage_Schule[@[Warmwasserbedarf_W_m2]]*DHW_demand_schoolsec_kWhm2/DHW_default_schoolsec_kWhm2/1000
DHW_schoolprim_kW	DHW	HEAT DEMAND per usage	kW	=NFA_schoolprim*Usage_KIGA[@[Warmwasserbedarf_W_m2]]*DHW_demand_schoolprim_kWhm2/DHW_default_schoolprim_kWhm2/1000
DHW_retailsupermarket_kW	DHW	HEAT DEMAND per usage	kW	=NFA_retailfood*Usage_Handel_Food[@[Warmwasserbedarf_W_m2]]*DHW_demand_retailfood_kWhm2/DHW_default_retailfood_kWhm2/1000
DHW_retailother_kW	DHW	HEAT DEMAND per usage	kW	=NFA_retailother*Usage_Handel_NonFood[@[Warmwasserbedarf_W_m2]]*DHW_demand_retailother_kWhm2/DHW_default_retailother_kWhm2/1000
DHW_other_kW	DHW	HEAT DEMAND per usage	kW	0
DHW_storage_losses_1	DHW	DHW Losses	kW	=DHW_losses_1*DHW_storage_1_liter*cp_water/1000
DHW_storage_losses_2	DHW	DHW Losses	kW	=DHW_losses_2*DHW_storage_2_liter*cp_water/1000
DHW_heat_demand_1_kW	DHW	DHW Use	kW	=DHW_1_is_used*([@[DHW_storage_losses_1]]+DHW_1_incl_distribution_factor*SUMPRODUCT(sim[@[DHW_residential_kW]:[DHW_other_kW]],DHW_system1_shares))
DHW_heat_demand_2_kW	DHW	DHW Use	kW	=DHW_2_is_used*([@[DHW_storage_losses_2]]+DHW_2_incl_distribution_factor*SUMPRODUCT(sim[@[DHW_residential_kW]:[DHW_other_kW]],1-(DHW_system1_shares)))
Tdhw1_passive_losses	DHW	DHW Use		=[@[Tdhw1_0]]-IFERROR([@[DHW_heat_demand_1_kW]]*1000/DHW_storage_1_liter/cp_water,0)
Tdhw2_passive_losses	DHW	DHW Use		=[@[Tdhw2_0]]-IFERROR([@[DHW_heat_demand_2_kW]]*1000/DHW_storage_2_liter/cp_water,0)
Qdhw_1_min	DHW	HEAT DEMAND Minimum	Wh/m²NFA	=MAX(0,(DHW_Tmin-[@[Tdhw1_passive_losses]])*DHW_storage_1_liter*cp_water)*per_NFA
Qdhw_2_min	DHW	HEAT DEMAND Minimum	Wh/m²NFA	=MAX(0,(DHW_Tmin-[@[Tdhw2_passive_losses]])*DHW_storage_2_liter*cp_water)*per_NFA
Qdhw_min	DHW	HEAT DEMAND Minimum	Wh/m²NFA	=SUM(sim[@[Qdhw_1_min]:[Qdhw_2_min]])

var_name	Domain	Measure	Unit	Formula
Qdhw_1_flexpotential	DHW	HEAT DEMAND Flexibel	Wh/m²NFA	=flex_dhw_use*DHW_1_is_electric*MAX(0,(DHW_Tmax-[@[Tdhw1_passive_losses]])*DHW_storage_1_liter*cp_water*per_NFA-[@[Qdhw_1_min]])
Qdhw_2_flexpotential	DHW	HEAT DEMAND Flexibel	Wh/m²NFA	=flex_dhw_use*DHW_2_is_electric*MAX(0,(DHW_Tmax-[@[Tdhw2_passive_losses]])*DHW_storage_2_liter*cp_water*per_NFA-[@[Qdhw_2_min]])
Edhw_1_min_el	DHW	HEAT DEMAND Flexibel	Wh/m²NFA	=[@[Qdhw_1_min]]*(DHW_conversion_1*DHW_1_is_electric+DHW_1_el_aux)
Edhw_2_min_el	DHW	HEAT DEMAND Flexibel	Wh/m²NFA	=[@[Qdhw_2_min]]*(DHW_conversion_2*DHW_2_is_electric+DHW_2_el_aux)
EVd_res	E-Mobility	EV Anzahl Vor Ort	Cars	=EV_count_residential*mob[@[p_res_d]]
EVd_work	E-Mobility	EV Anzahl Vor Ort		=EV_count_work*mob[@[p_work_d]]
EVd_retail	E-Mobility	EV Anzahl Vor Ort		=EV_count_retail*mob[@[p_retail_d]]
EV_SOC0_d_res	E-Mobility	EV State-of-Charge	%	0.5
EV_SOC0_d_work	E-Mobility	EV State-of-Charge	%	0.5
EV_SOC0_d_retail	E-Mobility	EV State-of-Charge	%	0.5
EV_SOC0_a_res	E-Mobility	EV State-of-Charge		0.5
EV_SOC0_a_work	E-Mobility	EV State-of-Charge		0.5
EV_SOC0_a_retail	E-Mobility	EV State-of-Charge		0.5
EV_maxpower	E-Mobility		Wh/m²	=MIN(SUM(sim[@[EVd_res]:[EVd_retail]]),EV_charging_stations)*EV_charging_station_power*1000*per_NFA
Eev_Cmin_res	E-Mobility	Charge Minimum	Wh/m²	=MIN(MIN([@[EVd_res]],EV_charging_stations)*EV_charging_station_power,[@[EVd_res]]*EV_battsize_kWh*MIN(EV_max_charging_power_ratio,MAX(EV_soc_minimum-[@[EV_SOC0_d_res]],0))*1000*per_NFA)
Eev_Cmin_work	E-Mobility	Charge Minimum	Wh/m²	=MIN([@[EV_maxpower]]-[@[Eev_Cmin_res]],[@[EVd_work]]*EV_battsize_kWh*MIN(EV_max_charging_power_ratio,MAX(EV_soc_min_work-[@[EV_SOC0_d_work]],0))*1000*per_NFA)

var_name	Domain	Measure	Unit	Formula
Eev_Cmin_retail	E-Mobility	Charge Minimum	Wh/m²	=MIN([@[EV_maxpower]]-[@[Eev_Cmin_res]]-[@[Eev_Cmin_work]],[@[EVd_retail]]*EV_battsize_kWh*MIN(EV_max_charging_power_ratio,MAX(EV_soc_min_retail-[@[EV_SOC0_d_retail]],0))*1000*per_NFA)
Eev_Cmin	E-Mobility	Charge Minimum	Wh/m²	=SUM(sim[@[Eev_Cmin_res]:[Eev_Cmin_retail]])
Eev_Cflex_pot_res	E-Mobility	Charge Maximum	Wh/m²	=MIN([@[EV_maxpower]]-[@[Eev_Cmin]],[@[EVd_res]]*EV_battsize_kWh*MAX(MIN(1-[@[EV_SOC0_d_res]],EV_max_charging_power_ratio-MAX(EV_soc_minimum-[@[EV_SOC0_d_res]],0)),0)*1000*per_NFA)
Eev_Cflex_pot_work	E-Mobility	Charge Maximum	Wh/m²	=MIN([@[EV_maxpower]]-[@[Eev_Cmin]]-[@[Eev_Cflex_pot_res]],[@[EVd_work]]*EV_battsize_kWh*MAX(MIN(1-[@[EV_SOC0_d_work]],EV_max_charging_power_ratio-MAX(EV_soc_min_work-[@[EV_SOC0_d_work]],0)),0)*1000*per_NFA)
Eev_Cflex_pot_retail	E-Mobility	Charge Maximum	Wh/m²	=MIN([@[EV_maxpower]]-[@[Eev_Cmin]]-[@[Eev_Cflex_pot_res]]-[@[Eev_Cflex_pot_work]],[@[EVd_retail]]*EV_battsize_kWh*MAX(MIN(1-[@[EV_SOC0_d_retail]],EV_max_charging_power_ratio-MAX(EV_soc_min_retail-[@[EV_SOC0_d_retail]],0)),0)*1000*per_NFA)
Eev_Cflex_pot	E-Mobility	Charge Maximum	Wh/m²	=SUM(sim[@[Eev_Cflex_pot_res]:[Eev_Cflex_pot_retail]])
Eev_Dflex_pot_res	E-Mobility	Discharge Maximum	Wh/m²	=ev_bidirectional_use*[@[EVd_res]]*EV_battsize_kWh*MIN(MAX([@[EV_SOC0_d_res]]-EV_soc_min_discharge,0),EV_max_charging_power_ratio)*1000*per_NFA
Eev_Dflex_pot_work	E-Mobility	Discharge Maximum	Wh/m²	=FALSE*[@[EVd_work]]*EV_battsize_kWh*MIN(MAX([@[EV_SOC0_d_work]]-EV_soc_min_discharge,0),EV_max_charging_power_ratio)*1000*per_NFA
Eev_Dflex_pot_retail	E-Mobility	Discharge Maximum	Wh/m²	=FALSE*[@[EVd_retail]]*EV_battsize_kWh*MIN(MAX([@[EV_SOC0_d_retail]]-EV_soc_min_discharge,0),EV_max_charging_power_ratio)*1000*per_NFA
Eev_Dflex_pot	E-Mobility	Discharge Maximum	Wh/m²	=SUM(sim[@[Eev_Dflex_pot_res]:[Eev_Dflex_pot_retail]])
Eev_S_res	E-Mobility	Discharge Drive	Wh/m²	=-EV_mileage_residential*EV_demand_kWhpkm*mob[@[Anteil JPkm]]*1000*per_NFA
Eev_S_work	E-Mobility	Discharge Drive	Wh/m²	=-EV_mileage_work*EV_demand_kWhpkm*mob[@[Anteil JPkm]]*1000*per_NFA
Eev_S_retail	E-Mobility	Discharge Drive	Wh/m²	=-EV_mileage_retail*EV_demand_kWhpkm*mob[@[Anteil JPkm]]*1000*per_NFA*mobility_is_included
Eev_Cext_res	E-Mobility	Charge Offsite	Wh/m²	=(EV_count_residential-[@[EVd_res]])*EV_battsize_kWh*MIN(EV_max_charging_power_ratio*10%,MAX(EV_soc_min_ext-[@[EV_SOC0_a_res]],0))*1000*per_NFA
Eev_Cext_work	E-Mobility	Charge Offsite	Wh/m²	=(EV_count_work-[@[EVd_work]])*EV_battsize_kWh*MIN(EV_max_charging_power_ratio*10%,MAX(EV_soc_min_ext-[@[EV_SOC0_a_work]],0))*1000*per_NFA

var_name	Domain	Measure	Unit	Formula
Eev_Cext_retail	E-Mobility	Charge Offsite	Wh/m²	=(EV_count_retail-[@[EVd_retail]])*EV_battsize_kWh*MIN(EV_max_charging_power_ratio*10%,MAX(EV_soc_min_ext-[@[EV_SOC0_a_retail]],0))*1000*per_NFA
Eev_min_intake	E-Mobility	Charge Minimum	Wh/m²	=[@[Eev_Cmin]]*EV_charging_losses_surcharge_factor
Eev_flex_intake	E-Mobility	Charge Maximum	Wh/m²	=[@[Eev_Cflex_pot]]*EV_charging_losses_surcharge_factor
BATT_cap_0	BESS	BESS	Wh/m²	=Batt_SOC_init*Batt_cap_Wh_per_NFA
Batt_auto_discharge	BESS	BESS	Wh/m²	=[@[BATT_cap_0]]*Batt_auto_discharge_factor
Batt_cap_after_losses	BESS	BESS	Wh/m²	=Batt_is_used*([@[BATT_cap_0]]-[@[Batt_auto_discharge]])
Batt_max_energy_input	BESS	BESS	Wh/m²	=Batt_is_used*(Batt_cap_Wh_per_NFA-[@[Batt_cap_after_losses]])
Eh_min_wasteheat	Electricity Min demands	Heating Minimum	Wh_el/m²	=(QH_aux_wasteheat)*[@[Qh_min_wasteheat]]
Eh_min_1el	Electricity Min demands	Heating Minimum	Wh_el/m²	=(QH_generation_eff_1el+QH_aux_el_to_th_1el)*[@[Qh_min_1el]]*(1+QH_distr_loss_1el)
Eh_min_2th	Electricity Min demands	Heating Minimum	Wh_el/m²	=QH_aux_el_to_th_2th*[@[Qh_min_2th]]*(1+QH_distr_loss_2th)
Eh_min_3el	Electricity Min demands	Heating Minimum	Wh_el/m²	=(QH_generation_eff_3el+QH_aux_el_to_th_3el)*[@[Qh_min_3el]]*(1+QH_distr_loss_3el)
Eh_min_4th	Electricity Min demands	Heating Minimum	Wh_el/m²	=QH_aux_el_to_th_4th*[@[Qh_min_4th]]*(1+QH_distr_loss_4th)
Eh_min	Electricity Min demands	Heating Minimum	Wh_el/m²	=SUM(sim[@[Eh_min_wasteheat]:[Eh_min_4th]])
Ec_min_freecooling	Electricity Min demands	Cooling Minimum	Wh_el/m²	=[@[Qc_min_0fc]]*QC_aux_fc
Ec_min_1el	Electricity Min demands	Cooling Minimum	Wh_el/m²	=$I$4*[@[Qc_min_1el]]*(1+QC_distr_losses_1el)*(QC_to_EC_1+QC_aux_1el)
Ec_min_2th	Electricity Min demands	Cooling Minimum	Wh_el/m²	=$I$4*[@[Qc_min_2th]]*(1+QC_distr_losses_2th)*(QC_aux_2th)
Ec_min_3el	Electricity Min demands	Cooling Minimum	Wh_el/m²	=$I$4*[@[Qc_min_3el]]*(1+QC_distr_losses_3el)*(QC_to_EC_3+QC_aux_3el)
Ec_min	Electricity Min demands	Cooling Minimum	Wh_el/m²	=SUM(sim[@[Ec_min_freecooling]:[Ec_min_3el]])
Ev_residential	Electricity Min demands	Ventilation Minimum	Wh_el/m²	=Ev_scale_residential*Usage_Wohnen[@[Lüfterstrom_W_m2]]*NFA_residential*per_NFA
Ev_office	Electricity Min demands	Ventilation Minimum	Wh_el/m²	=Ev_scale_office*Usage_Büro[@[Lüfterstrom_W_m2]]*NFA_office*per_NFA

var_name	Domain	Measure	Unit	Formula
Ev_edusec	Electricity Min demands	Ventilation Minimum	Wh_el/m²	=Ev_scale_school_sec*Usage_Schule[@[Lüfterstrom_W_m2]]*NFA_schoolsec*per_NFA
Ev_eduprim	Electricity Min demands	Ventilation Minimum	Wh_el/m²	=Ev_scale_school_prim*Usage_KIGA[@[Lüfterstrom_W_m2]]*NFA_schoolprim*per_NFA
Ev_retfood	Electricity Min demands	Ventilation Minimum	Wh_el/m²	=Ev_scale_retail_food*Usage_Handel_Food[@[Lüfterstrom_W_m2]]*NFA_retailfood*per_NFA
Ev_retail	Electricity Min demands	Ventilation Minimum	Wh_el/m²	=Ev_scale_retail_other*Usage_Handel_NonFood[@[Lüfterstrom_W_m2]]*NFA_retailother*per_NFA
Ev_otherusage	Electricity Min demands	Ventilation Minimum	Wh_el/m²	0
Ev_min	Electricity Min demands	Ventilation Minimum	Wh_el/m²	=SUM(sim[@[Ev_residential]:[Ev_otherusage]])
Eaux	Electricity Min demands	Aux	Wh_el/m²	=Nutzungsprofil[@[Aufzug, Control etc._W_m2]]
Elight_office	Electricity Min demands	Beleuchtung	Wh_el/m²office	=lighting_factor_office*IF(Usage_Büro[@[Betriebszeit_-]]>0,1,0)*(Plight_min_office*(1-daylightcontr_office)+daylightcontr_office*IF([@[Irr_horizontal]]*100*Daylightcoefficient_office>illuminance_min_office,0,Plight_max_office))
Elight_schoolsec	Electricity Min demands	Beleuchtung	Wh_el/m²schoolsec	=lighting_factor_schoolsec*IF(Usage_Schule[@[Personen_Pers_m2]]>0,1,0)*(Plight_min_schoolsec*(1-daylightcontr_schoolsec)+daylightcontr_schoolsec*IF([@[Irr_horizontal]]*100*Daylightcoefficient_schoolsec>illuminance_min_schoolsec,0,Plight_max_schoolsec))
Elight_schoolprim	Electricity Min demands	Beleuchtung	Wh_el/m²schoolprim	=lighting_factor_schoolprim*IF(Usage_KIGA[@[Personen_Pers_m2]]>0,1,0)*(Plight_min_schoolprim*(1-daylightcontr_schoolprim)+daylightcontr_schoolprim*IF([@[Irr_horizontal]]*100*Daylightcoefficient_schoolprim>illuminance_min_schoolprim,0,Plight_max_schoolprim))
Elight	Electricity Min demands	Beleuchtung	Wh_el/m²	=per_NFA*([@[Elight_office]]*NFA_office+[@[Elight_schoolsec]]*NFA_schoolsec+[@[Elight_schoolprim]]*NFA_schoolprim+Usage_Handel_Food[@[Lighting_W_m2]]*lighting_factor_retailfood*NFA_retailfood+Usage_Handel_NonFood[@[Lighting _W_m2]]*lighting_factor_retailother*NFA_retailother)
Edhw_min	Electricity Min demands	DHW Minimum	Wh_el/m²	=[@[Edhw_1_min_el]]+[@[Edhw_2_min_el]]
Ehvac_min	Electricity Min demands	HKLS Minimum	Wh_el/m²	=[@[Eh_min]]+[@[Ec_min]]+[@[Ev_min]]+[@[Edhw_min]]
E_plugAuxLight	Electricity Min demands	Nutzerstrom	Wh_el/m²	=Nutzungsprofil[@[Plugloads_W_m2]]+[@Eaux]+[@Elight]

var_name	Domain	Measure	Unit	Formula
Eev_min	Electricity Min demands	E-Cars	Wh_el/m²	=[@[Eev_min_intake]]
Ed_min	Electricity Min demands		Wh_el/m²	=[@[Ehvac_min]]+[@[E_plugAuxLight]]+[@[Eev_min]]
Eh_el1_flex_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich	Wh/m²	=(QH_generation_eff_1el+QH_aux_el_to_th_1el)*[@[Qh_flex_1el_potential]]*(1+QH_distr_loss_1el)
Eh_el3_flex_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich	Wh/m²	=(QH_generation_eff_3el+QH_aux_el_to_th_3el)*[@[Qh_flex_3el_potential]]*(1+QH_distr_loss_3el)
Ec_el1_flex_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich	Wh/m²	=$I$4*[@[Qc_flex_1el]]*(1+QC_distr_losses_1el)*(QC_to_EC_1+QC_aux_1el)
Ec_el3_flex_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich	Wh/m²	=$I$4*[@[Qc_flex_3el]]*(1+QC_distr_losses_3el)*(QC_to_EC_3+QC_aux_3el)
Edhw_1_flex_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich	Wh/m²	=DHW_1_is_electric*[@[Qdhw_1_flexpotential]]*(DHW_1_el_aux+DHW_conversion_1)
Edhw_2_flex_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich	Wh/m²	=DHW_2_is_electric*[@[Qdhw_2_flexpotential]]*(DHW_2_el_aux+DHW_conversion_2)
Eev_flex_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich	Wh/m²	=[@[Eev_flex_intake]]
Ebatt_charge_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich	Wh/m²	=IFERROR(Batt_is_used*MIN(Batt_max_power_specific,[@[Batt_max_energy_input]]/Batt_eff_factor_charge),0)
Etotal_flex_potential	FLEX Storage Potential	Electricity Flexibel Maximal zusätzlich		=SUM(sim[@[Eh_el1_flex_potential]:[Ebatt_charge_potential]])
PV_yield	PV Usage	Ertrag	Wh/m²	=PV_is_used*PV_scale*PV_efficiency*pv_profiles[@[Gewählte Variante]]*1000*per_NFA
PV_to_user	PV Usage	Direct use	Wh/m²	=MIN([@[PV_yield]],[@[E_plugAuxLight]])
PV_to_Eh_min	PV Usage	Direct use		=MIN([@[Eh_min]],[@[PV_yield]]-[@[PV_to_user]])
PV_to_Ec_min	PV Usage	Direct use		=MIN([@[Ec_min]],[@[PV_yield]]-SUM(sim[@[PV_to_user]:[PV_to_Eh_min]]))
PV_to_Edhw_min	PV Usage	Direct use		=MIN([@[Edhw_min]],[@[PV_yield]]-SUM(sim[@[PV_to_user]:[PV_to_Ec_min]]))

var_name	Domain	Measure	Unit	Formula
PV_to_Ev_min	PV Usage	Direct use		=MIN([@[Ev_min]],[@[PV_yield]]-SUM(sim[@[PV_to_user]:[PV_to_Edhw_min]]))
Valid_PV_direct_HVAC_use	PV Usage	Direct use		=IF(SUM(sim[@[PV_to_Eh_min]:[PV_to_Ev_min]])-[@[PV_to_HVAC_min]]<0.00000001,1,0)
PV_to_HVAC_min	PV Usage	Direct use	Wh/m²	=MIN([@[PV_yield]]-[@[PV_to_user]],[@[Eh_min]]+[@[Ec_min]]+[@[Edhw_min]]+[@[Ev_min]])
PV_to_Eev_min	PV Usage	Direct use	Wh/m²	=MIN([@[PV_yield]]-[@[PV_to_user]]-[@[PV_to_HVAC_min]],[@[Eev_min]])
PV_total_direct_use	PV Usage	Direct use	Wh/m²	=[@[PV_to_user]]+[@[PV_to_HVAC_min]]+[@[PV_to_Eev_min]]
PV_surplus	PV Usage	Storage charge	Wh/m²	=[@[PV_yield]]-[@[PV_total_direct_use]]
PV_to_Eh_flex_1el	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used*MIN([@[PV_surplus]],[@[Eh_el1_flex_potential]])
PV_to_Eh_flex_3el	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used*MIN([@[PV_surplus]],[@[Eh_el3_flex_potential]])
PV_to_Ec_flex_1el	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used*MIN([@[Ec_el1_flex_potential]],[@[PV_surplus]]-SUM(sim[@[PV_to_Eh_flex_1el]:[PV_to_Eh_flex_3el]]))
PV_to_Ec_flex_3el	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used*MIN([@[Ec_el3_flex_potential]],[@[PV_surplus]]-SUM(sim[@[PV_to_Eh_flex_1el]:[PV_to_Eh_flex_3el]]))
PV_to_Edhw1_flex	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used*MIN([@[Edhw_1_flex_potential]],[@[PV_surplus]]-SUM(sim[@[PV_to_Eh_flex_1el]:[PV_to_Ec_flex_3el]]))
PV_to_Edhw2_flex	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used*MIN([@[Edhw_2_flex_potential]],[@[PV_surplus]]-SUM(sim[@[PV_to_Eh_flex_1el]:[PV_to_Ec_flex_3el]]))
PV_to_Eev_flex	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used*MIN([@[Eev_flex_potential]],[@[PV_surplus]]-SUM(sim[@[PV_to_Eh_flex_1el]:[PV_to_Edhw2_flex]]))
PV_to_Batt	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used*Batt_is_used*MIN([@[Ebatt_charge_potential]],[@[PV_surplus]]-SUM(sim[@[PV_to_Eh_flex_1el]:[PV_to_Eev_flex]]))
PV_to_Storage	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used
PV_to_epatron	PV Usage	Storage charge	Wh/m²	=FLEX_PV_is_used
PV_total_flex_use	PV Usage	Storage charge		=SUM(sim[@[PV_to_Eh_flex_1el]:[PV_to_epatron]])
PV_to_Egrid	PV Usage	Grid feed in	Wh/m²	=[@[PV_surplus]]-SUM(sim[@[PV_to_Eh_flex_1el]:[PV_to_epatron]])
Batt_discharge_potential	BESS Usage	BESS useMin demands	Wh/m²	=Batt_is_used*NOT(AND(Batt_is_not_used_during_signals,[@Signal]))*MIN(Batt_max_power_specific,[@[Batt_cap_after_losses]]*Batt_eff_factor_discharge)
Batt_to_user	BESS Usage	BESS useMin demands	Wh/m²	=MIN(([@[E_plugAuxLight]]-[@[PV_to_user]]),[@[Batt_discharge_potential]])*Batt_is_used_for_plugloads
Batt_to_Eh_min	BESS Usage	BESS useMin demands		=MIN(([@[Eh_min]]-[@[PV_to_Eh_min]]),[@[Batt_discharge_potential]]-[@[Batt_to_user]])*Batt_is_used_for_HVACminimum
Batt_to_Ec_min	BESS Usage	BESS useMin demands		=MIN(([@[Ec_min]]-[@[PV_to_Ec_min]]),[@[Batt_discharge_potential]]-SUM(sim[@[Batt_to_user]:[Batt_to_Eh_min]]))*Batt_is_used_for_HVACminimum

var_name	Domain	Measure	Unit	Formula
Batt_to_Edhw_min	BESS Usage	BESS useMin demands		=MIN(([@[Edhw_min]]-[@[PV_to_Edhw_min]]),[@[Batt_discharge_potential]]-SUM(sim[@[Batt_to_user]:[Batt_to_Ec_min]]))*Batt_is_used_for_HVACminimum
Batt_to_Ev_min	BESS Usage	BESS useMin demands		=MIN(([@[Ev_min]]-[@[PV_to_Ev_min]]),[@[Batt_discharge_potential]]-SUM(sim[@[Batt_to_user]:[Batt_to_Edhw_min]]))*Batt_is_used_for_HVACminimum
Batt_to_HVAC_min	BESS Usage	BESS useMin demands	Wh/m²	=SUM(sim[@[Batt_to_Eh_min]:[Batt_to_Ev_min]])
Batt_to_Eev_min	BESS Usage	BESS useMin demands	Wh/m²	=MIN(([@[Eev_min]]-[@[PV_to_Eev_min]]),[@[Batt_discharge_potential]]-[@[Batt_to_user]]-[@[Batt_to_HVAC_min]])*Batt_is_used_for_EV
Batt_total_discharge	BESS Usage	BESS useMin demands	Wh/m²	=IFERROR(([@[Batt_to_user]]+[@[Batt_to_HVAC_min]]+[@[Batt_to_Eev_min]])/Batt_eff_factor_discharge,0)
Signal				=INDEX(Signals,ROW()-ROW(Signals[#Headers]),flex_Signals_selected_column)
VRGrid_potential	Flexible Grid use	Potential	Wh/m²	=FLEX_is_used*FLEX_grid_maxpower_Wm2*[@Signal]
VRGrid_to_user	Flexible Grid use	Direct use	Wh/m²	=FLEX_is_used_for_plugloads*MIN([@[VRGrid_potential]],[@[E_plugAuxLight]]-[@[PV_to_user]]-[@[Batt_to_user]])
VRGrid_to_Eh_min	Flexible Grid use	Direct use		=FLEX_is_used_for_HVAC_min*MIN([@[Eh_min]]-[@[PV_to_Eh_min]]-[@[Batt_to_Eh_min]],[@[VRGrid_potential]]-[@[VRGrid_to_user]])
VRGrid_to_Ec_min	Flexible Grid use	Direct use		=FLEX_is_used_for_HVAC_min*MIN([@[Ec_min]]-[@[PV_to_Ec_min]]-[@[Batt_to_Ec_min]],[@[VRGrid_potential]]-[@[VRGrid_to_user]]-[@[VRGrid_to_Eh_min]])
VRGrid_to_Edhw_min	Flexible Grid use	Direct use		=FLEX_is_used_for_HVAC_min*MIN([@[Edhw_min]]-[@[PV_to_Edhw_min]]-[@[Batt_to_Edhw_min]],[@[VRGrid_potential]]-SUM(sim[@[VRGrid_to_user]:[VRGrid_to_Ec_min]]))
VRGrid_to_Ev_min	Flexible Grid use	Direct use		=FLEX_is_used_for_HVAC_min*MIN([@[Ev_min]]-[@[PV_to_Ev_min]]-[@[Batt_to_Ev_min]],[@[VRGrid_potential]]-SUM(sim[@[VRGrid_to_user]:[VRGrid_to_Edhw_min]]))
VRGrid_to_HVAC_min	Flexible Grid use	Direct use	Wh/m²	=SUM(sim[@[VRGrid_to_Eh_min]:[VRGrid_to_Ev_min]])
VRGrid_to_Eev_min	Flexible Grid use	Direct use	Wh/m²	=FLEX_is_used_for_ev_min*MIN([@[VRGrid_potential]]-SUM(sim[@[VRGrid_to_user]:[VRGrid_to_Ev_min]]),[@[Eev_min]]-[@[PV_to_Eev_min]]-[@[Batt_to_Eev_min]])
VRGrid_total_min_use	Flexible Grid use	Direct use	Wh/m²	=[@[VRGrid_to_user]]+[@[VRGrid_to_HVAC_min]]+[@[VRGrid_to_Eev_min]]
VRGrid_to_Eh_flex_1el	Flexible Grid use	Storage charge	Wh/m²	=MIN([@[VRGrid_potential]]-[@[VRGrid_total_min_use]],[@[Eh_el1_flex_potential]]-[@[PV_to_Eh_flex_1el]])
VRGrid_to_Eh_flex_3el	Flexible Grid use	Storage charge	Wh/m²	=MIN([@[VRGrid_potential]]-[@[VRGrid_total_min_use]],[@[Eh_el3_flex_potential]]-[@[PV_to_Eh_flex_3el]])

var_name	Domain	Measure	Unit	Formula
VRGrid_to_Ec_flex_1el	Flexible Grid use	Storage charge	Wh/m²	=MIN([@[Ec_el1_flex_potential]]-[@[PV_to_Ec_flex_1el]],[@[VRGrid_potential]]-SUM(sim[@[VRGrid_total_min_use]:[VRGrid_to_Eh_flex_3el]]))
VRGrid_to_Ec_flex_3el	Flexible Grid use	Storage charge	Wh/m²	=MIN([@[Ec_el3_flex_potential]]-[@[PV_to_Ec_flex_3el]],[@[VRGrid_potential]]-SUM(sim[@[VRGrid_total_min_use]:[VRGrid_to_Eh_flex_3el]]))
VRGrid_to_Edhw1_flex	Flexible Grid use	Storage charge	Wh/m²	=MIN([@[Edhw_1_flex_potential]]-[@[PV_to_Edhw1_flex]],[@[VRGrid_potential]]-SUM(sim[@[VRGrid_total_min_use]:[VRGrid_to_Ec_flex_3el]]))
VRGrid_to_Edhw2_flex	Flexible Grid use	Storage charge	Wh/m²	=MIN([@[Edhw_2_flex_potential]]-[@[PV_to_Edhw2_flex]],[@[VRGrid_potential]]-SUM(sim[@[VRGrid_total_min_use]:[VRGrid_to_Ec_flex_3el]]))
VRGrid_to_HVAC_flex	Flexible Grid use	Storage charge		=SUM(sim[@[VRGrid_to_Eh_flex_1el]:[VRGrid_to_Edhw2_flex]])
VRGrid_to_Eev_flex	Flexible Grid use	Storage charge	Wh/m²	=MIN([@[Eev_flex_potential]]-[@[PV_to_Eev_flex]],[@[VRGrid_potential]]-[@[VRGrid_total_min_use]]-[@[VRGrid_to_HVAC_flex]])
VRGrid_to_Batt	Flexible Grid use	Storage charge	Wh/m²	=Batt_is_gridcharged*MIN([@[Ebatt_charge_potential]]-[@[PV_to_Batt]],[@[VRGrid_potential]]-[@[VRGrid_total_min_use]]-[@[VRGrid_to_HVAC_flex]]-[@[VRGrid_to_Eev_flex]])
VRGrid_total_flex_use	Flexible Grid use	Storage charge	Wh/m²	=SUM(sim[@[VRGrid_to_HVAC_flex]:[VRGrid_to_Batt]])
VRGrid_to_building			Wh/m²	=[@[VRGrid_total_min_use]]+[@[VRGrid_total_flex_use]]-[@[VRGrid_to_Eev_min]]-[@[VRGrid_to_Eev_flex]]
Eev_discharge_potential	Vehicle-to-Building	Potential	Wh/m²	=SUM(sim[@[Eev_Dflex_pot_res]:[Eev_Dflex_pot_retail]])*EV_charging_efficiency
Eev_to_user	Vehicle-to-Building	Direct use	Wh/m²	=-MIN([@[Eev_discharge_potential]],[@[E_plugAuxLight]]-[@[PV_to_user]]-[@[Batt_to_user]]-[@[VRGrid_to_user]])
Eev_to_Eh_min	Vehicle-to-Building	Direct use	Wh/m²	=-MIN([@[Eh_min]]-[@[PV_to_Eh_min]]-[@[Batt_to_Eh_min]]-[@[VRGrid_to_Eh_min]],[@[Eev_discharge_potential]]-[@[Eev_to_user]])
Eev_to_Ec_min	Vehicle-to-Building	Direct use	Wh/m²	=-MIN([@[Ec_min]]-[@[PV_to_Ec_min]]-[@[Batt_to_Ec_min]]-[@[VRGrid_to_Ec_min]],[@[Eev_discharge_potential]]-[@[Eev_to_user]]-[@[Eev_to_Eh_min]])
Eev_to_Edhw_min	Vehicle-to-Building	Direct use	Wh/m²	=-MIN([@[Edhw_min]]-[@[PV_to_Edhw_min]]-[@[Batt_to_Edhw_min]]-[@[VRGrid_to_Edhw_min]],[@[Eev_discharge_potential]]-[@[Eev_to_user]]-[@[Eev_to_Eh_min]]-[@[Eev_to_Ec_min]])
Eev_to_Ev_min	Vehicle-to-Building	Direct use	Wh/m²	=-MIN([@[Ev_min]]-[@[PV_to_Ev_min]]-[@[Batt_to_Ev_min]]-[@[VRGrid_to_Ev_min]],[@[Eev_discharge_potential]]-SUM(sim[@[Eev_to_user]:[Eev_to_Edhw_min]]))
Eev_to_HVAC	Vehicle-to-Building	Direct use	Wh/m²	=SUM(sim[@[Eev_to_Eh_min]:[Eev_to_Ev_min]])

var_name	Domain	Measure	Unit	Formula
Eev_discharge_total	Vehicle-to-Building	Direct use	Wh/m²	=(-[@[Eev_to_HVAC]]-[@[Eev_to_user]])/EV_charging_efficiency
Grid_to_user	grid use	Plugloads	Wh/m²	=[@[E_plugAuxLight]]-[@[PV_to_user]]-[@[Batt_to_user]]-[@[VRGrid_to_user]]+[@[Eev_to_user]]
Grid_to_Eh_min	grid use	HKLS		=[@[Eh_min]]-[@[PV_to_Eh_min]]-[@[Batt_to_Eh_min]]-[@[VRGrid_to_Eh_min]]+[@[Eev_to_Eh_min]]
Grid_to_Ec_min	grid use	HKLS		=[@[Ec_min]]-[@[PV_to_Ec_min]]-[@[Batt_to_Ec_min]]-[@[VRGrid_to_Ec_min]]+[@[Eev_to_Ec_min]]
Grid_to_Edhw_min	grid use	HKLS		=[@[Edhw_min]]-[@[PV_to_Edhw_min]]-[@[Batt_to_Edhw_min]]-[@[VRGrid_to_Edhw_min]]+[@[Eev_to_Edhw_min]]
Grid_to_Ev_min	grid use	HKLS		=[@[Ev_min]]-[@[PV_to_Ev_min]]-[@[Batt_to_Ev_min]]-[@[VRGrid_to_Ev_min]]+[@[Eev_to_Ev_min]]
test_grid_hvac	grid use	HKLS		=[@[Ehvac_min]]-[@[PV_to_HVAC_min]]-[@[Batt_to_HVAC_min]]-[@[VRGrid_to_HVAC_min]]+[@[Eev_to_HVAC]]
Grid_to_HVAC_min	grid use	HKLS	Wh/m²	=SUM(sim[@[Grid_to_Eh_min]:[Grid_to_Ev_min]])
Grid_to_Eev_min	grid use	EV		=[@[Eev_min]]-[@[PV_to_Eev_min]]-[@[Batt_to_Eev_min]]-[@[VRGrid_to_Eev_min]]
Grid_to_building_min	grid use	Gebäude		=[@[Grid_to_HVAC_min]]+[@[Grid_to_user]]
E_grid	grid use	Insgesamt		=SUM([@[Grid_to_user]],[@[Grid_to_HVAC_min]],[@[Grid_to_Eev_min]])
Eh_flex_1el_final	Speicherladung (aus PV + Netzsignal)	HKLS	Wh/m²	=[@[PV_to_Eh_flex_1el]]+[@[VRGrid_to_Eh_flex_1el]]
Eh_flex_3el_final	Speicherladung (aus PV + Netzsignal)	HKLS	Wh/m²	=[@[PV_to_Eh_flex_3el]]+[@[VRGrid_to_Eh_flex_3el]]
Ec_flex_1el_final	Speicherladung (aus PV + Netzsignal)	HKLS	Wh/m²	=[@[PV_to_Ec_flex_1el]]+[@[VRGrid_to_Ec_flex_1el]]
Ec_flex_3el_final	Speicherladung (aus PV + Netzsignal)	HKLS	Wh/m²	=[@[PV_to_Ec_flex_3el]]+[@[VRGrid_to_Ec_flex_3el]]
Edhw1_flex_final	Speicherladung (aus PV + Netzsignal)	HKLS	Wh/m²	=[@[PV_to_Edhw1_flex]]+[@[VRGrid_to_Edhw1_flex]]

var_name	Domain	Measure	Unit	Formula
Edhw2_flex_final	Speicherladung (aus PV + Netzsignal)	HKLS	Wh/m²	=[@[PV_to_Edhw2_flex]]+[@[VRGrid_to_Edhw2_flex]]
Eev_flex_final	Speicherladung (aus PV + Netzsignal)	EV	Wh/m²	=([@[PV_to_Eev_flex]]+[@[VRGrid_to_Eev_flex]])*EV_charging_efficiency
Eev_flex_final_res	Speicherladung (aus PV + Netzsignal)	EV	Wh/m²	=IFERROR([@[Eev_flex_final]]*[@[Eev_Cflex_pot_res]]/[@[Eev_Cflex_pot]],0)
Eev_flex_final_work	Speicherladung (aus PV + Netzsignal)	EV	Wh/m²	=IFERROR([@[Eev_flex_final]]*[@[Eev_Cflex_pot_work]]/[@[Eev_Cflex_pot]],0)
Eev_flex_final_ret	Speicherladung (aus PV + Netzsignal)	EV	Wh/m²	=IFERROR([@[Eev_flex_final]]*[@[Eev_Cflex_pot_retail]]/[@[Eev_Cflex_pot]],0)
Batt_total_charge	Speicherladung (aus PV + Netzsignal)	BESS	Wh/m²	=([@[PV_to_Batt]]+[@[VRGrid_to_Batt]])*Batt_eff_factor_charge
Qh_min_excl_distr_losses	Heizenergie	HWB Raumseitig	Wh/m²	=SUM(sim[@[Qh_min_wasteheat]:[Qh_min_4th]])
Qh_flex_wasteheat_final	Heizenergie	HWB Raumseitig		=[@[Qh_wasteheat_flex]]
Qh_flex_1el_final	Heizenergie	HWB Raumseitig	Wh/m²	=IFERROR([@[Qh_flex_1el_potential]]*[@[Eh_flex_1el_final]]/[@[Eh_el1_flex_potential]],0)
Qh_flex_3el_final	Heizenergie	HWB Raumseitig	Wh/m²	=IFERROR([@[Qh_flex_3el_potential]]*[@[Eh_flex_3el_final]]/[@[Eh_el3_flex_potential]],0)
Qh_total_final	Heizenergie	HWB Raumseitig	Wh/m²	=SUM(sim[@[Qh_min_excl_distr_losses]:[Qh_flex_3el_final]])
Qh_u	Heizenergie	HWB Raumseitig	Wh/m²uncooled	=IFERROR([@[Qh_min_excl_distr_losses]]*[@[Qh_to_room_uncooled_min]]/[@[Qh_to_room_min]],0)+IFERROR(SUM(sim[@[Qh_flex_1el_final]:[Qh_flex_3el_final]])*[@[Qh_to_room_flex_u]]/[@[Qh_to_room_flex]],0)
Qh_c	Heizenergie	HWB Raumseitig	Wh/m²cooled	=IFERROR([@[Qh_min_excl_distr_losses]]*[@[Qh_to_room_cooled_min]]/[@[Qh_to_room_min]],0)+IFERROR(SUM(sim[@[Qh_flex_1el_final]:[Qh_flex_3el_final]])*[@[Qh_to_room_flex_c]]/[@[Qh_to_room_flex]],0)
Qheb_1el	Heizenergie	Heat Load (inkl. Distr. losses)	Wh/m²	=([@[Qh_min_1el]]+[@[Qh_flex_1el_final]])*(1+QH_distr_loss_1el)
Qheb_2th	Heizenergie	Heat Load (inkl. Distr. losses)	Wh/m²	=[@[Qh_min_2th]]*(1+QH_distr_loss_2th)

var_name	Domain	Measure	Unit	Formula
Qheb_3el	Heizenergie	Heat Load (inkl. Distr. losses)	Wh/m²	=([@[Qh_min_3el]]+[@[Qh_flex_3el_final]])*(1+QH_distr_loss_3el)
Qheb_4th	Heizenergie	Heat Load (inkl. Distr. losses)	Wh/m²	=[@[Qh_min_4th]]*(1+QH_distr_loss_4th)
Qheb_total	Heizenergie	Heat Load (inkl. Distr. losses)	Wh/m²	=SUM(sim[@[Qheb_1el]:[Qheb_4th]])
QH_distr_losses	Heizenergie	Heat Load (inkl. Distr. losses)	Wh/m²	=[@[Qheb_total]]-[@[Qh_total_final]]
Eh_aux	Heizenergie	Heat Load (inkl. Distr. losses)		=[@[Qheb_1el]]*QH_aux_el_to_th_1el+[@[Qheb_2th]]*QH_aux_el_to_th_2th+[@[Qheb_3el]]*QH_aux_el_to_th_3el+[@[Qheb_4th]]*QH_aux_el_to_th_4th
Night_schedule	Kühlenergie	Nachtlüften		TRUE
Night_use	Kühlenergie	Nachtlüften		=IFERROR([@[Night_schedule]]*[@Ti0uncooled]>Tsetcool_max,FALSE)
ACH_nightvent	Kühlenergie	Nachtlüften	m³/h	=[@[Night_use]]*IF(-[@[dT_uncooled]]>2,ACH_night_m3,0)
Qv_nightvent	Kühlenergie	Nachtlüften	Wh/m²uncooled	=[@[ACH_nightvent]]*cp_air*[@[dT_uncooled]]*per_NFA_uncooled
Qc_min_excl_losses	Kühlenergie	Cooling demand raumseitig	Wh/m²cooled	=-SUM(sim[@[Qc_min_0fc]:[Qc_min_3el]])
Qc_flex_1el_final	Kühlenergie	Cooling demand raumseitig	Wh/m²cooled	=-IFERROR([@[Qc_flex_1el]]*[@[Ec_flex_1el_final]]/[@[Ec_el1_flex_potential]],0)
Qc_flex_3el_final	Kühlenergie	Cooling demand raumseitig	Wh/m²cooled	=-IFERROR([@[Qc_flex_3el]]*[@[Ec_flex_3el_final]]/[@[Ec_el3_flex_potential]],0)
Qc_flex_excl_losses	Kühlenergie	Cooling demand raumseitig	Wh/m²cooled	=[@[Qc_flex_1el_final]]+[@[Qc_flex_3el_final]]
QC_total_final	Kühlenergie	Cooling demand raumseitig	Wh/m²cooled	=[@[Qc_min_excl_losses]]+[@[Qc_flex_excl_losses]]
Qced_1el	Kühlenergie	Cooling load(inkl. Distr. losses)	Wh/m²cooled	=([@[Qc_min_1el]]-[@[Qc_flex_1el_final]])*(1+QC_distr_losses_1el)
Qced_2th	Kühlenergie	Cooling load(inkl. Distr. losses)	Wh/m²cooled	=[@[Qc_min_2th]]*(1+QC_distr_losses_2th)
Qced_3el	Kühlenergie	Cooling load(inkl. Distr. losses)	Wh/m²cooled	=([@[Qc_min_3el]]-[@[Qc_flex_3el_final]])*(1+QC_distr_losses_3el)
Qced_total	Kühlenergie	Cooling load(inkl. Distr. losses)		=SUM(sim[@[Qced_1el]:[Qced_3el]])
Qc_distr_losses	Kühlenergie	Cooling load(inkl. Distr. losses)		=[@[Qced_total]]+[@[QC_total_final]]

var_name	Domain	Measure	Unit	Formula
Ec_aux	Kühlenergie	Cooling load(inkl. Distr. losses)		=[@[Ec_min_freecooling]]+[@[Qced_1el]]*QC_aux_1el+[@[Qced_2th]]*QC_aux_2th+[@[Qced_3el]]*QC_aux_3el
Qdhw_1_flex2	Warmwasser	DHW	Wh/m²	=IFERROR([@[Edhw1_flex_final]]*(1-DHW_1_el_aux)/(DHW_conversion_1*DHW_1_is_electric+DHW_1_el_aux),0)
Qdhw_2_flex	Warmwasser	DHW	Wh/m²	=IFERROR([@[Edhw2_flex_final]]*(1-DHW_2_el_aux)/(DHW_conversion_1*DHW_1_is_electric+DHW_1_el_aux),0)
Qdhw_1_total	Warmwasser	DHW	Wh/m²	=[@[Qdhw_1_min]]+[@[Qdhw_1_flex2]]
Qdhw_2_total	Warmwasser	DHW	Wh/m²	=[@[Qdhw_2_min]]+[@[Qdhw_2_flex]]
Qdhw_total	Warmwasser	DHW	Wh/m²	=SUM(sim[@[Qdhw_1_total]:[Qdhw_2_total]])
EV_SOCc_d_res	E-Mobility	E-Mobility Beladung		=IFERROR([@[EV_SOC0_d_res]]+([@[Eev_Cmin_res]]+[@[Eev_flex_final_res]]-[@[Eev_discharge_total]])/1000*NFA_total/[@[EVd_res]]/EV_battsize_kWh,0)
EV_SOCc_d_work	E-Mobility	E-Mobility Beladung		=IFERROR([@[EV_SOC0_d_work]]+([@[Eev_Cmin_work]]+[@[Eev_flex_final_work]])/1000*NFA_total/[@[EVd_work]]/EV_battsize_kWh,0)
EV_SOCc_d_retail	E-Mobility	E-Mobility Beladung		=IFERROR([@[EV_SOC0_d_retail]]+([@[Eev_Cmin_retail]]+[@[Eev_flex_final_ret]])/1000*NFA_total/[@[EVd_retail]]/EV_battsize_kWh,0)
EV_SOCc_a_res	E-Mobility	E-Mobility Beladung		=IFERROR([@[EV_SOC0_a_res]]+([@[Eev_S_res]]+[@[Eev_Cext_res]])/(EV_count_residential-[@[EVd_res]])/EV_battsize_kWh/1000*NFA_total,0)
EV_SOCc_a_work	E-Mobility	E-Mobility Beladung		=IFERROR([@[EV_SOC0_a_work]]+([@[Eev_S_work]]+[@[Eev_Cext_work]])/(EV_count_work-[@[EVd_work]])/EV_battsize_kWh/1000*NFA_total,0)
EV_SOCc_a_retail	E-Mobility	E-Mobility Beladung		=IFERROR([@[EV_SOC0_a_retail]]+([@[Eev_S_retail]]+[@[Eev_Cext_retail]])/(EV_count_retail-[@[EVd_retail]])/EV_battsize_kWh/1000*NFA_total,0)
Ti_final_uncooled	States	Thermal Storage	°C	=[@[Ti_passive_uncooled]]+([@[Qh_u]]+[@[Qv_nightvent]])/heat_cap_eff_uncooled_m2
Ti_final_cooled	States	Thermal Storage	°C	=[@[Ti_passive_cooled]]+([@[Qh_c]]+[@[QC_total_final]])/heat_cap_eff_cooled_m2
Tdhw_1_final	States	Thermal Storage	°C	=IFERROR([@[Tdhw1_passive_losses]]+[@[Qdhw_1_total]]*NFA_total/DHW_storage_1_liter/cp_water,0)
Tdhw_2_final	States	Thermal Storage	°C	=IFERROR([@[Tdhw2_passive_losses]]+[@[Qdhw_2_total]]*NFA_total/DHW_storage_2_liter/cp_water,0)
Batt_final_Whm2	States	BESS	Wh/m²	=[@[Batt_cap_after_losses]]-[@[Batt_total_discharge]]+[@[Batt_total_charge]]
SOC_preheat_u	States	Thermal Storage		=IFERROR(([@[Ti_final_uncooled]]-Tsetheat_min)/flex_heat_dT,0)
SOC_preheat_c	States	Thermal Storage		=IFERROR(1-(Tsetheat_flex-[@[Ti_final_cooled]])/flex_heat_dT,0)
SOC_precool_c	States	Thermal Storage		=IFERROR(1-([@[Ti_final_cooled]]-Tsetcool_flex)/flex_cool_dT,0)
SOC_dhw1	States	Thermal Storage		=IFERROR(1-(DHW_Tmax-[@[Tdhw_1_final]])/(DHW_Tmax-DHW_Tmin),0)
SOC_dhw2	States	Thermal Storage		=IFERROR(1-(DHW_Tmax-[@[Tdhw_2_final]])/(DHW_Tmax-DHW_Tmin),0)

var_name	Domain	Measure	Unit	Formula
EV_SOC_d_res	States	E-Mobility		=((mob[@[p_res_d]]-mob[@[p_res_leave]])*[@[EV_SOCc_d_res]]+mob[@[p_res_return]]*[@[EV_SOCc_a_res]])/(mob[@[p_res_d]]+mob[@[p_res_return]]-mob[@[p_res_leave]])
EV_SOC_d_work	States	E-Mobility		=((mob[@[p_work_d]]-mob[@[p_work_leave]])*[@[EV_SOCc_d_work]]+mob[@[p_work_return]]*[@[EV_SOCc_a_work]])/(mob[@[p_work_d]]+mob[@[p_work_return]]-mob[@[p_work_leave]])
EV_SOC_d_retail	States	E-Mobility		=((mob[@[p_retail_d]]-mob[@[p_retail_leave]])*[@[EV_SOCc_d_retail]]+mob[@[p_retail_return]]*[@[EV_SOCc_a_retail]])/(mob[@[p_retail_d]]+mob[@[p_retail_return]]-mob[@[p_retail_leave]])
EV_SOC_a_res	States	E-Mobility		=((mob[@[p_res_a]]-mob[@[p_res_return]])*[@[EV_SOCc_a_res]]+mob[@[p_res_leave]]*[@[EV_SOCc_d_res]])/(mob[@[p_res_a]]-mob[@[p_res_return]]+mob[@[p_res_leave]])
EV_SOC_a_work	States	E-Mobility		=((mob[@[p_work_a]]-mob[@[p_work_return]])*[@[EV_SOCc_a_work]]+mob[@[p_work_leave]]*[@[EV_SOCc_d_work]])/(mob[@[p_work_a]]-mob[@[p_work_return]]+mob[@[p_work_leave]])
EV_SOC_a_retail	States	E-Mobility		=((mob[@[p_retail_a]]-mob[@[p_retail_return]])*[@[EV_SOCc_a_retail]]+mob[@[p_retail_leave]]*[@[EV_SOCc_d_retail]])/(mob[@[p_retail_a]]-mob[@[p_retail_return]]+mob[@[p_retail_leave]])
EV_SOC_res	States	E-Mobility	is this performant?	=[@[EV_SOC_d_res]]*mob[@[p_res_d]]+[@[EV_SOC_a_res]]*mob[@[p_res_a]]
EV_SOC_work	States	E-Mobility		=[@[EV_SOC_d_work]]*mob[@[p_work_d]]+[@[EV_SOC_a_work]]*mob[@[p_work_a]]
EV_SOC_retail	States	E-Mobility		=[@[EV_SOC_d_retail]]*mob[@[p_retail_d]]+[@[EV_SOC_a_retail]]*mob[@[p_retail_a]]
SOC_Batt				=[@[Batt_final_Whm2]]/Batt_cap_Wh_per_NFA
EUIh_2th	Thermal Energy carrier	Heizen	Wh/m²	=[@[Qh_min_2th]]*(1+QH_distr_loss_2th)*QH_generation_eff_2th
EUIh_4th	Thermal Energy carrier	Heizen	Wh/m²	=[@[Qh_min_4th]]*(1+QH_distr_loss_4th)*QH_generation_eff_4th
EUIc_2th	Thermal Energy carrier	Kühlen	Wh/m²	=[@[Qc_min_2th]]*(1+QC_distr_losses_2th)*QC_generation_eff_2th
EUIdhw_1th	Thermal Energy carrier	DHW	Wh/m²	=NOT(DHW_1_is_electric)*[@[Qdhw_1_min]]*DHW_conversion_1

var_name	Domain	Measure	Unit	Formula
EUIdhw_2th	Thermal Energy carrier	DHW	Wh/m²	=NOT(DHW_2_is_electric)*[@[Qdhw_2_min]]*DHW_conversion_2
cf_PEI_grid	Primary energy	Conversion factor	-	=INDEX(fPE[@],,fPE_grid_column)
cf_PEI_flex_grid	Primary energy	Conversion factor		=[@[cf_PEI_grid]]*fPE_flex_factor
cf_PEI_flex_gridsub	Primary energy	Conversion factor		=[@[cf_PEI_grid]]-[@[cf_PEI_flex_grid]]
cf_PEI_PV	Primary energy	Conversion factor		0
cf_PEI_PV_gridsub	Primary energy	Conversion factor		=[@[cf_PEI_grid]]-[@[cf_PEI_PV]]
PEI_el_user	Primary energy	Demand		=[@[cf_PEI_grid]]*[@[E_plugAuxLight]]
PEI_el_hvac	Primary energy	Demand		=[@[cf_PEI_grid]]*SUM([@[Ehvac_min]],sim[@[Eh_flex_1el_final]:[Edhw2_flex_final]])
PEI_district_heating	Primary energy	Demand		=SUMPRODUCT(sim[@[EUIh_2th]:[EUIdhw_2th]],TRANSPOSE(is_district_heating),TRANSPOSE(pe_conversion_factors_thermal))
PEI_natural_gas	Primary energy	Demand		=SUMPRODUCT(sim[@[EUIh_2th]:[EUIdhw_2th]],TRANSPOSE(is_natural_gas),TRANSPOSE(pe_conversion_factors_thermal))
PEI_biomass	Primary energy	Demand		=SUMPRODUCT(sim[@[EUIh_2th]:[EUIdhw_2th]],TRANSPOSE(is_biomass),TRANSPOSE(pe_conversion_factors_thermal))
PEI_other	Primary energy	Demand		=SUMPRODUCT(sim[@[EUIh_2th]:[EUIdhw_2th]],TRANSPOSE(is_other_thermal_pe),TRANSPOSE(pe_conversion_factors_thermal))
PEI_mob_el	Primary energy	Demand		=[@[cf_PEI_grid]]*([@[Eev_min]]+[@[Eev_flex_final]])
PEI_mob_ext	Primary energy	Demand		=[@[cf_PEI_grid]]*SUM(sim[@[Eev_Cext_res]:[Eev_Cext_retail]])
PEI_el_demand	Primary energy	Demand	WhPE/m²	=[@[cf_PEI_grid]]*[@[E_grid]]
PEI_flex_grid	Primary energy			=[@[cf_PEI_flex_grid]]*([@[VRGrid_total_min_use]]+[@[VRGrid_total_flex_use]])
PEI_flex_grid_substitution	Primary energy			=[@[cf_PEI_grid]]*([@[VRGrid_total_min_use]]+[@[VRGrid_total_flex_use]])
PEI_PV_direct	Primary energy			=[@[cf_PEI_grid]]*[@[PV_total_direct_use]]
cf_GHG_grid	☁ THG	Conversion factor	kgCO2eq./WhEE	=INDEX(fGHG[@],,fGHG_grid_column)
GHG_plugloads	☁ THG	Plugloads	gCO2eq./m² NGF	=[@[cf_GHG_grid]]*([@[Grid_to_user]]+[@[VRGrid_to_user]])
GHG_hvac	☁ THG	HKLS	gCO2eq./m² NGF	=[@[cf_GHG_grid]]*SUM([@[Grid_to_HVAC_min]],[@[VRGrid_to_HVAC_min]],sim[@[VRGrid_to_Eh_flex_1el]:[VRGrid_to_Edhw2_flex]])
cost_E_grid	Kosten Grid use			=[@[E_grid]]*prices[@[Bezug '[€']]]/1000
cost_VRGrid_flex	Kosten Grid use			=([@[VRGrid_total_min_use]]+[@[VRGrid_total_flex_use]])*prices[@[Bezug '[€']]]/1000

var_name	Domain	Measure	Unit	Formula
cost_PV_to_Egrid	Kosten Grid use			=[@[PV_to_Egrid]]*prices[@[Einspeisung '[€']]]/1000

A3. Simulation Outputs

Tool Shema Version 1.11.4

Column	Meaning
var_name	internal model variable
domain	model domain
measure	calculation group
formula	implemented formula for calculation or lookup
unit	physical unit
Temporal scope	Timeframe such as winter, summer, annual, life cycle

var_name	domain	unit	formula
Ta_annual_avg	weather	°C	MITTELWERT(sim[Ta])
Ta_avg_heating_period	weather	°C	MITTELWERTWENN(is_heating_period;1;sim[Ta])
Ta_avg_cooling_period	weather	°C	MITTELWERTWENN(is_cooling_period;1;sim[Ta])
UED_plugloads	useful_energy	kWh/m²NGFa	EUI_plugAuxLight-UED_lighting-UED_auxiliary
UED_lighting	useful_energy	kWh/m²NGFa	SUMME(sim[Elight])/1000
UED_auxiliary	useful_energy	kWh/m²NGFa	SUMME(sim[Eaux])/1000
UED_heating	useful_energy	kWh/m²NGFa	QH
UED_cooling	useful_energy	kWh/m²NGFa	QC
UED_dhw	useful_energy	kWh/m²NGFa	SUMME(sim[[DHW_residential_kW]:[DHW_other_kW]])*per_NFA
UED_ventilation	useful_energy	kWh/m²NGFa	EUIv_el
UED_mim_electric	useful_energy	kWh/m²NGFa	UED_mob_el_total
UED_mim_fossile	useful_energy	kWh/m²NGFa	EUI_mob_fossile
QT	heat_balance	kWh/m²NGFa	SUMME(sim[QT_u] * NFAfrac_u; sim[QT_c] * NFAfrac_c) * -0,001
QV	heat_balance	kWh/m²NGFa	QV_u * NFAfrac_u + QV_c * NFAfrac_c
QVn	heat_balance	kWh/m²NGFa	NFAfrac_u * QVn_u
QS	heat_balance	kWh/m²NGFa	QS_u * NFAfrac_u + QS_c * NFAfrac_c
QI	heat_balance	kWh/m²NGFa	QI_u * NFAfrac_u + QI_c * NFAfrac_c
QH	heat_balance	kWh/m²NGFa	QH_u * NFAfrac_u + QH_c * NFAfrac_c
QC	heat_balance	kWh/m²NGFa	QC_c * NFAfrac_c
QV_heatrecovery	heat_balance	kWh/m²NGFa	QV_heatrecovery_u * NFAfrac_u + QV_heatrecovery_c * NFAfrac_c
QT_winter	heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[QT_u])*NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[QT_c])*NFAfrac_c) * -0,001
QV_winter	heat_balance	kWh/m²NGFa	QV_inf_winter + QV_window_winter + QV_mechvent_winter
QVn_winter	heat_balance	kWh/m²NGFa	SUMMENPRODUKT(sim[is_heating_period]; sim[Qv_nightvent]) * NFAfrac_u * -0,001
QS_winter	heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[QS_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[QS_c]) * NFAfrac_c) * 0,001
QI_winter	heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[QI_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[QI_c]) * NFAfrac_c) * 0,001
QH_winter	heat_balance	kWh/m²NGFa	QH_min_winter + QH_flex_winter

QC_winter		heat_balance	kWh/m²NGFa	QC_min_winter + QC_flex_winter
test_heat_balance_winter	☑	heat_balance	FALSCH	(QT_winter+QV_winter+QVn_winter+QC_winter) - (QS_winter+QI_winter+QH_winter)
QT_wall_winter		heat_balance	kWh/m²NGFa	QT_winter * transmittance_walls_Wm2NFA / transmittance_Wm2
QT_roof_winter		heat_balance	kWh/m²NGFa	QT_winter * transmittance_roof_Wm2NFA / transmittance_Wm2
QT_ground_winter		heat_balance	kWh/m²NGFa	QT_winter * transmittance_fundament_Wm2NFA / transmittance_Wm2
QT_window_winter		heat_balance	kWh/m²NGFa	QT_winter * transmittance_windows_Wm2NFA / transmittance_Wm2
QT_psi_winter		heat_balance	kWh/m²NGFa	QT_winter * transmittance_heatbridge_Wm2NFA / transmittance_Wm2
QV_inf_winter		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[Qvinf_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[Qvinf_c]) * NFAfrac_c) * -0,001
QV_window_winter		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[Qvwindow_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[Qvwindow_c]) * NFAfrac_c) * -0,001
QV_mechvent_winter		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[Qvmechvent_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[Qvmechvent_c]) * NFAfrac_c) * -0,001
QV_heatrecovery_winter		heat_balance	kWh/m²NGFa	SUMMENPRODUKT(sim[is_heating_period]; sim[dT_uncooled]; sim[ACH_mechvent_u]- sim[ACH_mechvent_therm_u])*cp_air*per_NFA_uncooled * -0,001 * NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[dT_cooled]; sim[ACH_mechvent_c]- sim[ACH_mechvent_therm_c])*cp_air*per_NFA_cooled * -0,001 * NFAfrac_c
QH_min_winter		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[Qh_to_room_uncooled_min]) * NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[Qh_to_room_cooled_min]) * NFAfrac_c) * 0,001
QH_flex_winter		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[Qh_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_heating_period]; sim[Qh_c]) * NFAfrac_c) * 0,001 - QH_min_winter
QC_min_winter		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[Qc_min_excl_losses]) * NFAfrac_c) * -0,001

QC_flex_winter		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_heating_period]; sim[Qc_flex_excl_losses]) * NFAfrac_c) * -0,001
QH_flexshare_winter		heat_balance	ratio	WENNFEHLER(QH_flex_winter/QH_winter;0)
QC_flexshare_winter	F	heat_balance	ratio	WENNFEHLER(QC_flex_winter/QC_winter;0)
QT_summer		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[QT_u])*NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[QT_c])*NFAfrac_c) * -0,001
QV_summer		heat_balance	kWh/m²NGFa	QV_inf_summer + QV_window_summer + QV_mechvent_summer
QVn_summer		heat_balance	kWh/m²NGFa	SUMMENPRODUKT(sim[is_cooling_period]; sim[Qv_nightvent]) * NFAfrac_u * -0,001
QS_summer		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[QS_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[QS_c]) * NFAfrac_c) * 0,001
QI_summer		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[QI_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[QI_c]) * NFAfrac_c) * 0,001
QH_summer		heat_balance	kWh/m²NGFa	QH_min_summer + QH_flex_summer
QC_summer		heat_balance	kWh/m²NGFa	QC_min_summer + QC_flex_summer
test_heat_balance_summer	☑	heat_balance	FALSCH	(QT_summer+QV_summer+QVn_summer+QC_summer) - (QS_summer+QI_summer+QH_summer)
QT_wall_summer		heat_balance	kWh/m²NGFa	QT_summer * transmittance_walls_Wm2NFA / transmittance_Wm2
QT_roof_summer		heat_balance	kWh/m²NGFa	QT_summer * transmittance_roof_Wm2NFA / transmittance_Wm2
QT_ground_summer		heat_balance	kWh/m²NGFa	QT_summer * transmittance_fundament_Wm2NFA / transmittance_Wm2
QT_window_summer		heat_balance	kWh/m²NGFa	QT_summer * transmittance_windows_Wm2NFA / transmittance_Wm2
QT_psi_summer		heat_balance	kWh/m²NGFa	QT_summer * transmittance_heatbridge_Wm2NFA / transmittance_Wm2
QV_inf_summer		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[Qvinf_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[Qvinf_c]) * NFAfrac_c) * -0,001
QV_window_summer		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[Qvwindow_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[Qvwindow_c]) * NFAfrac_c) * -0,001
QV_mechvent_summer		heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[Qvmechvent_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[Qvmechvent_c]) * NFAfrac_c) * -0,001

QV_heatrecovery_summer	heat_balance	kWh/m²NGFa	SUMMENPRODUKT(sim[is_cooling_period]; sim[dT_uncooled]; sim[ACH_mechvent_u]-sim[ACH_mechvent_therm_u])*cp_air*per_NFA_uncooled * -0,001 * NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[dT_cooled]; sim[ACH_mechvent_c]-sim[ACH_mechvent_therm_c])*cp_air*per_NFA_cooled * -0,001 * NFAfrac_c
QH_min_summer	heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[Qh_to_room_uncooled_min]) * NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[Qh_to_room_cooled_min]) * NFAfrac_c) * 0,001
QH_flex_summer	heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[Qh_u]) * NFAfrac_u + SUMMENPRODUKT(sim[is_cooling_period]; sim[Qh_c]) * NFAfrac_c) * 0,001 - QH_min_summer
QC_min_summer	heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[Qc_min_excl_losses]) * NFAfrac_c) * -0,001
QC_flex_summer	heat_balance	kWh/m²NGFa	(SUMMENPRODUKT(sim[is_cooling_period]; sim[Qc_flex_excl_losses]) * NFAfrac_c) * -0,001
QH_flexshare_summer	heat_balance	ratio	WENNFEHLER(QH_flex_summer/QH_summer;0)
QC_flexshare_summer	heat_balance	ratio	WENNFEHLER(QC_flex_summer/QC_summer;0)
test_heat_balance	☑ heat_balance	WAHR	(QT+QV+QVn+QC)-(QS+QI+QH)
QT_u	heat_balance	kWh/m²NGFungekühlta	-0,001*SUMME(sim[QT_u])
QV_u	heat_balance	kWh/m²NGFungekühlta	QV_inf_u+QV_window_u+QV_mechvent_u
QVn_u	heat_balance	kWh/m²NGFungekühlta	-0,001*SUMME(sim[Qv_nightvent])
QS_u	heat_balance	kWh/m²NGFungekühlta	0,001*SUMME(sim[QS_u])
QI_u	heat_balance	kWh/m²NGFungekühlta	0,001*SUMME(sim[QI_u])
QH_u	heat_balance	kWh/m²NGFungekühlta	QH_min_u+QH_flex_u
QT_wall_u	heat_balance	kWh/m²NGFungekühlta	QT_u*IN!N110/transmittance_Wm2
QT_roof_u	heat_balance	kWh/m²NGFungekühlta	QT_u*IN!N111/transmittance_Wm2
QT_ground_u	heat_balance	kWh/m²NGFungekühlta	QT_u*IN!N112/transmittance_Wm2
QT_window_u	heat_balance	kWh/m²NGFungekühlta	QT_u*IN!N113/transmittance_Wm2
QT_psi_u	heat_balance	kWh/m²NGFungekühlta	QT_u*IN!N114/transmittance_Wm2
QV_inf_u	heat_balance	kWh/m²NGFungekühlta	-SUMME(sim[Qvinf_u])/1000
QV_window_u	heat_balance	kWh/m²NGFungekühlta	-SUMME(sim[Qvwindow_u])/1000

QV_mechvent_u	heat_balance	kWh/m²NGFungekühlta	-SUMME(sim[Qvmechvent_u])/1000
QV_heatrecovery_u	heat_balance	kWh/m²NGFungekühlta	SUMMENPRODUKT(sim[dT_uncooled];sim[ACH_mechvent_u]-sim[ACH_mechvent_therm_u])*cp_air*per_NFA_uncooled * -0,001
QH_min_u	heat_balance	kWh/m²NGFungekühlta	SUMME(sim[Qh_to_room_uncooled_min])/1000
QH_flex_u	heat_balance	kWh/m²NGFungekühlta	SUMME(sim[Qh_u])/1000-QH_min_u
QH_flexanteil_u	F heat_balance	ratio	WENNFEHLER(QH_flex_u/QH_u;0)
QT_c	heat_balance	kWh/m²NGFgekühlta	-0,001*SUMME(sim[QT_c])
QV_c	heat_balance	kWh/m²NGFgekühlta	QV_inf_c+QV_window_c+QV_mechvent_c
QVn_c	heat_balance	kWh/m²NGFgekühlta	
QS_c	heat_balance	kWh/m²NGFgekühlta	SUMME(sim[QS_c])/1000
QI_c	heat_balance	kWh/m²NGFgekühlta	0,001*SUMME(sim[QI_c])
QH_c	heat_balance	kWh/m²NGFgekühlta	QH_min_c+QH_flex_c
QC_c	heat_balance	kWh/m²NGFgekühlta	QC_min_c+QC_flex_c
QT_wall_c	heat_balance	kWh/m²NGFgekühlta	QT_c*IN!N110/transmittance_Wm2
QT_roof_c	heat_balance	kWh/m²NGFgekühlta	QT_c*IN!N111/transmittance_Wm2
QT_ground_c	heat_balance	kWh/m²NGFgekühlta	QT_c*IN!N112/transmittance_Wm2
QT_window_c	heat_balance	kWh/m²NGFgekühlta	QT_c*IN!N113/transmittance_Wm2
QT_psi_c	heat_balance	kWh/m²NGFgekühlta	QT_c*IN!N114/transmittance_Wm2
QV_inf_c	heat_balance	kWh/m²NGFgekühlta	-SUMME(sim[Qvinf_c])/1000
QV_window_c	heat_balance	kWh/m²NGFgekühlta	-SUMME(sim[Qvwindow_c])/1000
QV_mechvent_c	heat_balance	kWh/m²NGFgekühlta	-SUMME(sim[Qvmechvent_c])/1000
QV_heatrecovery_c	heat_balance	kWh/m²NGFgekühlta	SUMMENPRODUKT(sim[dT_cooled];sim[ACH_mechvent_c]-sim[ACH_mechvent_therm_c])*cp_air*per_NFA_cooled * -0,001
QH_min_c	heat_balance	kWh/m²NGFgekühlta	SUMME(sim[Qh_to_room_cooled_min])/1000
QH_flex_c	heat_balance	kWh/m²NGFgekühlta	SUMME(sim[Qh_c])/1000-QH_min_c
QC_min_c	heat_balance	kWh/m²NGFgekühlta	-SUMME(sim[Qc_min_excl_losses])/1000
QC_flex_c	heat_balance	kWh/m²NGFgekühlta	-SUMME(sim[Qc_flex_excl_losses])/1000
QC_flexanteil	F heat_balance	ratio	WENNFEHLER(QC_flex_c/QC_c;0)
QH_flexanteil_c	F heat_balance	ratio	WENNFEHLER(QH_flex_c/QH_c;0)
Tu_avg_winter	comfort	°C	MITTELWERT(MITTELWERTWENNS(sim[Ti_final_uncooled];month;winter_months))
Tc_avg_winter	comfort	°C	MITTELWERT(MITTELWERTWENNS(sim[Ti_final_cooled];month;winter_months))

Tu_avg_summer		comfort	°C	MITTELWERT(MITTELWERTWENNS(sim[Ti_final_uncooled];month;summer_months))
Tc_avg_summer		comfort	°C	MITTELWERT(MITTELWERTWENNS(sim[Ti_final_cooled];month;summer_months))
dTu_winter		comfort	K	MITTELWERT(MITTELWERTWENNS(sim[Ti_final_uncooled];month;winter_months))-Tsetheat_min
dTc_winter		comfort	K	MITTELWERT(MITTELWERTWENNS(sim[Ti_final_cooled];month;winter_months))-Tsetheat_min
dTu_summer		comfort	°C	Tsetcool_max - MITTELWERT(MITTELWERTWENNS(sim[Ti_final_uncooled];month;summer_months))
dTc_summer		comfort	°C	Tsetcool_max- MITTELWERT(MITTELWERTWENNS(sim[Ti_final_cooled];month;summer_months))
EUI_plugAuxLight		electricity_end_use	kWh/m²NGFa	PV_to_plugloads+Batt_to_plugloads+VRGrid_to_plugloads+Grid_to_plugloads+Eev_to_plugloads
EUI_plugloads		electricity_end_use	kWh/m²NGFa	SUMME(sim[E_plugAuxLight])/1000-EUI_auxiliary-EUI_lighting
EUI_auxiliary		electricity_end_use	kWh/m²NGFa	SUMME(sim[Eaux])/1000
EUI_lighting		electricity_end_use	kWh/m²NGFa	SUMME(sim[Elight])/1000
EUIh_el		electricity_end_use	kWh/m²NGFa	PV_to_Eh_min+PV_to_Eh_flex+Batt_to_Eh_min+VRGrid_to_Eh_min+VRGrid_to_Eh_flex+Eev_to_Eh_min+Grid_to_Eh_min
EUIh_el_aux		electricity_end_use	kWh/m²NGFa	SUMME(sim[Eh_aux])/1000
EUIc_el		electricity_end_use	kWh/m²NGFa	PV_to_Ec_min+PV_to_Ec_flex+Batt_to_Ec_min+VRGrid_to_Ec_min+VRGrid_to_Ec_flex+Eev_to_Ec_min+Grid_to_Ec_min
EUIc_el_aux		electricity_end_use	kWh/m²NGFa	SUMME(sim[Ec_aux])/1000
EUIdhw_el		electricity_end_use	kWh/m²NGFa	PV_to_Edhw_min+PV_to_Edhw+Batt_to_Edhw_min+VRGrid_to_Edhw_min+VRGrid_to_Edhw_flex+Eev_to_Edhw_min+Grid_to_Edhw_min
EUIv_el		electricity_end_use	kWh/m²NGFa	PV_to_Ev_min+Batt_to_Ev_min+VRGrid_to_Ev_min+Eev_to_Ev_min+Grid_to_Ev_min
EUIdhwdirect_el		electricity_end_use	kWh/m²NGFa	PV_to_Edhw_direct
Batt_total_charge_input		electricity_end_use	kWh/m²NGFa	PV_to_Batt+VRGrid_to_Batt
EUIev_el		electricity_end_use	kWh/m²NGFa	PV_to_Eev_min+PV_to_Eev_flex+Batt_to_Eev_min+VRGrid_to_Eev_min+VRGrid_to_Eev_flex+Grid_to_Eev_min
EUI_el_total	**O**	electricity_end_use	kWh/m²NGFa	EUI_plugAuxLight+EUIh_el+EUIc_el+EUIdhw_el+EUIv_el+EUIdhwdirect_el+EUIev_el
PV_total		electricity_dispatch	kWh/m²NGFa	PV_total_direct+PV_total_flex+PV_to_Egrid
PV_to_plugloads		electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_user])/1000

PV_to_Eh_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_Eh_min])/1000
PV_to_Ec_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_Ec_min])/1000
PV_to_Edhw_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_Edhw_min])/1000
PV_to_Ev_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_Ev_min])/1000
PV_to_Eev_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_Eev_min])/1000
PV_total_direct	electricity_dispatch	kWh/m²NGFa	PV_to_plugloads+PV_to_Eh_min+PV_to_Ec_min+PV_to_Edhw_min+PV_to_Ev_min+PV_to_Eev_min
PV_to_Eh_flex	electricity_dispatch	kWh/m²NGFa	SUMME(sim[[PV_to_Eh_flex_1el]:[PV_to_Eh_flex_3el]])/1000
PV_to_Ec_flex	electricity_dispatch	kWh/m²NGFa	SUMME(sim[[PV_to_Ec_flex_1el]:[PV_to_Ec_flex_3el]])/1000
PV_to_Edhw	electricity_dispatch	kWh/m²NGFa	SUMME(sim[[PV_to_Edhw1_flex]:[PV_to_Edhw2_flex]])/1000
PV_to_Edhw_direct	electricity_dispatch	kWh/m²NGFa	
PV_to_Batt	electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_Batt])/1000
PV_to_Eev_flex	electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_Eev_flex])/1000
PV_total_flex	electricity_dispatch	kWh/m²NGFa	PV_to_Eh_flex+PV_to_Ec_flex+PV_to_Edhw+PV_to_Edhw_direct+PV_to_Batt+PV_to_Eev_flex
PV_to_Egrid	electricity_dispatch	kWh/m²NGFa	SUMME(sim[PV_to_Egrid])/1000
Batt_to_plugloads	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Batt_to_user])/1000
Batt_to_HVAC	electricity_dispatch	kWh/m²NGFa	SUMME(K156:K159)
Batt_to_Eh_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Batt_to_Eh_min])/1000
Batt_to_Ec_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Batt_to_Ec_min])/1000
Batt_to_Edhw_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Batt_to_Edhw_min])/1000
Batt_to_Ev_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Batt_to_Ev_min])/1000
Batt_to_Eev_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Batt_to_Eev_min])/1000
Batt_charging_losses	electricity_dispatch	kWh/m²NGFa	Batt_total_charge-Batt_total_discharge
VRGrid_to_plugloads	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_user])/1000
VRGrid_to_Eh_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_Eh_min])/1000
VRGrid_to_Ec_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_Ec_min])/1000
VRGrid_to_Edhw_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_Edhw_min])/1000
VRGrid_to_Ev_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_Ev_min])/1000
VRGrid_to_Eev_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_Eev_min])/1000
VRGrid_to_min	electricity_dispatch	kWh/m²NGFa	VRGrid_to_plugloads+VRGrid_to_Eh_min+VRGrid_to_Ec_min+VRGrid_to_Edhw_min+VRGrid_to_Ev_min+VRGrid_to_Eev_min
VRGrid_to_Eh_flex	electricity_dispatch	kWh/m²NGFa	SUMME(sim[[VRGrid_to_Eh_flex_1el]:[VRGrid_to_Eh_flex_3el]])/1000

VRGrid_to_Ec_flex	electricity_dispatch	kWh/m²NGFa	SUMME(sim[[VRGrid_to_Ec_flex_1el]:[VRGrid_to_Ec_flex_3el]])/1000
VRGrid_to_Edhw_flex	electricity_dispatch	kWh/m²NGFa	SUMME(sim[[VRGrid_to_Edhw1_flex]:[VRGrid_to_Edhw2_flex]])/1000
VRGrid_to_Batt	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_Batt])/1000
VRGrid_to_Eev_flex	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_Eev_flex])/1000
VRGrid_to_flex	electricity_dispatch	kWh/m²NGFa	VRGrid_to_Eh_flex+VRGrid_to_Ec_flex+VRGrid_to_Edhw_flex+VRGrid_to_Batt+VRGrid_to_Eev_flex
VRGrid_to_building	electricity_dispatch	kWh/m²NGFa	SUMME(sim[VRGrid_to_building])/1000
Eev_to_plugloads	electricity_dispatch	kWh/m²NGFa	-SUMME(sim[Eev_to_user])/1000
Eev_to_HVAC	electricity_dispatch	kWh/m²NGFa	SUMME(K182:K185)
Eev_to_Eh_min	electricity_dispatch	kWh/m²NGFa	-SUMME(sim[Eev_to_Eh_min])/1000
Eev_to_Ec_min	electricity_dispatch	kWh/m²NGFa	-SUMME(sim[Eev_to_Ec_min])/1000
Eev_to_Edhw_min	electricity_dispatch	kWh/m²NGFa	-SUMME(sim[Eev_to_Edhw_min])/1000
Eev_to_Ev_min	electricity_dispatch	kWh/m²NGFa	-SUMME(sim[Eev_to_Ev_min])/1000
Eev_discharge_loss	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Eev_discharge_total])/1000-K187
Eev_to_district	electricity_dispatch	kWh/m²NGFa	Eev_to_plugloads+Eev_to_Eh_min+Eev_to_Ec_min+Eev_to_Edhw_min+Eev_to_Ev_min
Grid_to_plugloads	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Grid_to_user])/1000
Grid_to_Eh_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Grid_to_Eh_min])/1000
Grid_to_Ec_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Grid_to_Ec_min])/1000
Grid_to_Edhw_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Grid_to_Edhw_min])/1000
Grid_to_Ev_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Grid_to_Ev_min])/1000
Grid_to_building_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Grid_to_building_min])/1000
Grid_to_Eev_min	electricity_dispatch	kWh/m²NGFa	SUMME(sim[Grid_to_Eev_min])/1000
Grid_to_min	electricity_dispatch	kWh/m²NGFa	Grid_to_plugloads+Grid_to_Eh_min+Grid_to_Ec_min+Grid_to_Edhw_min+Grid_to_Ev_min+Grid_to_Eev_min
Eev_ext_charge	electricity_dispatch	kWh/m²NGFa	EV_charging_losses_surcharge_factor*SUMME(sim[Eev_Cext_res];sim[Eev_Cext_work];sim[Eev_Cext_retail])/1000
EUIh_district_heating	energy_dispatch	kWh/m²NGFa	0,001*(heat_th2_is_dh*SUMME(sim[EUIh_2th])+heat_th4_is_dh*SUMME(sim[EUIh_4th]))
EUIdhw_district_heating	energy_dispatch	kWh/m²NGFa	0,001*(dhw1_is_dh*SUMME(sim[EUIdhw_1th])+dhw2_is_dh*SUMME(sim[EUIdhw_2th]))
EUI_district_heating	energy_dispatch	kWh/m²NGFa	EUIh_district_heating+EUIdhw_district_heating
EUIh_natural_gas	energy_dispatch	kWh/m²NGFa	0,001*(heat_th2_is_ng*SUMME(sim[EUIh_2th])+heat_th4_is_ng*SUMME(sim[EUIh_4th]))

EUIdhw_natural_gas	energy_dispatch	kWh/m²NGFa	0,001*(dhw1_is_ng*SUMME(sim[EUIdhw_1th])+dhw2_is_ng*SUMME(sim[EUIdhw_2th]))
EUI_natural_gas	energy_dispatch	kWh/m²NGFa	EUIh_natural_gas+EUIdhw_natural_gas
EUIh_biomass	energy_dispatch	kWh/m²NGFa	0,001*(heat_th2_is_bio*SUMME(sim[EUIh_2th])+heat_th4_is_bio*SUMME(sim[EUIh_4th]))
EUIdhw_biomass	energy_dispatch	kWh/m²NGFa	0,001*(dhw1_is_bio*SUMME(sim[EUIdhw_1th])+dhw2_is_bio*SUMME(sim[EUIdhw_2th]))
EUI_biomass	energy_dispatch	kWh/m²NGFa	EUIh_biomass+EUIdhw_biomass
EUIh_other	energy_dispatch	kWh/m²NGFa	0,001*(heat_th2_is_other*SUMME(sim[EUIh_2th])+heat_th4_is_other*SUMME(sim[EUIh_4th]))
EUIc_other	energy_dispatch	kWh/m²NGFa	0,001*cool_th2_is_other*SUMME(sim[EUIc_2th])
EUIdhw_other	energy_dispatch	kWh/m²NGFa	0,001*(dhw1_is_other*SUMME(sim[EUIdhw_1th])+dhw2_is_other*SUMME(sim[EUIdhw_2th]))
EUI_other	energy_dispatch	kWh/m²NGFa	EUIh_other+EUIc_other+EUIdhw_other
EUI_mob_fossil	energy_dispatch	kWh/m²NGFa	PEI_mob_fossile/pe_conversion_factor_gasoline
EUIh_2th	energy_dispatch	kWh/m²NGFa	SUMME(sim[EUIh_2th])/1000
EUIh_4th	energy_dispatch	kWh/m²NGFa	SUMME(sim[EUIh_4th])/1000
EUIc_2th	energy_dispatch	kWh/m²NGFa	SUMME(sim[EUIc_2th])/1000
EUIdhw_1th	energy_dispatch	kWh/m²NGFa	SUMME(sim[EUIdhw_1th])/1000
EUIdhw_2th	energy_dispatch	kWh/m²NGFa	SUMME(sim[EUIdhw_2th])/1000
Grid_total_flexandnotflex	final_energy	kWh/m²NGFa	VRGrid_to_building+Grid_to_min
context_factor_density	primary_energy_balance	kWhPEges./m²NGFa	-MIN(cfd_fPE*(cfd_A/(FSI+cfd_dx)-cfd_EUI);cfd_cutoff)
context_factor_mobility	primary_energy_balance	kWhPEges./m²NGFa	per_NFA*(cfm_budget_residential*NFA_residential+cfm_budget_office*NFA_office+cfm_budget_school*(NFA_schoolprim+NFA_schoolsec)+cfm_budget_retail*(NFA_retailfood+NFA_retailother+NFA_other))
context_factor_renovation	primary_energy_balance	kWhPEges./m²NGFa	cfr_budget*SUMMENPRODUKT(Renovation_shares;NFA_shares)
PEI_demand	primary_energy_balance	kWhPEges./m²NGFa	PEI_el_plugloads+PEI_el_hvac+PEI_district_heating+PEI_natural_gas+PEI_biomass+PEI_other
PEI_el_plugloads	primary_energy_balance	kWhPEges./m²NGFa	SUMME(sim[PEI_el_user])/1000
PEI_el_hvac	primary_energy_balance	kWhPEges./m²NGFa	SUMME(sim[PEI_el_hvac])/1000
PEI_district_heating	primary_energy_balance	kWhPEges./m²NGFa	SUMME(sim[PEI_district_heating])/1000
PEI_natural_gas	primary_energy_balance	kWhPEges./m²NGFa	SUMME(sim[PEI_natural_gas])/1000
PEI_biomass	primary_energy_balance	kWhPEges./m²NGFa	SUMME(sim[PEI_biomass])/1000
PEI_other	primary_energy_balance	kWhPEges./m²NGFa	SUMME(sim[PEI_other])/1000
PEI_mob_total	primary_energy_balance	kWhPEges./m²NGFa	PEI_mob_fossile+PEI_mob_el
PEI_mob_fossile	primary_energy_balance	kWhPEges./m²NGFa	EUI_mob_fossile*pe_conversion_factor_gasoline

PEI_mob_el	primary_energy_balance	kWhPEges./m²NGFa	UED_mob_el_target*EV_charging_losses_surcharge_factor*MITTELWERT(sim[cf_PEI_grid])
PEI_storage_losses	primary_energy_balance	kWhPEges./m²NGFa	0,001*(SUMMENPRODUKT(sim[Batt_total_charge];sim[cf_PEI_flex_gridsub])/Batt_eff_factor_charge-SUMMENPRODUKT(sim[Batt_total_discharge];sim[cf_PEI_flex_gridsub])*Batt_eff_factor_discharge)
PEI_cf_density	primary_energy_balance	kWhPEges./m²NGFa	context_factor_density
PEI_cf_mobility	primary_energy_balance	kWhPEges./m²NGFa	mobility_is_included*context_factor_mobility
PEI_cf_renovation	primary_energy_balance	kWhPEges./m²NGFa	context_factor_renovation
PEI_sub_PV	primary_energy_balance	kWhPEges./m²NGFa	SUMMENPRODUKT(sim[PV_yield];sim[cf_PEI_grid])/1000
PEI_sub_flex	primary_energy_balance	kWhPEges./m²NGFa	SUMME(sim[PEI_flex_grid_substitution])/1000
PEI_balance	primary_energy_balance	kWhPEges./m²NGFa	PEI_cf_renovation+PEI_cf_mobility+PEI_cf_density+PEI_sub_flex+PEI_sub_PV-PEI_demand-PEI_mob_total-PEI_storage_losses
PEI_grid_import	pe_import_export_balance	kWhPEges./m²NGFa	0,001*SUMME(sim[cf_PEI_grid]*sim[E_grid])
PEI_flex_import	pe_import_export_balance	kWhPEges./m²NGFa	SUMMENPRODUKT(sim[cf_PEI_flex_grid];sim[VRGrid_total_min_use]+sim[VRGrid_total_flex_use])/1000
PEI_evExtCharge_import	pe_import_export_balance	kWhPEges./m²NGFa	0,001*SUMMENPRODUKT(sim[cf_PEI_grid];sim[Eev_Cext_res]+sim[Eev_Cext_work]+sim[Eev_Cext_retail])*EV_charging_losses_surcharge_factor
PEI_import	pe_import_export_balance	kWhPEges./m²NGFa	PEI_district_heating+PEI_natural_gas+PEI_flex_import+PEI_biomass+PEI_other+PEI_mob_fossile+PEI_grid_import+PEI_evExtCharge_import
PEI_pv_export	pe_import_export_balance	kWhPEges./m²NGFa	SUMMENPRODUKT(sim[PV_to_Egrid];sim[cf_PEI_grid])/1000
PEI_evOtherTravel_export	pe_import_export_balance		UED_mob_el_other*EV_charging_losses_surcharge_factor*MITTELWERT(sim[cf_PEI_grid])
PEI_export	pe_import_export_balance	kWhPEges./m²NGFa	PEI_pv_export+PEI_evOtherTravel_export
PEI_saldo_project	pe_import_export_balance	kWhPEges./m²NGFa	PEI_export-PEI_import
PEI_saldo_target	pe_import_export_balance	kWhPEges./m²NGFa	-PEI_cf_density-PEI_cf_mobility-PEI_cf_renovation
PEI_importExport_balance	pe_import_export_balance	kWhPEges./m²NGFa	PEI_saldo_project-PEI_saldo_target
fPE_grid			MITTELWERT(sim[cf_PEI_grid])
fPE_eff			(K249+K250+K251)/(Grid_to_min+VRGrid_to_min+VRGrid_to_flex+Eev_ext_charge)

GWP_ee_walls_fossile	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_walls_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_walls_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_walls_biogenic	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_walls_name;GWP_Katalog;5;FALSCH)*MAX(SVERWEIS(GWP_walls_name;GWP_Katalog;7;FALSCH);55%)*SVERWEIS(SVERWEIS(GWP_walls_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_walls	☁ embodied_emissions	Jahre	WENNNV(SVERWEIS(GWP_walls_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_walls_fossil	embodied_emissions		GWP_ee_walls_fossile*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_walls; 0))
GWP_ee_lc_walls_biogenic	embodied_emissions		GWP_ee_walls_biogenic*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_walls; 0))
GWP_ee_lc_walls	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_walls_fossil+GWP_ee_lc_walls_biogenic
GWP_ee_windows_fossile	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_windows_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_windows_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_windows_bigenic	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_windows_name;GWP_Katalog;5;FALSCH)*MAX(SVERWEIS(GWP_windows_name;GWP_Katalog;7;FALSCH);55%)*SVERWEIS(SVERWEIS(GWP_windows_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_windows	☁ embodied_emissions	Jahre	WENNNV(SVERWEIS(GWP_windows_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_windows_fossile	embodied_emissions		(GWP_ee_windows_fossile)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_windows; 0))
GWP_ee_lc_windows_biogenic	embodied_emissions		(GWP_ee_windows_bigenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_windows; 0))
GWP_ee_lc_windows	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_windows_fossile+GWP_ee_lc_windows_biogenic
GWP_ee_roof_fossile	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_roof_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_roof_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_roof_biogenic	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_roof_name;GWP_Katalog;5;FALSCH)*MAX(SVERWEIS(GWP_roof_name;GWP_Katalog;7;FALSCH);55%)*SVERWEIS(SVERWEIS(GWP_roof_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)

GWP_life_roof	☁ embodied_emissions	Jahre	WENNNV(SVERWEIS(GWP_roof_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_roof_fossile	embodied_emissions		(GWP_ee_roof_fossile)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_roof; 0))
GWP_ee_lc_roof_biogenic	embodied_emissions		(GWP_ee_roof_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_roof; 0))
GWP_ee_lc_roof	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_roof_fossile+GWP_ee_lc_roof_biogenic
GWP_ee_ground_fossile	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_ground_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_ground_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_ground_biogenic	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_ground_name;GWP_Katalog;5;FALSCH)*MAX(SVERWEIS(GWP_ground_name;GWP_Katalog;7;FALSCH);55%)*SVERWEIS(SVERWEIS(GWP_ground_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_ground	☁ embodied_emissions	Jahre	WENNNV(SVERWEIS(GWP_ground_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_ground_fossile	embodied_emissions		(GWP_ee_ground_fossile)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_ground; 0))
GWP_ee_lc_ground_biogenic	embodied_emissions		(GWP_ee_ground_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_ground; 0))
GWP_ee_lc_ground	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_ground_fossile+GWP_ee_lc_ground_biogenic
GWP_ee_ceilings_fossile	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(COMP_name_ceiling;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(COMP_name_ceiling;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_ceilings_biogenic	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(COMP_name_ceiling;GWP_Katalog;5;FALSCH)*MAX(SVERWEIS(COMP_name_ceiling;GWP_Katalog;7;FALSCH);55%)*SVERWEIS(SVERWEIS(COMP_name_ceiling;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_ceilings	☁ embodied_emissions	Jahre	WENNNV(SVERWEIS(COMP_name_ceiling;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_ceilings_fossile	embodied_emissions		(GWP_ee_ceilings_fossile)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_ceilings; 0))
GWP_ee_lc_ceilings_biogenic	embodied_emissions		(GWP_ee_ceilings_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_ceilings; 0))

GWP_ee_lc_ceil	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_ceilings_fossile+GWP_ee_lc_ceilings_biogenic
GWP_ee_general_fossile	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_general_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_general_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_general_bigonenic	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_general_name;GWP_Katalog;5;FALSCH)*MAX(SVERWEIS(GWP_general_name;GWP_Katalog;7;FALSCH);55%)*SVERWEIS(SVERWEIS(GWP_general_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_general	☁ embodied_emissions	Jahre	WENNNV(SVERWEIS(GWP_general_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_general_fossile	embodied_emissions		(GWP_ee_general_fossile)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_general; 0))
GWP_ee_lc_general_biogenic	embodied_emissions		(GWP_ee_general_bigonenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_general; 0))
GWP_ee_general	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_general_fossile+GWP_ee_lc_general_biogenic
GWP_ee_other_fossile	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_other_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_other_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_other_biogenic	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_other_name;GWP_Katalog;5;FALSCH)*MAX(SVERWEIS(GWP_other_name;GWP_Katalog;7;FALSCH);55%)*SVERWEIS(SVERWEIS(GWP_other_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_other	☁ embodied_emissions	Jahre	WENNNV(SVERWEIS(GWP_other_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_other_fossile	embodied_emissions		(GWP_ee_other_fossile)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_other; 0))
GWP_ee_lc_other_biogenic	embodied_emissions		(GWP_ee_other_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_other; 0))
GWP_ee_lc_other	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_other_fossile+GWP_ee_lc_other_biogenic
GWP_ee_direct_fossile	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_direct_fossile
GWP_ee_direct_biogenic	☁ embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_direct_biogenic*GWP_direct_biogenic_share
GWP_life_direct	☁ embodied_emissions	Jahre	GWP_direct_life

GWP_ee_lc_direct_fossile		embodied_emissions		(GWP_ee_direct_fossile)*(1+ABRUNDEN(GHG_LCA_timeframe_years/K310; 0))
GWP_ee_lc_direct_biogenic		embodied_emissions		(GWP_ee_direct_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/K310; 0))
GWP_ee_lc_direct	☁	embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_direct_fossile+GWP_ee_lc_direct_biogenic
GWP_ee_con_build	☁	embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_walls_fossile+GWP_ee_walls_biogenic+GWP_ee_windows_fossile+GWP_ee_windows_bigenic+GWP_ee_roof_fossile+GWP_ee_roof_biogenic+GWP_ee_ground_fossile+GWP_ee_ground_biogenic+GWP_ee_ceilings_fossile+GWP_ee_ceilings_biogenic+GWP_ee_general_fossile+GWP_ee_general_bigonenic+GWP_ee_other_fossile+GWP_ee_other_biogenic++GWP_ee_direct_fossile+GWP_ee_direct_biogenic
GWP_ee_rep_build	☁	embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_construction-GWP_ee_con_build
GWP_ee_lc_fossil		embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_walls_fossil+GWP_ee_lc_windows_fossile+GWP_ee_lc_roof_fossile+GWP_ee_lc_ground_fossile+GWP_ee_lc_ceilings_fossile+GWP_ee_lc_general_fossile+GWP_ee_lc_other_fossile+GWP_ee_lc_direct_fossile
GWP_ee_lc_biogenic		embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_walls_biogenic+GWP_ee_lc_windows_biogenic+GWP_ee_lc_roof_biogenic+GWP_ee_lc_ground_biogenic+GWP_ee_lc_ceilings_biogenic+GWP_ee_lc_general_biogenic+GWP_ee_lc_other_biogenic+GWP_ee_lc_direct_biogenic
GWP_ee_lc_construction	☁	embodied_emissions	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_walls+GWP_ee_lc_windows+GWP_ee_lc_roof+GWP_ee_lc_ground+GWP_ee_lc_ceil+GWP_ee_lc_general+GWP_ee_lc_other+GWP_ee_lc_direct
GWP_ee_tga_pv_fossile	☁		GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_pv_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_pv_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_tga_pv_biogenic			GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_pv_name;GWP_Katalog;5;FALSCH)*SVERWEIS(SVERWEIS(GWP_pv_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_tga_pv			Jahre	WENNNV(SVERWEIS(GWP_other_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_pv	☁		GWP 100A (kgCO2equiv/m²BGF)	(GWP_ee_tga_pv_fossile+GWP_ee_tga_pv_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_tga_pv; 0))
GWP_ee_tga_boreholes_fossile	☁		GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_boreholes_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_boreholes_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)

GWP_ee_tga_boreholes_biogenic		GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_boreholes_name;GWP_Katalog;5;FALSCH)*SVERWEIS(SVERWEIS(GWP_boreholes_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_tga_boreholes		Jahre	WENNNV(SVERWEIS(GWP_boreholes_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_boreholes	☁	GWP 100A (kgCO2equiv/m²BGF)	(GWP_ee_tga_boreholes_fossile+GWP_ee_tga_boreholes_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_tga_boreholes; 0))
GWP_ee_tga_ventilation_fossile	☁	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_ventilation_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_ventilation_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_tga_ventilation_biogenic		GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_ventilation_name;GWP_Katalog;5;FALSCH)*SVERWEIS(SVERWEIS(GWP_ventilation_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_tga_ventilation		Jahre	WENNNV(SVERWEIS(GWP_ventilation_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_ventilation	☁	GWP 100A (kgCO2equiv/m²BGF)	(GWP_ee_tga_ventilation_fossile+GWP_ee_tga_ventilation_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_tga_ventilation; 0))
GWP_ee_tga_solarthermal_fossile	☁	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_solarthermal_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_solarthermal_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_tga_solarthermal_biogenic		GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_solarthermal_name;GWP_Katalog;5;FALSCH)*SVERWEIS(SVERWEIS(GWP_solarthermal_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_solarthermal		Jahre	WENNNV(SVERWEIS(GWP_solarthermal_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_solarthermal	☁	GWP 100A (kgCO2equiv/m²BGF)	(GWP_ee_tga_solarthermal_fossile+GWP_ee_tga_solarthermal_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_solarthermal; 0))
GWP_ee_tga_battery_fossile	☁	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_battery_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_battery_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_tga_battery_biogenic		GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_battery_name;GWP_Katalog;5;FALSCH)*SVERWEIS(SVERWEIS(GWP_battery_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_battery		Jahre	WENNNV(SVERWEIS(GWP_battery_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_battery	☁	GWP 100A (kgCO2equiv/m²BGF)	(GWP_ee_tga_battery_fossile+GWP_ee_tga_battery_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_battery; 0))

GWP_ee_tga_storage_fossile	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_storage_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_storage_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_tga_storage_biogenic	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_storage_name;GWP_Katalog;5;FALSCH)*SVERWEIS(SVERWEIS(GWP_storage_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_storage	Jahre	WENNNV(SVERWEIS(GWP_storage_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_storage	GWP 100A (kgCO2equiv/m²BGF)	(GWP_ee_tga_storage_fossile+GWP_ee_tga_storage_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_storage; 0))
GWP_ee_tga_general_fossile	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_tga_general_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_tga_general_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_tga_general_biogenic	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_solarthermal_name;GWP_Katalog;5;FALSCH)*SVERWEIS(SVERWEIS(GWP_tga_general_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_tga_general	Jahre	WENNNV(SVERWEIS(GWP_tga_general_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_tga_general	GWP 100A (kgCO2equiv/m²BGF)	(GWP_ee_tga_general_fossile+GWP_ee_tga_general_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_tga_general; 0))
GWP_ee_tga_other_fossile	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_tga_other_name;GWP_Katalog;4;FALSCH)*SVERWEIS(SVERWEIS(GWP_tga_other_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_ee_tga_other_biogenic	GWP 100A (kgCO2equiv/m²BGF)	WENNNV(SVERWEIS(GWP_tga_other_name;GWP_Katalog;5;FALSCH)*SVERWEIS(SVERWEIS(GWP_tga_other_name;GWP_Katalog;3;FALSCH);refArea;2;FALSCH);0)
GWP_life_tga_other	Jahre	WENNNV(SVERWEIS(GWP_tga_other_name;GWP_Katalog;6;FALSCH);GHG_LCA_timeframe_years+1)
GWP_ee_lc_tga_other	GWP 100A (kgCO2equiv/m²BGF)	(GWP_ee_tga_other_fossile+GWP_ee_tga_other_biogenic)*(1+ABRUNDEN(GHG_LCA_timeframe_years/GWP_life_tga_other; 0))
GWP_ee_con_tga ☁	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_tga_pv_fossile+GWP_ee_tga_pv_biogenic+GWP_ee_tga_boreholes_fossile+GWP_ee_tga_boreholes_biogenic+GWP_ee_tga_ventilation_fossile+GWP_ee_tga_ventilation_biogenic+GWP_ee_tga_solarthermal_fossile+GWP_ee_tga_solarthermal_biogenic+GWP_ee_tga_battery_fossile+GWP_ee_tga_battery_biogenic+GWP_ee_tga_storage_fossile+GWP_ee_tga_storage_biogenic+GWP_ee_tga_general_fossile+GWP_ee_tga_general_biogenic+GWP_ee_tga_other_fossile+GWP_ee_tga_other_biogenic

GWP_ee_rep_tga	☁	GWP 100A (kgCO2equiv/m²BGF)	K356-K354
GWP_ee_lc_tga	☁	GWP 100A (kgCO2equiv/m²BGF)	GWP_ee_lc_pv+GWP_ee_lc_boreholes+GWP_ee_lc_ventilation+GWP_ee_lc_solarthermal+GWP_ee_lc_battery+GWP_ee_lc_storage+GWP_ee_lc_tga_general+GWP_ee_lc_tga_other
GWP_ee_mob_fossile	☁	GWP 100S (kgCO2equiv/m²BGF)	GWP_miv_count_fossile*GWP_mobility_construction_fossil*1000/GFA_total
GWP_ee_mob_ev	☁	GWP 100S (kgCO2equiv/m²BGF)	GWP_miv_count_ev*GWP_mobility_construction_ev*1000/GFA_total
GWP_ee_lc_mob	☁	GWP 100S (kgCO2equiv/m²BGF)	GWP_ee_mob_fossile+GWP_ee_mob_ev
GWP_oe_battery_charge	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMMENPRODUKT(sim[cf_GHG_grid];sim[VRGrid_to_Batt])/1000/NFA_to_GFA_ratio
GWP_emInt_grid_avg	☁	gCO2eq/kWh	MITTELWERT(sim[cf_GHG_grid])*1000
GWP_emInt_grid	☁	gCO2eq/kWh	GWP_oe_grid_build*NFA_to_GFA_ratio/(Grid_to_min-Grid_to_Eev_min)*1000
GWP_emInt_flex	☁	gCO2eq/kWh	GWP_oe_flex_build*NFA_to_GFA_ratio/(VRGrid_to_flex-VRGrid_to_Eev_flex+VRGrid_to_min-VRGrid_to_Eev_min)*1000
GWP_emInt_batt_charge	☁	gCO2eq/kWh	GWP_oe_battery_charge/Batt_total_charge_input*1000*NFA_to_GFA_ratio
GWP_emInt_ev_charge		gCO2eq/kWh	GWP_oe_mob_ev/EUIev_el_total*1000*NFA_to_GFA_ratio
GWP_emInt_PV_feedin	☁	gCO2eq/kWh	GWP_oe_grid_feedin*NFA_to_GFA_ratio/PV_to_Egrid*1000
GWP_oe_grid_build	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMMENPRODUKT(sim[cf_GHG_grid];sim[Grid_to_building_min])/1000/NFA_to_GFA_ratio
GWP_oe_flex_build	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMMENPRODUKT(sim[cf_GHG_grid];sim[VRGrid_to_building])/1000/NFA_to_GFA_ratio
GWP_oe_district_heating	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMMENPRODUKT(EUI_th;is_district_heating;ghg_conversion_factors_thermal)/NFA_to_GFA_ratio
GWP_oe_natural_gas	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMMENPRODUKT(EUI_th;is_natural_gas;ghg_conversion_factors_thermal)/NFA_to_GFA_ratio
GWP_oe_biomass	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMMENPRODUKT(EUI_th;is_biomass;ghg_conversion_factors_thermal)/NFA_to_GFA_ratio
GWP_oe_other	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMMENPRODUKT(EUI_th;is_other_thermal_pe;ghg_conversion_factors_thermal)/NFA_to_GFA_ratio
GWP_oe_grid_feedin	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	-SUMMENPRODUKT(sim[cf_GHG_grid];sim[PV_to_Egrid])/1000/NFA_to_GFA_ratio

GWP_oe_building	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMME(K379:K385)
GWP_oe_mob_ev	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	SUMMENPRODUKT(sim[cf_GHG_grid];sim[Grid_to_Eev_min]+sim[VRGrid_to_Eev_min]+sim[VRGrid_to_Eev_flex]+EV_charging_losses_surcharge_factor*(sim[Eev_Cext_res]+sim[Eev_Cext_work]+sim[Eev_Cext_retail]))/1000/NFA_to_GFA_ratio
GWP_oe_mob_ev_export	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	-GWP_emInt_grid_avg*UED_mob_el_other*EV_charging_losses_surcharge_factor/1000/NFA_to_GFA_ratio
GWP_oe_mob_fossile	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	mobility_is_included*K461*GWP_mobility_car_gpPKm/1000/GFA_total
GWP_oe_mob	☁	GWP pro Jahr (kgCO2equiv/m²BGFa)	GWP_oe_mob_fossile+GWP_oe_mob_ev+GWP_oe_mob_ev_export
GWP_oe		GWP pro Jahr (kgCO2equiv/m²BGFa)	GWP_oe_mob+GWP_oe_building
GWP_oe_lc_grid_build	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_grid_build*GWP_rcpi_grid
GWP_oe_lc_flex_build	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_flex_build*GWP_rcpi_grid
GWP_oe_lc_district_heating	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	NFA_to_GFA_ratio*SUMMENPRODUKT(EUI_th;is_district_heating;ghg_conversion_factors_thermal;GWP_rcpis_thermal)
GWP_oe_lc_natural_gas	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	NFA_to_GFA_ratio*SUMMENPRODUKT(EUI_th;is_natural_gas;ghg_conversion_factors_thermal;GWP_rcpis_thermal)
GWP_oe_lc_biomass	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	NFA_to_GFA_ratio*SUMMENPRODUKT(EUI_th;is_biomass;ghg_conversion_factors_thermal;GWP_rcpis_thermal)
GWP_oe_lc_other	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	NFA_to_GFA_ratio*SUMMENPRODUKT(EUI_th;is_other_thermal_pe;ghg_conversion_factors_thermal;GWP_rcpis_thermal)
GWP_oe_lc_grid_feedin	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_grid_feedin*GWP_rcpi_grid_substition
GWP_oe_lc_building	☁ operational_emissions		SUMME(K396:K402)
GWP_oe_lc_emission	operational_emissions		SUMME(K396:K401)
GWP_oe_lc_emission_savings	operational_emissions		-GWP_oe_lc_grid_feedin
GWP_oe_lc_mob_ev	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_mob_ev*GWP_rcpi_grid
GWP_oe_lc_mob_export	☁ operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_mob_ev_export*GWP_rcpi_grid

GWP_oe_lc_mob_fossile	☁	operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_mob_fossile*GWP_rcpi_mob_fossile
GWP_oe_lc_mob	☁	operational_emissions	GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_lc_mob_ev+GWP_oe_lc_mob_export+GWP_oe_lc_mob_fossile
GWP_lc_total	☁		GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_lc_mob+GWP_oe_lc_building+GWP_ee_lc_mob+GWP_ee_lc_tga+GWP_ee_lc_construction
GWP_lc_OE_total	☁		GWP 100S (kgCO2equiv/m²BGF)	GWP_oe_lc_mob+GWP_oe_lc_building
GWP_lc_EE_total	☁		GWP 100S (kgCO2equiv/m²BGF)	+GWP_ee_lc_mob+GWP_ee_lc_tga+GWP_ee_lc_construction
flex_signal_ratio				ZÄHLENWENN(sim[Signal];1)/8760
flex_signal_ratio_winter				SUMMENPRODUKT((sim[Signal]=1) * ISTZAHL(VERGLEICH(month;winter_months;0))) / 8760 * 2
flex_signal_ratio_summer				SUMMENPRODUKT((sim[Signal]=1) * ISTZAHL(VERGLEICH(month;summer_months;0))) / 8760 * 2
flex_GSRU	"			VRGrid_to_min/EUI_el_total
flex_GSRI				VRGrid_to_flex/EUI_el_total
flex_GSR				K426+K427
PV_own_consumption_direct				PV_total_direct/PV_total
PV_own_consumption_flex				PV_total_flex/PV_total
PV_own_consumption	S			PV_own_consumption_direct+PV_own_consumption_flex
EUI_self_sufficiency				(PV_total_direct+PV_total_flex)/EUI_el_total
StatPAX		mobility	StatPAX	K449+K445+K444
mob_annual_milage_district		mobility	km/a	SUMME(K453:K456)
mob_annual_mileage_PAX		mobility	km/PAX/a	mob_annual_milage_district/StatPAX
UED_mob_el_target			kWh/m²NGFa	mobility_is_included*K460*EV_demand_kWhpkm*per_NFA
EUI_mob_fossile			kWh/m²NGFa	mobility_is_included*K461*fossile_demand_kWhpkm*per_NFA
EV_FEC				WENNFEHLER(SUMME(sim[Eev_Cmin];sim[Eev_flex_final])/K471;0)
UED_mob_el_total			kWh/m²NGFa	mobility_is_included*K466*EV_share*EV_demand_kWhpkm*per_NFA
UED_mob_el_other			kWh/m²NGFa	mobility_is_included*(K473-UED_mob_el_target)
EUIev_el_total				EUIev_el+Eev_ext_charge
Edhw_th		dhw	kWh/m²NGFa	Edhw_1_th+Edhw_2_th
Edhw_el		dhw	kWh/m²NGFa	Edhw_1_el+Edhw_2_el
Edhw_aux_el		dhw		Edhw_1_aux_el+Edhw_2_aux_el
Edhw_1_th		dhw	kWh/m²NGFa	SUMME(sim[Qdhw_1_total])/1000
Edhw_1_el		dhw	kWh/m²NGFa	SUMME(sim[Edhw_1_min_el];sim[Edhw1_flex_final])/1000
Edhw_1_aux_el		dhw	kWh/m²NGFa	Edhw_1_th * DHW_1_el_aux

Edhw_2_th	dhw	kWh/m²NGFa	SUMME(sim[Qdhw_2_total])/1000
Edhw_2_el	dhw	kWh/m²NGFa	SUMME(sim[Edhw_2_min_el];sim[Edhw2_flex_final])/1000
Edhw_2_aux_el	dhw	kWh/m²NGFa	Edhw_2_th * DHW_2_el_aux
Eh_th	heating	kWh/m²NGFa	SUMME(sim[Qheb_total]) * 0,001
E_th_total	final_energy	kWh/m²NGFa	Edhw_th+Eh_th
Ec_th	cooling	kWh/m²NGFa	SUMME(sim[Qced_total]) * 0,001

Annex B. Published Works

B.1. A Quantitative Positive Energy District Definition with Contextual Targets

Publication type: Journal paper

Authors: Simon Schneider, Thomas Zelger, David Sengl, and José Baptista.

Original publication:

Schneider, Simon, Thomas Zelger, David Sengl, und José Baptista. „A Quantitative Positive Energy District Definition with Contextual Targets“. *Buildings* 13, Nr. 5 (2023): 5. https://doi.org/10.3390/buildings13051210.

Contribution to the thesis:

PED balance target Contextualization method for density and mobility contexts and Context Calibration to national decarbonization pathways.

buildings

MDPI

Article

A Quantitative Positive Energy District Definition with Contextual Targets

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Abstract: This paper presents the goals and components of a quantitative energy balance assessment framework to define Positive Energy Districts (PEDs) flexibly in three important contexts: the context of the district's density and local renewable energy supply (RES) potential, the context of a district's location and induced mobility, and the context of the district's future environment and its decarbonized energy demand or supply. It starts by introducing the practical goals of this definition approach: achievable, yet sufficiently ambitious, to be inline with Paris 2050 for most urban and rural Austrian district typologies. It goes on to identify the main design parts of the definition—system boundaries, balancing weights, and balance targets—and argues how they can be linked to the definition goals in detail. In particular, we specify three levels of system boundaries and argue their individual necessity: operation, mobility, and embodied energy and emissions. It argues that all three pillars of PEDs, energy efficiency, onsite renewables, and energy flexibility, can be assessed with the single metric of a primary energy balance when using carefully designed, time-dependent conversion factors. Finally, it is discussed how balance targets can be interpreted as information and requirements from the surrounding energy system, which we identify as a "context factor". Three examples of such context factors, each corresponding to the balance target of one of the previously defined system boundaries, operation, mobility, and embodied emissions, are presented: density (as a context for operation), sectoral energy balances and location (as a context for mobility), and an outlook on personal emission budgets (as a context for embodied emissions). Finally, the proposed definition framework is applied to seven distinct district typologies in Austria and discussed in terms of its design goals.

Keywords: Positive Energy District; PED definition; context factors; PED assessment; energy transition; energy balance assessment; sustainable districts; key performance indicators

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1. Introduction

The need for a Positive Energy District (PED) definition stems from both EU and individual project levels: on the level of the European Union (EU), it is necessary to uniformly define PEDs to measure the success of the strategic SET Plan mission of bringing a hundred PEDs underway by 2025 [1], and at the same time individual projects need criteria that can be met to be referenced and possibly certified as PEDs. As a consequence, PED's definition has been discussed extensively [2–6], and although there is recognition of the need for a common definition, none could be arrived at as of yet [5–7]. Many approaches have been put forward that do not or do not exclusively employ a quantitative assessment scheme [8–10], but many researchers and practitioners agree that the definition of a PED must ultimately entail the evaluation of an energy balance, whose result must be positive [1,2,4–6,11].

The differences between existing definitions that employ such a positive energy balance as a sufficiency criterion can thus be outlined: firstly, the question of which energy

services should be considered. There exist many approaches [6], but most converge on the minimum for the operation of the district's heating, ventilation, and air-conditioning (HVAC) systems [9] and sometimes also user electricity such as plug loads [2,12]. Mobility and embodied energy, on the other hand, are less prevalent due to the apparent negative effects on a positive energy balance or the lack of suitable assessment methodologies in theory and practice. Other approaches forego a uniformly quantifiable definition altogether and instead make this determination a project-specific process [2,3].

The second divergence is the balance metric or key performance indicators (KPI) to be used: energy end-use or flexibility KPIs [13], total or non-renewable primary energy [2,12,14], or greenhouse gas (GHG) emissions [15] or a combination of the above and others [8,13]. In the latest Annex 83 review of the International Energy Agency (IEA), however, most definitions use a primary energy indicator, with notable exceptions [16]. However, differences in primary energy conversion can cause drastically different balance assessments [17–19].

Thirdly, there is divergence in the kind of system boundaries the balance is evaluated on, although "the majority of PEDs in Europe apply the dynamic-PED concept, with geographical boundaries" [6] (p.13). This can be further obfuscated by the fact that many projects do not readily achieve their positive balance without some form of "offsite subsidies", be it in the form of RES credits or outright including these resources in the PED boundary [2,9], but not necessarily clear rules as to how this inclusion must be performed. Temporally, most use an annual balancing period of an operation year [16].

What is typically not discussed at length is the goal and scope of the definition: is it suitable for green or brown field developments, any climate zone, and density? Should it be possible for any district, and of which ambition level to achieve a "PED"? As with other European standardization processes such as the Energy Performance of Buildings directive (EPBD [20]), one could separate what a PED is in different levels of regionality from European, national, and municipal all the way down to project-specific. In practice, however, most definitions are developed in international or national projects but used mostly by the districts within that project. This further adds to the conundrum of which definitions can and should be used for which PED projects, and most importantly: why. In fact, using multiple different definitions on different levels of regionality and detail, from broad European frameworks through national standardizations to project-level specification could be a promising way forward. In the meantime the need to bridge the gap between a standardized and unambiguous definition for EU reporting on one side and the flexibility to account for local contexts and feasibility on the other has become abundantly clear [4,7]. This paper aims to add to the theory underpinning PED definitions by introducing a conceptual view on the positive energy balance, in which the assumed target of positivity can be calibrated by the use of so-called "context factors" (CF for short). These are virtual balance components that need to be designed district-independently to maintain PED feasibility in the desired context.

2. Methods and Approach: PED Definition as a Design Problem

It can be argued that creating a definition is in itself not a scientific process but rather a design problem [21]. A definition cannot be observed, theorized upon, and validated. Instead, it specifies and regulates the appropriate use of language. In this case, "appropriate" is very open for interpretation (compare e.g. [22]). If the definition is to be useful, it needs to show exactly how. For that reason, it is our understanding that the most important part of the PED definition is the argument as to **why** it was designed in any particular way, and it must be very articulate about the goals it is aimed at. Most definitions indeed solve many of the varying stakeholder problems, but they often do so implicitly rather than stating these plainly. Only if the aims of the definition are declared alongside, can it be tested if it is indeed suited to fulfil them. The following figure illustrates this taken design approach starting with the goals and deriving the criteria and their operationalization thereafter:

After the definition's goals, the actual components of a PED definition were identified as a quantitative balance assessment that can be structured in three main parts (in accordance with [21]): (1) system or balance boundaries, (2) a balance weighting system, and (3) a balance target, which we will proceed to construct by means of the aforementioned "context factors" (CF), which, as the name implies, are derived project-independently from a national PED context. This is illustrated in Figure 1.

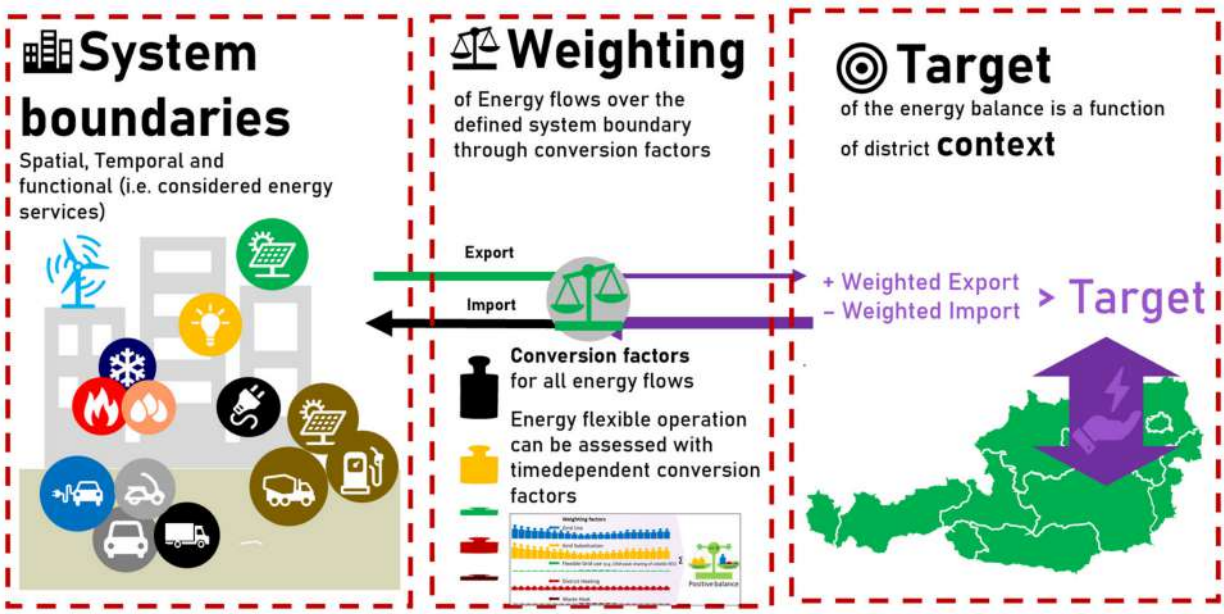


Figure 1. The three parts of a PED definition via an energy balance assessment.

These three areas of the district system boundary, system balance weighting, and balance targets correspond to the three questions a quantitative PED definition via an energy balance needs to address in unison as the definition design problem. From the start, the balance target is not necessarily positive or zero, but can, in principle, be a function of any set of parameters deemed relevant. Thus, the definition of an energy balance target can include both project-intrinsic and project-extrinsic factors. On the one hand, this is an additional challenge, but at the same time it is an opportunity: With this, dynamic external requirements can be related to project-specific proposed solutions.

This might be surprising, as it seems to contradict the prevailing understanding of a hard-set target of "greater than zero". However, it can be argued that targets can be reformulated as context factors and, as such, be included as virtual demands and supplies in a still formally positive balance. In fact, many current definitions already use some form of virtual demands and supplies to modulate the balance feasibility. Most notably, not including certain energy services in the balance, such as mobility and embodied energy, is equivalent to instead including a virtual supply of equal size from somewhere to offset the demand. The goal of this paper is to provide a formalism in which this somewhere can be defined more transparently and uniformly.

3. Goals of the PED Definition

The PED definition cannot be separated from its goals and implications in practice. The presented definition's goal is: to be achievable, yet sufficiently ambitious to be compatible with Paris 2050 for any urban and rural Austrian district typology. This is in line with the EU Commission's statement that the ambition of PEDs is to "go well beyond what is already requested in the Energy Performance of Buildings Directive" [23], but also aims at the sufficiency question as well. The definition development of the approach was therefore done under these guiding objectives:

1. The PED definition contains all relevant features of a future 100% renewable energy system. Such PEDs anticipate future requirements by the precautionary principle and must take into consideration their future surroundings.
2. The PED definition is achievable in both rural and urban contexts, or areas of low and high building density (technically, legally, and economically). Lower densities should not be implicitly favored by the PED definition.
3. The PED definition is achievable for different types of usage mixes with comparable ambition, not just for uses with low energy demand or good temporal alignment between supply and demand.
4. The PED definition's achievability is not dependent on the incidental but uncommon availability of local renewables such as local (industrial) waste heat, hydro, or wind power.
5. The PED definition links the national climate goals (i.e., a decarbonized future energy supply) with the local targets of a district in a comprehensive quantitative system.
6. The PED definition is compatible with the definition developed at the European level by the Alignment Task-Force JPI UE Framework Definition [1].
7. The PED definition has directional stability and consistency for all process phases: From project development to implementation or monitoring (zoning, architectural competition, planning, execution, and operation). This ultimately means a stable definition operationalization and accompanying assessment framework as part of a nationally accredited standardization and certification scheme.
8. The PED definition concept should be flexible and extendable, from operation (PED Alpha) to mobility (PED Beta) to the entire life cycle (PED Omega), and should lend itself to transparent reparameterization in the future.

In this paper, the introduction of a quantitative PED definition assessment scheme using context factors is aimed at facilitating these goals. In particular, the density context-factor addresses Goals 1–3 and 5 by connecting national PV capacity targets (Goal 1 and 5), depending on density (Goal 2), and is comparably achievable for different usages (Goal 3), as shown in Section 7. Goal 4 is facilitated by introducing other means of achieving a nominally positive energy balance. The proposed is also compatible with the JPI UE Framework definition (Goal 6), as the latter also mandates a positive energy balance but also expects some form of contextualization. This paper also tries to further Goal 7 by introducing a formal definition framework that can subsequently be parametrized for national standardization and certification. Goal 8 is addressed by the differentiation of the definition into three possible perspectives through the introduction of three possible system boundaries (Alpha, Beta, Omega).

The goal of this definition is to envisage the PED as part of a future decarbonized energy system and to quantitatively relate the PED balance targets to the achievement of these national and international climate goals. This requires a quantitative method of allocation and effort sharing, which was presented in [24] and will further be detailed in Sections 4–6.

The aim of the definition is to make the flexibility of the district quantifiable, such that higher energy flexibility simultaneously has a positive impact on the achievement of the PED balance target. This is important to quantitatively link all three energy dimensions of a PED, i.e., energy efficiency, local renewable generation, and energy flexibility. This also allows different districts to prioritize and realize their respective potentials for achieving the PED definition according to their local circumstances without the need for further definitional additions. This is addressed as part of the balance weighting system in Section 5.

Non-Goals

The focus of the PED definition design is encapsulating the climate neutrality requirements of the built environment through the means of a positive energy balance. Therefore, an explicit design choice is made to not concern the definition with possible additional—albeit

important—dimensions and aspects of district development and assessment. Instead, the definition should be compatible and work in conjunction with already existing assessment and certification systems that consider these aspects. Additionally missing in this definition are criteria and specifications for the district development and planning process, as different stakeholders and ownership structures (such as new construction versus existing) and spatial and urban planning organizations require different planning processes. Prescribing a specific process or awarding points complicates and distracts from the main design goals of this PED definition. Instead, districts can use dedicated systems for this, such as the Austrian klima:aktiv standard for districts and neighborhoods [25], Leadership in Energy and Environmental Design (LEED [26]), or the Building Research Establishment Environmental Assessment Method (BREEAM [27]).

Another non-issue is the definition of important aspects of social, economic, and environmental dimensions. This is not to say that these aspects of district development such as social inclusion, safety and comfort, community development, and the creation of a sustainable ecosystem are not equally, if not more, important than the energy assessment. However, they should be better covered by assessment and certification systems focused on them and complementary to achieving the PED definition, such as Total Quality Building (TQB), klima:aktiv [28], local requirements of subsidized housing, and other regulatory and legal instruments [29].

This approach also foregoes the definition of minimum or sufficiency criteria for individual aspects such as energy efficiency, renewable production, and flexibility: Because districts have very diverse typologies, the definition of accurate individual criteria is correspondingly complex. Sufficiency criteria, e.g., defined in the building code or other standards, are helpful but not necessary in the context of a PED definition: A district should be free to decide *how* to best use its potential and how it contributes to climate neutrality. Complementary systems could, again, be passive house standards [30] and the local building code [31], instruments of zoning and development plans, urban planning framework agreements [32], etc.

4. System Boundaries

System boundaries are considered in a spatial, temporal, and functional sense in accordance with the fundamentals of PED energy modeling described in [11]. The definition of system boundaries is required to enable the balancing of flows over these defined boundaries. As such, it is necessary to distinguish and define all three different types:

Spatial means an actual physical boundary of included energy services and supplies. In other approaches, this boundary is sometimes used to make “offsite” RES party possible. In the presented approach, the available “offsite RES” is instead addressed as part of every PED’s surrounding in the form of a balance-target-adjusting context factor, introduced in Section 6 and not subject to spatial boundaries.

Temporal system boundaries can be interpreted as the balancing period and are typically set to one operational year.

Functional system boundaries are used to identify specific energy functions, uses, or demands to be included or excluded according to function, rather than spatial proximity. Functional system boundaries can be further differentiated into renewable energy supply within the system boundary, referred to as “onsite”, and energy services to be accounted for in the balance. Note that “onsite” here does not necessarily mean spatially onsite but rather “within the system boundary”, counting positively towards the energy balance.

The functional system boundaries and the included energy services can be roughly grouped into three regimes of increased responsibility: (1) operational energy and user electricity, (2) mobility, and (3) embodied energy and emissions. This approach defines three variants, or shells, from PED Alpha in the innermost part considering only the operating energy, over PED Beta including private everyday mobility, up to PED Omega in the outermost part, where the embodied energy of district construction, maintenance, repair, and mobility are also considered. The three system boundaries reflect three areas

of quantitative assessment: operation, mobility induced by the location, and embodied Energy, respectively. Each is associated with increasing effort and greater uncertainty than the last. The expandability of the system boundary is important because, on the one hand, different data and information are available in the course of the project, and on the other hand, there are already partly considerable differences between projects in the available data and the objectives. At the same time, appropriate data are necessary for simulation and verification. The system boundaries are illustrated in Figure 2 and are detailed in the following subsections.

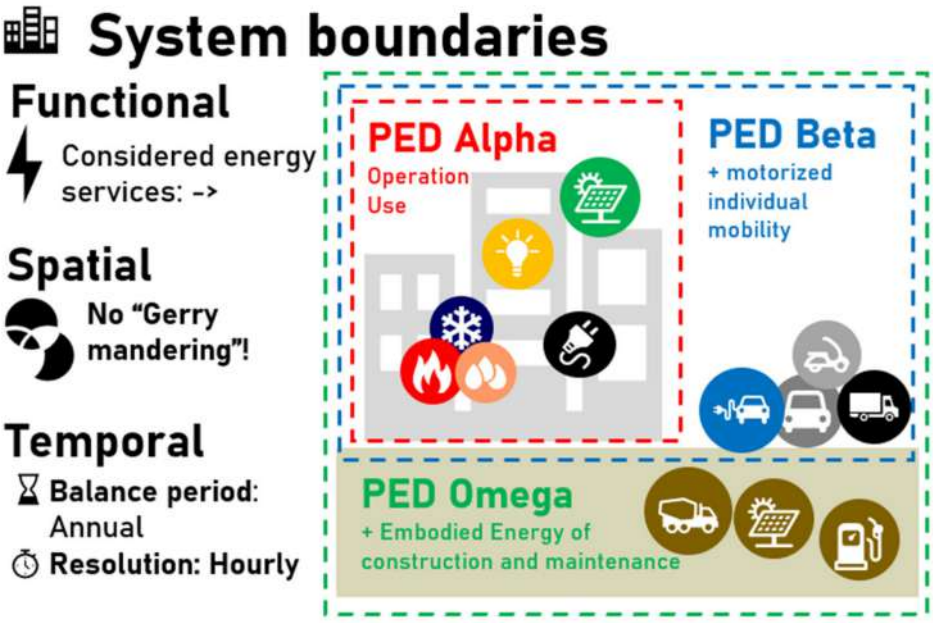


Figure 2. Types and extent of defined system boundaries.

4.1. Functional System Boundary: Considered Energy Services

The definition considers all energy demand for building operation, domestic hot water, lighting, and building services as well as the energy demand for living, working, and services (e.g., appliances, computers). Process energy is not considered directly, but indirectly through the crediting of national surpluses or deficits. Everyday individual mobility and embodied energy are also included in their respective boundaries, as listed in Table 1.

4.2. Spatial System Boundary

The spatial system boundary coincides with the physical district boundary. All definition components refer to the area that is necessary for the full use of the district. A "Gerrymandering" of the spatial boundaries for targeted inclusion or exclusion for energy and emission balance reasons must be avoided. District boundaries should be as convex as possible and as concave as necessary. Within these spatial boundaries, all local renewable energy sources are permissible regardless of the conversion technology and can be used to cover the energy balance as long as the assumption holds that its use within the district does not limit the usability for areas outside the district. This is particularly important to consider for the thermal and electrical use of flowing water for cooling and small hydropower. Further than that, it does not seem practical to generalize the validity and shape: The spatial system boundary should be drawn in such a way that the immediately surrounding areas do not suffer any obvious disadvantage from becoming PEDs themselves.

Table 1. Considered energy services within PED boundaries Alpha, Beta, and Omega.

Energy Services		Alpha	Beta	Omega	Implicit *
Building operation	Heating	✓	✓	✓	-
	Cooling	✓	✓	✓	-
	Humidification and dehumidification	✓	✓	✓	-
	Ventilation	✓	✓	✓	-
	Auxiliary power of the building services system	✓	✓	✓	-
	General power and lift	✓	✓	✓	-
	Lighting	✓	✓	✓	-
District operation Industry, agriculture	Power requirements of users (plug loads)	✓	✓	✓	-
	Operating power (office, retail, school)	✓	✓	✓	-
	Process heat	-	-	-	✓
	Process cooling	-	-	-	✓
	Electricity demand for industrial production processes	-	-	-	✓
	Electricity demand for general use (incl. services)	✓	✓	✓	-
Mobility	Motorized private transport	-	✓	✓	-
	Public transport	-	-	-	✓
	Other mobility	-	-	-	-
Embodied Energy	Components of the Austrian energy certificate	-	-	✓	-
	Accessory components (cellars, underground parking, garages, carports, bicycle storage areas, balconies and terraces, other outbuildings)	-	-	✓	-
	Building and energy equipment	-	-	✓	-
	Vehicles and infrastructure for mobility	-	-	✓	-
	Public transport	-	-	-	✓

(✓) Included in system boundary, (-) not included in system boundary. * **Implicit**: Not included in any system boundary but instead part of the district context of the surrounding energy scenario that in turn influences the balance target as context factors.

5. Balance Weighting System

The weighting and the associated evaluation of energy flows is a central and controversially discussed topic of the PED definition. The approach presented here focuses less on the physical self-sufficiency or autonomy of the district but rather on the assessment of the district's contribution to the climate neutrality of the overarching national energy system. Specific weighting objectives therefore are:

1. Linking to planning practice and existing literature: The use of total primary energy and GHG emissions by means of conversion factors from the current building code or, in the case of district heating, county-specific regulations.
2. Mapping of seasonal differences: Monthly conversion factors based on Austrian building codes [31], renewable feed-in during summer and grid import in winter are weighted differently due to their different grid support and substitution alternatives.
3. The evaluation of energy flexible, grid-serving, i.e., time-sensitive, grid use and feed-in: Otherwise unavailable energy in the surrounding system is weighted with a conversion factor of zero.
4. Biomass use is possible, but not implicitly preferred due to low conversion factors in the building code: Instead, an average of total and non-renewable primary energy is used. If only the first were used, biomass would mostly be infeasible, and if only the latter were used, biomass systems would easily outperform electricity-based systems. These goals led to the conversion factors presented in Table 2, which are further discussed in Section 8.

5.1. Energy-Flexible Grid Use through Demand-Side Management (DSM)

Although other indicators such as the Grid Support Coefficient [33] or the Smart Readiness Indicator [34–36] facilitate more in-depth assessments of energy flexibility and

grid serviceability, they can also increase the complexity of modeling and verification. Instead, this PED definition aims to assess a district's capabilities for energy-flexible and grid-serving operation through a time-dependent weighting of the energy flows instead of mandating an additional KPI. The resulting energy balance reflects energy efficiency, local renewable generation, and energy flexibility measures. This also allows different districts different approaches and strategies in achieving a positive energy balance.

Table 2. Weighting factors for energy flows over the defined system boundaries.

Energy Flow	PED Alpha, PED Beta	PED Omega	Source
	Total Primary energy	CO_2-equiv.	
Uncontrolled grid use and feed-in	Monthly conversion feed-in sign-inversed	Monthly conversion feed-in sign-inversed	National building code [31]
Energy-flexible grid use (DSM)	Zero	Zero	Section 5.1
Biomass	100% renewable + 50% non-renewable primary energy	CO_2-equiv.	
Other energy carriers	Total primary energy	CO_2-equiv.	National building code [31]
Fuels (mobility)	Total primary energy	CO_2-equiv.	

In operationalizing the time-dependence of the weighting factors, a distinction is made between "generally available regional renewables" and "situational RES", which are useable only by an appropriately energy-flexible district—i.e., the energy flexibility provided by the district. Only the latter is considered favorably in the balance. The former is not disregarded but handled as a context factor, as shown in Section 6. The flexibility is assessed by weighting with a primary energy conversion factor of zero in the balance, but is subject to operational constraints: This temporally available renewable electricity is situational and would not be useable without corresponding strategies and regulations. Enabling this integration is a key goal of energy flexibility PEDs and is therefore treated differently in terms of methodology in principle. For a 100% renewable Austria, expansion rates of wind power by a factor of 5 and PV by a factor of 20 are needed, depending on the scenario, which means that strong seasonal fluctuations have to be compensated.

How can this energy flexibility be provided? The physical principles presented in IEA EBC Annex 67 on energy-flexible buildings [37] (p.67) apply: In particular, the thermal storage masses of buildings and districts with good thermal insulation and heavy building components can lead to significant displacement periods of several days, during which the system can freely choose the timing of its energy use for heating and cooling [38]. The principle is schematically summarized in the following Figure 3. The DSM operation is included and assessed in the energy balance by lowering the associated weighting of energy import during times of regionally available surpluses.

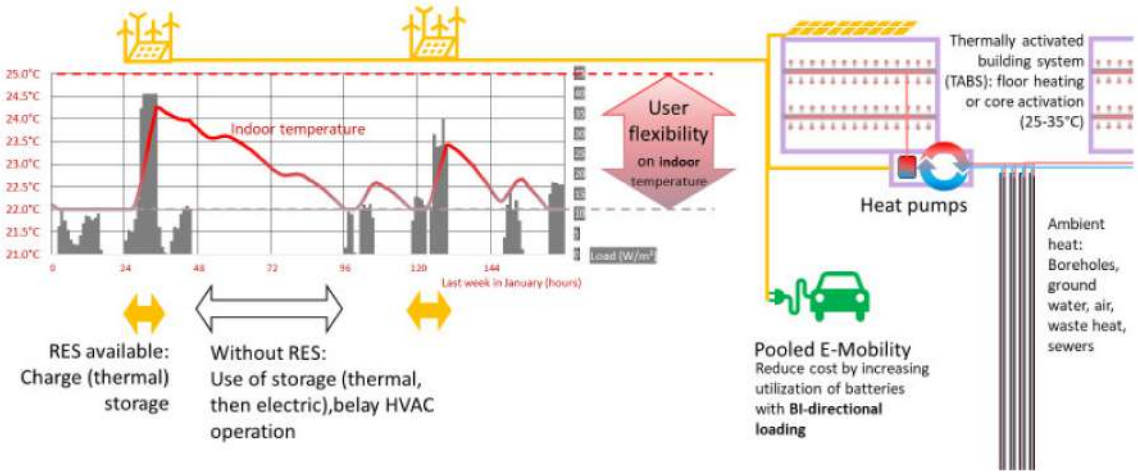


Figure 3. Example illustration of flexible DSM in a PED to maximize RES utilization.

6. Balance Targets (Are Reversed Context Factors)

By now it has become apparent that the typically assumed target of the energy balance, to be positive or above zero, needs further clarification and, as pointed out above, also justification. By rephrasing balance targets as contextual factors in a strictly positive balance, the targets can be parametrized to reflect any desired context. The balance target value becomes a target function: A key feature of the PED definition approach is that the target value is a variable quantity and does not have to be positive, per se. A variable target value for the balance across the system boundary can be considered as a virtual credit for the district balance. The quantitative target value is equivalent to an external credit/debit of the opposite sign:

$$\text{Primary Energy Balance} = \text{Weighted Exports} - \text{Weighted Imports} > \text{Balance target} \quad (1)$$

$$\text{Sum of Context factors} = \sum_i CF_i = -\text{Balance target (in a context)} \quad (2)$$

$$PED\ Balance = Weighted\ Exports - Weighted\ Imports + \sum_i CF_i > 0 \quad (3)$$

(1) is a specification of the more general description in [23], with a weighting of primary energy conversion and a general balance target. In (2), generic context factors are introduced in CF_i, denoting an arbitrary context factor, and the sum of these to be the negative of a balance target. (3) combines both (1) and (2), showing how the energy balance can maintain the nominal target > 0 while including context-sensitive targets.

Figure 4 illustrates and compares this approach with context factors in a situation, where the feasibility of the PED balance is accomplished by other means. It is an important consequence of this definition that any balance definition can be mapped to any other with an appropriate context factor. In other words, any inclusion or exclusion of energy services in the energy balance can be realized with the inclusion of an appropriate context factor. This can be leveraged to shift the discussion of system boundaries to a discussion of targets and appropriate context factors, which can offset balance components.

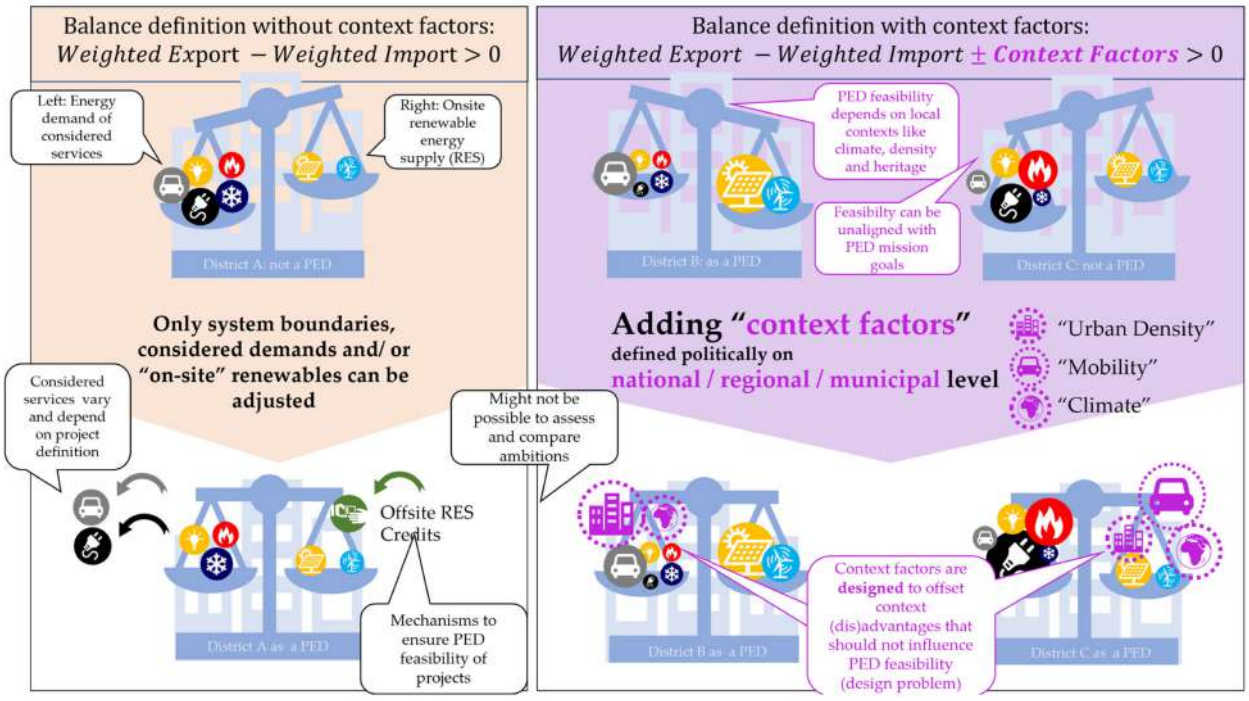


Figure 4. Illustration of the conceptual differences without (**left**) and with (**right**) explicit context factors.

Using this paradigm, a quantitative PED definition can be designed with a positive balance target using the context factors for the appropriate system boundaries depicted in Table 3. The inclusion of these ultimately virtual factors in the balance could be designed in an arbitrary number of ways. Their use must be rooted in its comprehensibility and link to the definition goals, which are examined in the following subsections, that can be quantified.

Table 3. Balance targets.

System Boundary	Scope	Balance	Context Factors	Target	KPI
Alpha	Operation, use	Primary Energy Exports–Imports *	±CF ∗∗ Density	> 0	kWh PE tot./m^{2}NFA/a
Beta	Operation, use, individual motorized mobility	Primary Energy Exports–Imports *	±CF Density ± CF Mobility	> 0	kWh PE tot./m^{2}NFA/a
Omega	Operation, mobility, and embodied emissions	GHG Emission Imports – Exports *	−CF Emissions	≤ 0	kg CO_2eq./m^{2}NFA/a

* Energy flows into the district are counted negatively (e.g., grid electricity purchases and district heating, Table A1) and energy exports across the system boundary are accounted for positively (e.g., PV surpluses). The emission balance must be negative, the local and imported emissions are offset by export (external emission prevention) and a CF Emission, which represents an emission budget per reference area. ** CF—context factor.

6.1. PED Alpha Context: Density and the Feasibility for PED Districts in Urban Contexts

One of the main results of preliminary projects was to link the target value of the energy balance of a sustainable district to its building density, expressed by the floor area ratio (FAR). The approach was motivated by the observation that the energy balance depends significantly on the floor area ratio, i.e., the ratio between gross floor area (GFA) and plot size. This is because energy demands predominantly correlate with GFA, whereas potentials for local renewable energy generation generally are proportional to plot size [14]. The inverse proportionality between district density and achievable energy balance is both empirically evident [39] and analytically derivable. A definition that is to be comparably achievable in an urban context must therefore take this into account and refrain from a static target value. There are two aspects to consider here. First, this is simply a consequence of the physical facts, as detailed in [14,29,40]. However, there is a second point: in principle, the PED concept should not be opinionated about when and where it can be achievable and how easily. This is ultimately a political design question that can only be answered by quantitatively embedding the definition in the appropriate political context.

For Austria, this context is that the available building land reserves have been declining sharply, and, subsequently, land-use conflicts will become ever more prevalent due to geographic constraints [41]. Therefore, efficient land use has been a recurring political topic in recent years and has been called for with good reason. In Austria, therefore, special care must be taken to ensure that a PED definition does not encourage inefficient land use by making it easier to achieve in lower densities. Figure 5 illustrates the relationship graphically: the development on the left creates more usable space on the same lot and requires less infrastructure per person but would require infeasible amounts of PV (as the only commonly available source of local renewable electricity).

A 2021 study on the PED potential of urban typologies in Vienna, Austria found that only detached housing districts can achieve a positive annual energy balance (for heat and power) of 110%, whereas more dense typologies fail to achieve the criteria, with an annual balance ranking between 61% and 97% [40].

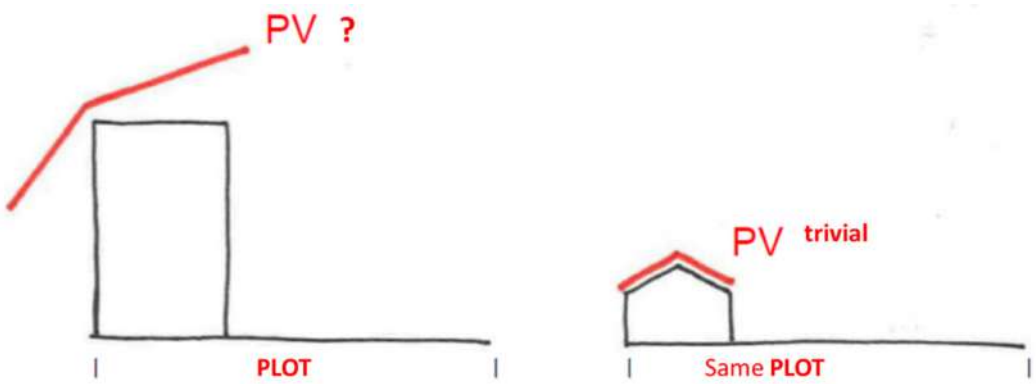


Figure 5. Illustration of the physical difference between high- and low-density districts in balancing their high and low energy use, respectively, with onsite renewables (represented by required PV area).

Figure 6 shows example energy balances of a number of districts in Austria that also show the impact of building density on the spread of the primary energy balance of districts. The vertical spread is caused by the different variants of a district (of constant density), from conventional variants without local renewable generation and efficiency measures at the lower end of the spectrum to "maximum" variants at the upper end, which feature highly efficient thermal hulls and HVAC systems with heat pumps for low-temperature heating and cooling and DHW heating and consider energy-flexible DSM and the maximum technical PV potential. The latter was determined on a project-specific basis, which inhibits a direct comparison between the maximum and minimum district variants. However, the representation of all analyzed district variants as one point on this FAR-to-PE balance diagram shows the connection with the building density and the target value derived from it. The red line represents the proposed context factor density as a modified target.

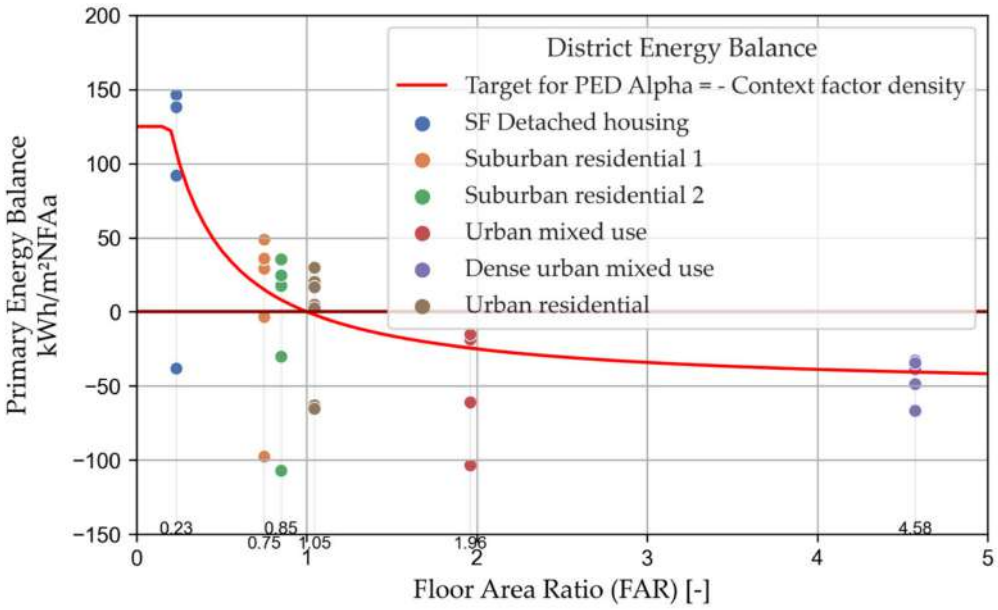


Figure 6. Primary energy balances of example districts over their density (FAR).

This correlation was also reported in similar studies [39] and comes as no surprise. The empirical relation between density and the possible PED balance can be also derived analytically. Based on a balance of energy demand and local renewable generation, the dependency on building density (FAR) is derived as follows:

$$\text{BALANCE} = \text{RES} - \text{ED} \quad (4)$$

$$\text{BALANCE} = \ f_{RES}\ \text{Plot Area} - f_{ED}\text{Floor Area} \tag{5}$$

with the onsite renewable energy supply (RES) and the local energy demand (ED), which both can be expressed as a product of the specific supply and demand per reference area (f_{RES} per available plot area and f_{ED} per useable floor area, see also Table 4). The division of the entire equation with the useable floor area changes the balance from absolute to floor-area-specific (kWh/m^{2}NFA) and reveals the analytical dependency on the floor area ratio, as it is defined as FAR = Floor Area/Plot Area:

$$\text{BALANCE}(\text{FAR}) = f_{RES}\frac{1}{\text{FAR}} - f_{ED}\ \left[\text{kWh}_{PE}/\text{m}^2_{GFA}\right] \tag{6}$$

Table 4. Formula abbreviations.

Variable	Unit	
CF_{Type}	$\frac{\text{kWh}_{PE}}{\text{m}^2_{GFA}}$	Context factor of a given type (density, mobility)
f_{RES}	$\frac{\text{kWh}_{EEU}}{\text{m}^2_{PA}}$	Potential electricity yield per buildable plot area
f_{ED}	$\frac{\text{kWh}_{EEU}}{\text{m}^2_{GFA}}$	Electricity demand per gross floor area
dx	$\frac{\text{m}^2_{GFA}}{\text{m}^2_{PA}}$	Sensitivity parameter floor area ratio (FAR)
Cutoff	$\frac{\text{kWh}_{PE}}{\text{m}^2_{GFA}}$	Maximum resulting energy balance

Since the two parameters of the balance function f_{RES} and f_{ED} are conceptual variables, a scientifically deterministic determination is not expedient. Instead, they are considered control variables for the effort sharing between sparsely and densely built-up districts. They are determined politically within the framework of technical and economic feasibility according to the provision principle. For operational purposes, the formula for the density context factor was extended to include a cutoff and an offset in the *x*-axis (dx):

$$\text{Context Factor Density} = CF_{Type}(\text{FAR}) = -\text{Balance Target for PED Alpha Operation} = -\min\begin{cases}\frac{f_{RES}}{\text{FAR}+dx} - f_{ED}\\ \text{cutoff}\ (\max value)\end{cases} \tag{7}$$

A comparison with Table 5 and Figure 7 shows that the factors were subsequently parameterized with approximately

$$f_{RES}\ \approx 30.4\frac{\text{kWh}_{PE}}{\text{m}^2_{plot}}, f_{ED} \approx 26.4\frac{\text{kWh}_{PE}}{\text{m}^2_{GFA}} \tag{8}$$

Table 5. Energy balance performance approximation of Austrian district typologies.

District Type	Share	Potential Electricity Yield f_{RES}	Electricity Demand f_{ED}	dx	Cutoff	Electricity Balance
		$\frac{kWh_{EEU}}{m^2_{PA}}$	$\frac{kWh_{EEU}}{m^2_{GFA}}$	$\frac{m^2_{GFA}}{m^2_{PA}}$	$\frac{kWh_{PE}}{m^2_{GFA}}$	*PJ/a*
Unrefurbished	0%	−1.0	50	0.15	-	0
Thermal refurbishment	40%	0.0	38.5	0.15	-	−42.16
Refurbishment with minimal PV	20%	30.4	35	0.15	13	−3.69
PED Refurbishment	20%	30.4	26.4	0.15	62	+11.55
PED New construction	20%	30.4	26.4	0.15	62	+11.55
Total						−22.7

It is important to note that the parameterization is not a quantitative representation of the technical potential of a district as a function of the FAR. In fact, the parameterization is motivated

by the technical potential, but the level of the factors is significantly lower than the actual technical potential and demand. Their level is derived from the resulting effort-sharing in the Austrian building sector. It is important to keep this in mind: the parameterization of the PED Alpha target through an equivalent context factor for "density" only operationalizes a small part of the technical potential difference. As a result, in the comparison of possible district configurations, some project variants are able to exceed the target value—sometimes significantly. These are mostly variants with technical but not economic feasibility.

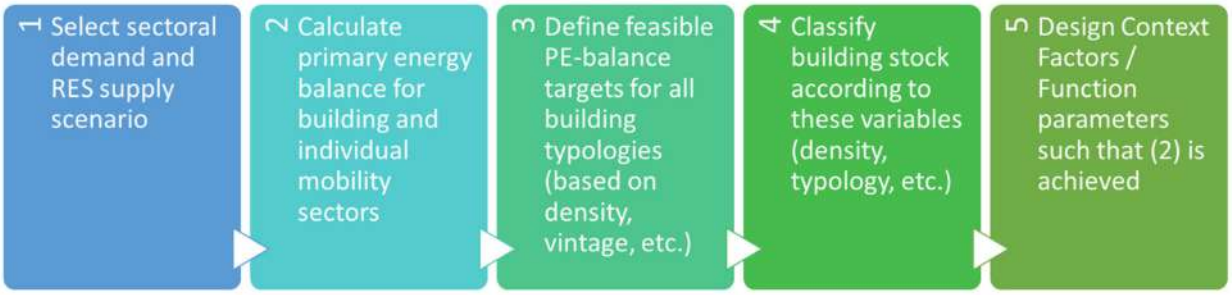


Figure 7. Parametrization steps of district density context factor.

The parameterization is carried out through the steps outlined in Figure 8. The target of the building sector is determined from the top-down consideration of the 100% renewable energy scenario Austria 2040 and the distribution according to the balance sheet allocation of generation and demand sectors presented at the beginning. With the assumption of the following refurbishment rates and classes, this results from Table 6 in the following parameterization across all FARs of the building sector:

$$\text{Balance Target Building sector} = \sum_{\text{Type}} \sum_{\text{FAR}=0}^{\infty} \text{CF}_{\text{Type}}(\text{FAR}) = -22.7\ \text{PJ/a} \quad (9)$$

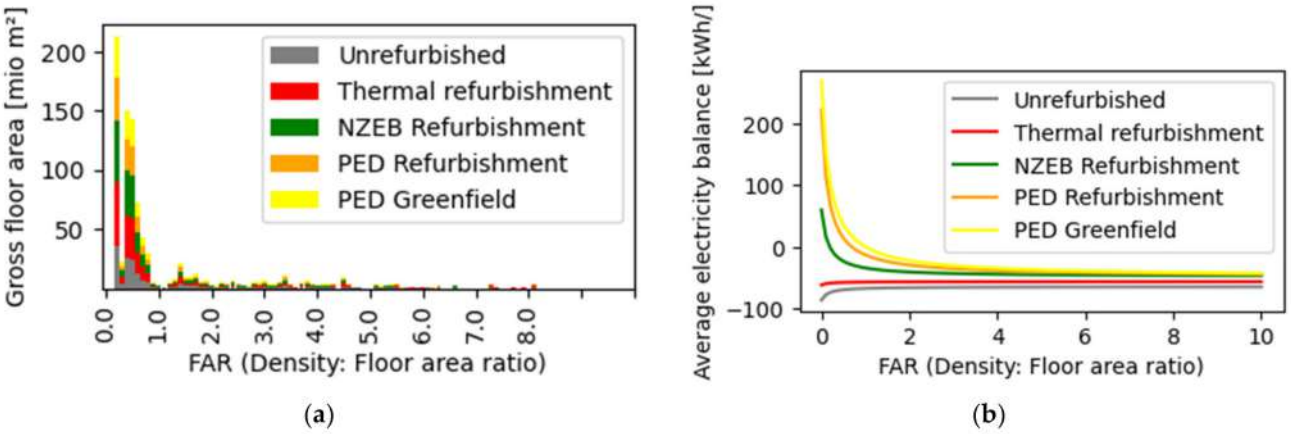


Figure 8. (**a**) Distribution of district typology gross floor area over the floor area ratio (FAR). (**b**) Distribution of district typology energy balance over the floor area ratio (FAR).

Table 6. Assumptions for the surrounding energy system.

Energy Demands and Supplies 2040		**Source**
Electricity demand of the building sector 2040 for operation and MIT after sectoral allocation	137.6 PJ/a	[24] based on [42]
Technical potential of the building sector energy demand 2030	48.2 PJ/a	[43]
Allocation scenario		
Photovoltaics target 2040 (allocation to buildings)	114.8 PJ/a	[24] based on [42]
Electricity balance target of the building sector	−22.7 PJ/a	

As a next step, the share of PEDs in the building sector and the other district typologies are classified and their performance is estimated depending on density, as is shown in Table 5. Note that this is a first approximation and can and should be detailed as necessary to reflect the desired context.

Figure 8 shows the size and distribution of the defined district typologies depending on their density (a) and their respective energy balance target (b). This allocation can be tweaked to reflect other scenarios of building sector performance in a future energy system by changing the balance target of −22.7 PJ/a. It can also serve as grounds for discussion as to the relative performance and effort sharing of different parts of the building stock.

6.2. PED Beta Context: Mobility and the Surrounding Energy System

Nationally and internationally, there are conflicting opinions as to whether and in what form mobility should be included as an energy service in the system boundary of a PED. The main arguments are listed in the following Table 7.

The PED Beta system boundary is an attempt to expand the PED definition to include everyday mobility in the sense of the above-mentioned arguments without decreasing the feasibility of reaching a positive balance outright. For this purpose, the following approach was developed, which can be applied to all districts and neighborhoods in Austria at the beginning of project development without significant effort. Specifically, this is done by considering two additional components in the primary energy balance:

1. The mobility energy demand induced by the individual motorized mobility of the district as a statistical approximation. This is operationalized to depend on the public transport connection of the location, as well as the mix of uses in the district, which results in a district-specific mobility profile and the associated energy demand.
2. A project-extrinsic mobility energy budget, or context factor, from the surrounding renewable energy system, derived as the surplus from the regional renewable supply, which is allocated to the district via its share of useable floor space in the building sector.

Table 7. Arguments to include mobility in the PED energy balance.

PRO Mobility Inclusion	AGAINST Mobility Inclusion
• It corresponds to a more complete balancing of all energy and emission loads. • The quality of the location in terms of the everyday mobility induced by it can be assessed. • Concrete measures to reduce everyday mobility or the emissions caused by it, such as mobility sharing offers, charging infrastructure for e-cars, etc., should be quantitatively assessable. • Synergies of e-mobility charging infrastructure through the dynamic consideration of actual charging times and PV surpluses in the district and the advantages of energy-flexible districts can thus be mapped.	• Lack of data and methods to reliably determine the energy balances and emissions of transport without great effort and uncertainty. • The plus-energy standard is made considerably more difficult or even impossible (in the sense of a strictly positive energy balance). • The scope of action for developers is significantly limited. Instead, it is primarily the municipality or city that must set specifications and, if necessary, take measures outside the building site.

Together with the density context factor of PEQ Alpha, PEQ Beta thus fulfills a key requirement of this PED definition approach: the link to the surrounding renewable energy system and the mapping of effort sharing within Austria's building sector with the assessment target of a PED's energy balance.

Particularly in dense urban environments, it is questionable to mandate complete energy autonomy. Conversely, rural areas are more likely to have a surplus of renewables such as large wind and hydropower plants in close proximity. The design question is how these "regionally available renewables" can be allocated to and assigned to specific energy needs, such as in a district. Different approaches exist: the use of virtual system boundaries, which allows a form of offsetting (e.g., through the acquisition of credits); the distribution of the energy to districts through a provisioning quota (i.e., per inhabitant or floor area); or direct contracting with the operators of external RE plants.

All considerations ultimately lead to an allocation problem that should be openly addressed considering the wider energy system surrounding a district. Regionally available RES must be regionally balanced. Its availability must be allocated on a balance sheet to prevent individual actors from overusing the available resources, thereby creating a more difficult situation for the rest. The use

of available RES must be divided between the sectoral needs of industry and agriculture, mobility, and buildings through appropriate effort sharing. This has to happen on a superordinate level. Figure 9 shows an example allocation and resulting quantifiable provisioning of available RES that can be allocated to each district or building—not just PEDs. This "surplus" can be distributed per capita or floor area to the entire building sector with system boundary PEQ Beta, i.e., including individual motorized mobility. The quantification of this surplus is presented in [24].

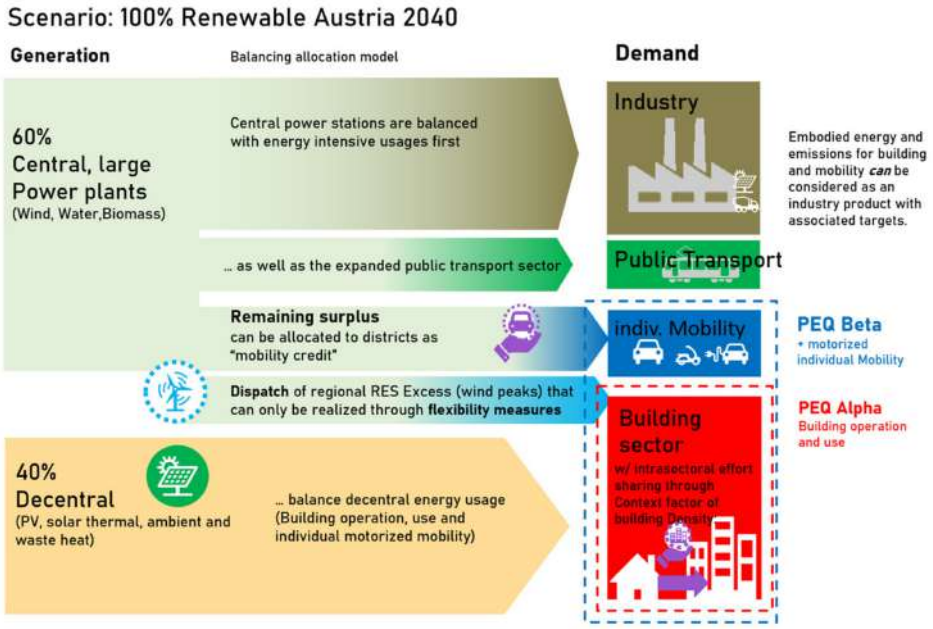


Figure 9. Schematic balancing of a national 100% renewable energy system 2040.

It is important to note that in this approach, no a priori allocation of external regional renewable energies is required for the building sector—and thus especially not for PEDs. This means that by and large, the building sector is responsible for its own renewable energy supply by the use of decentralized generation (e.g., PV, solar thermal, etc.).

Derivation of the Mobility Context Factor of a PED

The integration of everyday mobility into the PED balance impedes its positivity potential unless there is a form of offsetting that does not have to be created directly at the site. We again propose to design a virtual context factor to fulfill this role. However, what mobility offsetting budget can be allocated to every district? We try to answer by looking at the district in the context of the Austrian national energy system "top-down". Assuming, according to current legislation and political aims, that Austria will be 100% renewable by 2040, a sectoral budget can be identified for projected energy supply and consumers that can be used per person to cover private everyday mobility. The following figure illustrates the national allocation as modeled in [24].

The surplus (or deficit) from large-scale renewable power plants is allocated to the entire resident population. In Austria, most scenarios of future 100% renewable energy supply result in a surplus from central power plants and thus a bonus for the individual districts. In other systems, there may just as well be a deficit, making the increased use of decentralized RES plants necessary to also account for more of its mobility energy demands. Assuming that these are also equally distributed to the building sector, this results in an additional malus or negative context factor. A PED Beta must then not only cover its local individual everyday mobility but also support the surrounding energy system with an additional local renewable surplus. However, the allocation illustrated in Figure 9 results in an electricity surplus from the regional "central large-scale power plants" of 6.3 TWh EE/a and a surplus from biomass of 0.36 TWhEE/a, or a total primary energy surplus of 10.7 TWhPE/a for use for MIV in the built environment, as shown in Table 8.

Table 8. Central surplus as budget to cover private everyday mobility in Austria per person.

Large-Scale Renewable Power Plants		
Electric (wind, hydropower)	6.30	*TWh/a*
Biomass	0.36	*TWh/a*
Primary Energy	10.68	*TWh* $PE_{tot.}/a$

This surplus is distributed to all settlement and district areas in Austria on a pro rata basis according to the Austrian average share of this use in destination traffic, as shown in Table 9. About 50% of all Austrians' journeys home are for residential use. Accordingly, these journeys receive a 50% share of the total credit.

Table 9. Allocation of the central surplus budget by usage share of destination traffic and the distribution of usable NFA into a context factor per m^2NFA.

Usage	AT: Share of Destination Traffic [44]	Budget Share *TWh* $PE_{tot.}/a$	Usable NFA AT *mio* m^2_{NFA}	CF* Mobility *kWh* $PE_{tot.}/m^2_{NFA}/a$
Residential	50%	5.30	375.6	14.11
Office and commercial	21%	2.20	53.1	41.47
Education	3%	0.29	22.5	12.93
Retail and other	27%	2.89	96.5	29.97
Total	100%	10.68	547.8	

* CF: context factor

At the same time, the energy demand of motorized individual mobility must be determined by means of an Austria-wide statistical allocation of usable areas to inhabitants and their average annual trip kilometers, as outlined in [25,44] and shown in Figure 10 for a range from a hundred percent EV use in usage color to a hundred percent fossil vehicles in gray. It shows that, on average, the budget—or mobility context factor—in blue is not sufficient to cover the energy demand of private everyday mobility for most locations in Austria. In general, PED Beta is thus more difficult to achieve than PED Alpha and more ambitious. In particular, however, the energy demand depends on the remaining share of private transport via the location-dependent public transport quality and can deviate by up to 50% upwards and downwards from the Austrian mean value. This allocation has the effect that in "rural" areas with poor public transport accessibility, the target value is more difficult to achieve than in "urban" areas with higher public transport density.

6.3. Context Outlook: Emission Budgets and Embodied Emission Context

The PED Omega system boundary represents the final shell of this methodology. It aims to enable to balance of the entire climate change-related environmental impacts of a district and make them comparable with a target value or context-sensitive budget. This enables a statement on whether the district is compatible with the demands of a future climate-friendly, emission-neutral society or to which extent it would require further measures and possibly retrofitting down the line. Despite or precisely because of the methodological complexity, it is now more necessary than ever to start with the quantitative linking of (inter)national and individual climate goals of each person and to locate them and make them visible where there is also concrete scope for action for this—as in the case of district planning—and where it is possible to set the course for achieving the 2040 climate goals. The context factor or credit for PED Omega is currently under development and consists of three parts: Firstly, a basic personal credit that represents the emission target of 800 kg CO_2equiv/Pers/a, which represents the share that is allocated for the operation of the building and its production and repair. In addition, it includes two credits, which are calculated analogously to the first two system boundaries PED Alpha and PED Beta and result from a conversion of the primary energy credits for building density and a mobility budget into GHG emission equivalents. The detailed methods and parametrizations will be published once finalized and tested.

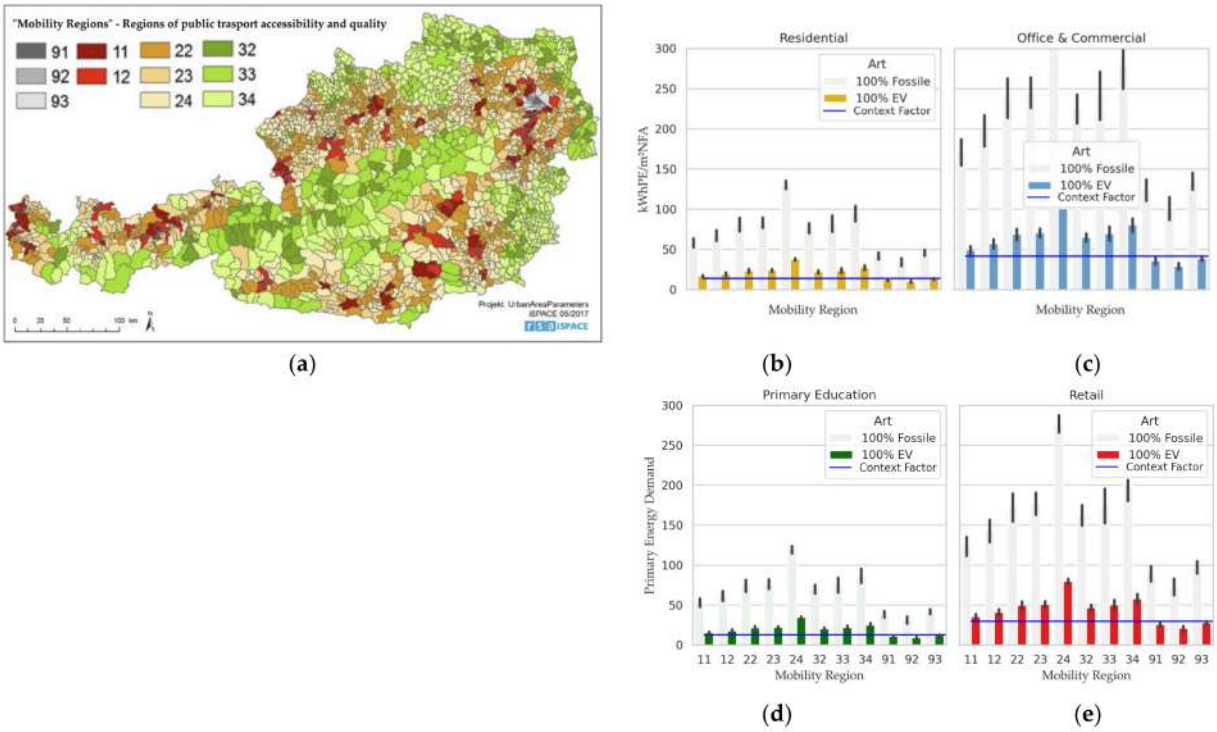


Figure 10. (**a**) AT "Mobility Regions" [44] (**b**–**e**) Energy demand for motorized individual mobility (bars) in these mobility regions for 100% fossil (grey), 100% EV shares (color), and offsetting context factor (blue line). (**b**) Residential (orange). (**c**) Office and commercial (blue). (**d**) Primary education (green). (**e**) Retail (red).

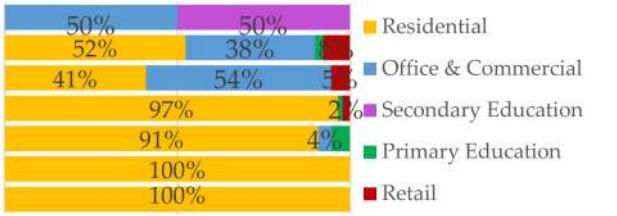
7. Definition Application Examples

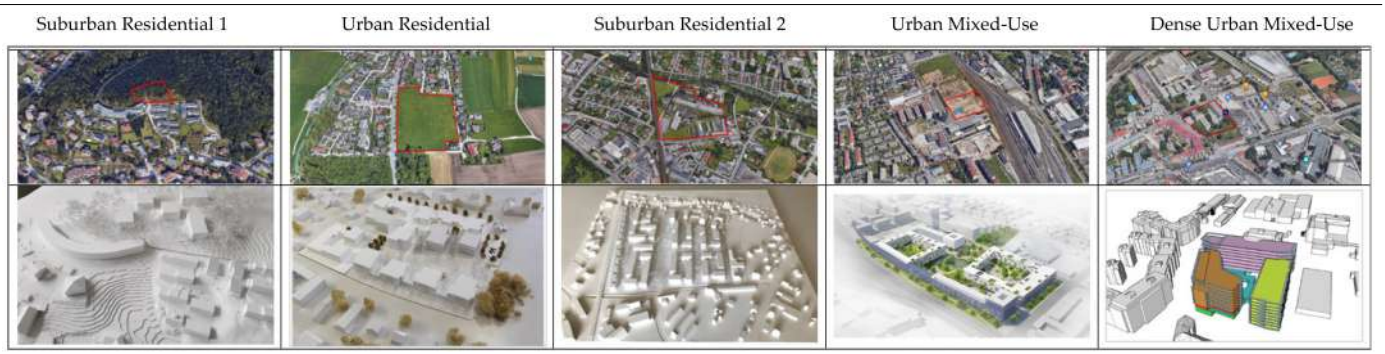
This section gives a brief overview of the example assessments of seven different Austrian district typologies. As shown in Table 10, these districts represent different usage patterns and cover a wide range of different densities from suburban and rural detached single-family homes to dense urban typologies of different uses with up to seven stories. The definition was designed to be feasible for all typologies with extensive additional measures of onsite RES, energy-flexible operation, and partial electrification and reduction in private motorized mobility. The resulting energy balances are shown in Figure 11. All points represent a district configuration with energy demand on the x-axis and renewable energy supply on the y-axis. The points above the dashed 45° line are considered PEDs, following the established convention of supply exceeding demand. Districts show the following configurations, which each add to the last: (1) baseline (in grey, project-specific, varying ambition); (2) with additional measures (in orange, maximum ambition of the project); (3) with consideration of the density context factor (in red)—note that it adds both virtual demand (horizontally) or supply (vertically) depending on the district density (FAR); (4) included energy for individual motorized mobility (in black, varying share of EV, around 50% of all car trips); and (5) including the context factor designed to offset the mobility energy in the balance (in blue).

Note how the positivity of the project balance without context factors correlates with density: higher FAR districts cannot achieve a positive balance even with the most ambitious measures, whereas districts of lower density could (over)achieve a positive balance, the lower the density the easier, without ambitious measures.

Table 10. Overview of example districts.

District	NFA m^2	FAR	Type	Usage
Dense Urban Education	25,009	5.80	Refurbishment	
Dense Urban Mixed-Use	26,805	4.58	Green field	
Urban Mixed-Use	43,778	1.96	Green field	
Urban Residential	40,383	1.05	Green field	
Suburban Residential 1	19,838	0.85	Green field	
Suburban Residential 2	4098	0.75	Green field	
SF Detached housing	124	0.23	Greenfield	





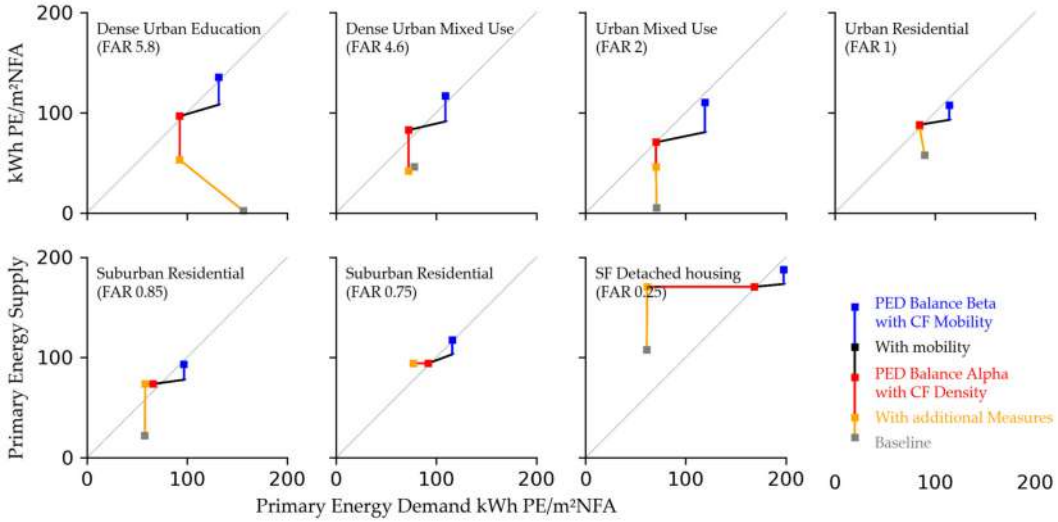


Figure 11. Primary energy balance of seven example districts.

8. Discussion

8.1. Context Factors Compared to Other Offsetting Mechanisms Such as RES Credits

The main difference is not in its effect but rather in its conceptual perspective and its resulting comparability. A context factor is just a framework to include an arbitrary design goal into the quantitative definition and formulate it as explicitly as possible. In fact, practically anything can be mapped into a context function and, as such, included in a given quantitative framework. Arguably, "context function" might be a more appropriate term, as it almost always is also dependent on the project's internal and external variables.

This may even proof useful for the comparison of different quantitative definitions themselves. The resulting balance for a district under a certain assessment can be reinterpreted as its own context function and compared. With enough such PED balance assessments, it could even be useful to compare the effects of different system boundaries and weighting systems in this manner. This also means that, in contrast to, e.g., [4] (p.13), we see the framework of context factors capable of flexibly

encapsulating enough contexts of different districts, so that the other parts of the balance definition, i.e., the system boundaries and weighting systems, can omit their current task of modulating feasibility and instead be defined uniformly and comparatively.

8.2. Why Is the Context Factor for Mobility Allocated per Usable NFA and Not per Person?

The advantage of this approach is that neither the resulting credit nor the mobility energy demand induced by the district depends on the actual occupancy density and number of users in the district. The approach is, therefore, more uniform and not dependent on the actual density of people in the district, which proves difficult to determine in practice.

8.3. Why Is Mobility Only Taken to Include Individual Motorized Mobility?

A guiding principle of the PED definition is that of subsidiarity, according to which the smallest possible unit in a system should have the greatest possible autonomy in dealing with tasks. A district can affect individual forms of mobility through its location, design, and measures, which is why it should be made partly responsible in the context of a PED definition. Public transport, on the other hand, is a superordinate mode of transport that primarily serves to connect districts with each other and with other means of transport. The accessibility of a district or settlement can strongly depend on the available public transport infrastructure; conversely, district developers cannot necessarily, and actually only in exceptional cases, influence the design of public transport. Moreover, more energy-intensive infrastructure is necessary for public transport, which also serves the general public beyond the district. For these reasons, the energy and emission provisions for public transport should be situated at a higher than district level. These provisions could, in turn, be allocated to the district, which then gives rise to the non-trivial question of the allocation method, which is even more complex for public transport than for individual everyday mobility, because the occupancy density and the allocation of trips to floor usage are less clear.

The district development timeline also plays a crucial role here, as district projects and their development with public transport do not always take place at the same time and in accordance with the initial plan.

8.4. Consideration of Delivery and Other Occupational Traffic

Delivery and occupational traffic can be considered, given an appropriate dataset, which was not available in this definition. Namely, if the underlying annual transport survey includes occupational and delivery journeys differentiated by targeted space use, i.e., home deliveries and occupational deliveries. However, again, inclusion will likely only serve the PED definition design goals if district measures can be quantified and linked to a derivative target from the surrounding.

8.5. Non-Everyday Mobility and Air Traffic

This is excluded from consideration for similar reasons: There is no methodology to assess the effect of district locations and measures on non-commonplace mobility. On top of that, it is predominantly influenced by individual lifestyles and it is not clear if and how energy and emission targets can be derived here.

8.6. Hourly Weighting of Energy Flows

Instead of monthly weighting, which improves the accuracy compared to annual primary energy conversion factors, it is considered to use hourly conversion factors. Providers such as ElectricityMap [45] already provide both historical and real-time data for emission intensities. Conversion to primary energy content is in principle possible but has not yet been standardized. In addition, forecasts of hourly PE and GHG intensities are of particular interest in the case of corresponding demand and generation developments.

8.7. Timeframe: Current, Future, or Cumulative?

Based on the difference in primary energy conversion today and in the future, the question of different observation periods arises. What statements and consequences do these different considerations and standards have on the measures and projects derived from them? What are the advantages and disadvantages and how can they be combined? In view of the advancing climate catastrophe, not only the statistically annual but also the cumulative consideration of emissions up to 2040, taking into account the time of emission, is relevant, even if this would pose additional challenges for operationalization.

8.8. Existing Districts and Refurbishment

The renovation sector was largely excluded from the analysis, although it will of course play the most important role in the coming years. In principle, the definition and operationalization presented here can also be applied to the refurbishment of existing buildings and, as initial studies show, can sometimes be achieved. However, it is clear that especially the PED Alpha system boundary with the relatively high implicit requirements for energy efficiency and local renewable energy production will not be easily—if at all—achievable for all existing quarters. Here, apart from the building density, the credit must also be examined and, if necessary, parameterized depending on additional parameters such as the building age or the settlement typology.

Although the presented definition can be used for both green and brown field development, there currently is no specific context factor considering potentially lower balance feasibility for refurbishments. As the first two boundaries, Alpha and Beta, do not include embodied emissions, they are easier to achieve for green field developments, whereas the opposite is true for brown field and refurbishments, where the embodied emissions due to materials are lower and thus favor the last definition, Omega. It is unclear as of yet if this distinction is sufficient in theory and practice, as the sample size of refurbishment PEDs is still small and only now expected to expand with the investigation of the second round of JPI UE PED projects.

8.9. Data Availability and Accessibility

One potential challenge of the presented approach is that of data availability and accessibility, as it partly relies on data that are not readily available for all buildings and districts in Austria, especially in brown field developments. Amongst these, the most critical are (1) hourly data on external grid flexibility requirements and normative methods to obtain them, (2) time-sensitive primary energy conversion weighting factors in general, and (3) mobility data. Further research must yield possible data sources, and normative standardization processes must formalize a standardized dataset for certification.

8.10. Possible Implications of the Proposed Definition in Shaping Future Policies

As one of the design goals of the definition is to lead to a national PED certification, it is important to reflect on the possible impacts and implications of the proposed definition. First, the density context of the definition shifts the ambition pressure to the side of low-density developments, which might be a position not justifiable by regulation and legislation. Second, the introduction of a certifiable PED definition with a purely technical character might set wrong incentives for district developments to forego other certifications that have more emphasis on social and ecological assessments that should be used in conjunction with the proposed definition. Third, the rigidity, ambition, and complexity of the framework might deter potential PED districts from pursuing such a standard.

9. Conclusions

PED definition must be understood as a design problem and cannot be detached from the goals and aims to be furthered by it. Indeed, it is only these goals and aims that make the definition useful. The positive energy balance is the unique feature of the PED concept, and these three design choices on boundaries, weighting, and targeting can be used to define PEDs and assess some aspects of energy efficiency, RES, and even flexibility together within the energy balance. Using only a single indicator also lends itself to standardization and certification. Crucially, the inclusion of dynamic balance targets allows a comprehensive link between balance assessment and definition goals. However, such a balance cannot be used to assess all relevant aspects of PED development by itself and needs to be accompanied and complemented by other assessment and certification systems, be it on a regulatory or project level.

In line with its PED definition design goals, this paper introduced a theoretical framework of a quantitative energy balance that can be defined with relative uniformity while still leaving room for contextual targeting. With this framework, it is possible to contextualize PED balance assessments on different frames of regionality, from international to municipal. The paper shows that setting PED balance targets is analogous to quantifying context impacts, which allows for a formalization of PED contexts and can improve comparability between projects and other quantitative balance definitions. This general framework was used to construct three distinct PED definitions that are needed for PED planning and certification practice in Austria: PED Alpha for operation and use, PED Beta for included private mobility, and PED Omega for embodied emissions from construction and mobility.

The effects of this PED definition design were exemplified by seven district assessments that show its feasibility in various green field contexts. The use of a context factor for density allows both district typologies of very high (FAR > 3) and low (FAR < 1) density to achieve a positive energy balance for operation with comparable ambition for energy efficiency, energy flexibility, and onsite renewable generation measures. Furthermore, the use of a context factor for mobility derived as a credit from the surrounding energy system allocated by usable floor area can be used to offset the energy demand induced by including individual motorized mobility in the energy balance. With this, it is feasible for both urban and rural districts and neighborhoods to achieve a positive energy balance with, again, similar ambition in terms of reduced motorized mobility demand and a switch to electric vehicles.

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Abbreviations

AT	Austria
CF	Context factor
DHW	Domestic hot water
DSM	Demand-Side Management
EU	European Union
GHG	Greenhouse gas
GFA	Gross floor area
JPI UE	Joint Programme Initiative Urban Europe
KPI	Key Performance Indicator
MIT	Motorized individual transport
PA	Area of buildable plot or parcel

Appendix A

Table A1. Grid electricity conversion factors.

Month	Primary Energy kWh/kWh	GHG-Emissions $kgCO_2eq./kWh$
January	1.80	0.304
February	1.79	0.304
March	1.72	0.264
April	1.58	0.211
May	1.47	0.167
June	1.46	0.163
July	1.44	0.163
August	1.48	0.167
September	1.58	0.208
October	1.71	0.260
November	1.77	0.282
December	1.79	0.291
Average	1.63	0.231

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Contribution to the thesis:

Conceptual framing of PED definitions as instruments linking European climate objectives to local implementation. Identification of PED definition trilemma and of definition regionality levels. Assigning Method of normative target setting to the national level while preserving project-level flexibility.

Definition of Positive Energy Districts

Cities4PEDs WP2 Deliverable

What is a Positive Energy District?

> *If you can't explain it simply, you don't understand it well enough*
>
> *- Albert Einstein*

Simon Schneider
University of Applied Sciences Vienna
May 2023

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How to use this document

This report is not a step-by-step guide on how to develop a PED. For the most part it is not aimed at stakeholders and developers of local districts, but instead addresses national, municipal and European policy makers and researchers: It discusses the different aspects and approaches of PED definition from a theoretical perspective and highlights the challenges and requirements for bridging the gap to a practical and operational definition.

The document is split into four parts, that contain

1) INITIAL OBSERVATIONS and results from the definition development process itself
2) the resulting CONSENSUS on PED definition within the project consortium, and
3) AN EXAMPLE of a possible national Definition Implementation and Operationalization and example Assessment of the C4P districts according to this definition.
4) FINAL OBSERVATIONS

	Chapter 1 APPROACHING a PED DEFINITION	**Chapter 2** DEFINITION CONSENSUS	**Chapter 3** EXAMPLE DEFINITION and ASSESSMENTS	**Chapter 4** OBSERVATIONS
				
Inside	**Observations on the way to a PED definition** **Challenges** **The definition from different perspectives** **Requirements of a definition**	**Minimum consensus definition** **Common Ground of the three demo sites on PED definition**	**Example assessments with quantitative definition of positive energy balance** **Primary Energy Balance to assess efficiency, local renewables and flexibility ambitious, yet achievable**	**Discussion of observations and outlook**

The following pages give you a more comprehensive look at what you will find in this deliverable.

1: Definition approach

This chapter recounts the approach to the definition development process taken in the Cities4PEDs Project WP 2. It gives an account of the steps taken as well as the reasons behind it and their main findings. This in itself constitute main results of WP2: Exchanging positions and approaches towards PEDs between different Countries, Cities and Projects and investigate the commonalities and differences, as well as determining the reasons for what works and what does not. It presents the heterogeneous practical requirements identified from the districts and compares them with a theoretical framework that should be able to give guidance on how to approach the definition PEDs. This first sections contents are structured as follows:

Sub-Chapter	What's inside?
Identify Goals and requirements of a PED definition	In this first step, ten goals were identified, that any possible PED definition should be entail to be deemed useful from the perspective of the three local PED sites.
Finding Common denominators between countries	Analysis of the differences and commonalities in terms of the dichotomies of "quantitative vs qualitative" and "Energy and Emissions vs Social Aspects"
Splitting the Definition into three regimes of Regionality: EU, National and Local	How to distribute the definition responsibilities between three appropriate levels of regionality to untangle the identified differences between districts while retaining the identified commonalities, in particular of common definition goals. This goes in-depth on the problems of definitions that remain to vague, as well why definitions must not be too specific without appropriate field testing
Communicating and using the PED definition on the ground	This chapter gives insights into the requirements and problems that the project stakeholders individually identify when dealing with the PED definition. This gives insights into the main pain points and threats to the implementation and adoption of PEDs as well as the perceived opportunities when using a PED concept on the ground

2: C4P PED Definition

The second part of this deliverable summarizes the formal PED definition as developed in the Cities4PEDs project:

Sub-Chapter	What's inside ?
Definition Goals and Requirements	1. The PED definition should support high quality implementations. 2. PED definition takes regional context and potentials into account. 3. The PED definition should cover all relevant phases but not set strict requirements for the process itself. 4. The PED definition should make PED projects comparable transnationally. 5. The PED definition focuses on the measurability of quantitative and qualitative targets. 6. The PED definition links district targets to regional/national targets. 7. PED definition enables the implementation in urban and rural areas. 8. The PED definition should allow some form of distinction between newly built and existing districts. 9. PED definition is neutral towards different technology and innovation levels. 10. PED definition takes regional context and potentials into account.
Common Definition Aspects	Positive Energy Balance System boundary includes: Operation and Plug loads of the Buildings, as well as mobility Weighting System: Nationally defined conversion factors (Energy End use, Primary Energy, Carbon) Balance Target includes context factors context factors climate, heritage, flexibility
Different and Open Definition Aspects	• Social and other qualitative Aspects (if and how to include them) • Weighting system: Which system should be used? • National definition of "onsite RES" • Quantification of Climate context factor • Quantification of Heritage context factor • Weighting Factors depicting grid-supportivness

3: Example Definition Operationalization and Assessment

The third section of this report is dedicated to showing how a theoretical framework can be used to draft a fully operational and assessable PED definition that achieves the previously identified practical goals and requirements. It is in the nature of this approach that this definition is targeting a single nation, in this example Austria. It also includes a comparative assessment of the three PED sites adhering to this definition.

This can be used as a blueprint to draft similar definitions with enough flexibility to account for national, regional and project specific contexts while still adhering to the same principles – an important feature, as will be discussed later.

4: Observations

Finally, main discussion points are summarized:

DISCUSSION	⌂ A common definition between projects satisfying all stakeholder needs is most likely not feasible ⌂ A common definition might only be necessary on a broad EU level without operationalization ⌂ Social and other qualitative aspects might be better defined and assessed by other means such as certifications, standards and city planning instruments



1 Definition Approach

The European Union wants to create 100 Positive Energy Districts (PEDs) by 2025 as part of its Strategic Energy Technology (SET) plan. The EU has set out a broad framework definition to describe what PEDs are:

"Positive Energy Districts are energy-efficient and energy-flexible urban areas which produce net zero greenhouse gas emissions and actively manage an annual local or regional surplus production of renewable energy. They require integration of different systems and infrastructures and interaction between buildings, the users and the regional energy, mobility and ICT systems, while optimizing the livability of the urban environment in line with social, economic and environmental sustainability."

This gives a first idea of the objectives and the way in which such a PED can be realized. But in order to apply the definition, it is necessary to make the various aspects of it more concrete.

Cities4PEDs is one of the four research projects from the first JPI Urban Europe Pilot Call focusing on PEDs. The consortium consists of municipalities, experts, research institutions and civil society organizations from Brussels, Stockholm and Vienna. With this consortium we aim to contribute to a unified PED definition on a European level from the perspective of our own local contexts. As a consortium, we consider a PED as follows: it is the process of transformation or implementation of a district by means of instruments, tools, methods, collaborations, etc. towards ambitious objectives on the level of a positive energy balance, energy efficiency, energy flexibility, integration between systems and infrastructures, integration between users, livability, social sustainability, economic sustainability, environmental sustainability, etc. The definition is therefore twofold: it concerns both the process of transformation or implementation and the objectives themselves. In our research, we will examine both aspects. The methods, tools, instruments and collaborations needed to operationalize PEDs are investigated and structured in the PED Atlas, based on a series of cases. The objectives, criteria and targets are put central in this document.

The development of a common PED definition was undertaken in a collaborative process. Within work sessions with the whole consortium, an outline of the goals and relevant aspects were screened, discussed and sorted out. This was developed further by setting two different workgroups around two specific sides of the PED definition: the technical aspects and the process and social aspects. In this way it was possible to use everyone's expertise as efficiently as possible and have a more in-depth way of working on these specific aspects of the definition. In the technical deep dive sessions, it has been possible to start working on the nitty-gritty part of the definition, while other partners outlined how the definition could also include social dimensions and a step-by-step approach in becoming a PED enabling local processes of transformation. Newt to that, partners within our consortium joined the EU discussions and EU alignment task force on the PED definition. We harvested the resulting observations in this document.

1.1 Identify Goals and requirements of a PED definition

The PED idea can be interpreted and subsequently conceptualized in any number of ways. It then follows that the definition should cover all relevant aspects to that conceptualization.

With the definition development we started in the same way as in an actual district development: With the goals. The guiding question of the process was:

- What does the PED definition need to do?
- What do we want to achieve with it?
- What is it supposed to ensure, promote or prohibit?



Figure 1 Schematic approach of the development a PED definition in the Workshops (WP.2.2)

After clarifying that we can set out to develop a framework that delivers just that. As can be seen in chapter 2.1 however, the goals and requirements raised at the definition were quite extensive and heterogenous, to the point that it became infeasible and contradictory. For example, a definition required to include precise calculatory prescriptions can still be also designed in such a way that it is easy to communicate. But adding the requirements that the definition should also be attainable with similar effort in different climates and densities ban be unreconcilable. This can be illustrated with the following trilemma of common definition requirements: simplicity, determinism and achievability: Achieving 2 of 3 is straightforward, but all three simultaneously seems difficult.

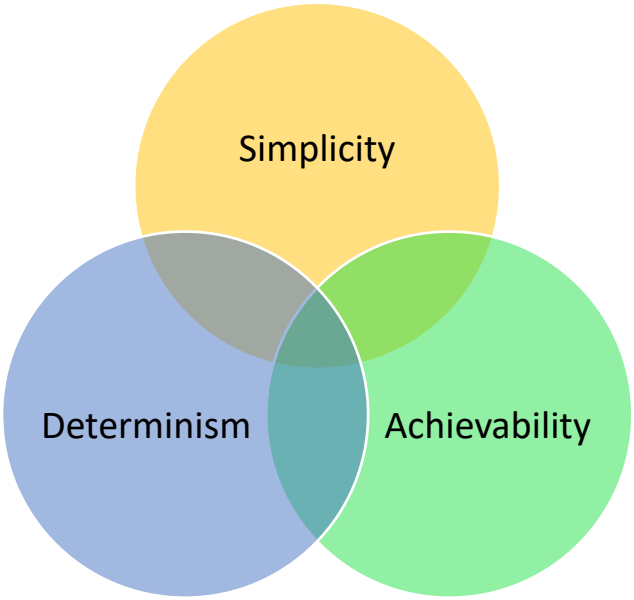


Figure 2 The PED Definition Trilemma: Simplicity, Determinism and Achievability

Table 1 Example PED definition framings for pairs of definition dimensions

Combination	Example Framing	Resulting Problem or Challenge
Simplicity + Determinism	"A PED has a Positive Energy Balance for its operation over a year"	Might not be achievable for dense urban districts compared to rural areas, very ambitious, but dense projects do not achieve this arbitrary balance
Simplicity + Achievability	Each district defines their own goals with their own process. "It's about the journey now, not the foggy destination"	No frame of reference for the existence and sufficiency of actual, quantifiable goals such as climate neutrality and necessary emission reductions lead to relative improvements while collectively failing
Achievability + Determinism	Rigorous technical definition of balance components with complex calculatory offsetting mechanisms (such as context factors) to quantify effort sharing on societal climate goals	Requires more models and assumptions that in turn need to be linked and calibrated. Not simple to communicate and requires more effort to operate

This constitutes a crucial problem of the PED definition: How to solve this Trilemma? Or, atleast, where do we position ourselves in this Venn diagram and why? What is possible? What is practicable?

1.2 Finding Common denominators between countries

In the development of WP2 Definition the consortium originally set out to find a common definition of what a PED actually is that the three demo cases in each city can adhere to. This process culminated in the observations and results on the ***purpose and goals of the definition***, which were also published as a distinct whitepaper.

However, it was gradually becoming clearer that a PED definition needed to accommodate the differences between the districts – both physical and temporal. To this end, a strictly common and unified definition was deemed either inadequate or outright impossible to devise without concrete examples of operationalization. Instead, it was accepted and anticipated that ***PEDs would possibly need different definitions in different contexts.***

1.2.1 Commonalities

The first step, then, was to identify the commonalities between these contexts present in the C4P project:

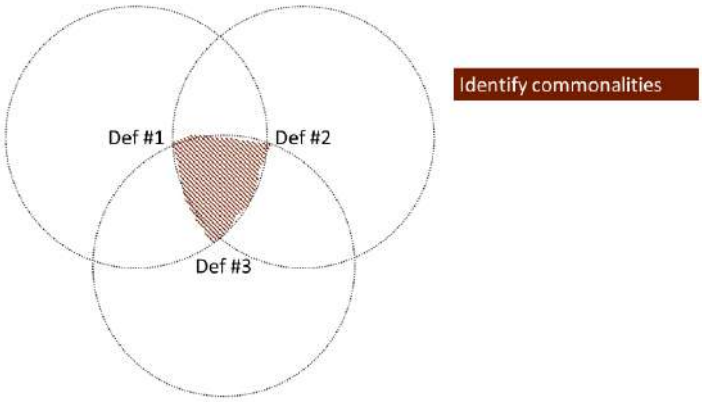


Figure 3 Different PED Definitions can have commonalities but also diverging aspects

These commonalities would form the core of the C4P Common Definition and included the determination of the following

- Key Performance Indicators
- Included Energy services

Also, among the consortium, several indicators on a technical level were brought forward as relevant, among which: (i) energy demand and consumption, (ii) renewable energy sources local production, and (iii) GHG emission reductions. During the discussions with the Brussels partners regarding the social and organizational indicators, the ones brought forward as most relevant are: (i) stakeholder engagement/co-ownership, (ii) funding instruments, (iii) financial impact on the inhabitants (which was not initially in the catalogue), (iv) improved quality of life (which was not initially in the catalogue).

1.2.2 Differences

At this stage it has become clear that there are sizeable national differences that would need to be addressed in the definition, but independently. The main areas of these differences include but are not limited to:

- Norms and Standards regarding Calculation of KPIs such as Energy End Use, Primary Energy Balance, GHG Emission Balance
- Energy and Emission Conversion Factors: Allocation and accounting schemes have a great impact on the resulting balance, but vary significantly as they reflect different political agendas and landscapes
- Impact of climate in the forms of heating demand, available solar radiation for renewable generation, etc.
- Types and quality of established instruments on
  - Spatial Energy Planning on a district scale
  - District development process criteria (e.g. through building code or certifications)
  - Qualitative criteria (e.g. through certifications, labels, municipal planning, etc)

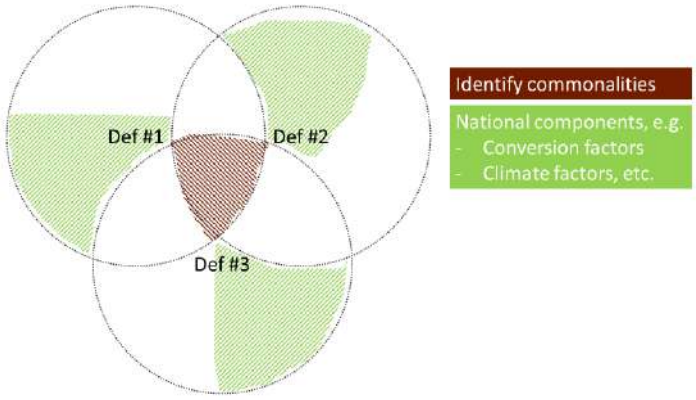


Figure 4 National definition differences are caused by national differences in operationalization and gaps in existing tools

On top of these national differences, there is a further layer that distinguishes regional and municipal differences that might require different definition considerations:

- Connection / compatibility with city instruments / processes and targets on emissions, participation, affordability, etc.

These can also result in individual aspects only relevant to a certain project.

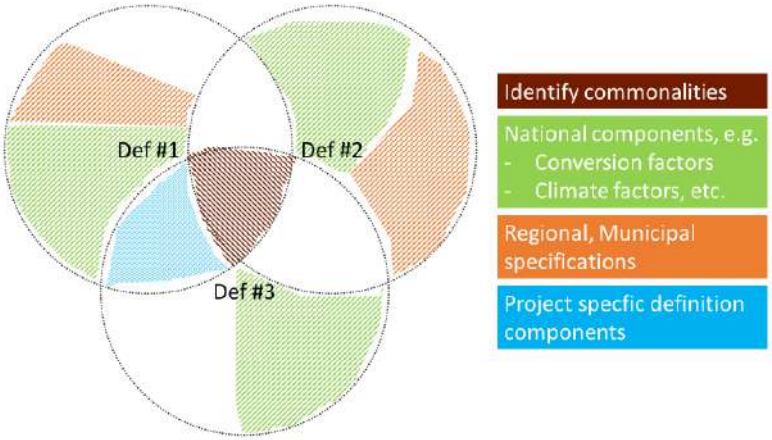


Figure 5 Definition aspects can include common (brown), national (green), municipal/regional (green) and project specific (blue) aspects

As the goals and motivations behind the usage of PEDs vary between stakeholders of different background, so does their requirements for the content and form of various definition aspects, again in terms of simplicity, determinism and achievability. This also means that when talking about the "PED definition" we are actually talking about definitions on at least four different levels: A definition that holds for all PEDs in Europe, one that defines all PEDs of a Member State, a region or municipality and finally a level of definition only applicable to a specific district.

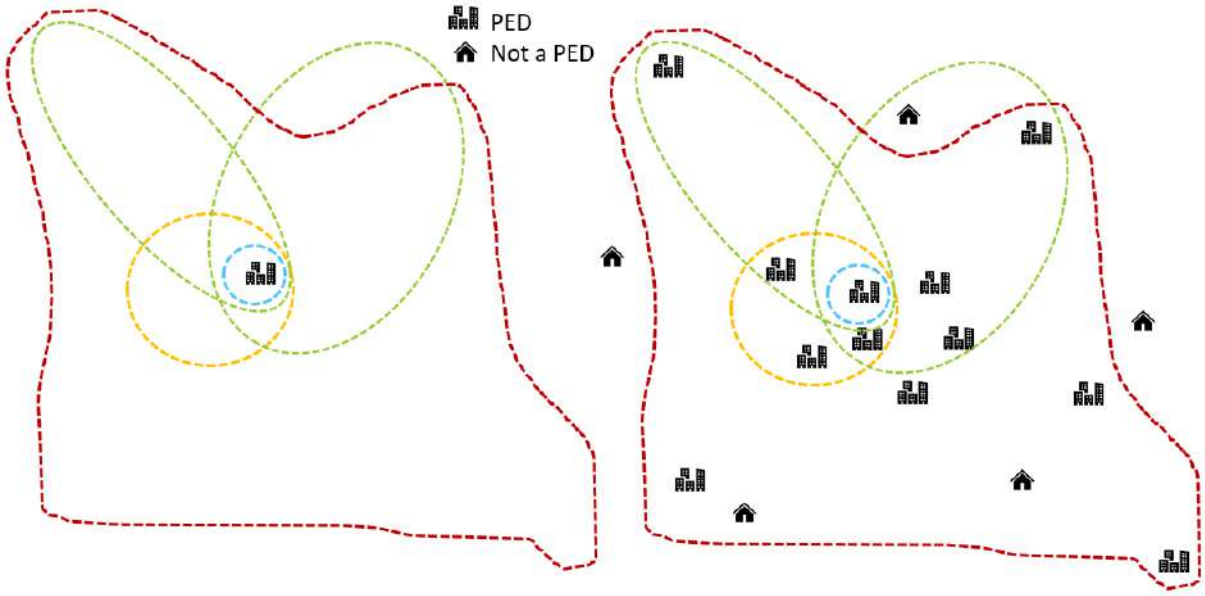


Figure 6 PEDs and other districts separated by various definitions (dashed lines)

The above figure shows dashed lines representing different possible definitions. The Icons represent districts that should or should not be considered PED. This further depends on who you ask. And explains why different definitions cover different districts as they are typically tailored to them: A single district can of course adhere to and fulfil different yet intersecting definitions. But as the number of districts that a definition should cover grows, so does the aforementioned trilemma of simplicity, achievability and determinism. The above schema illustrates this: The dashed lines represent possible definition designs. Their simplicity can be described as their "roundness", their achievability as their area (as it is more achievable to be inside a larger enclosed space) and their determinism by how many non-PEDs can be found within "by accident". With a single District it can be very straight forward to design a suitable

definition. But as the number of districts that should fit within the definition grows, the trilemma becomes visible. But it also hints at a possible solution: Just use more than one definition to account for the different definition needs at an appropriate place.

Note how this is different from saying: "Every PED designs their own definition themselves". This *can* be a part of a definition, but it will not be the *only* one; there will also be other definitions on European and most importantly national and municipal level, that the PED needs to adhere to. This means splitting the definition into four different definitions of increasing determinism, complexity and regionality, that each district need to fulfil simultaneously: *European*, *National*, *Municipal/Regional* and *Project Specific*. This allows for a minimal common denominator between all European PEDs while still ensuring that more localized requirements can be detailed, assessed and enforced.

1.3 Splitting the Definition into three regimes of Regionality: EU, National and Local

It becomes apparent that the struggle towards a PED definition is rooted in the apparent divergence of two if its main goals: uniformity and locality which leads to problems when designing a common definition:

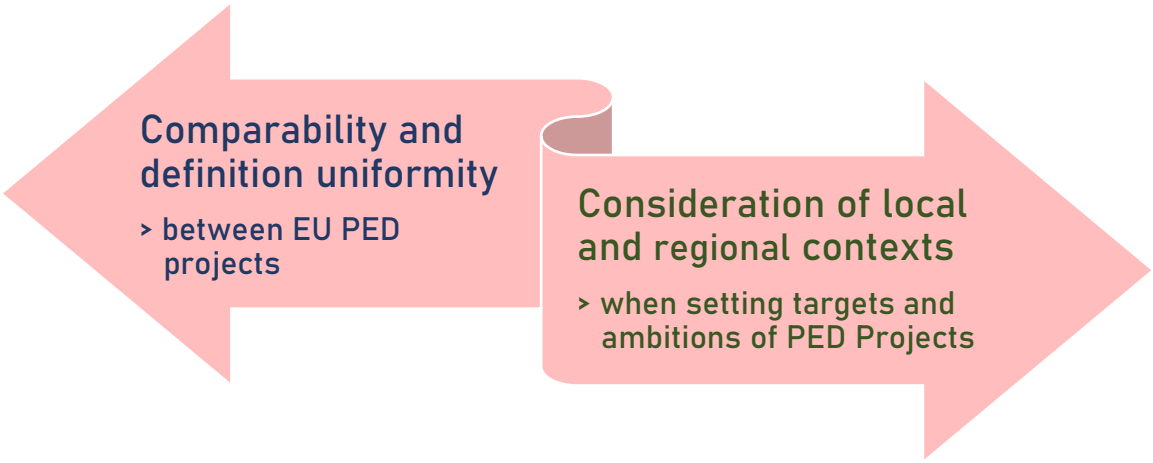


Figure 7 The divergence between two main goals: Uniformity and Locality

Analysis of the definition process within the project as well as in exchange with the other JPI UE projects furthermore showed that the results of the definition efforts are arbitrary to the extent as to which stakeholders and perspectives are most well represent. What a PED is, will always be a matter of interpretation, but it remains to be decided for *whose interpretation* that is.

In C4P we try to avert this "di/trilemma" by looking at all definition aspects that have been raised and try to compartmentalize them into different groups of regionality, that seem most suited to define this particular aspect with sufficient accuracy and detail. This means that we spread the definition aspects across multiple layers of different size from individual project level all the way to a pan-European level. The *definition items* and *aspects* can then be addressed to varying depth on different levels. This is an attempt at adding nuance to the current dichotomy between the definitions encompassing all EU PED projects[1] which must be very general in nature and project specific definitions that are partly operationalized in great detail, but can typically not be generalized beyond its specific project context. To

[1] Such as the attempt from JPI UE

put it more concretely: The proposed definition is divided into different *aspects* which in turn are defined to increasing degree of concreteness at every level from European to national to regional to municipal to project specific.

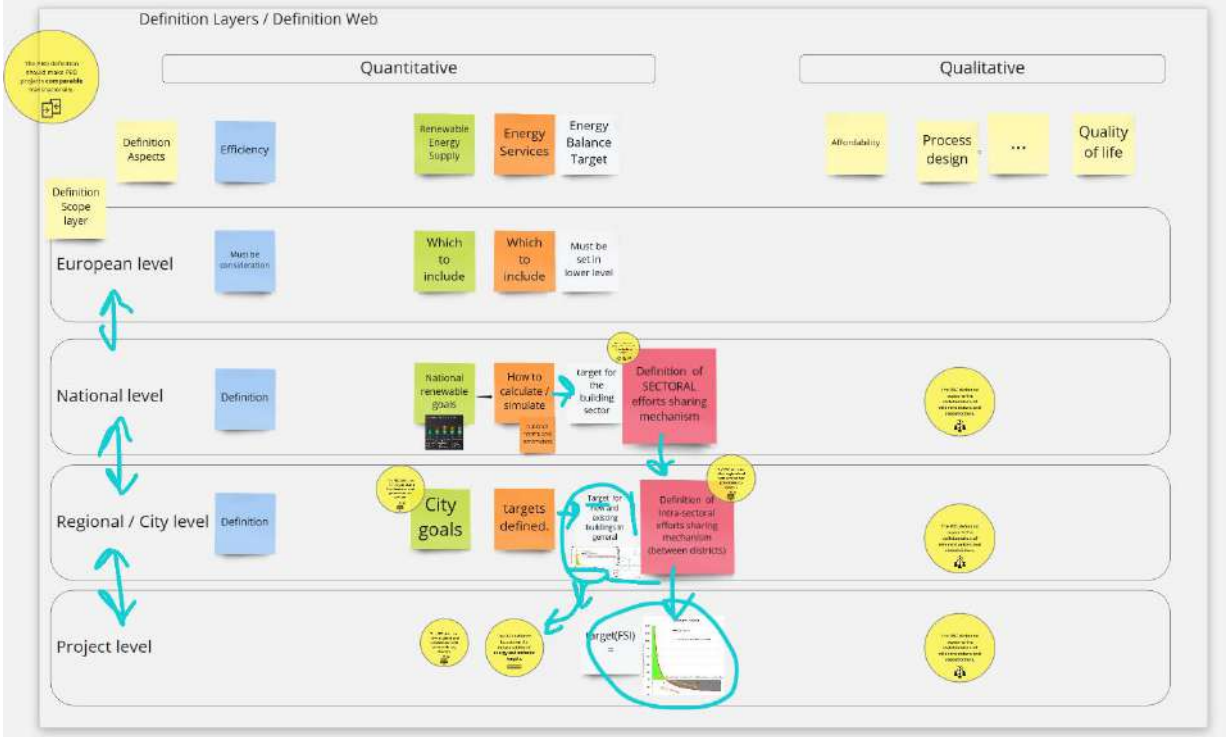


Figure 8 Schematic of definition aspects and "context factors" being defined at different levels

A consequence of this is that at each level the definition content and detail changes: From vague mission statement and intent at the European level to operationalized standardization scheme on a national level to a monitorable and communicable implementation plan on project level.

The advantage of this approach that it allows *for as much cohesion as possible* by pulling every aspect that turns out as common between projects as high up as the European level as necessary. And at the same time allows for *as much context flexibility as necessary* as projects seem fit to implement their local PEDs.

It also is clear that the mediation of national stakeholders such as funding agencies and standardization bodies is required to separate the PED definition aspects into nationally common and project individual aspects. It is ultimately up to them to decide how open or narrow the definition should be developed in the respective country.

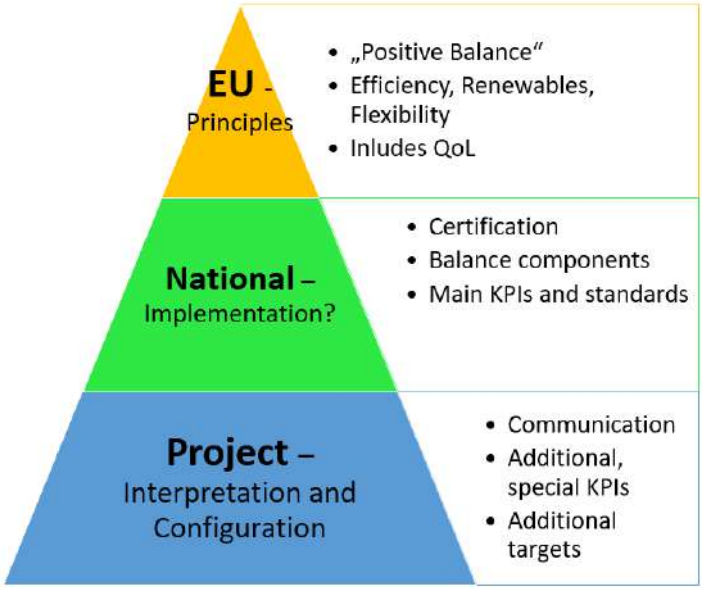


Figure 9 Definition aspects split into three layers of regionality

This becomes necessary due to different adjustment requirements of PEDs in different countries, cities and districts to ensure feasibility in different *densities, heritages and climates.* It is however important to decide which level is responsible for which context factors.

Example: A urban density context factor should be defined at **national or regional level**, but not uniformly at EU level to allow for country building density differences (e.g. Sweden vs Belgium), and also not on district level to avoid "cherry picking". If e.g. Urban Density context factor definition is indeed decided to be national responsibility, every country could employ their own suitable method of calculation their factor, including *not* using it at all. However, Regions / Cities and Districts would not be allowed to define their own context factors anymore.

1.4 Communicating and using the PED definition on the ground

There is a divide between the concerted efforts for definition development that adheres to as many goals and requirements on the one hand, and on the other hand being manageable and communicable, or just plain useful, for the actors involved at the ground, actually implementing said PEDs. Their requirements to the definition are much more focused on practicability, assessability and communicability. The following table gives a broad cluster of needs and their most likely associated stakeholders:

Table 2 Main definition requirements

Definition Requirement	Important for Stakeholder
Scientific rigidity	Scientific community, Practitioners
Certifiability	Project Developers, Administrations
Achievability	Project Developers, Governmental Bodies, Practitioners
Comparability	Transnational and national stakeholders: JPI UE, European Union, Researchers
Comprehensibility	Governing bodies, project developers, users

Since a PED definition needs to satisfy all requirements, it is paramount to fully understand them: where they originate, what their purpose is and if and how they can be compartmentalized into different regimes of responsibility. The following results therefore show the consortium's own perception on key questions regarding the usefulness of the PED concept and the PED definition at the last half year of the project.

Table 3 Reflection Questions on the PED Concept and the PED Definition from the Perspective of the C4P stakeholders

Area	Question	Purpose
PED Concept	**How or why is it useful to you?**	This shows, what the goals and possibilities of the PED concept are, maybe already partly realised. It also aims to identify the perceived *requirements* for its usefulness
	Threats to usefulness	The last two questions can be seen as somewhat complementary. The difference in threat perception and usefulness requirements can be seen in the relative risk of it failing. Greater focus is therefore warranted for the threats instead of the more easily attainable requirements
	Biggest Pains in use	The last two questions can be seen as somewhat complementary. The difference in threat perception and usefulness requirements can be seen in the relative risk of it failing. Greater focus is therefore warranted for the threats instead of the more easily attainable requirements
PED Definition	**Requirements and reasons behind them**	Get a better understanding of the importance and purposes behind the definition requirements

The questions were posed in an mutual interview setting with 24 responses from the expert fields of the C4P consortium. The following word cloud gives a qualitative impression of the present occupations and roles:



Figure 10 Background and Role of the C4P stakeholder participants in the reflective interview [2]

The following sections summarize the answers and insights into the posed questions:

[2] larger words have been named more often

1.4.1 Usefulness of the PED concept

The first question was split into two possible answers, that indicate the usefulness ("because…") and requirements ("if…"). As can be seen in the following graph, the usefulness of the PED is perceived to lie in two distinct fields: The communication of various PED aspects and the promotion of certain key activities and goals:

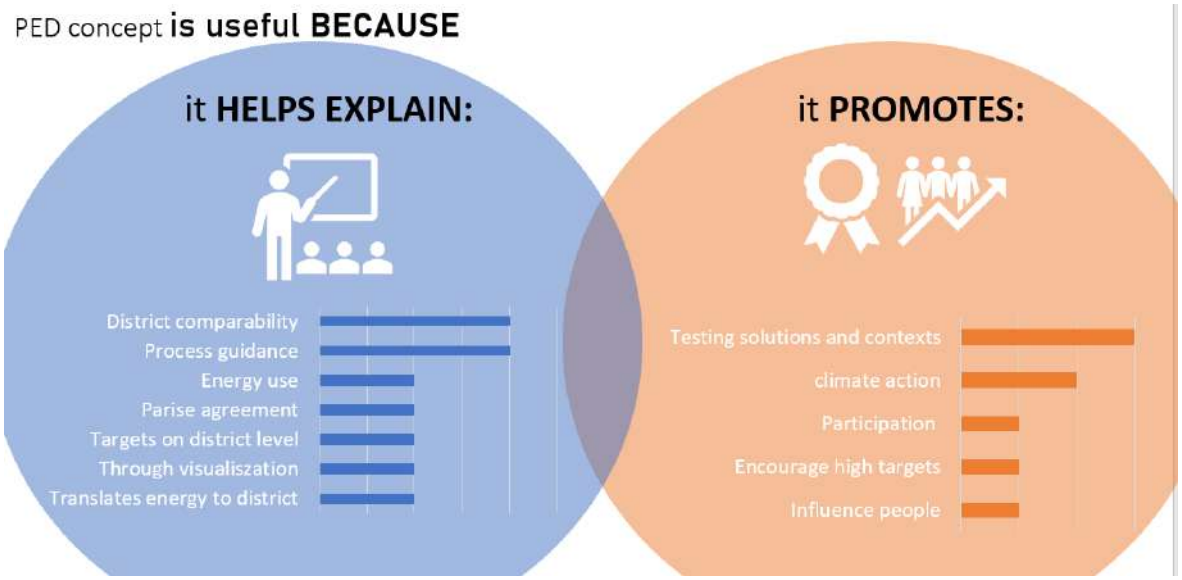


Figure 11 The two main uses of the PED concept in the eyes of the C4P consortium

1.4.2 Requirements to the usefulness of PED concepts

A reflection on the requirements for the PED concept itself – independently of the question of a PED definition – yields interesting results. As the figure below illustraties, the answers can be clustered in 4 general areas of requirements to the PED concept:

1. First are specific goals
2. Requirements to the PED targets themselves
3. Standardization and (inter)national recognition
4. Communicability to important facilitating stakeholders



Figure 12 Requirements for the PED concept in the eyes of the C4P consortium

Interestingly, there is a strong sense that the PED concept itself is not useful without simple, concrete and concise, yet attainable targets.

1.4.3 Usefulness of the PED definition

Usefulness of the PED definition can be divided into two main categories: The description of what a definition ***is*** and what the definition ***has*** to be useful, which can be roughly seen as definition *attributes* and definition *requirements* respectively:

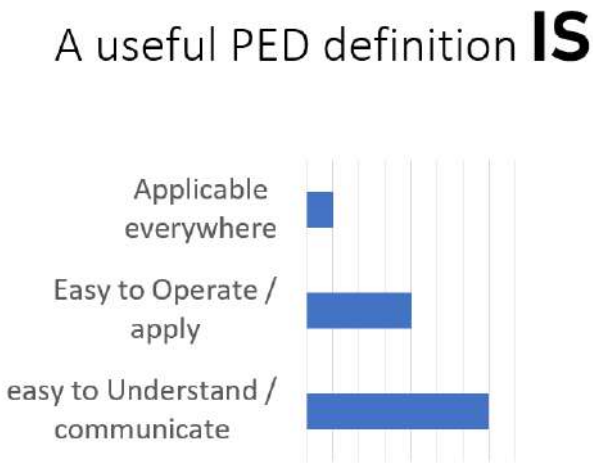


Figure 13 Main Attributes of a useful PED Definition n the eyes of the C4P consortium

Here it can be seen that the main focus of the involved practitioners is a*pplicability* and c*ommunicability*. This shows a rather cohesive outlook on the definition. When it comes to the requirements and attributes a definition should *have,* there is more diversity but nonetheless a clear picture.

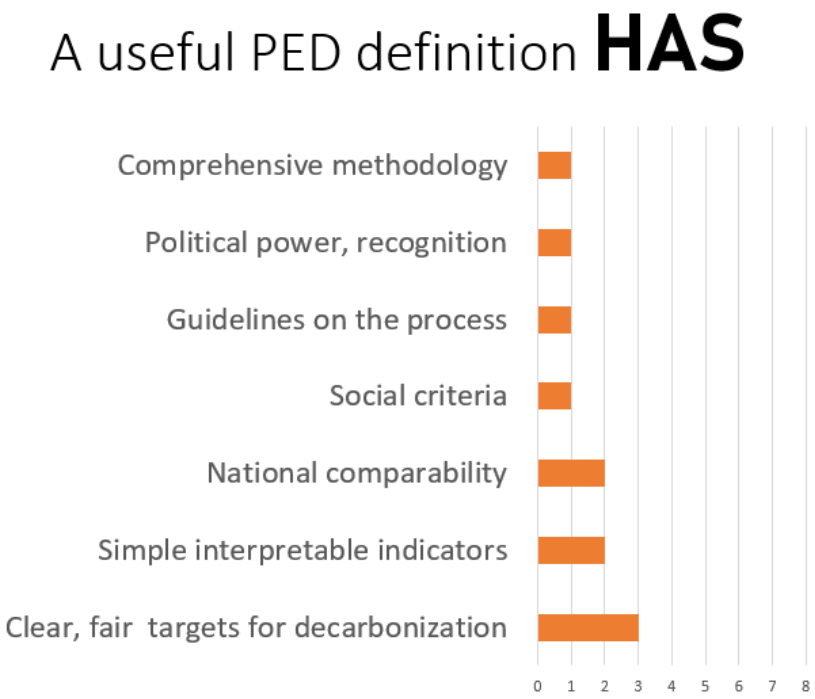


Figure 14 Main Requirements of a useful PED Definition n the eyes of the C4P consortium

What is interesting here is in particular what does not show up: Note that from the practitioners view, the definition does not necessarily has transnationally common features. It does not have unified specifications on system boundaries, energy balancing systems or the like. Although this is not particularly surprising, it is important to keep in mind that local stakeholders have very different requirements and from transnationally and nationally operating actors. The PED definition needs to be able to reflect and address these different needs.

1.4.4 Threats to the PED Concepts

The last two questions of the interview can be seen as inverts of the questions before. It is intended to identify the biggest threats and most pressing concerns, that should be addressed moving forward:



Figure 15 Main Threats to the PED concept in the eyes of the C4P consortium

The risks to loosing support for the PED concept are seen in several different areas, such as *lacking ambition*, but at the same time lacking *general achievability*. These can potentially be diverging, if not opposing threats, similar to the *lacking speed of the definition process* and the *lacking clarity, precision*. Ultimately, this snapshot is reflective of the diverse group of stakeholders and their diverging interests with the PED concept in the C4P consortium.

When it comes to the greatest perceived pains with PEDs currently, the majority of stakeholders identify two main aches:

1. The Vagueness of the PED concept and the amount of undefined parts and
2. That they don't understand what a PED exactly is themselves.

Some of the other mentioned pains can also be seen as more concrete or symptomatic pains rooted in one of the above causes:

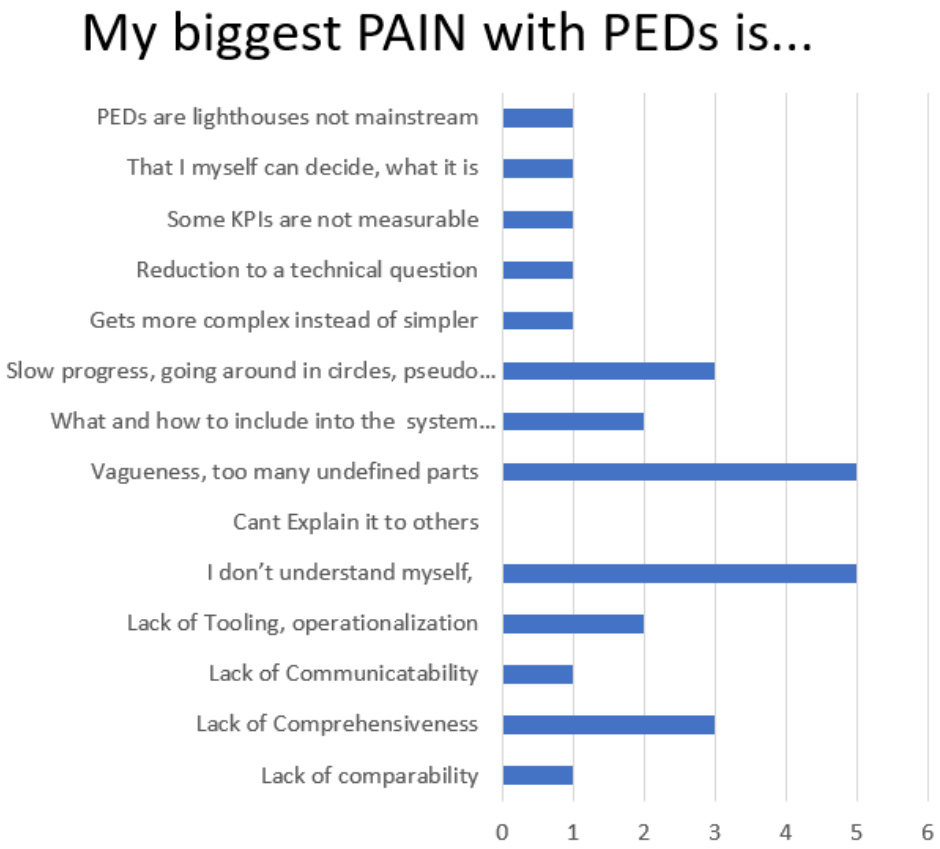


Figure 16 Main Pain points of working with the PED concept in the eyes of the C4P consortium

Despite their statistical insignificance, these results nonetheless give insight of how key people involved with PEDs feel about the most pressing issues: It is probably safe to subsume that this research design setting bringing together experts and stakeholders from all areas of PED research, implementation and support is not equipped to adequately address the definition question by themselves.

As pointed out before, it becomes transparent how different stakeholders value different properties of PEDs for a variety of reasons. This carries over to their perceived PED definition requirements – which diversify quite extensively as a consequence. This is a rather remarkable observation, given that it could be argued that many concept benefits could be obtained without having to explicitly define them. Many of the perceived benefits of the PED concept of communicating and mobilizing ambitious projects for example would arguably be still valid, if the definition was not easy to operate or communicate, but the concept still was, as they are two different domains with different stakeholders involved. Instead, the fact that the PED concept and the PED definition are seen in this close interlink could possibly have more to do with the questionnaire setting putting them in proximity in the first place. That, and the fact that everybody in the project team was questioned on the topic of the PED definition, regardless of whether their role would actually put them in contact with it or not. This further highlights two important unanswered questions:



Which stakeholders are exposed to which part of a PED definition?



Who is responsible for designing the interfaces between different parts of a PED definition?

This could be connected to another perceived missing key: This are stakeholders that have a mandate to define the legal frameworks such as municipal and national building codes. Such an authority was felt missing in addressing all PED definition requirements in their entirety.

2 Commonalities in PED Definition of the C4P Partners



The following chapter describes the commonalities and consensus on common goals and aspects of a PED definition. A common operationalization is not part of this picture, but an example operationalization that adheres to these principles can be found in the later chapter 3. The extent – or rather the lack thereof – of concrete definition aspects and how they are to be operationalized in three different countries and cities is a testament first and foremost to the differences of the district themselves and the purpose the PED concept is envisaged to play in their development. It is perhaps one of the most important insights of this project, that they cannot yield unified and operationalizable definitions across a wide range of district applications. Even though the need for such a definition from the practitioner's side – as shown in chapter 1.3. is undeniable, the focus of the C4P research project was in exchange rather than unification. This is also indicative of the call structure with a limited number of demo cases comparing and exchanging results and methods.

As such, the obtained definition is lacking and too vague for some and too tight and irrelevant for others. Nevertheless, it underpins the common ground of understanding and ambition, and hints at possible implementation solutions down the line, that – nevertheless – have to be developed in a more thorough - and most likely nationally individual – process.

Table 4 Overview of common definition Aspects and design decision

Definition aspects	Design decision	Unit
Spatial balance boundary	Geographical district boundary	
Temporal balance boundary	Annual balance	
Functional System boundary	See below	
Indicators	**Total primary energy balance**	[kWh/m²a]
	Green house gas emissions	[kg/m²a]
	Life cycle differential costs	[€/m²30a]
Energy Balance Target (over spatial and temporal system boundary)	**Positive**	
Context factors to be included in the Energy Balance	**No common design decision, to be decided individually per PED Site / Country**	

2.1 Definition Goals

The question we start with is: why do we need a more concrete definition? In the end, every contribution made to the energy transition is a step in the right direction, and it is up to each country or city to draw the lines for such a policy. Then, why is it necessary to set criteria that are the same for the whole of Europe? From the perspective of city administrations and politics, the need for such a unified definition is underlined. In fact, it will be a tool that allows and pushes for high-quality implementations at the district

level and serves as a lever for capacity building by including indications on the development process and a defined step-by step approach.

At the same time, and because it is a Europewide definition, it should make PED projects comparable transnationally. It therefore allows for the measurability of the set targets, and further for the connection of district targets to the supra-local (regional and rational) targets. Coming from different geographical backgrounds, and not wanting to exclude PEDs based on their location, the definition should be applicable in different types of areas, such as urban and rural areas, and existing as well as newly built areas. In order to achieve this, it is important that the definition enables the collaboration of relevant actors and stakeholders to have a broadly supported PED with more chances of success, and covers all relevant phases of a PED development, from setting up the process until after realization.

2.1.1 Common Criteria

The following illustration shows the ten main goals or criteria identified for the C4P common definition. They illustrate core concerns of the consortium and were later refined into a whitepaper on common goals of a definition, which is available as a separate deliverable.

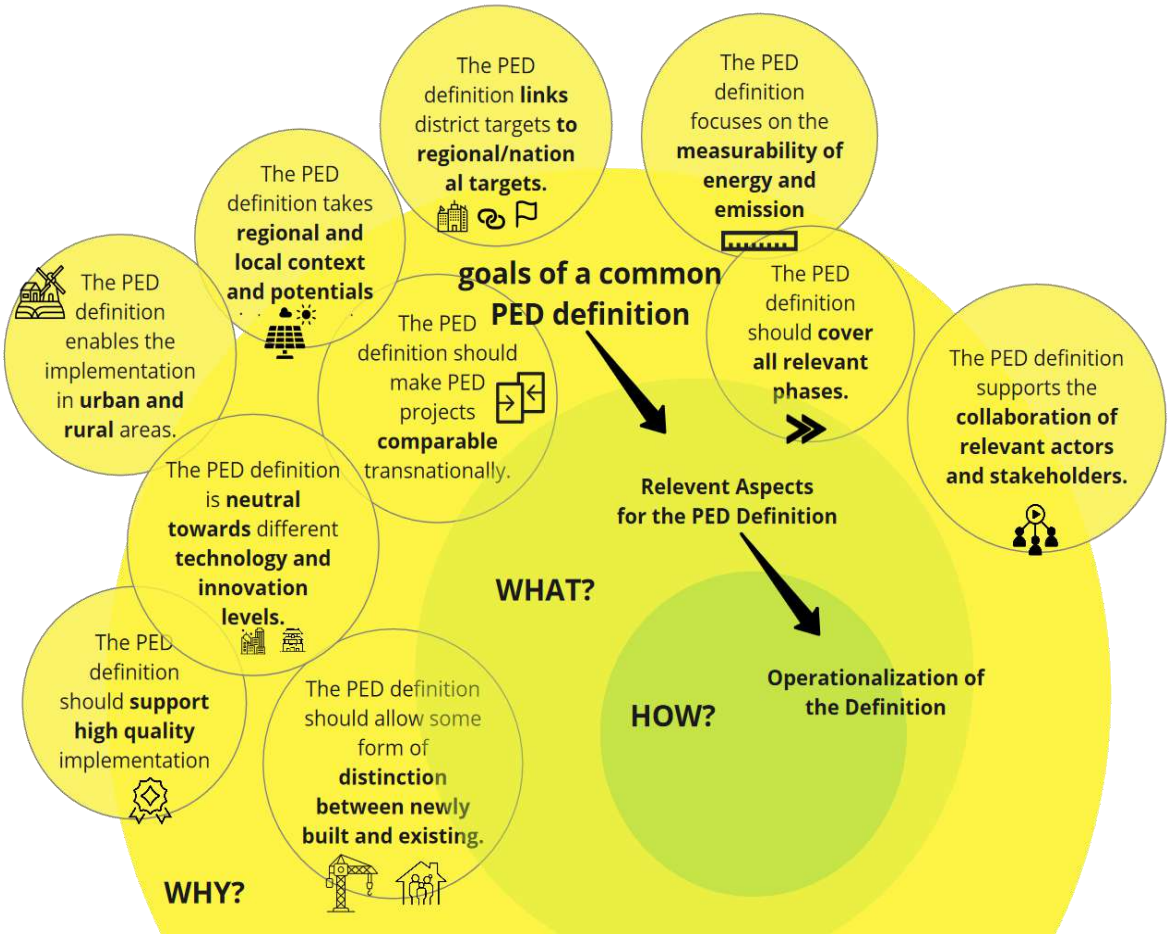


Figure 17 Main Goals and Criteria for the PED definition from the perspective of the C4P consortium

Based on this preliminary identification of PED definition goals, the following five goals were agreed to be most important to the Cities4PEDs consortium at large:

The PED definition should allow and push for high quality implementations.

The more sharply defined criteria for a PED should allow districts to push for high quality transformations and developments. In order to achieve the ambitious targets by 2025, 2030 and 2050, our districts will have to meet far-reaching requirements. The PED criteria provide a framework for cities and their partners to make this quality measurable. On this basis, cities and others can award a PED label and encourage PEDs financially, legally, etc.

The PED definition should translate national or regional goals to a district level

The framework of PEDs will allow the translation of European, national and regional targets into measurable objectives at the district level, which will then lead to a feasible project scale. In this way, the often vague, large-scale objectives are broken down into achievable projects. The criteria should therefore take into account the specific national or regional objectives.

The PED definition should make PED projects comparable transnationally.

One of the main reasons for a shared definition on an EU scale is that local, context-specific aspects of districts can be compared. We can measure whether districts perform high in comparison to each other, but also this can allow to build a basis for exchange at the European levels on learning and success factors around the PEDs implantation, initiating process of acceleration. To do so in a meaningful way, we need to go beyond a "one-size-fits-all" approach.

The PED definition should make PED projects in different contexts comparable

It is very difficult to compare the development of districts in different contexts. In rural areas, for instance, much more open space is available for energy production via wind, whereas in inner-city districts, more energy is needed and far fewer options exist. The shared criteria for a PED definition would allow the deployment of these different contexts to be compared in a meaningful way

The PED definition should enable cities and stakeholders to initiate and engage in the cooperation towards PEDs

A shared PED definition can also be very mobilizing towards a broad group of stakeholders. A European definition provides a framework for a process of cooperation and stimulates local commitment. It is a recognition that can attract different actors to participate and it gives the process a degree of credibility.

Definition Non-Goals

Non-goals were not identified and collected in a structured way, but a number of concerns and non-goals popped up from various stakeholders such as:

- Unified Definition – is it necessary or desired? And if so which parts are unified?
- Economic considerations

2.2 Common Definition Aspects

Based on the above-mentioned definition goals, the consensus was to define a PED through the achievement of a positive energy balance. The following common determinations regarding system boundaries and considered energy services have been made:

2.2.1 Considered Energy services

The energy demand for room conditioning (heating and cooling), domestic hot water, lighting and building services as well as the energy demand for living, working and services (e.g. appliances, computers) are taken into account in the PED definition.

Table 5 Considered energy services in different boundary definition with related assessment. The energy balance calculations is carried out in hourly steps.

	Energy services		Possible assessment method
Building Operation	Heating	✓	Dynamic building simulation methods, stationary monthly methods, or forecasting methods
	Cooling	✓	
	De-/humidification	✓	
	Ventilation	✓	
	Auxiliary power for heating system	✓	
	General electricity & elevators	✓	
	Lighting	✓	
Industrial Operation	Process heating	×	Not included, as energy is used to serve a wider range than the district itself and should therefore also share the burden of energy supply
	Process cooling	×	
	Electricity demand for production	×	
	Electricity demand for general building use and services	✓	Hourly load profiles are needed

2.2.2 System Boundary

Within Cities4PEDs the ***functional system boundary*** was decided on district Operation and user electricity aka plug loads. There was agreement to *not* include the embodied energy and emissions in the functional system boundary for two main reasons:

- Required effort for LCA assessments
- The notion, that balance could not be positive when including embodied energy. (This could be addressed with appropriate context factors, as exemplified in (Schneider et al., 2023)[3])

[3] S. Schneider, T. Zelger, D. Sengl, und J. Baptista, „A Quantitative Positive Energy District Definition with Contextual Targets", *Buildings*, Bd. 13, Nr. 5, Art. Nr. 5, Mai 2023, 10.3390/buildings13051210.

For the *spatial boundary* we propose geographical boundary of the district up to building height and down to 300 m below the ground. This is a rather arbitrary distinction and should not make or break the system. In fact, it is of lesser concern than defining the functional system boundary, and can therefore be left for individual project definition.

One important rule however should be that drawing the *geographical system boundaries* must not make it harder for other districts to later also become a PED ("Gerrymandering")

2.3 Existing Definitions: Are they suitable?

In an effort to find suitable definition approaches for the identified definition design goals, a broad literature review was conducted. The focus was to cluster and compare different PED definition approaches along the following characteristics:

Table 6 Description of analyzed characteristics of PED definitions found in literature

Characteristic	Description
Definition Designation	Name of the definition / concept
Focus and Approach	Interpretation of the main focus compared to the other identified methods
Target/Indicators	Which and how many KPIs are used?
Target Values Characteristics	Are there sufficiency requirements or targets defined for the KPIs in use and if so for which KPIs and which targets?
Ambition Control	How does the definition deal with differently challenging situations, how can a district modulate or even choose its ambition level?
Energy Services in Balance	Which energy services such as building operation, user plug loads, mobility and even embodied energy for construction and maitenance are included in the assessment of the energy balance
System boundary used / defined	How is the system boundary for the energy flows defined? (Physical functional or vitual, or other
Target calculation methods	Is there a calculation methodology for energy balance targets defined?
Temporal resolution	Is there a specific temporal resolution or aggregation period defined, such as hourly, monthly or annualy for modelling accuracy and resolution of flexibility and other timedependent grid interactions
Analysis period	Which period is used for balancing

The results are shown in Figure 18. In a second step, the identified PED definitions were compared with the previously identified PED definition design goals and requirements of the Cities4PEDs consortium. This analysis is presented in Figure 19. It shows that some definition goals can readily be met with most definitions, but there are a number of goals that can hardly be satisfied by any existing definition. The most challenging goals are:

- "The PED definition should make PED projects comparable transnationally": Most definitions do not consider context specifics such as climate or density when setting the target of a positive energy balance. As a consequence, districts of different location and density can achieve a positive balance with very different ambition levels, this making the projects incomparable transnationally from an action and ambition perspective (rather than that of a nominally positive balance).
- "The PED definition links district targets to regional/national targets": This definition requirement is not readily satisfied in any definition except the one specifically designed for it (Schneider et al., 2023)



Figure 18 Characterization of PED definitions found in literature



Figure 19 Compatibility of PED definitions found in literature with the identified PED definition goals and requirements in Cities4PEDs

Table 7 Investigated PED Definitions

	Definition	References
0	JPI Urban Europe Framework Definition	(JPI Urban Europe, 2019)
1	Syn.IKIA	(Salom and Tamm, 2020)
2	Making Cities	(Gabaldón Moreno et al., 2021)
3	ZEN Norway	(Fufa et al., 2016; Wiik et al., 2019, 2018)
4	IEA EBC Energy Positive Neighbourhoods	(Hedman et al., 2021)
5	PED Switzerland	(U. Nyffenegger, 2018)
6	Zukunftsquartier Austria	(Schneider et al., 2023, 2019a; Schöfmann et al., 2020)
8	EU JRC PED	(European Commission. Joint Research Centre., 2020)
9	CityXChange	(Dahlen et al., 2020)

2.4 Differing and Open definition aspects

The following aspects were also identified as relevant criteria to be dealt with in the definition in one way or another. For these, however, there was consensus found on how it should be included in a uniform definition, with a certain level of ambiguity or incoherence remaining. Nevertheless, these aspects can be addressed and operationalized, as exemplified in the last chapter, but it might not be purposeful to try and solve this uniformly across Europe.

The PED definition should cover all relevant phases but not set strict requirements for the process itself.

It remained unclear how this could be satisfied, since "relevant phases" varies between green- and brownfield development, and even between green field developments in different settings such as urban or rural developments. It was also not clear who would be responsible in enforcing or monitoring the adherence to a PED definition in different phases of district development and if district developers or their could or should be held accountable for violations.

This possibly hints at a mix-up between requirements of the PED definition and the PED implementation itself. It might be easier to have the PED definition include that their IMPLEMENTATION CONCEPT should cover all relevant phases without the definition explicitly prescribing the process itself.

The PED definition focuses on the measurability of quantitative and qualitative targets.

This was not uniformly shared. As laid out in the previous chapter, focusing on getting the quantitative targets a specific meaning in a municipal, regional and national context can be seen as one of the great opportunities of the PED concept. In practice this is not necessarily the core concern of many involved stakeholders and as such this aspect also receives little support by some.

PED definition enables the implementation in urban and rural areas.

As the stakeholders in the C4P project all come from cities, the feasibility in rural areas did not receive great consideration and did not have any particular advocates.

The PED definition should allow some form of distinction between newly built and existing districts.

This was widely accepted as requirement, but unclear as to how this could be solved in a unified way across the participating countries and cities. Again, a way forward would be to uphold the requirement, but reduce its uniformity to a national level, e.g. by different MS adopting different context factors for vintage or heritage, that reflect the composition of their building sector and the relative targets of the groups within it.

PED definition is neutral towards different technology and innovation levels.

Again, this aspect was agreed upon but not operationalized due to the differences in the participating cities. To satisfy this requirement however, a PED definition must simply not make specifications technologies or innovations required or prohibited.

3 Example PED Definition and assessments



This chapter has two parts: The first is addressing the principles of using a positive energy balance as a sole definition criterion for PEDs and introducing its conception primarily as a design question. It theoretically introduces "context factors" as a way of allocating an external, possibly national effort-sharing scheme into the energy balance requirements of a PED to ensure feasibility in the desired urban contexts.

Second, this quantitative definition approach is exemplified by exercising it on the three districts of the Cities4PEDs project: The Stockholm Royal Seaport, the Brussels North district and the Vienna Seestadt Aspern.

3.1 How to Use a Positive Energy balance to assess the main PED goals of energy efficiency, flexibility and local renewable production

Even though many different definition and assessment frameworks have been put out over the last few years ((Ala-Juusela et al., 2016; Alpagut et al., 2019; Civiero et al., 2021; Gabaldón Moreno et al., 2021; JPI Urban Europe, 2019; Koutra et al., 2018; Marszal et al., 2011; Salom and Tamm, 2020; Schneider et al., 2019b; Wiik et al., 2018), it is widely accepted that at the core of a PED is the sufficiency criterium of a positive energy balance. This is widely regarded as the common ground and smallest denominator of the PED idea. Subsequently though, expectations and approaches quickly diverge amongst participants:

The positivity condition is simple to state, yet hard to specify.

A basic principle in science is that a measurement consists of two parts: One, a precise definition of the quantity being measured and second, the measurement result itself. Obviously, measurement results are only useful if the quantity being measured is controllably invariant and different measurements can be taken on the same quantity, making it comparable and possible to check the validity of the measuring process. Once this process is established, then and only then can there also be meaning to a *threshold* of the measurement.

A zero positive energy balance is such a threshold: namely on the measurement of the energy balance of a district. Contrary to other well-defined metrics such as heating demand or energy end use however, this is not clearly defined: It is like saying, something on a scale should weigh no more than zero kilograms without specifying which items the weighing needs to include, and the balloons and items representing the onsite renewable energy supply and the accountable energy demands and services in the district respectively).

It is therefore insufficient to assess that PEDs "achieve a positive energy balance". What that really means is that one of many different possibilities to draw the system boundaries has a positive energy balance. This is important, because it means that it is in principle arbitrarily achievable to reach a positive energy balance by using appropriate system boundaries, inclusions and exclusions from the balance. In practice this enables projects to "achieve a positive energy balance", but only covering heating and cooling energy demands while other include mobility demands except flights or including purchasable offsite renewable

generation credits, etc. The ambition of these projects and whether are in line with their embedding climate goals cannot be derived in general terms but for each project specifically at most.

This does not mean that these project approaches are not valid, not ambitious or flawed in any way: They simply reflect local conditions in the aspiration to adhere to the PED principles of having a "positive effect" on the surrounding energy system. They can in fact be very ambitious, but these achievements can hardly be put in perspective due to the lack of comparability and assessment standardization.

There is no one to blame for this either: In the absence of a recognized standard definition, projects currently do not have a choice but to creatively draw their own system boundaries to fulfil an arbitrary condition of "something must be > 0" so as to adhere to the overarching PED concept.

On the other hand, definitions with rigorous assessment standards and detailed operationalizations - as required by practitioners - that could in theory provide comparability suffer from a trade-off of different nature: Either they are tailored to a specific situation that is only applicable to a small subset of possible districts of a certain characteristic and fail to request achievable results in other cases. Or if they aim to include a broad variety of districts, the methodology needs to ensure their achievability over a wide range of configurations, which means that the calculations need to depict that with various edge cases and generally become more convoluted. With this in mind, the question of PED definition transforms into a whole other question:

- Which districts should be able to achieve adherence to a PED definition?
- Is it just very ambitious individual lighthouse projects or
- should the definition be a framework that most of the building sector can aspire to?

The argument can be made either way, but the main feature will arguably feasibility: A definition framework can adopt multiple layers of goalposts, targets, or levels of achievement for different groups of districts.

3.1.1 Possible Approaches

Approaches to defining Positive Energy Districts can be distinguished based on two key characteristics:

- Whether quantitative target values are used for definition or not and
- how the derivation of these target values is carried out methodically (bottom-up project-specific or top-down across projects).

As (Shnapp et al., 2020) elaborates, there are the following methodological options to determine target values for a district:

- Derivation from monitoring and measurement data relative to a baseline
- Derivation from target values of reference buildings.
- Determination by means of modelling and simulation of corresponding district energy systems
  - Either with static energy balance targets greater zero or
  - Dynamic targets based on additional typological parameters

But independent of the methodological approach of determining and arriving at a target for the PED energy balance, there is an arguably more fundamental distinction between different sets of PED targets or criteria. PED definitions often employ both simultaneously.

Table 8 Types of PED assessment targets / criteria

Criterion	Interpretation	Example in the context of the definition of Positive Energy Districts
Sufficiency	➔ A concrete goal **must** be achieved. Positive energy balance	⌂ Positive energy balance ⌂ Positive emission balance
Maximization	➔ Shall be achieved as far as possible ➔ Shall be maximized	⌂ As much local renewable generation as possible ⌂ As much energy efficiency as possible ⌂ As much energy flexibility as possible ⌂ As low cost of living as possible ⌂ As high quality of life as possible

The key difference between these two types of criteria is mainly that only the first, i.e. sufficiency criteria, can distinguish different groups, such as the group of "plus-energy districts" and that of non-plus-energy districts through verifiable quantification.

Is that even the goal? Many approaches take a different path and interpret the quantification criteria of a PED according to what is technically and economically feasible and goes beyond the existing state of the art. These approaches can be characterized by focusing on the development process of the PED and individually suitable criteria themselves. This is certainly more accessible to a wide range of districts and particularly helpful for districts where there is a lack of focus and instruments to address the development and planning process of a district itself.

However, it can be argued that the greatest opportunity of the Positive Energy District concept is precisely that the definition already implies a quantifiable balance target and since targets may be aimed towards a goal, PED targets could be interpreted to *quantify the contribution to climate neutrality*. This important point is often not considered in the development and discussion of possible draft definitions, so it should be explicitly pointed out again here. The opportunity lies not only in a uniform framework for certifying greater quality of buildings and renovations in the European Union, but also in the quantification of the sufficiency of climate neutrality of districts. The claim should be to define as generally valid as possible, and as concretely as necessary.

Regardless, to form any definition framework the following questions must be answered:

- ⌂ Which criteria are considered?
- ⌂ Are they "sufficiency criteria" or "maximization criteria" or neither?
- ⌂ How is the evaluation method of the criteria determined in order to assess the fulfilment of sufficiency or the quantification of maximization?
- ⌂ And for sufficiency criteria: How is the target value at which "sufficiency" is achieved determined?

In principle, a definition can include any number of these criteria and can also include a combination of sufficiency and maximization criteria. The original "PED Framework Definition" by (JPI Urban Europe, 2019) can be understood as one such definition:

> *„A positive Energy District has an annual positive energy balance by maximizing local renewables, energy efficiency and energy flexibility* "

This constitutes one sufficiency criterion and three maximization criteria:

- Sufficiency
  - Energy Balance greater zero
- Maximization
  - Local renewable energy generation
  - Energy Efficiency
  - Energy Flexibility

The lack of concrete statements on how these imperatives are to be evaluated has resulted in a whole range of different interpretations being developed, each of which has been suitable for the associated district(s). However, the above questions must be answered unambiguously for each dimension of the assessment at the latest when specific projects are evaluated. In practice, concrete answers and operationalization interpretations to some of the above questions are typically only made based on individual research or development projects that can inherently only include a handful of districts at most. This, again, means that PEDs run the risk of being largely incomparable in terms of their goal achievement and ambition. As stated before, this need not automatically be a problem, if for example the use of PEDs is primarily in their exploration of conceptual and technological frameworks from which lighthouse projects are to emerge. However, if general quantifiability towards climate neutrality and comparison of districts is of concern, then clear and uniform regulatory and quantitative specifications must be made on a higher than individual project level. At this point, the following important fact should also be pointed out:

> *To clearly distinguish between two groups, such as PEDs and non-PEDs at least one clearly operationalized sufficiency criterion is required as part of a definition*

In this context, the sufficiency criteria, i.e. the target values, can also be defined individually for each project, as suggested in a number of approaches. However, this generally precludes comparability between districts.

Note that most approaches to defining PEDs today are quantitative in nature, which seems to be rooted in the fact that the name "Positive Energy District" alone implies a sufficiency criterion that must by definition be quantifiable. Nevertheless, there are also a number of approaches that call for qualitative criteria, usually in addition. In doing so, the human being, the inhabitant, the user, the worker, etc. moves into the center and qualitative criteria of the district evaluation almost inevitably result, such as

- Affordability
- Safety and security
- Social and physical inclusion
- Social mix
- Sufficient green and open space
- Space for positive social gathering and expression
- Holistic and environmental sustainability
- Reduction of urban heat islands
- Et cetera

Here, it is generally more difficult to formulate cross-project target values and to make statements about their fulfilment or non-fulfilment.

There is no shortage of categorizations of these evaluation dimensions in literature. The fact that these criteria are repeatedly and increasingly becoming the focus of attention is good and important. Not least because they are seen by many stakeholders and experts as essential for implementing a PED: The

categories governance, incentives, process, market, technology, social and context are attested by more than 50% of the experts to be important to very important for the implementation of a PED (Steemers et al., 2022).

The question is however how these qualitative criteria should be included in a PED definition and to what extent. This depends on who you ask, and it could also not be answered by the C4P team with certainty. However, as the positive energy balance as a quantifiable sufficiency criterion is such a prime feature and opportunity of the PED concept, one should be careful not to "overburden" it with considerations and quality insurance that might be better addressed elsewhere. At the end of the day, nothing prevents PEDs from also adhering and aspiring to other standards and goals apart from a climate neutral, positive energy balance.

3.1.2 Anatomy of a quantitative definition using a positive energy balance

The definition of the energy balance can be broken down into three major parts, of which only the first two typically receive plenty of attention: (1, left) the system boundaries and subsequent energy sources and demands considered within it, (2, middle) the weighting /assessment scheme for different energy flows in and out of the system boundary and their cumulative balance and finally (3, right) the target of that balance in possible adherence to an allocation or offsetting mechanism to take the district's context into account without generalizing this context.

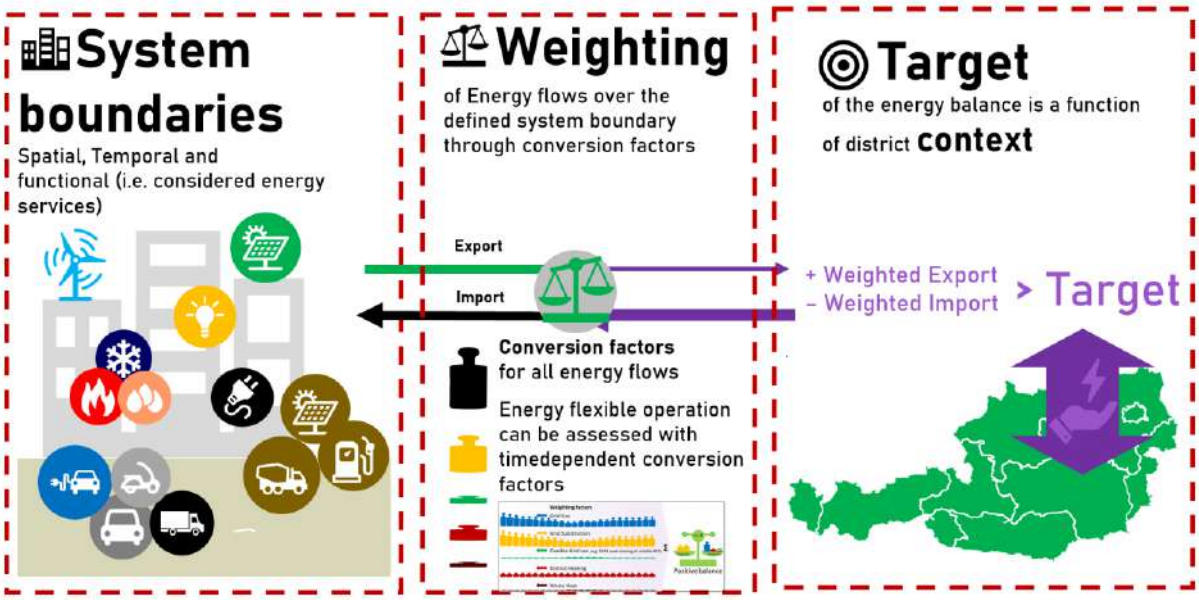


Figure 20The three main design elements of a positive energy balance assessment (Schneider et al., 2023)

These three areas of district system boundary, system balance weighting and balance targets correspond to the three questions a quantitative PED definition via the energy balance needs to address in unison distinct design problem. The balance target is not necessarily positive or zero, but can in principle be a function of any set of parameters deemed relevant. Thus, the definition through an energy balance target can include both project-intrinsic and project-extrinsic factors. On the one hand, this is an additional challenge, but at the same time it is the great opportunity: With this, dynamic external requirements can be related to project-specific proposed solutions.

But to achieve this, a definition must be able to answer the following three questions:

1. Where is the functional system boundary of the balancing drawn?

- Which energy services have to be covered? Is this selection district-specific or more generally defined?
- Which renewable energy sources are considered "local", i.e. within the system boundary?

2. Weighting
   - How are the energy flows over this system boundary weighted in the balance?
   - Are all flows treated equally as physical quantities?
3. Allocation of external resources expressed as a target function
   - What energy or GHG budgets are available to the project from external sources?
   - Which local and external contexts need to be considered?

The following pictures is an adaptation of (Sartori et al., 2012) and summarizes these questions:

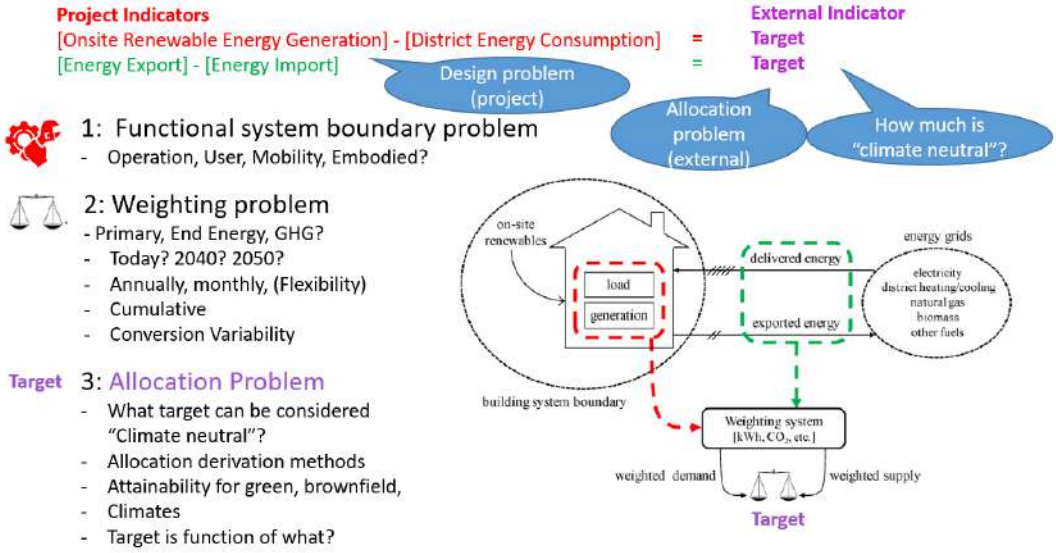


Figure 21 The three problems of defining a positive energy balance, based on (Sartori et al., 2012)

The advantage of this lies in the possibility to separate *project-specific* planning topics on the one hand and *project-independent* target values not only as conceptual topics but also in the practical derivation and operationalization. This would also be true when the PED target value is set to a fixed zero. However then, only implicitly and without further coordination between the targets and the surrounding energy system. Note that the definition of an externally derived balance target value is equivalent to the definition of an external renewable energy budget that can be used in the district and added to within a "virtual system boundary". This can also be seen the previous illustrations.

Determining the externalities of a district - via an allocation budget or the equivalent target value - is necessarily part of a balance-based PED definition.

This analogy is explained in detail in the section on the balancing targets. From this perspective, almost all quantitative definition approaches today use some form of mechanism to adjust the ambitions, i.e. the relationship between the energy services considered, their weighting and their balance target value to the respective project circumstances. They mostly do this only implicitly by setting certain system boundaries or weighting systems without much consideration whether this approach is also feasible for different districts with different contexts. The typical alternatives to the above allocation approach of external budgets and thus definition of a target value is summarized in the following table, alongside typical possibly problematic consequences:

Table 9 Alternatives to context-specific RES allocation

Alternative to context specific RES allocation, i.e. balance targets	Possibly problematic consequences
No allocation of external resources	Achievability is more difficult Positive energy balance is often only achievable by not considering many energy services (such as plug loads, mobility, and embodied energy), Non-consideration in turn diminishes statement on climate neutrality achievement
The usability of external renewable energies is proven individually by projects: For example, through certificates, purchase contracts, or similar.	Does the project then overuse external resources? Will there be enough external energy left over for the rest? Does the RE balance add up on a municipal or national level if every district adopts this approach? These questions usually cannot be answered beyond doubt or cannot be answered in the same way for a large number of districts
Flexible or individual definition of system boundaries and considered energy services	The term "Positive Energy District" then loses the quantitative dimension of comparability: One "PED" may consider only heating and cooling, but another also operating energy, mobility and embodied energy, a third excludes energy-intensive uses of the industrial plant from the consideration, a fourth extends the system boundary to remote wind power plants.

The following image illustrates the differences between a project-specific bottom-up approach to defining system boundaries, balancing, and goal setting on the left with the top-down method presented here through the uniform allocation of credits, shown here as "context factors" on the right. By individually defining system boundaries, energy services considered, and "local" renewables (left), the definition must be made anew for each district. On the other hand, locally different districts can meet the same quantitative definition if it includes an allocation approach that assigns external budgets according to district context (so-called "*context factors*") This quantitative allocation approach is necessarily project-independent (right).

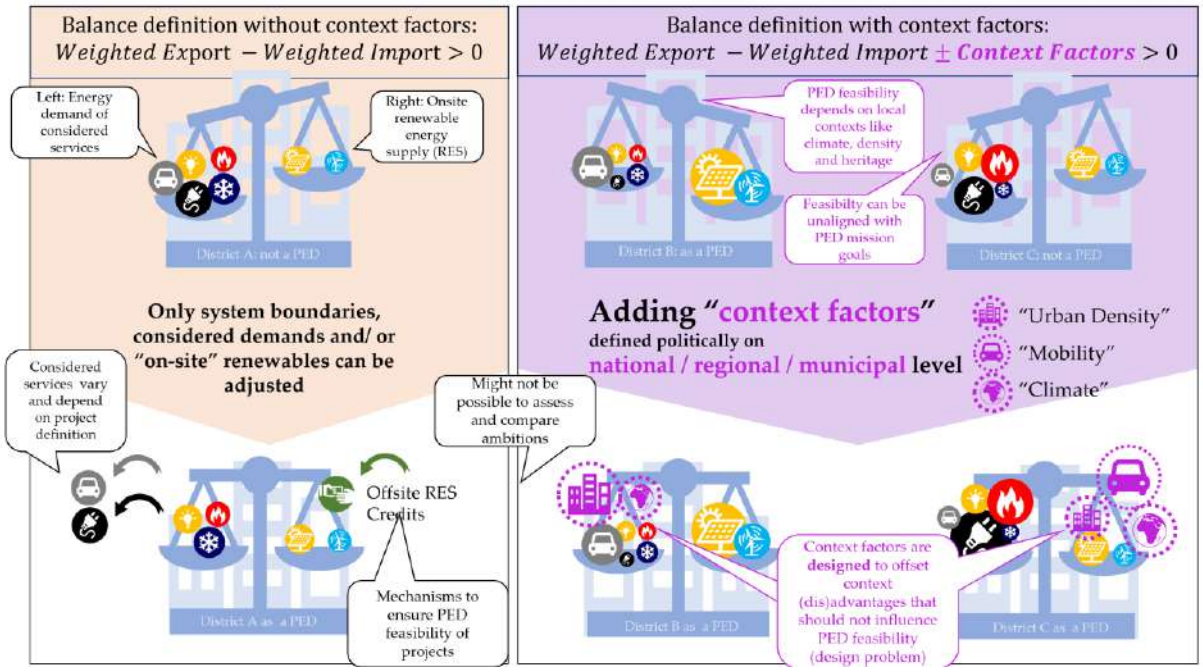


Figure 22 How can different districts in different local contexts meet a quantitative PEQ definition? (Schneider et al., 2023)

In the section 3.1.5 the allocation problem and how to derive possible context-sensitive external budgets are described in more detail, in particular:

1. A "Density factor": An allocation based on the building density of the district to compensate for practical differences in the generation potential of local renewable energy relative to the energy intensity of the land use. Density is one of the main governing factors for reaching any energy balance target
2. A "Sectoral credit of the national renewable energy system": Nationally available RES for the building sector allocated to the district according to its share of usable space created of the national building stock. This also covers and compensates for the mobility needs arising from the use of the district.

3.1.3 System Boundaries

The definition of system boundaries is required to enable balancing of flows *over* these defined boundaries. Typically, and as used here, this refers to boundaries of the ENERGY SYSTEM and the associated energy flows. Nevertheless, when defining the system boundaries, it is important to distinguish between different types: *spatial, functional, and temporal:*

Spatial means that there is an actual physical boundary that can be defined and in theory, all energy flows over this boundary can be measured. In practice, this approach quickly becomes cumbersome or unspecific when faced with the need for a more nuanced distinction between different energy services.

Functional system boundaries are more flexible, as they identify specific energy functions, uses or demands to be included or excluded according to function, rather than spatial proximity. Functional system boundaries can be further differentiated in renewable energy supply within the system boundary, often referred to as on-site, and energy services to be accounted for in the balance. Note that "on-site" here does not necessarily mean spatially on-site but rather "within the system boundary", counting positively towards the energy balance.

The functional system boundaries and the included energy services can be roughly grouped into three regimes of increased responsibility:

- Operational energy and user electricity
- Energy demand for everyday mobility
- Embodied energy

The following figure illustrates three different functional system boundaries with their included energy services and RES potentials respectively. They also include rough indications of the spatial system boundaries

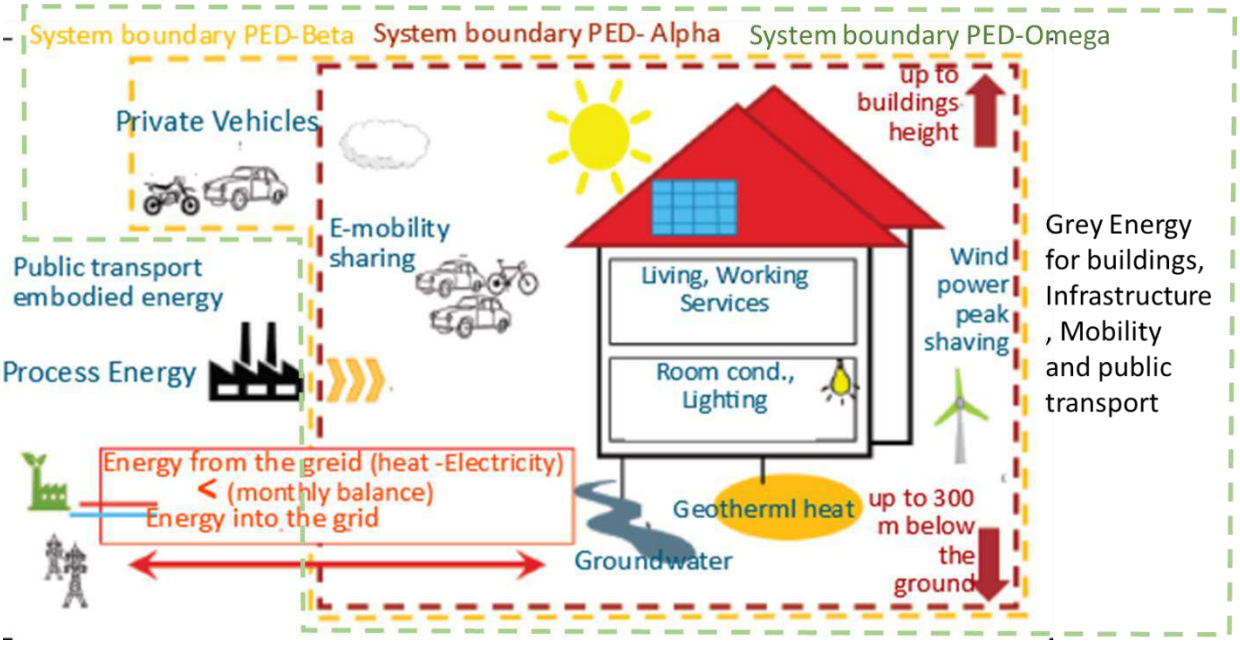


Figure 23: Three EXAMPLE system boundaries of a PED energy system

Finally, *temporal* system boundaries are often not specifically addressed but instead implicitly assumed to be one operational year without further distinction for degradation, maintenance et cetera. This typically only becomes relevant once trying to include embodied energy and emissions in the balance, at which point typically a number of years, or life span, is defined as a temporal system boundary.

3.1.4 Weighting System

One of the fundamental questions in the PED definition is the weighting of the different energy flows in a PED energy balances. This is sometimes framed as deciding between different predefined energy measures such as final energy or renewable, non-renewable or total primary energy, or greenhouse gas emissions in terms of CO_2 equivalents. But this extends to the more general question of which physical energy flow is assigned which weight in the balance, and perhaps more importantly, why. As illustrated in the figure, every weighting system entails inherent consequences on the achievement of the resulting PED balance, independent of the actual energy flows and services considered. Or put differently: Each district can find a weighting system under which the energy balance is positive. It is therefore important to not just *choose* a weighting system when defining the energy balance of a PED, but also lay out the reasoning behind it and assess the feasibility of different districts.

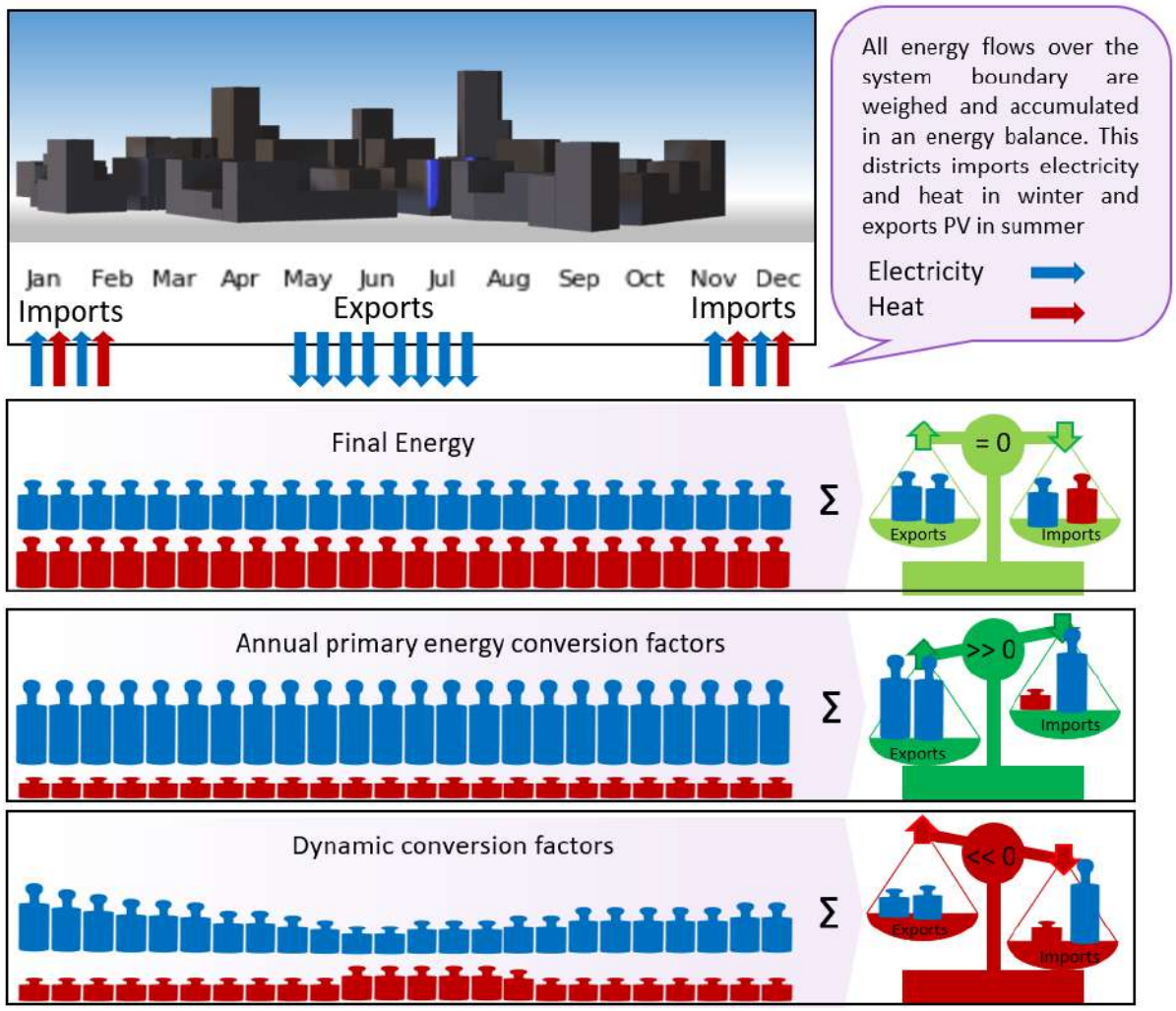


Figure 24 Weighting systems assign each energy flow of a district a weight by multiplying with a weighting factor. The resulting weighted energy flows are aggregated in the energy balance (right). For the same district, different weighting systems (three boxes on bottom) lead to different balances and with it different assessments.

Final energy or primary energy?

If equal weights of 1 are adopted across the board, they correspond to balancing the final energy flows over the system boundary. However then, the amount of energy generated and consumed must be equal. This prohibits a qualitative comparison of different energy carriers, particularly electrical and thermal energy. For example, a district could import 10 MWh of grid electricity and feed in 11 MWh of thermal energy to the surrounding to achieve a positive final energy balance. Whether this thermal energy is actually useful, in particular AS useful as the electricity imported from the grid, is highly situational and therefore not suited for general use.

Using a primary energy weighting instead allows the consideration of waste energy through the whole process chain, as well as overall energy efficiency, and actual energy impact of the district depending on the energy sources. Consequently, countries that generate significant energy more from renewable sources profit from this energy balance calculation. This also allows the comparison and balancing of embodied energy, operational energy, thermal, and electrical energy by converting all types of energy to primary energy.

Internationally, a number of different methodologies with their own strengths and weaknesses are used, particularly for factors related to the conversion of final energy to primary energy and greenhouse gas

emission equivalents[4]. A public, unified dataset does not exist; instead, there are a number of national and international[5] methodologies and datasets, some of which are outdated or not publicly available, or only represent specific energy conversions, often only for specific regions.

Assessing Energy Flexibility through balance weighting factors

The main concern with temporal resolution of the energy balance is with enabling the quantification of energy flexibility provided by the PED. The key to achieving economic, environmental, and social objectives of PED is designing a flexible and sustainable energy system in the district. The main part of flexible energy system is the smart grid, that decides best time to store, use and extract energy. Smart grid provides the possibility to use the storage capacities of the system as much as possible and adapt the energy demand without a negative impact on user comfort.

There are a number of ways to quantify the energy flexibility of an energy system, which all come with different requirements and limitations, such as simulation of different baseline and flexibly controlled systems under various grid scenarios. Apart from notable exceptions[6], these approaches require higher temporal resolution of the energy flows, which in turn necessitates dynamic building simulation and/or monitoring data.

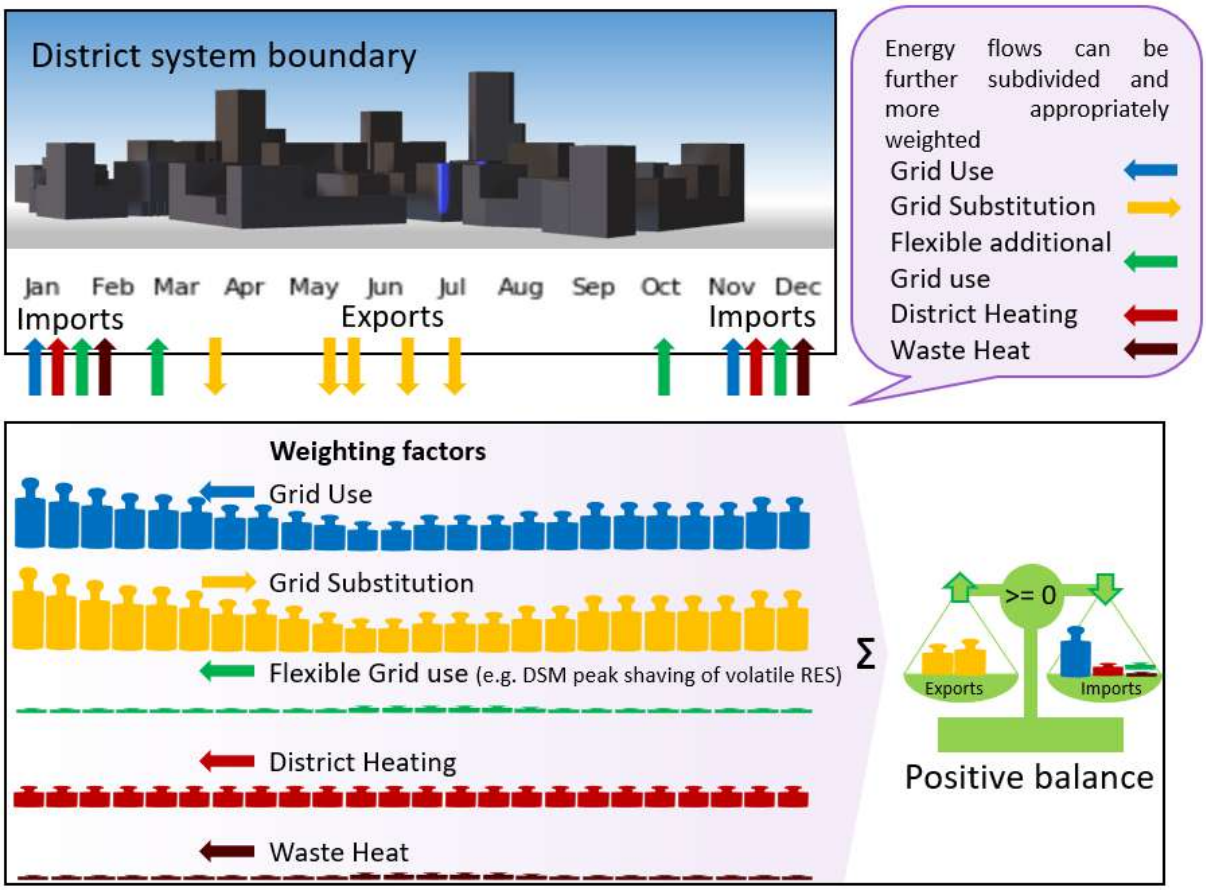


Figure 25 Example of time-dependent weighting factors of energy flows

[4] See (Hamels et al., 2021)
[5] For example, the CEN (prEN17423) and ISO (52000) standards differ substantially from one another
[6] See e.g. (Märzinger and Österreicher, 2019)

The above figure illustrates how time-dependent weights can be used to assess energy flexibility in the energy balance: The Energy balance is calculated by cumulating all incoming and outgoing energy flows of the district with the specific weight assigned to that energy flow at that time. These weights depending on time and type of energy reflect their relative importance and availability (or scarcity) in the surrounding energy system in a way appropriate to it (e.g. economic, ecologic, exergy content or else).

District Energy flexibility can be divided into two categories corresponding with the associated district internal and external effects: The first and immediate effects for a district through flexibility are mainly derived from synergy concurrency and efficiency effects, namely the reduction of peak power demands and increased load balancing, which in turn translate to economic benefits

The second and more intermediate effect of flexible district operation can lead to benefits for the surrounding energy system, such as providing buffer storages and load shifting potential, in turn leading to lower system costs, but also possibly the inclusion of a higher share of volatile renewables in the grid.

The external requirements of energy flexibility need to be communicated to the district energy system though. Although in principal any kind of signal would be possible, the most predominant in research and practice was the use of price signals. Their main advantage is typically that they are readily available, easy to use and communicate. The downside however is that prices reflect only the flexibility demand and not the flexibility supply potential of the district itself. This means that without additional care of interpreting the price signals and modulating such that it considers the district energy system and its limitations, it can lead to unintentional behavior and suboptimal performance from the perspective of the PED conceptual ideas.

Within the PED concept and it's idea of a positive energy balance, there is the opportunity to also include an assessment of the energy flexibility in it: The idea of assessing energy flexibility in the energy balance is simple: Through appropriately designed weighting factors the energy imports and exports of the districts are weighted with a set of temporally highly resolved weighting factors, typically hourly, that reflect the usefulness of the energy import / export at that given time for the external grid.

In this context, energy flexibility is a service of the district to its surrounding to incorporate more renewables into the overall energy system. As such, it is important that the assessment weighting scheme is therefore also a determination of the surrounding energy system and not a design choice of the district. Instead, it should communicate flexibility demand and supply between district and the "hinterland". This can have a number of implications in terms of *operationalization*, which will be exemplified in detail in chapter 3, particularly section 3.2:

- *Transient Simulation* of energy flows including e-mobility (at least hourly)
- *Hourly load balancing* with appropriate weighting factors
- Inclusion of *energy flexible control schemes and DSM* to increase utilization of volatile RES and increase PED target score
- Inclusion of *building thermal storage potential* to increase utilization of volatile RES and increase PED target score

3.1.5 Balance Targets (context factors)

As shown before, most definitions use some form of virtual offsetting and crediting system that change the requirements of the energy balance in one way or another. The European alignment task force call these mechanisms "context factors". They can be defined on a project level, but should really be taken care of on at least municipal, regional or even better yet national level.

With the introduction of context factors, considering more energy services such as mobility and embodied emissions does not necessarily increase the difficulty to achieve a positive balance, but rather have increasing scope and increasing number of "context factors" attached. With that, though, comes increasing modelling and calculation effort. There are two immediate parts to a district's energy balance and its feasibility to reach a numerical net zero or positivity target, as illustrated in the following Figure:

- The choice of accountable Energy services and onsite RES to be included within the balance and
- The weights or "conversion factors" given to the resulting energy flows over the system boundary

The colored cogs show three prime examples of adjusting a PED balance. Selective design of any of these parts can have a great impact on the resulting energy balance. Due to the lack of a uniform definition of system boundaries, which energy services to consider and how to convert to the weighting balance, every PED project is required to do this –implicitly or explicitly – to show a positive energy balance. It is not surprising that a given district can show a net zero or positive energy balance for a given definition and a negative balance for a different configuration.

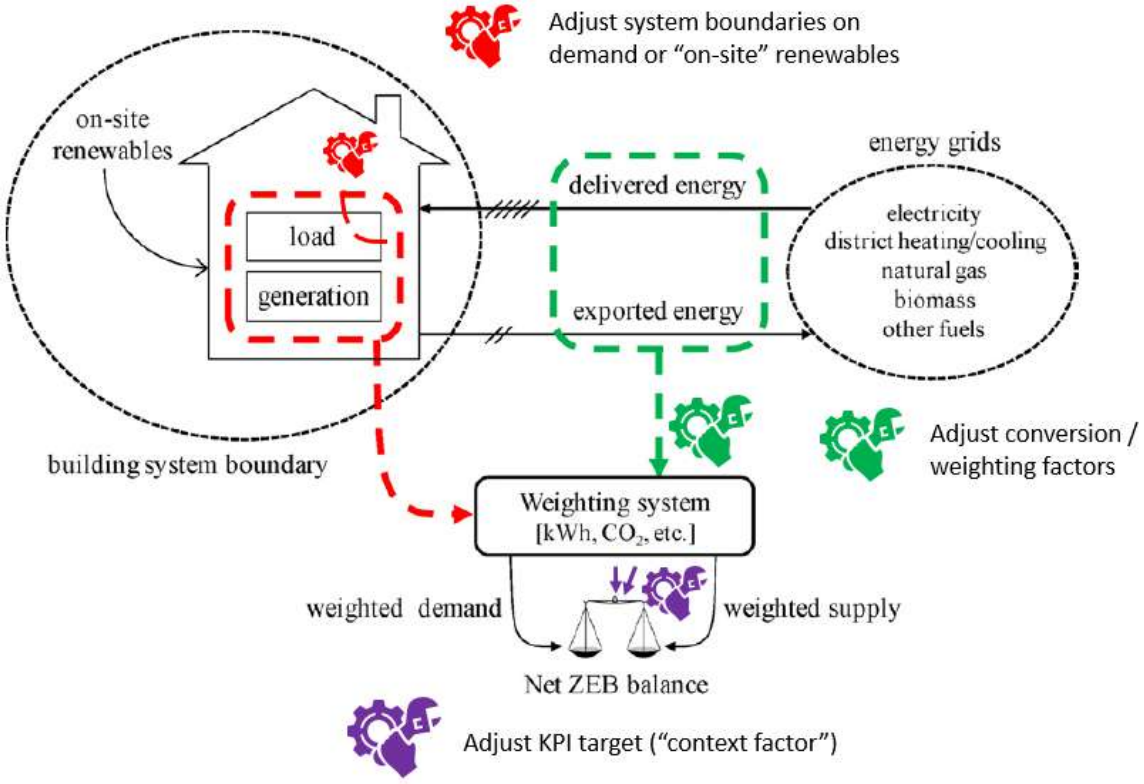


Figure 26 Positive Energy District Assessment framework adapted from (Sartori et al., 2012)

It is important to point out that the different definitions factor in different contexts implicitly by designing their system boundaries and considered energy services and onsite renewables according to considerations appropriate to their districts. Therefore, definitions *implicitly* include "*context factors*".

What are they? In the following, "*context factors*" are introduced as virtual factors included in the energy balance to deliberately offset certain effects of local contexts on the energy balance. This means that instead of changing the system boundary or the considered energy services or onsite renewables, a context factor is changed and included in a otherwise unchanged balance instead. This allows to replace context specific definitions of an energy balance, where the considered supply and demand are variable and incomparable like

$$Positive\ Balance = Supply\ (variable\ definition) - Demand(variable\ definition) > 0$$

to a more general and uniform definition of an energy balance with fixed system boundaries and considered energy services, where only the context factors vary:

$$PE\ Balance = Supply(fixed)\ - Demand(fixed) + Context\ Factors(variable) > 0$$

Note that in the second definition, what is fixed is the kind of energy services and supplies considered in the balance, not the actual flows themselves. The consideration of the local and regional context is explicitly moved away into its own term of the context factor, to separate concerns: The ones that the district can actively shape and the ones that it has no control over but are important to inform the district's goals in its respective surrounding.

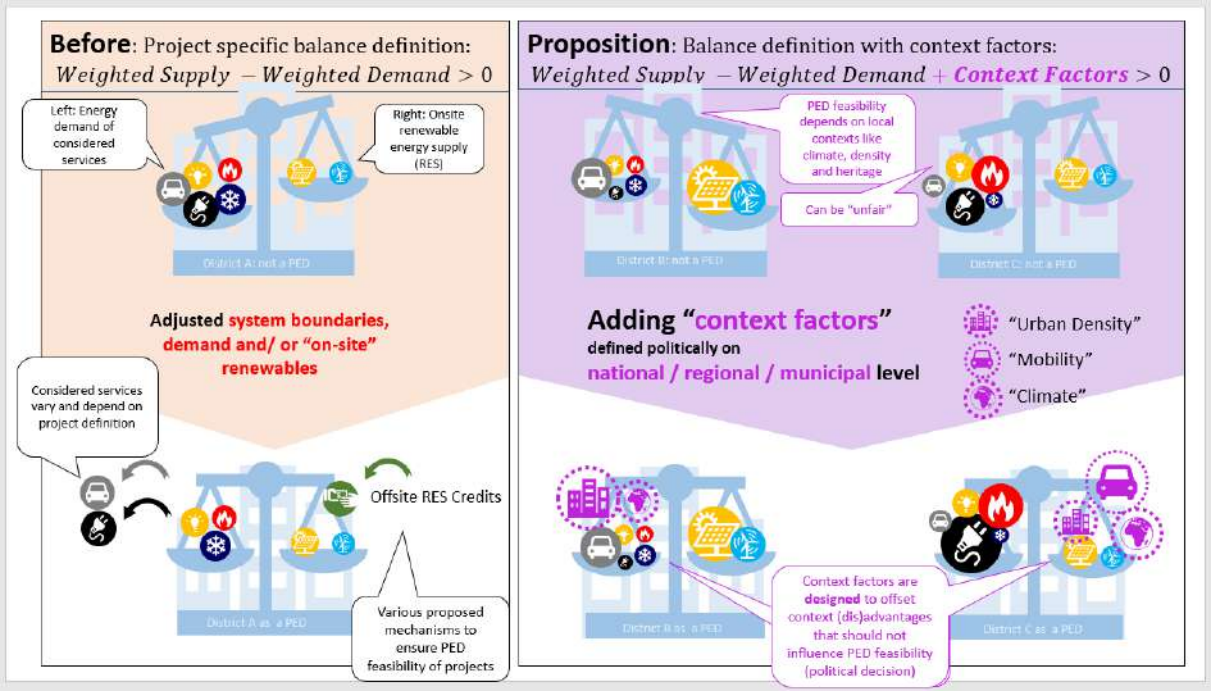


Figure 27 Comparison of offsetting strategies for achieving a positive energy balance

Although both lead to quantitatively positive energy balance for some districts, achieving this by the means of context factors offers the following advantages:

- Districts can still adhere to and achieve the same definition
- Differences are explicit (in form of the context factor) rather than implicit (in the form of different definitions)
- Comparability within a given context

The downside is that we need increasingly complex context factors if we want to increase the dimensions over which projects should be comparable but are actually not (density, climate, vintage, etc.). The principle idea of the context factor is to determine the *inverse* of the context effect and apply that to the assessment as a correction factor.

There are a number of possible contexts put out there. The following figure illustrates them and also hints at the difference between countries when dealing with them. Some, such as density might be more important to countries and cities with large spreads in district and city densities, whereas the factor might not be as important for countries and communities with fairly homogeneous densities. Likewise, climates are more different between countries and within them, making this context more important for

transnational comparability, but still relevant for tweaking the overall level of energy balance performance required within a country given a certain climate.

Another important aspect is that of heritage and the context of renovation. Generally, if renovations of buildings can achieve the same energetic performance as that of green field developments, it is only with additional effort and resources. Except for special edge cases, refurbishing the existing building stock to achieve a positive energy balance is virtually impossible for all but the most limited system boundaries. At the same time for the achievement of the Paris goals it is critical to conduct swift and broad renovations to a sufficiently high energy performance standard. The height of this energy performance standard in turn depends on the national energy systems at large and their envisioned demand and supply structure compatible with Paris 2015. This can give an indication on how to derive an energy balance target for existing districts that is connected to the rest of the building sector and the future energy system and considers context specific renovation potential. This could further be detailed by considering the vintage and construction materials of the district.

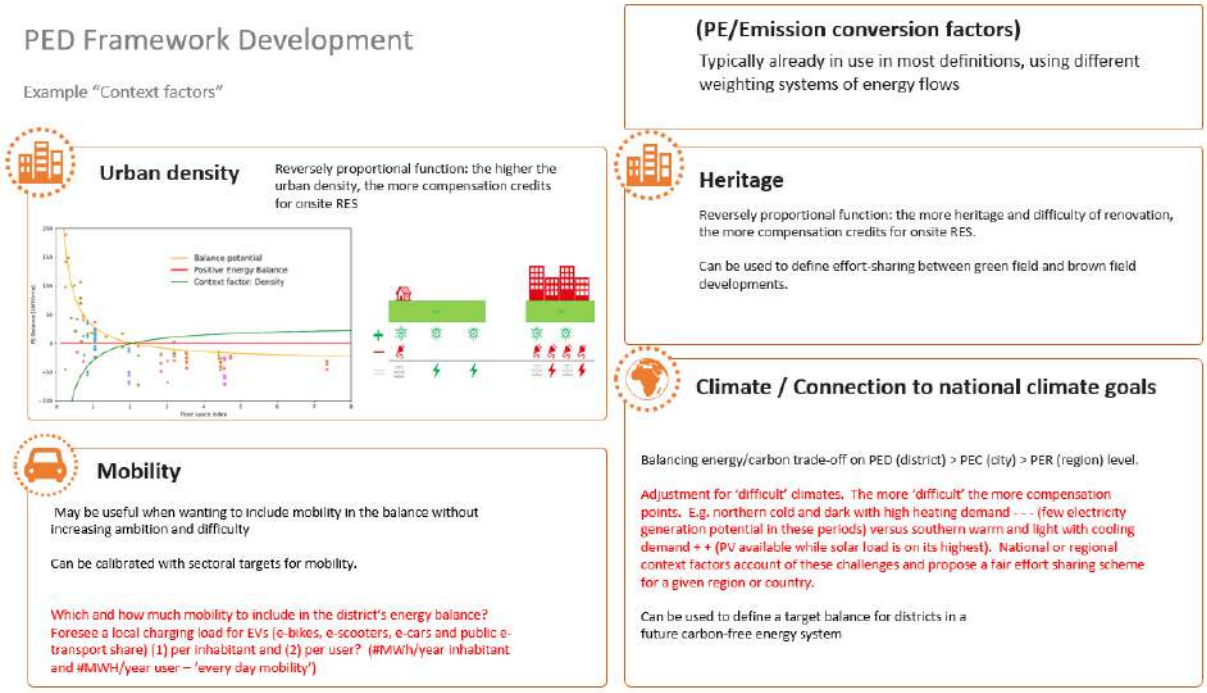
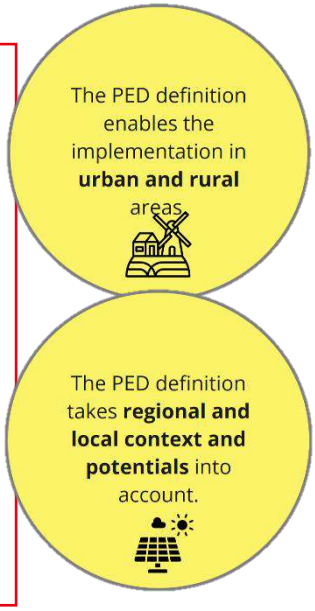


Figure 28 Examples of possible contexts that influence the energy balance and could be accounted for by designated factors

3.1.6 Context factor 1: District density and urban feasibility

The following is an example for how local potentials – in this case district density – can be accounted for in a definition fit for a wide range of uses. Please also note that the parametrization of the PE-balance target function was done for Austria, but could also be adapted for other EU member state.

- Every PED definition inadvertently also defines a position on Effort Sharing by defining relative achievability: This needs to be done explicitly rather than implicitly:
- *Local renewable energy potential (RES)* is determined by available land and plot size, *energy demand* on the other hand by useable floor area: A classic static positive target independent of density leads to disproportionate effort sharing between sparse and dense districts.
- Available land is a scarce resource, its use must be fair and efficient: Inefficient use of available land (e.g. SF detached housing) must not be incentivised by requiring less RES per land used to achieve PEDs.
- Intensity of local renewables should balance out with intensity of useable floor area per land use.



Ceteris paribus, districts with a low building density have an inherent advantage as they have relatively large space per person available for cost effective on-site supply of renewable energy. In contrast, dense districts struggle to meet their own energy demand on-site. This is simply due to the difference in available renewable energy versus accountable energy demands as illustrated in the following figure. A possible indicator for this is the *floor area ratio* (FAR = gross floor area per plot area): The share of renewable energy production of low FAR districts can be more than the share of renewable energy production of high FAR buildings as they have less place to produce renewable energy (thermal and PV) on site.

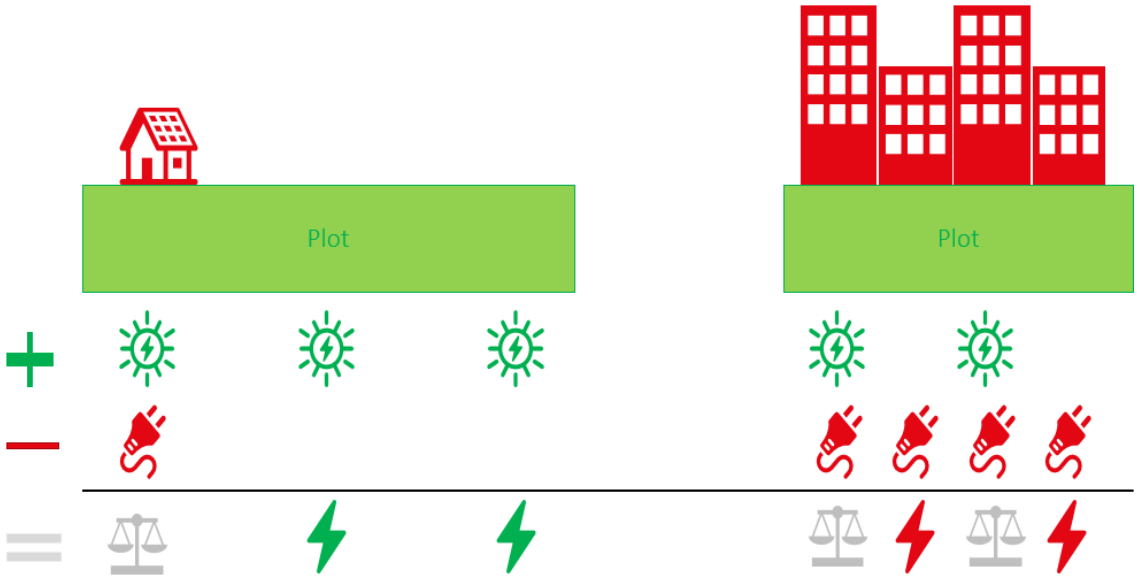


Figure 29 Schematic comparison of energy supply and demand of low- and high-density district typologies

As illustrated here, ceteris paribus, district density can be the defining element for the feasibility of a positive energy balance. To take these unequal conditions into account in structured and transparent way, *a function of density is introduced for the energy target of the district*. This can be done by analyzing the primary energy balance of a district, which in its simplest form can be characterized like this:

$$PEB = RES - OE, \quad with\ RES\ \ldots Renewable\ energy\ supply\ within\ the\ System\ boundary,$$
$$OE\ \ldots Operational\ energy\ of\ the\ district$$

The physical RES potential is depending on the available plot area, as it determines the amount of available irradiation and environmental heat. Operational energy on the other hand is proportional to the energy reference floor area. Substituting for their area specific energy intensities, the above formula expands to:

$$PEB = f^*_{RES} A_{plot} - f^*_{OE} A_{floor}$$

$PEB \dots Primary\ Energy\ balance\ \left[\frac{kWh}{a}\right]$

$f^*_{RES} \dots Primary\ Energy\ potential\ of\ renewables\ onsite\ per\ buildable\ plot\ \left[\frac{kWh}{m^2_{plot} a}\right]$

$f^*_{OE} \dots Primary\ energy\ demand\ for\ district\ operation\ per\ floor\ area\ \left[\frac{kWh}{m^2_{floor} a}\right]$

$A_{plot} \dots Available\ plot\ area\ \left[m^2_{plot}\right]$

$A_{floor} \dots Reference\ floor\ area\ of\ the\ building\ \left[m^2_{floor}\right]$

$FAR \dots floor\ area\ ratio, = \frac{A_{floor}}{A_{plot}}$

Dividing the above formula by A_{floor} and expressing the result in terms of the floor area ratio gives the specific primary energy balance on the left side as inversely proportional to the floor area ratio:

$$PEB(FSI) = f^*_{RES} \frac{1}{FAR} - f^*_{OE} \quad \left[\frac{kWh}{m^2_{floor} a}\right]$$

This can now be used as a context factor for the primary energy balance:

$$Context\ Factor: CF_{density} = f^*_{RES} \frac{1}{FAR} - f^*_{OE}$$

$$PEB_{in\ context} = \mathrm{PEB_{classic}} - \mathrm{CF_{density}} > 0 \quad \left[\frac{kWh}{m^2_{floor} a}\right]$$

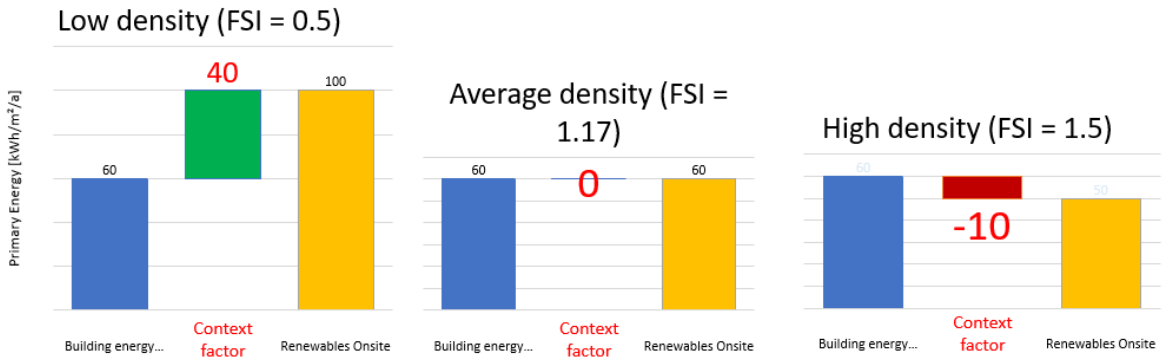


Figure 30 Example of district density context factor offsetting the balance gaps between supply and demand due to density

The context factor communicates a general sense of achievability *within a certain climatic and technological frame of reference and must be specified on a national or municipal level*. In this regard, they are similar to legal heating demand requirements as a function of compactness[7]. This also represents a physical

[7] In Austria, the maximum allowed heating demand is a function of the characteristic length of buildings, with less compact buildings having higher thresholds.

dependency being linked with technically feasible target values within a certain climatic and technological frame.

The below function shows the context factor depending on district density, which is equal to the primary energy balance target BEFORE applying the factor. Below a FAR of nearly 1.2, the energy balance must be above zero (green), while at higher FSIs it may be below (grey). Applying this definition, a positive balance can be achieved across a much wider range of districts and districts of different densities.

3.1.7 Context factor 2: Inclusion of mobility

In much the same way, the above approach can be carried out, with the inclusion of a portion of the mobility into the accountable energy services of a district and thus within the district system boundary. The context factor can be calculated by determining the required energy demand and deciding to what extent the district has to cover it by itself: This can range from "Not at all" to "100%" and anything in between. "Not at all" corresponds to a context factor equal to the required energy demand, making it a zero sum position. "100%" corresponds to a context factor of Zero. In between could be a context factor as shown in the figure below: A "mobility energy budget per capita" derived from sectoral effort sharing in a Paris 2015 compatible future scenario.

However, the context factor does not necessarily be derived from complex system analysis: A municipality could simply *define* a "mobility budget per capita" that reflects their desired typical modal split. Districts that can show that they will not induce more than this traffic have a zero-sum game: The context factor exactly offset the energy demand for the mobility. Districts with higher than municipal desired mobility energy demand need to either implement additional mobility options or other measures such as increased efficiency or RES to offset the deficit.

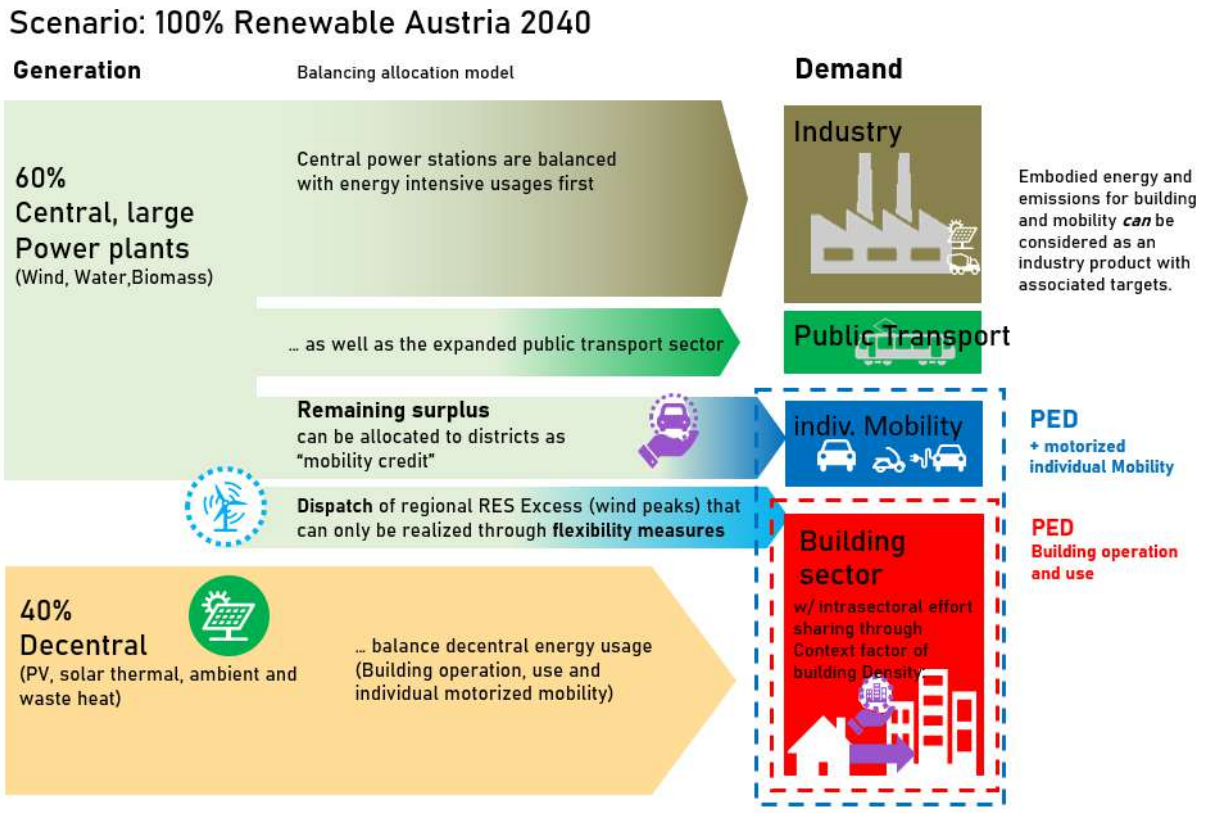


Figure 31 Example allocation of national energy supply and demand scenarios to the building and mobility sector informing district energy performance targets

Motorized individual mobility (MIM) can and should be included in the PED energy balance, if there is an allocation mechanism as depicted here in the center of a national supply and demand matching: It requires both supply and demand targets for each sector that are in line with a sustainable energy system. A concrete operationalization is published in (Schneider et al., 2023) and exemplified for the Cities4PEDs demo sites in section 3.2 of this report.

3.1.8 Difference between green and brown field development as a context factor

Although the topic has been increasingly researched in recent years, one key trend emerges across most approaches: The definitions do not make a quantitative connection between national climate goals and their analogous district target values to a renewable energy balance and GHG balance reduction.

Most approaches do not include a statement about the allocation of the overarching GHG and energy budgets that climate change scenarios and plans indicate are available to achieve the ambitious climate goals. In the authors' view, however, this is one of the most pressing problems. Without a quantitative model for effort sharing, positive energy districts are an arbitrary communicative concept. The approach presented here therefore focuses less on "What is possible?" and more on "What is needed?"

This question can only be answered by considering the scenario of a future climate-neutral overall energy system. The central question is how the effort sharing required for this can be divided up and mapped within the PED definition. From the authors' point of view, such approaches, which determine project-specific potentials and target values, are only useful if they also allow a statement about the effort sharing.

For a country, region or municipality it can be useful to define a desired energy balance of their entire building sector and many state corresponding targets in their climate strategies. This allows for further quantification of the effort sharing within that building sector: The higher energy balance of the green field developments, the lower the renovation targets can be and vice versa. This is a specific case of the general approach outlined in the section above.

3.2 Example PED definition through Energy balance assessment

The following section gives an overview of the assessment results of the three Cities4PEDs districts when using a quantitative PED definition with a primary energy balance with context factors for density and mobility.

3.2.1 Stockholm Royal Seaport

The Stockholm Royal Seaport Loudden district model is illustrated in the following site plan with residential usage (white), educational usage (rose), office and commercial usage (blue) and retail usage (yellow) indicated:

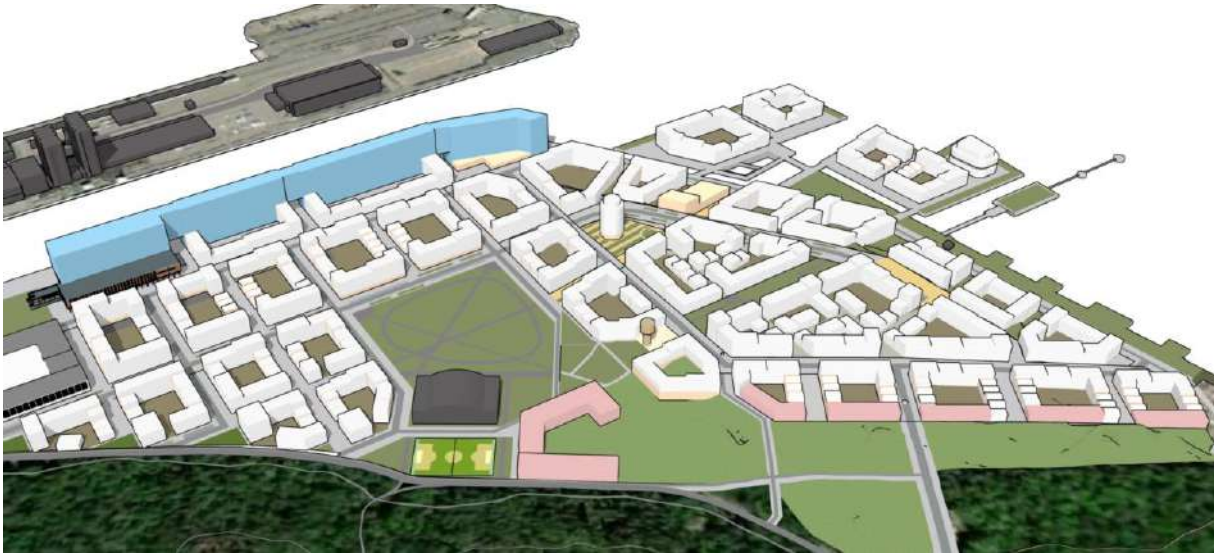


Figure 32 Stockholm Royal Seaport Loudden site plan

The model yields the following reference areas and usage distributions:

Table 10 District characteristics

Reference Areas		
District Area	55.56	ha
Gross Floor Area	647 331	m²
District Plot Area	236 471	m²
Building Storeys (avg)	5.2	
Floor Area Ratio [FAR]	2.74	FAR
Net to Gross Floor Ratio	85%	

Table 11 District Usages

Space Use			
Residential	80%	516 717	m² GFA
Commercial	14%	92 043	m² GFA
Highschool	2%	14 524	m² GFA
KIGA & Primary School	1%	4 785	m² GFA
Retail	3%	19 262	m² GFA

The following site plan shows the net construction areas of each plot that is used to determine the floor area ratio by dividing the gross floor area by it. It is important to distinguish this area from the total district area, that also includes spaces for traffic and green and blue infrastructure, which is typically not available for renewable solar energy generation.



Figure 33 Net construction areas (crimson) of the district is the reference area for density calculations

The following table shows the density context factor of the district. According to the example definition operationalization, this factor needs to be included in the primary energy balance of the district to assess whether it achieves a PED positive energy balance. Another way to think about this is that due to its density, the district needs to only achieve a primary energy balance of negative -27.5 kWh PE per square meter gross floor area. Note that this is only the case for dense urban districts. For neighborhoods with low density (with FAR < 1) the density context factor is negative, meaning their energy balance must be that much more positive.

Table 12 Density context factor Stockholm Royal Seaport

PED Alpha	
Context Factor due to Density	
27.5	kWh/m²GFAa
32.3	kWh/m²NFAa

As introduced in Section 3.1 in principle and 3.1.6 in particular, this context factor represents an effort-sharing allocation from low-density districts to high-density districts, reducing the energy balance target of the latter by the given amount of primary energy.

PV Model

The PV Modelling was carried out in 2 steps: First, a theoretical PV system application with a maximum (100%) utilization of the available roof areas was modelled as can be seen in the figure below. Second, the resulting yields were incorporated in the district energy simulation in 3 variations, each scaling the yield of the 100% example roof to the total district roof area with a given utilization rate: The analyzed variations were:

- **25%** of district roof areas utilized by PV Systems
- **70%** of district roofs areas

- 90% of district roofs areas

PV System parameters can be found in the Appendix. PED Assessment Results

Results were obtained investigating 3 groups of variations according to the three PV utilization levels: 25%. 70% and 90% of gross roof areas respectively:

Baseline (BL):	70% Roof PV:	90% Roof PV Utilization
- 25% Roof PV Utilization - SRS Hulls - GS Heatpumps	- as BL - with thermal Flexibility of - 0.5 K - 2 K - w/ 50% wastewater recuperation	- as before - 0.5 thermal flexibility

The Baseline scenario included *25% of roof utilization for PV,* as well as the thermal parameters stated above with a *groundsource heatpump system* for heating and cooling of the district.

The second scenario included PV utilization on 70% of the districts roof areas with 2 notable extensions:

1. Introduction of thermal flexibility: At times of excess renewables in the district, the setpoint temperature is raised during heating and lowered during cooling by a given delta (0.5 and 2 Kelvin respectively) to "save" some of the excess renewables from the PV system as heating/cooling in the very well insulated and thermally inert building mass via the HVAC system. As can be seen I the following diagram, even a thermal flexibility of just 0.5K, which is typically unnoticeable by users as it has minimal effect on indoor air temperature, can increase PV Self-consumption from 15.2 to 16.6 kWh/m²NFA and reduce the PV excess fed into the grid by a quarter
2. Recuperation of thermal energy of waste water, which in this model has an insignificant effect on the energy end use intensity.

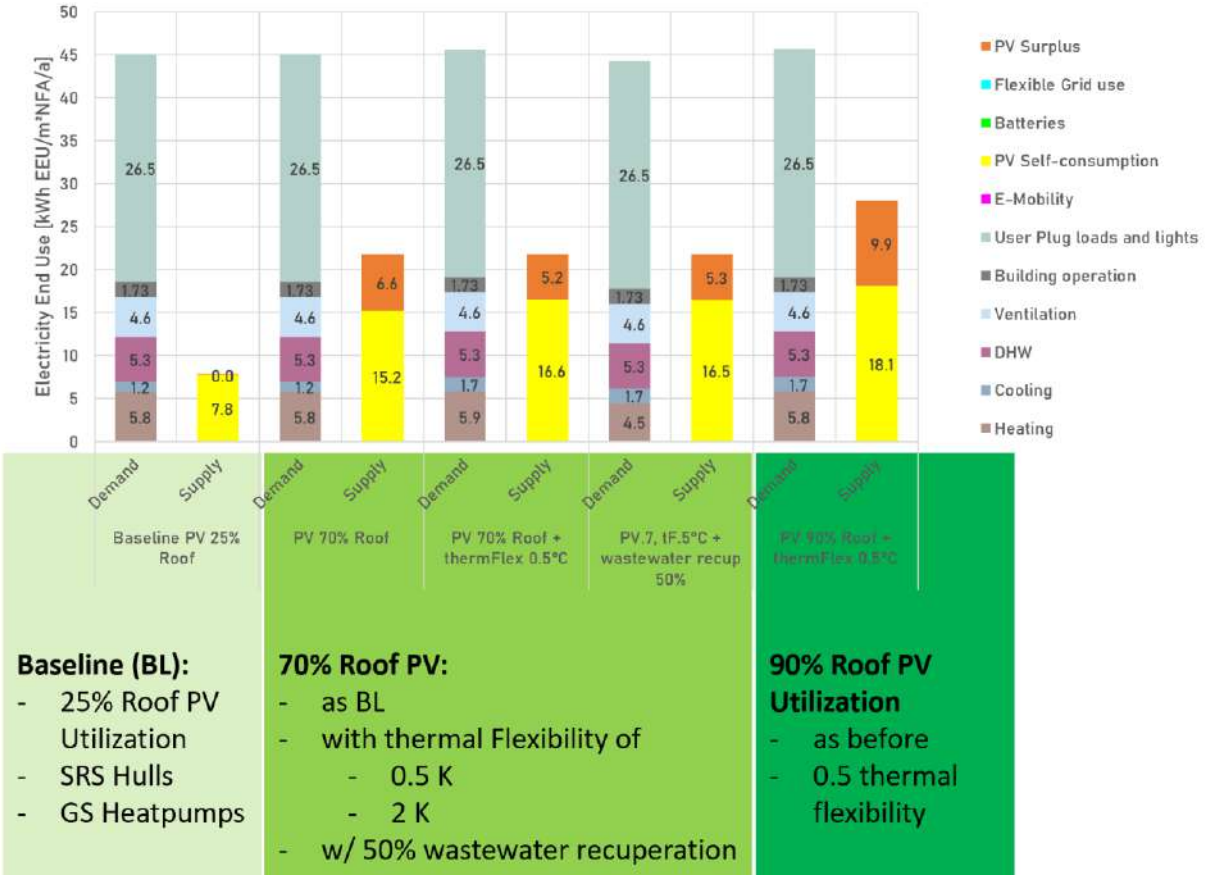


Figure 34 Electricity end use

As can be seen 70% PV roof utilization is sufficient to cover the district's HVAC energy demands, but insufficient in covering the user plug loads and lighting demands. The latter is actually not possible onsite at all with only roof mounted PV systems and heat pump systems.

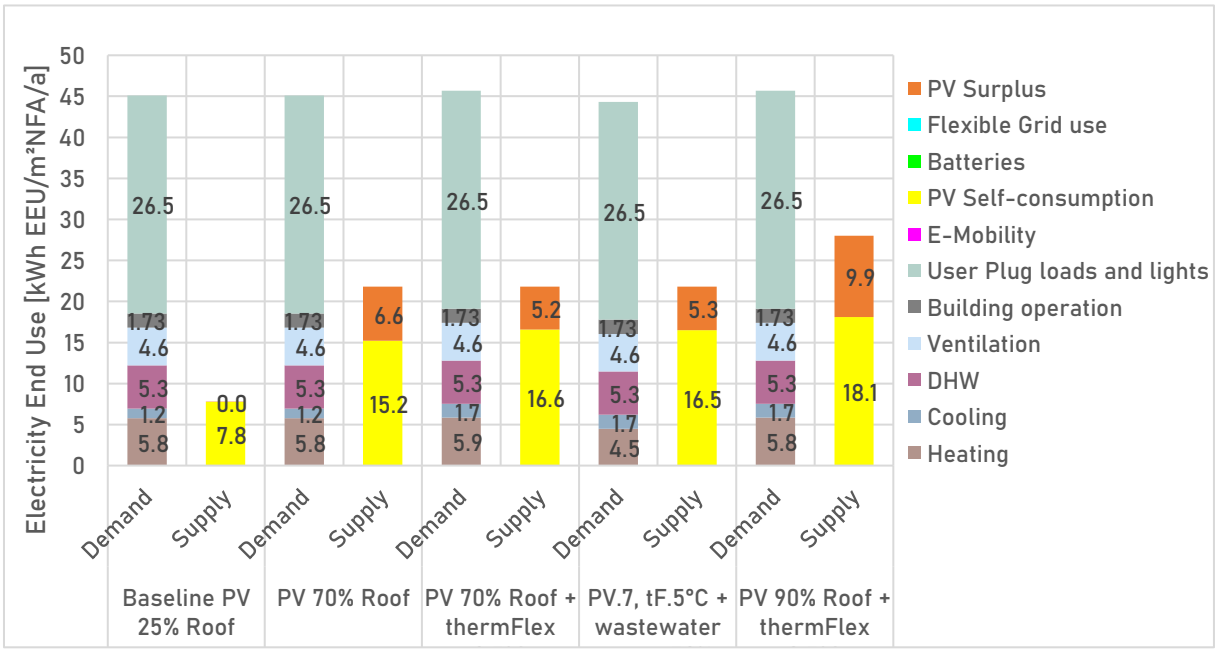


Figure 35 Energy End use balane Stockholm Royal Seaport

The following figure illustrates the resulting Primary Energy Balance, which needs to be in equilibrium between supply and demand to fulfil the PED requirement. On the right side of each scenario, the supply of the districts energy balance consists of three parts: the direct PV utilization on site (yellow), the credit of indirect PV utilization by PV feed in (orange, typically with a lower conversion factor during summer time) and the context factor due to district density (hatched red). Note that without the context factor, the energy balance cannot be positive with only roof mounted PV installation in any scenario. Even façade installation short of completely enveloping the buildings will not change this.

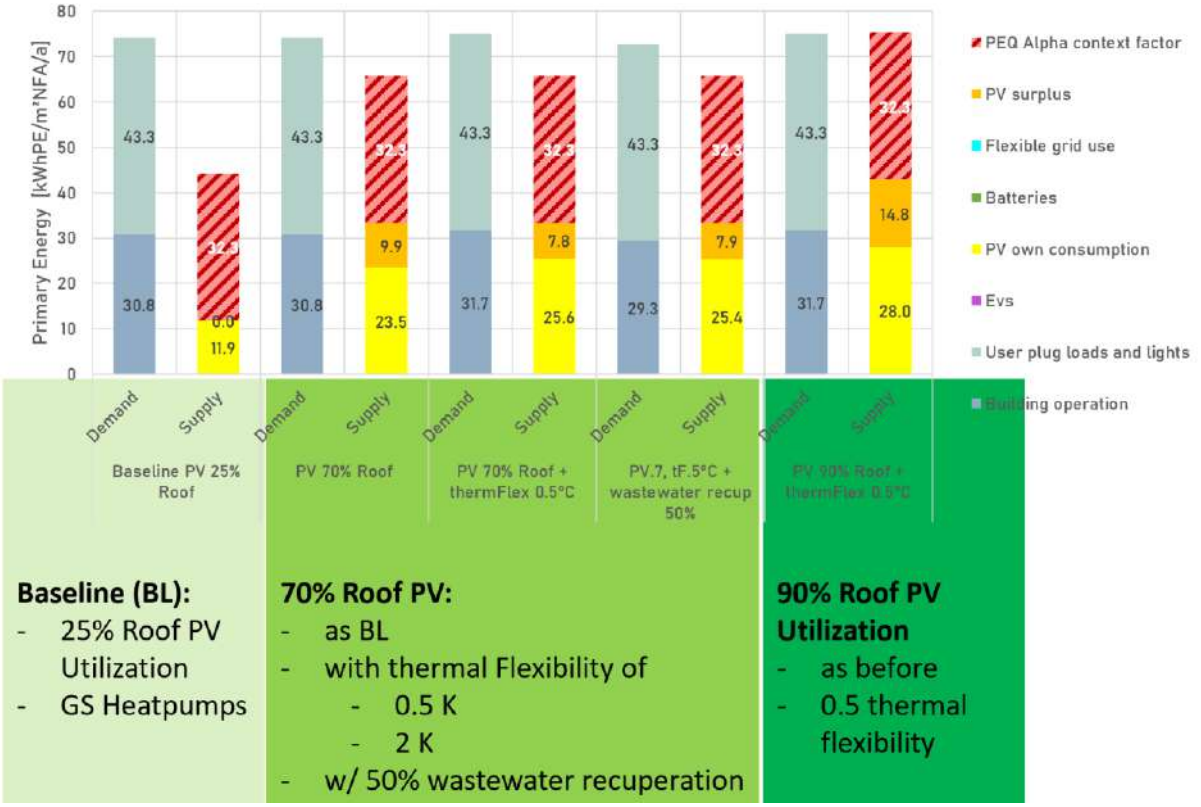


Figure 36 Primary energy balance for PED Alpha (Operation) Stockholm Royal Seaport

The conversion factors used are summarized here:

Table 13 Conversion factors Stockholm Royal Seaport

Conversion factors		
Average Electricity Primary Energy	1.6	kWhPE/kWhUE
Average Electricity GHG	0.146	kgCO2eq/kWhUE
District Heating var1	1.1	kWhPE/kWhUE
District Heating var2	0.33	kWhPE/kWhUE

Comparing the chosen HP system to a Supply via District heating depends on the chosen conversion factor: If the district heating conversion factor is below 0.4 kWhPE/kWhUE, the district heating has a slightly lower primary energy demand, above that it is higher. With typical factors of above one, the district heating scenario results in a doubling of primary energy demand.

Mobility

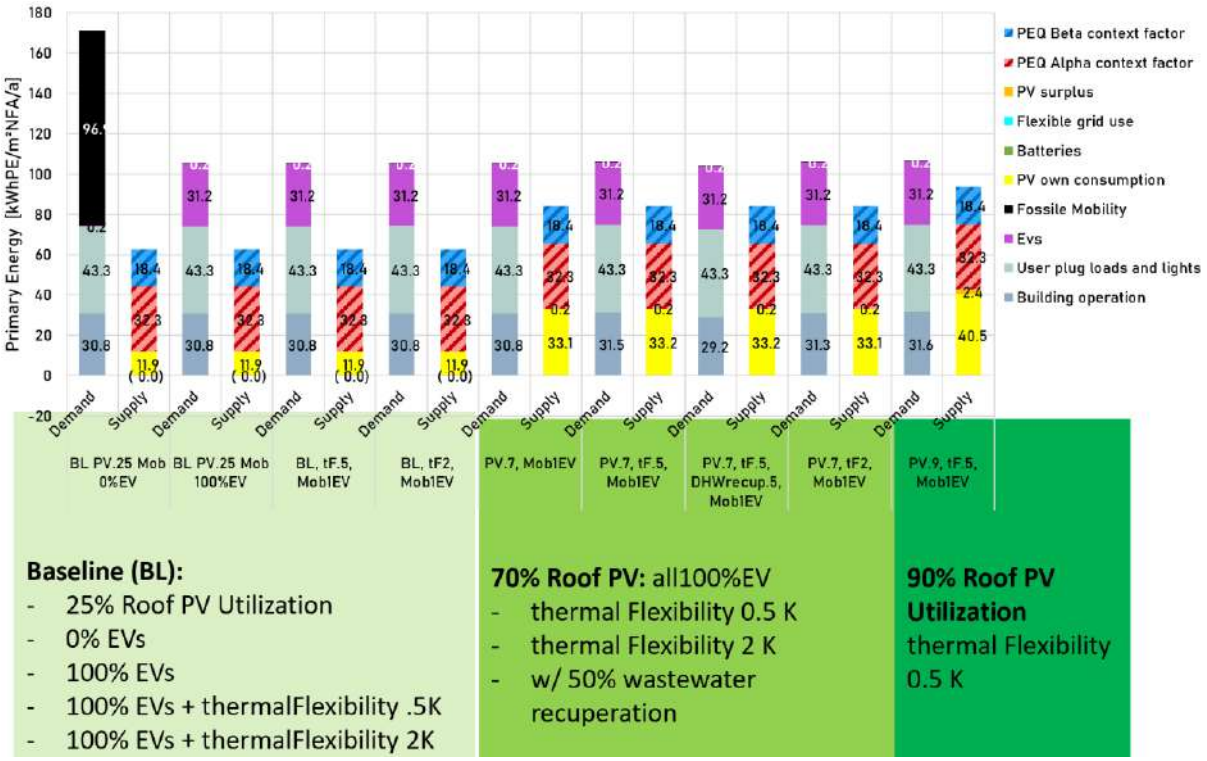
The results presented above were only considering the first layer of the PED definition operationalization covering District operation. In the following section, the approach is extended to include energy demand and supply context for everyday mobility in the district as well.

Table 14 Mobility Parameters

Mobility parameter		
Residents	**10334**	**PAX**
EVs (on site, domestic and incoming)	**3000**	
Annual individual motorized mobility	**11000**	**Person-km/a**

The following figure presents the same scenarios as discussed above, with the addition of energy demand of motorized individual mobility in two variations:

- Only first, leftmost demand: All individual motorized mobility is realized with fossile fuels
- All other variations utilize 100% Eletric vehicles (purple)



The blue bar on the supply side is the context factor for mobility, that can be derived from analyzing the surrounding Swedish energy system and allocating renewable energy budgets to the sectors industry, public transport from centralized sources (Wind, Water, biomass), resulting in a credit per floor area for district mobility coverage. Note that in this allocation approach, mobility of both public transport as well as industry and commerce is already accounted for in the corresponding sectoral allocation. One motivation for this is that the district is not the main beneficiary or cause of this type of mobility.

3.2.2 Brussels North District

The Austrian simulation model and definition framework was tested on the Brussels case in a simplified assessment, using four example blocks representing major typologies present at the district. They show the quantitative differences in the primary energy balance between the Austrian green field developments and the possible decarbonization paths of the Brussels North District. It can be used as a starting point for an operational definition of the energy and emissions quantification within the Brussels definition.

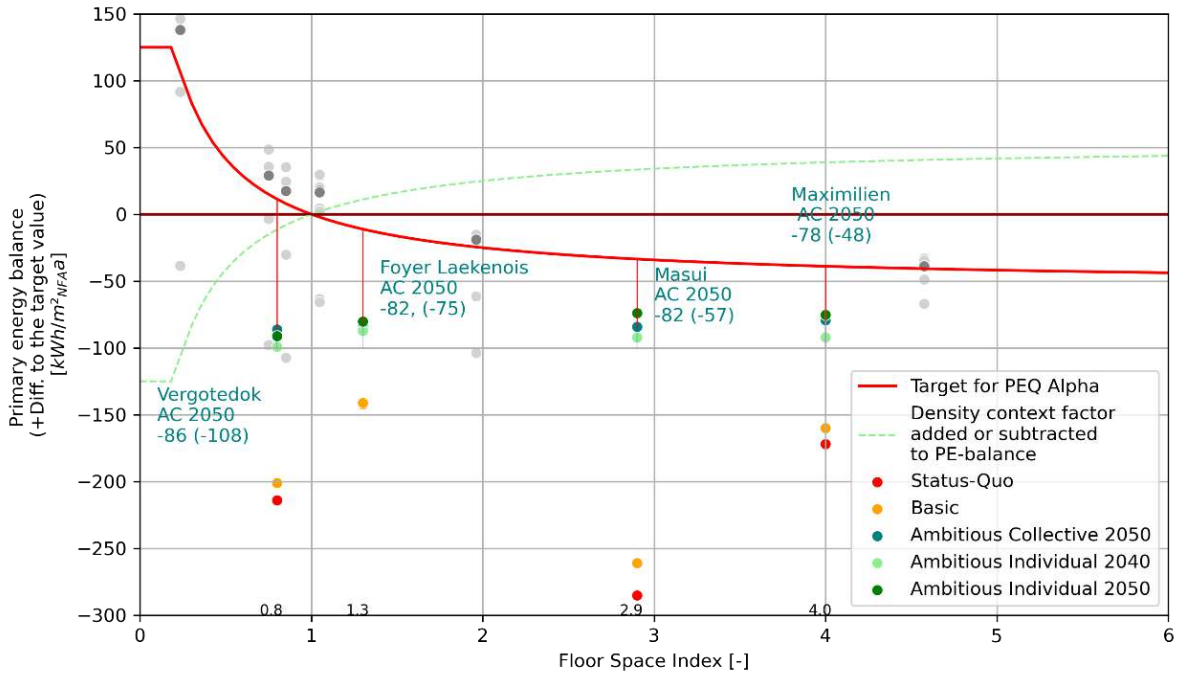


Figure 37 Primary Energy Import-Export Balance for four homogeneous areas of the Brussels North district as a function of the area density (lower to higher from left to right).

The red vertical line and numbers in brackets denote the difference from reaching the "PED Alpha" Definition of a PED, including only building operation and user electricity, excluding mobility and embodied energy. It can serve as a first indication of a possible context-factor of "heritage" as an allocated budget, offsetting the diminished energy balance potential due to the nature of the brown field development.

3.2.3 Vienna Seestadt Aspern

Following illustration gives a bird eye view of the Vienna Seestadt Aspern North District (grey). The example plot F12 investigated in this assessment is colored in white.



Figure 38 Vienna Seestadt Aspern North District (grey).

It is important to distinguish the buildable plot area from the total district area, that also includes spaces for traffic and green and blue infrastructure, which is typically not available for renewable solar energy generation. The district characteristics are summarized in the following table:

Table 15 District characteristics

Reference Areas		
District Area	102	ha
Gross Floor Area	1 909 056	m²
District Plot Area	759 345	m²
Building Storeys (avg)	5.3	-
Floor Area Ratio [FAR]	2.70	-
Net to Gross Floor Ratio	85%	-

As can be seen, the district is predominated by two parts residential and one-part office and residential usages, with the remaining ten percent mostly used by retail. The entire district is comprised of almost two million square meters gross floor area. The density of the district is rather high with a FAR of 2.7 and an average building storey of just over five.

As the entire district is too large to be evaluated with the existing PED assessment operationalization, only one block – F12 – is modelled instead pars pro toto. It has the same FAR and compactness as the entire district and can represent the district's envelope and shape most suitably. The block is modelled to have the same usage share as the entire district. The results obtained as such are related to the gross/net floor

area and can serve as an estimate for the entire district. The Distribution of plot densities in Seestadt Aspern Nord measured as floor area ratio (i.e. gross floor area over buildable plot area) are also shown in the following figure. As can be seen, most blocks congregate between 2 and 3.5 FAR. The average FAR for the entire district is 2.7.

Table 16 District Usages

District usage	Share		Aspern Nord Total	Representative Block F12	
Residential	60.1%		1 146 475	14 570	m² GFA
Office and Commercial	31.1%		594 648	7 557	m² GFA
Primary School	0.9%		17 303	220	m² GFA
Secondary School	0.0%		-	-	m² GFA
Retail Supermarket	2.4%	30%	45 189	574	m² GFA
Retail other	5.5%	70%	105 441	1 340	m² GFA
Total	100.0%		1 909 056	18809	m² GFA
Density (FAR)			2.7	2.7	-

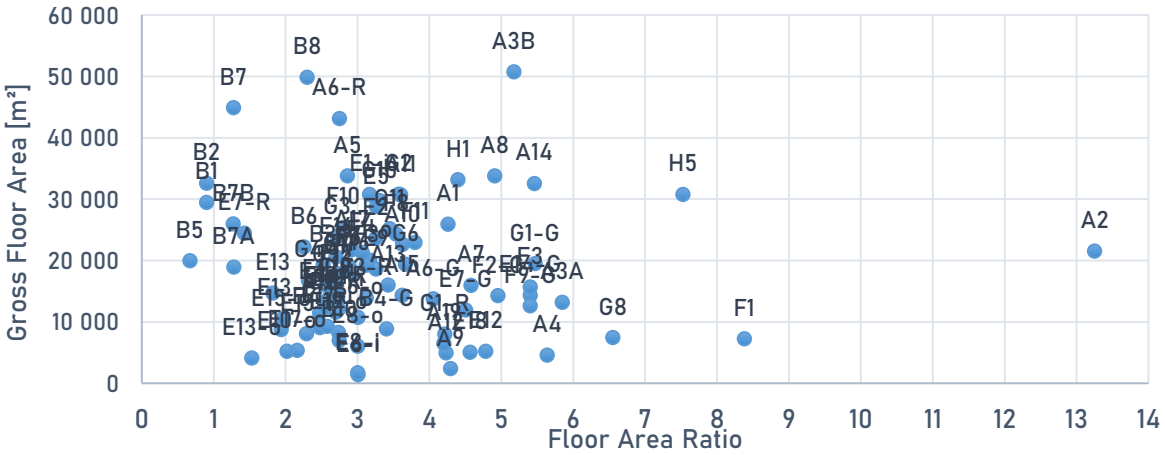


Figure 39 Distribution of plot densities in Seestadt Aspern Nord

The following table shows the density context factor of the district. According to the example definition operationalization, this factor needs to be included in the primary energy balance of the district to assess whether it achieves a PED positive energy balance. Another way to think about this is that due to its density, the district needs to only achieve a primary energy balance of negative -27.2 kWh PE per square meter gross floor area. Note that this is only the case for dense urban districts. For neighbourhoods with low density (with FAR < 1) the density context factor is negative, meaning their energy balance must be that much more positive.

Table 17 Density context factor Vienna Seestadt Aspern

PED Alpha		
Context Factor due to Density		
	27.2	kWh/m²GFAa
	32.1	kWh/m²NFAa

As introduced in Section 3.1 in principle and 3.1.6 in particular, this context factor represents an effort-sharing allocation from low-density districts to high-density districts, reducing the energy balance target of the latter by the given amount of primary energy.

PV Model

The PV Modelling was carried out in 2 steps: First, a theoretical PV system application with a maximum (100%) utilization of the available roof areas was modelled as can be seen in the figure below. Second, the resulting yields were incorporated in the district energy simulation in 3 variations, each scaling the yield of the 100% example roof to the total district roof area with a given utilization rate: The analyzed variations were:

- 25% of district roof areas utilized by PV Systems
- 70% of district roofs areas, or façade mounted PV with equivalent yield
- 90% of district roofs areas, or façade mounted PV with equivalent yield
- 105% of district roof areas, implying additional façade mounted PV with equivalent yield
- 120% of district roof areas, implying additional façade mounted PV with equivalent yield

PED Assessment Results

Results were obtained for both assessment boundaries Alpha (district operation and use) and Beta (also including individual private mobility) separately in a number of variating system configurations. The configurations are shown in the following table and can be characterized by four main differences:

- PV Utilization ("PV")
- Thermal Flexibility; characterized by the permitted indoor temperature("ΔT")
- "Flexible use of offsite RES"
- "Other Features"; specified individually

Table 18 Investigated system configurations

Variation Name	PV[8]	ΔT[9]	Flexible use of offsite RES	Other features
Baseline PV 25% Roof	25%	0	-	
BL + thermFlex 2°C	25%	2K	-	
BL + 2°C + WPS	25%	2K	✓	
BL + thermFlex 0.5°C	25%	0.5K	-	
BL + 0.5°C + WPS	25%	0.5K	✓	
PV 70% Roof	70%	0	-	
PV 70% Roof + thermFlex 2°C	70%	2K	-	
PV 70% Roof + 2°C + WPS	70%	2K	✓	
PV 70% Roof + thermFlex 0.5°C	70%	0.5K	-	
PV 70% Roof + 0.5°C + WPS	70%	0.5K	✓	
PV.7, tF.5°C + wastewater recup 50%	70%	0.5K	-	50% wastewater heat recuperation
PV.7, tf.5°C + 0.3kWh/kWp Battery	70%	0.5K	-	Electric Battery: 0.3 kWh/kWp
PV 90% Roof + thermFlex 0.5°C	90%	0.5K	-	
PV 105% Roof + thermFlex 0.5°C	105%	0.5K	-	

[8] Percentage of Gross Roof Area coverage with East-West facing 15° inclined modules

[9] Thermal Flexibility provided by pre-heating or pre-cooling to 0.5 or 2 Kelvin above/below the minimum required setpoint temperature of 22°C in winter and 25°C in summer. Note that this control always overfulfils the setpoint, and never undershoots the desired setpoints (warmer than set in winter, and summer vice versa)

PV 120% Roof + thermFlex 0.5°C	**120%**	**0.5K**	-

The Baseline scenario included *25% of roof utilization for PV,* as well as the thermal parameters stated above with *a ground source heat pump system* for heating and cooling of the district.

The second scenario included PV utilization on 70% of the districts roof areas with 2 notable extensions:

- Introduction of ***thermal flexibility***: At times of excess renewables in the district, the setpoint temperature is raised during heating and lowered during cooling by a given delta (0.5 and 2 Kelvin respectively) to "save" some of the excess renewables from the PV system as heating/cooling in the very well insulated and thermally inert building mass via the HVAC system. As can be seen I the following diagram, even a thermal flexibility of just 0.5K, which is typically unnoticeable by users as it has minimal effect on indoor air temperature, can increase PV Self-consumption from 17.7 to 9.4 kWh/m²NFA and reduce the PV excess fed into the grid in half. A larger thermal flexibility of 2K has no significant further increase in self utilization, as it is already with 0.5K at 90% at a very high ratio.
- Introduction of ***flexible use of offsite RES***: Leads to a further decrease of regular grid electricity from around 27 kWh/m²NFA/a by a quarter (7-8 kWh/m²NFA/a in flexible grid use, shown in the following diagram in cyan).

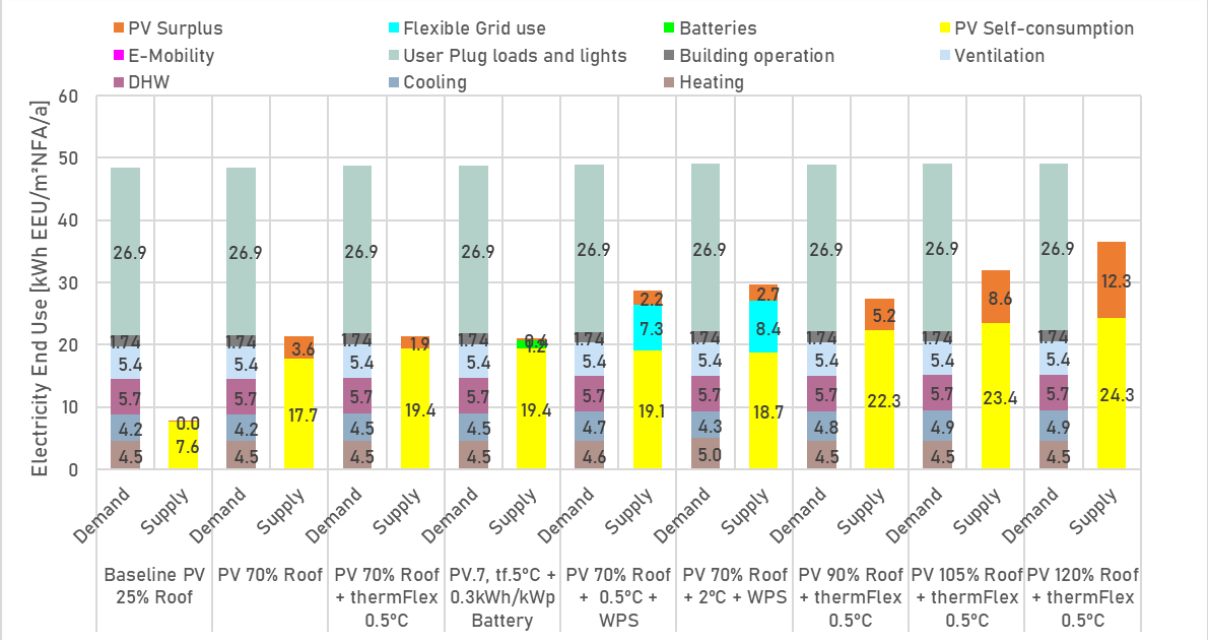


Figure 40 Electricity end use Vienna Seestadt Aspern

As can be seen 70% PV roof utilization is sufficient to cover the district's HVAC energy demands, but insufficient in covering the user plug loads and lighting demands. The latter is actually not possible onsite at all with only roof mounted PV systems and heat pump systems. Additional façade mounted PV systems amounting to approx. 13 kWh/m²NFA/a would be necessary.

The following figure illustrates the resulting Primary Energy Balance, which needs to be in equilibrium between supply and demand to fulfil the PED requirement. On the right side of each scenario, the supply of the districts energy balance consists of three parts: the direct PV utilization on site (yellow), the credit of indirect PV utilization by PV feed-in (orange, typically with a lower conversion factor during summer time) and the context factor due to district density (hatched red). Note that without the context factor, the energy balance cannot be positive with only roof mounted PV installation in any scenario. Even façade installation short of completely enveloping the buildings will not change this.

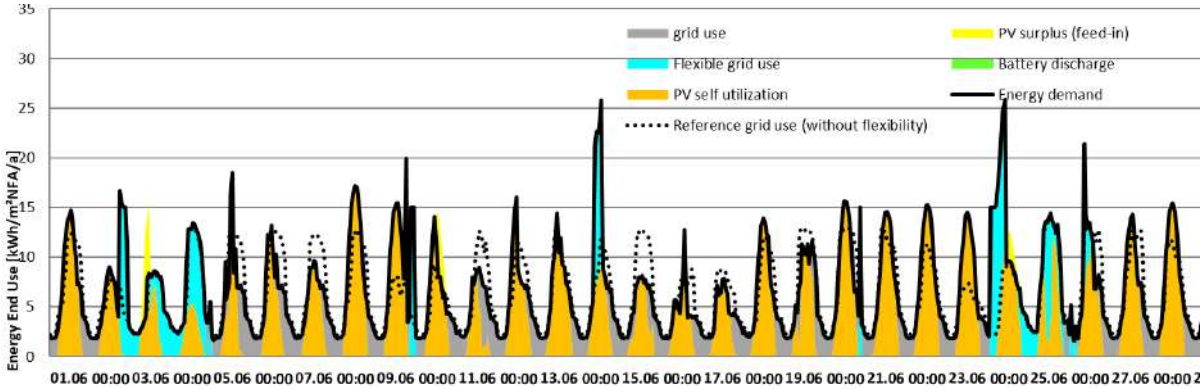


Figure 41 Temporal Illustration of electricity energy end use in June.

The above figure illustrates the flexibility of the PED in action during a sunny summer month: Flexible grid support events, where the PED uses additional offsite RES such as wind peaks to reduce energy demand relative to a reference (dotted line) later are shown in cyan. The thermal flexibility also enables higher direct utilization rates of PV (in orange) compared to the baseline (dotted line), as can be seen e.g. on the 6th, 22nd and 23rd of June Both onsite (PV in orange) and offsite (wind peaks in cyan). RES is flexibly used and mostly stored thermally in the building mass, consequently reducing the energy demand down the line (compared to the reference energy demand indicated as a dotted line). As can be seen, with this flexible control even in clear sunny conditions and high roof utilization rates of 70%, the PV surplus can be held to a minimum.

A positive energy balance for operation (PED Alpha) can be achieved in two main ways:

- 105% of roof PV utilization or higher (or equivalent façade mounted systems)
- 70% roof PV Utilization and flexible grid use

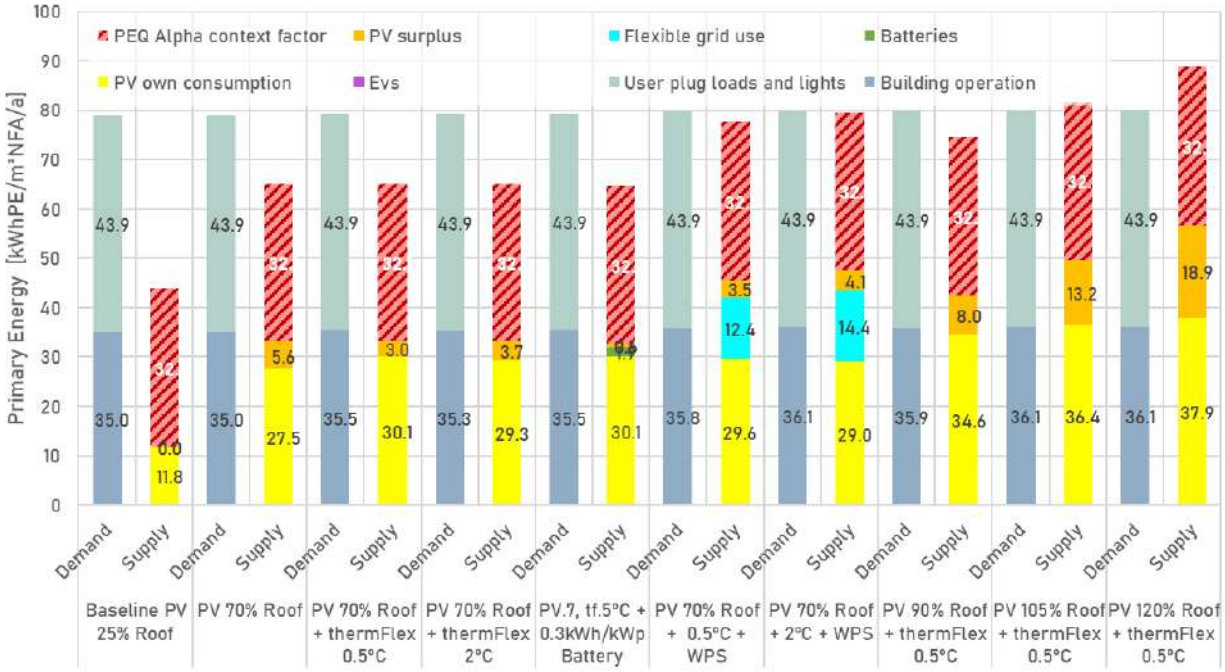


Figure 42 Primary Energy Balance Alpha (Operation) for selected system configurations

The conversion factors from energy end use to primary energy, as well as the context factors used for the balance are summarized in the following table:

Table 19 Conversion and Context Factors used in the Primary Energy Balances Alpha and Beta

Conversion factors		
Electricity Primary Energy (annual average [10])	1.630	kWhPE/kWhUE
Electricity GHG (annual average)	0.231	kgCO2eq/kWhUE
Context factors		
Alpha (Density)	32.06	kWh/m²NFA
	27.25	kWh/m²GFA
Beta (Mobility)	23.87	kWh/m²NFA
	20.29	kWh/m²GFA

Mobility

The results presented above were only considering the first layer of the PED definition operationalization covering district operation (PED Alpha). In the following section, the approach is extended to include energy demand and supply context for everyday mobility in the district as well. The following figure presents the same scenarios as discussed above, with the addition of energy demand of motorized individual mobility in two variations:

- All individual motorized mobility is realized with fossil fuels (in black; only for the first, leftmost configuration)
- 100% Electric vehicles (purple)
- 50% Fossil (black), 50% EV (purple) split (rightmost configuration)

The following figure illustrates the feasibility of achieving a PED including mobility with fossil and electric motorized individual traffic and a mix. Note that apart from cutting primary energy demands more than in half due to higher vehicle efficiency, the use of EVs has the additional advantage of providing large amounts of grid flexibility through their battery charging (cyan): If the pool of EVs in the district is large enough, they can be almost exclusively charged at times of renewable oversupply in the grid.

[10] The assessment is carried out using monthly conversion factors

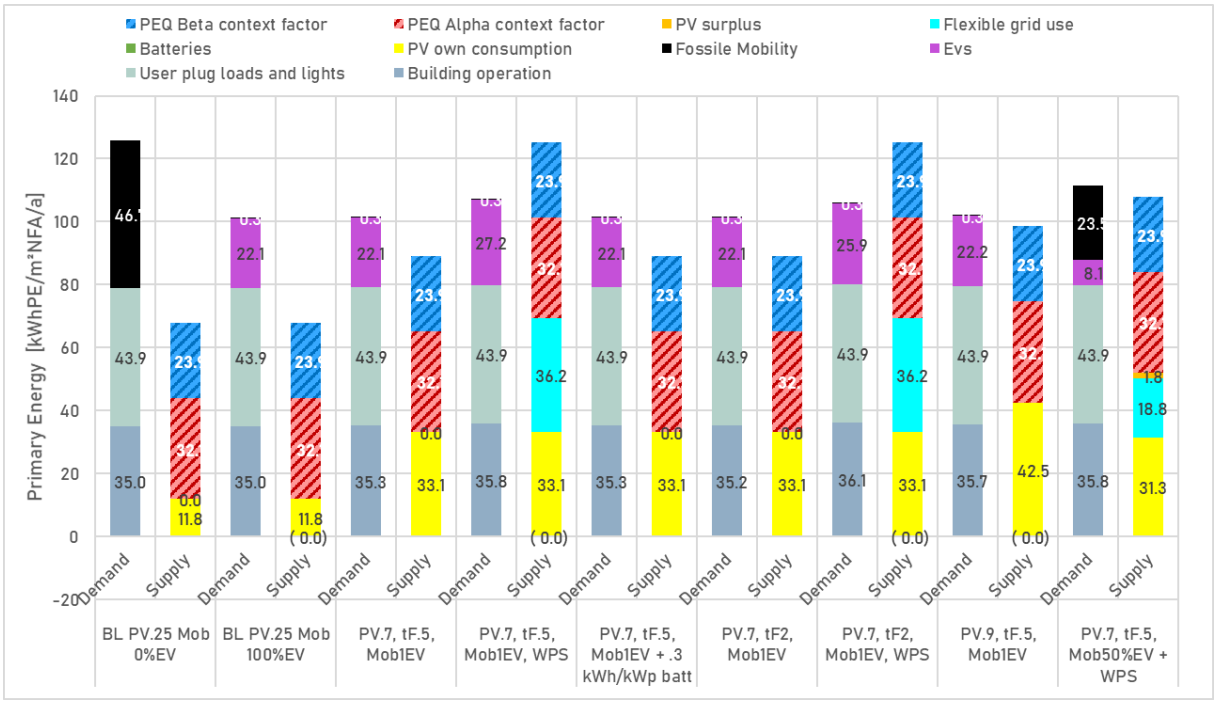


Figure 43 Primary Energy Balance Beta (Operation and Mobility) for selected system configurations

The model shows that with the very highly efficient system configurations a positive energy balance can only be achieved with EV shares of 60% or higher without additional measures such as reducing the annual volume of motorized individual mobility, efficient user appliances, other forms of onsite renewable generation such as waste water recuperation, etc.

The blue hashed bar on the supply side represents the context factor for mobility, that was derived from analyzing the surrounding Austrian energy system and allocating renewable energy budgets to the sectors industry, public transport from centralized sources (Wind, Water, biomass), resulting in a credit per floor area for district mobility coverage[11]. Note that in this allocation approach, mobility of both public transport as well as industry and commerce is already accounted for in the corresponding sectoral allocation.

[11] As described in (Schneider et al., 2023)

4 Observations



This section aims to give an overview of the discussions around PED definitions and the observations on selected key questions raised.

4.1 Which districts readily achieve a positive energy balance without context factors?

The previous example section illustrates that all three districts do not achieve either a positive primary energy balance or balance of energy end use and local supply, whether mobility is included or not. This is mostly due to the high density and as many stakeholders agree this needs a mechanism to allow dense urban districts to achieve a PED. As discussed at length in (Schneider et al., 2023), here is a rundown of the most common perspectives. It mostly comes down to your idea of general achievability of "PED". Which districts should achieve it?

If common balances must simply be positive -> generally mostly low density green field developments

With the example assessments of the previous section it is safe to subsume that in the absence of uncommon local potentials, only green field developments with low density can achieve a positive energy balance of their space use. Mobility is typically too energy intensive to be included in the balance without offsetting considerations.

This holds also true for the inclusion of embodied energy, for both green and brown field developments, although in direct comparison the latter can significantly close the gap to the former, or even perform better, depending on how renewable building materials are accounted for (i.e. how much carbon they can offset due to storage effects).

This illustrates the paradox of the arbitrary positive balance: Even though including embodied energy into the assessment would be desired as it more accurately measures what we care about – climate neutrality -, in practice it can not be done because it seems unattainable or required customized tricks.

Districts that unlock their unique potential

Many PED demo sites pursue ambitious plans to maximize the utilization of local renewable energy - as they should, which leads to innovative measures such as using industrial waste heat or residual heat from waste water as free energy source that enables a positive energy balance. (Hedman et al., 2021) As these measures are not typically generalizable and readily available for districts that are not situated in proximity to a heaty industry or a main collection sewer, they are also excluded from reaching the same energy balance, even though they apply the same solar utilization and efficiency measures.

This might just be a labelling issue, but from the perspective of a zero carbon mission it seems counterproductive to offer a special label "PED" to districts that unlock their unique potentials and deny it to those that just do the "regular" measures very ambitiously but ultimately not to the same balancing effect.

But it can also be argued the other way around: That it is important to utilize these unique local energy potentials and awarding districts that do so the district PED label.

In terms of climate neutrality on the scale of the building sector in general however, it is clear that doing "the existing boring measures" such as insulation and PV ambitiously on large scale has a much bigger effect than realizing situational potentials.

If all built environments should have a way of achieving PED

Although not all or even many stakeholders agree, there is an argument to be had that a PED definition should encapsulate the effort sharing for the entire building sector with Paris-compatible quantitative targets for all typologies within in. There are two ways in which this can be achieved:

1. Appropriately parameterized context factors that encode achievability and "set in context quantitatively"

Or

2. Individual goal setting on a district by district level, i.e. some form of processual approach and definition

The fact, that this requirement is not typically present at the start of a PED definition process, but rather emerges organically with the question on PED definition purpose and overarching goals. This indicates that the initial PED definition framework (JPI Urban Europe, 2019) is only really viable for frontrunner lighthouse projects but not suitable for follow-up replication and wide-spread application.

4.2 The PED as a Process

The development of a district is a delicate and complicated process. The care going into planning and organizing this process is one of the biggest predictors of a successful development. It is therefore important to not let arbitrary definitions dictate goals while losing track of the process itself.

It is however also important to not pit dynamic process design and normative goal setting against each other. Instead, they should be seen as two separate areas that both play important roles, but differently so in the interface between the various stakeholders:

When developers and planners interface with users, inhabitants and stakeholders of a certain district, quantitative energy goals, with their scientific derivations on positivity and climate neutrality are often not just unimportant but an outright obstacle in communicating vision and onboarding key participants.

This changes when you consider the interface between city and state administrations, legal bodies and district developers: Communicability of achievements is still important and should not be obfuscated by elaborate definitions. But it is also important to connect districts achievements and aspirations to energy and climate goals on a higher level in preparation for legislative normalization and certification. This need not concern the district developers "on the ground", but incidentally they are the ones currently involved in the PED projects explored to date.

4.3 Why different and open definition aspects is a good thing

If one thing has become clear about PED definition it is that there is no one uniform definition that can satisfy all stakeholder needs. Rather, PED definition must reflect the different areas and levels of target design and district planning. It seems clear now, that striving for a common definition is not a viable way forward. Instead, the differences in definition needs and perspectives must be embraced and studied in terms of classification: Different aspects of PED definition have already underwent this scrutiny, such as

system boundaries and calculation methodology of the energy and emission balances and the categorization of KPIs into thematical groups such as environmental, societal, economic, legal, etc.

It seems useful to expand this research onto perspectives of regionality and as instruments of normative goals such as the Paris Agreement and their national, regional and municipal interpretations.

4.4 What about...

Comparing this document with existing research and project reports on PED definition, the main focus of this report is obviously on the observations on the PED definition approach itself and a quantitative definition of PEDs through a positive energy balance. This is to not reiterate already established findings and work but rather contributing new aspects to the discussion. Nevertheless, there are a number of key topics that we want to address here in all brevity:

4.4.1 ...Economic aspects

Although important for project feasibility and implementation, both resource and cost efficiency were never explicitly discussed within the Cities4PEDs project PED definition discussion. In hindsight it might have been beneficial to include this dimension into the considerations as it could give directions and thresholds of ambitions defined by cost or resource efficiency.

4.4.2 ...Qualitative Aspects

The quality of district development is hardly defined by its energy performance. As such, there is also an argument for the definition of PEDs to be centered on the assessment of KPIs regarding quality of life and other social and inclusivity aspects, as other projects have shown. Although a valid approach, one must keep in mind that there is a difference between taking certain aspects into assessable account in the development of a specific district on one side, and taking that aspect as a definitory criterion into the definition itself that should be imposed on all districts.

4.4.3 ...Existing buildings and renovation

The renovation sector was largely excluded from the analysis, although it will of course play the most important role in the coming years. In principle, the definition and operationalization presented here can also be applied to refurbishment of existing buildings and, as initial studies show, can sometimes be achieved. However, it is clear that especially the PEQ Alpha system boundary with the relatively high implicit requirements for energy efficiency and local renewable energy production will not be easily - if at all - achievable for all existing quarters. Here, apart from the building density, the credit must also be examined and, if necessary, parameterized depending on additional parameters such as the building age or the settlement typology.

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8.1 PED Concept Stakeholder Interview Questions

Question 1

			Fill in the Answer of your interview partner
As a	*role/profession*	**the PED concept is useful to me**	because... if...

Question 2

	The role / profession of your interview partner		
As a	*role/profession*	**, i expect from a PED definition that because**	*fill in the answer of your interview partner* ...

Question 3

I will stop using or pursuing the PED concept if	...

Question 4

My greatest pain dealing with PEDs right now is	...

8.2 Assessment model parametrization

8.2.1 Stockholm Royal Seaport Loudden

PV Model

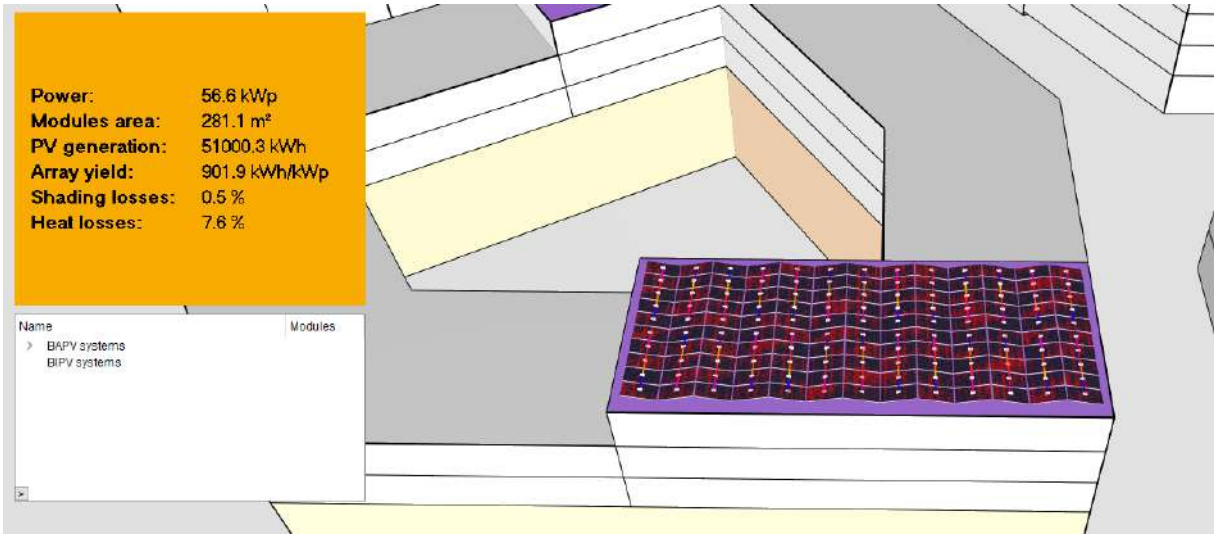


Figure 44 Example of a PV system application with a **100%** roof utilization using east-west oriented modules with 15° of inclination.

Table 20 PV system configuration and parametrization

Parameter		
Available Roof Area	124.313	m²
Weather	Stockholm Bromma 2020	
Source	meteonorm 8	
Annual Global Horizontal Irradiation	988	kWh/m²
Module	Sunpower SPR-435	
	1046 x 2067	mm
Annual Yield	901	kWh/kWp
Max Utilization (100% Roof areas)	19	MWp
	138	kWh/m²Roof
	26.5	kWh/m²GFA
Actual Utilization	25%	
Installed Power	4.754	MWp
PV Yield	6.6	kWh/m²GFA
	7.8	kWh/m²NFA

Energy Model Parameters

The following tables give an overview of the assumed properties of the buildings and the energy system used in the thermal and electric energy system simulation:

Table 21 Model parameters Stockholm Royal Loudden

Thermal Hull				
Thermal Transmittance		Wall	0.15	W/m²K
		Windows	0.87	W/m²K
		Roof	0.1	W/m²K
		Floor	0.16	W/m²K
Visible Transmittance			0.7	-
Thermal active mass			204	Wh/m²K
Ventilation				
Heat recovery			90%	
Energy Supply	Heat Pump (ground source, water)			
Heating	COP		4.25	
	Transmission losses		5%	
	Temp Set Point		22	°C
Cooling	COP		4.75	
	Transmission losses		5%	
	Temp Set Point		25	°C
DHW	COP		3	
Energy Use				
Heating			**23.2**	kWh/m²a
Cooling			**5.4**	kWh/m²a
DHW			**11.1**	kWh/m²a
Plug Loads	District avg		**26.5**	kWh/m²a
(per usage)	Residential		26.7	kWh/m²a
	Office		19.4	kWh/m²a
	Secondary and tertiary education		14.1	kWh/m²a
	Primary education and Kindergarten		6.1	kWh/m²a
	Retail Supermarket		30.8	kWh/m²a
	Retail		4.4	kWh/m²a

8.2.2 Vienna Seestadt Aspern

PV Model

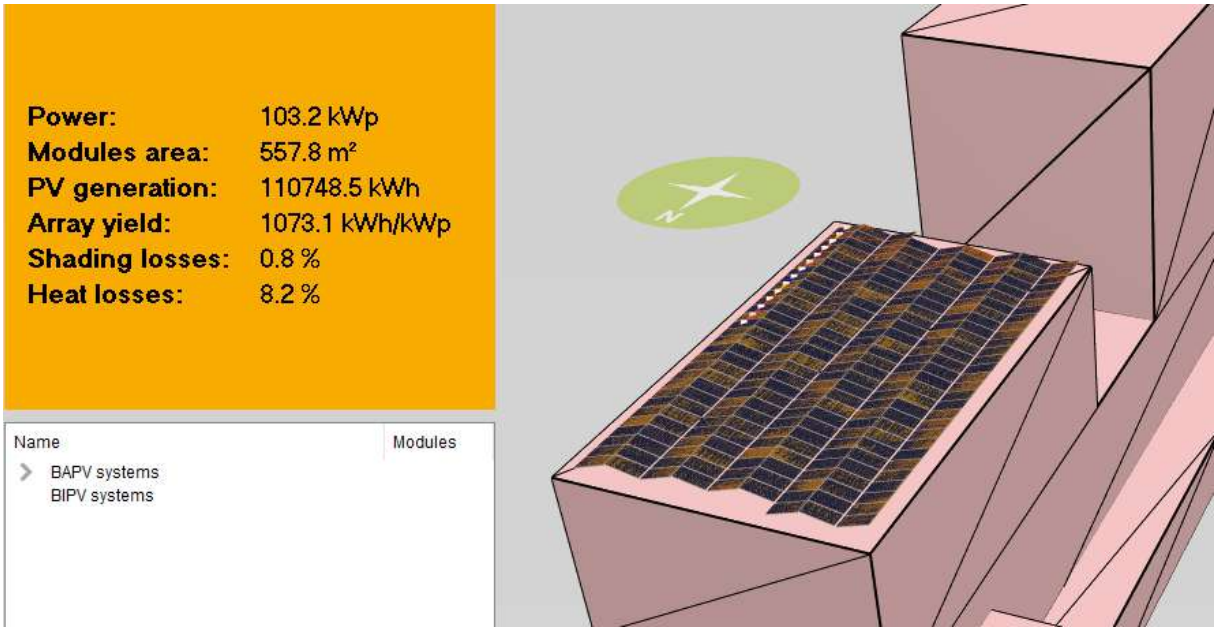


Figure 45 Example of a PV system application with a **100%** roof utilization using east-west oriented modules with 15° inclination.

The following table summarizes the PV system modelling parameters and resulting yields

Table 22 PV System parameters

PV Parameter		
Available Roof Area	311 394	m²
Weather	Vienna Hohe Warte 2020	
Source	meteonorm 8	
Annual Global Horizontal Irradiation	1199	kWh/m²
Module	Sunpower SPR-435	
	1046 x 2067	mm
Annual Yield	1071.85	kWh/kWp
Max Utilization (100% Roof areas)	45	MWp
	155	kWh/m²Roof
	25.7	kWh/m²GFA
Actual Utilization	25%	
Installed Power	18	MWp
PV Production	6.32	kWh/m²GFA
	5.37	kWh/m²NFA

Energy Model Parameters

The following tables give an overview of the assumed properties of the buildings and the energy system used in the thermal and electric energy system simulation:

Table 23 Model parameters

Thermal Hull			
Thermal Transmittance	Wall	0.15	W/m²K
	Windows	0.87	W/m²K
	Roof	0.10	W/m²K
	Floor	0.16	W/m²K
Visible Transmittance		0.7	-
Thermal active mass		204	Wh/m²K
Ventilation			
Heat recovery		90%	
Energy Supply	**Heat Pump (ground source, water)**		
Heating	COP	4.25	
	Transmission losses	5%	
	Temp Set Point	22	°C
Cooling	COP	4.84	
	Transmission losses	5%	
	Temp Set Point	25	°C
DHW	COP	3	
Energy Use			
Heating		18.2	kWh/m²a
Cooling		20.4	kWh/m²a
DHW		11.9	kWh/m²a
Plug Loads	District avg	24.7	kWh/m²a
(per usage)	Residential	26.7	kWh/m²a
	Office	19.4	kWh/m²a
	Secondary and tertiary education	14.1	kWh/m²a
	Primary education and Kindergarten	6.1	kWh/m²a
	Retail Supermarket	30.8	kWh/m²a
	Retail	4.4	kWh/m²a

B.3. Allocation of National Renewable Expansion and Sectoral Demand Reduction Targets to Municipal Level

Publication type: Journal paper

Authors: Simon Schneider, Edit Paráda, David Sengl, José Baptista, und Paulo Moura Oliveira.

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Contribution to the thesis:

Top-down allocation methodology translating national renewable expansion and sectoral demand reduction targets to the municipal level, preserving consistency with national decarbonization pathways.

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Allocation of national renewable expansion and sectoral demand reduction targets to municipal level

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Despite the ubiquitous term "climate neutral cities", there is a distinct lack of quantifiable and meaningful municipal decarbonization goals in terms of the targeted energy balance and composition that collectively connect to national scenarios. In this paper we present a simple but useful allocation approach to derive municipal targets for energy demand reduction and renewable expansion based on national energy transition strategies in combination with local potential estimators. The allocation uses local and regional potential estimates for demand reduction and the expansion of renewables and differentiates resulting municipal needs of action accordingly. The resulting targets are visualized and opened as a decision support system (DSS) on a web-platform to facilitate the discussion on effort sharing and potential realization in the decarbonization of society. With the proposed framework, different national scenarios, and their implications for municipal needs for action can be compared and their implications made explicit.

KEYWORDS

decision support system, energy transition, renewable expansion allocation, allocation models, energy transition visualization

1 Introduction

The starting point of this research is a simple question, often posed by municipal stakeholders in energy transition efforts: "If we [as municipality X] achieve certain decarbonization and energy balance targets, is that *enough*?" What is meant by "enough" can often vary, but typically is referring to *sufficient* action for nationally achieving the goals of the Paris agreement. As the scope is different (i.e., municipal and national), it cannot readily be answered for a single municipality without additional assumptions about the surrounding environment, i.e., the national energy system and the ambitions and achievements of the other municipalities (Krog, 2019). From this national perspective, not all municipalities need or are able to achieve a net zero greenhouse gas (GHG) or a completely renewable energy balance individually, but only collectively (Muñoz et al., 2023). If some—typically bigger—municipalities will not be able to cover their energy demands renewably themselves, there must be other municipalities to supply them. Municipal energy transition goals should in aggregate match or exceed national goals, and their renewable energy balances must be accordingly collectively balanced. But there is little to no framework to ensure this (Leal and Azevedo, 2016); Most research is focused on climate neutral cities—its carbon accounting is considering the interplay with the surrounding energy system predominantly in terms of individual offsetting credits, which are known to require further work (Huovila et al., 2022) and are often found to be insufficient (Reckien et al., 2018; Salvia et al., 2021).

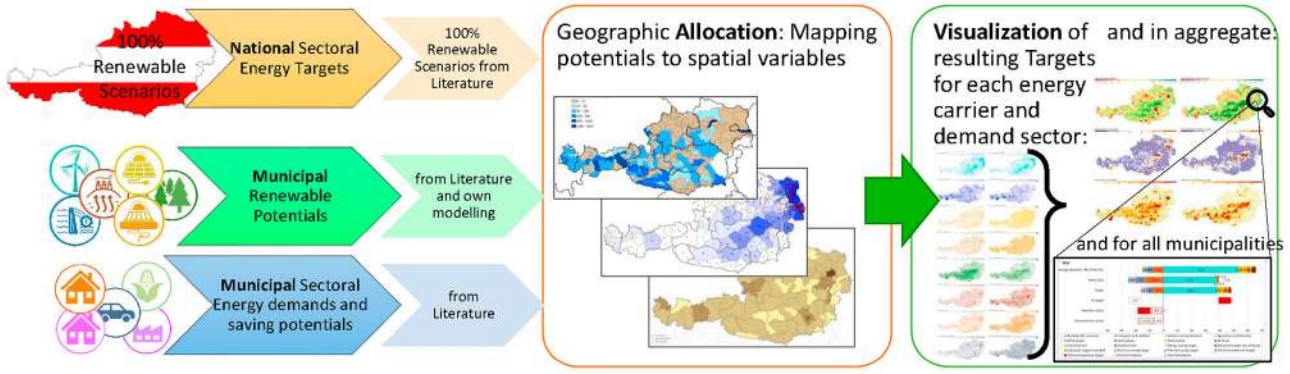


FIGURE 1
Method for sector-specific municipal targets.

Much research is focused on evaluating different municipal plans and on developing and applying methods to benchmark incremental progress rather than normative targets (Zhang, 2023). In fact, carbon accounting of municipalities varies widely in both scope and method, rendering resulting accounting balances mostly incompatible. Carbon accounting per capita e.g., can give a good indication for individuals but does not help municipalities in identifying climate goals that sufficiently contribute to achieving Paris 2050. The same is true for energy balances, where there are virtually no quantifiable municipal targets considering national energy transition scenarios with certain effort sharing and allocation mechanisms, with notable exceptions: (Bertoldi et al., 2018; Farghali et al., 2020). This is in stark contrast to the widely available literature on effort sharing *between* nations on reaching Paris 2050 (Brown et al., 2018; van den Berg et al., 2020; Vatalis et al., 2022). Instead, the public debate on if and which renewables should be expanded is often split into a local and a national discussion, with a strong presence of NIMBY sentiments ("not in my backyard"). What is lacking in public discussions on effort sharing and allocation of expansion and reduction targets is the visualization of effects and consequences of political allocation decisions. For example, if a national region does not realize their wind power potential, which many national decarbonization scenarios depend upon, due to political reasons, the remaining transition scenario components are negatively affected; The country would need to realize this renewable energy potential elsewhere or with different carriers or save more energy accordingly. The resulting differences in allocation and effort sharing should be readily available for visualization and inspection to ground this much needed discussion in quantifiable facts.

This paper aims to contribute to this by providing a theoretical structure and example implementation for Austria for the following items:

1. A flexible allocation structure for all energy carriers and energy demands based on available reference data connecting national to municipal potentials.
2. Quantitative targets for all energy balance components for each municipality in Austria, depending on these relative potentials for energy savings and installation of additional renewable energy sources.
3. A comprehensive visualization for both the municipal energy balance target composition, and a national comparisons of all municipal balance components to aid local development paths and facilitate discussion on decarbonization effort sharing.

To the knowledge of the authors, such an allocation of sectoral demand reduction potentials and renewable expansion targets for Austria has been calculated only on county level (NUTS-3) in the work of Baumann et al. (2021), but not on the level of individual municipalities, where it is most effective and needed (Pasimeni et al., 2014).

2 Methods and materials

The general method for deriving municipal energy transition targets involves several steps as outlined in Figure 1. The first three steps involve data acquisition and homogenization, then mapping of energy data to geographic attributes and finally results visualization.

2.1 Data acquisition and homogenization

First, a national baseline for renewable energy production and sectoral energy demand is established that should be achieved collectively by all municipalities. This is done by mapping existing energy transition scenarios for a 100% renewable energy system in Austria to a uniform balance structure. From there, any scenario can be selected for further study. If the categorization is not sufficient in scope or detail, additional supply and demand categories can be introduced by establishing the same geographical allocation associations as for the others. The selected scenario subsequently provides national renewable energy supply and demand targets and serves as a guideline for the transition to a 100% renewable country. Due to the uniformly defined data structure of the scenarios, they can also be readily exchanged for one another and compared. *Second*, the total national

renewable potential, as well as its geographic distribution is researched from literature where available and modeled based on geographic characteristics where necessary. *Third*, data on sectoral energy demand is collected, and saving potentials are identified through literature review. These steps determine the underlying potential for energy saving and energy demand for any given scenario.

The determination of renewable potentials at the municipal level is carried out in one of two ways: In general, the potential for renewable energy technologies is calculated using municipal level data. This covers photovoltaics (PV), solar thermal (ST) and biomass utilization. Only in cases where the required municipal level data is missing, the potentials are estimated by distributing the potentials only quantified at county level or above according to their relative overall size. This was done for wind power, hydropower, and ambient heat. A detailed account of all allocations used is given in Section 3.2. Municipal energy balances then require the juxtaposition of these renewable energy potentials with energy consumption and future demand data by key sectors. This also requires targets for energy savings and future energy demands within each sector. The necessary consumption and demand data structure was categorized into five sectors: Household (HH), Agriculture and forestry, Industry and production, Services, Transport, and mobility.

Table 1 provides an overview of the main materials and literature sources used to obtain the results. They will be referenced in their detailed sections.

2.2 Geographic allocation of renewable energy expansion and demand saving potentials via spatial variables

The energy supplies and demands—and their respective expansion and reduction potentials—are allocated geographically to municipal level by correlating them to spatial variables in a set of *allocation scenarios*. For example, a national scenario requires a certain energy target $T_{n,BM}$ of biomass utilization. This renewable supply is then associated with the geographic attribute of *forested area* $A_{n,\,F}$ on a national level. To derive the *municipal* target for biomass utilization $T_{\mathbf{m},BM}$, the national target $A_{n,\,F}$ is allocated according to the municipal share of the associated national variable, in this case the forested area. This ensures a balanced distribution of target contribution across different regions in respect to their relative potentials regarding renewable energy implementation, energy-saving opportunities, and other energy demands specific to each region such as industries. It also ensures that the corresponding targets of the underlying national scenario are collectively met. The main design parameters of these *allocation scenarios* are the choices of spatial variables and associated allocation functions. They must be transparent, easy to understand and readily changeable to ensure that allocations are comprehensible and can be discussed and compared. Table 2 introduces the three main resulting energy balance quantities: The current Status-Quo ***S***, the technically achievable potential ***P***, and the resulting annual Target ***T***. Each of these quantities are available for all levels of geographic aggregation ***g***, all renewable energy supply carriers ***e***, as well as all sectoral demands ***D***. These designations are consistently used throughout this paper. The *geographical* indices and the nomenclature of the energy supply carriers ***e*** and sectoral energy demands ***D*** is also summarized in Table 2 and specified in the corresponding sections.

The allocation employs the principle of equal effort, wherein the municipal target for any component of the energy balance (renewable supplies and demands) is equal in share to its potential. The municipal potential is determined through two distinct methods (Equations 1, 2).

$$P\frac{Target\ \mathbf{T}_{municipality\ \mathbf{m},\quad energy\ carrier\ \mathbf{e}}}{Potential\ \mathbf{P}_{municipality\ \mathbf{m},\quad energy\ carrier\ \mathbf{e}}} = \frac{Target\ \mathbf{T}_{national\ \mathbf{n},\quad energy\ carrier\ \mathbf{e}}}{Potential\ \mathbf{P}_{national\ \mathbf{n},\quad energy\ carrier\ \mathbf{e}}} \tag{1}$$

$$\mathbf{T}_{n,e} = \sum_{m} \mathbf{T}_{m,e} \tag{2}$$

When municipality-specific data is accessible, the potentials of renewable energy generation are calculated according to specific energy intensities, which link them to the designated reference area (Equation 3):

$$Potential\ \mathbf{P}_{\mathbf{m},\mathbf{e}} = Reference\ Area\ \mathbf{A}_{m,e} * Energy\ intensity\ \mathbf{f}_{e} \tag{3}$$

In cases where such data are not available, area ratios between the municipality and county level are used to derive the potential from the county level (Equation 4):

$$\mathbf{P}_{\mathbf{m},\mathbf{e}} = \frac{\mathbf{A}_{m,e}}{A_{c,e}} * P_{c,e} \tag{4}$$

The future sectoral energy demands are calculated using a sector-specific reduction factor (that can be linked to a reference area, Equation 5):

$$Demand\ \mathbf{D}_{\mathbf{m},\mathbf{e}} = Consumption\ \mathbf{C}_{m,e} * reduction\ factor\ \mathbf{f}_{e} \tag{5}$$

2.3 Results visualization

Finally, the results of the geographic allocation including the municipally distributed targets are visualized through interactive balance charts and maps on the web as a decision support system (DSS). This allows a comprehensive and visually appealing representation of the allocated potentials and targets, providing stakeholders and decision-makers with a clear understanding of the spatial distribution and progress toward achieving climate-neutral objectives. By leveraging the power of data visualization, the approach enhances transparency, facilitates communication, and fosters informed decision-making in the pursuit of a sustainable and climate-neutral future. The visualizations are published under open source policy and is available under https://observablehq.com/@editp/allocation-of-national-decarbonization-targets-to-municipal-level-in-austria.

3 Results

This section presents three intermediary results required to estimate the resulting energy balance targets on municipal level:

TABLE 1 Overview of materials and data used.

Data type	Granularity				Renewables						Sectors				Specification	References
	National	County (NUTS-3)	Municipality (LAU)	Per reference area	Wind power	Hydropower	Photovoltaic	Solar Thermal	Biomass	Ambient heat	Building	Mobility	Industry	Agriculture		
Energy balance targets	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100% RES without growth 100% RES with growth	Streicher et al., 2010
Energy balance targets	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100% RES w/growth w/o deep geothermal	Schneider et al., 2020
Renewable supply potentials		✓			✓										Technical Potentials	Winkelmeier et al., 2019
Renewable supply potentials		✓				✓									Technical and economic potentials	Fuchs, 2018
Renewable supply potentials	✓			✓			✓	✓							By building type	Fechner, 2020
Renewable supply potentials	✓			✓					✓						*Energiemosaik,* Based on forested area	Abart-Heriszt et al., 2020
Renewable supply potentials		✓								✓						Stanzer et al., 2010
Status Quo		✓			✓											Winkelmeier et al., 2019
Status Quo		✓				✓										Fuchs, 2018
Status Quo		✓					✓	✓	✓	✓						Statistics Austria, 2022
Status Quo	✓	✓	✓								✓	✓	✓	✓	*Energiemosaik*	Abart-Heriszt et al., 2020
Sectoral saving potentials				✓							✓				Four building types, ten constructions	Institute for Housing and Environment, 2017
Sectoral saving potentials				✓								✓			Mobility	Mair am Tinkhof, 2020
Sectoral saving potentials				✓									✓		Industry	Dehli, 2020; OHoE, 2021
Geographic referencing area	✓	✓	✓			✓				✓					Total area	Statistics Austria, 2022
Geographic referencing area	✓	✓	✓		✓										Available cultured land	Statistics Austria, 2022
Geographic referencing area	✓	✓	✓	✓			✓	✓							Roof areas	Statistics Austria, 2022
Geographic referencing area	✓	✓	✓	✓					✓						Forested area	Statistics Austria, 2022

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TABLE 2 Designation of resulting key energy quantities and indices.

Result	Description		
$S_{g,e}$	Current (2019) annual energy supply for renewable energy carrier *e* at geographic aggregation level *g*		
$C_{g,e}$	Current (2019) annual energy consumption of sector *e* at geographic aggregation level *g*		
$D_{g,e}$	Future annual energy demand of sector *e* at geographic aggregation level *g*		
$P_{g,e}$	Technical potential annual energy yield for renewable energy carrier *e* at geographic aggregation level *g* ("Potential")		
$T_{g,e}$	Target for annual energy yield for renewable energy carrier *e* or energy demand for sector *e* at geographic aggregation level *g* ("Target")		
$A_{g,e}$	Spatial variable reference Area for geographic level *g* and energy carrier or demand *e*		
Subscript		NUTS classification	Example
g,_	"Geographic level": Generic placeholder for any level of aggregation	–	–
m	Municipality	LAU	Gols
c	County	NUTS-3	Neusiedl am See
s	State	NUTS-2	Burgenland
n	Country/National	–	Austria
Energy subscript_,*e*		Description	
WI	Supplies	Wind power	
HY		Hydropower	
PV		Photovoltaics	
ST		Solar thermal	
BM		Biogenic energy	
AH		Ambient heat	
FF		Fossil fuels (natural gas, oil, coal etc.).	
IM		Imports	
HH	Demands	Households (residential building operation and use)	
SE		Services (commercial building operation and use)	
TM		Transport and mobility	
IP		Industry and production	
AF		Agriculture and forestry	

National sectoral targets, renewable supply potentials and energy consumption and saving potentials on municipal level.

3.1 National sectoral targets

The following literature was identified to include future renewable energy scenarios for Austria. The method presented in this paper is exemplified by these national scenarios, but not limited to it: With the scenario and energy balance structure defined in Section 2, it can be used to represent any given distribution of energy carriers and demands. The literature review yielded several sources that described a (nearly) hundred percent renewable energy supply of Austria, as shown in Figure 2. For demonstration purposes of this paper, the latest 100% scenario of Schneider et al. (2020) was selected, which in turn is based on Streicher et al. (2010). The original study by Streicher et al. (2010) analyzed the "Energy self-sufficiency for Austria 2050" and aimed to investigate the feasibility of achieving a 100% energy supply in Austria by the year 2050 through self-generated renewable energies. Two main scenarios were investigated: One without additional societal growth and consumption rates of 2008, and Second with an average growth rate of 0.8% p.a., resulting in a 40% increase by 2050 compared to 2008. Other studies, such as Krutzler et al. (2017) and Veigl (2017) were not considered as examples, as their scenarios featured non-renewable fossil-fuel components of 10 and 6%, respectively, due to their analysis being focused on energy savings and cost-optimal decarbonization rather than 100% renewable supply. Schneider et al. (2020) aimed to represent a scenario of 100% renewables with growth with the notable omission of deep geothermal exploitation from Streicher et al. (2010), instead substitution with additional wind and PV and recent mobility assumptions from Krutzler et al. (2017), as elaborated in Schneider et al. (2020). Figure 2 illustrates these future scenarios and their respective target balances. The required sectoral energy demand is shown on the left. On the right are the targets for the different energy utilizations.

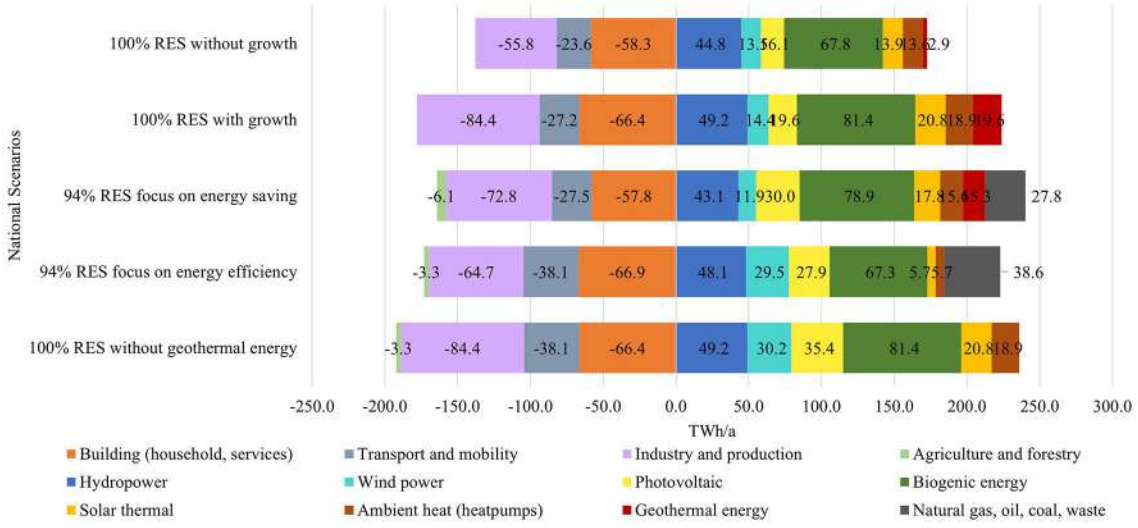


FIGURE 2
The targets for 2050 according to the studies in TWh per annum. **Left**: energy demand, **right**: energy generation from renewable and fossil energy technologies.

TABLE 3 **National expansion potentials for renewable energy technologies in TWh per annum.**

	Expansion potential (TWh/a)	
	Scenario electric priority	Scenario thermal priority
Wind power	69.7	69.7
Hydropower	56.1	56.1
Photovoltaics	11.4	0
Solar thermal	0	156.4
Biomass	93.8	93.8
Ambient heat	35.5	35.5
Total (electric)	137.2	125.8
Total (thermal)	129.3	285.7

3.2 Renewable supply potentials

Table 3 summarizes the potentials of renewable energies from photovoltaic and solar thermal systems. Due to their competition for space, the maximum expansion potentials are separately listed in two distinct scenarios, one with electric and one with thermal expansion priority, respectively. For now, a simple contravariance was assumed, meaning that PV and ST are considered exclusively. There is of course possibility for additional scenarios with different distributions and combination of systems.

3.2.1 Wind power

The wind power potential is derived from the study Winkelmeier et al. (2019), in which the technical wind power potential was assessed at county level (NUTS-3), which requires downscaling to municipal level. This was done pro rata for the available agricultural areas on municipal and county level, respectively. The potential assessment itself was considering these main aspects: Based on planned land use (area reserved for settlements and infrastructure), protected areas (national parks, nature reserves, and Natura 2000), unsuitable terrain conditions and land cover, available wind resources and finally technology and costs of installations. Depending on the wind conditions and achievable yield, locations in Austria were classified into three quality classes: A, B, and C. According to the study, Austria is assumed to have a wind power potential of 32,939 MW, which is divided down to the county level. The wind power potential for quality class A is also separately available at the county level and amounts to 10,209 MW. The potentials for quality classes B (8,214 MW) and C (14,516 MW) are available only as national aggregates. Therefore, the total potential has been separated into quality class A and quality classes B + C in aggregated and calculated on county level. For municipal level, a specific wind potential in MW/m² is calculated using the agricultural area as spatial variable. The municipal potential (Equation 6) calculated by multiplying the specific wind power potential of the county with the municipal agricultural areas, where the different quality classes refer to different full-load hours t_f. Table 4 summarizes the information on the quality classes and the expected annual potential. On the basis of the calculations, the annual generation potential of wind power plants in Austria is 69.7 TWh.

$$P_{m,wi} = \frac{A_{m,cultivated}}{A_{c,cultivated}} * P_{c,wi} * t_f \quad (6)$$

To determine the remaining potential at the municipality level after existing installations, it was assumed that 3,018 MW of currently installed wind power capacity are located exclusively in quality class A sites. This means that the entire potential of quality class B + C is remaining as untapped potential and only roughly two thirds of the

TABLE 4 Annual potential for wind power generation in Austria based on Winkelmeier et al. (2019, p. 20).

Quality classes	Expansion potential $P_{n,wi}$ (MW)	Full load hours t_f (h)	Generation potential (TWh/a)
A	10,209	2,609	26.6
B	8,214	2,174	17.9
C	14,516	1,739	25.2
Total			69.7

quality class A potential remain unclaimed. With this remaining potential, a specific remaining wind power potential is calculated for each county based on the county's agricultural area, following a similar approach as for the total potential. The remaining wind power potential is then scaled up for each municipality based on its agricultural area.

3.2.2 Hydropower

Data on the technically economic hydropower potential were obtained from Fuchs (2018, p. 12), which defines the potential as the technical-economic hydropower potential that can be reasonably realized under the given technical and economic conditions. For the assessment of the potential, sub-areas were classified with reference to their suitability for hydropower utilization. The study gives a value for Austria's hydropower potential of 56.1 TWh. This data on hydropower potential is available at county level. Municipal potentials were derived by Equation (7), using the total areas of municipalities and counties for allocation.

$$P_{m,hy} = \frac{A_m}{A_c} * P_{c,hy} \tag{7}$$

The determination of the **remaining** potential was obtained by subtracting the installed power plant capacity from the total technical-economic potential, resulting in a remaining potential of 16,000 GWh (Fuchs, 2018, p. 35). The estimation of the municipal potentials is analogously scaled to the total potential by the share of municipal to county land area.

3.2.3 Photovoltaics

The PV potential was estimated using a study Fechner (2020) with a focus on identifying suitable areas for PV systems within various space use categories such as buildings, including commercial and industrial areas, transportation and military areas, as well as agricultural and forestry areas. The objective of the study was to determine the potential of these areas for generating energy through photovoltaic systems and to evaluate the feasibility of implementing this potential by 2030. It analyzed various aspects, ranging from theoretical/physical considerations to technical, economic, ecological, and social factors, to identify the barriers that need to be overcome during the implementation of the PV potential. These barriers could include technological limitations, economic constraints, environmental considerations, and social acceptance. Table 5 shows the potential of PV from the physical-theoretical to the economic/social potential based on the study (Fechner, 2020). However, for this paper, the assessment of the potential is limited to the building sector only.

This results in an estimated PV potential of 3.5 TWh/a when excluding façade utilization. However, to meet the national energy and climate targets, specified in the Renewable Expansion Act [Nationalrat, 2022, §4 (4)] to an annual PV generation of 11 TWh/a, additional technically feasible potential must be utilized. Excluding façade areas, the resulting annual **11.4 TWh/a** would be sufficient to cover the national target. The municipal allocation of this target was carried out according to the share of roof areas in each category, which were obtained from the Austrian national building stock register count in 2011 (Statistics Austria, 2022) by approximating the roof area to the gross floor areas over the number of floors in each category. Equations (8) and (9) show how the total potential is determined using specific PV yield values and roof area.

$$P_{m,pv,\,cat} = \frac{A_{m,roof,cat}}{A_{n,roof,cat}} P_{n,pv,\,cat} \tag{8}$$

$$P_{m,pv} = \sum_{Category} P_{m,pv,\,cat} * A_{m,roof,cat} \tag{9}$$

To estimate the overall potential at the municipal level, three different scenarios are provided: (i) The potential based on the original study (Fechner, 2020), (ii) the potential with increased specific yields (kWh/m^2_{roof}), (iii) specific utilization of pitched roofs [single-family houses (SFH)], flat roofs [multi-family houses (MFH)], and industrial roofs, as well as specific PV yield in kWh/m^2_{PV}. Potential (i) on municipality level is calculated via the determined roof area. The survey results from the consideration of east-west oriented roofs, which were not used for occupancy in the potential study. To compensate for the resulting inadequate specific yield values, a surcharge factor of 1.67 (due to five instead of three orientations) and a discount factor of 0.93 (due to the reduced solar irradiation for east and west orientations compared to the south-facing modules) were considered in scenario (ii). The municipal potential is determined using the determined roof area and specified yields as well as percentages for PV occupancy for steep, flat, and industrial building roofs (Equation 10). Default values for specific yield values is 170 kWh/m^2_{PV}, as well as 35% for steep, 55% for flat and 70% for industrial building roofs, leading to scenario (iii), which was used for demonstration purposes.

$$P_{m,pv} = A_{m,roof} * f_{specific\ yield,pv} \tag{10}$$

Table 6 summarizes the collected roof area, specific yield values, and the sum of the options for the total PV potential in Austria.

The estimation of the **remaining potential** involved leveraging data on subsidized installations at the municipality level, obtained

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TABLE 5 Potential of photovoltaics: from physical/theoretical to economic/social potentials (data in TWh) based on Fechner (2020, p. 26).

Category	Physical/theoretical potential	Technical potential	Economic potential	Social/ecological potential	Assumed Potential
	With a 5% efficiency improvement by 2030 (average 2.5%), subtracting the already realized potential.	Structural considerations (10% of single- and multi-family homes, and 30% of industrial buildings unsuitable) Protection of historical and ensemble buildings, competition with solar thermal, roof and façade greening (10%)	Second homes, vacant objects, industrial sites (18% of total area). Economic downsizing (55% for single-family homes, 10% for multi-family homes, 25% for industrial buildings, 50% for façades)	Social factors (age, lack of investment capital), bureaucracy, effort, payback periods	
Single-family houses	6.1	5.4	2.4	1.2	5.4
Multi-family houses	1.7	1.2	1.1	0.5	1.2
Industrial buildings	7.9	4.8	3.6	1.8	4.8
Façades	2.9	2.0	1.0	0.5	0
Total	18.6	**13.4**	8.1	4.0	11.4

TABLE 6 The collected roof area, specific yield values, and the total PV potential in Austria.

	Roof area (km²)	Specific yield ($kWh/m^{2*}_{PV}a$) scenario		
		(i)	(ii)	(iii)
SFH	279.6	19.3	30.0	59.5
MFH	65.1	18.4	28.6	93.5
Industry	256.5	18.7	29.0	119.0
Total (TWh/a)		11.4	17.7	53.2

from Statistics Austria (2022). This data provided information on the **exploited installed capacity** of subsidized projects within each municipality. The total installed capacity of these projects amounted to 1.78 GW_p. To contextualize this information, it was compared to the total installed capacity in Austria, which was reported to be 2.5 GW_p (Wien Energie GmbH, 2021). The ratio of the sum of subsidized installations to the total installed capacity, was used to allocate the proportion of the total capacity attributable to each municipality. This allows for an assessment of the distribution of installed capacity among municipalities and facilitates the estimation of the remaining potential for further development in those areas.

3.2.4 Solar thermal

For the ST potential the same roof areas were used as in the PV case, but there is competition for space between the two technologies. It is assumed that the heat and domestic hot water demand in industrial buildings is the highest, which is why ST installations are mainly expected on industrial roofs. When specifying the utilization rates of these roof areas, the PV potential is correspondingly reduced (Equation 11).

$$P_{m,st} = A_{m,roof} * f_{specific\ yield,st} \tag{11}$$

The total ST potential for Austria amounts to 156.4 TWh of thermal energy per year. The current energy production of ST systems in Austria was obtained through statistical data (Statistics Austria, 2022). As the available data was at the states level, the energy production quantities were scaled down to the municipality level based on the ratio of municipality roof area to state roof area.

3.2.5 Biomass

The biomass potential was estimated based on forested Areas. According to Abart-Heriszt et al. (2020), Austria has a total forest area of 4.88 million hectares. The average annual growth is around 8.9 solid cubic meters (scm) per hectare (Federal Ministry of Agriculture Forestry, Regions and Water Management, 2019, p. 8). Solid cubic meters are converted to stere (st) with a factor from literature of ~1.8 st/scm. With an average calorific value of 1,500 kWh/st, the maximum potential, assuming the utilization of the

entire annual growth, amounts to 93.85 TWh/a:

$$P_{m,bm} = A_{m,forest}\ [ha] * f_{growth}\left[\frac{scm}{ha}\right] * 1,8\ \left[\frac{st}{scm}\right] * f_{calorific}\left[\frac{kWh}{st}\right]$$

The current biomass utilization is estimated based on the forest area of each municipality. For the maximum potential, 100% of the annual growth is allocated, while for the current utilization, it is assumed that 90% of the annual growth is harvested and only 30% of that harvested biomass is used for combustion (Equation 12):

$$S_{m,bm} = P_{m,bm} * 0.9 * 0.3 \tag{12}$$

3.2.6 Ambient heat

According to this study (Stanzer et al., 2010, p. 131), the geothermal heat potential in Austria is estimated to be 35,506 GWh per annum and available at county level. This potential correlates with the heat demand, which was derived as follows: (i) The number of buildings for each type, that the study uses (i.e., four building types and six construction periods, resulting in 24 distinct types) was determined on a county level, (ii) determination of the residential or usable areas for all building types per county, (iii) for each building type, a heating energy indicator from literature is assigned according to the construction periods and building types, (iv) the indicator for the heat demand for domestic hot water was assumed to be 3,600 kWh per household. In addition, a minimum efficiency requirement was introduced that ruled out heat pump systems for space heating for building types constructed before 1981. Finally, the municipal potentials are allocated from county level based on the total municipal and county areas (Equation 13).

$$P_{m,ah} = \frac{A_m}{A_c} * P_{c,ah} \tag{13}$$

The data on **current energy production** from near-surface geothermal energy was obtained from Statistics Austria (2022) at the county level (NUTS-3). The energy production at the municipality level was scaled based on the ratio of municipality area to county land area. This approach allowed for a more detailed analysis of near-surface geothermal energy production, considering the specific characteristics and sizes of municipalities within each region. By scaling down the regional data to the municipality level, a more accurate picture of near-surface geothermal energy generation across Austria was achieved. The **remaining potential** for the municipalities, as well as the overall remaining potential for the country, is determined by calculating the difference between the total potential and the already developed potential.

3.3 Energy consumption, saving potentials and future demands

The final energy consumption status quo from Abart-Heriszt et al. (2020) for Austria in 2019 and its sectoral distribution was used. In the following sections, the sectors are presented individually, outlining their components, and how they were integrated to fit into the structure of the future scenarios. These consumption data are available on municipal level, further differentiated in subgroups with one or two dimensions, which can be found in the Appendix.

The energy saving potential of municipalities is estimated from so-called reduction factors that are an estimate for the attainable energy demand savings of any given sector and current energy consumption. The energy saving potential is based on municipal data, but the calculation of the Reduction factor depends on the sector. It should be noted that the projected energy demand represents only one possible scenario, which may vary depending on changes in the reduction factors.

Table 7 complies the initial current final energy consumption for the year 2019 and the future final energy demand based on calculations involving reduction factors applied to the current energy consumption values.

3.3.1 Building (household and services)

In the case of buildings, a reduction factor represents energy savings for energy efficiency measures such as renovation and optimization of heating and hot water systems. This was operationalized using two parameters readily available in municipal statistical data: the year of construction and a rough building type (single-family houses and multi-family or multi-storey houses from TABULA) for each built area. The TABULA (Typology Approach for Building Stock Energy Assessment) Webtool features a set of energy performance indicators (energy costs, CO_2 emissions, heat demand, primary energy demand) and provides online calculations of exemplary buildings, showcasing their energy characteristics and the potential energy savings through the implementation of renovation measures. The package of measures includes the **modernization of the thermal envelope and the heating supply system** (Institute for Housing and Environment, 2017). The national building typology in TABULA is defined by a "Building Type Matrix," with four main building typologies defined: single-family houses, terraced houses, multi-family houses, and apartment blocks.

The rows of the matrix correspond to different construction year classes, which align with the years available in *Energiemosaik*. This alignment allows for a seamless combination of data between TABULA and *Energiemosaik*, facilitating comprehensive analyses and assessments of the building sector's energy characteristics.

To obtain the reduction factor, the following approach was employed: Data on energy consumption before and after renovation for each of the four building size classes [SFH, terraced houses (TH), MFH, and apartment blocks (AB)] was collected. The difference between the energy consumption before and after the renovation was then determined and referred to as the energy-saving potential, and the corresponding fraction as the Reduction factor.

Subsequently, the four building size classes were mapped onto the two available categories for energy consumption from *Energiemosaik* (Abart-Heriszt et al., 2020): "single-family houses and terraced houses" and "multi-family houses and apartment blocks" by using the average reduction for each category. The projected energy demand of municipalities in the household sector is then determined by multiplying the individual reduction factors with the respective final energy consumption. Table 8 displays

TABLE 7 Austrian final energy consumption in 2019 and the achievable future energy demand per sector.

	Final energy consumption in 2019 (TWh/a)		Achievable future energy demand (TWh/a)	
	$\mathbf{C_{n,*}}$	$\mathbf{D_{n,*}}$		
Buildings (households, services)	$\mathbf{C_{n,HH}}$	111.9	$D_{\mathbf{n,HH}}$	76.8
Transport & mobility	$\mathbf{C_{n,TM}}$	61.3	$\mathbf{D_{n,TM}}$	38.7
Industry & production	$\mathbf{C_{n,}}_{IP}$	90.9	$\mathbf{D_{n,}}_{IP}$	37.2
Agricultural & forestry	$\mathbf{C_{n,}}_{AF}$	6.1	$\mathbf{D_{n,}}_{AF}$	6.1
Total	$\sum_i \mathbf{C_{n,i}}$	270.2	$\sum_i \mathbf{D_{n,i}}$	158.8

the reduction rates for consumption sourced from the TABULA website, alongside the corresponding calculated relative reduction factors. Additionally, it provides the net floor areas for the buildings based on data extracted from *Energiemosaik*.

The Equation (14) represents the total energy demand for the household sector. It is calculated by sum the energy demand for each building category (a) across all construction periods (t).

$$D_{hh} = \sum_{a,t} D_{a,t} \tag{14}$$

In the case of sectoral services, the focus is primarily on renovating the buildings themselves, while the optimization of processes related to the specific services is not considered. To determine the reduction factor for this sector, the average value of the housing category was calculated and used as the reduction factor for services (Equation 15).

$$f_{se} = \frac{\sum_y f_{t,1}}{|t|} \tag{15}$$

The end energy consumption of the sector services is then multiplied by reduction factors. The results obtained in this way lead to the future final energy demand (Equation 16).

$$D_{se} = C_{se} * f_{se} \tag{16}$$

3.3.2 Transport and mobility

To estimate the energy saving potential of daily mobility, the spreadsheet tool described in Mair am Tinkhof (2020) was used. The tool provides calculation of various indicators related to everyday mobility in settlements and neighborhoods, considering planned mobility concepts and local conditions and offers diverse options for influencing the calculation of everyday mobility. To develop a comprehensive methodology with a broad applicability, it was decided that in future scenarios all passenger cars will be electric vehicles charged with green electricity. Based on these assumptions, the current and future GHG emissions and the primary energy demands were calculated. The ratio between the two was used as reduction factor f_{tm} for the transport and mobility sector. The energy consumption data from *Energiemosaik* is multiplied by the f_{tm} to calculate the energy demand of the transport and mobility sector in the future (Equation 17). This gives an estimate of the achievable energy demand reduction in the sector based on the assumed improvements in energy efficiency and the transition to electric vehicles charged with green electricity.

$$D_{tm} = C_{tm} * f_{tm} \tag{17}$$

3.3.3 Industry and production

In the industry and production sector, energy savings were assumed to be obtained through process optimization and automation, as well as the use of energy-efficient equipment and lighting. Technological improvements and automation offer a straightforward and effective path to enhancing energy efficiency. According to Dehli (2020), meaningful opportunities for increasing energy efficiency include:

- Enhanced insulation in technical facilities.
- Utilization of both central and decentralized combined heat and power systems.
- Implementation of energy management systems.
- Deployment of energy-efficient components, such as pumps, fans, refrigeration systems, electric motors, air technology, devices, and lighting techniques.

An overview of end energy consumption in different application areas within the industrial sector is also provided in Dehli (2020): It clearly illustrates that the most significant application areas are process heat, space heating, and domestic hot water, which together account for over 70% of the total end energy consumption. Mechanical energy comes second, constituting 23% of the overall consumption. The underlying data for this analysis again is sourced from *Energiemosaik* (Abart-Heriszt et al., 2020). By weighting the shares of application areas and their corresponding saving potentials, an overall $\boldsymbol{f_{ip}}$ is derived. To obtain the projected energy demand of the sector, the end energy consumptions from *Energiemosaik* data are multiplied by this f_{ip} (Equation 18).

$$D_{ip} = C_{ip} * f_{ip} \tag{18}$$

3.3.4 Agricultural and forestry

A reduction factor for the agriculture and forestry sector was not considered as it constitutes only a minor share of 2.3% in the total end energy demand and does not significantly impact the overall analysis. This can be addressed in future research by defining the desired allocation function and spatial variable as necessary.

TABLE 8 Reduction factors and living spaces of household buildings.

	Category (a)				Category (a)	Category (a)				
	SFH					MFH				
Age	TABULA (red. factor)		*Energiemosaik* SFH and TH			TABULA (red. factor)		*Energiemosaik* MFH and AB		
	SFH	TH	Average reduction f_c	Net floor area $A_{m,NFA}$	$D_{SFH,t}$ (TWh/a)	MFH	AB	Average reduction f_c	Net floor area $A_{n,NFA}$	$D_{MFH,t}$ (TWh/a)
Prior 1919	62.7%	58.5%	60.6%	31.7 km²	4.63	55.1%	54.2%	54.7%	34.4 km²	3.41
1919–1944	71.3%	65.6%	68.4%	14.8 km²	2.37	67.3%	64.1%	65.7%	11.2 km²	1.41
1945–1960	64.9%	54.2%	59.6%	23.7 km²	3.48	60.5%	60.2%	60.4%	15.3 km²	1.81
1961–1970	63.0%	60.6%	61.8%	30.2 km²	3.92	64.6%	57.2%	60.9%	22.1 km²	2.09
1971–1980	63.0%	60.6%	61.8%	37.8 km²	4.94	64.6%	57.2%	60.9%	23.5 km²	2.24
1981–1990	50.2%	49.3%	49.7%	35.5 km²	3.10	40.2%	16.7%	28.4%	16.9 km²	0.75
1991–2000	16.4%	20.9%	18.6%	29.7 km²	1.18	20.8%	15.7%	18.3%	18.2 km²	0.53
2001–2010	23.4%	30.8%	27.1%	33.9 km²	1.09	13.8%	20.9%	17.3%	20.2 km²	0.34
2011–2019	50.0%	37.8%	43.9%	24.6 km²	1.25	51.1%	43.0%	47.1%	21.2 km²	0.80
Sum				261.9 km²	25.96				183.0 km²	13.38

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3.4 Target allocation

As part of the result visualization, several figures were created, that can be accessed via a spreadsheet tool and the DSS website. The website also presents heatmaps showing the results in a clear and comprehensive manner. The heatmaps cover renewable energy sources, energy consumption, energy-saving potentials, and the future energy demand of the sector. Additionally, the website features a heatmap for the energy balances of all Austrian municipalities. The diagrams are interactively generated, and this interaction allow users to easily visualize and compare the results.

3.4.1 Map of energy supply expansion and demand reduction targets

The following Figure 3 visually depicts the expansion targets for renewable energies across municipalities left in GWh per annum and right in MWh per capita per annum. Intensity of color indicates the magnitude of required expansion for each municipality. Darker shades indicate a higher obligation for the municipality to augment their renewable energy capacity, suggesting the presence of greater potential. This is aimed to discern the distribution of expansion goals, guiding municipalities toward targeted efforts in harnessing their specific renewable energy potential and fostering sustainable energy development. It can be seen that installation targets of wind power is higher toward the eastern and central parts of the country, whereas the mountainous west generally has lower potential and subsequent targets, with notable exceptions. There is no notable change in distribution between absolute and relative targets per capita. The distribution of expansion targets across municipalities shows that about 11% of the expansion target is distributed among 1,260 municipalities, each having a target of $<$10 GWh, or roughly two modern wind turbines. But 30% of the expansion target is achieved by only sixty municipalities, which hold goals exceeding 100 GWh based on their high wind potential. The hydropower expansion targets mainly highlight the remaining potentials of the western region. The illustration of the distribution of hydropower expansion targets among municipalities shows that 11% of municipalities lack significant hydropower potential, and another 37% of municipalities only account for 6% of the national target. In contrast, 30% of the expansion target is achieved by only sixty-six municipalities with goals exceeding 100 GWh/a each. The Photovoltaic and Solar Thermal heatmaps highlight the need for action in larger cities and urban areas.

Note that in this context, the PV and Solar thermal potential is not solely dependent on solar radiation; rather, it relies on available roof areas for photovoltaic installations and industrial rooftops for solar thermal installations. Evidently, the municipalities with the greatest potentials are primarily urban centers such as Linz, Salzburg, and districts within Vienna, which is consistent with their larger built environments and roof areas. A comparison with the heatmap per capita shows a different perspective, where the obligations of urban areas is in fact much smaller when looking at the per capita targets than for rural settlements. The expansion target for biomass can be located in the forested middle of the country and extends to the southern and western regions. The distribution per capita shows a slight deviation, due to the varying population density of these rural and mountainous regions. Note that the potential refers to biomass use of wood only, which means that the potential depends on the size of the municipality's forest area. As such, seventeen municipalities have no biomass potential, and another 711 municipalities have such a low potential that their cumulative contribution would be only 4% of the target. However, due to the mostly large forest areas in middle, south, and west of Austria, about half of the expansion target could be reached by 222 municipalities. In the case of ambient heat, the larger cities are again highlighted due to the assumption of predominant use for space heating and cooling. The specific per capita breakdown on the right again contrasts the high absolute targets with low relatives.

Finally, the reduction targets for energy consumption for buildings and mobility sectors are shown with color intensity corresponding to the magnitude of required energy savings for each municipality. In both building and mobility sectors, larger municipalities can be seen to have a greater potential for energy savings due to higher building and traffic density. The reduction potentials per capita unsurprisingly show the opposite trend.

The aggregate of the different renewable expansion targets and sectoral reduction targets are presented as an integrated energy balance target for all Austrian municipalities in Figure 4 in GWh per annum and in MWh per Capita per annum. The colors represent the following: Municipalities marked in red have negative energy balance targets, indicating either a limited potential for renewable energy sources, high energy consumption with minimal saving potential or a combination of both. Naturally, their contribution to the national target will be lower. On the other hand, green-shaded municipalities have positive energy balance targets, indicating higher potential and subsequent targets for renewable energy expansion than their achievable energy demand requiring significant expansion to help offset the deficits of the red-marked regions. Yellow-colored municipalities fall somewhere in between, indicating a more balanced position with regards to their renewable energy potential and consumption levels. This visually informative representation assists in identifying and addressing disparities, promoting a more equitable distribution of efforts to achieve sustainable energy objectives across all municipalities.

Next, Figure 4 show the distribution of municipalities based on the size of their absolute target balance. They illustrate that many municipalities in fact have a target balance in the range between $-$200 and 200 GWh/a, as well as the quantified difference between negatively balanced urban areas and positively balanced hinterland. The balance target provides an important quantitative threshold—how to reach it is a different story, which should take into consideration local contexts and political decisions. Therefore, apart from the detailed targets presented before, it is also useful to group the results into concrete needs for action: Figure 4 also depict these needs as electrical and thermal actions, respectively, represented on the left in GWh per annum and on the right in MWh per capita per annum. Values below zero indicate that the municipality has already sufficiently exploited its potential for renewable energies and/or exhibits low energy consumption. Conversely, higher positive values indicate the additional energy capacity required for the municipality to meet its targets. Crucially, this can be used to quantify other forms of renewable expansion or savings than what the initial allocation yielded. For example,

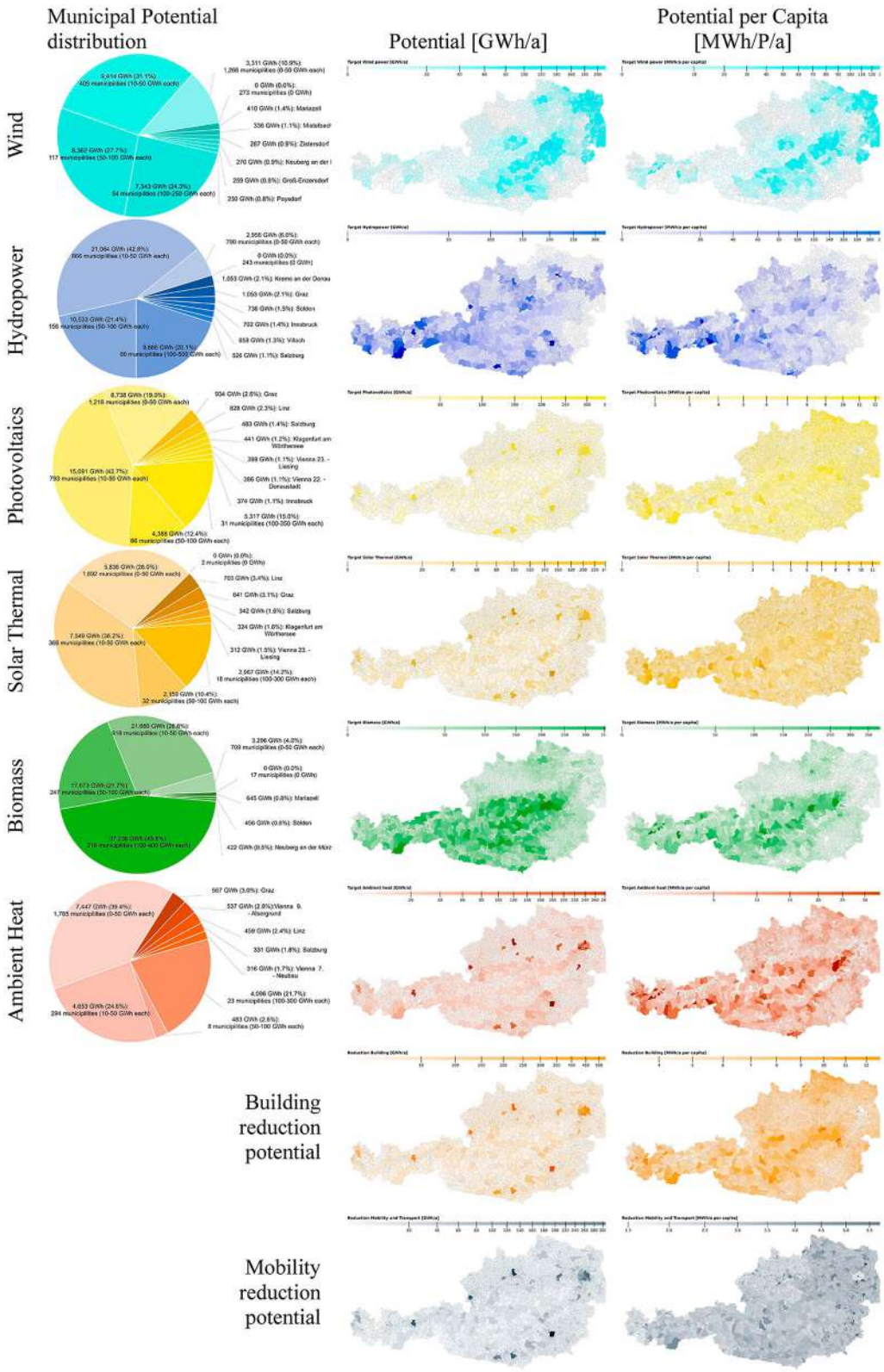


FIGURE 3
Renewable Expansion target distribution **(left)**, RES expansion and reduction targets per energy carrier and sector in GWh/a **(center)** and in MWh/a per capita **(right)**.

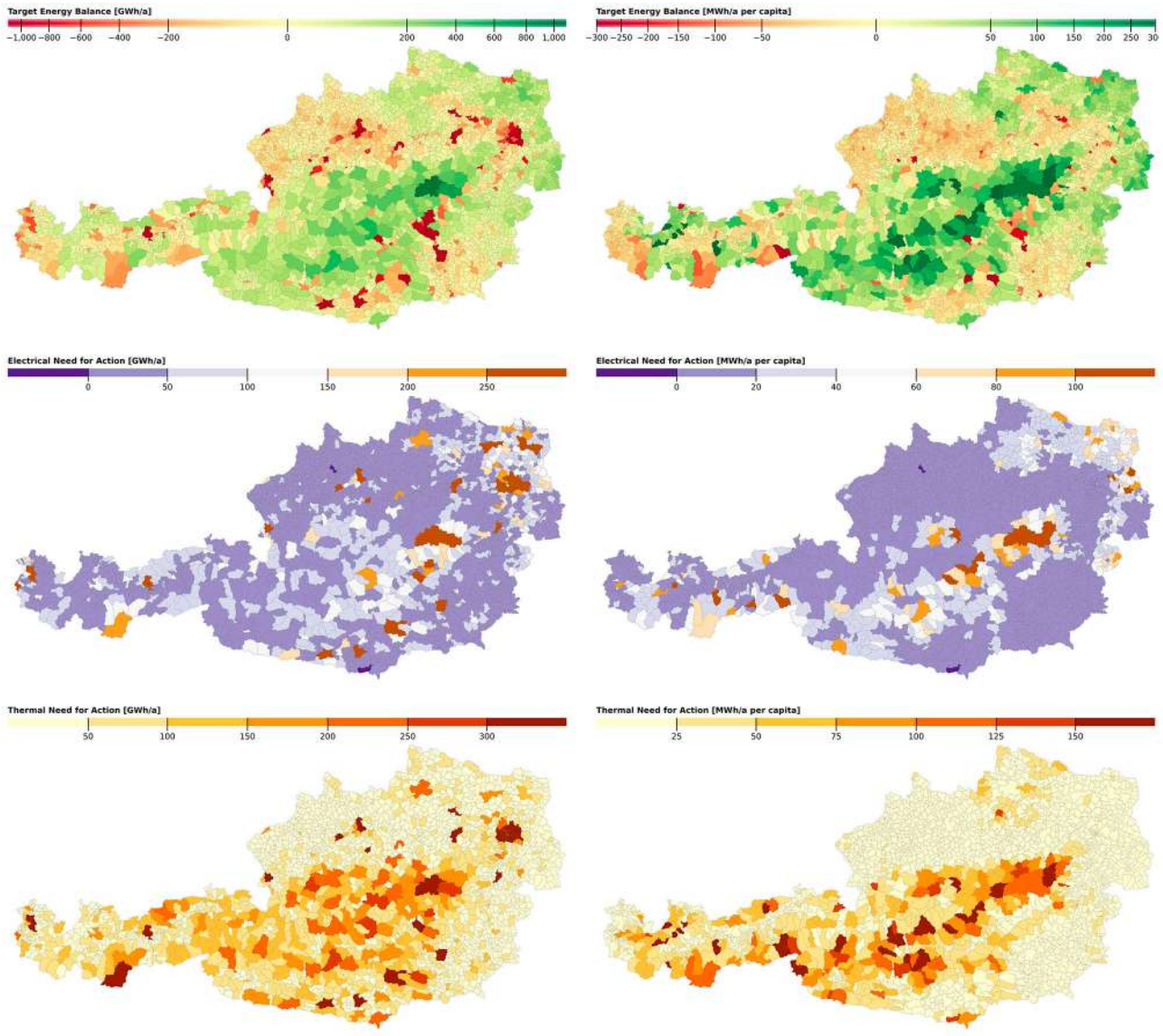


FIGURE 4
Municipal energy balance target **(top)**, electrical **(center)**, and thermal need for action **(bottom)** in GWh/a **(left)** and MWh/a per capita **(right)**.

a municipality could opt to realize their electrical need for action resulting from a high wind potential with increased PV installation and additional demand reductions beyond the initially targeted thresholds instead.

3.4.2 Energy balances and needs for action on county level (NUTS-3)

The results can also be aggregated on an intermediary level of geographic granularity. This section presents the resulting renewable exploitation and sectoral saving targets aggregated by federal states. Figure 5 shows the resulting energy balances on the level of Austrian federal states (NUTS-3) from left to right: Demand Status-Quo (2019), Demand Targets, RES Expansion targets and Status-Quo (2019). The excess RES potential *not* required for expansion is depicted in green, the "reduction gap" between current energy demand and target levels in red on top of the demand target column, and the "expansion gap" between current RES and target levels in gray in top of the RES Status Quo column. According to the allocation approach, a federal state is expected to contribute more to the overall target if it has a greater potential from a particular renewable energy source. For instance, the third state from the left—"Burgenland", a western state with high wind potential—is projected to generate significantly more energy from wind power, the alpine eastern Tyrol more energy from hydropower, and the forested state of Styria in the south more energy from biomass. This allocation strategy aims to capitalize on the specific strengths and resources of each region, optimizing the use of renewable energy sources according to their availability.

The figure illustrates current energy consumption and corresponding energy demand targets for individual federal states. The approach is characterized by a clear allocation strategy. It

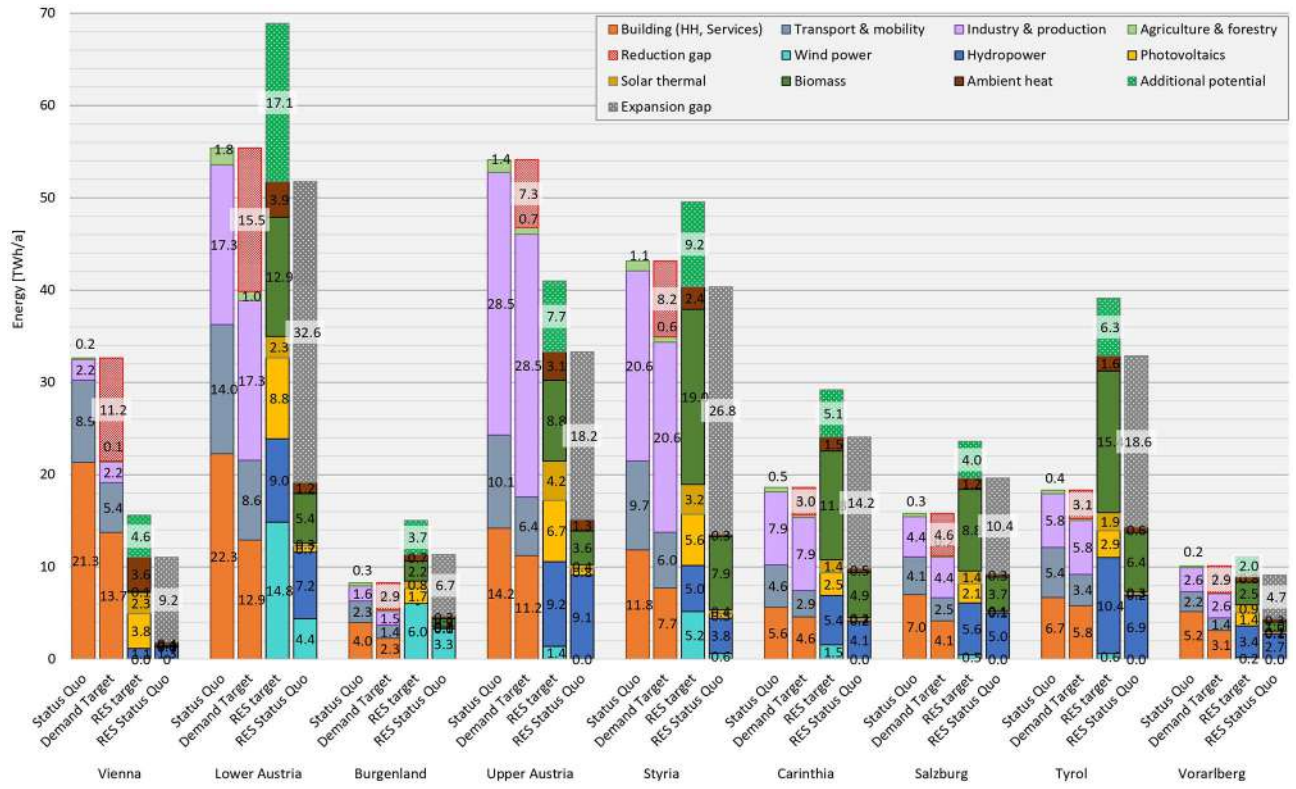


FIGURE 5
Resulting energy balances on the level of Austrian federal states (NUTS-3) from left to right: Demand Status-Quo (2019), Demand Targets, RES Expansion targets, and Status-Quo (2019).

considers the particularities of each federal state, such as varying proportions of buildings in need of renovation, and higher energy consumption in the transport and mobility sector. Consequently, the targets are adjusted to address these specific circumstances. This customized allocation method recognizes the heterogeneity of energy consumption patterns and building stocks across different regions of Austria. By setting tailored targets for each federal state based on their unique characteristics, the energy efficiency measures can be more precisely targeted to address the specific requirements and challenges of each area. The allocation of targets in this manner ensures an equitable and balanced contribution from all parts of the country toward achieving the overarching national objectives in energy efficiency and sustainability.

The figure also presents a comparison between the target final energy demand and the expansion target for renewable energies across different federal states. The graph highlights the varying degrees of individual agency or discretion that each federal state possesses. Notably, states like Lower Austria and Styria exhibit significant untapped potential for renewable energy expansion. Conversely, federal states such as Vienna or Upper Austria, which possess renewable energy expansion potentials lower than their current end energy consumption, should complement expansion efforts with energy efficiency measures, including building renovations and energy-saving initiatives.

3.4.3 Municipal energy balances and needs for action

The following presents the resulting energy balance component targets for three example municipalities of decreasing size: *Steyr* (38 thousand inhabitants, motor industry), *Hollabrunn* (twelve thousand inhabitants), and *Gols* (four thousand inhabitants, high wind power installation). The left side of the balance figure show energy demand and consumption, the right-side renewable energy sources.

- The first bar on the left represents achievable energy demand through energy-saving measures, while the right side shows the technically possible total potential of renewable energies.
- The second bar illustrates the current state, indicating the end energy consumption from 2019 and the already developed potential of renewable energies (up to 2020).
- The third bar outlines future targets as derived from the allocation mechanism.
- The "to target" section represents the differences between the current state and the targets. Here, both consumption and expansion aspects are separately shown.
- The "need for action" section further breaks down these differences, separating them into electrical and thermal energy needs.

- The last bar illustrates the total need for action, adding thermal and electric energy.

Figure 6 illustrates the results in a comparison. The first and largest municipality of **Steyr** exemplifies a pronounced industrial sector, which in this allocation scenario has no saving targets (16.9 kWh/Person/a). The focus of the need for action is split into one third energy savings (−8.3 kWh/Pers/a) and two thirds on renewable expansion of 16.2 kWh/Pers/a. The diagram shows that PV, ST, and ambient heat are the largest untapped potentials.

In contrast, the central bar chart shows the smaller municipality of **Hollabrunn**. Here, the industrial sector is insignificant, and the need for action is instead constituted predominantly by the need for expansion of renewables. The bottom bars show the nominal energy balance of the smallest municipality, which is already significantly positive due to the extensive development of wind power by 2019, as depicted in the first two bars of the Status Quo. Nevertheless, the potential-dependent allocation process still indicates a further need for action in *Gols*, both in terms of renewable expansion and energy savings of −9.2 and 8.8 kWh/Pers/a respectively.

4 Discussion

4.1 Regional imbalances in renewable energy supply potential

The results show local and regional imbalances in the potential for renewable energy supply. Unsurprisingly, urban areas act as "energy sponges": Despite ambitious efficiency measures and the use of limited roof space for PV, they will not have sufficient onsite renewables to supply their future energy demand. Rural areas, especially those with high wind potential and low industrial demand, will easily be able to cover their local energy demand, often exceeding it by far, becoming potential regional suppliers of renewable energy. This highlights the importance of coordinating the effort sharing between regional energy systems in becoming renewably powered. Analyzing local energy systems in terms of their renewable self-sufficiency is an important step but it is potentially setting the wrong incentives to municipalities and ending an important conversation prematurely: How is the effort-sharing of the energy transition allocated between regions and municipalities? One solution would be to determination of legally binding energy targets to federally developed energy spatial plans, that are negotiated between municipal and regional authorities. A roadmap of milestones in achieving these renewable energy targets on municipal level could then be incorporated in the negotiation of the so-called "Finanzausgleich" (german: "financial equalization"), the Austrian financial instrument determining tax allocation and transfer payments between federal, state, and municipal authorities.

4.2 Different data sources

It is possible to include other sources to model different baselines, scenarios, and target allocations. The biggest challenge still lies in the required homogenization and structuring of the data. Often, when data structures do not entirely match, additional assumptions have to be introduced to bridge this gap. For example, in the building sector future demand calculation, projections were available for four types of buildings, but consumption data only for two distinct types, which necessitated merging two of the former four types together to match the latter two by using an area-weighted average.

The less data sources are directly connectable via structure, the more additional supplementary data is generally required to glue the datasets together, generally resulting in some data and details being lost or rendered unassignable. Future research would greatly profit from deciding and adhering to uniform data structures when reporting similar findings such as scenarios and renewable and savings potentials.

4.3 Other ways of achieving 100% renewable energy supply

The differentiation of energy targets into detailed potentials and targets for each renewable energy carrier and demand sector facilitates important discussions. Which results should be realized to which degree? By aggregating the targets into a supply target, a reduction target, an electrical and thermal target as well as an overall balance target, municipalities can take an informed decision, which potentials they want to commit to politically. If e.g., the realization of wind is politically infeasible, the framework shows how much energy would need to be saved or renewably produced by other means. This should further the typical discussions arising in municipalities around NIMBY (not in my backyard), by quantifying the alternatives, which might be costlier.

4.4 Different scenarios

The presented work was only operationalized for a specific scenario and the results reflect that. Different scenarios can be used as baseline for national targets, which can simply be accomplished by changing the data in Figure 2. Apart from that, it is also possible to construct different allocation scenarios, meaning different ways of allocating the national targets (and potentials) to higher geographic resolution. This can be done by either introducing different referencing area base datasets or using a different allocation formula, or any combination of both. The main limiting factor to the complexity of the allocation mechanism apart is the availability of data with appropriate format and structure.

4.5 Accuracy and uncertainty of the model

Given the many assumptions in connecting the different datasets, the resulting uncertainty of the municipal potentials and targets is naturally high. The key to increasing result reliability lies in the compatibility of the required datasets for renewable and saving potentials and their reference to geographic attributes and allocation parameters.

Another uncertainty lies in the use of building footprints to estimate energy demands for buildings

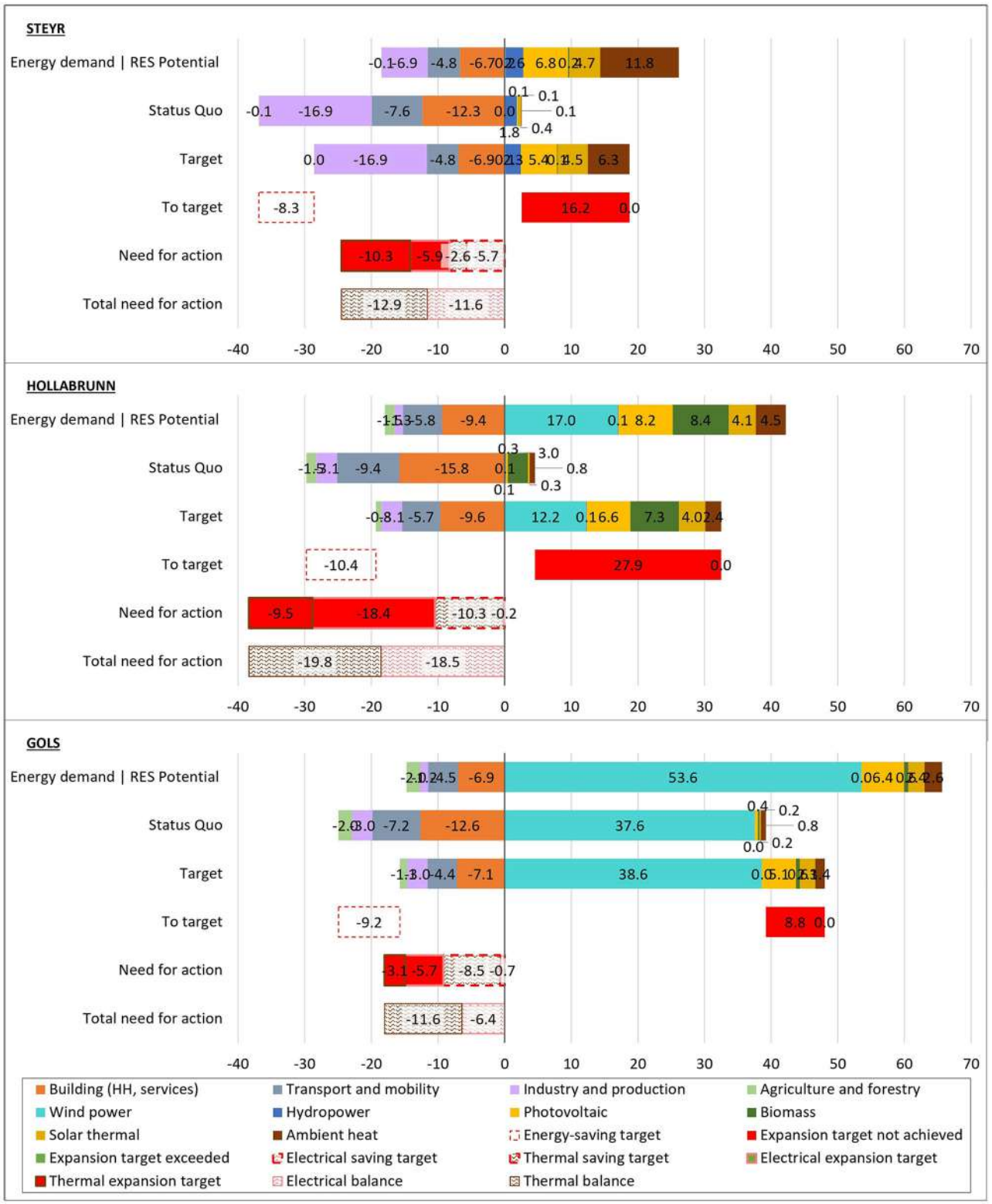


FIGURE 6
Municipal results comparison of three municipalities: Steyr, Hollabrunn, and Gols. **Left**: energy consumption, **right**: local renewable energy generation in kWh/Pers/a.

in general, which does not take into consideration, which buildings are actually in use, and with which frequency and intensity. Also, the used typology is rather generic and does not account for the diversity and extremes of the building sector at large.

4.6 Result usability

Working with three municipalities for testing showed the results to be generally well received. One caveat is that the resulting municipal energy balance visualizations were difficult to comprehend, even for some experts on the field. Especially the inclusion of aggregate "Municipal action required" for all renewable potential targets, saving targets and for thermal and electric balance targets in addition to all three main results, i.e., Current Status-Quo (S), Renewable and Savings Potentials (P), the resulting Targets for each (T) was overwhelming. It might be sensible to further reduce the main results and drop the aggregates, as they always required additional briefing.

The nationwide availability of these data offers a holistic view of energy efficiency and the potential of renewable energies in Austria. This is crucial for developing and implementing well-informed energy policy measures at both national and local levels. The heatmaps thus serve as valuable tools for analyzing the status and development opportunities in the energy sector in Austria and for driving a sustainable energy transition forward. The data obtained can also be visualized in several different ways: Using reference areas to specify the quantities can be interesting.

5 Conclusion

Consistently localizing energy and emission targets from national to municipal level is an important challenge in tackling the accountability problem of climate neutrality ambitions. In addition to individual bottom-up target formulation and disconnected climate accounting, models and visualizations for effort sharing of the required renewable expansion and energy savings are required. The key model property is consistency of the individual municipal with national goals and targets.

The approach presented here is simple yet comprehensive. It can deliver municipal targets for energy demand reduction and renewable expansion based on national decarbonization strategies and local potential estimators. It uses an allocation of local and regional potential estimates for demand reduction and the expansion of renewables and scales these KPIs according to specified reference areas to the desired national targets.

Crucially, both the national target scenarios and the allocation strategies using reference areas can be specified freely. Thus, the model offers versatility to take different modeling and scenario assumptions into account. The results are spatially differentiated on municipal level and functionally divided into typical energy demand sectors and renewable energy carriers. They can provide a detailed look into possible municipal effects of national climate targets: 100% Renewable supply is not necessarily required for municipalities with large industry sectors or little renewable potential. Municipalities with high renewable potentials on the other hand might need to overachieve a renewable energy balance by up to a hundred percent, meaning they must provide twice as much renewable energy than they demand themselves. The results can be used to analyse and discuss different municipal potentials and needs of action both in terms of general balance targets and in detailed sub-targets of sectoral expansion and saving.

The resulting target maps and energy balances are also visualized and interactively explorable online (https://observablehq.com/@editp/allocation-of-national-decarbonization-targets-to-municipal-level-in-austria). This facilitates dissemination, discussion and analysis on effort sharing and potential realization in the decarbonization of society: With the proposed framework, different national scenarios, and their implications for municipal needs for action can be compared and their implications made explicit. The resulting targets provide quantifiable and meaningful municipal decarbonization goals. Nevertheless, it must be stated that the presented approach is hampered by insufficient and patchy data availability, both for potential estimates and for suitable scaling options in terms of reference areas. The latter can only provide a coarse estimate of applicability and can only give a first indication of more detailed local methods.

Data availability statement

The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found below: https://observablehq.com/@editp/allocation-of-national-decarbonization-targets-to-municipal-level-in-austria.

Author contributions

SS: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. EP: Writing – original draft. DS: Writing – review & editing. JB: Supervision, Writing – review & editing. PO: Writing – review & editing.

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/frsc.2023.1294361/full#supplementary-material

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B.4. Digital Positive Energy Districts: A Scalable Strategy for Urban Heat and Power Transition?

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Authors: Simon Schneider, Albert Treytl, Stefan Wilker, Michael Schmidthaler, Thomas Zelger, and Elisabeth Kerschbaum.

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Contribution to the thesis:

Integration of PED criteria into urban development competitions and early-stage project evaluation.

DIGITAL POSITIVE ENERGY DISTRICTS: A SCALABLE STRATEGY FOR URBAN HEAT AND POWER TRANSITION?

Simon SCHNEIDER[1], Albert TREYTL[2], Stefan WILKER[3], Michael SCHMIDTHALER [4], Thomas ZELGER[1], Elisabeth KERSCHBAUM[1]

Abstract

The “DigiPEQ” project (FFG No. 880562) explored the requirements and potentials of digital Positive Energy Districts (PEDs) in contributing to national and municipal decarbonization efforts and energy and climate goals. PEDs are identified as key in urban planning for energy sustainability, generating more energy than they consume through local renewables and digital technologies.

Insights into ten thematic modules were jointly developed with innovative practitioners of sustainable building and district development, user-oriented and sustainable cooperation and business models, Innovative digital technologies for integrated, user-oriented energy systems and the establishment of replicable planning and operational management processes of Positive Energy Districts.

Focusing on fundamentals, system boundaries, and framework conditions of digital, livable Positive Energy Districts, the project investigated dynamic **assessments** of energy and emission flows in PEDs, focusing on various system boundaries and conversion factors and how they can be quantitatively linked to municipal and national climate goals and scenarios.

Several key strategies for enabling **energy flexibility**, load shifting, grid services, sector coupling, and maintaining user comfort and acceptance, as well as aspects of load prediction and user behavior are investigated, highlighting the importance of adaptable control algorithms and the consideration of grid status in PEDs.

The project further addressed the **energy economic, legal, and technical environment**, emphasizing the need for regulatory frameworks to support PED development. Considerable legal and administrative barriers to PED implementation and near-term scalability were identified.

The project underscored the significance of **participative design** and information management in promoting sustainable behaviors in PEDs.It also explored the application of **new digital technologies** like ML, AI, and Web3 in PEDs.

Finally, the project highlighted the relevance of PED topics to **practitioners** and outlined strategies for knowledge and skill transfer in PED development, and a **criteria catalogue** is presented to steer and assess early PED development and planning processes in practice.

In conclusion, DigiPEQ demonstrates the promise of PEDs in urban energy transitions, offering insights for future district design and the role of digital technologies in this process.

Introduction

Positive Energy Districts (PEDs) are emerging as a pivotal solution in urban planning to achieve energy efficiency and sustainability. These districts generate more energy than they consume, utilizing local renewable sources and innovative digital technologies. The relevance of PEDs lies in their potential to reduce carbon emissions, enhance energy security, and promote sustainable urban development. The research in this area is focused on developing strategies for energy exchange, stakeholder engagement, implementation of energy concepts, and assessment methodologies to foster the realization of PEDs [1], [2]. They are neighborhoods or several connected buildings, often with diverse space uses, meeting

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the highest standards of energy efficiency. They cover their entire energy needs locally from renewable sources, enable flexible operation, and create the highest possible comfort and quality of life on-site through the involvement and participation of users [3], [4]. These districts provide high-quality usable spaces, promote the formation of energy communities, and play a crucial role in local and regional energy supply [5], [6], [7].

Objectives and Methodological Approach

The challenges and opportunities offered by PEDs, including energy flexibility, load shifting, grid services, and sector coupling, as well as aspects of user integration and their requirements for livable districts, were examined by developing and executing an "Innovation Course", developed and conducted for and with all relevant stakeholders and practitioners of an innovative and integrative planning process, i.e. developers, building and utility operators, architects, participation and communication specialists, building, mobility, building automation and energy engineers. Figure 1 shows the specific fields included and combined. This approach enabled the integration of practical insights with current scientific findings, discussing and combining them: Particularly, legal, and administrative conditions and barriers in implementation were successfully identified, and concrete proposals for their elimination were jointly formulated.



Figure 1 Digital Positive Energy Districts were investigated in ten topical Modules and

Results and Insights

The project demonstrates that PEDs, especially newly built have significant potential due to mixed-use and property-wide consideration of supply and demand curves. The vision of a connected energy system through infrastructural linking (heating/cooling/electricity network) and the establishment of organizational frameworks, such as local energy communities, offers a scalable, efficient way to achieve the energy and climate goals of 100% renewable electricity supply by 2030 and 100% total energy consumption by 2040 in an urban context. However, these challenges are more complex in retrofitting existing buildings. Key insights and topics are summarized in the following sections:

Positive Energy District Performance Assessment through Energy and Emission Balancing

The project investigated how districts' energy and emission flows can be modeled and simulated to obtain dynamic assessments on the interplay of local renewable energy production and flexible district operation. Over the last years, several approaches have been put forward [4], [5], [6], [7], [8], [9], [10], [11], that can be classified by the key considerations illustrated in the following figure: System boundaries, weighting and conversion factors of energy flows in the assessment balance, and the target of the balance itself. As the name implies, the balance target of a Positive Energy District is generally accepted to be positive. This alone is however not sufficient to define a PED, as the scope of this positive balance varies greatly for different **functional, spatial and temporal system boundaries**: Functional system boundaries define which energy services are to be included in the balance and can range from covering HVAC operation, user plug loads, district street lights, individual motorized mobility, public transport, to embodied energy and emissions for construction, maintenance, and even food and holiday air travel of inhabitants. The inclusion of local renewable energy sources can change for different spatial

boundaries: It is not uncommon to include “external” potentials such as large PV-roofs of neighboring industrial uses or contractually secured offsite wind power into so-called “virtual” system boundaries, leading to “virtual PEDs”[12]. Finally, although not universally accepted, the PED balance is typically assessed in terms of Primary Energy flowing over the system boundary, but different weighting factors are known to play a significant impact role in deciding the final result [11], [13], [14].

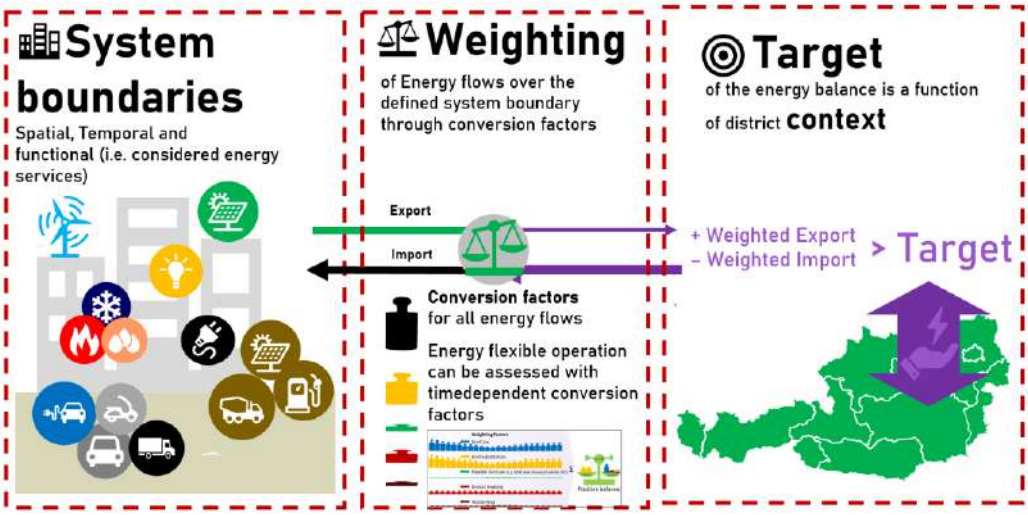


Figure 2 Three main consideration for quantitative Positive Energy District assessment [15]

In summary, the positive balance target of PEDs can not be considered a concrete standard, but rather a vehicle for instigating ambitious and holistic district development practices and communicating quantitative effects of sustainability measures. As such the approaches to PED assessment can be divided in one of the two: First, **process oriented approaches** acknowledge the diversity in context, challenges and ambitions between district development projects and thus favor flexibility in assessment scope and or targets [9], [16], [17], [18]. In Austria e.g., the **“klimaaktiv Siedlungen und Quartiere”** standard offers a step-by-step guide including certification for this process to interested district developers[5].

Normative approaches on the other hand aim to define PEDs through the use of quantitative targets for the district energy and emission balance [15], [19], [20]. This can be used to connect to municipal and national decarbonization goals, as it allows to bridge the gap between Top-Down goals and scenarios with Bottom-Up assessment models such as energy building performance certificates (EBPCs). In the case of DigiPEQ, the main approach investigated focuses on district balance targets linked to national and sectoral energy and emission scenarios [21]: The energy and emission balance model was examined as a quantitative evaluation system for positive energy districts and applied considering energy services for operation, everyday mobility, construction, and maintenance, and also the urban context [15]. The assessment is based on an annual balance of primary energy and emission import and export with monthly conversion factors and an hourly simulation resolution. Here, balance targets crucially depends on the context of the district’s density: Dense urban districts have a lower than zero balance target, whereas the balance targets for scattered settlements increases with one over the density, expressed as the floor area ratio. Analyzing green-field districts of different density in seven Austrian counties showed that all districts can achieve their density-specific balance target with high, but comparable ambition in terms of energy efficiency of thermal hulls and HVAC system, use of energy flexibility and high utilization of photovoltaics. In this approach, flexibility can also be assessed as part of the primary energy balance by assigning grid-supportive or flexible use of energy a discounted weighting in terms of its primary energy conversion factor. Similarly, seasonal mismatches are assessed by using hourly and/or monthly conversion factors for grid use and grid feed-in.

[5] See https://www.klimaaktiv.at/gemeinden/qualitaetssicherung/Siedlungen.html

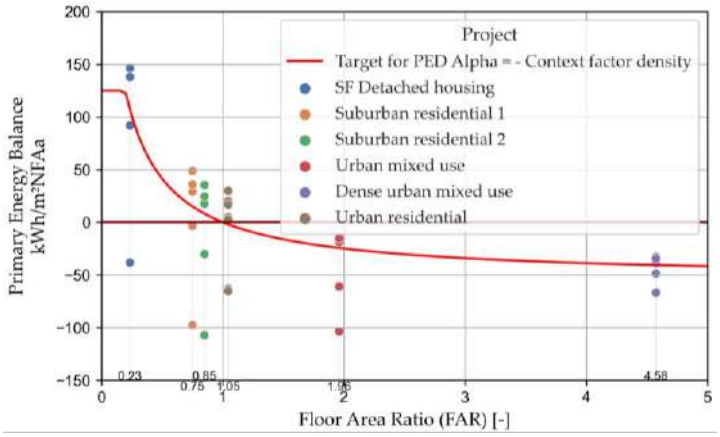


Figure 3 Austrian Target (red line) and modelled Primary Energy Balance for seven districts of different density (colored dots). Floor Area Ratio is defined as the gross floor area over the building plot area

Prediction of Loads and User Behavior

The project further investigated how districts' energy and emission flows can be modeled and simulated in practice to dynamically assess the interplay of local renewable energy production and flexible district operation and sector coupling: Whereas most of today's systems are designed and dimensioned on a balance and billing system only, where the matching of demand and supply is inherently done by the physics of the system only, cross sector coupling or integrated energy systems demands for more coordinated ways to exchange energy. A prerequisite of current installations is that demand is much greater than supply, so that no curtailment needs to be applied. Yet, PEDs can use renewable energies much more efficient if there are control algorithms applied that actively match supply and demand. Applying such control algorithms enables PEDs to serve as load even for largescale access power from wind parks or large PV power plants. Recent scientific work is investigating these issues to avoid curtailment in production, prevent, increase the number of renewable production volume, prevent supply shortfall and congestions. E.g., [22], [23] uses so called active community controllers for coordination and optimal usage of flexibility using electric and thermal storage capacities in buildings. Other projects such as [24], [25] use flexibilities from electric vehicles having the advantage that batteries in electric vehicles have a dual use reducing CO2 footprint and giving an important co-financing especially when used in car sharing operation. Another option is the storage of electricity via heat pumps in thermally activated building systems (TABS). With high thermal mass and insulation, these systems can block their heating and cooling operation to 6-12 hours with increased load in a timeframe of three to seven days without energy demand. Recent studies have concluded that user comfort is not negatively impacted for strictly positive temperature deltas of up to 3K (e.g. heating to 25°C instead of 22°C when a flexibility signal is present and withholding operation until the minimum setpoint of 22°C is reached again). Figure 4 illustrates these flexibility strategies in PEDs:

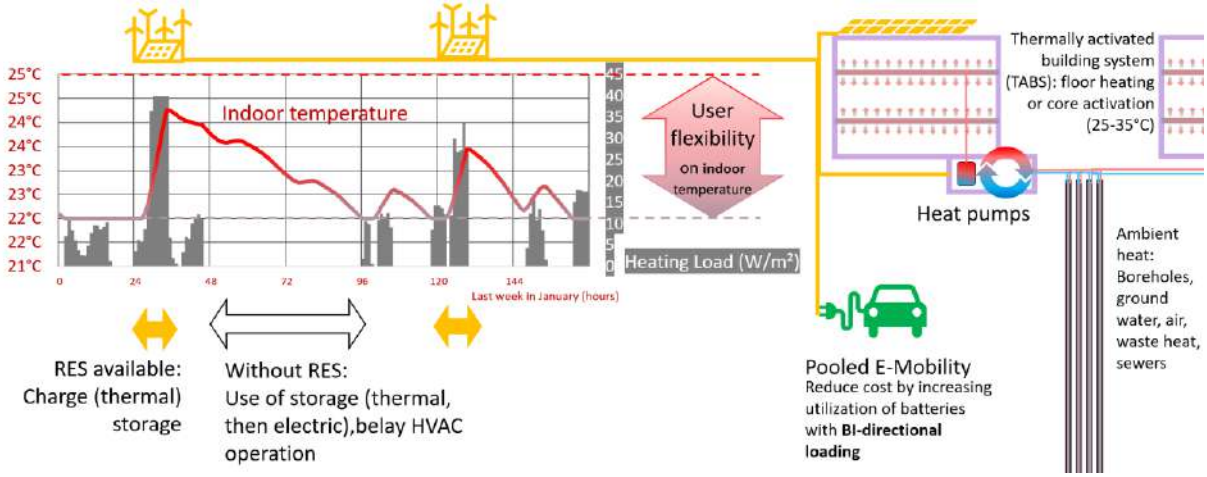


Figure 4 Example of flexible heating and cooling with Thermally Activated Building Systems and User flexibility of up to 3K increase in indoor air temperature during heating season

An important issue for such concepts is the prediction of load and consumption, which was a hot topic among the stakeholder community in the “innovation course”. Standard load profiles are the state-of-the-art to predict users’ load behavior. These profiles leverage the fact that with an increasing number of consumers spurious effects annihilate due to statistic effects. Figure 5 shows the effect of this standardization of load profiles depending on the number of users.

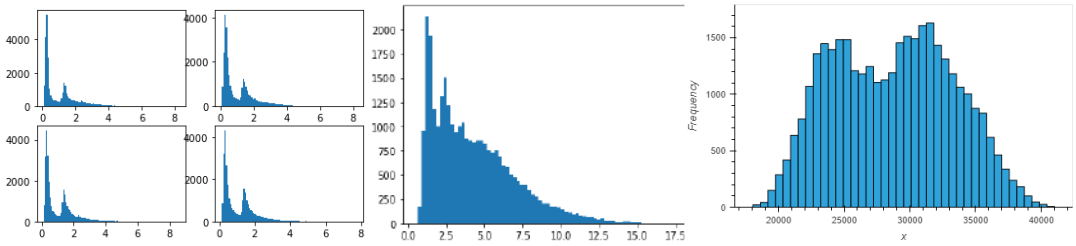


Figure 5: Load profiles for single households (left), aggregation of 4 households (middle), fully integrated standard load profile over all users (right) (figures based on data from [26])

On the base of practical installations, the participants of the innovation course investigated what is the effect of load profile assumptions on typical projects. Although this requires further research, the results show important effects. The participants investigated a residential home complex with 30 apartments and 1 shop by using the smart metering data from the housing units. They classified the housing units by the number of inhabitants and the life circumstances (e.g., working couple – no kids, family with one employed person, retired persons, etc.). Alternatively, [27] defined more individual personas including a more detailed background or [28] used machine learning to retrieve energy consumptions of individual production steps in a factory. In general, it is expected that due to uncertainties in the tenant structure (changes during the lifetime of a building need to be accounted for) a simple characterization will not provide large drawbacks and offer planning flexibility.

With a developed tool the participants have been able to create individual load profiles for differently used flats and aggregate them according to the building usage based only on the assumption of a yearly overall consumption. In particular for buildings with similar social background (e.g., only retired people) the outcomes deviated significantly from assumptions based on standard load profiles. This is an important fact to be considered in energy business models and resulted, e.g., in different ROI for the investment of roof-top PV installations. Outcomes are important for smaller installations such as renewable energy communities (REC) since local effects have a higher influence. Nevertheless, it might also be applicable to citizen energy communities (CEC) with a few but large consumers.

In PED additionally the status of the grid has to be considered in automation and optimization algorithms, since the amount of exchangeable energy is limited by grid capacity. It was identified that mechanisms are required to achieve the goal of a positive energy balance of the district. Although technical and organizational rules (e.g. grid codes) are in place to stabilize frequency and voltage forward-looking and anticipatory signalization is required to use volatile renewable energy sources, especially, in urban PED with limited space for production including spatial conflicts with recreation and ecosystem services. Within the “innovation course” a traffic light system (see Figure 6) was presented to allow planning and optimization in the 5 minutes range relieving reserve energy provisioning and giving consumers and prosumers a tool to act in a grid friendly way.



red	green	blue	Interpretation	Response
0	0	0	no light ⇒ no signal received	act as usual
1	0	0	grid access is diconnected	island mode if possible
1	0	1	energy supply is scarce	insert more & use less now if feasible
0	0	1	energy is "expensive"	insert more & use less if convenient
0	1	1	ambivalent situation	act as usual
0	1	0	energy is "cheap"	insert less & use more if useful
1	1	0	green curtailment is enforced	insert less & use more now if feasible
1	1	1	ambiguous signal ⇒ ignore it	act as usual

Figure 6: Multi-Level Traffic Light Signals Integrating Energy Market and Grid Needs [29]

Energy Economic, Legal, and Technical Environment

From an energy economic perspective, the contributions of Positive Energy District primarily is to balance load and consumption imbalances in general and to grid services in particular were examined. One objective was to actively motivate users to grid-friendly operating modes, thereby providing energy flexibility resources.

Potential economic welfare gains of PEDs are found to be significant, both in terms of monetary dimension as well as with regards to positive environmental and societal effects [30].

The regulatory framework plays a decisive role in the development and implementation of positive energy districts. With regards to potential hindering blocks for rapid implementation, several legal frameworks require particular attention. Especially the rights of and for consumers play a vital role.

This entails rules specific for Austria, such as the heating cost accounting law, which entails a very strong legal position in favor of tenants and homeowners. Furthermore, legal frameworks will most likely be modified once positive energy districts attain increased significance.

Thus, in summary, though PEDs offer a promising solution for the sustainable energy supply of residential and commercial areas, specific problems require solving before they reach further practical significance. The regulatory characteristics of the Austrian energy system which entail specificalities such as feed-in tariffs and laws abiding the integration into the electricity grid, are decisive for the viability and scalability of positive energy districts. Close cooperation between politicians, energy suppliers and the regulatory authority (Energie Control Austria) will be required to drive forward their development and implementation and thereby support a sustainable energy transition.

Information Management and Use of Participative Design Approaches

One significant objective driving these contributions is facilitating user empowerment to cultivate sustainable awareness. Incorporating gamification approaches and providing attractive information are critical strategies for achieving this objective. As applied in sustainable development, gamification involves integrating game elements into non-game contexts to enhance engagement and encourage positive behaviors. This approach has shown promise in various domains, including education, health, and environmental conservation.

Research indicates that gamification can effectively engage users and motivate sustainable behaviors. For instance, a study by [31] highlights the positive impact of gamification on user engagement, emphasizing its potential to enhance motivation and participation. By integrating game-like elements, such as challenges, rewards, and competitions, into information management systems, users are more likely to participate in sustainability initiatives and adopt environmentally friendly practices actively.

Participative design approaches further amplify the impact of information management in plus-energy districts. These approaches involve actively involving end-users in the design process, ensuring their perspectives, needs, and preferences are considered. Research by [32] emphasizes the importance of co-design in creating solutions that resonate with users, fostering a sense of ownership and commitment. In the context of plus-energy districts, participative design enhances the relevance of

information and promotes a sense of community and collective responsibility. For embedding the participative design even further, the concept of personas has been used in the creation process of the information management. Personas are fictional characters that shall represent different future user types within the target audience, hereby especially in regards to ecological behavior, environmental awareness and their sustainable or not-so sustainable attitudes. With the help of personas, significantly more life situations have been able to be covered in the design to avoid bias and blindness to everyday situations of future inhabitants.

In conclusion to this point, the convergence of information management and participative design approaches holds immense potential for promoting community-friendly climate behavior in PEDs. By incorporating gamification strategies, the use of personas and ensuring the accessibility of information, stakeholders can enhance user engagement and foster sustainable awareness for plenty of target groups. Additionally, the integration of participative design approaches empowers residents, democratizes decision-making processes, and aligns information management strategies with the community's specific needs and avoids bias that could lead to exclusion.

Consideration and Application Possibilities of New Digital Technologies

The contemporary landscape of digital technologies has undergone a transformative evolution, with an escalating adoption of cutting-edge innovations over the last decade. The emergent Web3 technologies such as Blockchain, Ethereum and other protocols, Machine Learning (ML) for data processing, and Artificial Intelligence (AI)-based generators for image and text generation, are of particular significance. In this era of technological proliferation, the discourse on the consideration and application possibilities of these new digital technologies becomes paramount, especially in the context of PEDs, where the intersection of sustainability and technological advancement holds the promise of revolutionary developments.

ML algorithms, with their capacity to analyze vast datasets and discern intricate patterns, contribute significantly to optimizing user experiences by personalizing content recommendations and enhancing predictive functionalities [33] the context of PEDs, ML applications extend to energy management systems, enabling adaptive and intelligent control mechanisms that enhance energy efficiency. Within the project, applications were investigated that can manage the information flow from data acquisition at sensor level via preprocessing to maintaining consistent data storages. After a basic introduction, various company data was analyzed and directions given on how to gather, collect, clean and store the data setup necessary to work with ML. Yet, without opening the confidential company data to the public, following conclusions have been drawn:

- There is nothing, a script cannot do (with the prerequisite, access rights and connectivity to data points is given)
- Regular testing and plausibility checks on automated structures need to be done.

Certain problem types can be handled with toolchains such as Great Expectations under Apache-2.0 License, however they give mostly little freedom for adaption to be embedded into existing company toolchains [34]. In the context of Web3, also the tokenization (the virtual representation of a predefined unit, for example, a kWh) of energy assets through blockchain, coupled with using digital tokens as rewards, offers a tailored and effective incentive mechanism for motivating energy-saving behaviors in the unique context of plus-energy districts. By aligning with behavioral economic principles and emphasizing positive reinforcement, this design approach encourages personas and future residents to actively contribute to their PEDs surplus energy generation and overall sustainability goals.

Finally, there is to say, that the use of multiple new digital technologies might lead to exclusion of certain, most typically older, user-groups. With participatory design as explained in the previous paragraph, the complexity of understanding and representing such mechanisms shall be minimized in order to achieve the desired results. Needless to say, the most sophisticated technology is of no use, if there are no users willing or able to use it.

Perceived relevance of PED topics by practitioners

When bringing innovative concepts such as PEDs to the ground, the barriers and challenges perceived by practitioners and stakeholders of concrete district developments should be considered. The following

figure summarizes the feedback of participating practitioners, that are all already involved in implementing PEDs as part of their day-to-day activities, in terms of most interesting, most novel (new) topics, and which topics they could take-away the most: The results showcase the importance of cross-cutting activities and breaking up existing silos and highlight certain key topics such as green financing, energy flexibility modelling, ecological and advanced district scale energy modelling, but also possible applications of AI and the importance of site visits and learning from real-world examples.

Table 1 Practitioners feedback when being asked what topics were most "new", most "interesting" and had the highest "take-away" for their practical work. Grades rank from 1 (highest) to 5 (lowest)

Rank	Ten best "New" Modules	Grade	Ten most "Interesting" Modules	Grade	Ten greatest "Take-Aways"	Grade
1	Green Financing	1,0	Übung "Raus aus Gas - Rein in PEQ"	1,0	"Raus aus Gas - Rein in PEQ"	1,0
2	"Raus aus Gas - Rein in PEQ"	1,3	Übung anhand Smart Block Geblergasse	1,0	Übung anhand Smart Block Geblergasse	1,2
3	Advanced Machine Learning	1,4	Flexibilität	1,1	Workshop Vernetzung u Reflexion	1,4
4	Exkursion Smart Block Geblergasse	1,5	Green Financing	1,1	Ökologische Bewertungsmethoden	1,4
5	Gruppenarbeit Usability	1,6	Impulsvorträge und Diskussion	1,1	(Tages)Lichtversorgung	1,4
6	Simulation mit IDA Districts/ICE	1,6	Gruppenarbeit Usability	1,2	Exkursion Co-living Seeparq	1,4
7	Vereinfachte Verfahren in der Quartiersimulation	1,6	Ökologische Bewertungsmethoden	1,2	Impulsvorträge und Diskussion	1,4
8	Smart Block Geblergasse	1,7	Energiemärkte im Umbruch	1,2	Energiemärkte im Umbruch	1,4
9	Inbetriebnahme-management	1,7	(Tages)Lichtversorgung	1,3	Verkehrsentstehung	1,4
10	Model Predictive Control	1,7	Erschließung von Flexibilitäten in aktiven Energiegemeinschaften	1,3	Exkursion Smart Block Geblergasse	1,5

The importance of cross-collaboration is also showcased in these projects designed to transfer new knowledge and skills on PED development into the field by a joint effort from practitioners and researchers. It highlights the concrete gaps and opportunities for the improvement of planning infrastructure, methods and result:.

Table 2 Knowledge and Skill Transfer projects for practitioners of PED development

	Transfer project	Description
	Supergrätzel	Designing a combination of energy communities and 'Supergrätzel' utilizing existing networks with a focus on social aspects (energy poverty).
	Anonymous Measured Profiles	Analysis of electricity consumption data from different households in a multi-family house as a basis for Plus-Energy planning.
	Energy Flow in Small Communities	Access to energy production and consumption data to create an energy flow diagram with a focus on heat consumption.
	Planning Data Pool	Collecting and recording electricity and heat consumption data to develop characteristic values for the planning and monitoring of Plus-Energy districts.
	From PED to AD (autonomous district)	Creating a roadmap for the development (or extension) of a simulation tool for Positive-Energy districts focusing on the degree of energy autonomy.

	Data Quality Monitoring	Checking data sets before they are used for further processing.
	OBENAUF - District with a Cherry on Top	Optimization of roof areas at the district level using an existing densification project (on the top, Währingerstraße) - measures and feasibility.
	Onboarding Tool EEG	Development of a user journey for the digital founding of a Renewable Energy Community for implementation on the team4.energy platform.
	Integral Planning Freudenau	Construction kit for a development concept for a future renovation district (Freudenau).

In a “PEQathon” format, the complex topic was finally successfully tested in an interdisciplinary planning workshop using concrete urban development areas in Vienna as examples. The results were adopted by the City of Vienna: The PED consideration and the resulting balance goals for energy supply and emissions in operation, mobility, and the establishment of a climate-neutral district are now incorporated into the development and implementation plan of the “WieNeu+” and “RausAusGas” district “Am Tabor”.

The PEQathon results were assessed by a jury using the following point-based catalogue, that tries to encapsulate and weigh the most important aspects of PED development at an early planning stage:

Table 3 “PEQathon” PED early planning process assessment catalogue

Task		Criterion	Weight
Task A: District and Neighborhood		Achievement of quantitative PED Targets (Primary Energy Balance)	10
		Design of the Development Process	5
		Climate Resilience, Avoidance of Heat Islands, Rainwater Management	5
		Energy Flexibility and Storage	5
		Suitability for Integration of Digital Components	5
		Sector Coupling and Grid Services	5
		Fair Effort Sharing for the Transformation of the Entire District	5
		Integration of Stakeholders and Users	10
Task B: Micro-Districts		Proof of PED Targets	20
		Greening Concept	10
		Economic Efficiency over the Life Cycle	5
		Probability of Implementation	5
Task C		Improvement of Tender Documents	10

Final Considerations and Outlook

The added value of the PED consideration lies in the context-specific, but holistic assessment of building operation, everyday mobility, and embodied energy at the district level, thus bridging the gap between individual building code and assessments to overarching climate and energy goals of municipalities and cities. This also includes important questions of "Effort Sharing" and how the burdens and responsibilities of the transition can be distributed.

The collaborative exploration in ten thematic modules revealed that digital, livable PEDs can significantly contribute to national and municipal energy goals. Key insights include the importance of user-oriented

and sustainable models, innovative digital technologies, and the need for replicable planning and operational processes in PEDs. The project underscored the challenges and opportunities in new constructions and retrofitting existing buildings, highlighting the complexities of energy flexibility, load shifting, and grid integration. The research also revealed the critical role of participative design and information management in fostering sustainable behaviors within PED communities. The exploration of new digital technologies such as ML, AI, and Web3, offered insights into future possibilities in energy management and user engagement. The project's findings are crucial for stakeholders, practitioners, and policymakers, providing a roadmap for implementing PEDs effectively.

In conclusion, DigiPEQ underlines the transformative potential of PEDs in achieving sustainable urban development, enhancing energy efficiency, and advancing towards climate neutrality.

It should also be noted that the term "**Positive Energy District**" is frequent cause for confusion and debate as to what exactly it refers to. It should be replaced by the more comprehensible term "**Climate Neutral District**" in future research.

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B.5. PEExcel: A Fast One-Stop-Shop Assessment and Simulation Framework for Positive Energy Districts

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Contribution to the thesis:

Lightweight, spreadsheet-based PED simulation framework enabling rapid, hourly PED assessment consistent with a quantitative definition.

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PEExcel: A fast one-stop-shop Assessment and Simulation framework for Positive Energy Districts

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Abstract

We present a simulation and assessment framework *PEExcel* that allows researchers and practitioners to quickly model and assess the primary energy and emission balance of operation, use and mobility of a district beginning in an early planning phase as a decision support system (DSS) for the Austrian klima**aktiv** label "climate neutral Positive-Energy-District-" (ger. "klimaneutrales Plus-Energie-Quartier").

The framework provides a one-stop-shop spreadsheet-based tool for input gathering, simulation, analysis, and assessment. It provides a quick and comprehensive overview on user-specified scenarios that achieve the required balances targets by the standard and features simulation times below ten seconds per district scenario on regular laptop machines.

It implements an hourly thermoelectric simulation model and a built-in energy-flexible DSM controller for thermally activated building systems (TABS), hot water buffer tanks, electrochemical and EV batteries as well as offsite flexibility signals.

It also serves as a common receiver and merger for most common assessment results: Timestep data for onsite renewable heat and electricity generation from generic sources, load and occupancy profiles, weather and climate data, conversion factors, etc. The framework combines this data in a ready-made analysis and assessment toolbox that includes a dashboard, interactive views of input and simulation timestep data (loads, States-of-charge, etc.), and formatted results pages featuring energy balances, load profiles, energy flows and assessment results.

Introduction

Positive Energy Districts (PED) represent large and complex systems, their implementation requires integrated planning in a holistic manner across multiple disciplines and domains ("Ten questions concerning positive energy districts | Elsevier Enhanced Reader," n.d.). City planners believe that addressing PED implementation challenges in the early conceptual phase through a broad stakeholder commitment approach can help prevent many problems by incorporating these intentions into the master planning directives (Nielsen et al., 2019).

However, it is apparent that for a PED definition focusing on a holistic and integrated energy assessment, including building operation and use, mobility and embodied energy, current dynamic models are unfeasible to deploy in early planning phases (Hedman et al., 2021).

The lack of early integration of energy planning in the district development is mostly due to the lack of data to feed more intricate energy simulation frameworks (Koutra et al., 2018): Even if sufficient data is available and modelling possible, using district simulation models is typically too resource-intensive to deploy early, as the planning will inevitably change in the process, rendering detailed simulation results invalid. On the other hand, simpler decision support systems (DSS) employing stationary methods can lack resolution of key flexibility features of PEDs or lack coverage of all required energy flows.

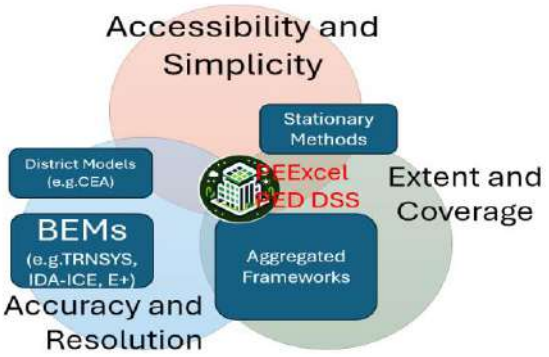


Figure 1 District energy models (DEMs) trade-off between accessibility & simplicity of modeling, accuracy, resolution and speed of simulation results, and coverage of energy services (from HVAC, DHW, plug-loads, mobility to embodied energy)

Aims and Design principles

The framework design is based on district implementation observations: PED feasibility is investigated in early district planning with many project data still unknown, and building mass and geometries can be subject to frequent change. This work aims to provide a simulation and assessment framework that is **simple to use**, **highly iterable** and **covers all aspects** of the klima**aktiv** "Positive-Energy-District Standard" of district operation and use, mobility and embodied energy and emissions with sufficient temporal and spatial resolution to serve as a DSS in an early planning stage. The theoretical framework of the underlying PED definition is based on (Schneider et al., 2023).

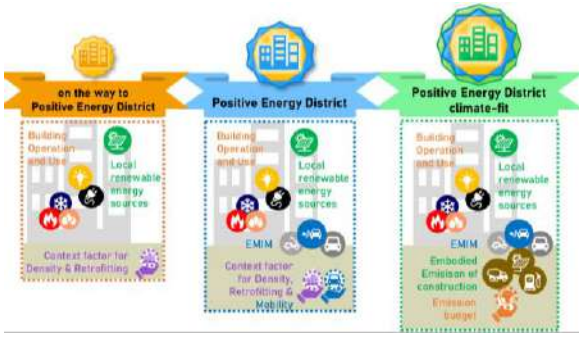


Figure 2 Available klima**aktiv** certifications for three system boundaries from operation and use (left) to individual motorized mobility (center) to embodied emissions (right)

This PED assessment necessitates hourly modelling of energy flows including e-mobility to resolve effects of sector-coupling and flexible operation and integration of electrical (batteries and EVs) and thermal storage (TABS and DHW tanks).

In addition to these requirements, the design of the framework pursues the following goals: 1) **Reduce cost of model creation**: "Default first, input later": Reduction of required input data by extensive provisioning of defaults from Austrian building code and scientific literature (Institute for Housing and Environment, 2017; Österreichisches Institut für Bautechnik, 2019). 2) **Reduce cost of data transformation**: Provision of copy-paste interfaces for data from Energy Performance Certificates (EPC) and Passive House Planning Packages (PHPP). 3) Simple creation and fast iteration through different district **scenarios**. 4) Simple **iterability** within the planning process. 5) **Accessibility and transparency**: No use of proprietary tools and software. Open access of the algorithms and data, transparency of calculations for practitioners without skills or experience in programming or scripting. 7) Simple **extensibility** and **customizability** of the simulation model for practitioners. 8) One-Stop-Shop for planning, evaluating, quality assurance and verification, reduction of data transfer requirements to a minimum.

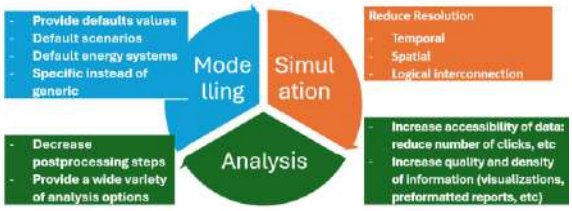


Figure 3 Design goals of the framework for three key activities of PED planning

Approach and Structure

Due to the design goals, the framework is implemented as an Excel Spreadsheet, and the simulation model is exclusively implemented in cell formulas, allowing for immediate tracing of errors, as well as edits and additions without resorting to VBA macros, or other types of scripting (i.e. with Python or MATLAB). This also ensures that virtually all stakeholders can receive, open, use, change and send the framework by themselves. The framework presentation is structured in the following three sections:

1. Model input structure and interfaces
2. Simulation model
3. Analysis and Assessment

Model input structure and interfaces

The implementation as a spreadsheet tool realizes numerical inputs in two ways:

1. Independent, atomic parameters are input in cells.
2. Semantically connected parameters (e.g. timeseries data or catalogue components) are grouped into catalogues stored in 1d or 2d cell ranges and selected by selector cell inputs.

More complex interfaces supported by Excel such as links to local or remote files or databases were foregone for the sake of accessibility and reduced external points of failure.

All model inputs can be categorized as mandatory or optional. Mandatory inputs have been reduced as much as possible and provisioned with default parameters and catalogues drawn from Austrian legal code and best-practice examples (Passive house components, etc.). The following table overviews the number and type of parameters for district modelling, requiring at a minimum nine numeric values, five catalogue selections and five

Boolean selections, which specify reference areas and usage mix, type of hull, HVAC system and flexibility strategy. In addition, 153 numeric values, 15 catalogue and 7 Boolean selections are provisioned with default values but can be specified if known by the user. Although the framework provides default timeseries data for both weather data and PV yields of a handful of locations in Austria that can be used as a starting point, TSD are typically required to be provided from other sources. But all required timeseries such as weather, usage profiles and onsite renewable production through PV can also be used from default settings. The following table shows mandatory and optional inputs per model category.

Table 1 Types of input parameters: **Mandatory** inputs are bolded, +optional inputs with defaults are preceded by a +, (N) Numeric Values, (B) Boolean, (C) Catalogue Selections, Annual Timestep Data (TSD),

Model input category	**Mandatory** +optional inputs
Reference Areas: Gross floor areas per usage	**6N**+2N
Thermal Hull	**4N1C** +31N
HVAC system	+11C,35N,6B
Space Use	+11N,66TSD
Local Renewables	**1B**+2N,1B,2TSD
Mobility	**1B**+1C,48N
Flexible Control, DSM	**3B,3C**+12N,1TSD
Weather	**1C**+6TSD
Embodied GWP	+11N
Balance Assessment	**1N**+1N,2C,3TSD
Total mandatory and +optional model inputs	**9N,5C,5B** +153N,15C,7B, +76TSD

Input and Output serialization

All non-timeseries, atomic input data (numeric, Boolean and catalogue selections), both mandatory and optional is serialized into an input vector with 194 values, representing a district scenario that together with a 295d vector of annual outputs can be represented by a single row of tabular data. Such scenario rows can subsequently be easily created, copied, compared, and reloaded for later analysis and refinement. This also facilitates the unsupervised serial simulation of batches of dozens to hundreds of scenarios configuration without additional user input.

Multiple Input layers

The serialization of model inputs also facilitates the separation of concerns within the same excel spreadsheet: Separate, user specific sheets can expose certain inputs and outputs selectively, including interfaces that the users are familiar with (e.g. PHPP, EPC forms, for architects, for bachelor and masters students), filling in the rest according to user group specific assumptions. Researchers also have access to the simulation model for verification and modification purposes and maintainers actively manage these views with permission management.

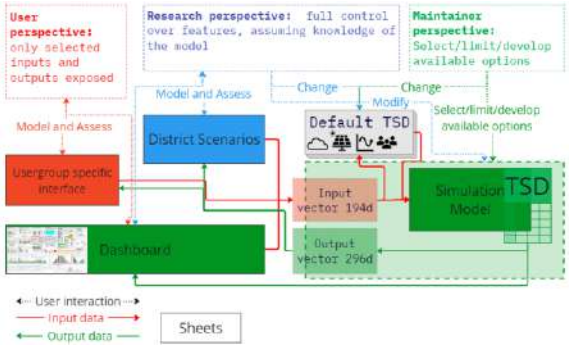


Figure 4 Model inputs and data interface structure of the PEExcel PED DSS framework.

User interface

In addition to relying on Excel's own application interface, the framework implements a custom UI ribbon offering shortcuts for menu and sheet navigation, selecting and editing scenarios.



Figure 5 Custom UI ribbon of the PEExcel PED DSS framework.

Simulation Model

The simulation model is designed to assess district scenarios in regard to their achievement of the positive energy balance targets of the klima**aktiv** PED certification standards, which necessitate the following requirements: 1) Dynamic simulation of energy flows on an hourly basis, 2) Representation of all energy services and flows for operation, use of the district's buildings, then individual motorized mobility and scope 3 embodied energy and emissions and 3)

Use of standardized defaults for space use profiles and primary energy conversion factors (Österreichisches Institut für Bautechnik, 2019). To achieve this, the simulation model is designed to trade low spatial resolution for high temporal resolution (hourly) and functional coverage over an entire year. It implements a simple single thermal zone with an RC-model for the entire district.

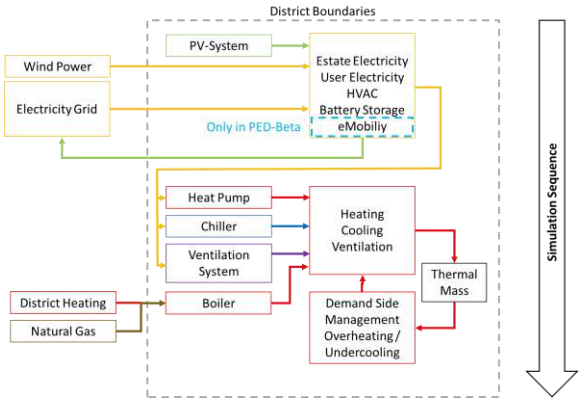


Figure 6 Model energy flows and nodes.

As can be seen in Figure 6, the model can include a boiler and a heat pump system. For comparison reasons, districts are simulated by default with four common heating systems shown in Table 2, main difference being the conversion efficiencies, transmission losses and PE- and emission conversion factors for the balance assessment.

Table 2 Default energy supply options

Energy Supply	Heating, DHW	Cooling
Natural gas	Combi boiler	VCR
District heating	DH station	VCR
Heat pump without DSM	Heat pump	Heat pump
Heat pump with DSM	Heat pump	Heat pump

Flexibility

Flexible district operation through demand side management (DSM) is modelled by using TSD of a standardized signal to control the charging and discharging of district storages, that include both thermal and electric options: TABS, DHW buffer tanks, EVs and electric batteries. Key element of the usage of building thermal mass as storage is for users to consent to a potential overachievement of their temperature setpoints, e.g. 24°C instead of 22°C, but never a reduction of setpoints. Similarly, EV, Electric battery and DHW buffer tank state-of-charge (SoC) are controlled between a minimum and maximum setpoint storing local and regional renewable surpluses whenever a corresponding signal is present.

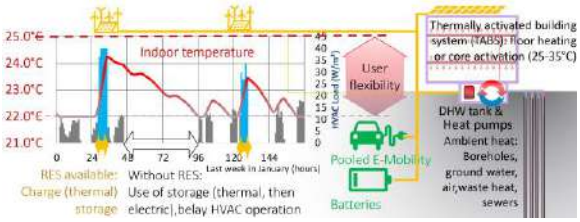


Figure 7 Flexible operation of district thermal and electric energy storages.

The DSM signal governing times of opportune grid consumption can be user defined but is representing offsite wind peak events in Austria 2015 by default for certification purposes.

One drawback of spreadsheet-based models is that dynamic and branching logical structures are hard to implement, debug and compute thanks to the required linearization into a calculation chain. This prohibits an arbitrary configuration of energy dispatch between sources and sinks. Instead, a fixed dispatching order is implemented, whose individual flows can be calibrated or turned off, but not reordered or redirected. In this linear chain, the first energy supply dispatched is from local renewables, followed by battery discharge, then from volatile regional sources (e.g. wind peak shaving) and only then regular grid electricity is used. On the demand side, user plug loads are covered first, followed by HVAC and E-mobility demand for regular operation reaching minimum specified requirements (i.e. room and DHW temperatures, aggregated EV SoC). At this point any available local or regional renewable surpluses are used for DSM, i.e. to charge the available thermal and electric storages (TABS, DHW buffer tank, EV and other batteries.



Figure 8 Model system boundary and energy flow dispatch strategy between model components for each hourly timestep.

Mobility

Key feature of the PEExcel DSS framework is the quantitative model for energy and emissions associated with everyday motorized individual

mobility (EMIM) from the district. It uses annual statistical mileage per modal split and regional location based on the Austrian Mobility (Tomschy et al., 2016).

The association of personal EMIM mileage to the district reference areas is achieved by a top-down allocation approach for all usages, allocating the statistical EMIM mileage with a certain usage destination (home, work, shopping) evenly across all floor areas in Austria of that type (residential, office, etc.) As described in (Schneider et al., 2023), this virtual balancing approach only considers energy demands from destination travel. Actual charging and fuel demand can be modelled by increasing the ratio of source travel also taken into the energy balance. It also includes a virtual sectoral mobility energy budget derived from national decarbonization scenarios.

This allows the assessment of changes in the energy and emission balance based on 1) Share of EVs of total district EMIM, 2) Temporal EV charging profile, 3) EV battery parameters and SOC thresholds, 4) bidirectional loading but also other effects such as reduced EMIM mileage due to alternative mobility measures such as cargo-bike and bike sharing, reduced parking space, and similar measures.

Analysis and Assessment

The *PEExcel* framework gathers the simulation data in a ready-made analysis and assessment toolbox that includes a dashboard, interactive views of input and simulation timestep data (loads, States-of-charge, etc.), and formatted results pages featuring energy balances, load profiles, energy flows and assessment results.

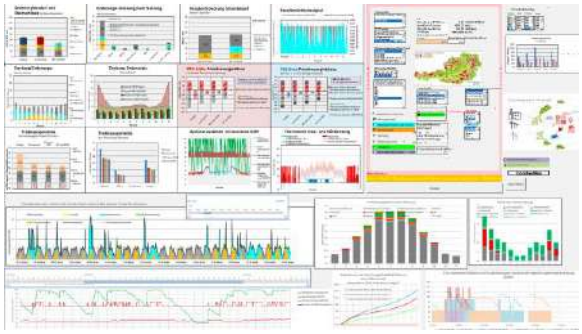


Figure 9 Screenshot of the PEExcel DSS framework Dashboard indicating main scenario selections and results.

Here, the following results are compared for 4 supply scenarios: Natural Gas, District Heating, and Heatpumps with and without flexible operation:

- Annual Energy end use balance
- Annual primary energy balance with and without mobility
- GHG Emission balance over the lifecycle including embodied emissions

Additionally, for the currently investigated scenario, the following results are displayed:

- Annual demand coverage distribution for each energy service
- Monthly Energy End Use
- Monthly Primary Energy demand, renewable and non-renewable
- Hourly Flexibility use
- Hourly storage SOC (electric, e-cars, thermal TABS and DHW tank)
- Hourly heating and cooling loads
-

The main assessment metric for positive energy districts is a positive annual total primary energy balance of energy imports and exports over the system boundary. One key element to achieve this is the inclusion of virtual context factors (Schneider et al., 2023), which depend on the district's density and the surrounding energy system for mobility and which are automatically included in the assessment balance:

The effects of flexible DSM effects can be analysed by a dynamic view of the hourly timeseries that can be panned and zoomed to a specific timeframe. Here, the effects of available local renewable surpluses and regionally available excess wind power and other external grid signals on the district storage SOC of TABS, DHW Buffer tanks, Electric Batteries and EV can be analysed. This flexible operation and its effect on the balance through dynamic weighting of the primary energy conversion factor is described in (Schneider et al., 2023), and its effect on the resulting balance can be assessed in an hourly and monthly view

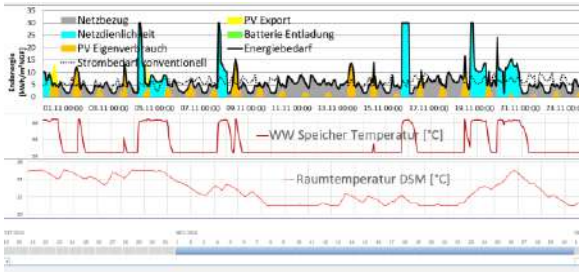


Figure 10 Interactive view of flexible load and storage SOC TSD

Scenario comparison

One major goal of the PEExcel DSS framework is to cover a large area of the PED solution space quickly and facilitate modelling, assessment, and comparison of many scenarios. This is achieved by dynamic diagrams for the energy end use

balance, and the primary energy balance with and without mobility that can display selected simulated district scenarios, as shown below.

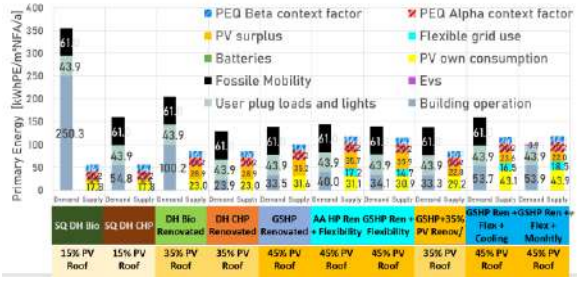


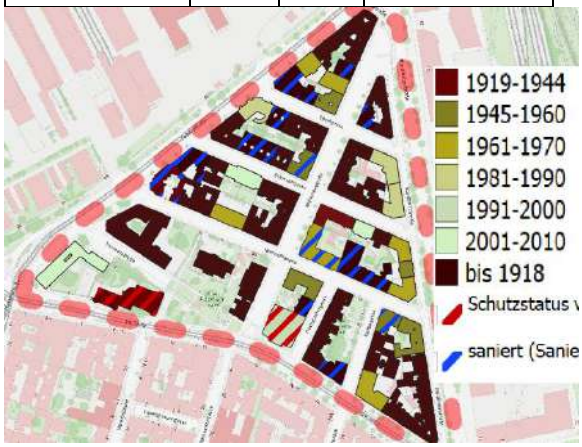
Figure 11 Example comparison of ten different scenarios in primary energy balance for operation and mobility.

Comparison to other DEMs: City Energy Analyst (Fonseca et al., 2016)

A comparison of simulated heating demands for 68 buildings in a historic Viennese district with 90% residential, a wide range of building types and a 20% renovation rate yielded the following results:

Table 3 Example district “Alliiertenviertel” in Vienna

Construction period	Buildings <Original / Renovated>		Gross Floor area [m²]
until 1918	32	11	90 554
1919-1944	2	1	8 466
1945-1960	4	2	16 669
1961-1970	6	1	21 768
1981-1990	3	0	9 846
1991-2000	1	0	3 947
2001-2010	3	0	12 230
Total	**53**	**15**	**180 492**



The assessment framework was compared in two scenarios: As-is (“stock”) and with all buildings renovated to current building code. The results in the following figure show both models with identical building geometry and building specific properties (U-Values, etc) and have an overall good correlation between the excel tool and CEA, especially for the existing building stock. Here, the red line indicates statistically typical heat demands for buildings of that age, renovation and usage. For the renovated cases, our tool results are higher and closer to Austrian renovation benchmarks (Richard Büchele et al., 2021), CEA results were too low in most instances (heating demands below 20 for regular Gründerzeit renovations), which is mostly attributed to default values of occupancy and internal loads not matching Austrian

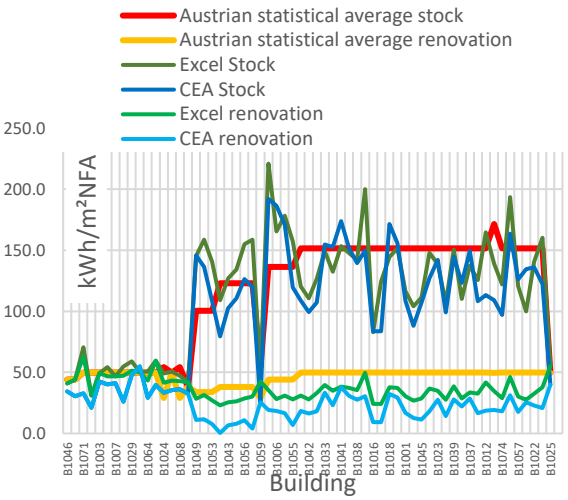


Figure 12 Heat demand for 68 buildings for scenario “stock” (current situation) and “renovation” for our simulation framework “Excel” and the City Energy Analst (CEA) compared to Austrian building type benchmarks

Discussion

This section discusses the advantages and disadvantages of the spreadsheet-based model.

Model Limits

The model design imposes the following that can be differentiated in soft limits, which can be overcome by minor changes in input data or calculation logic, and hard limits that are imposed by the model structure and cannot be circumvented in the current framework. Hard limits are:

- **Usage mix**: The number of different uses (governing load profiles, occupancy, air exchange rates, etc.) can not exceed the 6 predefined uses, however defaults of existing uses can be altered to reflect a different use type altogether.
- **Thermal LOD**: Indoor temperature in the entire district is equally distributed, no

zonal differences can be resolved. Also at any given time, the entire district uses the same temperature setpoint.
- The dispatch sequence cannot be changed or added on to simulate different controls or energy flows, only the existing steps representing individual predefined energy flows in Figure 6 can be enabled and calibrated.
- **Geometry LOD**: Geometric information is merged into eight aggregated will areas facing in eight cardinal directions, and two areas for roofs and ground floors and cellars. Detailed geometric effects such as orientation-based insolation and shading cannot be modelled.
- **Fixed hourly temporal resolution**

Additionally in the default framework the following "soft" limits exist, that can be circumvented if necessary:

- **Monovalent energy system**: Although 4 different systems are simulated simultaneously by default (gas, DH, HP w/&w/o DSM), it is not immediately possible to mix energy systems to rely on two or more energy systems at the same time (excl. PV).
- **Multiple different construction and hull types** in the district cannot be modelled directly but need to be merged by taking averages weighted by the corresponding reference areas. E.g. for thermal transmittances, thermally activated building mass, etc.)

Simulation Speed and Coverage

Simulation speeds of BEMs are heavily dependent on model LOD and size and can reach from seconds to hours and days for large district models. One approach to limiting run time is to limit such simulations to a smaller timeframe such as typical days or weeks. By contrast, the simulation times of our model is independent of district size and is reliably below 10s on regular laptop machines. This cheapness enables the heavy use of variational analysis iterating through 600 district scenarios per hour.

Accessibility: Model creation

The cost of model creation is critical to its accessibility, determined by the extent of manual steps and the availability and quality of default data. Our framework restricts the potential to model any kind of district energy system (DES) to a high LOD and instead aims to make modelling of typical decarbonized DES as cheap as possible with suitable default parameters and catalogue options. It trades a lack of customizability, flexibility and LOD in the solution space for quicker model creation and iteration.

Accessibility: Deployment

The tool is designed as a DSS in the early planning phases for district development stakeholders such as city planners, architects and engineers that are not necessarily familiar with expensive BEMs and DEMs.

Accessibility: Open source

The spreadsheet and supplementary material is publicly available as open source[1]. Furthermore, all calculations and algorithms are directly accessible as cell formulas and can be inspected and changed at will.

Functional Coverage

Although it may be possible in principle, modifying existing frameworks to cover additional energy services, such as user plug-loads, and in particular EMIM and embodied energy, typically also incurs additional modelling costs to supply required data and processing models to the framework. Compared to simple stationary methods, the tool provides higher temporal resolution and accuracy for critical KPIs such as energy self-sufficiency, ratio of own-consumption, and temporally weighted primary energy and emission conversions.

Algorithm programming with VBA

MS Excel files can be modified programmatically with Visual Basic for Applications (VBA). The framework makes use of this, but only for convenience functions such as navigation, loading and saving of scenario configurations and results and other utility functions for development, maintenance, and debugging. In contrast, the calculation logic was deliberately implemented in sheet formulas exclusively to ensure that users can trace calculations and potential errors with readily available excel functions.

Version control

Although limited options exist, a major disadvantage of spreadsheets is their inherent inaccessibility of a kind of version control commonly found in software development (Macedo et al., 2019). The combined structure of data, algorithms and user interface in a single file severely hinders any form of parallel activities, in particular collaborative development of features and version control. As such, structural integrity and validity of the framework implementation is

[1] at https://github.com/simonschaluppe/peexcel

naturally low and development cycles need to be carefully orchestrated.

Testing and Validation

Ensuring the validity of the simulation model is a main concern of the DSS framework. This is typically done by both unit tests and integration tests, testing individual functional units and the integration into the whole system respectively. For a spreadsheet-based solution, unit tests can hardly be realized, because the control flow of the calculations is hardcoded and cannot be changed dynamically. Instead, the framework is forced to rely on integration tests only, which are realized as hundreds of district scenario test cases with known results.

Managing file size

The entire spreadsheet tool of the PED DSS framework consists of 25 sheets. The hourly simulation consists of 375 energy flows, calculated for 8760 timesteps each. When dealing with this amount of structured data in an unstructured data format such as Excel, it is very important to carefully manage cross references between cells and sheets: If done incorrectly, files can gain orders of magnitude in size due to redundant and lengthy links. In general, calculation on any given sheet should only reference previous ones to the left and then to the top, avoiding references to the bottom and right and especially cross-sheet lookups as much as possible. With these optimizations, the file size is 75 MB for regular .xlsm and 48 MB when stored in a binary .xlsb format.

Conclusion

This paper presents a simulation and assessment framework for the Austrian klima**aktiv** certification of climate neutral Positive Energy Districts. It allows researchers and practitioners to quickly model and assess the primary energy and emission balance of operation and mobility of a district and provides a quick yet detailed overview on user-specified scenario configurations that achieve the energy and emission balances required by the certification standard. The framework provides a one-stop-shop spreadsheet-based tool for input gathering, simulation, analysis, and assessment. It serves as a common receiver and merger for most common assessment results: Timestep data for onsite renewable heat and electricity generation from generic sources, stationary load profiles, weather and climate data, conversion factors, etc.

The hourly thermoelectric simulation enables dynamic assessment of energy flexibility operation but due to the limitations of spreadsheets the flexibility strategy itself is fixed.

Its strengths lie in its fast deployment and low cost of model creation, quick iterability and high number of investigable scenarios due to its fast simulation times of five to twenty seconds per run on regular laptop machines.

However, these capabilities come at the expense of the spatial resolution and the flexibility of the energy model itself, limiting the application of the tool to its designed use cases.

Acknowledgements

This research was partly funded by the Austrian Research Promotion Agency (FFG) research project “Zukunftsquartier Österreich” (Schöfmann et al., 2020) and the federal ministry for climate.

Data and Tool Availability Statement

The PEExcel PED DSS framework is publicly available at https://github.com/simonschaluppe/peexcel.

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B.6. Can Urban Retrofitting Achieve a Positive Energy Balance? a Case Study of Four European Positive Energy Districts

Publication type: Conference paper

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Contribution to the thesis:

Systematic, multi-context application of the density-contextualized PED energy balance framework to real European retrofitting districts.

Can urban retrofitting achieve a positive energy balance? A case Study of four European Positive Energy Districts

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***Abstract*—** Urban retrofitting has emerged as a key strategy in the transition towards sustainable cities, with Positive Energy Districts (PEDs) serving as a model for achieving energy-positive urban environments. This paper explores the potential for urban retrofitting to achieve a positive energy balance through a case study of four existing districts in European Municipalities: Settimo Torinese (Italy), Großschönau (Austria), Amsterdam (Netherlands), and Resita (Romania). The analysis leverages energy balance simulations, considering various retrofitting scenarios, including building insulation, photovoltaic (PV) installations, and the adoption of flexible grid usage. The findings indicate that while achieving a PED is challenging, it is attainable through a combination of aggressive retrofitting measures, renewable energy integration, and smart energy management. The study highlights the importance of context-specific strategies, as climatic and urban characteristics significantly influence the outcomes. It aims to add to the ongoing discourse on sustainable urban development by providing empirical insights into the pathways and challenges of achieving PEDs through urban retrofitting.

Keywords — Positive Energy Districts, PED, retrofitting, Climate-Neutral Districts

I. Introduction

Positive Energy Districts (PEDs) represent an advanced step beyond Net-Zero-Energy Buildings, focusing on not just reducing energy consumption but actively generating surplus energy. Rooted in the evolution of sustainable building practices, such as the passive house, PEDs align with the European Union's increasingly stringent Energy Performance of Buildings Directive [1]. However, as the urgency for decarbonization intensifies, PEDs aim to achieve a positive environmental impact by enhancing energy efficiency, utilizing local renewable energy sources, and promoting energy flexibility. The EU SET-Plan 3.2 underscores the importance of PEDs, setting ambitious goals for their widespread adoption by 2025 [2]. Nonetheless, challenges in defining and operationalizing the PED concept must be addressed to guide development and funding effectively While recognized as a key strategy for decarbonizing cities and improving residents' quality of life, the practical implementation of PEDs in various contexts such as retrofitting, different climates and surrounding energy systems remains underexplored [3], [4], [5], [6], [7], [8].

In contrast to greenfield developments, urban retrofitting presents additional challenges: Constraints related to heritage protection, which can limit the implementation of technologies aimed at reducing thermal losses, such as insulation and air tightness measures. Similarly, the installation of photovoltaic (PV) systems is more complex on existing roofs with small, tessellated surfaces [9], [10], [11].

This paper examines whether urban retrofitting can achieve a positive energy balance by analyzing four European districts. These districts – in Settimo Torinese, Italy, Großschönau, Austria, Amsterdam, Netherlands, and Resita, Romania – differ significantly in their climatic conditions, urban fabric, and energy profiles. The study applies a normative approach to simulate energy balance scenarios under different retrofitting strategies, focusing on insulation improvements, photovoltaic (PV) panel installations, and the implementation of flexible energy management systems,

II. Methodology

The methodology employed in this study is based on the district energy balance simulation and assessment framework [12], [13], which considers both demand-side and supply-side measures to achieve a positive energy balance. Its distinguishing features are an hourly temporal resolution to assess the effects of flexible renewable utilization rates and dynamic conversion factors which is recognized as being crucial for PED assessment methodologies [14], [15], [16], [17] and the contextualization of the district energy balance target as a function of density as argued for in [6], [12], [13], [18], [19], [20].

The assessment concept is summarized in Figure 1 and consists of specifications on system boundaries, the weighting of the energy flows in the balance and the contextualization of balance targets depending on district density. The analysis involved the following steps of data gathering and preparation, which are visualized in Figure 2:

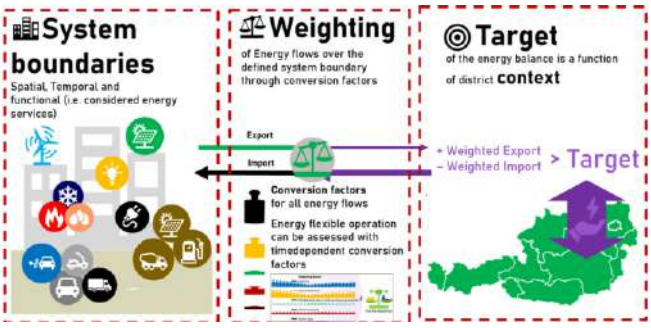


Fig. 1. PED Assessment system

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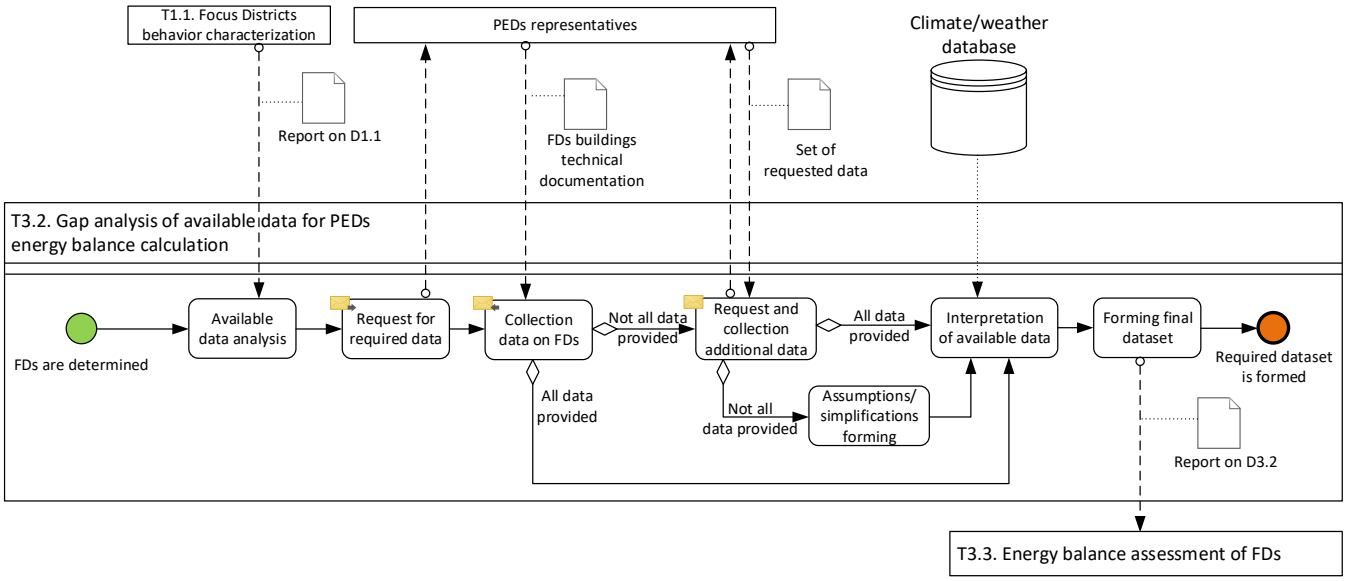


Fig. 2. Scenario Development and Data Preparation process

A. District Characterization

Each district is characterized based on its physical and energy infrastructure, climatic conditions, and urban fabric. This includes the identification of building typologies, energy usage patterns, and renewable energy potentials and is summarized in Table I.

B. Scenario Development

For each district, multiple retrofitting scenarios are developed. These scenarios include various combinations of building insulation, window replacement, roof insulation, and the installation of PV panels. Additionally, the potential for integrating flexible grid usage and renewable energy technologies is assessed. All districts are investigated by a number of different **scenarios** that each add measures and features compared to the previous one:

Baseline represents the status-quo without any additional measures. Next, an initial local renewable energy scenario is added by including **roof-mounted PV (+X%PV),** where X denotes the share of roof areas covered by PV modules. Next, a **realistic renovation scenario (RRS)** implements thermal renovation measures feasible to local practitioners, as shown in Table II. This is typically followed by another **PV scenario**. Second to last are measures representing the **maximum technical potential for thermal energy savings (TRS). Flexibility measures (Flex)** are considered according to the specifications in Table I.

Finally, these scenario measures are grouped into a **Realistic** scenario and a **Technical maximum scenario (TMS)**

C. Energy Balance Calculation

The energy balance for each scenario is calculated according to [12] by comparing the total energy demand of the district with the energy generated from renewable sources. The simulation accounts for seasonal variations in energy demand and generation, as well as the efficiency of the implemented technologies. **System Boundaries** for energy balance calculations include energy demands for building operation and user plug-loads and renewable energy produced within the district.

TABLE I. DISTRICT CHARACTERIZATION

Parameter	***Unit***	***IT, Settimo Torinese***	***AT, Großschönau***	***NL, Amsterdam***	***RO, Resita***
District Area	ha	19.0	705.0	3.0	47.0
Gross Floor Area	m² GFA	213 937	40 161	98 941	130 700
District Buildable Plot Area	m² PA	146 000	391 000	30 035	420 859
Floor Area Ratio (FAR)	-	1.47	0.15	3.29	0.31
Site coverage Ratio (SCR)	%	26	7	78	18
Net to Gross Floor Area Ratio	%	80	70	80	80
Building storeys (avg)		5.6	1.5	4.2	4.9
Residential usage	%	98.5	82.2	19.1	70.0
Commercial usage	%	0.2	11.7	29	3.9
Primary School Usage	%	0.0	5.0	0.0	0.0
Secondary School Usage	%	0.0	0.0	0.0	9.8
Others (Retail) Usage	%	1.4	1.1	51.9	9.8
Heating setpoint	°C	22	22	22	22
Heating COP Heat pump (Flex)	-	3.5 (4.5)	3.5 (4.5)	3.5 (4.5)	3.5 (4.5)
Heating degree days	°Cd	4786.0	3483.1	3917.9	3877.8
Cooling Setpoint	°C	26	26	26	26
Cooling COP Heat pump (Flex)	-	2.5 (5.0)	2.5 (5.0)	2.5 (5.0)	2.5 (5.0)
Cooling degree days	°Cd	0.4	6.9	1.1	16.4
Primary Energy Conversion factors	Source	[2]	[3]	[4]	[5], [6]

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Assessing flexible grid use is achieved by Primary Energy conversion using time dependent conversion factors.

D. *Contextual Analysis*

The results are analyzed in the context of each district's specific characteristics, including climatic conditions, building density, and energy infrastructure. This analysis provides insights into the effectiveness of different retrofitting strategies in achieving a positive energy balance. The **realistic** and **TMS** scenarios were subsequently analyzed with the following Energy balances and contexts:

Building use balance (BUB) includes operation and plug-loads. It is the default balancing boundary and typically most negative.

Building operation balance (BOB) excludes user plug-loads, making it easier to achieve a positive balance.

Including a **context factor for density (CFD)** as a virtual balance component as a function of the floor area ratio (FAR):

$$CFD = MIN\left(\left(\frac{61.94}{FAR+0.15}\right) - 53.79\right), 125) \quad (1)$$

CFD*2: The initial contextualization is suitable for high density urban districts but leads to unrealistically high expectations for local renewable energy supply in low density neighborhoods. Here, it must be assumed that only parts of the plot area can be used for energy generation. This is operationalized by introducing an additional factor of "RES plot utilization", that is bound by two times the site cover ratio (**SCR**) up to 100% of the available plot.

$$CFD = MIN\left(\mathbf{2\ SCR}\left(\frac{*61.94}{FAR+0.15}\right) - 53.79\right), 125) \quad (2)$$

CFD*3: Works the same as CFD*2, except with a higher factor of site cover utilization of three. This is a compromise between assuming the underlying RES potential to correlate with plot area and roof area respectively.

$$CFD = MIN\left(\mathbf{3\ SCR}\left(\frac{*61.94}{FAR+0.15}\right) - 53.79\right), 125) \quad (3)$$

CFR: Finally, a simpler approach was analyzed by adding a flat 15 kWhPE/m²NFA discount to the energy balance.

III. Model Assumptions

The energy balance simulations for the four European districts considered in this study were conducted under a set of standardized assumptions that account for both the current state and potential retrofitting scenarios. These assumptions presented in **Table 1** and are crucial in ensuring consistency across the different districts and in providing a realistic assessment of the energy performance of each district under various retrofitting measures. The key assumptions related to heating and cooling systems, PV installations, and other relevant parameters are described below.

A. *Heating and Cooling Systems*

Heating System: The primary heating systems considered in the model are heat pumps, which are either air-source or ground-source. The coefficient of performance (COP) for these systems varies depending on the specific scenario, with higher COP values representing more efficient systems. The baseline scenarios use existing heating systems, while the retrofitted scenarios assume the installation of modern heat pumps with varying COPs.

Cooling System: All scenarios following the baseline include the introduction of cooling systems, which operate with setpoints between 26°C and 23°C. This range is designed to balance occupant comfort with energy efficiency, minimizing cooling demand while maintaining acceptable indoor temperatures.

Flexible Heating and Cooling Setpoint Temperature: The heating and cooling setpoint temperatures are set at 22°C and 26°C respectively. In scenarios where flexible pre-heating and cooling is employed, these setpoints may increase to 24°C for heating and 23°C for cooling during peak renewable energy availability, allowing for optimized energy usage and greater integration of renewable energy sources at the cost of marginally increasing overall energy demand due to increased thermal losses.

B. *Photovoltaic (PV) Systems*

PV Installation: The retrofitting scenarios include the installation of PV systems on the roofs of buildings within the districts. The percentage of roof area covered by PV panels varies across the scenarios, reflecting different levels of investment and ambition. The table below presents the assumed percentage of roof area allocated to PV installations for each scenario.

C. *Climate*

Weather data was used from the Meteonorm software for the reference year 2020. Monthly average temperatures are presented in Figure 3.

IV. Case Study Analysis

This section presents the primary energy demand and supply balance for the investigated scenarios of each district, as shown in Figure 8. Here, all districts show a significant potential for energy savings in heating and cooling of 40 to 80%. The local renewable potential is realized by PV roof installations, ranging up to 90 kWh primary energy per square meter net floor area. For all districts, flexible HVAC operation can achieve an additional RES use of up to 20 kWhPE/m²NFA instead of regular grid use.

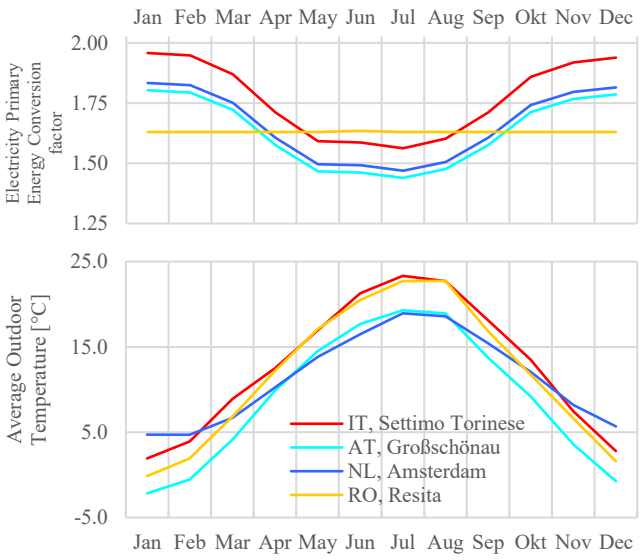


Fig. 3. Monthly electricity primary energy conversion factor and average outdoor temperature

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A. *Settimo Torinese, Italy*

Settimo Torinese, located in northern Italy, represents a typical Mediterranean urban district. The district's retrofitting potential was evaluated under scenarios involving the insulation of walls and roofs, window replacement, and the installation of PV panels on 80% of the roofs. The simulation results indicate that while the district can significantly reduce its energy demand through these measures, achieving a positive energy balance requires the integration of flexible grid usage and additional renewable energy sources.

B. *Großschönau, Austria*

Großschönau is a rural municipality in northern Austria, characterized by low building density and a predominantly residential building stock. The district's cold climate necessitates a focus on reducing heating demand through insulation and efficient heating systems. The retrofitting scenarios show that while insulation improvements and PV installations can reduce the district's energy deficit, achieving a PED status would require the widespread adoption of energy-efficient technologies and potentially the expansion of renewable energy generation beyond the district's boundaries.

C. *Amsterdam, Netherlands*

The Amsterdam focus district represents a small, historical area within the city, with significant restrictions on building modifications due to heritage protection laws. Despite these constraints, the simulation suggests that targeted retrofitting, particularly in improving insulation and installing PV panels where permissible, can bring the district closer to a positive energy balance. However, the district's high energy demand, particularly in commercial buildings, poses a significant challenge.

D. *Resita, Romania*

Resita's Lunca Pomostului district is a mixed-use area with a diverse building stock. The district's retrofitting scenarios focus on insulation improvements and PV installations on residential and educational buildings. The results indicate that while these measures can substantially reduce energy consumption, the district's high user plug loads and limited renewable energy potential hinder the achievement of a PED. The integration of flexible grid usage and further reductions in energy demand are necessary to approach a positive energy balance.

V. Discussion

The results illustrate that meaningful decarbonization targets must quantifiably include district density. Notably, achievability of a positive energy balance depends more on district density than on the amount of heating and cooling days determined by climate. Differences in PV yields due to different climate has an effect, but also to a much lesser degree. This is highlighted by the fact that the district with the coldest climate and most heating degree days, AT, Großschönau, achieves the best, almost positive energy balance, even though the PV roof utilization rate of 58% is similar if not lower when compared to the other districts.



Fig. 4. IT, Settimo Torinese Focus Distrcit



Fig. 5. AT, Großschönau Focus District

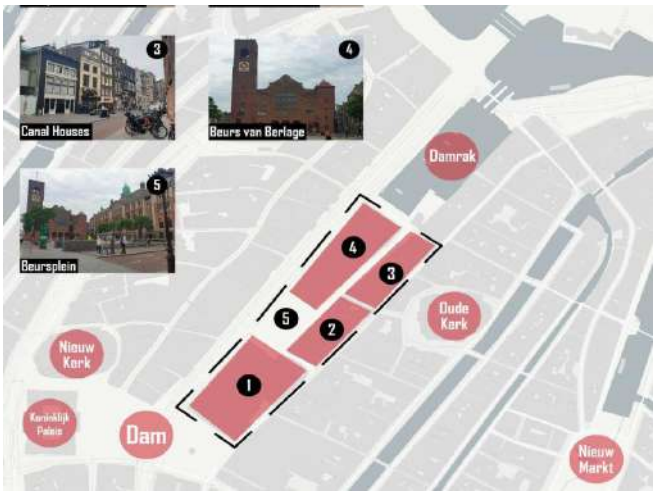


Fig. 6.Amsterdam Focus District



Fig. 7. Ro, Resita Focus District

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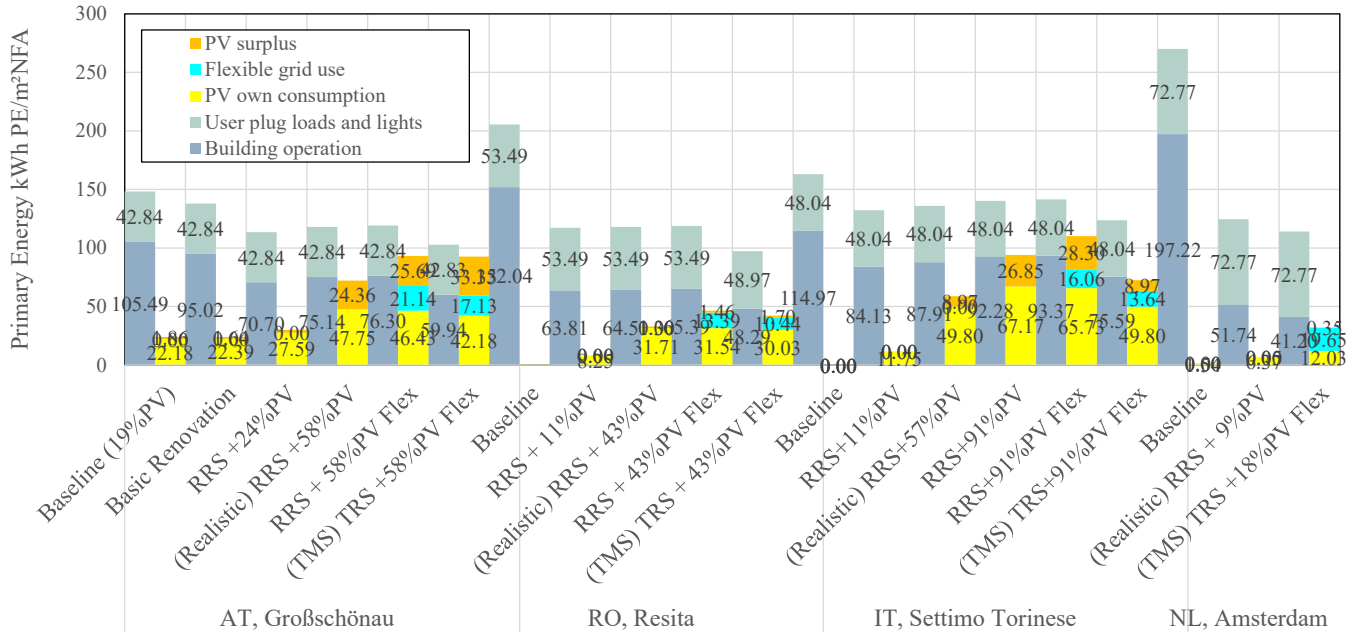


Fig. 8 Primary Energy balance of investigated scenarios, grouped by district (Sections), demand (left columns) and supply (right columns)

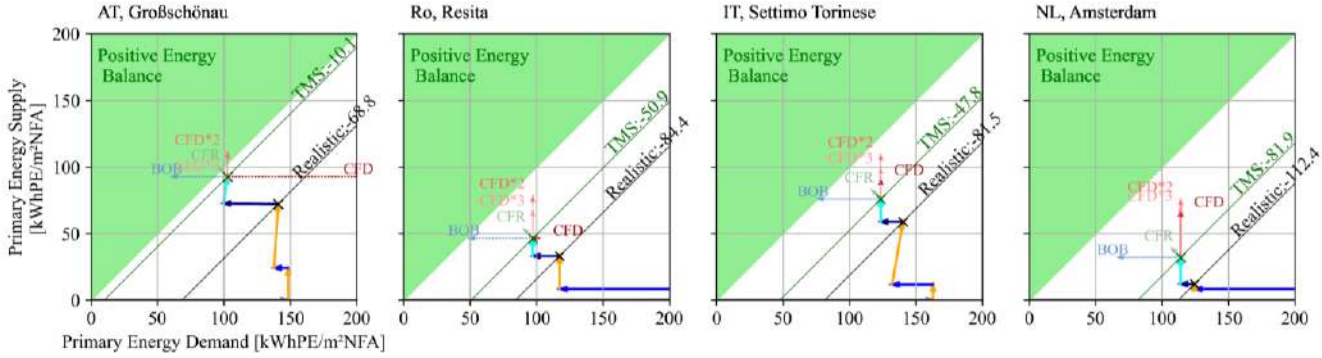


Fig. 9. Primary Energy supply (y-axis) against demand (x-axis) for consecutive measures represented by solid arrows indicating energy savings (dark blue), additional PV installation (orange) and flexible HVAC operation (cyan). The 45° lines indicate the energy balance of the realistic (black) and technical maximum scenario (TMS, green) respectively. The TMS balance point (green x) is origin to the contextual balance components BOB (lightblue), CFD (crimson), CFD*2 (lighter red), CFD*3 (light red) and CFR (teal)

Use of the already established context factor density (CFD)[12] can compensate for dense urban context's low potential, but is not feasible for rural retrofitting, as can be seen by the overwhelmingly negative balance shift for AT, Großschönau with FAR=0.1 and the negligible to moderate effects for the districts of medium density, RO, Resita (FAR=0.9) and IT, Settimo Torinese (FAR=1.5).

Instead, a combined density context factor including the site coverage ratio (SCR) as employed with CFD*2 and CFD*3 can maintain a density offsetting effect for urban areas while prevent unreasonably high requirements of PV installation for low density districts by decoupling it from the plot area as sole variable. With CFD*2, restricting the potential PV area to twice the roof area significantly dampens the balance target for all densities so that the initial effort sharing effect between high and low densities diminishes.

When combined with a small context factor for renovation (CFR), the application of CFD*3 is sufficient to ensure PED feasibility for all densities while retaining a strong density offsetting effect.

Nevertheless, heritage protected buildings pose additional challenges to a positive energy balance that cannot and, due to their nature as significant buildings, should not be handled by the application of standardized virtual balance components.

The case studies highlight the variability in the potential for urban retrofitting to achieve a positive energy balance across different European contexts. The results emphasize the importance of tailored retrofitting strategies that consider the specific climatic, urban, and regulatory conditions of each district. In colder climates like Großschönau, insulation and heating efficiency are critical, while in sunnier regions like Settimo Torinese, PV installations play a more significant role. In historical districts like Amsterdam, regulatory constraints necessitate innovative approaches to retrofitting.

The study also underscores the importance of integrating flexible energy management systems, which can optimize the

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use of renewable energy and enhance the district's energy balance. However, the path to achieving a PED is complex and requires a holistic approach that combines technical, regulatory, and behavioral changes.

VI. Conclusion

Urban retrofitting can significantly contribute to achieving a positive energy balance in European districts, but its success depends on a combination of factors, including climatic conditions, building characteristics, and regulatory frameworks. The case studies of Settimo Torinese, Großschönau, Amsterdam, and Resita demonstrate that while PEDs are attainable, they require aggressive retrofitting measures, the integration of renewable energy sources, and advanced energy management systems. The findings provide valuable insights for policymakers and urban planners aiming to promote sustainable urban development through retrofitting.

Acknowledgment

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B.7. Lessons Learned from Four Real-Life Case Studies: Energy Balance Calculations for Implementing Positive Energy Districts

Publication type: Journal paper

Authors: Helmut Bruckner, Svitlana Alyokhina, Simon Schneider, Manuela Binder, Zain Ul Abdin, Rudi Santbergen, Maarten Verkou, Miro Zeman, Olindo Isabella, Marco Pagliarini, Cristiana Botta and Ana Streche.

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Contribution to the thesis:

Sensitivity analysis on density context operationalization.

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Lessons Learned from Four Real-Life Case Studies: Energy Balance Calculations for Implementing Positive Energy Districts

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Abstract: Positive Energy Districts (PEDs) are integral to achieving sustainable urban development by enhancing energy self-sufficiency and reducing carbon emissions. This paper explores energy balance calculations in four diverse case study districts within different climatic conditions—Fiat Village in Settimo Torinese (Italy), Großschönau (Austria), Beursplain in Amsterdam (Netherlands), and Lunca Pomostului in Reşiţa (Romania)—as part of the SIMPLY Positive project. Each district faces unique challenges, such as outdated infrastructure or heritage protection, which we address through tailored strategies including building renovations and the integration of renewable energy systems. Additionally, we employ advanced simulation methodologies to assess energy performance. Simulation results highlight the significance of innovative technologies like photovoltaic-thermal (PVT) systems, application of demand-side actions, and flexible grid usage. Furthermore, mobility assessments and resident-driven initiatives demonstrate the critical role of community engagement in reducing carbon footprints. This study underscores the adaptability of PED frameworks across varied urban contexts and provides actionable insights for scaling similar strategies globally, supporting net-zero energy targets.

Keywords: positive energy district; energy balance; photovoltaic-thermal system; mobility; demand-side actions; sustainable development

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1. Introduction

The global transition toward sustainable urban development has become a critical priority, motivated by the urgent need to combat climate change and achieve climate neutrality in line with international frameworks such as the Paris Agreement. Positive Energy Districts (PEDs) embody a transformative concept within this transition, focusing on empowering urban areas to generate more renewable energy than they consume. A PED is seen as a district with annual net-zero energy import and net-zero CO2 emissions, working towards an annual local surplus production of renewable energy [1]. They are seen as a tool for achieving climate-neutral cities, embedded in integrated urban strategies providing liveable, sustainable, and inclusive urban neighbourhoods. By leveraging energy-efficient buildings, smart grids, and innovative renewable energy technologies, PEDs significantly contribute to net-zero carbon goals while enhancing urban resilience. A PED Programme

has been aligned with the Driving Urban Transition (DUT) Partnership and has become one of the three DUT Pathways. The programme will continue to expand its activities, with the aim of initiating 100 PEDs in Europe by 2025. [2]

However, implementing PEDs, particularly in established urban locales, presents numerous challenges. These challenges include the lack of universally accepted methodologies for assessing energy balances, the complexities of integrating renewable energy systems into dense and often outdated urban infrastructures, and the pressing need for multi-stakeholder collaboration to secure local acceptance and ensure equitable development processes [3,4].

This paper highlights findings from the research project "Supporting Innovative and Ambitious Cities and Municipalities on Their Pathway to Positive Energy Districts through Easy, Clear, and Understandable Guidelines, Targets, and Strategies" (SIMPLY Positive) [5]. Its main goal is to provide a practical overview and outline possible strategies for energy balance calculation within the focus district, which is a key aspect of implementing PEDs.

2. Materials and Methods

2.1. Focus Districts Description

The project considers four focus districts: Fiat Village in Settimo Torinese (Italy), Großschönau (Austria), Beursplain in the centre of Amsterdam (Netherlands), and Lunca Pomostului in Reşiţa (Romania).

The main parameters of focus districts are presented in Table 1.

Table 1. Focus district parameters.

Parameter	Unit	IT, Fiat Village, Settimo Torinese	AT, Großschönau	NL, Beursplain, Amsterdam	RO, Lunca Pomostului Reşiţa
District Area	ha	19.0	705.0	3.0	47.0
Population	-	3200	450	300 [6]	9900
Gross Floor Area	m^2 GFA	213,937	401,060	98,941	130,700
District Buildable Plot Area	m^2 PA	146,000	391,000	30,035	420,859
Floor Area Ratio (FAR)	-	1.47	0.15	3.29	0.31
Site Coverage Ratio (SCR)	%	26	7	78	18
Net to Gross Floor Area Ratio	%	80	70	80	80
Building Storeys (avg)	-	5.6	1.5	4.2	4.9
Residential Usage	%	98.4	82.2	19.1	76.5
Commercial Usage	%	0.2	11.7	29.0	3.9
Primary School Usage	%	0.0	5.0	0.0	0.0
Secondary School Usage	%	0.0	0.0	0.0	9.8
Others (Retail) Usage	%	1.4	1.1	51.9	9.8
Heating Setpoint	°C	22	22	22	22
Heating COP Heat Pump (Flex)	-	3.5 (4.5)	3.5 (4.5)	3.5 (4.5)	3.5 (4.5)
Heating Degree Days	°C·d	4786.0	3483.1	3917.9	3877.8
Cooling Setpoint	°C	26	26	26	26
Cooling COP Heat Pump (Flex)	-	2.5 (5.0)	2.5 (5.0)	2.5 (5.0)	2.5 (5.0)
Cooling Degree Days	°C·d	0.4	6.9	1.1	16.4
Primary Energy Conversion Factors	Source	[7]	[8]	[9]	[10,11]

In Settimo Torinese, the district Fiat Village is primarily residential, with most buildings over 50 years old, requiring extensive renovations to enhance energy performance. Goals include establishing a Renewable Energy Community to facilitate local energy production and consumption and implementing energy management systems to optimize energy use across both public and private sectors.

Großschönau, a small rural municipality in Austria, is committed to sustainability and known for its long-standing environmental initiatives. Plans are in place to attain the Positive Energy Municipality status by 2030, focusing on energy self-sufficiency and innovative projects, including small-scale wind turbines and a centralized biomass heating network.

The selected area Beursplain in Amsterdam is characterized by old buildings under heritage protection. Some buildings already feature photovoltaic (PV) systems, with the primary goal being to achieve a positive energy balance.

In Reşiţa, the focus Lunca Pomostului district combines residential, commercial, and institutional spaces, although much of its infrastructure is outdated. Efforts in this district aim to reduce emissions by 20% compared to 1990 levels. Current initiatives focus on urban redevelopment, including the modernization of public spaces, the addition of green infrastructure, and improved mobility options such as cycling lanes and pedestrian pathways.

These four districts represent diverse climatic conditions and energy challenges. For instance, Reşiţa experiences a temperate climate with Mediterranean influences, while Settimo Torinese and Amsterdam share moderate climates with varying precipitation patterns. Großschönau has a relatively dry climate with cold winters. Despite these differences, all districts prioritize the integration of renewable energy systems, with Amsterdam and Settimo Torinese emphasizing centralized heating solutions, and Großschönau showcasing leadership in decentralized renewable energy projects.

2.2. Methodology Description

The core of the definition of Positive Energy Districts (PEDs) lies in establishing a clear rationale behind their design, which articulates the objectives they aim to achieve. This design approach [12] begins by defining specific goals and subsequently deriving criteria for their operationalization based on these objectives.

System boundaries are established from spatial, temporal, and functional perspectives, in line with the principles outlined in the fundamentals of PED energy modelling [13]:

1. Spatial boundaries: these refer to the geographical limits enclosing included energy services and supplies;
2. Temporal boundaries: these represent the balancing period, typically defined as one operational year;
3. Functional boundaries: these identify specific energy functions, uses, or demands that are included or excluded based on their purpose rather than proximity.

The functional boundaries and included energy services can be categorized into three main areas:

1. Operational energy and user electricity;
2. Mobility aspects;
3. Embodied energy and emissions.

This approach defines three variants or layers of PEDs:

- PED operation (innermost level): focuses solely on operational energy;
- PED mobility: expands to include private daily mobility;
- Climate-neutral PED (outermost layer): incorporates embodied energy related to district construction, maintenance, repair, and mobility.

Each layer adds complexity and introduces uncertainty compared to the previous one. Access to adequate data is critical for simulation and verification, and these system boundaries are visually represented in Figure 1.

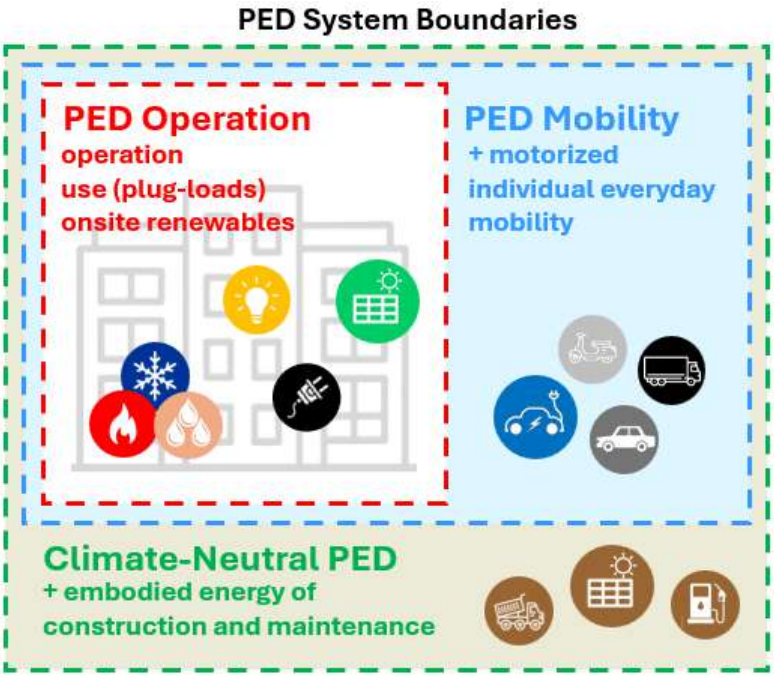


Figure 1. Three expanding functional system boundaries of PED: from operation and use (red) to including everyday individual motorized mobility (blue) to also include energy and emissions from construction and maintenance (green).

Under the SIMPLY Positive project, the first step in defining PEDs involved determining the type of each focus district along with its respective system boundaries. The spatial boundaries for each focus district align with the geographical borders of the selected regions. The sizes of these districts vary from the large city of Amsterdam to the small village of Großschönau. Despite these differences, nearly all self-set district goals include energy-related key performance indicators (KPIs).

The Romanian focus district set itself the goal of reducing CO_2 emissions. This can be reframed as increasing the share of renewable energy, which contributes directly to lowering CO_2 emissions and aligns it with the objectives of other districts. In the Italian focus district, two of the three self-set goals are socially oriented, making them more challenging to quantify; these can be classified as supportive objectives.

Temporal boundaries for all focus districts are set at one operational year, as there are no additional local or project-specific requirements. The functional boundary type of each focus district categorizes them primarily as residential areas with a minor proportion of social buildings (e.g., schools and offices).

The energy simulation was carried out to obtain the annual primary energy balance KPI and utilizes a simplified model of a single-zone thermal building and is calculated as follows:

$$PEB = CF + \sum_i \sum_j f_{ij} E_{ij}$$

with E_{ij} the energy flow j at hour i over the system boundary (outwards positive) weighed with f_{ij} the time-dependant primary energy conversion factor of that energy flow and CF representing "Context Factors", which are virtual balance components derived from top-down model parametrization described in [12]. The context factors depend on district density, expressed as the Floor Area Ratio (FAR) between the district's gross floor area (GFA) and the underlying buildable plot area (PA):

$$FAR = \frac{GFA}{PA}$$

The context factors used for analysis were operationalized according to the following [13]:

$$CFD = Min\left(2\left(\frac{61.94}{FAR + 0.15}\right) - 53.79\right), 125\left[\frac{kWh_{PE}}{m^2_{GFA}}\right]$$

$$CFR = 15\left[\frac{kWh_{PE}}{m^2_{GFA}}\right]$$

$$CFR + D = CFD + CFR$$

The functional system boundary of the energy balance includes building use of operation and plug-loads (BUB). A second boundary only includes building operation (BOB) and is calculated by subtraction of user electricity and plug-loads (UE) from the use balance:

$$BOB = BUB - UE$$

Electrical power needs are primarily met by a photovoltaic (PV) system. Any surplus energy generated can either be redirected to domestic hot water systems as part of demand-side management (DSM) strategies or fed back into the grid to offset overall energy consumption, with the inverted conversion factor for grid electricity at that time. Unmet energy demands are supplemented by grid-supporting sources, such as wind peak shaving from nearby wind farms, which are modelled by the use of external wind availability data (>40% of installed capacity) at which point the primary energy conversion is set to zero, effectively discounting grid use during these hours. The effect of these measures is separately shown as "Flexibility Measures".

Thermal energy demand is defined by the target temperature settings for heating, cooling, and domestic hot water (DHW) systems. This demand can be satisfied through electrical solutions, such as heat pumps, or other methods like district heating and natural gas boilers. Each energy system is evaluated using specific primary energy conversion factors, which may be obtained from national or regional standards, measurement data, or projections for future energy systems.

Any excess energy generated through demand-side management can be stored in the building's thermal mass, which helps to reduce peak load demands, particularly during winter nights. The simulation method is illustrated schematically in Figure 2 and described in more detail in [14].

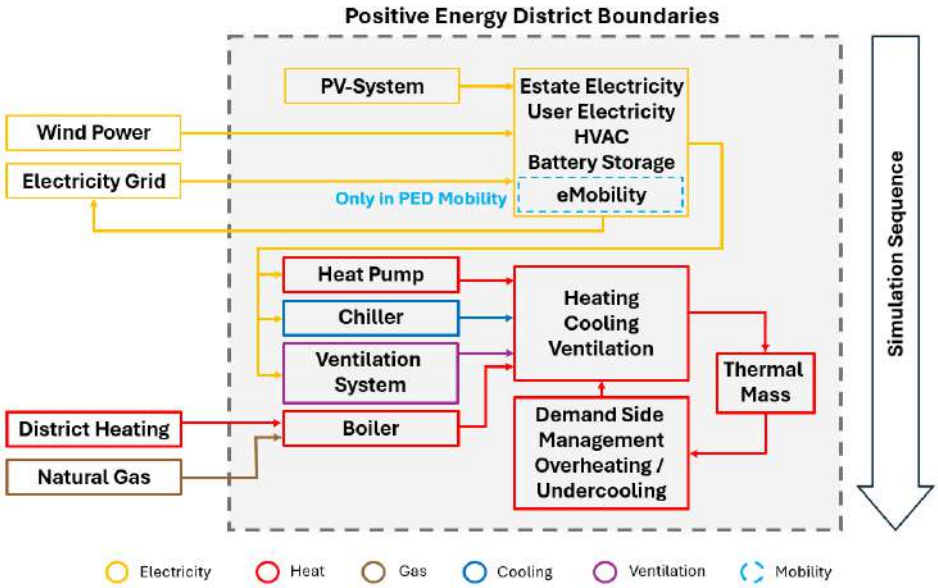


Figure 2. Applied simulation components of the PED establishment.

3. Energy Balance Calculation Results

For all the focus districts (FDs) considered, the development scenarios aimed at achieving Positive Energy Districts (PEDs) emphasize two primary strategies: building renovations (including insulating walls, roofs, and basements, as well as replacing windows) and the installation of photovoltaic (PV) panels. In regions with relatively cold climates, such renovations are crucial for minimizing heat loss during winter months. Conversely, in southern regions with high solar irradiation, PV installations are particularly effective, although these areas typically face increased energy demands for cooling during hotter months.

The most effective development scenarios combine a variety of available strategies, such as flexible grid usage and the integration of renewable energy technologies. Since primary energy is a critical factor in assessing PEDs, the effectiveness of these scenarios can be illustrated through a comparison of primary energy demand and supply, as shown in Figure 3.

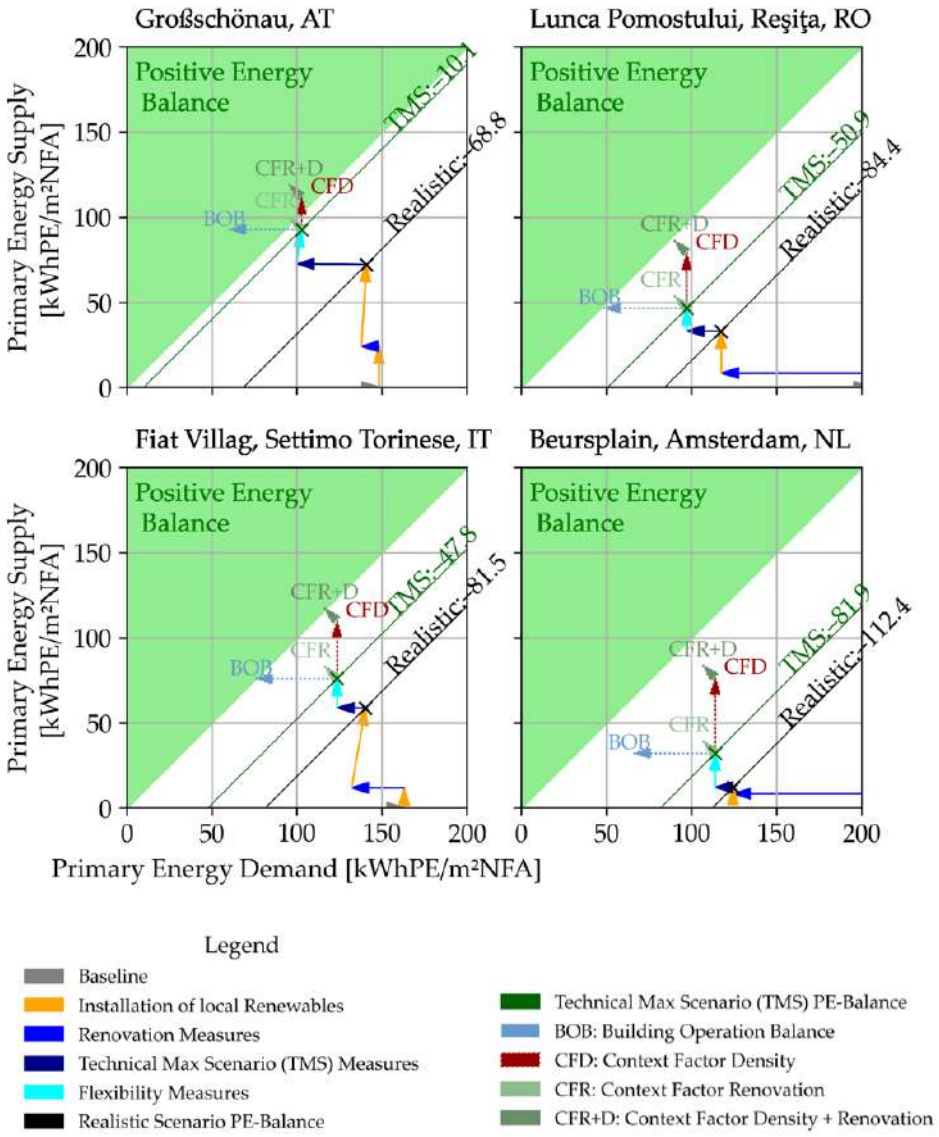


Figure 3. Primary energy in focus districts: demand vs. supply with and without accounting for contexts. BOB (blue): building operation balance; CFR (light green): context factor renovation; CFD (red): context factor density; CFR+D (green): combination of CFD and CFR. District scenarios with a final primary energy balance in the green shaded area are considered PEDs in Operation.

The dashed grey line in Figure 3 indicates the balance point, where energy demand equals supply—this represents the baseline requirement in the conventional definition of PEDs.

However, achieving a comprehensive understanding of PEDs also necessitates considering contextual factors [12].

In Figure 3, the blue line depicts PED pathways that do not account for these contextual factors, while the orange line illustrates pathways that include such considerations. The scenarios progress from the current state (represented by the rightmost points) through various stages, including the following

- Building renovation: initial energy efficiency improvements;
- Renovation with PV installation: increasing the integration of PV systems within districts;
- Implementation of flexible grid systems: enhancing management and distribution of energy.
- Adoption of all measures: combining renovations, PV installations, flexible grid systems, and more efficient energy equipment in the focus districts (represented by the leftmost points).

By considering these development scenarios, we can better understand the pathways toward achieving PEDs in diverse urban contexts.

4. Discussion on Further Improvement Areas of FDs Towards PEDs

In this research, focus districts were considered at the PED level of building operation and used to minimize complexity and uncertainty related to initial data. Several potential improvements could enhance the study's outcomes, including the following:

- Integrating a photovoltaic-thermal (PVT) simulation model to better align district goals with PED standards;
- Elevating the focus of the study to PED Mobility levels;
- Evaluating demand-side actions by residents to reduce energy consumption and carbon footprints within the districts.

4.1. Improvement of Simulation Results by Integrating PV/PVT Model in PED

One way to enhance the energy balance calculation within the focus districts is to utilize a more detailed model that accounts for various scenarios incorporating photovoltaic (PV) systems. In this work, we specifically compare photovoltaic-thermal (PVT) systems with conventional photovoltaic (PV) and solar thermal (ST) systems. While a PV module generates electricity, an ST module produces heat. In contrast, a PVT system integrates both PV and ST functionalities into a single unit, allowing for the simultaneous production of electricity and heat.

The modelling of these systems is grounded in energy conservation principles and considers heat exchanges—conductive, convective, and radiative—across each collector component. Several assumptions are made, including neglecting pressure losses, edge losses, dust, and partial shading. Additionally, we assume uniform temperature across components, with the thermo-physical properties of materials treated as temperature independent. Each collector area is approximately 2 m^2. To simplify annual simulations across the district, we apply a rapid and efficient reduced temperature T_{red} approach [15], obtaining unique coefficients by solving ordinary differential equations (ODEs), and these highly detailed models are integrated into the PVMD Toolbox [16]:

$$T_{red} = \frac{T_{in} - T_{am}}{G}, \quad (1)$$

This approach characterizes the thermal and electrical performance of the system as functions of operational conditions, which include fluid inlet temperature (T_{in}), ambient

temperature (T_{am}), and solar radiation (G). The reduced temperature also influences the efficiency of photovoltaic (PV) systems, as cell temperature is dependent on operating conditions; however, this impact on overall PV performance is minimal. The thermal efficiency of the solar thermal (ST) collector, considering unique coefficients, can be expressed as follows:

$$\eta_{th} = 0.75 - 3.67\ T_{red} \quad (2)$$

Similarly, the electrical and thermal efficiencies of the photovoltaic-thermal (PVT) collector are based on the PVMD Toolbox [16] calculated as follows:

$$\eta_{el} = 0.195 - 0.33\ \mathrm{T}_{red}, \quad (3)$$

$$\eta_{th} = 0.47 - 10\ \mathrm{T}_{red} \quad (4)$$

These equations define the efficiencies of the solar collectors, considering several operational parameters, including baseline efficiency and temperature-dependent loss coefficients. Both electrical and thermal efficiencies are calculated on an hourly basis, which allows us to estimate the electrical and thermal yield of each collector type effectively.

To evaluate the potential for large-scale implementation of PVT systems, we applied this modelling approach to the Dutch focus district, the city centre of Amsterdam. We utilized openly available height data [17], processed as a rasterized digital elevation model (DEM) with a resolution of 0.5 metres. The footprint data from the cadastre [18] were used to clip the data points corresponding to each building within their respective polygon shapes.

Subsequently, we employed the Random Sample Consensus (RANSAC) algorithm [19] to detect suitable planar surfaces of at least 10 m². The layout of modules on these surfaces was optimized using a heuristic algorithm designed to fit them in rectangular formations on sloped rooftops [20], utilizing both portrait and landscape orientations. For flat rooftops, a maximum fit strategy was used, along with a simple economic model that determined whether an east–west or south-facing layout was more beneficial. If a building hosted at least four modules, our skyline-based approach [21] was implemented to estimate solar irradiance as an hourly profile spanning an average year. The required climate data were procured from the Royal Netherlands Meteorological Institute (KNMI) [22] over the past ten years.

4.1.1. Key Results

- Buildings and footprint: There are a total of 16,452 buildings in the focus district, encompassing a combined footprint area of 2.76 km^2. This area is further divided into the following:
  - 1.09 km^2 detected as flat surfaces;
  - 1.17 km^2 categorized as sloped rooftops.
- Module installation potential: on these surfaces, it is feasible to install 333,145 modules across 11,300 separate systems.

Furthermore, Table 2 presents the results of the geospatial mapping of PV, PVT, and ST modules throughout the focus district.

4.1.2. Energy Yield Comparison

- Photovoltaic modules: yield only electricity, with no thermal yield;
- Solar thermal modules: yield only heat, with no electrical yield;
- PVT systems: achieve a combined total yield of 359.4 GWh/year, translating to 130 kWh/year/m^2 when divided by the building footprint area.

While the thermal yield of PVT systems is lower than that of ST systems, which generate a specific yield of 164 kWh/year/m^2, PVT systems are often the most efficient approach for roofs in terms of energy generation.

Table 2. Results of the geospatial mapping of PV, PVT and ST modules on the focus district.

	Module Count	Nominal Power (MW)	Electric Energy (GWh/yr)	Thermal Energy (GWh/yr)	Total Energy (GWh/yr)
PV	333,145	123.3	117.3	0	117.3
ST	333,145	123.3	0	452.6	452.6
PVT	333,145	123.3	121.9	237.5	359.4

4.2. Assessing Mobility in PED

To assess the electricity demand for private e-mobility within the Positive Energy District (PED), we developed a simulation model based on the parking-based approach described in the Horizon Europe INCIT-EV Project [23]. This model estimates both the overall annual energy demand and the temporal distribution of energy requests throughout the year [24].

The parking-based approach focuses on understanding drivers' habits and behaviours, particularly the timing and duration of parking. Notably, this method does not require knowledge of the origin-destination trip matrix; instead, it emphasizes modelling parking and charging patterns to determine energy demand.

4.2.1. Model Inputs

The model utilizes three main categories of inputs:

1. Car fleet characteristics:
    - Average number of electric vehicles (EVs) travelling within the PED (internal), originating from the PED (outgoing), and heading into the PED (incoming) each day;
    - Typical daily travel distances.
2. Driver behaviour:
    - Preferred times and locations for charging;
    - State of charge (SOC) thresholds that trigger a charging event (initial SOC) and stop the charging operation (final SOC);
    - These behaviours are represented in the model using probability distributions.
3. Charging infrastructure:
    - Number and power of charging points available within the district;
    - Temperature data can also be incorporated to account for variations in energy consumption due to seasonal temperature changes, which affect battery efficiency.

Once the inputs are defined, the model runs an hourly simulation for one entire year, processing each vehicle sequentially. Daily average values are used for all days of the year. It evaluates charging decisions based on the vehicle's current SOC, its travel or parked status, and the availability of charging stations. When all charging requirements are met, the vehicle's status changes to "charging," and its energy demand is added to the current timestep and cumulative annual demand.

4.2.2. Model Outputs

The outputs of the model include the following:

- Maximum power demand: the peak power demand on the local electricity grid and the specific time it occurs;
- Total annual energy demand: this provides essential insights regarding the additional energy supply requirements needed to meet district-wide demand;
- The model also tracks disservice events where charging requests cannot be fulfilled due to all stations being occupied.

As an example, we present the results of the simulation conducted for the focus district in Settimo Torinese (Italy), illustrated in Figure 4. The results are shown monthly to highlight potential annual patterns in energy demand.

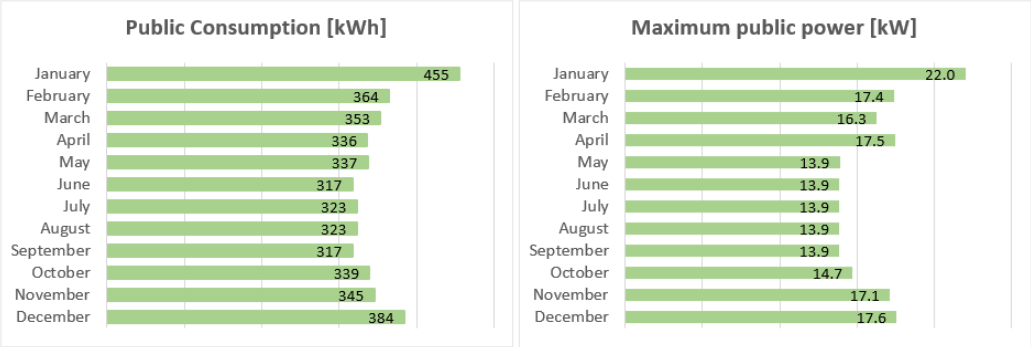


Figure 4. Annual public consumption and maximum public power in Fiat Village, Settimo Torinese.

4.2.3. Findings

- Seasonal trends: Winter months exhibit higher consumption due to reduced battery efficiency in cold temperatures. On average, the energy required monthly is approximately 400 kWh, comparable to the consumption of two average Italian-sized families [25].
- Maximum power: The maximum power requested from the grid is 22 kW, which the current infrastructure can easily accommodate. No disservice events were recorded, indicating that the existing charging infrastructure meets demand effectively.

These results were instrumental in completing the energy balance for the focus district of Settimo Torinese, providing a more comprehensive overview of total energy consumption.

4.3. Demand-Side Actions of Residents

Demand-side actions by residents—such as behavioural changes and the adoption of sustainable technologies—are essential components in the transition towards a PED. A study conducted in the SIMPLY Positive project evaluates the potential for energy and greenhouse gas (GHG) reductions through resident-driven actions within the Focus District of Großschönau, Austria [26].

Utilizing both qualitative and quantitative methods, the study identified and ranked scalable actions that contribute to district-wide energy reduction and provide insights for urban energy planning. Data on energy usage patterns were collected through household interviews, examining factors such as heating methods, appliance usage, and mobility practices. Subsequently, energy consumption data and GHG emission metrics were analyzed, and buildings were sorted based on Austrian typology [27], to quantify the saving potential.

The following Table 3 ranks best-practice examples by replication potential, which were identified as the most effective ones, considering the district-wide energy-saving potential.

Table 3. Most effective identified demand-side actions by replication potential.

Demand-Side Actions	Energy Saving Potential for the Whole Focus District [kWh]	GHG Saving Potential Depending on Energy Mix [kg CO_2-Equivalents]
(A) *Simple to replicate for everyone and associated with low costs*		
Energy consultations in residential buildings	135,660	n/a
Regular ventilation of radiators	129,430	n/a
Adapting the room temperature to the usage and daytime	74,053	3566
Reduction in the room temperature by 1 °C	68,304	2139
Conversion to LED technology	55,200	683–15,710
(B) *Possible under certain circumstances and/or low investment required*		
Switch to e-bikes for going to work	346,090	112,020
Carpooling with at least 2 persons	341,360	110,940
Conversion of street lighting to LED	268,755	n/a
Correct ventilation of windows and shadowing instead of an air conditioner	196,890	1970–45,280
Conversion from combustion cars to public transport	78,000	24,540
(C) *Costly and/or complex technical adaptation required*		
Renovate private buildings to passive house standard	4,060,720	n/a
Thermal renovation as performed in reality	3,401,460	n/a
Renovate public buildings to passive house standard	529,132	95,240
Conversion from combustion cars to electric cars	421,231	143,900–173,900
Conversion from oil heater to heat pumps	247,823	11,357–17,232

4.3.1. Key Findings

The analysis reveals that significant district-wide energy reductions can be achieved through several key measures, including the following:

1. Thermal renovation: Upgrading building insulation and installing energy-efficient windows and doors can dramatically reduce heating and cooling needs. For instance, a thermal renovation case in Großschönau has been shown to decrease energy consumption by over 50%, reducing the energy index from 119 kWh/m^2a to 56 kWh/m^2a. With approximately 111 buildings identified as suitable candidates for renovation, district-wide implementation could result in annual savings of up to 3400 MWh [28];
2. Transition to electric vehicles: Replacing combustion-engine vehicles with electric cars is one of the most effective strategies for reducing GHG emissions in the transportation sector. The study found that a typical household could save about 7 MWh and 3 tons of CO_2 equivalents per year by switching to an electric vehicle powered by renewable energy. Scaling this across all inhabitants in Großschönau could yield annual savings of about 420 MWh and 174 tons of CO_2 equivalents;
3. E-bike and carpooling initiatives: Encouraging residents to shift from personal vehicles to e-bikes or carpooling can significantly reduce energy consumption and emissions. For example, regular carpooling can save around 6 MWh annually, while covering a daily commute of 16 km on an e-bike instead of using a combustion vehicle could save approximately 350 MWh and 110 tons of CO_2 equivalents per year [29,30].

These actions not only support environmental goals but also provide health benefits and reduced transportation costs for residents.

4.3.2. Best-Practice Examples

The following measures were identified as having the highest potential for replication and impact on district-wide energy savings (see Table 3 for details):

1. Thermal renovations;
2. Conversion from combustible cars to electric cars;
3. Conversion to e-bikes and carpooling.

A final ranking provides a roadmap for scalable and impactful strategies that balance environmental gains with ease of implementation. Some actions, which require minimal technical intervention, can be readily adopted by most residents and still significantly impact the district's energy balance. By increasing awareness of energy usage and identifying inefficient appliances or consumption patterns, residents can make sustainable choices without substantial financial burden [30].

These findings provide a replicable framework for other districts aiming to achieve PED status, serving as a decision-support tool for public incentives and awareness-raising measures.

4.4. Integrating the SECAP Methodology to Trigger PEDs Within a City

Given the complexity surrounding the development of Positive Energy Districts (PEDs), we developed a guideline offering a framework for potential developers, including municipalities and private sector entities, focused on leveraging the Sustainable Energy and Climate Action Plan (SECAP) methodology [31]. The guideline emphasizes synergies and interoperability between the two processes, concentrating on shared principles, compatible frameworks, common data management, and specific integration points.

The guideline was developed by identifying SECAP steps that facilitate the operationalization of PEDs during the creation of the SECAP within the focus district in Reşiţa, Romania. Key elements common to both SECAP and PED processes were determined for effective planning and monitoring of PEDs, aligned with the PED framework and key performance indicators.

A fundamental premise of these guidelines is that an SECAP should be initiated prior to PED development, as PEDs typically represent neighbourhoods within cities where the spatial boundaries of SECAPs may extend beyond those of individual PEDs.

As PED development does not lend itself to a rigid standardized approach, these guidelines provide users with a structured framework that can serve as a foundation for personalized PED development and operationalization. The framework is presented in four main phases:

1. Initiation phase
    a. Political commitment: securing the backing of political leaders to foster support for PED initiatives;
    b. Mobilizing relevant stakeholders: engaging all key players, including residents, businesses, and civic organizations, to ensure comprehensive participation;
    c. Building support: gaining momentum and endorsement from stakeholders to facilitate the project.
2. Planning phase
    a. Framework setting: establishing the context and guidelines for PED operationalization;
    b. Documenting policy synergies: identifying and aligning existing policies that complement PED objectives;

c. Elaboration of a PED action plan: developing a detailed strategy to implement and achieve PED goals.

3. Implementation phase

This phase integrates all planning efforts into actionable steps, ensuring that strategies are executed effectively.

4. Monitoring phase

Implementing monitoring processes to track progress toward objectives and assess the effectiveness of implemented actions.

Although some SECAP elements may not apply directly to the development and establishment of PEDs, the maturity of the SECAP methodology has effectively supported municipal decarbonization planning throughout Europe for over a decade. Furthermore, these guidelines reference the SIMPLY Positive deliverables, which offer practical approaches for defining PEDs, operational scenarios, energy balances, key performance indicators, and project monitoring.

By synthesizing the SECAP methodology, as applied in the focus district in Reşiţa, and the insights from the SIMPLY Positive project, the "Guidelines on How to Use the SECAP Methodology to Trigger Flagship Positive Energy Districts Within a City" are expected to significantly aid developers in their PED operationalization processes.

5. Conclusions

This study underscores the critical role of Positive Energy Districts (PEDs) in fostering sustainable urban environments. Through an analysis of energy balance calculations across four diverse case studies—Settimo Torinese, Großschönau, Amsterdam, and Reşiţa—this research highlights the adaptability and efficacy of tailored strategies across various urban and climatic contexts. Key findings include the following:

- Significant potential for positive energy balances: despite variations in infrastructure age, climate, and urban density, all focus districts demonstrated substantial potential to achieve positive energy balances through targeted interventions, including the integration of renewable energy sources;
- Scalable methodologies: The structured framework—with three expanding functional system boundaries—provides a scalable approach to addressing increasing complexities, from operational energy management to embodied emissions. This framework ensures that solutions remain feasible and impactful over the long term.
- Innovative technologies: innovative solutions such as photovoltaic-thermal (PVT) systems have shown effectiveness in optimizing rooftop utilization by producing both electrical and thermal energy, thus meeting diverse energy needs.
- Community engagement: active community involvement and enhancements in e-mobility contribute significantly to energy efficiency and emissions reduction, highlighting the essential role of resident engagement and behavioural change in the success of PEDs.

The findings also emphasize the replicability of the proposed methodologies, presenting a robust framework for other cities aspiring to establish their own PEDs. Comprehensive energy simulations, collaborative planning, and adaptive measures are crucial to the successful implementation of these districts.

This research can serve as a valuable resource for policymakers, urban planners, and stakeholders focused on developing sustainable, resilient, and energy-positive communities. Future work should prioritize the integration of emerging technologies and advanced simulation techniques to refine the design and functionality of PEDs.

6. Future Work

Future research should focus on the integration of emerging technologies and advanced simulation techniques to enhance the design and functionality of Positive Energy Districts (PEDs). This includes exploring the potential of smart grid technology to optimize energy distribution and consumption, as well as incorporating energy storage solutions to manage supply and demand fluctuations effectively. Additionally, investigating the role of advanced data analytics and machine learning can provide deeper insights into resident behaviours and energy usage patterns, leading to more personalized and effective demand-side management strategies. Expanding the scope of case studies to include a wider range of urban contexts and geographical settings will also allow for a comprehensive understanding of how different factors—such as socio-economic conditions and local policies—impact the implementation and success of PEDs. By focusing on these areas, future work can build a more resilient framework for urban energy planning and foster broader adoption of sustainable practices in cities worldwide.

Author Contributions: Conceptualization, H.B. and S.A.; methodology of energy balance calculation, S.S.; methodology of PV/PVT calculation, Z.U.A., R.S., M.V., M.Z. and O.I.; methodology mobility calculation M.P. and C.B.; methodology of demand-side evaluation, H.B. and M.B.; methodology SECAP applying, A.S.; investigation of PED energy balance, S.A. and S.S.; investigation of PV/PVT, Z.U.A., R.S., M.V., M.Z. and O.I.; investigation of mobility, M.P. and C.B.; investigation of demand-side actions, H.B. and M.B.; writing—original draft preparation, S.A.; writing—review and editing, H.B., S.S., M.B., Z.U.A., R.S., M.V., M.Z., O.I., A.S., M.P. and C.B; project administration, H.B. All authors have read and agreed to the published version of the manuscript.

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B.8. Annual Hourly E-Mobility Modelling and Assessment in Climate Neutral Positive Energy Districts

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Authors: Simon Schneider and José Baptista.

Original publication:

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Contribution to the thesis:

Annual, hourly e-mobility modelling approach integrated into district-scale energy balance assessment framework. Demonstrates the impact of charging behaviour, flexibility assumptions, and electrified mobility on PED feasibility.

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Annual hourly E-Mobility modelling and assessment in climate neutral Positive Energy Districts

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***Abstract*— This paper presents a full-year hourly district e-mobility model and its integration into a Positive Energy District simulation and assessment model including building operation, use and embodied energy and emissions. The aim of this work is to model the operation and energy flexibility potential of an EV fleet in a district through mono- and bi-directional charging and enable its assessment in terms of self-utilization of local and volatile regional RES surpluses. Results of example residential, office, school and supermarket use cases show an increase in self-utilization of local PV of up to 30% due to EV inclusion, even if PV installation size exceeds legal building code requirements by a factor of two to four. Bi-Directional charging can cut annual grid electricity by up to 30% but require an increase in battery full equivalent cycles of 20%.**

Keywords—electric mobility, positive energy districts, multimodal energy systems

I. INTRODUCTION

Emissions in the mobility sector are high and rising [1], marking their importance of conceptual and practical integration in decarbonization efforts in the built environment [2]. One promising area is that of vehicle to home (V2H) and vehicle to grid V2G, offering consumer benefits such as lower prices and independence from grid outages [3], [4]. Other potential use cases are that of vehicle to vehicle (V2V) charge transfer [5] and grid frequency stabilization [6]. But most importantly, EVs battery usage can support increased integration of RES by providing much needed intermittent storage capacity [2], [6], [7].

Positive Energy Districts are collections of buildings with sufficient energy efficiency, local renewable production and provision of energy flexibility, to be in line with national decarbonization goals in terms of energy and greenhouse gas balances for operation, mobility and embodied energy and emissions [8].

Including mobility as a service into the energy assessment of PEDs has several advantages, as outlined in the definition of Positive energy districts (PED) and the balance calculation in [8]: First, local energy surpluses can be passed on to mobility of residents and visiting vehicles [9]. Second, the assessability of sustainable mobility solutions as part of the building/district assessment promotes measures in the planning and implementation of buildings and districts [10], [11]. This means e.g. that it can be easier to achieve a PED if it quantifiably promotes bicycle instead of car usage.

Furthermore, the impact of district location on induced trips of everyday motorized individual mobility (MIM) can be quantified and assessed, showcasing the lack of sustainability of remote areas without access to public transport. Concrete measures to reduce everyday mobility or the emissions caused by it, such as mobility sharing offers, charging infrastructure for e-cars, etc., can be quantitatively assessed [10]. The inclusion of mobility in the BEM also maps synergies of e-mobility charging infrastructure through the dynamic consideration of actual charging times and PV surpluses in the district and the advantages of energy-flexible districts [12]. And finally, utilization of bidirectional EV charging to provide battery services to the district and building energy systems, reducing grid interaction and feed-in of PV surpluses can be modelled.

Despite the growing interest in integrating e-mobility into energy systems, significant challenges remain in accurately modeling the energy balances and emissions of transport at the district level in an hourly full-year energy simulation, which is required to resolute flexible dispatch: First, there is lack of data and methods to reliably determine the energy balances and emissions of transport without great effort and uncertainty [13]. Second, it is difficult to abstract aggregate district flexibility and storage effects of E-Mobility based on individual EV models and patterns [14]. Third, modelling individual mobility agents as part of a PED multi-Energy system is often infeasible due to size and performance constraints of the assessment model [15].

We propose a model for these three problems as part of our PED modelling and assessment framework [16] that include:

- Full-year flexible e-mobility model including hourly location, energy demand and state-of-charge of EVs
- Flexible mono- and bi-directional charging logic based on DSM signals from onsite and offsite regional volatile Renewable Energy Systems (RES, i.e. PV and Wind)

The following questions are addressed here:

- How much mobility demand can be expected from a district at a given location? Both annually and translated into hourly mileage profiles.
- What logic governs mono- and bi-directional charging strategies?
- How can district EVs be modelled collectively instead of individual agents?
- What effects on the multi-energy system (MES) can be observed from flexible (bi-)directional charging strategies for example districts of different usages?

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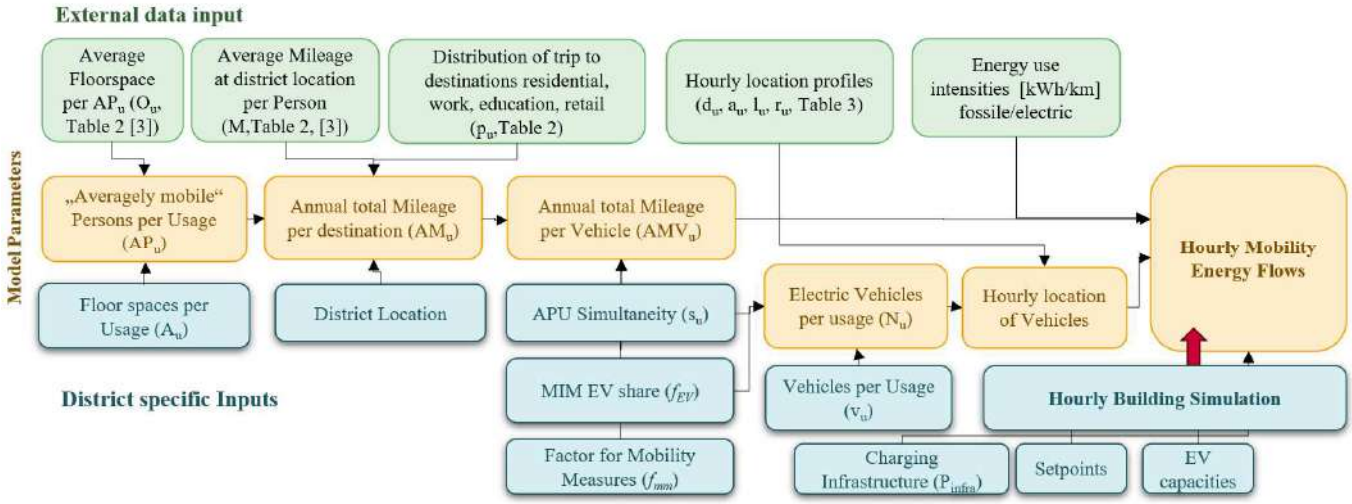


Fig. 1. Overview of Mobility modelling steps

This approach focuses on modelling everyday motorized individual mobility (MIM), other forms of everyday mobility are accounted for outside the PED system boundary in a national balancing approach described in [8], as is delivery and holiday trips.

TABLE I. NOMENCLATURE

Variable	Description	Unit
u	District floor usage type, can be (res)idential, (com)mercial, (edu)cation, (ret)ail	
A_u	Useable Floor Area of Usage type *u*	m²
O_u	Average Floor Space of Usage type *u* per AP_u	m²/Pers
AP_u	Number of averagely mobile Persons per Usage u	
AM_u	Annual total mileage per destination u	km/APu/a
s_u	Simultaneity factor: ratio between averagely mobile Persons of a given usage u and real Persons	%
f_{ev}	EV share of MIM vehicles	%
f_{mm}	Scaling factor of AM_u based on onsite mobility measures	%
N_u	Expected total number of unique EVs originating and visiting the district usage u	
v_u	Vehicle density per 100m² floor area of usage u	1/100m²
$d_u(t)$	Share of EVs of usage u being **onsite** at the district at hour t	%
$a_u(t)$	Share of EVs of usage u being **offsite**, not at the district, at hour t	%
$l_u(t)$	Share of EVs of usage u leaving the district at hour t	%
$r_u(t)$	Share of EVs of usage u returning to the district at hour t	%
m(t)	Share of EVs travelling at hour t	%
$SoC_{x,u}$	Average battery State-of-Charge of EVs of usage u (x=d…at district, onsite or a…away, offsite)	%
c	Specific battery capacity of EVs	kWh
$E_{s,u}(t)$	Hourly Energy demand for driving	Wh/m²
$E_{Cmin,u}(t)$	Hourly Energy demand for charging onsite EVs to minimum SoC	Wh/m²
$E_{Cflex,u}(t)$	Hourly Energy demand for charging onsite EVs to maximum SoC	Wh/m²
$E_{Cext,u}(t)$	Hourly Energy demand for charging offsite EVs to minimum offsite SoC	Wh/m²
$E_{D,u}(t)$	Hourly Energy supply from discharging onsite EVs to the district to a minimum SoC	Wh/m²

II. METHODS AND DATA

A. Annual travelers, trips and mileages

The model estimates expected travelers and travels for each usage type: residential, work and shopping and calculates their annual mileage (i.e. length of all trips) through these steps: First, for each usage in the district, the expected number of "averagely mobile" users ("APu", i.e. residents, workers, shoppers) are estimated based on census data of average floorspace per usage (Table II):

$$AP_u = A_u O_u^{-1} \quad (1)$$

The associated annual total MIM Mileage driven to these destinations (AMu) is selected from literature of annual mileage per location based on district location [17] (as shown in Figure 2 and Table II):

$$AM_u = AP_u p_u M(region) \quad (2)$$

The annual total mileage per vehicle of each usage (AMVu) is then determined factoring in EV share of MIM, a factor based on mobility measures in the district reducing MIM mileage, and a simultaneity factor for the ratio of real people to APUs (mainly for education, as the annual statistical mileage of 100 APUs contain 5% trips to education which in reality are concentrated on just 15 real Students (i.e. 15% simultaneity of the base population), each making 30% of trips there.

$$AMV_u = AM_u s_u f_{mm} f_{EV} \quad (3)$$

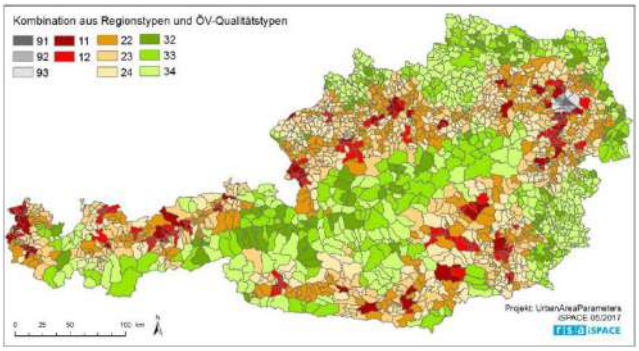


Fig. 2. Average Annual Mileage *M* per *region* per Person from [7]

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TABLE II. MOBILITY CENSUS DATA [7]

Region	*Average MIM Mileage per Person (km/P/a)*	Destination			
		Residential	*Work*	*Education*	*Retail*
11	5 527	51%	19%	2%	28%
12	6 456	50%	17%	3%	31%
22	7 810	48%	23%	3%	27%
23	7 846	47%	22%	3%	27%
24	11 831	49%	25%	2%	24%
32	7 204	55%	18%	3%	25%
33	8 038	47%	25%	2%	26%
34	9 125	49%	22%	3%	26%
91	4 062	52%	21%	2%	25%
92	3 414	48%	18%	4%	30%
93	4 291	50%	17%	3%	29%
Average Floorspace per AP_u (m^2/APu)		41.7	5.9	2.5	10.7
AP_U Simultaneity (s_u)		100%	80%	15%	100%

B. How many EVs are at the district at each hour?

The expected total number of unique EVs originating and visiting the district is dependent on location specifics such as parking lot availability, the specific vehicle intensity must be provided as district input:

$$N_u = v_u s_u f_{EV} \quad (4)$$

C. EV locations per hour: onsite or offsite

Vehicle location is tracked in aggregate, one for each usage types residential, work and retail. They represent the share of EVs currently onsite at the district, or offsite, i.e. driving or parked at other uses outside the district. Only onsite EVs may be bidirectionally charged, driving only happens with offsite EVs. The location of an average vehicle at any hour of the year is determined by weekly profiles from [18], given its relative probability of being onsite or offsite as shown in Figure 3. A vehicle is considered "present" at the district, if its location matches its originating usage, e.g. if an inhabitant's car is "at home" or a shopper's car is "at the shops". Otherwise, a vehicle associated with district usage u is considered "away" and cannot be charged/discharged at the district:

$$a_u(t) = 1 - d_u(t) \quad (5)$$

The total number of vehicles at the district $N_{d,u}$ or offsite $N_{a,u}$ is obtained by multiplying the above probability with the total EVu in the district. The share of vehicles of usage u **leaving from** or **returning to** the district at any given hour t is the difference of present vehicles between hours plus a time-depending exchange rate representing the fraction of onsite cars leaving just as others arrive:

$$l_u(t) = (d_u(t) - (1 + x(t))d_u(t+1))^+ \quad (6)$$

$$r_u(t) = (1 + x(t))d_u(t+1) - d_u(t))^+ \quad (7)$$

For destinations with typically diurnal stays such as residential and work, the turnaround per hour is low (5-10%), but for retail and shopping it is high (75% to 100%) effectively decreasing the average shopping stay at the district to one or two hours. This does not account for vehicles leaving twice an hour, whose stay would be too short to have a significant impact on annual charging and discharging in the district.

All vehicles not at any location are considered in transit:

$$m(t) = 1 - \sum_u d_u(t) \quad (8)$$

D. Hourly Vehicle SoC changes

Vehicle battery state-of-charge (SoC) is also tracked in two aggregates per usage: (1) vehicles present at the district $N_{d,u}$ and all vehicles currently away $N_{a,u}$, their respective $SoC_{d,u}(t)$ and $SoC_{a,u}(t)$ are:

$$SoC_{d,u}(t) = \varepsilon SoC_{d,u}(t-1) + (E_{Cmin,u} + E_{Cflex,u} - E_{D,u} + E_{r,u} - E_{l,u})c^{-1}N_{d,u}^{-1} \quad (9)$$

$$SoC_{a,u}(t) = \varepsilon SoC_{a,u}(t-1) - (E_{S,u} + E_{Cext,u} - E_{r,u} + E_{l,u})c^{-1}N_{d,u}^{-1} \quad (10)$$

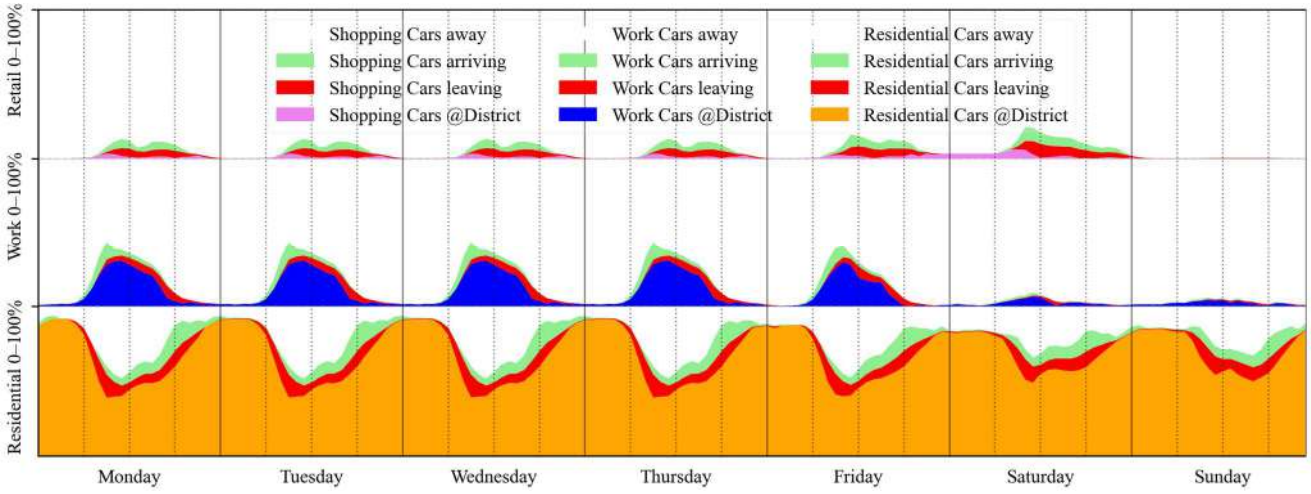


Fig. 3. Hourly probabilities of vehicle presence per usage: residential (orange, bottom), work (blue, middle) or retail/shopping (top, purple). Vehicles "leaving" (red) are considered onsite, switching to offsite at the end of the hour but can still be charged before, "returning" are offsite and arrive at the end of the current hour

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With the following hourly changes: **(1) battery self-discharge** of all vehicles proportional to their current SoC, given by an hourly charge retention rate ε,

(2) drive discharge, which is evenly distributed from the annual mileage assuming constant mileage per moving vehicle per hour:

$$E_{S,u}(t) = \eta AMV_u \frac{m(t)}{\sum_t m(t)} \quad (11)$$

(3) charging at the district is limited by the number of cars and charging stations onsite and the maximum charging power per EV and per charging station occurs in and occurs in two steps: (a) to reach the given minimum SoC per usage

$$E_{Cmin,u}(t) = \min\left(P_{infra}, cN_{d,u}(t)\min(P_{ev}, SoC_{min,u} - SoC_{d,u}(t-1)\right) \quad (12)$$

energy dispatch is used from these sources in order if available: local PV, local BESS, volatile offsite renewables and finally from the grid, as shown in the dispatch sequence in Figure 4. (b) If additional local PV or offsite renewables are available, they are used to charge EV battery to full capacity, given the remaining power after charging to minimum SoC:

$$E_{Cflex,u}(t) = \min\left(P_{infra} - \sum_u E_{Cmin,u}, cN_{d,u}P_{ev,remaining}, cN_{d,u}(1 - SoC_{d,u}(t-1))\right) \quad (13)$$

(4) discharging vehicles to the district occurs **only for residential usage**, and then **only** if (a) the $SoC_{d,res}$ is greater than the given minimum SoC, (b) there is uncovered electricity demand in the district for user plug loads or HVAC operation:

$$E_D(t) = \min\left(E_{D,pot}, E_{ev>pl} + E_{ev>hvac}\right) \quad (14)$$

(5) External vehicle charging away from the district occurs if the average SoC of offsite EVs drops below a given point:

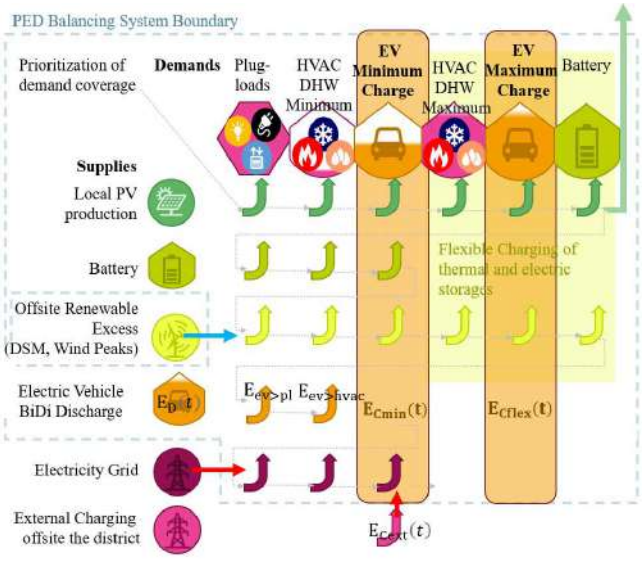


Fig. 4. Energy dispatch sequence in the district.

$$E_{Cext,u}(t) = cN_{a,u}\min(P_{EV}, \left(SoC_{min,ext} - SoC_{a,u}(t-1)\right)^+ \quad (15)$$

(6) Change due to **SoC carried by the vehicles** leaving from (El) and returning (Er) to the district, given as

$$E_{l,u}(t) = cN_{d,u}l_u SoC_{d,u}(t-1) \quad (16)$$

$$E_{r,u}(t) = cN_{a,u}r_u SoC_{a,u}(t-1) \quad (17)$$

E. Example Use Cases

Mobility energy flows are assessed for four different district usages of the same size: A group of ten single family detached houses (10 SFHs), an office, a school and a supermarket. All example buildings feature similar HVAC systems with heat pumps and similar properties of modern buildings constructed to code (thermal hull, etc.), key model parameters are summarized in Table III.

TABLE III. EXAMPLE BUILDING PARAMETERS

Variable	10 SFHs	Office	School	Super-market
Gross floor area (m²)	10x 240	2 400	2 400	2 400
Storeys	2	1	1	1
Unique visiting EVs	19	115	58	408
Vehicles per Area (1/100m²)	1	10	10	20
EV Share	100%	100%	30%	100%
Hourly EV Simultaneity	73%	10.1%	10.1%	3.4%
Min/Average /Max EVs per hour	9/14/18	0/12/40	0/5.8/58	0/14/49
Averagely mobile People	46	305	768	191
Annual EV Mileage (mio km)	0.44	2.97	0.9	1.74
PV (kWp) =100kWh/m²NFA	200	200	200	200
Flexible Grid use (up to x kW)	0	96	96	96

III. RESULTS

Figure 5 shows the resulting electricity balance for the four example usages, highlighting the energy demand for EV mobility ranging from 53 to 205 kWh/m² to cover MIM through EVs, which occurs for residential almost exclusively onsite, 50% offsite for office and retail, and 34% for educational use which only has 30% EV share.

This annual energy balances shown in Figure 5 highlight the potential for EV MIM inclusion to enable larger economic PV integration by pushing PV self-utilization much higher than just for building operation: Residential self-utilization of 60% appears low but is actually very high for specific PV yields of 100 kWh/m²NFA, which dwarf typical installation sizes by a factor two to four. With 100% EV integration, the supermarket and office buildings support a PV installation 10 times the size required by the building code (1kWp per 100m² Gross Floor Area) with a self-utilization rate of 90%.

In this example, non-residential uses also feature flexible dispatch through hourly flexibility signals from volatile

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regional RES, which occur predominantly in the winter due to wind power peak events and enable an increase in use of charging infrastructure by 30% for office, and almost 50% for supermarkets uses. The limiting factor here is the fractional time of EVs spent onsite charging: Even though onsite RES are sufficient to fully charge onsite vehicles, they represent only a fraction of all EVs visiting the district, having little effect on their average SoCs.

Using bi-directional charging for residential uses down to an average minimum SoC of 60% can reduce annual grid demand for heating and plug-loads by 28% from 29,3 to 21,3 kWh/m²NFA, at the cost of increasing annual full equivalent cycles (FEC) by 20% from 189 to 229 facilitating local utilization of large PV systems. This flexible discharge predominantly happens in the summer months after sunny days as can be seen in Figure 5 on the top right (in pink).

Figure 6 shows how the bidirectional EV integration into the same group of single-family houses with a smaller PV installation of 4 kWp per 100m² NFA (52kWh/m²NFAa) acts like a BESS, effectively increasing annual PV self-utilization from 37% without EV integration to 95% and reducing grid electricity demand in summer months to 5% of total electricity demand.

Figure 7 shows the hourly dynamism of the presented aggregate charging model and the difference between different usages, for example weeks in winter (left) and summer (right).

IV. Discussion

The method is very broad and has great uncertainties making it infeasible for concrete planning. In particular, the use of aggregate SoCs rather than tracking individual EV agents might oversimplify key constraints such as hourly charging potential based on actual SoCs, which may overlook peak demand issues and individual vehicle constraints.

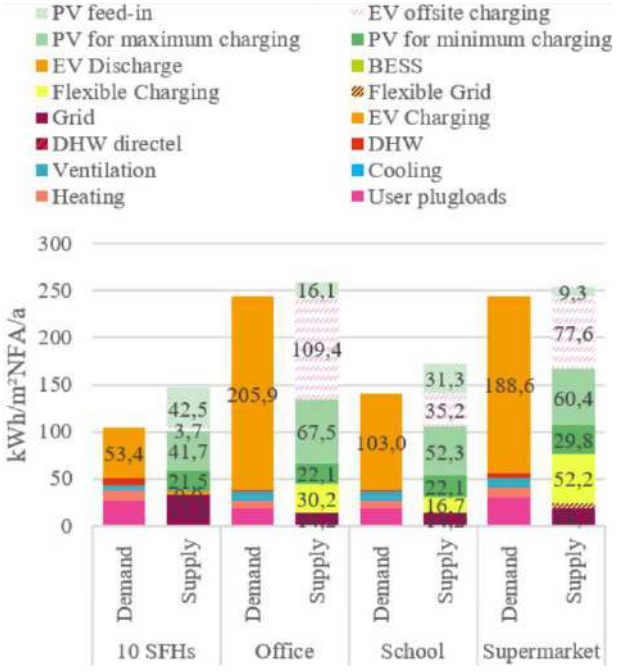


Fig. 5. Annual electricity balance of demand (left) and supply (right) for four 4 example usages (residential, work, education and retail) including energy services for heating, cooling, domestic hot water ventilation, user plug-loads, general and auxiliary electricity demand

However, it delivers an effective method for quickly assessing bidirectional EV operation over a wide range of use cases and a multitude of effects:

- Influence of Location and mobility measures increasing or reducing overall mileage
- Effect of EV penetration and motorization
- Effects of charging infrastructure and if it can be expected to be under- or oversized
- Combination of BESS, EVs and DSM flexibility
- Economic analysis of charging patterns
- Influence of usage mix onto expected MIM energy demand and energy balance

The presented model is parametrized for modelling any building and district in Austria, but the approach can be readily adopted to different situations, countries and climates by adjustment of annual mileages and the coupled energy building simulation.

This method is particularly interesting for non-residential and mixed usages, where the number of occupants and users is not well defined, does not align with typical residential vehicle use patterns and can vary greatly.

Modelling residential – and to a certain degree work related – mobility patterns is relatively straightforward due to immediate correspondence between floor area, vehicle usage and location. The same cannot be said for education and shopping-related mobility: Actual educational mobility patterns differ from statistical patterns, with only 15% of the population regularly visiting an educational facility, reducing the number of vehicles visiting accordingly. For shopping, the effect is reversed, as people travel to different shops during the week, the number of chargeable vehicles – and their corresponding charging potential – is significantly higher from the perspective of the shop location than just the average vehicles per shop. This is important for larger mixed-use districts, which typically curtail PV installation to self-consumption levels: Adding a supermarket to an energy

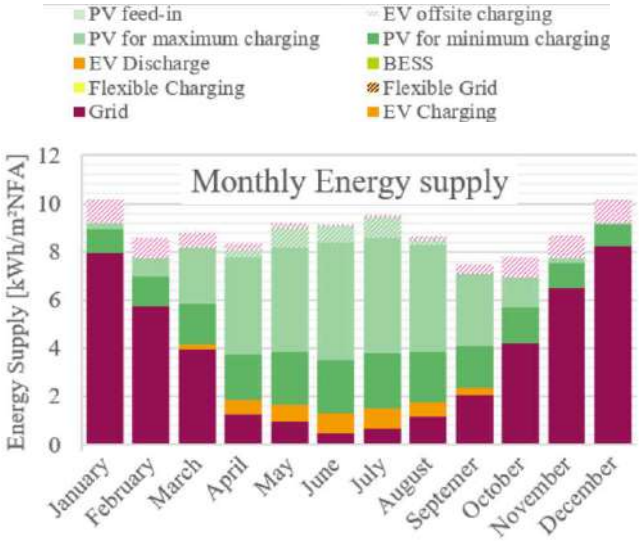


Fig. 6. Monthly electricity supply for 10 Single family houses with bidirectional EV charging and 100 kWp PV system.

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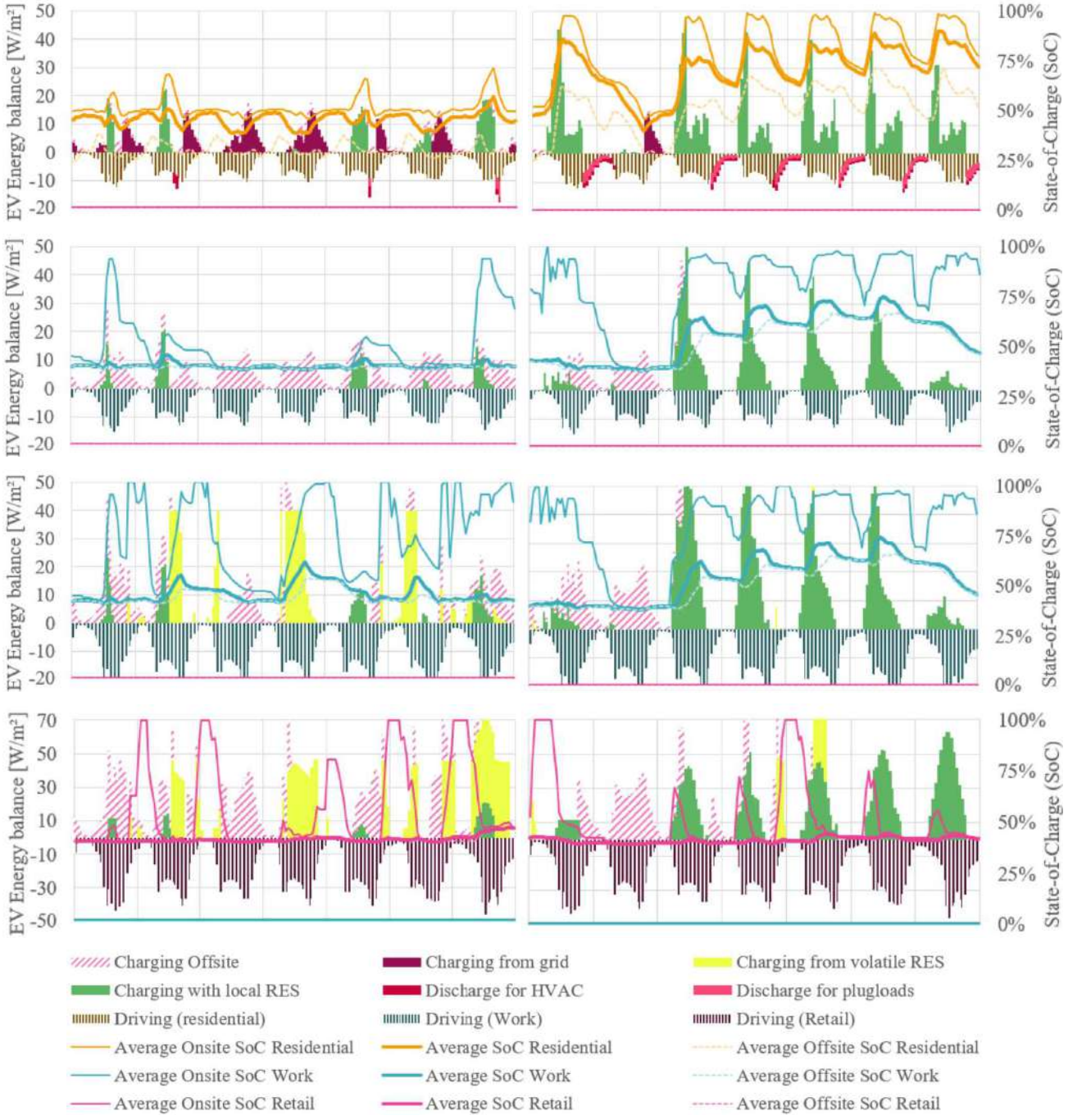


Fig. 7. Hourly mobility energy balance (left axis) and EV SoC (right axis) for four example usages residential (top, orange), office (second, blue), school (second to last, blue), supermarket (bottom, pink) for a week in winter (28.January.-3.February., left) and summer (8.-15.July, left)

community of low-density residential housing can increase the self-consumption of PV 70% to 85%.

V. Conclusions

The presented mobility modelling approach demonstrates the significant impact of EV integration on the energy balance and self-utilization of PV systems across various building types: The total energy demand for EV mobility ranges from 53 to 205 kWh/m²a for residential to office and supermarket uses respectively, which match 106% to 446% of the energy demand for building operation and use itself, highlighting the importance of including these expected energy demands in BEM and analysis.

By incorporating bi-directional charging, residential buildings can reduce grid demand by 28%, while non-residential uses benefit from increased flexibility, particularly during winter wind power peak events. Furthermore, a smaller PV installation paired with EV integration can increase self-utilization from 37% to 95%, reducing grid electricity demand in summer to just 5%. With 100% EV integration, the supermarket and office buildings support a PV installation 10 times the size required by the current building code.

These findings highlight the potential for large-scale PV integration by leveraging EVs for enhanced self-consumption and flexible energy use.

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B.9. Klimaneutrales Alliiertenviertel

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Contribution to the thesis:

Full-scale, real-world application of the PED assessment framework in a dense urban retrofit context under municipal planning constraints.

Klimaneutrales Alliiertenviertel



Stadt Wien | Technische Stadterneuerung

WieNeu+

Klimaneutrales Alliiertenviertel

Studie zu Dekarbonisierungsszenarien und Absenkpfade für den Endverbrauch und die THG-Emissionen der Gebäude im Alliiertenviertel.

Herausgeberin:

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MA 25 – Technische Stadterneuerung

Gruppe Programm WieNeu+ und Sonderprojekte

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Projektauftrag:

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Zusammenfassung / Executive Summary

Die vorliegende Studie analysiert unter welchen Rahmenbedingungen das gründerzeitliche Alliiertenviertel im zweiten Wiener Gemeindebezirk bis 2040 dekarbonisiert werden kann. Der Fokus liegt auf der Reduktion betriebsbedingter Treibhausgasemissionen im Gebäudebestand.

Grundlage der Modellbildung war eine umfassende Datenerhebung, bei der öffentliche Daten, Informationen städtischer Stakeholder, und Ergebnisse vorangegangener Forschungsprojekte sowie Erkenntnisse aus Vor-Ort-Begehungen verwendet wurden.

Die Ergebnisse stützen sich auf stündliche Simulationen von Einzelgebäude vor und nach Sanierungsmaßnahmen über ein Betriebsjahr, die zur weiteren Datenverarbeitung und Visualisierung als GIS-Datensätze zur Verfügung stehen und verwendet werden können.

Die bauliche Heterogenität der Gebäude im Quartier wurde nach Merkmalen gegliedert und durch 72 individuelle Gebäudetypen abgebildet, die sich in Bauperiode, Gebäudezustand und Energieversorgung unterscheiden und als Konkretisierung der Wiener Dekarbonisierungstypen dienen.

Ausgehend vom Status-Quo werden mehrere Entwicklungsszenarien untersucht: Eine Transition zu einem technisches Maximalszenario mit jährlichen Sanierungsraten von bis zu 4% pro Jahr auf Passivhausstandard, wie er im Quartier bereits demonstriert wurde, erreicht bis 2040 eine Reduktion des mittleren Heizwärmebedarfs (HWB) um bis zu 81%.

Das städtische Ziel des Wiener Klimafahrplans von 20% Endenergieeinsparung bis 2030 kann bereits mit einer mittleren Sanierungsrate von etwa 2,5% pro Jahr erzielt wird. Dabei führt bereits die Sanierung der ineffizientesten 17% des Gebäudebestands zu Einsparungen für Heizen, Kühlen und Warmwasser von 23 %.

Die laut Smart Klima City Strategie festgelegte Einsparung von 30% des Endenergiebedarfs bis 2040 ist durch eine jährliche thermische Sanierungsrate von 2% erreichbar: Die Sanierung der 28% ineffizientesten Gebäude im Quartier ermöglicht eine Einsparung von 35%. Eine Sanierung über das laut Wiener Bauordnung erforderliche Niveau, etwa auf Passivhaus- oder EnerPhit-Standard, ist dabei weniger ausschlaggebend als der frühestmögliche Zeitpunkt und die höchstmögliche Anzahl an Sanierungen, angefangen mit den ineffizientesten Gebäuden.

Ein vollständiger Ausstieg aus Gas in der Wärmeversorgung im Sinne des Wiener Klimaziels „Raus aus Gas“ erfordert die Umstellung der Heiz- und Warmwassersysteme in 80% der Gebäude bis 2040, was einer jährlichen Kesseltauschrate von etwa 5% entspricht.

Alle Szenarien wurden auch hinsichtlich des klima**aktiv** Standards für klimaneutrale Plus-Energie-Quartiere bewertet: Keines der betrachteten Szenarien erreicht eine ausreichende Primärenergie-Bilanz im Betrieb, wenn auch nur knapp: Mit 4% jährlicher Sanierung auf Passivhausstandard und 100% Ausnutzung der PV-Solardachpotentiale fehlt für einen nachweislich ausreichend klimaneutraler Betrieb 2040 nur ca. 20% der benötigten Versorgung mit Primärenergie. Potentiale dafür stehen im Quartier für zukünftige Betrachtungen ausreichend zur Verfügung: Nachverdichtung mit optimierter PV-Nutzung, Ausbau von E-Mobilität und Sektorkopplung, Schaffung von Energieflexibilität durch Kombination von Umweltwärme, Wärmepumpen und Fußbodenheizung.

Auch eine Erweiterung der bilanziellen Betrachtung um Alltagsmobilität, und Graue Energie und Emissionen der die Sanierungsmaßnahmen ist für die Bewertung als „Klimaneutral“ von Vorteil.

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1 Das Quartier „Alliiertenviertel“

Das Alliiertenviertel ist ein rund 8,7 ha großer Stadtteil im zweiten Wiener Gemeindebezirk. Das dreieckige Gebiet wird im Nordwesten von der Taborstraße, im Osten von der Nordbahnstraße und im Süden von Am Tabor begrenzt. Das Viertel war historisch von den zwei Bahnhofsarealen Wien Nord und Wien Nordwest eingegrenzt und in Richtung Osten und Nordwesten vom umliegenden Stadtraum abgeschnitten. Durch das Auflassen dieser beiden Frachtenbahnhöfe wurden und werden diese Barrieren abgebaut, eine energetische Anbindung an die Versorgung der Neubaugebiete wurde aber nicht forciert.

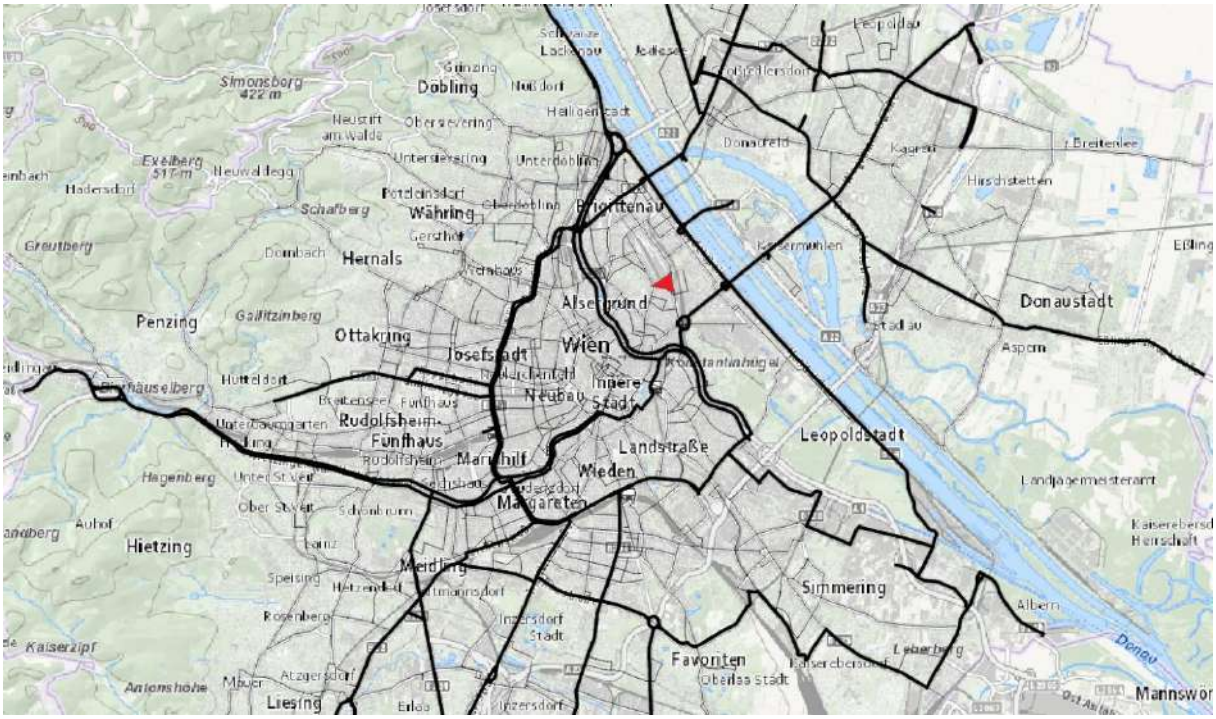


Abbildung 1-1 Das Alliiertenviertel befindet sich im zweiten Wiener Gemeindebezirk zwischen den Entwicklungsgebieten Nord- und Nordwestbahnhof und besteht aus 70 größtenteils gründerzeitlichen mehrgeschoßigen Wohnbauten.

Die namensgebende Alliiertenstraße weist mit rund 25 m einen breiteren Straßenquerschnitt als die querenden Gassen auf und bildet die zentrale Achse des Viertels. Das Gebiet ist überwiegend von einer 5-geschoßigen Blockrandbebauung geprägt, welche Richtung Süden besonders im Bereich der Trunnerstraße zunehmend aufgebrochen wird. Die Baublöcke weisen eine Seitenlänge von maximal 200 m auf, was eine hohe Durchlässigkeit ermöglich.

Die Gebäude der geschlossenen Bebauung gehen zu einem Großteil auf die Gründerzeit zurück, wobei die für diese Zeit charakteristischen Fassadenelemente nur mehr teilweise erhalten sind. Der Chopinhof (Am Tabor 1-3), die evangelische Verklärungskirche (Am Tabor 5) und das Wohngebäude in der Marinelligasse 6 heben sich als Solitäre von der vorliegenden Blockrandbebauung ab.

Das Viertel ist hauptsächlich von **Wohnnutzung** geprägt und beherbergt fünf Wohnhausanlagen der Gemeinde Wien („Gemeindebauten“). An den Hauptverkehrsstraßen am Rand des Viertels gibt es vermehrt gewerbliche Nutzung im Bereich des Erdgeschoßes, wobei eine Häufung in den höherfrequentierten Kreuzungsbereichen vorliegt.

Der motorisierte Individual-Verkehr verlagert sich hauptsächlich auf die begrenzenden Hauptverkehrsrouten, sodass das Verkehrsaufkommen im Viertel sehr reduziert ist, zusätzlich

herrscht eine Geschwindigkeitsbegrenzung von 30 km/h. Einige Straßen sind als Einbahn geführt, welche jedoch mit dem Fahrrad in beide Richtungen befahren werden können. Die starkbefahrenden Begrenzungsstraßen (Nordbahnstraße, Taborstraße, Am Tabor) verfügen über baulich getrennte Radwege. Die Anbindung an den öffentlich Personennahverkehr des Viertels ist über die Straßenbahnlinien 1, 2, 5 und O bzw. die Bus-Nightline N29 gegeben. Aktuell erfolgt der Neubau der Straßenbahnlinie 12. Die Stationen der genannten Straßenbahnen befinden sich jeweils an den Ecken des dreieckigen Viertels, also an den Kreuzungspunkten der begrenzenden Straßen.

Wichtige Grünräume im Alliiertenviertel sind der Else-Feldmann-Park und der Vilma-Neuwirth-Park, teilweise gibt es begrünte Innenhöfe. Die Alliiertenstraße verfügt über Baumbepflanzung auf der östlichen Straßenseite, die Taborstraße, die Nordbahnstraße und Am Tabor sind beidseitig mit Bäumen bepflanzt. Die genannten Parks verfügen über Parkbänke, im restlichen Viertel gibt es im öffentlichen Raum nur eine einzige Sitzgelegenheit in der Alliiertenstraße.

Die bautechnische Betrachtung des Quartiers zeigt ein zu großen Teilen gründerzeitlich geprägtes Quartier mit überwiegend erdgasversorgten Gebäuden. Hervorzuheben sind das Leuchtturmprojekt Eberlgasse 3[1] und einige räumlich verteilte thermische Sanierungen.

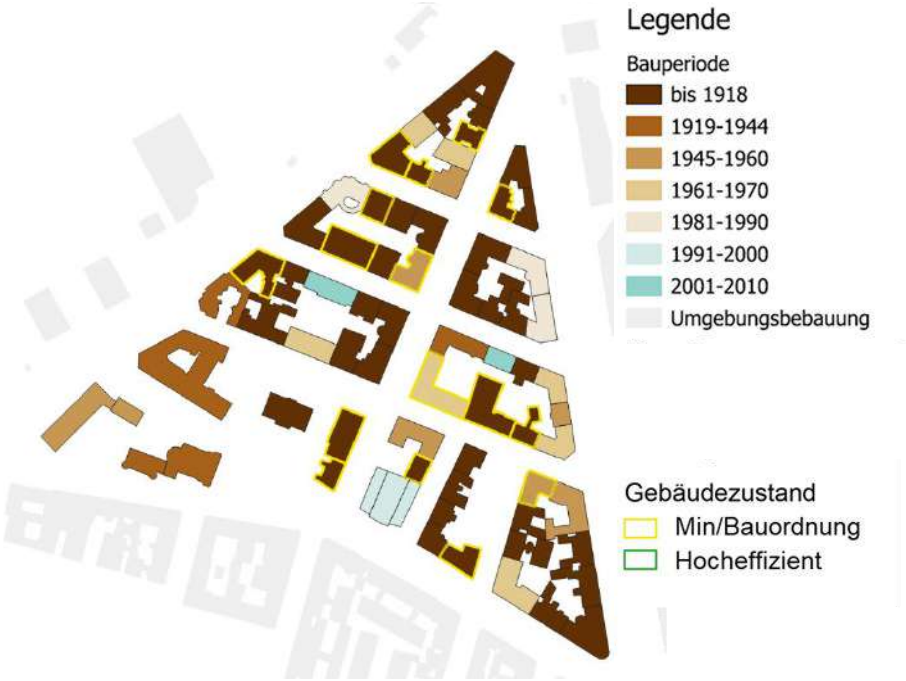


Abbildung 1-2 Status-Quo Alliiertenviertel: Bauperiode, Energieversorgung und Sanierungsstatus („Gebäudezustand“)

Die Analyse des Quartiers umfasst eine Betrachtung der architektonischen und baulichen Strukturen. Dazu gehören die bauliche Dichte und Kompaktheit als zentrale Parameter für die städtebauliche Entwicklung sowie allgemeine Erhebungen zur Gebäudestruktur, darunter die Anzahl der Gebäude nach Bauperiode und der Anteil sanierter Gebäude. Ergänzend wird die Nutzungsmischung untersucht, um die funktionale Zusammensetzung des Quartiers und deren Einfluss auf die gebaute Umwelt zu erfassen. Als Datengrundlage dienen neben Informationen der Stadt Wien eigene Erhebungen und Begehungen durch die Autor:innen.

Das Quartier weist eine heterogene Gebäudestruktur mit überwiegend historischer Bausubstanz auf. Von den insgesamt 72 Baukörpern stammen 44 aus der Bauperiode *vor 1918*, die damit die

[1] Siehe https://nachhaltigwirtschaften.at/de/hdz/projekte/sanierung-gruenderzeitgebaeude-eberlgasse-auf-passivhausstandard.php und https://www.wien.gv.at/umwelt/vorzeigeprojekt-eberlgasse

größte Gruppe ausmacht. Diese Gründerzeitgebäude weisen gleichzeitig den höchsten Sanierungsanteil auf (ca. 30 %). Die übrigen Bauperioden sind relativ gleichmäßig verteilt: *Zwischenkriegszeit* mit fünf Gebäuden (keines saniert), *Nachkriegszeit bis 1960* mit acht Gebäuden (25 % saniert), sowie sieben Gebäude aus den *1960er Jahren*, drei aus den *1980ern*, drei aus den *1990ern* und zwei aus den *2000ern*.

Hinsichtlich der Nutzung zeigt sich ein klarer Schwerpunkt auf Wohngebäude: 43 Baukörper werden ausschließlich zu Wohnzwecken genutzt. Die verbleibenden Gebäude weisen entweder eine *Nicht-Wohnnutzung* oder eine *Mischnutzung* auf, meist in Form einer Kombination aus Wohnen mit einer gewerblichen Sockelzone.

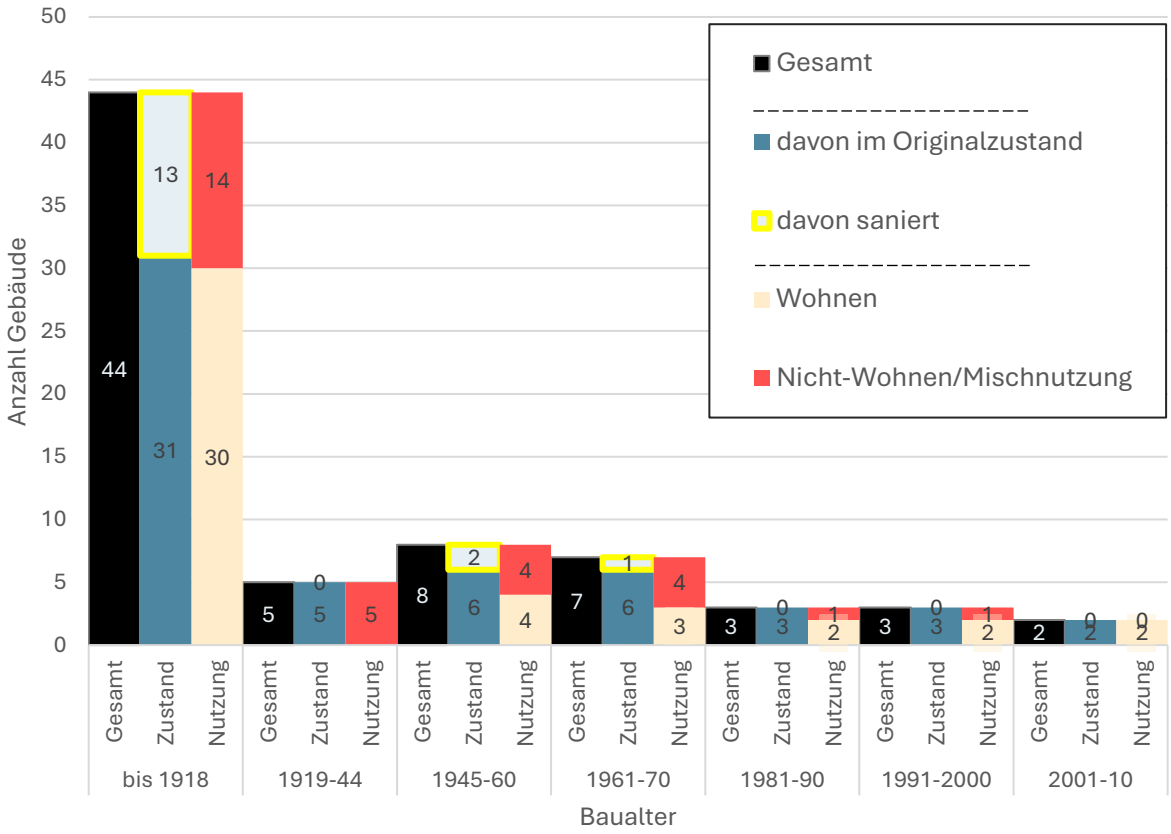


Abbildung 1-3: Übersicht Gebäude je Bauperiode, Zustand (original oder saniert) und grobe Einteilung Nutzung

Nutzungen

Da die **Nutzung der Quartiersflächen** einen maßgeblichen Einfluss auf den Energiebedarf und dessen zeitliche Verteilung hat, ist es wichtig zumindest zwischen Wohnen, Nicht-Wohnen und Mischnutzungen gebäudescharf zu unterscheiden. In Abbildung 1-4 ist die Nutzungsmischung des Quartiers ersichtlich.



Abbildung 1-4 Nutzungsmischung im Quartier, unkonditionierte Gebäude sind gelb gekennzeichnet, Blasengröße je nach Bruttogrundfläche der Gebäude

- Die identifizierten Nutzungen werden durch Nutzungsprofile der schweizerischen Norm für **Raumnutzungsdaten SIA 2024** abgebildet – die Profile sind im Anhang 6.2 dargestellt.
- Die **Sollwerttemperatur** wird von der Norm abweichend konstant auf **21°C im Heizfall** und im **Kühlfall 26°C** eingestellt.
- Auf eine Darstellung einzelner Detailnutzung, wie beispielsweise eines Yogastudios, wurde in der Modellierung zwecks Praktikabilität und der geringen Auswirkungen auf die Ergebnisse verzichtet. Stattdessen werden für spezielle Nutzungen Profile mit ähnlichen energetischen und zeitlichen Charakteristika verwendet.

Maß der baulichen Nutzung

Ein wesentlicher Faktor bei der Potentialanalyse von Quartiers-Energie- und THG-Bilanzen ist das Maß der baulichen Nutzung. Sie wird unter anderem durch die Geschoßflächenzahl (GFZ) ausgedrückt, die das Verhältnis aus Bruttogrundfläche zur Grundstücksfläche angibt. Folgendes Histogramm und Schwarzplan zeigen die GFZ der bebauten Grundstücke im Alliiertenviertel, die für Wien typische Werte im Bereich 2 bis 5 aufweisen, mit einem **Quartiersmittelwert von 3,75**.

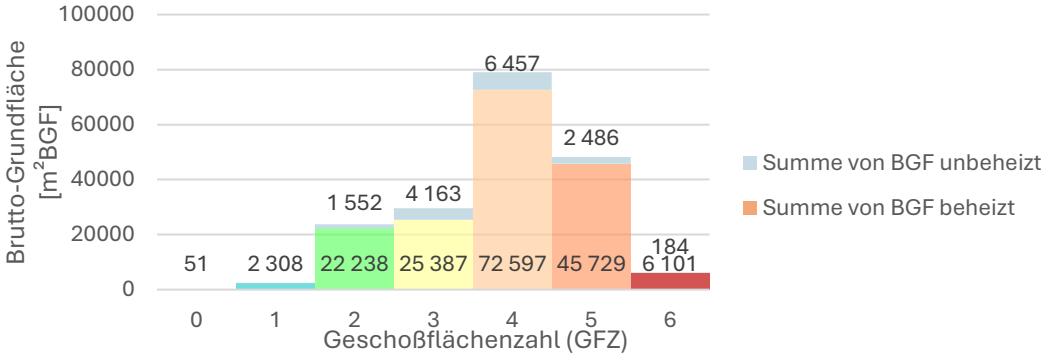


Abbildung 1-5 Brutto-Grundfläche je Geschoßflächenzahl (GFZ) im Quartier. Das Quartier weist im Durchschnitt eine GFZ von 3,75 auf und ist damit ein Quartier hoher baulicher Dichte.



Abbildung 1-6: Darstellung der baulichen Dichte in Form der Geschossflächenzahl (GFZ) je Grundstück.

In Abbildung 1-6 ist zu erkennen, dass nahezu alle Gebäude der klassischen Blockrandbebauungen eine Geschoßflächenzahl von 4 bis 6 aufweisen, nur die Gebäude im südwestlichen Teil, wie die Verklärungskirche, das Bezirksgericht und die Ecke Alliiertenstraße – Marinelligasse, weisen aufgrund zusätzlicher PKW-Abstellflächen oder privaten Gartenflächen geringere GFZ-Werte auf. Öffentliche Parkflächen wie der Else-Feldmann-Park oder der Gemeinschaftsgarten der Wiener Gebietsbetreuung haben keinen Einfluss auf die GFZ, da es sich nicht um bebaubare Grundstücke handelt.

Kompaktheit

Ein wichtiger Einflussfaktor für die Energieeffizienz von Gebäuden ist deren **Kompaktheit**, die durch das Verhältnis der wärmeabgebenden Hüllfläche (A) zum beheizten Volumen (V) angegeben wird. Je kleiner das sogenannte A/V-Verhältnis, desto geringer ist der spezifische Energiebedarf pro m^3 beheiztem Raum bei sonst gleichen Bedingungen. Das kleinste A/V-Verhältnis hat ein kugelförmiges Objekt, gefolgt von einem Würfel[2].

In Österreich wird oft der Kehrwert des A/V Verhältnis – die sogenannte „charakteristische Länge“ **lc** verwendet. Hier gilt das Gegenteil: Je höher die charakteristische Länge, umso kompakter das Gebäude und umso weniger Wärme wird pro Bezugsfläche zur Konditionierung benötigt.

Abbildung 1-7 zeigt, dass die charakteristische Länge **lc** im Alliiertenviertel bei einem Großteil der Gebäude in einem hohen Bereich von 2,5 bis 4 liegt. Zum Vergleich: Freistehende Einfamilienhäuser weisen typischerweise lc-Werte von 0,8-1,5 auf.

Die kompaktesten Gebäude des Quartiers sind das Bezirksgericht, das Eckgebäude am nördlichen Ende des Alliiertenviertels und vier Gebäude in der Schweidlgasse und Eberlgasse, die mit einem l_c größer 4 in Abbildung 1-7 gelb dargestellt sind. Diese Gebäude weisen die geringste Außenfläche aufgrund ihrer einfachen Quaderform mit angrenzenden Flächen an Nachbargebäude, und Gebäude mit besonders hoher Trakttiefe und damit umschlossenen Volumen dar.

[2] Quelle: Baunetzwissen (https://www.baunetzwissen.de/nachhaltig-bauen/fachwissen/planungsgrundlagen/gebaeudeform-662875 , aufgerufen am 17.4.2025)

Niedrige Kompaktheit mit einem lc<1,5 weisen nur wenige ein- und zweigeschoßige Gebäude und Gebäude mit besonders geringer Trakttiefe bzw. einem hohen Anteil an Außenflächen auf, die in Karte Dunkelblau bis Lila dargestellt sind.

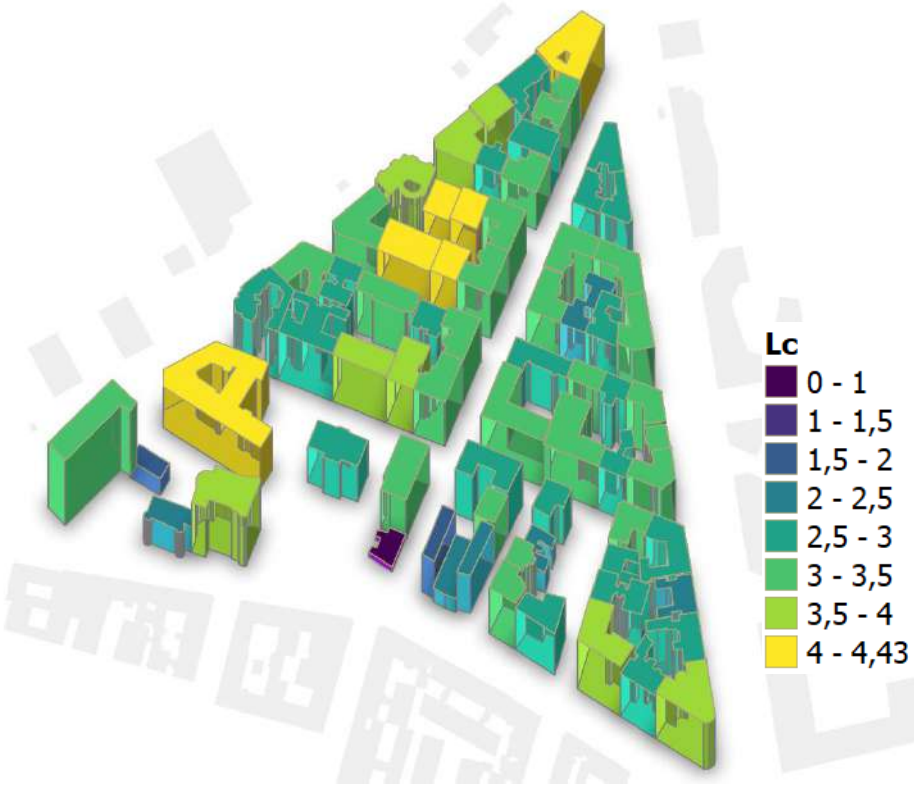


Abbildung 1-7 Die Charakteristische Länge (lc) ist der Kehrwert des Oberflächen-Volums-Verhältnis (A/V) und liegt für die meisten Baukörper im Bereich 2,5 - 4

2 Ziele der Studie

Wien will bis 2040 im Gebäudesektor gänzlich auf fossile Energiequellen verzichten. Ziel ist es, den Pro-Kopf-Endenergieverbrauch für Heizen, Kühlen und Warmwasser bis 2030 um 20% bzw. bis 2040 um 30% zu reduzieren, und die damit verbundenen CO_2-Emissionen zuerst um 55% und in weiterer Folge auf null zu reduzieren[3] – Aber wie? Die Entwicklung soll anhand von ausgewählten Quartieren wie dem gründerzeitlichen „Alliiertenviertel“ demonstriert und davon ausgehend repliziert werden.

Ziel der vorliegenden Studie ist die Ermittlung realistischer Dekarbonisierungsszenarien und Absenkpfade für den Energieverbrauch und die THG-Emissionen des Gebäudebetriebs im Quartier. Die erzielbaren Ergebnisse sollen auf Kompatibilität mit übergeordneten Zielsetzungen geprüft werden: Einerseits mit den Zielen der Stadt Wien (Smart Klima City Strategie) und andererseits mit den Zielen für „Plus-Energie-Quartiere“ des klima**aktiv** Deklarationssystems[4]. Ziel des Projekts ist konkret die Modellierung, Simulation und Bewertung von folgenden Szenarien:

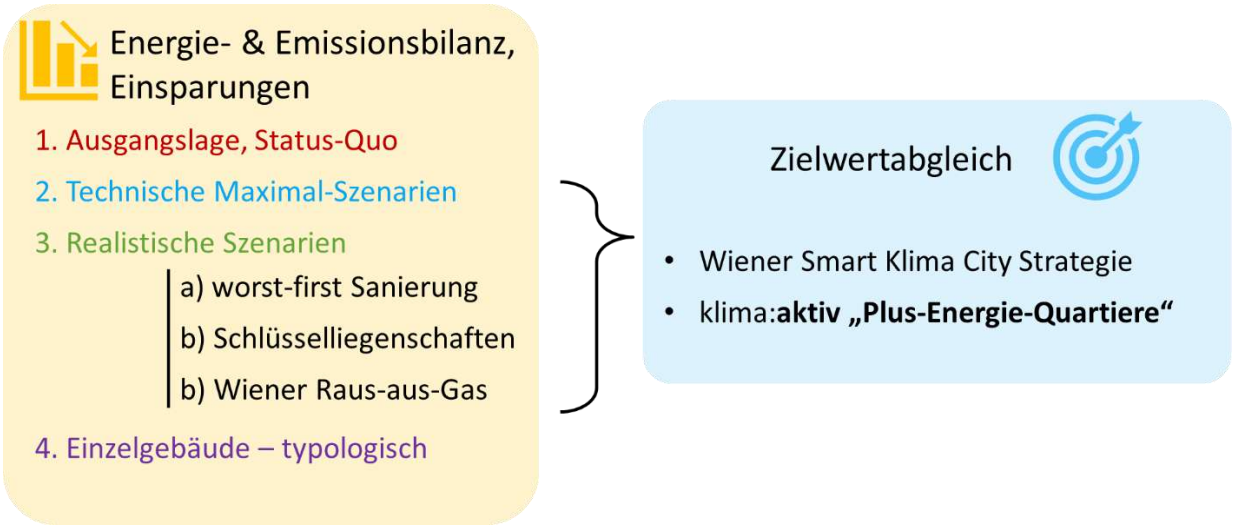


Abbildung 2-1 Übersicht der untersuchten Szenarien links und Ziele rechts

Nach einer Erhebung der „Ausgangslage“ 2040 und der maximalen Potentiale durch die sogenannten „technischen Maximal-Szenarien“ sollen „realistische Szenarien“ aufzeigen, wieviele Gebäude bis 2030 bzw. 2040 tatsächlich saniert sein müssten, um das Ziel der Reduktion des Endenergieverbrauchs um 20 % bis 2030 bzw. 30 % bis 2040 zu erreichen. Diese Szenarien gliedern sich in drei Betrachtungen:

- „**worst-first**“ zur Darstellung unterschiedlicher thermischer Sanierungsraten bei Priorisierung der Sanierung nach den ineffizienztesten Gebäuden zuerst. Dh. Priorisierung der Gebäude nach erwartetem Einsparpotential. Nur teilweise Sanierung des Quartiers mit Sanierungsraten von 1%, 2%, 3% und 4%p.a. unter Verwendung des hocheffizienten Standards führen bis 2040 zu unterschiedlichen Jahresbilanzen. Dabei bleibt der Kesseltausch an die thermische Sanierung geknüpft.

[3] Quelle: Wiener Klimaziele: Gebäude - Klimaschutz: Wien wird klimaneutral - Wiener Klimafahrplan
[4] Quelle: Klimaaktiv Deklaration klimaneutrale Plus-Energie-Quartiere

- „**Schlüsselliegenschaften**“: 15 von der MA25 als „Schlüsselliegenschaften“ definierten Gebäude werden auf einen hocheffizienten Gebäudestandard mit Fernwärmeversorgung saniert.
- „**Wiener Szenario**“ mit Entkopplung von thermischer Sanierungsrate und Kesseltauschrate: Die Kesseltauschrate wird auf 5%p.a. angehoben, sodass bis 2040 „Raus-aus-Gas“ zu 100% im Quartier erreicht ist, die thermische Sanierungsrate wird von derzeit etwa 1%p.a. auf 2%p.a. verdoppelt[5]

Zur besseren Einordnung der Ergebnisse werden alle Einzelgebäude im Quartier zusätzlich in allen typologischen Merkmalen simuliert und analysiert, um für zukünftige Studien mit ähnlichen Gebäuden eine Aussage auf Basis der typologischen Merkmale Kompaktheit, Bauperiode, Sanierungsumfang, und Energieversorgungsystem zu erleichtern.

[5] Annahmen zu Sanierungsraten in Anlehnung an: T. Steffl, J. Wahlmüller, C. Unterkircher, H. Eder, und S. Potuschak, „GLOBAL 2000 Wohnbaucheck 2021 - Wie klimafit sind Österreichs Bundesländer beim Heizen und Sanieren?“, GLOBAL 2000, Wien, 2021.

3 Methoden

Die Studie wurde in folgenden Schritten durchgeführt:

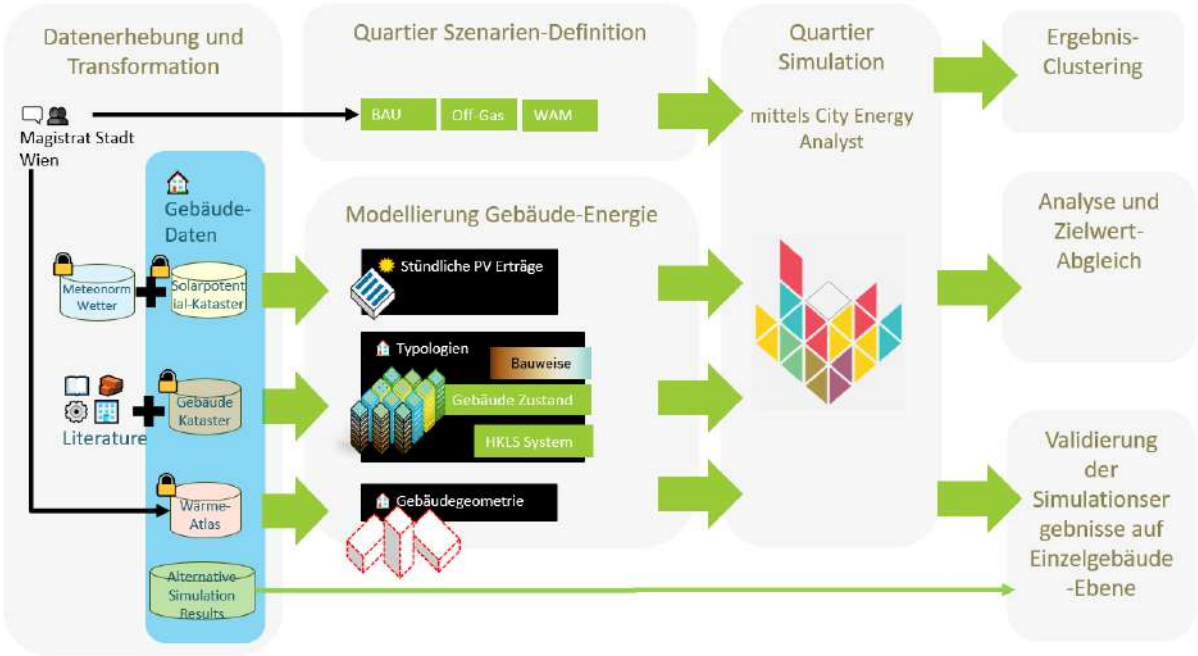


Abbildung 3-1 Methodik der Studie

Die **Datenerhebung** besteht aus der Sammlung und Erhebung verfügbarer Informationen (städtische Stakeholder, Begehung, etc.) und deren Strukturierung und Verarbeitung in GIS-Formate zur weiteren Verwendung.

Die **Modellierung des Quartiers** erfolgte auf Einzelgebäudebasis. Dazu wurden die einzelnen Gebäude des Quartiers nach deren übergeordneten Charakteristiken nach „Ähnlichkeit“ in typologische Gebäude eingeteilt, sodass die Einzelgebäude aufgrund dieser Typologien modelliert und simuliert werden können. Diese übergeordneten Charakteristiken sind die **Bauperiode** eines Gebäudes, der **Gebäudezustand** und die überwiegende **Energieversorgung**. Daneben werden (gebäudespezifisch) die **Baukörper** modelliert und die **Nutzungsmischungen** miteinbezogen.

Die **Simulation** der Energiebedarfe wurde mit der Open-Source-Software City Energy Analyst (CEA) der ETH Zürich durchgeführt[6], die je Gebäude ein thermisches 1-Zonen-Modell in stündlicher Auflösung verwendet. Eingangsgrößen umfassen Wetterdaten (Solarstrahlung, Außentemperatur), bauphysikalische Eigenschaften (Gebäudehülle, Transmissionsverluste), Nutzungsprofile (innere Lasten) und vereinfachte Anlagensysteme. Das thermische Modell bilanziert Heiz-, Kühl- und Lüftungsbedarfe auf Basis dieser Parameter, wie im Folgenden beschrieben. Dabei ist zu betonen, dass durch die Modellierungsannahmen der Autor:innen und die typologische Beschreibung der Einzelgebäude die Simulationsergebnisse auch nur typologische Geltung besitzen und **damit keine Rückschlüsse auf reale Verbrauchswerte oder adressspezifische Informationen der tatsächlichen Gebäude ermöglichen**. Die ermittelten Ergebnisse auf Ebene der Einzelgebäude sind als typische Kennwerte zu verstehen.

Die Analyse und der **Zielwertabgleich** konzentrieren sich auf die Einsparung des Endenergiebedarfs, sowie der Treibhausgasemissionen im Einklang mit den Klimazielen der Stadt

[6] Quelle: https://www.cityenergyanalyst.com/

Wien. Ergänzend werden weitere Kennzahlen wie die Geschossflächenzahl (GFZ) und die Kompaktheit des Gebäudes (A/V-Verhältnis) erhoben, um eine umfassende Bewertung des Quartiers durchzuführen. Diese Kennzahlen dienen u.a. als Grundlage für die Analyse und die Einordnung in den klimaaktiv Plus-Energie-Quartiers-Standard.

3.1 Datenquellen

Basis der Untersuchung bilden die städtischen Gebäude-Daten des **Wärmeatlas** aus dem GEL-SEP Projekt. Die im Datensatz verfügbaren Attribute und ihre Beschreibung sind im Anhang angeführt. Grundstücksfläche wird dem Grundstückskataster der Stadt Wien entnommen und entspricht mit geringen Abweichungen der Summe aus Außenflächen und Bruttogrundfläche. Folgende Daten wurden zur Verfügung gestellt und verwendet:

Tabelle 1 Übersicht Datenquellen und zugehörige Nutzung

Verfügbare Daten	Nutzung im Rahmen der Studie	Granularität	Quelle
OGD WFS Stadt Wien	Gebäudegrundrisse, Höhen, bebaubare Grundstücke	Grundstücke	Online[7]
3D Baukörpermodell	Prüfung der 3D Gebäudegeometrie	Baukörper	Online[8]
Open Street Map	Bebaute Flächen	Gebäude	Online[9]
Wiener Wärmeatlas	Gebäudehülle zu Umgebung und Adiabate/Nachbarumgebung, Statistischer Top-down Wärmebedarf	Gebäude	UIV
Gebietsanalyse	Allgemeine Informationen zum Quartier	Gebäude/ Grundstücke	MA25
Solarpotential-kataster	Größe der für eine PV-Anlage geeigneten Dachfläche, Qualität und potenzieller Ertrag	Dächer	Online[10]
PV Studie	PV Erträge Einzelliegenschaften, Validierung der stündlichen PV-Simulation der vorliegenden Studie	Beispiele	MA25
Flächennutzung der Sockelzonen	Nutzungsprofile, Belegungen, Interne Lasten/Gewinne, Nutzerstrom	Gebäude	Begehung

3.2 Energetische Modellierung des Quartiers mit CEA

Das Quartier wurde durch energetische Modellierung der Einzelgebäude abgebildet, die in folgenden Bereichen kategorisiert wurden:

a. Gebäudenutzung
b. Gebäudetypologie
c. Erneuerbare Potentiale in Form von Photovoltaik

Die **Gebäudegeometrie** wurde je Gebäude aus den verfügbaren Modelldaten des Wärmeatlas und des 3D-Stadtmodells abgeleitet und ist damit detaillierter als erwartet und in anderen Methoden, etwa den Wiener Dekarbonisierungstypen[11] abgebildet. Gebäude sind zusätzlich

[7] https://www.data.gv.at/katalog/dataset/stadt-wien_webfeatureservicewfswien
[8] https://www.data.gv.at/katalog/dataset/214a3e63-31aa-47de-a49c-5ea5a80330ed
[9] https://www.openstreetmap.org/search?query=alliiertenviertel#map=17/48.225766/16.388276
[10] https://www.wien.gv.at/stadtentwicklung/energie/themenstadtplan/solarpotenzial/
[11] https://www.wien.gv.at/stadtentwicklung/energie/pdf/waerme-und-kaelte-2040.pdf

durch eine Gebäudetype, Gebäudehöhe und Anzahl Geschosse, sowie die Nutzungsmischung im Gebäude charakterisiert:

- Gebäudetypen sind durch die ursprüngliche **Bauperiode**, den **Gebäudezustand** und die Art der überwiegenden **Energieversorgung** definiert.
- Gebäudehöhe, Anzahl Geschosse und Nutzflächen:
  - Es wurde pauschal **ein unkonditioniertes Untergeschoss** angenommen,
  - die Dachgeometrie wird durch Quader entlang der obersten Geschoßdecke vereinfacht modelliert,
  - als Verhältnis von Brutto- zu Nettogeschossfläche wird 80 % festgelegt.
- Nutzungsmischung: Zuordnung zu Norm SIA2024 Raumnutzungstypen in prozentuellen Anteilen gemäß Begehung

Im Folgenden werden die Details und Hintergründe der Gebäudemodellierung erläutert.

3.2.1 Gebäudegeometrie

- Das geometrische Gebäudemodell basiert auf den bebauten Flächen aus OpenStreetMap (OSM), deren Aufteilung in einigen Fällen von den Grundstücksdaten der Stadt Wien abweicht. Diese Unterschiede wurden manuell überprüft und entsprechend zugewiesen, um eine konsistente Modellierung zu gewährleisten (siehe Abbildung 3-3).
- Die Grundlage für die Ermittlung der **Hüllflächen** ist das 3D-Stadtmodell bestehend aus extrudierten Grundrissen der Einzelgebäude.
- Die **Geschoßanzahl** wurden je Gebäude aus dem Wärmeatlas der Stadt Wien übernommen und nach einer Validierung mittels Vor-Ort-Begehung bzw. aktueller Satellitendaten **manuell korrigiert**.
- Die brutto **Geschosshöhen** wurden mittels Division der Gesamtgebäudehöhe durch die Geschoßanzahl ermittelt. Halbgeschosse wie Dachgeschosse oder ähnliche Zwischenräume wurden nach Ermessen und in Abhängigkeit vom sichtbaren Anteil als Teil eines Vollgeschosses in das Modell integriert, um eine möglichst realistische Darstellung der Gebäudestruktur zu gewährleisten.

In Abbildung 3-2 ist das resultierende 3D-Modell dargestellt, das in der energetischen Quartierssimulation verwendet wurde.

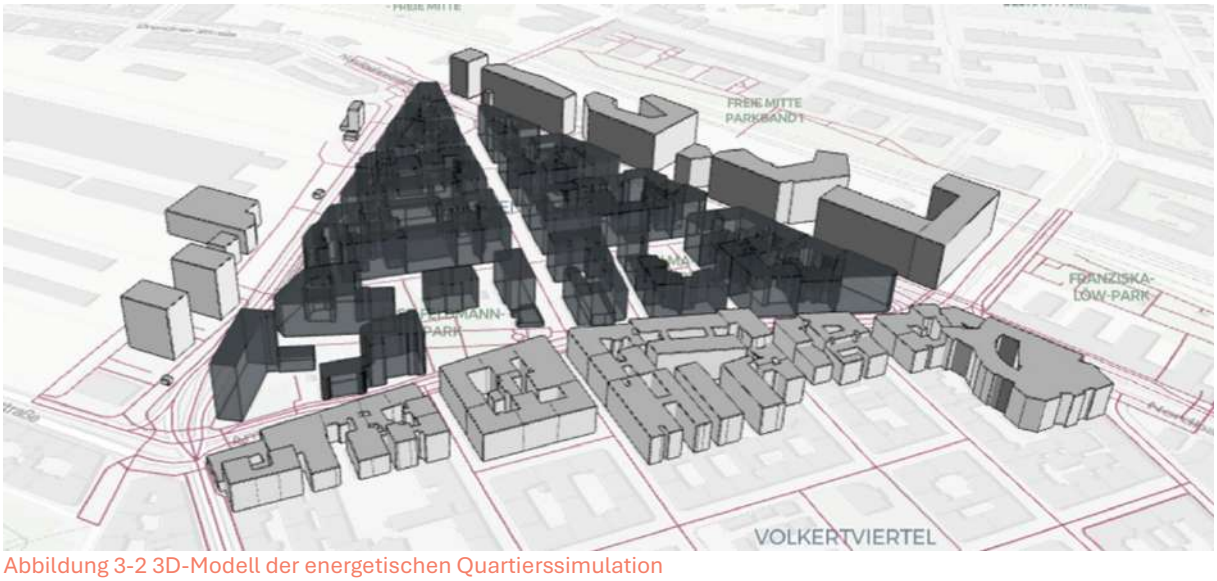


Abbildung 3-2 3D-Modell der energetischen Quartierssimulation

Das Quartier weist einige **nicht-konditionierte Flächen in den Innenhöfen** der Blockrandbebauungen auf, die Garagen, Schuppen oder ähnliche Sonderflächen darstellen und für die energetische Modellierung nicht berücksichtigt werden: Ein-geschossige Innenhofbebauungen, wie sie in einigen Bereichen vorkommen, werden als **nicht konditioniert** angenommen, da diese in der Regel zum Parken oder ähnliche Nutzungen verwendet werden und keinen Einfluss auf die energetische Simulation haben.



Abbildung 3-3 Vergleich Stadt Wien Grundstückskataster (links) und bebaute Flächen gemäß OSM (rechts). Kirchenräume (blau) wurden als nicht-konditioniert angenommen

Gebäuden mit speziellen Kubaturen wurden im Einzelfall händisch angepasst, wie die **katholischen Kirche** Am Tabor, deren Baukörper in drei Bereiche unterschiedlicher Höhe unterteilt und der mittlere Kirchenraum als unkonditioniert angenommen wurde. Die **evangelische Verklärungskirche** im Westen des Quartiers wurde ebenfalls als unkonditioniert angenommen.

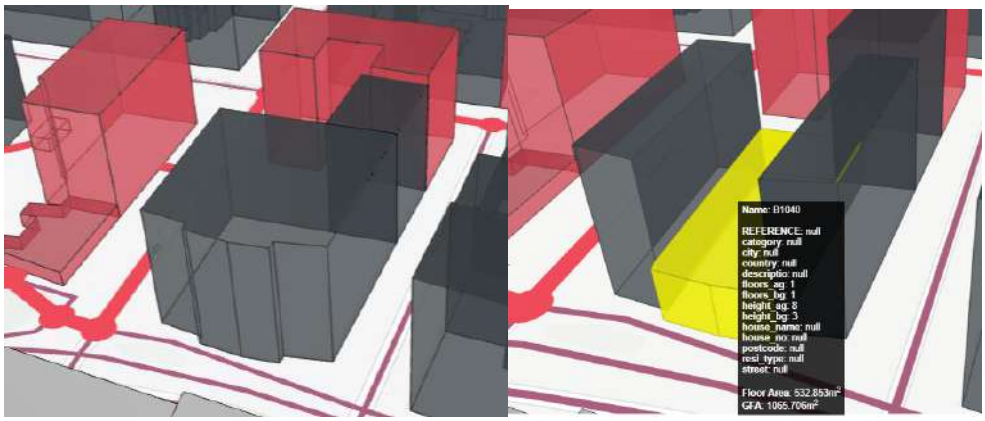


Abbildung 3-4 katholische Kirche links vor, rechts nach der Anpassung der Gebäudehöhe mit seitlich anschließenden Wohngebäuden

Trotz händischer Nachbearbeitung der Kubaturen bleiben Ungenauigkeiten hinsichtlich Nutz- und Hüllflächen, wie am Beispiel in Abbildung 3-5 ersichtlich: Die beiden farblich gekennzeichneten Gebäude am unteren Blockrand weisen im Modell nur eine Fläche mit einheitlicher Gebäudehöhe auf, während die realen Gebäude aus mehreren Teilbereichen unterschiedlicher Höhe und ggf. auch Konditionierung bestehen.

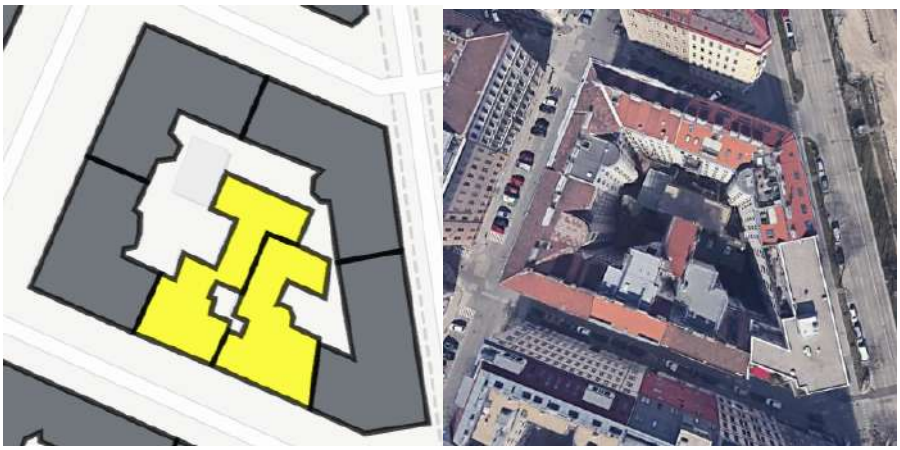


Abbildung 3-5 Ungenauigkeiten der geometrischen Gebäudemodellierung: Die beiden farblich gekennzeichneten Gebäude am unteren Blockrand weisen im Modell nur eine Fläche mit einheitlicher Gebäudehöhe auf, während die realen Gebäude aus mehreren Teilbereichen unterschiedlicher Höhe und ggf. auch Konditionierung bestehen.

3.2.2 Gebäudetypologien

Die Modellierung der nicht-geometrischen Gebäudeeigenschaften erfolgt je Gebäude und Szenario durch die Auswahl eines Typs in jeder der drei wesentlichen Eigenschaftsgruppen, die unabhängig voneinander wesentliche energetische Parameter des Gebäudes definieren:

Tabelle 2 Übersicht Datenquellen und zugehörige Nutzung

Eigenschaftsgruppe	Typen
Bauperiode	6 Typen
Energieversorgung	3 Typen
Gebäudezustand und Sanierung	4 Typen

Die Kombination der Kategorien aus den drei Eigenschaftsgruppen ergibt insgesamt 72 einzigartige Gebäudezustände oder „Archetypen“, die für die Modellierung des Quartiers genutzt werden. Abbildung 3-6 zeigt die Kategorien der 3 Eigenschaftsgruppen und beschreibt welche Parameter eines Gebäudes je Eigenschaft abgeleitet wird.

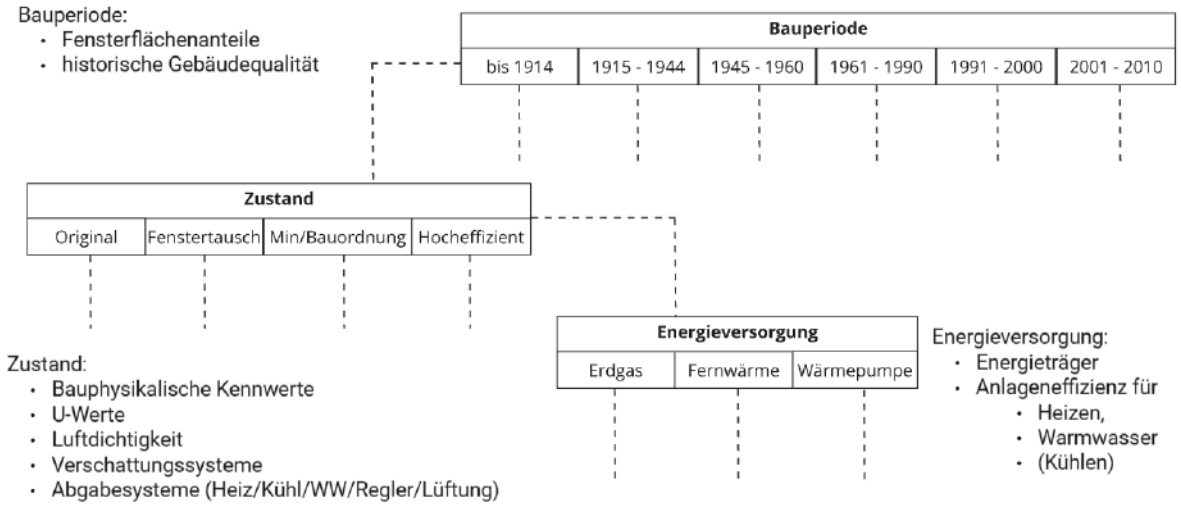


Abbildung 3-6 Die Auswahl eines Typs für jede der drei Eigenschaftsgruppen Bauperiode, Zustand und Energieversorgung führt zu 72 möglichen Gebäudetypen, die für jede Geometrie analysiert werden kann.

Folgende Tabelle gibt einen Überblick über die verwendeten Parametrisierung in CEA:

Tabelle 3 Verwendete Gebäude-Parameter im energetischen Quartiersmodell CEA

		Gebäudetyp Bestand je Bauperiode						Sanierungstyp		
Kategorie	**Parameter**	**bis 1918**	**1919-1944**	**1945-1960**	**1961-1990**	**1991-2000**	**2001-2010**	**Min/Bau-ordnung**	**Fenster-tausch**	**Hoch-effizient**
Luftdichtheit	n_{50}-Wert	4	4	4	4	2	2	1.5	*	1
Fenster	FF-Anteil	0.25	0.15	0.15	0.15	0.15	0.20	-	-	-
Verschattung	Faktor	1	1	1	1	1	1	1	*	0.15
Fenster	U-Wert	2.42	2.53	2.53	2.86	198	1.54	1.17	1.12	0.77
Dach/OGD	U-Wert	1.21	0.88	0.88	0.61	0.33	0.22	0.33	*	0.22
Außenwand	U-Wert	1.98	1.98	1.76	1.16	0.44	0.39	0.29	*	0.21
Boden/Keller	U-Wert	0.22	1.32	1.32	0.72	0.55	0.44	0.33	*	0.22
Heizen	Abgabe	R90/70	R90/70	R90/70	R90/70	R70/55	R70/55	R70/55	*	FH40/35
Heizen	Regelung	Keine	Keine	Keine	Keine	Keine	Keine	-	-	-
Kühlen	System	Keine	Keine	Keine	Keine	Keine	Keine	*	*	AC 6/15
Kühlen	Regelung	Keine	Keine	Keine	Keine	Keine	Keine	-	-	PICO
Warmwasser	System	HTWS	HTWS	HTWS	HTWS	HTWS	HTWS	*	*	*
Lüftung	System	Fenster	Fenster	Fenster	Fenster	Fenster	Fenster	*	*	Mech

***** … *je nach Bauperiode*
FF-Anteil … *Fensterflächenanteil an der Brutto Außenwand (für alle Himmelsrichtungen)*
R90/70 … *Hochtemperatur Radiator mit 90/70°C Vorlauf/Rücklauf Temperaturen*
R70/55 … *Radiator mit 70/55°C Vorlauf/Rücklauf Temperaturen*
FH40/35 … *Fußbodenheizung mit 40/35°C Vorlauf/Rücklauf Temperaturen*
FC18/21 … *Fußbodenkühlung mit 18/21°C Vorlauf/Rücklauf Temperaturen*
AC 6/15 … *Zuluftkonditionierung mit 6/15°C Vorlauf/Rücklauf Temperaturen*
HTWS … *Hochtemperatur Warmwasserbereitung 60/10°C Temperaturen*
Fenster … *Fensterlüftung*
Mech … *mechanische Lüftung mit Wärmerückgewinnung*
PICO … *PI Regler mit „Optimum Tuning“ in CEA*

Bauperiode und Gebäudezustand

Die Einteilung in sieben Bauperioden laut Wärmeatlas Wien sind in Abbildung 3-7 dargestellt, die sich wie folgt auf die Gebäudemodellierung auswirkt.



Abbildung 3-7 Ursprüngliche Bauperiode je Baukörper

Fensterflächen-Anteile

Die Bauperioden geben einen ersten Aufschluss auf die Fensterflächenanteile (FFA). Hier wird in der Modellierung zwischen den Bauperioden nur gering variiert: Der Zusatz Kostenoptimalität zur OIB Richtlinie 6 (Österreichisches Institut für Bautechnik, 2024) gibt je nach Bauweise (offen, geschlossen) für Mehrfamilienhäuser (MFH) Fensterflächenanteile von 17% und 24%, für Geschosswohnbau 21% und 18% an und für reine Dienstleistungsgebäude pauschal 32% an.

Tabelle 4 Fensterflächenanteile je Bauperiode

Bauperiode	Fensterflächenanteil *
bis 1918	25%
1919-1944	15%
1945-1960	15%
1961-1970	15%
1981-1990	15%
1991-2000	15%
2001-2010	20%

* Anteil der Brutto-Fenster-Fläche inkl. Rahmen an der Fassadenfläche

Über die Verbindung zur Bauperiode gibt es in der Literatur wenig Anhaltspunkte: Von vergleichsweise hohen Werten während der Gründerzeit sinken die Fensterflächenanteilen in der Zwischenkriegs- und Nachkriegszeit, bevor sie mit steigender Qualität der Fenster später wieder ansteigen, was sich in den Annahmen in Die Bauperioden geben einen ersten Aufschluss auf die Fensterflächenanteile (FFA).

Hier wird in der Modellierung zwischen den Bauperioden nur gering variiert: Der Zusatz Kostenoptimalität zur OIB Richtlinie 6 (Österreichisches Institut für Bautechnik, 2024) gibt je nach Bauweise (offen, geschlossen) für Mehrfamilienhäuser (MFH) Fensterflächenanteile von 17% und 24%, für Geschosswohnbau 21% und 18% an und für reine Dienstleistungsgebäude pauschal 32% an.

Tabelle 4 widerspiegelt. Eine Unterscheidung des FFA je Orientierung der Fassade wird aufgrund des urbanen Umfelds nicht angenommen.

Gebäudezustand und Sanierung

Im Modell werden zusätzlich zu den unsanierten Ausganszuständen der Gebäude entsprechend ihrer Bauperiode **zwei Sanierungsstandards** definiert, die unterschiedliche Effizienzstufen für die energetische Erneuerung von Gebäuden repräsentieren:

- Der **Rechtlicher Mindeststandard *(„Min/Bauordnung“)*** entspricht den gesetzlichen Anforderungen der Wiener Bauordnung (Einzelmaßnahmen an der Gebäudehülle Punkt b) und stellt eine grundlegende Verbesserung der Gebäudehülle sicher. Die Sanierung umfasst die Dämmung aller Hüllflächen und eine leicht reduzierte Vorlauftemperatur, ohne jedoch auf besonders ambitionierte Effizienzmaßnahmen einzugehen.
- Der zweite Standard ***(„Hocheffizient“)*** ist angelehnt an enerPHit[12], stellt einen **hocheffizienten Sanierungsstandard** für Bestandsgebäude dar und zielt auf einen stark reduzierten Energiebedarf ab. Er zeichnet sich durch besonders niedrige U-Werte der Gebäudehülle, eine hohe Luftdichtheit und eine mechanische Lüftung mit Wärmerückgewinnung aus. In der Praxis wurde dieser Standard bereits erfolgreich in dicht bebauten Stadtgebieten angewandt, wobei Gebäude mit ursprünglichem Altbaustandard nach der Sanierung Heizwärmebedarfe unter 15 kWh/m^2 erreichen konnten. Im Modell bedeutet eine Sanierung auf *Hocheffizient*-Niveau eine zusätzliche Verbesserung gegenüber dem Mindeststandard, insbesondere durch:
    - Erhöhte Luftdichtheit, wodurch Wärmeverluste minimiert werden.
    - Mechanische Lüftung mit Wärmerückgewinnung.
    - Optimierung der Heizwasserverteilung.
    - Kühlung über die Zuluft.

Die zwei Sanierungsstufen ermöglichen eine differenzierte Bewertung der energetischen und wirtschaftlichen Auswirkungen verschiedener Sanierungsszenarien. Während der rechtliche Mindeststandard eine breite Anwendbarkeit mit soliden Effizienzgewinnen bietet, ermöglicht der hocheffiziente enerPHit-Standard eine maximale Reduktion des Heizwärmebedarfs und eine deutlich verbesserte Gebäudeperformance. Die Festlegung des Gebäudezustands wurde auf Basis der Daten im Wärmeatlas folgendermaßen vorgenommen:

Tabelle 5 Zuordnung Gebäudezustand durch Sanierungsstatus laut Wärmeatlas

Sanierungsstatus laut Wärmeatlas	Anzahl Gebäude	Annahme Gebäudezustand Quartiersmodell
„Unbekannt“	2	Originalzustand unsaniert gemäß Bauperiode
„Saniert (Sanierungs-Energieausweis vorhanden)“	16	„Min/Bauordnung“

[12] Siehe https://passiv.de/de/03_zertifizierung/02_zertifizierung_gebaeude/04_enerphit/04_enerphit.htm

„Schutzstatus vorhanden (nicht saniert)“	2	Originalzustand unsaniert gemäß Bauperiode
„kein höherer Schutzstatus (Sanierung unbekannt)“	50	Originalzustand unsaniert gemäß Bauperiode

Abbildung 3-8 zeigt das Ergebnis der Begehung, die bei einigen Gebäuden Unsicherheiten bezüglich des im Wärmeatlas angegebenen Status als Unsaniert zeigten .



Abbildung 3-8: Gebäudezustand (Umrandung) und aufgetretene Datenunsicherheiten (rote Punkte)

Energieversorgung

Zur Energieversorgung urbaner Gebäudetypologien wird ein vereinfachtes Modell mit monovalenten Versorgungskonzepten angenommen. Eine Verwendung verschiedener Energieträger innerhalb eines Gebäudes (z. B. Heizen mit Erdgas und Warmwasser über Fernwärme) wird vereinfacht, um eine zu kleinteilige Maßnahmenbetrachtung zu vermeiden und den Datenerhebungsaufwand zu reduzieren. Die untersuchten Versorgungsoptionen sind Erdgas, Fernwärme und Wärmepumpe, wobei keine Unterscheidung zwischen zentraler und dezentraler Versorgung sowie zwischen verschiedenen Anwendungen (Heizen, Warmwasser) erfolgt – jedes Gebäude wird über eine einheitliche Energiequelle wie in Abbildung 3-9 dargestellt versorgt.

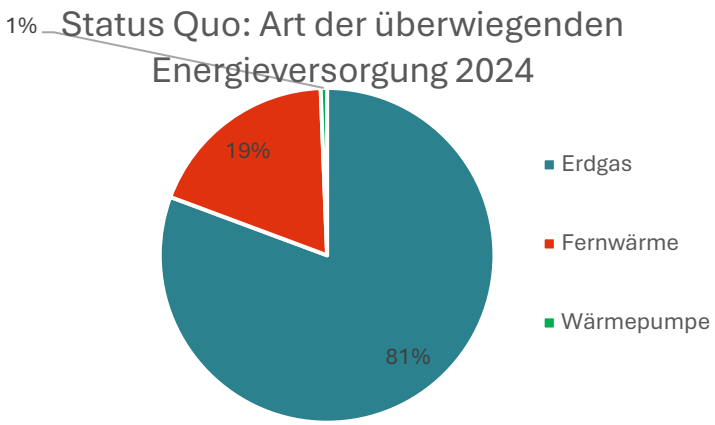


Abbildung 3-9: Status-Quo: Art der überwiegenden Energieversorgung je Gebäude (Anteile BGF)

Sofern eine aktive Kühlung erforderlich ist, wird für Gebäude mit Erdgas- oder Fernwärmeversorgung eine Kompressionskälteanlage (KKM) als Standardlösung angenommen. Gebäude mit Wärmepumpenversorgung nutzen keine separate Kälteanlage, da die Wärmepumpe potentiell auch zur Kühlung eingesetzt werden kann.

Tabelle 6 Annahmen des Energieträgers für Heizen/Kühlen/Warmwasser und Anlagenwirkungsgrade (EER) je System

Energiesystem	Heizen (EER)	Warmwasser (EER)	Kühlen (EER)
Erdgas	Erdgas (0,85)	Erdgas (0,85)	Wärmepumpe (2,5)
Fernwärme	Fernwärme (0,95)	Fernwärme (0,95)	Wärmepumpe (2,5)
Wärmepumpe	Wärmepumpe (4,5)	Wärmepumpe (4)	Wärmepumpe (4,5)

Gebäudetechnik und thermische Hüllqualität

Die Gebäudetechnik und Qualität der thermischen Hülle wird aus der Bauperiode und dem Gebäudezustand/Sanierungsstand abgeleitet:

Thermische Hülle:

- Die thermische Hülle der Gebäude ist in vier Bauteiltypen unterteilt: opake Außenwand, Dach, Boden gegen Erdreich bzw. Kellerdecke und Fenster.
- Für jeden Bauteiltyp werden Wärmedurchgangskoeffizienten (U-Werte) inklusive pauschale Berücksichtigung von Wärmebrücken in Höhe von 10 % (Tabelle 7) festgelegt.
- Eine thermische Differenzierung zwischen Außenwänden Richtung Innenhof und Straßenseite wird nicht berücksichtigt.
- Der Einfluss des Erdreichs bzw. eines unkonditionierten Kellers auf den Wärmetransfer wird durch einen angepassten Temperaturfaktor im Modell berücksichtigt.

Tabelle 7: U-Werte in W/m^{2}K je Bauteil und Bauperiode

	Bauperiode					
Bauteil	bis 1918	1919 - 1944	1945 - 1960	1961 - 1990	1991 - 2000	2001 - 2010
Fenster	2.42	2.53	2.53	2.86	1.98	1.54
Dach/ oberste Geschossdecke	1.21	0.88	0.88	0.605	0.33	0.22
Außenwand	1.98	1.98	1.76	1.155	0.44	0.385
Kellerdecke/ Fundament	0.22	1.32	1.32	0.715	0.55	0.44

Verschattung: Verschattungsvorrichtungen können in städtischen Gebäudetypologien nicht per se einer bestimmten Gebäudeart oder -periode zugeordnet werden, da sie häufig nachträglich und häufig nur wohnungsweise installiert oder nachgerüstet werden. Dies führt zu einer hohen Variabilität innerhalb einzelner Gebäudetypen. Um dieser Heterogenität Rechnung zu tragen, erfolgt die Modellierung von Verschattungsvorrichtungen nur für umfassende Sanierungen (und Neubauten). So kann der Einfluss moderner Verschattungssysteme auf den Energiebedarf und die thermische Behaglichkeit in standardisierten Szenarien berücksichtigt werden.

Lüftung: Die Art der Be- und Entlüftung ist ein wesentlicher Faktor in der Gebäudemodellierung, da sie sowohl den Energiebedarf als auch die Raumluftqualität maßgeblich beeinflusst. Im vorliegenden Fall kann die Art der Be- und Entlüftung nicht für alle Gebäude eindeutig erfasst werden. Stattdessen werden folgende Annahmen getroffen:

- Gebäude mit einer „hocheffizienten Sanierung“ verfügen über eine mechanische Be- und Entlüftung mit Wärmerückgewinnung
- In Bestandsgebäuden ohne Sanierung wird dagegen von einer manuellen Fensterlüftung ausgegangen

Diese Unterscheidung ermöglicht es, die energetischen und lüftungstechnischen Potenziale im Rahmen der Szenarien konsistent abzubilden und gleichzeitig die Heterogenität in der tatsächlichen Umsetzung zu berücksichtigen.

Kühlung: Die Integration (aktiver) Kühlungssysteme in Gebäuden ist, insbesondere in Zeiten zunehmender sommerlicher Überhitzung ein zentraler Aspekt der Gebäudemodellierung. Aktive Kühlsysteme, wie Split-Klimageräte und zentrale Kältesysteme, ermöglichen eine effektive Raumtemperierung, erhöhen jedoch den Strombedarf, abhängig von Systemeffizienz und Gebäudeparametern.

In der Modellierung wird davon ausgegangen, dass nur hocheffiziente Sanierungen mit aktiver Kühlung ausgestattet werden, um den thermischen Komfort auch bei höheren Außentemperaturen sicherzustellen. Für Bestandsgebäude ohne Sanierung wird keine aktive Kühlung angenommen.

3.3 Photovoltaik: Lokales Potential zur erneuerbaren Stromgewinnung

Die Ermittlung des PV-Potentials wurde mittels Stadt Wien Solardachkataster (SDK) durchgeführt: Dazu wurden alle Teilflächen mit Eignung von Sehr gut & Gut ausgewählt und einem von 5 stündlichen PV Profilen entsprechender Orientierung zugewiesen und gemäß der im Solardachkataster ausgewiesenen potentiellen Anlagenleistung skaliert.

Im Folgenden sind die Daten zur Verwendung aller Flächen, die im Solardachkataster der Stadt Wien als sehr gut oder gut geeignet eingestuft sind, tabellarisch dargestellt. Sie dienen zur Abschätzung der potentiellen Photovoltaikanlagenleistung und der benötigten.

Tabelle 8 Parameter der PV-Modellierung

Parameter	Wert	Einheit
Potentielle Anlagenleistung im Quartier (Flächen Sehr gut & gut)	1221	kWp
Modulleistung	460	Wp
Modulgröße	2.00	m^2
Modulleistung pro Quadratmeter	238.90	Wp/m^2PV
PV-Flächenbedarf pro kWp	4.34	m^2PV pro kWp
Gesamtdachfläche pro kWp	15.714	m^2Dach pro kWp
Belegungsdichte	28%	m^2 PV pro m^2 Dach

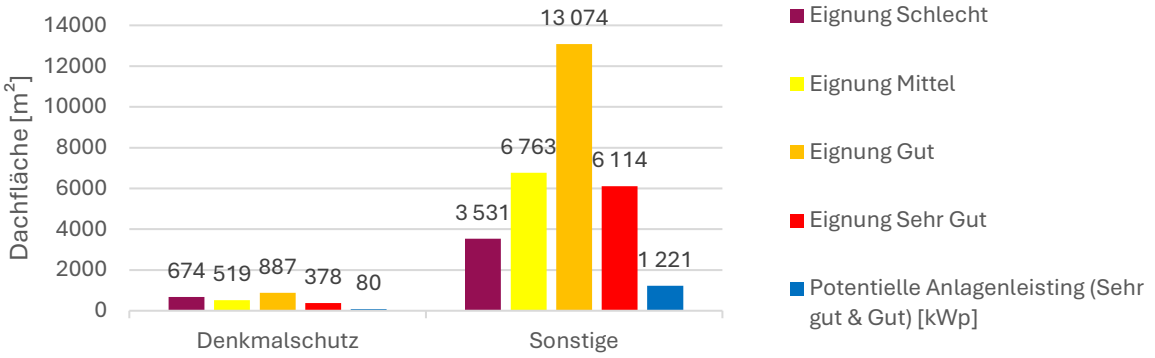


Abbildung 3-10 Anteile der Dachfläche im Quartier mit qualitativer Eignung zur PV-Belegung

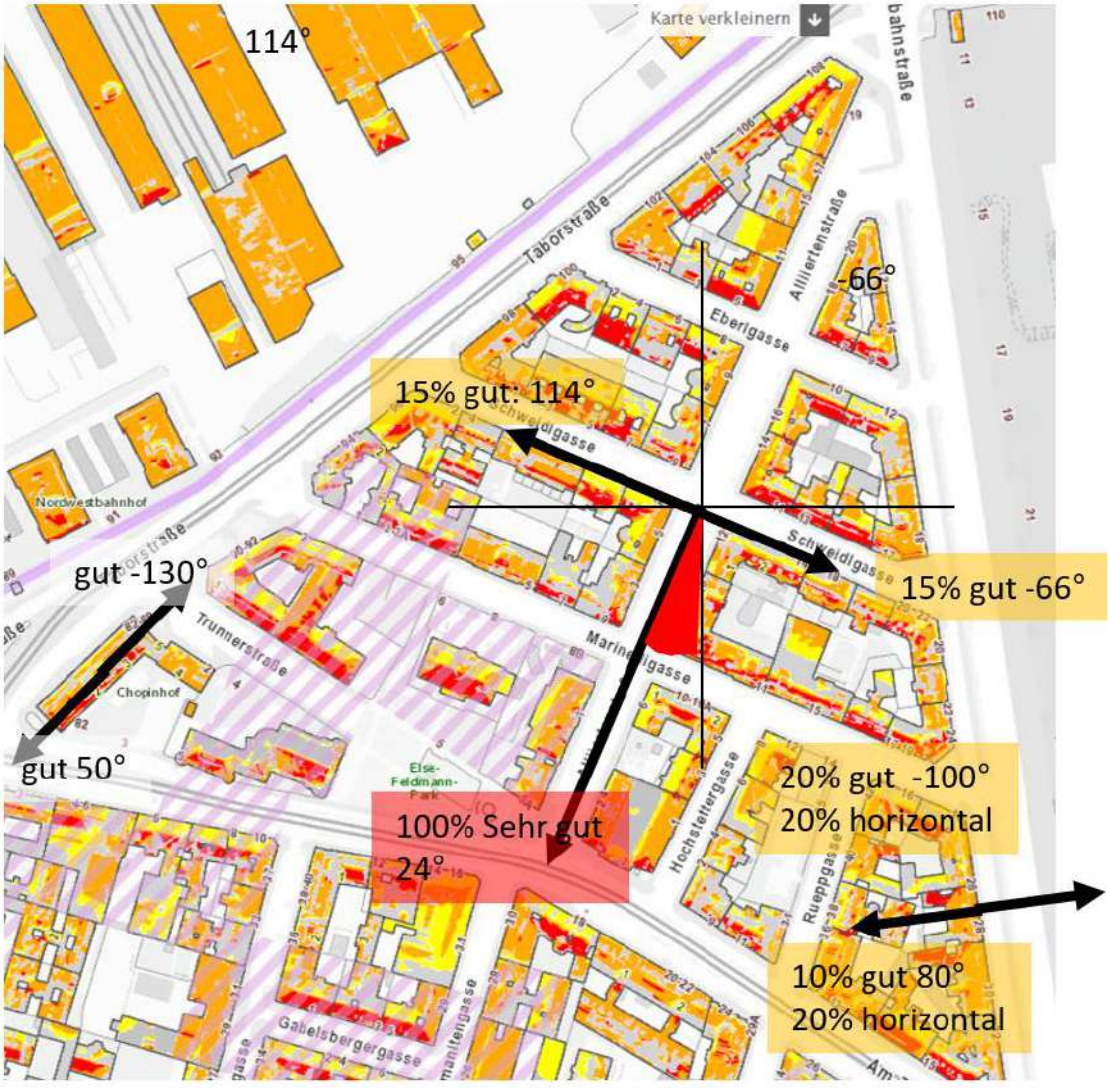


Abbildung 3-11 Falschfarbendarstellung der PV Eignung laut Solardachkataster sowie Ausrichtung und Anteil der fünf modellierten Orientierungen an den Eignungskategorien sehr gut & gut: Alle sehr gut geeigneten Flächen werden mit einer 24° S-SW Ausrichtung stündlich modelliert, die „gut“-geeigneten Flächen werden in sechs Bereiche unterschiedlicher Ost-West Orientierung unterteilt.

Aus den Abschätzungen der insgesamten Anlagenleistung wurden stündliche Ertragsprofile mittels folgender Schritte ermittelt:

1. Zuordnung der Flächen zu PV Systemen bestimmter Neigung. Wie im Solarpotentialkataster zu sehen, sind die vorherrschenden Ausrichtungen der Dachschrägen wenige Orientierungen entlang der Straßenzüge. Die PV-Neigung ist mit 27° angenommen, was der häufigsten Neigung der Dachflächen im Quartier entspricht.

Tabelle 9 Zuordnung von Solardachkataster (SDK) zu den Orientierungen

PV System	SDK „Sehr gut“	SDK „Gut“	PV Neigung	Azimuth*
Süd	100%		27°	+24°
Ost		15%	27°	-66°
West		15%	27°	+114°
Ost		20%	27°	-100°
West		10%	27°	+80°
Horizontal O		20%	15°	-100°
Horizontal W		20%	15°	+ 80°

* Winkel gegen Süden, im Uhrzeigersinn

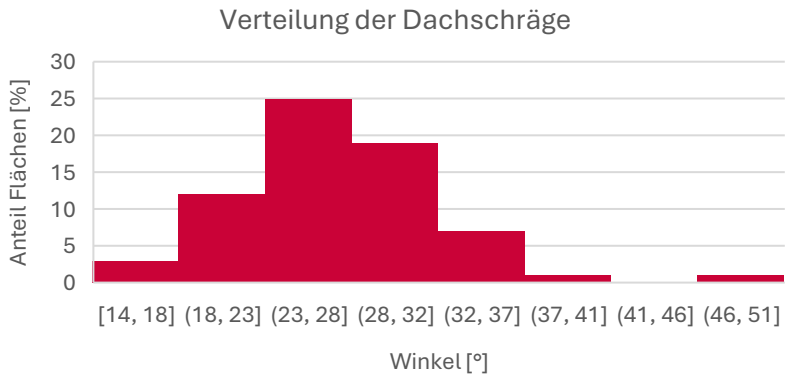


Abbildung 3-12 Statistische Verteilung der Dachflächenneigung im Quartier

2. Ermittlung der PV Erträge mittels BimSolar[13] mit folgenden Parametern:

Tabelle 10 Parameter der stündlichen PV Simulation mittels BIMsolar Software (PVSites, 2016)

PV System	Süd	Ost	West	Ost	West	Horizontal ost	Horizontal West
PV Modul	JinkoSolar TigerNeo						
Nennleistung	460 Wp						
Spez. Ertrag [kWh/kWp]	1299.3	1192.4	981.8	1056.1	1136.0	1101.9	1145.8
Installierte Leistung	**403.1**	**133.1**	**133.1**	**177.4**	**88.7**	**177.4**	**177.4**
Azimuth	24	-66	114	-100	80	-100	80
Neigung	27	27	27	27	27	15	15

3. Lineare Skalierung der resultierenden gebäudespezifischen Erträge auf Basis der Zuordnung (kWh/kWp) auf die Werte des Solardachkatasters

Damit ist eine 100%ige Übereinstimmung von Anlagenleistung (kWp) und spezifischem Ertrag der Anlagen (kWh/kWp) für das gesamte Quartier gewährleistet.

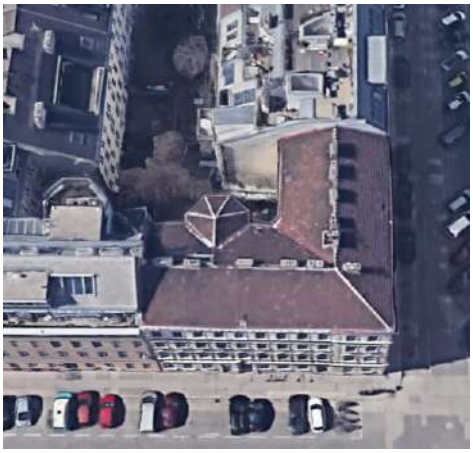
[13] https://www.bim-solar.com/

Änderung und Erweiterung: Änderungen der Nennleistung des Quartiers oder von Einzelgebäuden können bei Vorliegen entsprechender Daten ohne Weiteres vorgenommen werden. Es können auch zusätzliche PV Profile hinterlegt und einzelnen Gebäuden zugeordnet werden

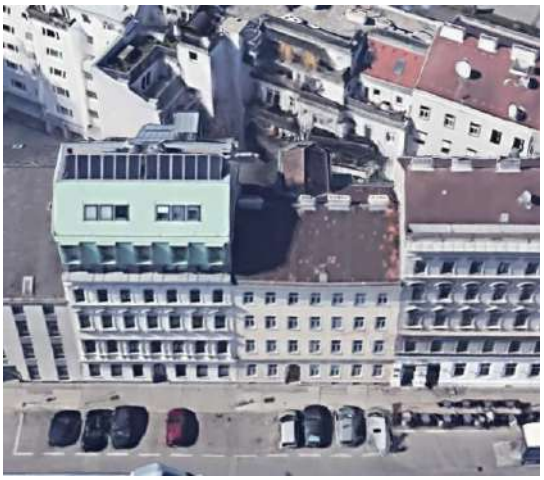
Verbesserungspotential: (Dynamische) PV Potentiale können auch Gebäudescharf zugeordnet werden, wenn Ausrichtungsinformation je „Gut“ und „Sehr gut“ Flächen vorliegt. zB. Einfach in Form einer zusätzlichen Spalte im zur Verfügung gestellten GIS Datensatz des PV Potentials

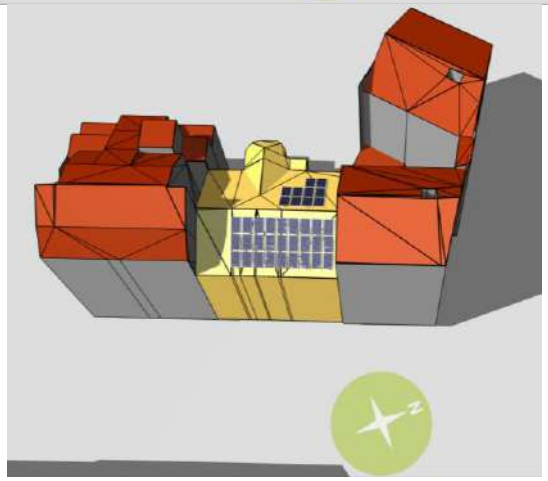
Das Potential laut Solarpotentialkataster wurde stichprobenartig auf Plausibilität getestet. Dazu wurden drei Gebäude mit der ausgewiesenen Anlagenleistung belegt. In Tabelle 11 sind die drei Gebäude dargestellt: Die modellierten PV-Anlagen über BIMSolar anhand des 3D-Modells der Stadt Wien stellen sich als durchaus plausibel heraus, lassen aber auch Erweiterungen zu; auch durch mögliche Nachverdichtung.

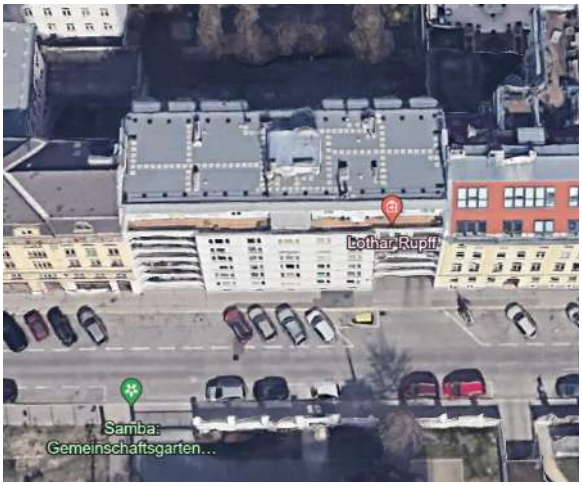
Tabelle 11 Satellitenbilder und LOD2.1 Stadtmodelle mit installierter PV-Kapazität laut Solarpotentialkataster

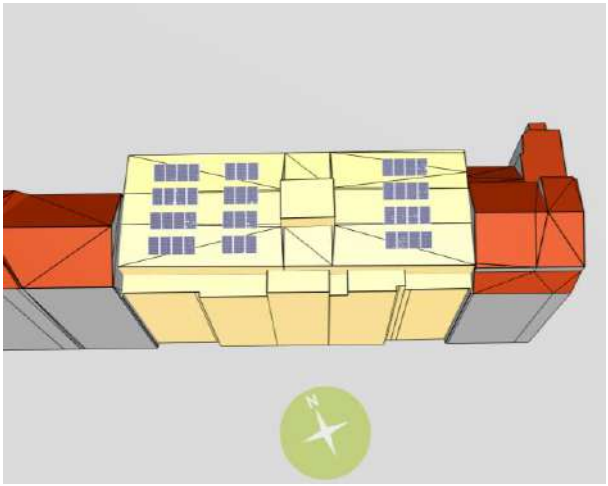












3.4 Energie- und THG-Bilanz

Das thermische Modell des City Energy Analyst (CEA) ermöglicht eine stündliche Simulation der Energieflüsse innerhalb eines Quartiers und liefert detaillierte Aussagen zum thermischen Verhalten der Gebäude. Es basiert auf einem thermischen 1-Zonen-Modell je Gebäude. Die Simulation verwendet bauphysikalische Eigenschaften der Gebäude wie die thermische Hüllqualität, Transmissions- und Lüftungsverluste, sowie Sollwerte für Raumtemperaturen und interne Lasten, die durch Personen, Geräte oder Beleuchtung entstehen. Klimatische Einflussgrößen wie Solarstrahlung und Außentemperatur werden über einen Wetterdatensatz für Wien 2020 in die Berechnungen integriert.

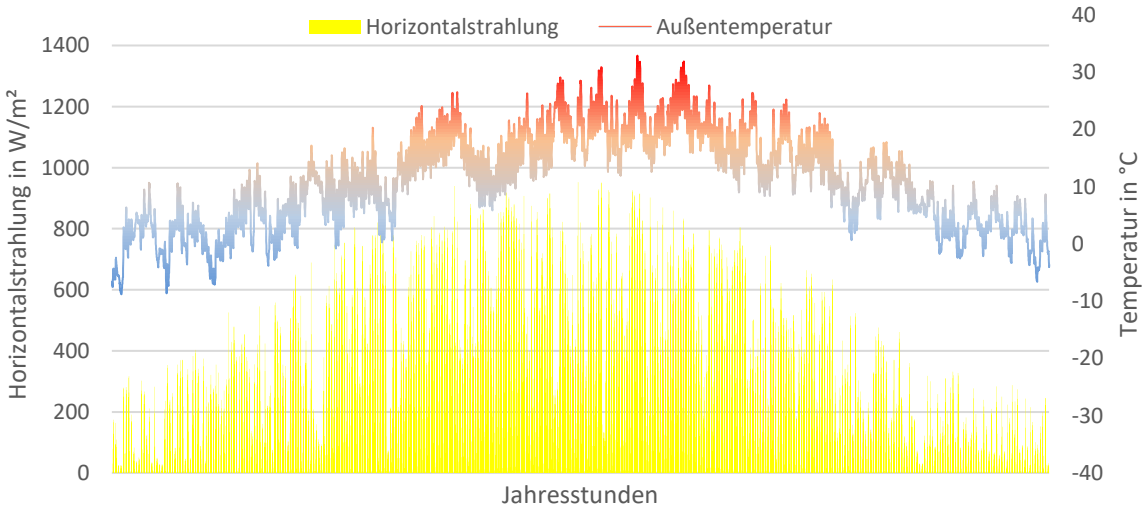


Abbildung 3-13: Stündliche Horizontalstrahlung (linke y-Achse) und Außentemperatur (rechte y-Achse) des verwendeten Wetterdatensatz „Wien - Hohe Warte 2020“

Das Modell berücksichtigt folgende Energieflüsse der Gebäude:

- **Transmissionswärmeverluste** durch den Wärmeaustausch über die thermische Hülle (Außenwände, Fenster, Dach, Kellerdecke) basierend auf den U-Werten und Temperaturdifferenzen
- **Ventilationswärmeverluste**, die sich aus kontrollierter und unkontrollierter Lüftung (mechanische Lüftung, Infiltration) ergeben

- **Solare Gewinne**, die aus der einfallenden Solarstrahlung, den g-Werten der Verglasung sowie Verschattungseffekten resultieren
- **Interne Gewinne** durch Abwärme von Personen, Geräten und Beleuchtung, modelliert anhand standardisierter Nutzungsprofile

Lüftung und Systemeffizienz: Das Modell integriert sowohl mechanische als auch natürliche Lüftungsstrategien, inklusive der Effekte von Wärmerückgewinnung und spezifischen Lüftungskonzepten wie Nachtlüftung. System- und Verteilverluste von luft- und wassergeführten Systemen werden ebenfalls abgebildet, einschließlich ihrer Abhängigkeit von Vor- und Rücklauftemperaturen.

Primärenergie und Treibhausgasemissionen

Die Umrechnung der Endenergiebedarfe in Primärenergie und Treibhausgasemissionen (CO_2eq) erfolgt mithilfe der Konversionsfaktoren der OIB-Richtlinie 6 (2023)[14].

- Der Primärenergiebedarf wird berechnet, indem der Endenergiebedarf mit dem Primärenergiefaktor des jeweiligen Energieträgers multipliziert wird. Dieser Faktor berücksichtigt den Energieaufwand für Gewinnung, Umwandlung und Verteilung
- Die Treibhausgas-Emissionen ergeben sich durch Multiplikation des Endenergiebedarfs mit dem entsprechenden Emissionsfaktor, der die direkten und indirekten Treibhausgasemissionen eines Energieträgers umfasst

Diese Faktoren ermöglichen eine vergleichbare und standardisierte Bewertung des Primärenergieeinsatzes und der CO_2-Emissionen von Gebäuden.

Tabelle 12 Konversionsfaktoren OIB Richtlinie 6 2023

Energieträger	$f_{PE, ges}$ [kWh/kWh]	f_{CO2eq} [g/kWh]
Fossile Brennstoffe gasförmig	1.1	201
Elektrische Energie (Liefermix)	1.76	156
Fernwärme (hocheffiziente KWK-Anlage)	0.59	67

3.5 Analyse und Zielwertabgleich

Vergleich mit Zielen der Stadt Wien Smart Klima City Strategie

Die **Smart Klima City Strategie** der Stadt Wien zielt darauf ab, bis 2040 Klimaneutralität zu erreichen und gleichzeitig eine hohe Lebensqualität für alle Bewohner*innen zu sichern. Ein besonderer Fokus liegt auf dem Gebäudesektor, der einen erheblichen Anteil des städtischen Energieverbrauchs und der CO_2-Emissionen verantwortet.

Bis 2030 soll der Endenergieverbrauch für Heizen, Kühlen und Warmwasser pro Kopf um **20 %** und bis 2040 um **30 %** im Vergleich zum Durchschnitt der Jahre 2005–2010 gesenkt werden. Gleichzeitig ist geplant, die damit verbundenen **CO_2-Emissionen pro Kopf bis 2030 um 55 %** zu reduzieren und bis 2040 auf **netto null** zu bringen. Das explizite Ziel „Raus aus Gas“ bis 2040 wird dabei qualitativ ergänzt. Die Nutzung von Gebäuden soll zudem auf eine maximale solare Energiegewinnung ausgerichtet werden. Ziel ist es, durch diese Maßnahmen den Energiebedarf zu minimieren und den verbleibenden Bedarf vollständig durch erneuerbare oder dekarbonisierte Energiequellen zu decken.

[14] https://www.oib.or.at/de/oib-richtlinien/richtlinien/2023/oib-richtlinie-6 (Zugriff 5.3.2025)

Seit 2005 konnten bereits bedeutende Fortschritte erzielt werden: Laut dem Wiener Energiebericht 2024 konnte der Endenergieverbrauch für Raumwärme und Warmwasser seit 2005-2010 bis 2022 bereits um 27 % reduziert werden. Die Treibhausgasemissionen (Heizen, Kühlen, Warmwasser) sanken im gleichen Zeitraum um 37 %, und die Erdgasnutzung in der Raumwärme wurde um 20 % verringert[15].

Dabei trugen verschiedene Teile der Wiener Bebauung unterschiedlich zu diesen Reduktionen bei: Ein guter Teil der Emissionseinsparung ist auf energieeffizientere und dekarbonisiertere Neubauten zurückzuführen. Aufgrund fehlender Daten und dem zusätzlichen Aufwand des „Rückrechnens“ ist der Bezugspunkt „Durchschnitt der Jahre 2005–2010“ für das Alliiertenviertel nicht zweckmäßig und wird hier durch das aktuelle Bezugsjahr 2024 ersetzt: Die hier ermittelten relativen Einsparungen beziehen sich auf den heute ermittelten Status-Quo, und nicht auf das unbekannte Niveau 2005-2010. Als Bestandsquartier ohne nennenswerte Neubautätigkeit und teils unbekannte Sanierungstätigkeit ist der Unterschied dabei erwartungsgemäß gering. Damit bietet die Methodik quantifizierte Dekarbonisierungspfade ab dem aktuellen Stand, kann aber die spezifischen historischen Fortschritte des Viertels seit 2005 nicht vollständig abbilden, da sie die (erwartbar kleinen) Einsparungen von 2005 bis heute nicht mitberücksichtigt.

Vergleich mit klimaaktiv Deklaration „Klimaneutrale Plusenergiequartiere“

Die Bewertung der Dekarbonisierungsszenarien wurde zusätzlich gemäß der österreichischen Deklaration für „Klimaneutrale Plus-Energie-Quartiere“ durchgeführt, das auf der Beurteilung der Primärenergie- und Treibhausgas (THG)-Bilanz von Quartieren basiert[16][17]. Dieses Label verfolgt einen **Top-Down-Ansatz**, um nationale sektorale Klimaziele – wie die Halbierung des Energieverbrauchs im Gebäudesektor und den Ausbau erneuerbarer Energien um den Faktor 5 bis 20 – auf einzelne Gebäude und Quartiere herunterzubrechen. Dabei werden Standort sowie Potenziale zur Energieeinsparung und lokalen erneuerbaren Energieproduktion berücksichtigt.

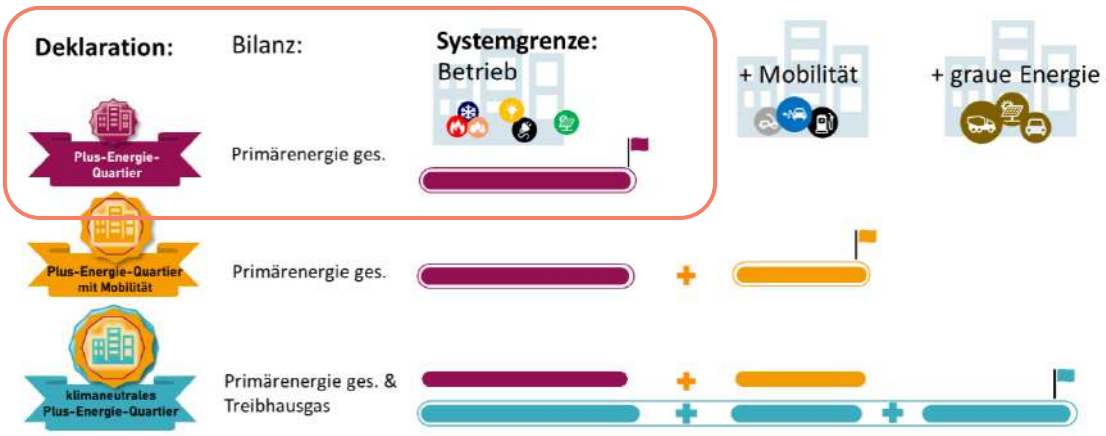


Abbildung 3-14 Für die Dekarbonisierungsstudie betrachtete Systemgrenze: klima**aktiv** „Plus-Energie-Quartier“ **Betrieb**

[15] Siehe https://www.wien.gv.at/spezial/energiebericht/files/Stadt_Wien_Energiebericht_2024-08-22.pdf
[16] https://klimaneutralestadt.at/resources/pdf/Broschuere_PEQ_klimaaktiv.pdf (Zugriff 5.3.2025)
[17] Siehe S. Schneider, T. Zelger, D. Sengl, und J. Baptista, „A Quantitative Positive Energy District Definition with Contextual Targets“, Buildings, Bd. 13, Nr. 5, Art. Nr. 5, Mai 2023, doi: 10.3390/buildings13051210. Und P. Schöfmann u. a., „Zukunftsquartier 2.0 Replizierbare, thermisch und elektrisch netzdienliche Konzeption von (Plus-Energie-) Quartieren im dichten urbanen Kontext“, Bundesministerium für Klimaschutz, Umwelt, Energie, Mobilität, Innovation und Technologie, Berichte aus Energie- und Umweltforschung 33/2023, Mai 2022.

Das Ziel der Deklaration ist die **Definition hinreichender Zielwerte für die Primärenergie- und THG-Bilanzen** von Gebäuden und Quartieren für die **österreichische Klimaneutralität 2040**. Die Zielwerte sind nehmen die Potentiale von Bestand und Neubau in Betracht und sind für urbane ebenso wie für ländliche Bebauungsformen erreichbar. In diesem Zusammenhang wird die Anforderung einer positiven Primärenergiebilanz – wie im vorliegenden Fall des **Alliiertenviertels** – durch sogenannte **Kontextfaktoren für urbane Dichte und Sanierung** angepasst, um realistische Zielsetzungen zu ermöglichen.

Zu erreichen ist eine positive Primärenergiebilanz, wobei Energiezuflüsse in das Quartier – wie Netzstrom oder Fernwärme – negativ gewertet und Energieabflüsse aus dem Quartier – wie ins Netz eingespeister Photovoltaikstrom – positiv gewertet werden. Es gibt keine spezifischen Schwellenwerte für Unterkategorien wie Heizwärmebedarf oder Photovoltaikflächen. Die Bewertungsmethode definiert die folgenden drei Hauptkomponenten:

1. **Systemgrenzen und bewertete Energiedienstleistungen**, abgestimmt auf die drei Kategorien: Betrieb, Mobilität und graue Energie.
2. **Gewichtung der Energieflüsse** in der Bilanz mittels monatlicher Umrechnungsfaktoren für Primärenergie und THG-Emissionen.
3. **Virtuelle Bilanzkomponenten**, die von Kontextfaktoren wie Bebauungsdichte, Standort, Art des Projekts (Neubau oder Sanierung) und verbleibendem THG-Budget abhängen.

Die Energieflüsse werden durch eine **stündliche Bilanzierung** von importierter und exportierter Energie erfasst. Flexible Energiemaßnahmen werden in der Bilanz je nach Netzfreundlichkeit und zeitlicher Nutzung unterschiedlich gewichtet.

Die Quartiersszenarien werden im Rahmen dieser Studie **nur für die erste Systemgrenze im Betrieb** ausgewertet, Mobilität und Graue Energie wird nicht betrachtet. Um die Erreichbarkeit des Standards für Neubaugebiete sowie Sanierungsquartiere mit niedriger und hoher Dichte sicherzustellen, wurden quartiersspezifische Potenziale und Unterschiede als „**Kontextfaktoren**“ quantifiziert, um eine faire Lastenteilung abzubilden. Durch die Integration dieser Faktoren als virtuelle Bilanzkomponenten kann der Standard sowohl von dicht bebauten Quartieren als auch von Sanierungsprojekten erfüllt werden, während gleichzeitig eine effiziente Nutzung der verfügbaren Flächen gefördert wird.

Die **Bebauungsdichte** ist dabei ein entscheidender Faktor für den Energiebedarf eines Quartiers: **Dicht bebaute Quartiere benötigen mehr Energie pro bebauter Fläche**, verfügen jedoch gleichzeitig über ein geringeres Potenzial zur Erzeugung erneuerbarer Energie vor Ort. Ohne eine entsprechende Anpassung wäre es deutlich einfacher, eine positive Energiebilanz in Quartieren mit vergleichsweise niedriger Bebauungsdichte zu erreichen. Auf Basis der nationalen Klimaziele wird daher der projektspezifische **Kontextfaktor der Dichte** berechnet, der als virtuelle Versorgungsgröße in die Bilanz einfließt.

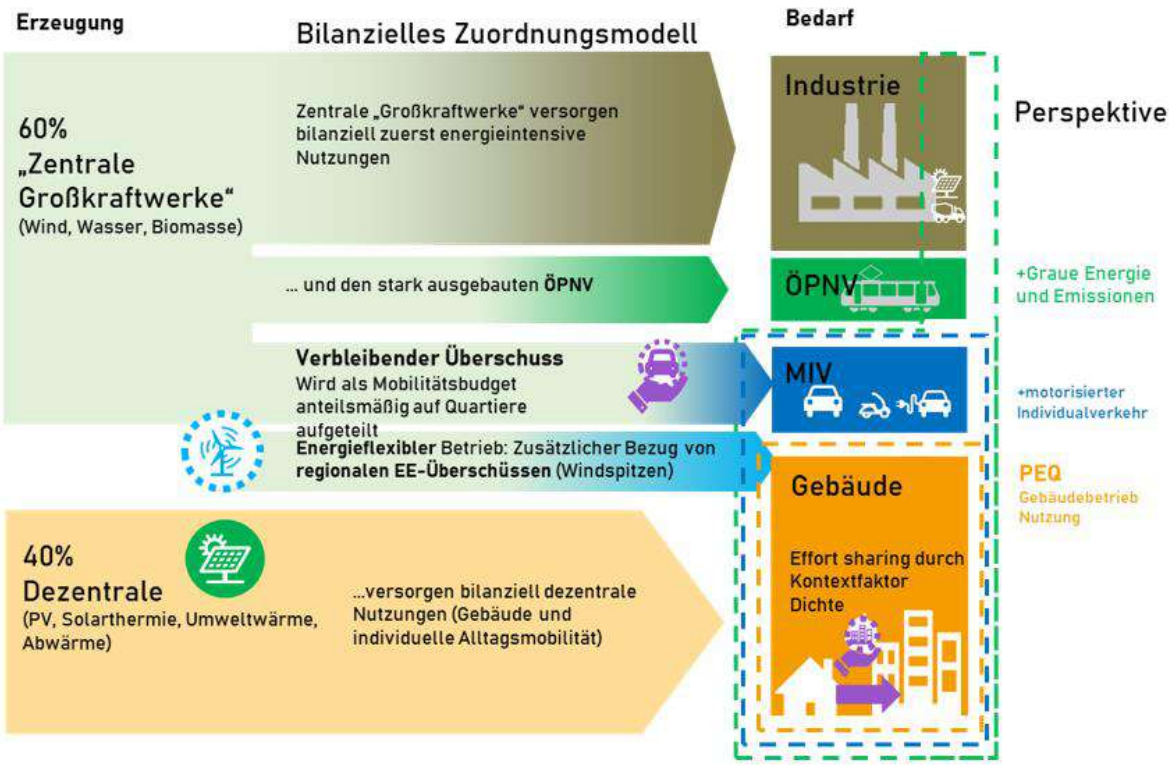


Abbildung 3-15 Nationales Bilanzmodell für ein klimaneutrales Österreich

4 Ergebnisse

Die Ergebnisse der vorliegenden Studie umfassen die allgemeine Analyse der Gebäudeeigenschaften, baulichen Dichte und Kompaktheit des Quartiers sowie die Energie- und THG-Bilanzen der Sanierungsszenarien auf Ebene des Gesamtquartiers und der Einzelgebäude (zusammengefasst nach Kompaktheit).

4.1 Potenzielle Heizenergieeinsparungen für Einzelgebäude nach Kompaktheit

Die gebäudescharfe Betrachtung einzelner Adressen wird durch eine typologische Analyse der Gebäude ersetzt, da die verfügbare Datenlage keine ausreichende Genauigkeit für eine direkte Kommunikation mit Bewohner*innen oder Eigentümer*innen gewährleistet. Die Unsicherheiten in den energetischen und baulichen Parametern auf Gebäudeebene sind zu groß, um verlässliche Aussagen für jedes einzelne Objekt treffen zu können und damit auf Bewohner*innen und Eigentümer*innen zuzugehen.

Stattdessen erfolgt eine typologische Untersuchung, die die Einsparpotentiale anhand der Gebäudecharakteristika wie Kompaktheit, Baualter und Zustand systematisch bewertet. Dabei werden alle Baukörper des Quartiers einmal mit jedem der 24 Typologien (Gebäudezustand und Bauperiode, ohne Variation der Energieversorgung) simuliert.

Abbildung 4-1 zeigt den resultierenden spezifischen Wärmebedarf aller simulierter Baukörper auf der y-Achse, aufgetragen gegen die charakteristische Länge (l_c) der x-Achse, modelliert für sechs anfängliche Bauzeiträume in separaten Teilplots, gekennzeichnet durch ihre Hintergrundfarbe, und für alle betrachteten Arten von Sanierungsmaßnahmen, gekennzeichnet durch die Markerfarbe: ursprünglicher Gebäudebestand (schwarz), mit Fenstertausch (blau), Mindestsanierung nach österreichischer Bauordnung (gelb) und hocheffizient (Passivhaussanierung) (grün). Wie erwartet korrelieren die Ergebnisse stark linear mit dem SVR: Der Effekt von Sanierungsmaßnahmen nimmt bei neueren Gebäuden ab, da diese bereits einen höheren Standard aufweisen.

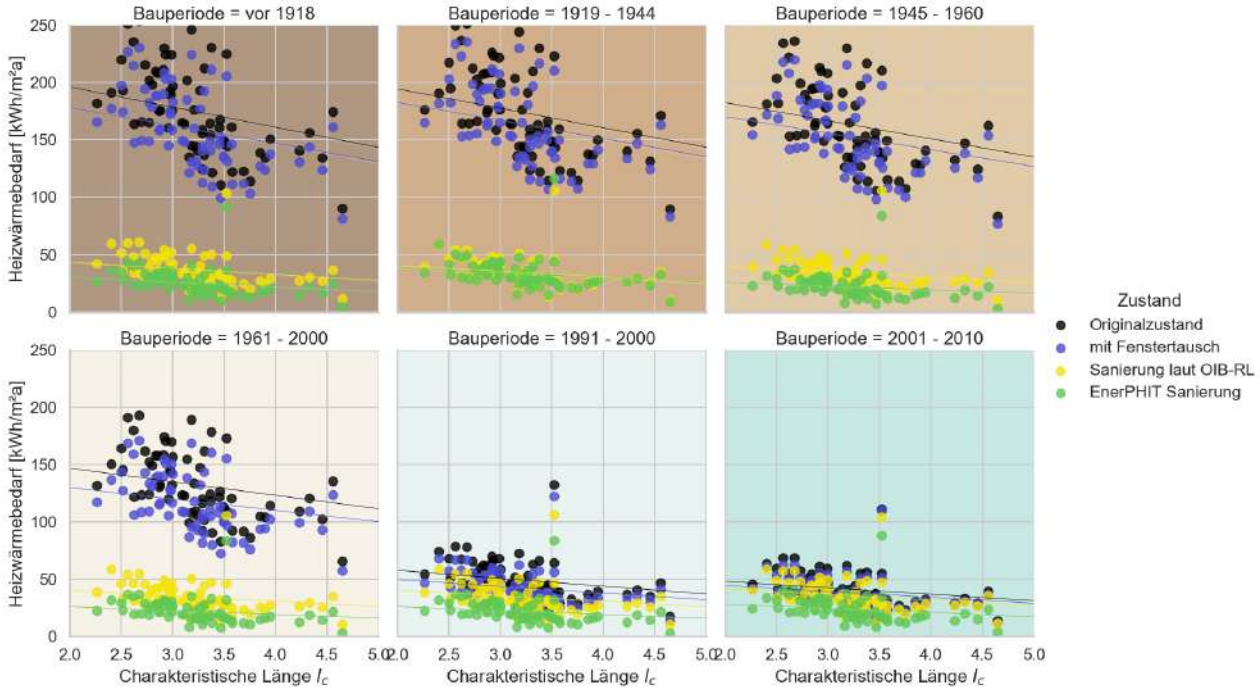


Abbildung 4-1 Heizwärmebedarf nach Bauperiode, Gebäudezustand und Oberflächen-Volumen-Verhältnis

Das Oberflächen-Volumen-Verhältnis kann ebenfalls in Kategorien und damit greifbare Gebäude des Quartiers überführt werden. In Abbildung 1-7sind die Baukörper farblich nach deren Kompaktheit eingeteilt.

Abbildung 4-2 zeigt die Bandbreiten des Heizwärmebedarfs nach Bauperiode, Gebäudezustand und Kompaktheit gebündelt. Davon ausgehend können übliche Heizwärmebedarfe und erwartete Einsparungen durch unterschiedliche Sanierungen aufgezeigt werden. Je älter und je geringer die Kompaktheit, desto höher das Potential der Einsparung. Die Auswirkung kompakter Bauweise ist hier hervorzuheben: Zwischen einer sehr hohen Kompaktheit und einer niedrigen Kompaktheit mit ansonsten den gleichen Rahmenbedingungen (z.B. Bauperiode vor 1918-Bestand) liegen knapp 100 kWh/m^{2}a.

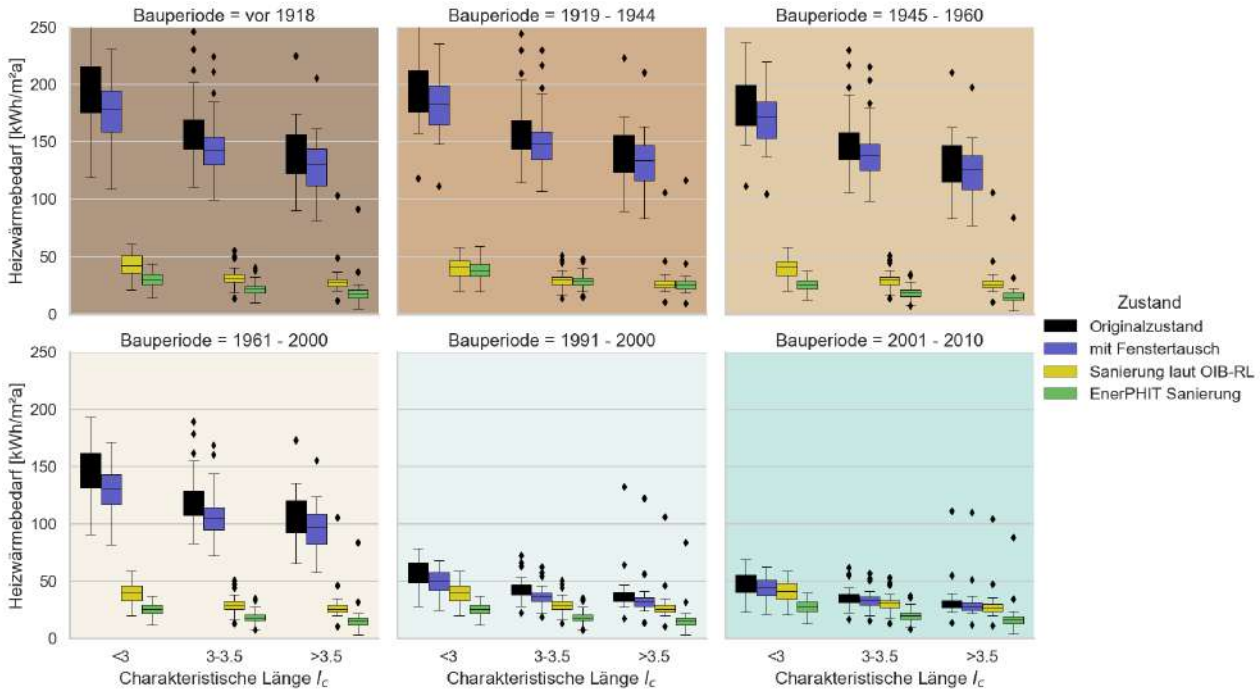


Abbildung 4-2 Bandbreite des spezifischen Heizwärmebedarfs je Bauperiode, Gebäudezustand und charakteristischer Länge des Gebäudetyps

Die HWB-Einsparungen durch Fenstertausch erscheinen hier auf den ersten Blick gering, sind aber mit ca. 10% vergleichbar mit Referenzwerten aus anderen Quellen (10% Einsparung bei 3-fach Verglasung bei Zweifamilienhäusern als Beispielgebäude im Handbuch Wegweiser für klima-positiven Gebäudebestand (DGNB, 2022)).

4.2 Szenarien

Die gewählten Dekarbonisierungsszenarien bilden die Bandbreite der Entwicklungsmöglichkeiten des Quartiers ab: Von technisch maximalen Einsparpotentialen bis hin zur Erreichung „realistischer“, stadtweiter Zielbilder. Den Szenarien liegen dabei übergeordnete Zielvorgaben wie Sanierungs- und Kesseltauschraten, Gebäudepriorisierungen oder die Qualität der Sanierung zu Grunde. Folgende Szenarien wurden untersucht:

Tabelle 13 Übersicht der untersuchten Szenarien

Szenario	Beschreibung
Ausgangslage	Quartiersmodell der Situation 2024
TMS (technisches Maximal-Szenario)	- Alle erdgasversorgten Gebäude steigen auf Fernwärme um - Alle nicht-sanierten, vor 2000 erbauten Gebäude werden auf folgenden typologischen Stand saniert:
„Bauordnung“	Alle Sanierungen auf den Standard „Min/Bauordnung“
„Hocheffizient“	Alle Sanierungen auf den Standard „Hocheffizient“
„Realistische“ Szenarien	
„Worst First“	Priorisierung der Gebäude nach erwartetem Einsparpotential. Nur teilweise Sanierung des Quartiers mit Sanierungsraten von 1%, 2%, 3% und 4%p.a. unter Verwendung des hocheffizienten Standards führen bis 2040 zu unterschiedlichen Jahresbilanzen.
„Schlüsselliegenschaften“	15 „Schlüsselliegenschaften“ werden auf den hocheffizienten Standard saniert
„Wiener“ Szenario	Eine thermische Sanierungsrate von 2%p.a. und eine „Kesseltauschrate“ (von Gas zu Fernwärme) von 5%p.a. bilden das „Wiener Szenario“ bis 2040

Die mittleren Nutzenergiebedarfe je Szenario sind in Abbildung 4-3 dargestellt. Im **technischen Maximalszenario (TMS) Hocheffizient** wird der geringste **mittlere Heizwärmebedarf** im Quartier erreicht. Gleichzeitig führt die mit dem „hocheffizienten Sanierungsstandard“ einhergehende aktive Kühlung zu einem **Kühlbedarf**, der bei etwa **5 kWh/m²a** liegt. Dies zeigt u. A. die Wechselwirkung zwischen Heiz- und Kühlenergiebedarf in hochgedämmten Gebäuden, insbesondere in urbanen Kontexten mit sommerlicher Überwärmung.

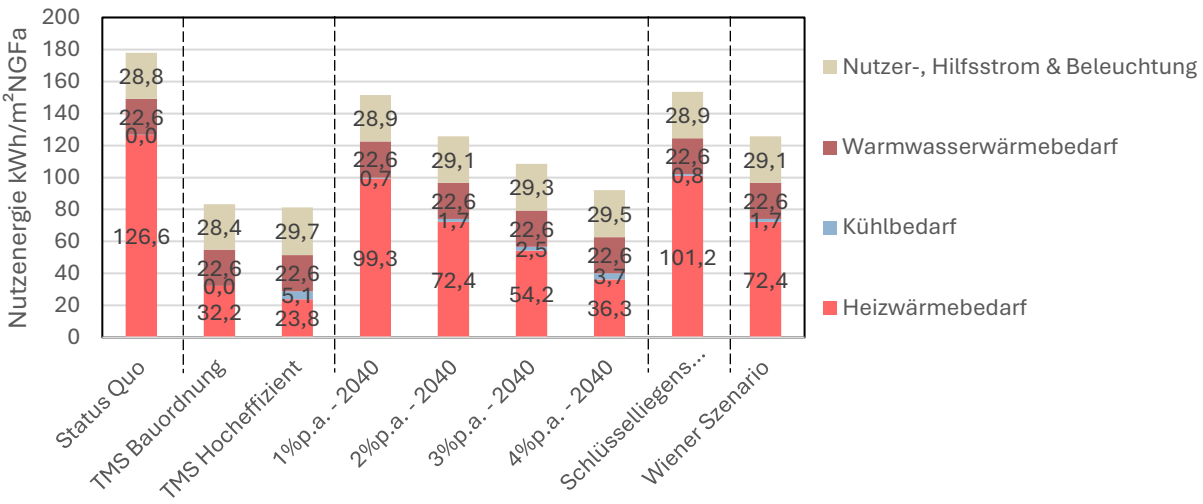


Abbildung 4-3: mittlere Nutzenergiebedarfe je Szenario. sortiert in Blöcken v.l.n.r Status Quo, Technische Maximalszenarien (TMS), Worst-First Szenarien, Schlüsselliegenschaften und Wiener Szenario. Worst-First-Szenarien durch gerundete Prozent der thermischen Sanierungsraten benannt.

Die Betrachtung der **Endenergie nach Energieträgern** (Abbildung 4-4) zeigt deutliche Unterschiede zwischen den Szenarien. In den Szenarien **TMS** und **Wiener Szenario** erfolgt eine vollständige Umstellung der Wärmeversorgung, sodass **100 % der Gebäude aus der Erdgasnutzung aussteigen**. Im Gegensatz dazu weist das **Szenario Schlüsselliegenschaften** die geringste Reduktion des Erdgasverbrauchs auf.

Die **gesamtquartiersbezogene Endenergieeinsparung** variiert je nach Szenario deutlich: Während im **Szenario Schlüsselliegenschaften** lediglich **16 %** eingespart werden, erreicht das **TMS Hocheffizient** eine Reduktion um **63 %**.

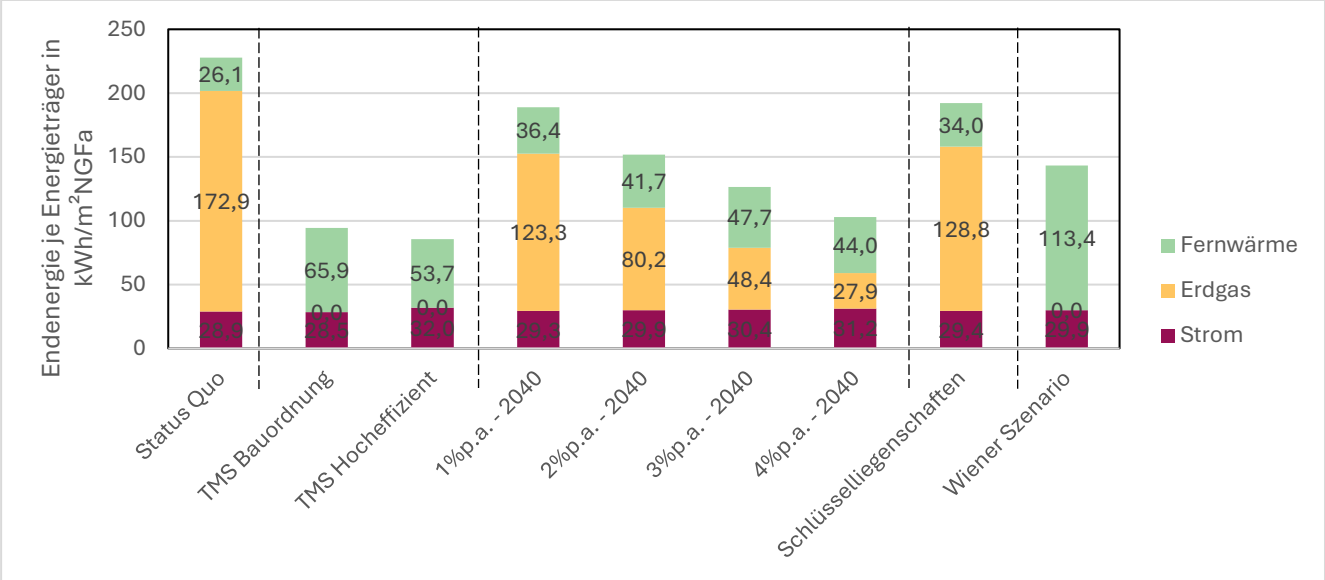


Abbildung 4-4 mittlere Endenergiebedarf je Energieträger und Szenario. sortiert in Blöcken v.l.n.r Status Quo, Technische Maximalszenarien, Worst-First Szenarien, Schlüsselliegenschaften und Wiener Szenario. Worst-First-Szenarien durch gerundete Prozent der thermischen Sanierungsraten benannt.

Der **Primärenergiebedarf** sinkt über die Szenarien hinweg um **17 % bis 66 %** und wird im ambitioniertesten Fall auf **87,9 kWh/m²a** reduziert. Die höchsten Einsparungen werden im **TMS hocheffizient** erzielt, während das **Szenario Schlüsselliegenschaften** aufgrund der begrenzten Sanierungsmaßnahmen die geringste Reduktion aufweist. Die **THG-Emissionen** zeigen ebenfalls eine große Bandbreite an Reduktionspotential, das von **20 % im Szenario Schlüsselliegenschaften** bis zu **79 % im TMS Hocheffizient** reicht. Im Folgenden sind die einzelnen Szenarien im Detail dargestellt.

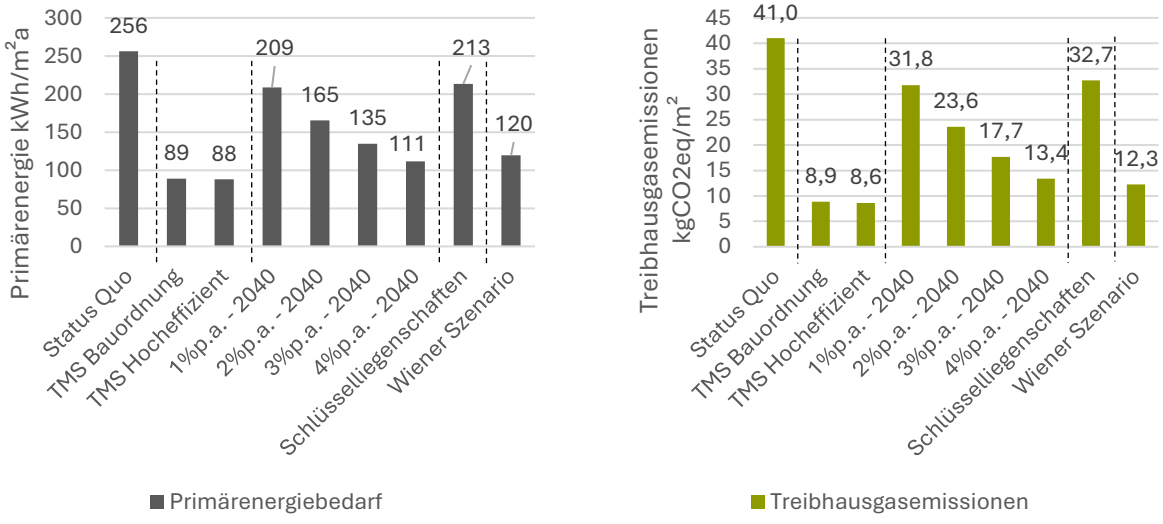


Abbildung 4-5 Übersicht der Szenarien in mittlerem Primärenergiebedarf und Treibhausgasemissionen

4.2.1 Ausgangslage, Status Quo

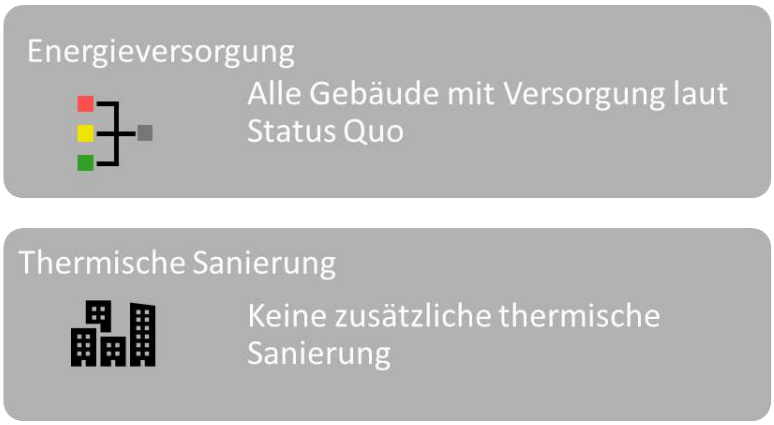
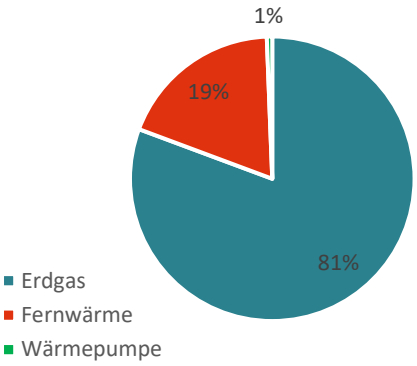
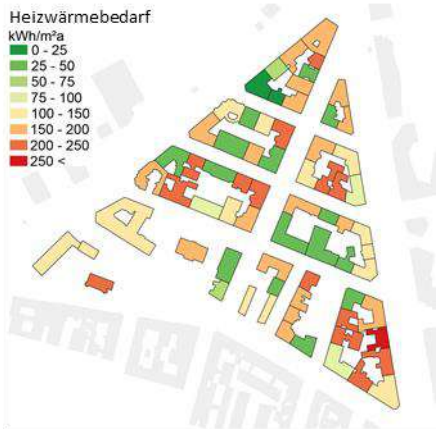


Abbildung 4-6 Übersicht Szenario Status Quo



Das Quartier weist derzeit eine gemischte Wärme-Versorgung durch Erdgas (60 Gebäude, 81% der Gebäude-Nutzflächen), Fernwärme (9 Gebäude, 19% der Gebäude-Nutzflächen), und Grundwassergespeiste Wärmepumpen (1 Gebäude, 1% der Gebäude-Nutzflächen) auf (siehe Grafik rechts oben). 23% der Gebäude weisen derzeit thermische Sanierungsmaßnahmen auf.

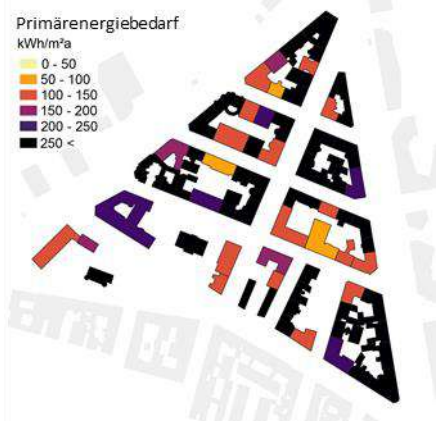


Der spezifische Heizwärmebedarf der Gebäude liegt im Bereich von 18 kWh/m²NGF bis 254 kWh/m² mit einem quartiersweiten Mittelwert von **127 kWh/m²**. Das entspricht einem jährlichen Heizenergiebedarf (inkl. WWWB) von 3,7 GWh Fernwärme und 24,8 GWh Gas. (siehe zweite Grafik rechts von oben).



Dazu kommt ein jährlicher Strombedarf für Beleuchtung, Geräte, Hilfsstrom und Belüftung von durchschnittlich 28,8 kWh/m² bzw. 4,1 GWh im Quartier.

Der Energiebedarf für den gesamten Gebäudebetrieb im Quartier führt 2024 zu einem jährlichen Treibhausgas-Ausstoß von 5800 Tonnen bzw. durchschnittlich 41 kgCO2eq/m²NGFa, wobei die Bandbreite der Einzelgebäude von ca. 6 bis 80 kgCO2eq/m²NGFa reicht (siehe zweite Grafik rechts von unten). Der durchschnittliche Primärenergieeinsatz im Quartier liegt bei 256 kWh/m²NGFa und reicht von etwa 70 bis 462 kWh/m²NGFa. Der Primärenergieeinsatz gesamt beläuft sich auf jährlichen 36,7 GWh.



4.2.2 Technische Maximalszenarien

Die **technischen Maximalszenarien** untersuchen das Dekarbonisierungspotential des Alliiertenviertel unter optimalen Bedingungen. Dabei wird angenommen, dass alle Gebäude, die vor dem Jahr 2000 errichtet wurden, auf einen hocheffizienten Standard saniert werden. Zusätzlich erfolgt die Umstellung der Wärmeversorgung: Gebäude, die derzeit mit Erdgas heizen und Warmwasser aufbereiten, werden an das Fernwärmenetz angeschlossen und künftig vollständig mit Fernwärme versorgt. Ziel ist es, die oberen Grenzen für Energieeinsparungen und Emissionsreduktionen aufzuzeigen. Die Ergebnisse des TMS hocheffizient sind in den vier Grafiken rechts kartografisch dargestellt. In Abbildung 4-7 sind neben den deskriptiven Bedingungen zum Szenario, die Sanierungs- und Kesseltauschraten bis 2040 dargestellt. Zusätzlich ist jeweils ausgeführt, welche Sanierungs- und Kesseltauschraten erforderlich wären, wenn das Szenario bis 2030 erreicht werden soll.

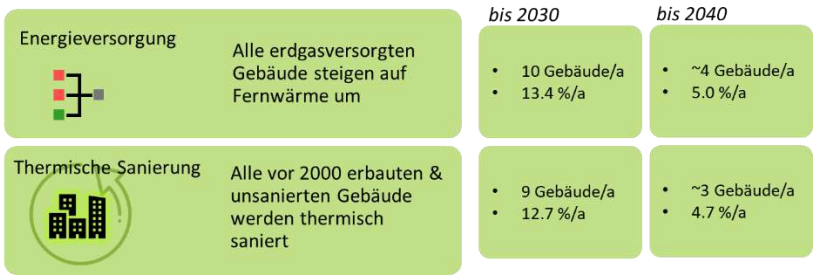


Abbildung 4-7 Überblick Szenario TMS

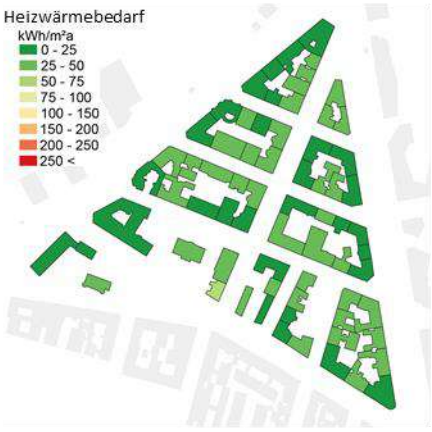
Das Quartier wird damit gänzlich ohne Erdgas, zu 99 % durch Fernwärme und zu 1 % über eine Grundwasser-Wärmepumpe versorgt. Nach Beendigung des Szenarios weisen zusätzliche 75 % der Gebäude thermische Sanierungsmaßnahmen auf.

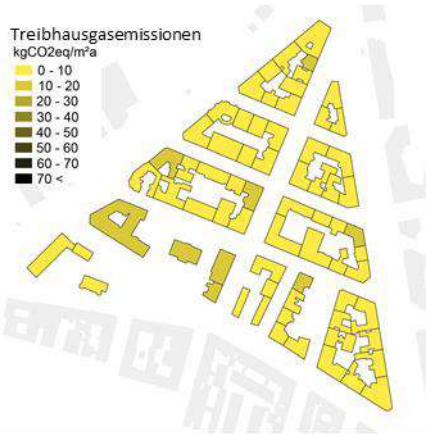
Der spezifische Heizwärmebedarf der Gebäude liegt im Bereich von 3 kWh/m^{2}NGF bis 53 kWh/m^2 mit einem quartiersweiten Mittelwert von **24 kWh/m^2**. Das entspricht einem jährlichen Heizenergiebedarf (inkl. WWWB) von 9,6 GWh Fernwärme. (Siehe zweite Grafik rechts von oben).

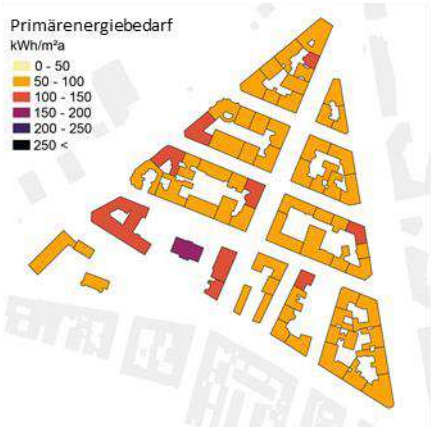
Dazu kommt neben dem jährlichen Strombedarf für Beleuchtung, Geräte, Hilfsstrom von durchschnittlich 29,7 kWh/m^2 bzw. 4,3 GWh, noch ein Anteil für die Kühlung (2,2 kWh/m^{2}NGFa und 322 MWh/a) im Quartier dazu.

Dies führt zu einem jährlichen Treibhausgas-Ausstoß von 1200 Tonnen bzw. durchschnittlich 8,6 kgCO2eq/m^{2}a und einem Primärenergieeinsatz von jährlichen 12,6 GWh bzw. durchschnittlichen 88 kWh/m^{2}NGFa.









4.2.3 „Worst First“ Szenarien

Das technische Maximalszenario (TM) wird durch vier Szenarien mit reduzierten thermischen Sanierungsraten ergänzt, um deren Einfluss auf die energetischen und ökologischen Potentiale des Quartiers abzubilden. Die Sanierung erfolgt dabei jeweils mit Raten von etwa 1%, 2%, 3% und 4% pro Jahr, wobei namensgebend für die Szenarien jeweils Gebäude mit höchstem spezifischem Heizwärmebedarf priorisiert werden. Im Zuge der thermischen Sanierung werden die Gebäude auch auf Fernwärme umgestellt.

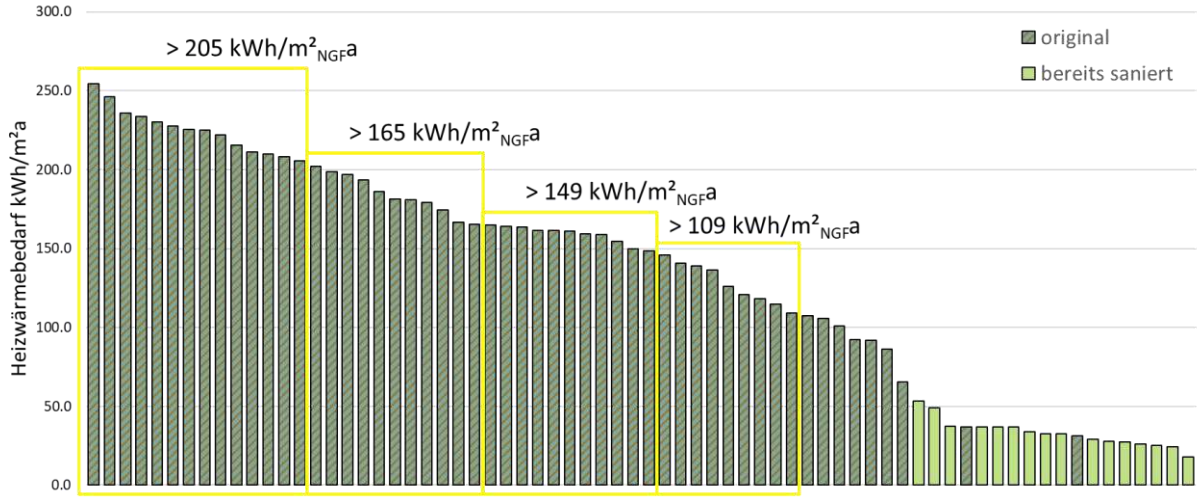


Abbildung 4-8 Szenario Worst First - bis zu welchem Heizwärmebedarf wird bei einer jährlichen Sanierungsrate von 1%, 2%, 3% und 4% bis 2040 saniert?

Bei einer jährlichen Sanierungsrate von 4 % wären bis 2040 alle Gebäude mit einem Heizwärmebedarf über 109 kWh/m²a saniert. Lediglich sieben Gebäude mit Bedarfswerten zwischen 105 und 65 kWh/m²a bleiben dabei unberührt. In einer Einzelfallbetrachtung würde eine Sanierung von Gebäuden mit bereits moderaten Bedarfswerten, etwa bei 65 kWh/m²a, wahrscheinlich nicht empfohlen.

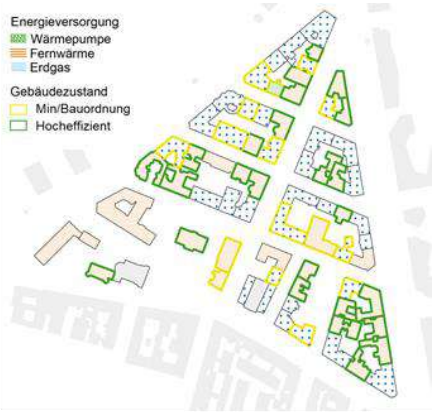
Tabelle 14 zeigt die exakten Änderungsraten und die Auswirkungen auf die Entwicklung des Quartiers über einen Zeitraum von 16 Jahren, bis 2040. Die Szenarien sind nach gerundeten thermischen Sanierungsraten benannt und unterscheiden sich in der Geschwindigkeit der Umsetzung.

Eine jährliche Sanierungsrate von **1 %** entspricht etwa **einem Gebäude pro Jahr**, während **2 %** rund **1,6 Gebäude pro Jahr**, **3 %** etwa **2,3 Gebäude** und **4 %** ca. **2,8 Gebäude** umfassen. Bei einer kontinuierlichen Sanierungsrate von **4 % pro Jahr** wären bis 2040 **83 % der Gebäude im Quartier** mit einer hocheffizienten thermischen Ausstattung versehen. Die Tabelle verdeutlicht damit den direkten Zusammenhang zwischen Sanierungsrate und Dekarbonisierungsgeschwindigkeit des Quartiers.

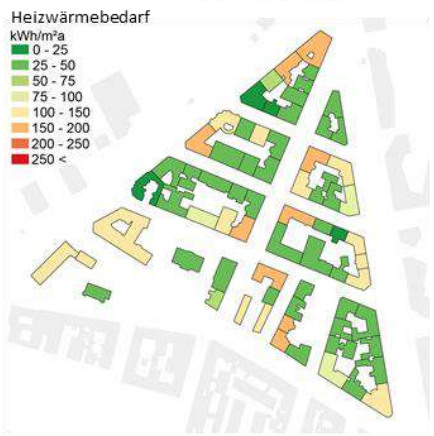
Tabelle 14 Übersicht Worst First Szenarien – was ändert sich im Quartier. Namensgebung der Szenarien nach gerundeten thermischen Sanierungsraten

Bis 2040	**1%**	**2%**	**3%**	**4%**
Jährliche Sanierungsrate von…	0.9%	1.8%	2.8%	3.7%
Anzahl Gebäude thermisch saniert…	14	25	36	45
Anteil am Quartier thermisch saniert…	14.5%	31.9%	45.2%	61.2%
Jährliche Kesseltauschrate von…	0.9%	1.8%	2.7%	3.3%
Anzahl Gebäude von Gas auf Fernwärme umgestellt…	14	25	35	42
Anteil am Quartier umgestellt…	14.5%	30.0%	43.0%	53.2%

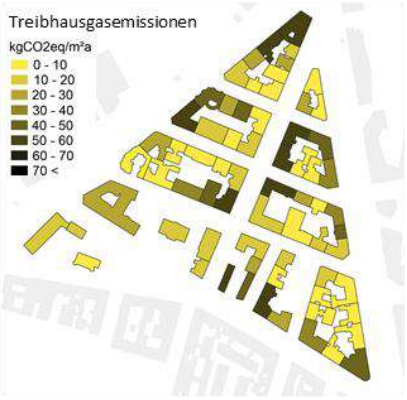
Das Szenario mit einer **jährlichen Sanierungsrate von 2 %** zeigt hier die energetische Verbesserung des Quartiers mit Fokus auf die Gebäude mit den höchsten Heizwärmebedarfen. Bis 2040 würden **32 % der ineffizientesten Gebäude saniert**, was eine Sanierung von **1 bis 2 Gebäuden pro Jahr** bedeutet. Dadurch wären am Ende des Betrachtungszeitraums **50 % der Gebäude nicht mehr mit Erdgas versorgt**.



Der **mittlere Heizwärmebedarf** reduziert sich um **43 %**, wobei das Einsparpotenzial in der Bandbreite zwischen **22 % (bei 1 % p. a.) und 71 % (bei 4 % p. a.)** liegt.



Durch die verbesserte Gebäudehülle und den Umstieg auf Fernwärme sinken die **Treibhausgas-Emissionen um 42 %** (Bandbreite 22 % bis 67 %), während die **Primärenergieeinsparung** je nach Sanierungsrate zwischen **19 % und 57 %** liegt.



In Abbildung 4-9 zeigen die Worst-First Szenarien die **Endenergie-bedarfe** je Energieträger. Für Heizung, Kühlung und Warmwasser explizit werden mögliche Einsparungen von **20 % bis 63 % aufgezeigt**.

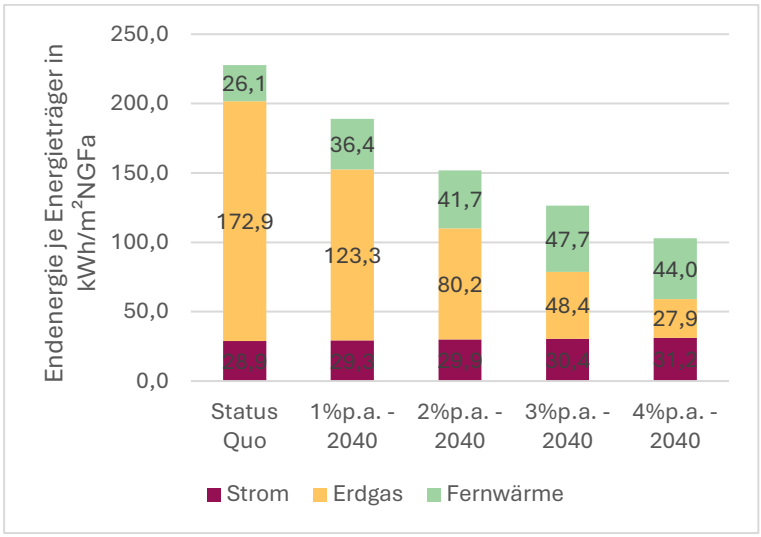
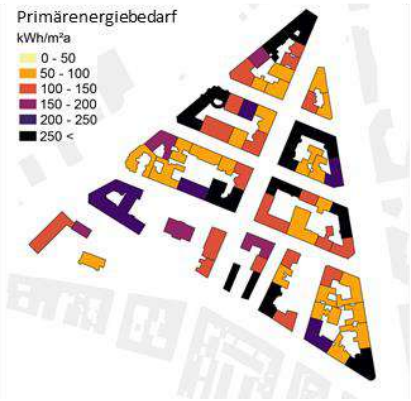


Abbildung 4-9 Endenergiebedarfe Worst-First Szenarien



4.2.4 „Wiener“ Szenario – „Raus aus Gas“

Das Wiener Szenario basiert auf dem Ziel der Wiener Stadtregierung, die Abhängigkeit von Erdgas zu beenden und Gebäude verstärkt an das Fernwärmenetz anzuschließen. Dabei wird angenommen, dass alle derzeit mit Erdgas beheizten Gebäude auf Fernwärme umgestellt werden. Parallel dazu erfolgt eine thermische Sanierung von etwa 2% der Gebäude pro Jahr. Priorität erhalten auch hier wieder Gebäude mit den höchsten energetischen Einsparpotenzialen. In Abbildung 4-10 sind neben den deskriptiven Bedingungen zum Szenario, die Sanierungs- und Kesseltauschraten bis 2040 dargestellt. Zusätzlich ist jeweils ausgeführt, welche Sanierungs- und Kesseltauschraten erforderlich wären, wenn das Szenario bis 2030 erreicht werden soll.

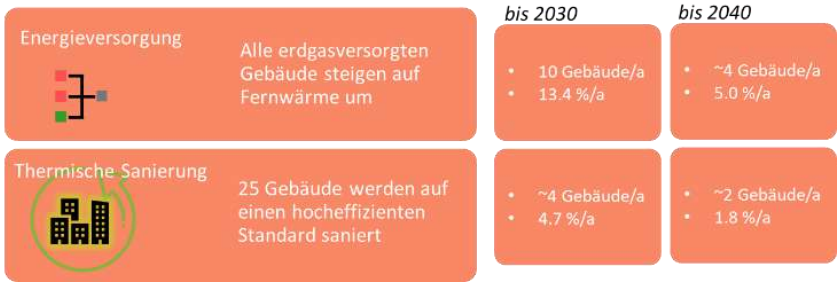


Abbildung 4-10 Überblick Wiener Szenario

Das Quartier wird damit gänzlich ohne Erdgas, zu 99% durch Fernwärme und zu 1 % über eine Grundwasser-Wärmepumpe versorgt.

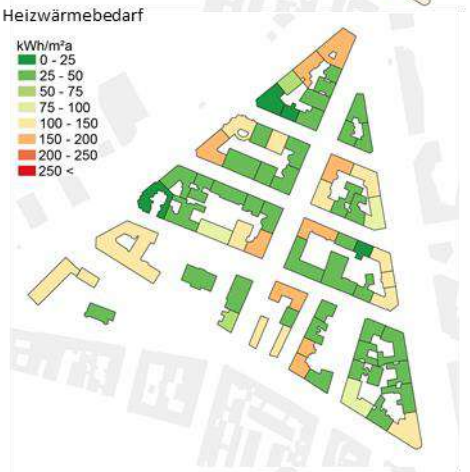
Im Jahr 2040 weisen demnach zusätzliche 32% der Gebäude thermische Sanierungsmaßnahmen auf.

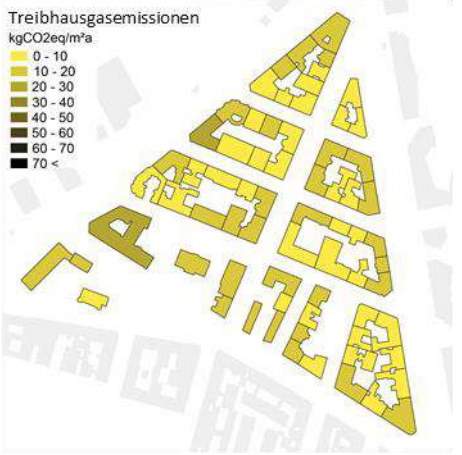
Der spezifische Heizwärmebedarf der Gebäude liegt im Bereich von 18 kWh/m^{2}NGF bis 164 kWh/m^2 mit einem quartiersweiten Mittelwert von **72 kWh/m^2.** Das entspricht einem jährlichen Heizenergiebedarf (inkl. WWWB) von 16,2 GWh Fernwärme. (Siehe zweite Grafik rechts von oben).

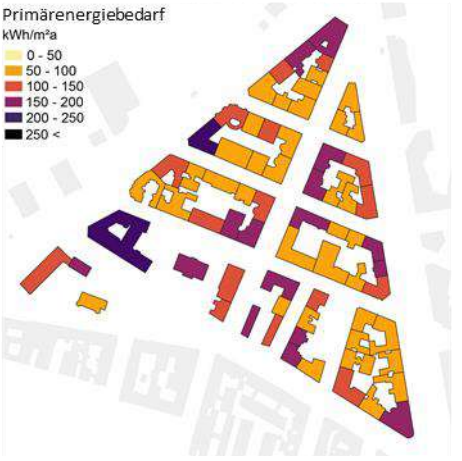
Dazu kommt neben dem jährlichen Strombedarf für Beleuchtung, Geräte, Hilfsstrom von durchschnittlich 29,1 kWh/m^2 bzw. 4,2 GWh, noch ein Anteil für die Kühlung (0,8 kWh/m^{2}NGFa und 111 MWh/a) im Quartier dazu.

Dies führt zu einem jährlichen Treibhausgas-Ausstoß von 1800 Tonnen bzw. durchschnittlich 12,3 kgCO2eq/m^{2}NGFa und einem Primärenergieeinsatz von jährlichen 17,2 GWh bzw. durchschnittlichen 120 kWh/m^{2}NGFa.



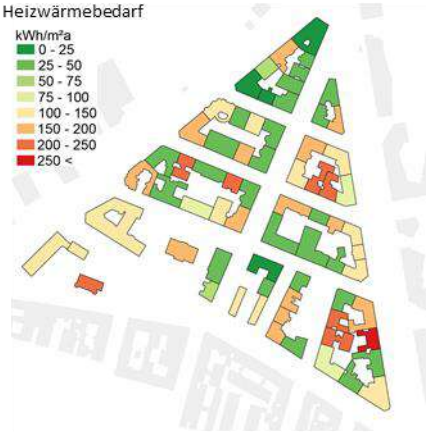






4.2.5 Schlüsselliegenschaften

Das „Szenario Schlüsselliegenschaften“ konzentriert sich auf die energetische Sanierung von 15 ausgewählten Gebäuden, die von der Stadt Wien (MA25) aufgrund ihrer potentiellen Vorbildwirkung, vergleichsweise einfachen Sanierungsoptionen und ihres hohen Reduktionspotentials identifiziert wurden. Diese Gebäude werden im Szenario auf einen hocheffizienten Sanierungsstandard gebracht, um maximale Energieeinsparungen und Emissionsreduktionen zu erzielen.

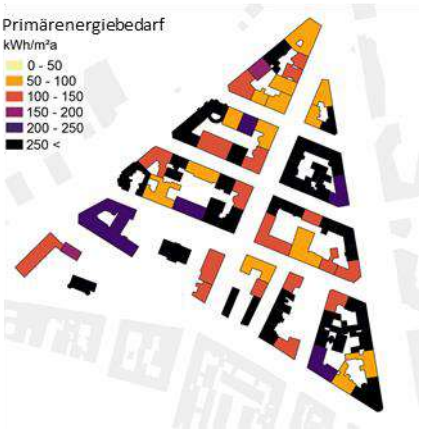


Für gasversorgte Gebäude wird zusätzlich die Wärmeversorgung auf Fernwärme umgestellt, um die Abhängigkeit von fossilen Energieträgern zu reduzieren. Dieses Szenario dient als Modellprojekt, das praxisnahe Umsetzungsstrategien und die Effektivität gezielter Maßnahmen zur Dekarbonisierung urbaner Quartiere aufzeigt.



		bis 2030	bis 2040
Energieversorgung	Schlüsselliegenschaften steigen auf Fernwärme um	• ~2 Gebäude/a • 2.8 %/a	• ~1 Gebäude/a • 1.0 %/a
Thermische Sanierung	Schlüsselliegenschaften werden auf hocheffizienten Standard saniert	• ~3 Gebäude/a • 3.1 %/a	• ~1 Gebäude/a • 1.2 %/a

Das Quartier wird dabei zu 65 % mit Erdgas, zu 34 % durch Fernwärme und zu 1 % über eine Grundwasser-Wärmepumpe versorgt. Im Jahr 2040 weisen zusätzliche 19 % der Gebäude thermische Sanierungsmaßnahmen auf.



Der spezifische Heizwärmebedarf der Gebäude liegt im Bereich von 18 kWh/m^{2}NGF bis 254 kWh/m^2 mit einem quartiersweiten Mittelwert von **101 kWh/**m^2. Das entspricht einem jährlichen Heizenergiebedarf (inkl. WWWB) von 23,4 GWh – 4,9 GWh Fernwärme und 18,5 GWh Erdgas.

Dazu kommt neben dem jährlichen Strombedarf für Beleuchtung, Geräte, Hilfsstrom von durchschnittlich 28,9 kWh/m^2 bzw. 4,2 GWh, noch ein Anteil für die Kühlung (0,4 kWh/m^{2}NGFa und 61 MWh/a) im Quartier dazu.

Dies führt zu einem jährlichen Treibhausgas-Ausstoß von 4700 Tonnen bzw. durchschnittlich 32,7 kgCO2eq/m^{2}NGFa und einem Primärenergieeinsatz von jährlichen 30,6 GWh bzw. durchschnittlichen 213 kWh/m^{2}NGFa.

4.2.6 Szenarien im zeitlichen Verlauf

Dieses Kapitel stellt den Absenkpfad der spezifischen Treibhausgasemissionen (THG) im Betrieb des betrachteten Quartiers anhand ausgewählter Beispielszenarien dar. Für alle Szenarien dient das Jahr 2024 als Ausgangspunkt. Der zeitliche Verlauf der Emissionsreduktion wird dabei durch unterschiedliche Annahmen zu Kesseltausch- und thermischen Sanierungsraten bestimmt, die in den Szenarien entkoppelt betrachtet werden. Priorität erhalten in jedem Szenario zunächst ineffiziente Gebäude.

Im *TMS-Szenario* wird durch eine maximale thermische Sanierungsrate (+ einhergehender Kesseltauschrate) bis 2040 auch die maximale Reduktion der spezifischen THG-Emissionen erreicht.

Demgegenüber steht das *Wiener Szenario*, das zwar ebenfalls bis 2040 den vollständigen Ausstieg aus der Gasnutzung vorsieht, aufgrund einer geringeren thermischen Sanierungsrate jedoch bis 2050 nicht das gleiche Niveau an THG-Reduktion wie das TMS-Szenario erreicht. Die zeitliche Verzögerung im Absenkpfad verdeutlicht, dass niedrigere Sanierungsraten den langfristigen Dekarbonisierungspfad signifikant beeinflussen.

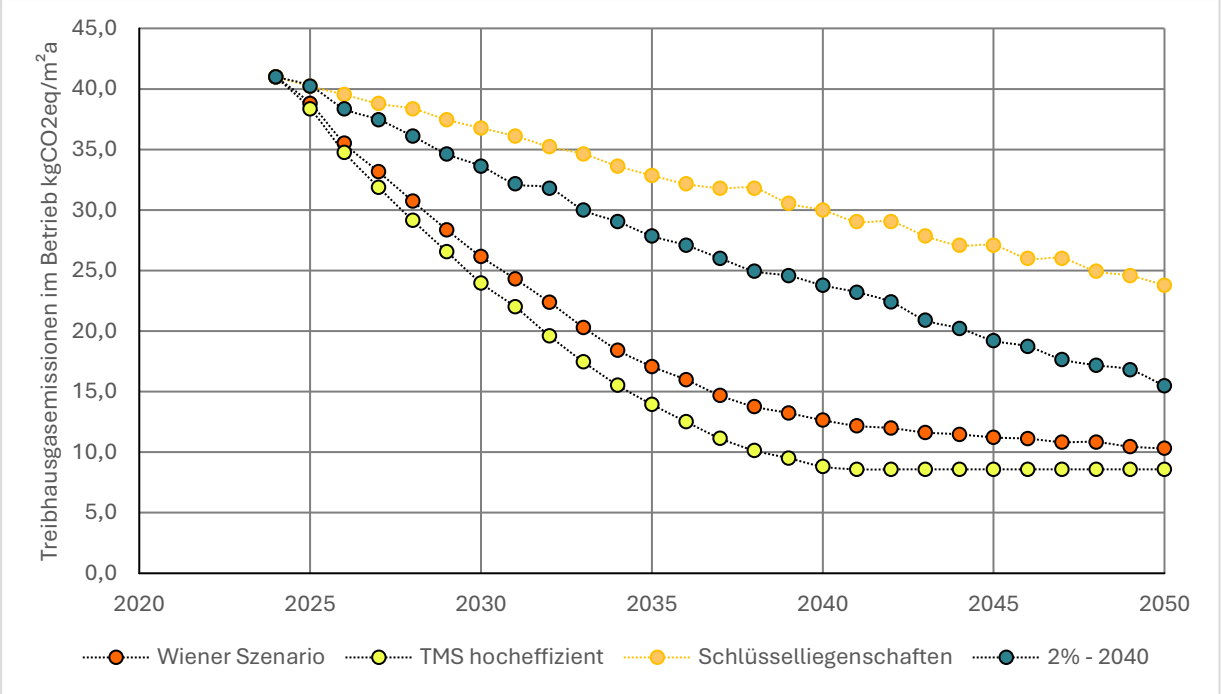


Abbildung 4-11 Absenkpfad spezifischer Treibhausgasemissionen im Betrieb für ausgewählte Szenarien

Im *TMS-Szenario* wird das Emissionsminimum szenariokonform im Jahr 2040 realisiert. Das *Wiener Szenario* (jährlich rund 2 Gebäude thermisch saniert und rund 4 Gebäude Raus aus Gas) sieht einen vollständigen Ausstieg aus der Erdgasnutzung bis 2040 vor, erreicht jedoch die maximale Emissionsreduktion erst bis zum Jahr 2063.

Das *„worst-first 2%“* Szenario mit einer jährlich 2% thermischen Sanierungsrate (jährlich rund 2 Gebäude thermisch saniert und rund 2 Gebäude von Gas auf Fernwärme umgestellt) erreicht die maximale Reduktion bis zum Jahr 2068. Die moderate Sanierungsrate führt zwar zu einer deutlichen Emissionsminderung, verzögert jedoch den vollständigen Rückgang der THG-Emissionen im Vergleich zum TMS-Szenario erheblich.

Im Szenario *Schlüsselliegenschaften*, das eine jährliche Kesseltauschrate von 1% (rund 1 Gebäude pro Jahr) sowie eine thermische Sanierungsrate von 1,2% (rund 1 Gebäude pro Jahr) annimmt, wird die maximale Reduktion der THG-Emissionen erst nach dem Jahr 2100 erreicht. Der langsame

Sanierungs- und Kesseltauschprozess führt zu einer erheblichen Verzögerung im Absenkpfad, was den langfristigen Dekarbonisierungsprozess stark beeinflusst.

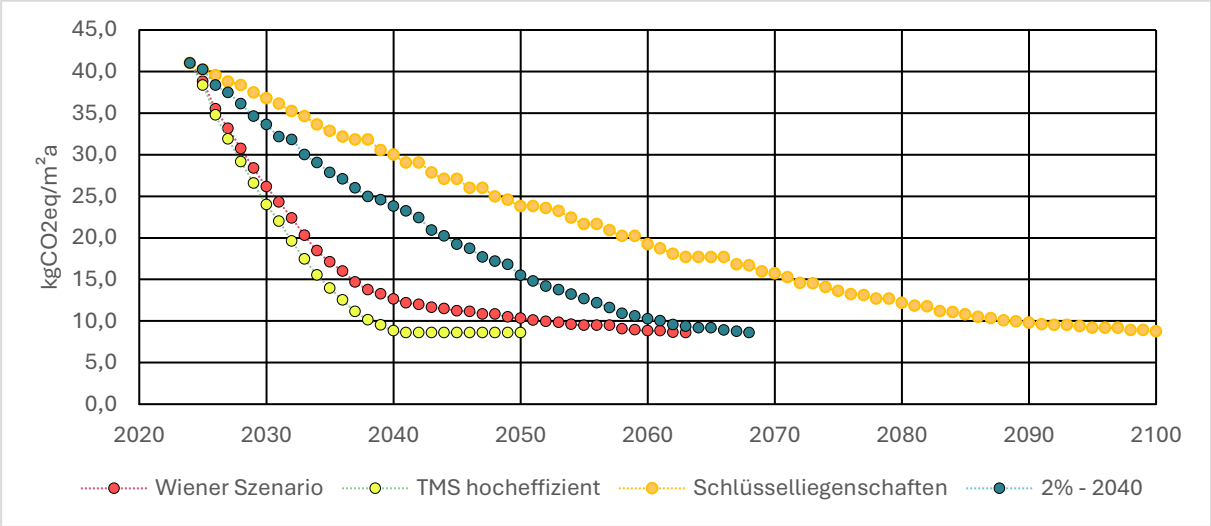


Abbildung 4-12 Zeithorizont jeweils zur Erreichung des Minimalwerts erweitert

4.3 Zielwert-Abgleich Smart Klima City Strategie

Die Gegenüberstellung der Szenarien ermöglicht eine fundierte Bewertung von Zielerreichung und Aufwand. Im Fokus steht, welche Zielwerte erreicht werden, welcher Aufwand erforderlich ist und wo sich Trends sowie Kompromisse zwischen Aufwand und Nutzen abzeichnen.

Die folgenden drei kategorischen Ziele werden den Szenarien gegenübergestellt. In Tabelle 15 sind die Zielwerte definiert.

- Energieträger – „100% Raus aus Gas“
- Reduktion Endenergie
- Reduktion CO2-Emissionen

Tabelle 15 Zielwerte zum Vergleich der Szenarienergebnisse. Für die Ziele Endenergie und CO2-Emissionen wird hier das Bezugsjahr 2024 statt dem Durchschnittswert der Jahre 2005–2010 herangezogen.

Ziele laut Wiener Klimafahrplan	Indikator	Ziel bis 2030	Ziel 2040
„Endenergiebedarf für Heizen, Kühlen und Warmwasser reduzieren“	Endenergiebedarf für Heizen, Kühlen und Warmwasser von Gebäuden pro Kopf	-20%	-30%
„CO2-Emissionen für Heizen, Kühlen und Warmwasser reduzieren“	Non-ETS-CO2-Emissionen für Heizen, Kühlen und Warmwasser von Gebäuden pro Kopf	-55%	-100%
„Bis 2040 gänzlicher Ausstieg aus der fossilen Wärmeversorgung“	Anteil Erdgas an Heizen & Warmwasser		-100%

Tabelle 16 zeigt, welche der untersuchten Ziele der Smart Klima City Strategie erreichbar sind und in welchem Umfang. Das Szenario *Worst First* mit einer jährlichen Sanierungsrate von 1% verfehlt sämtliche Zielvorgaben. Die technischen Maximalszenarien verdeutlichen die Grenzen des Machbaren, insbesondere im Hinblick auf die Ziele der Emissionsreduktion, die in dieser Betrachtung nicht vollständig erreicht werden können. Dabei ist zu berücksichtigen, dass der Referenzzeitraum für die THG-Emissionen von 2005 bis 2010 den Zielvorgaben für 2030 entgegenkommen könnte, da seit besagtem Referenzzeitraum bereits Sanierungsmaßnahmen sowie Umstellungen auf Fernwärme oder Wärmepumpen – wie das Beispiel der Eberlgasse 3 – erfolgten.

Ein Vergleich zwischen dem Szenario *TMS Hocheffizient* und *Worst First 4%/a* zeigt, dass im *TMS*-Szenario bis 2040 insgesamt 10 % mehr Endenergie eingespart wird. Allerdings führt die gezielte Priorisierung ineffizienter Gebäude im *Worst First*-Szenario dazu, dass bis 2030 eine um 3 % höhere Einsparung erzielt wird – trotz einer geringeren Anzahl sanierter Gebäude, die sich in einer etwa 1 % niedrigeren Sanierungsrate widerspiegelt.

Das Ziel „100% Raus aus Gas“ wird in den technischen Maximalszenarien und im „Wiener Szenario“ erreicht.

Tabelle 16 Einsparungen gegenüber 2024 Szenarien in Bezug auf Ziele der Smart Klima City Strategie. Erreichte Ziele in Grün

	Endenergiebedarf Heizen, Kühlen, Warmwasser		Treibhausgasemissionen Heizen, Kühlen, Warmwasser		**Erdgas Heizen & Warmwasser**	
	2030	*2040*	*2030*	*2040*	*2030*	*2040*
TMS Hocheffizient	-27%	-72%	-33%	-89%	-38%	-100%
TMS Bauordnung	-25%	-67%	-33%	-88%	-38%	-100%
Worst First 1%/a	-8%	-20%	-10%	-25%	-11%	-29%
Worst First 2%/a	-16%	-35%	-20%	-48%	-23%	-54%
Worst First 3%/a	-23%	-51%	-30%	-64%	-34%	-72%
Worst First 4%/a	-30%	-62%	-38%	-76%	-43%	-84%
Wiener Szenario	-16%	-43%	-30%	-79%	-38%	-100%
Schlüsselliegenschaften	-7%	-18%	-8%	-23%	-10%	-25%

Die Reduktionspotentiale je Indikator und dabei jeweils je zeitlichem Horizont sind in Abbildung 4-13, Abbildung 4-14 und Abbildung 4-15 dargestellt. Über die *Worst First* Szenarien sind die Unterschiede bzw. der Einfluss der Priorisierung nach Ineffizienz ersichtlich. Für die restlichen Szenarien wurde der 2030-Wert lediglich linear skaliert.

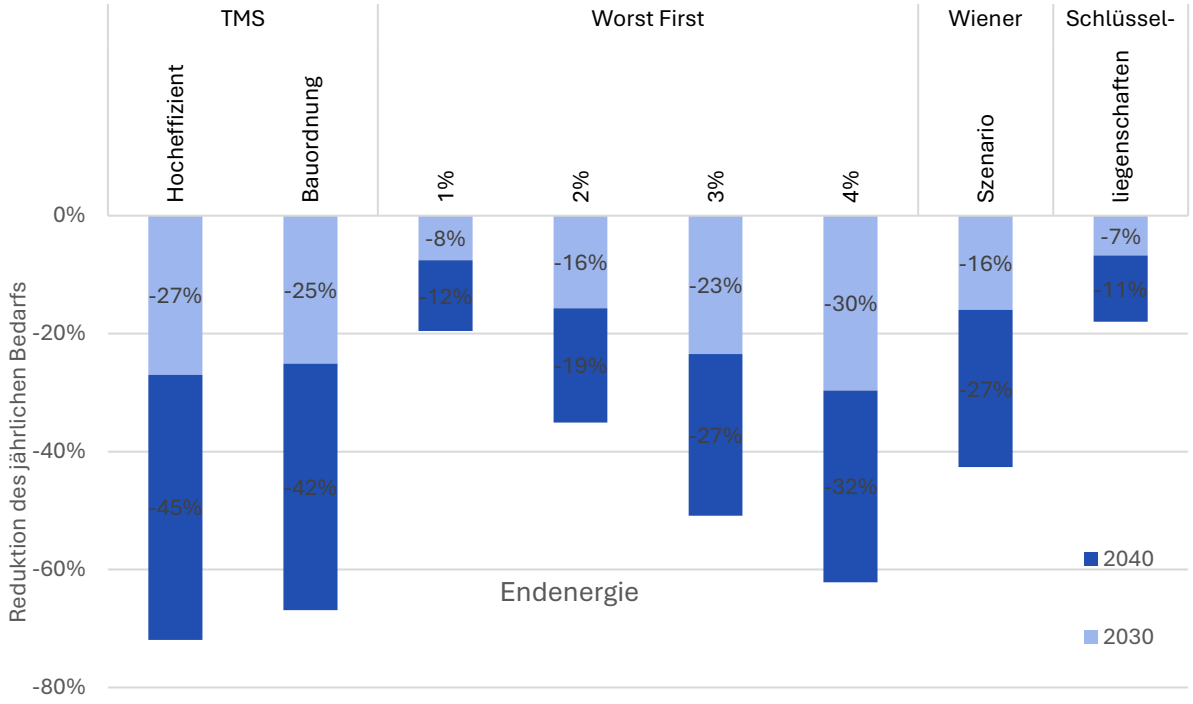


Abbildung 4-13 Reduktion des jährlichen Endenergiebedarfs für Heizen, Kühlen, Warmwasser je Szenario und Zeithorizont

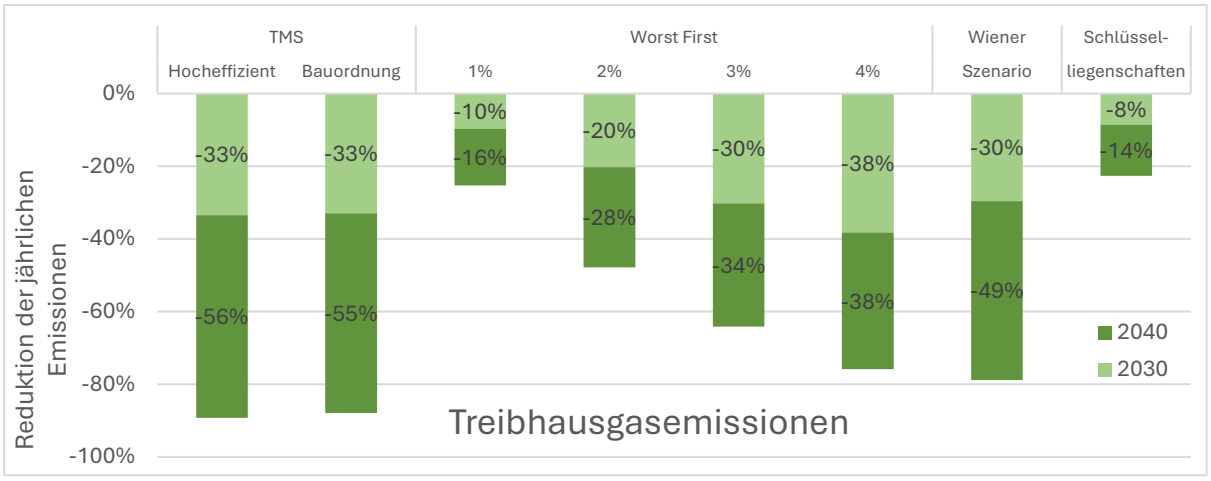


Abbildung 4-14 Reduktion der jährlichen THG-Emissionen für Heizen, Kühlen, Warmwasser je Szenario und Zeithorizont

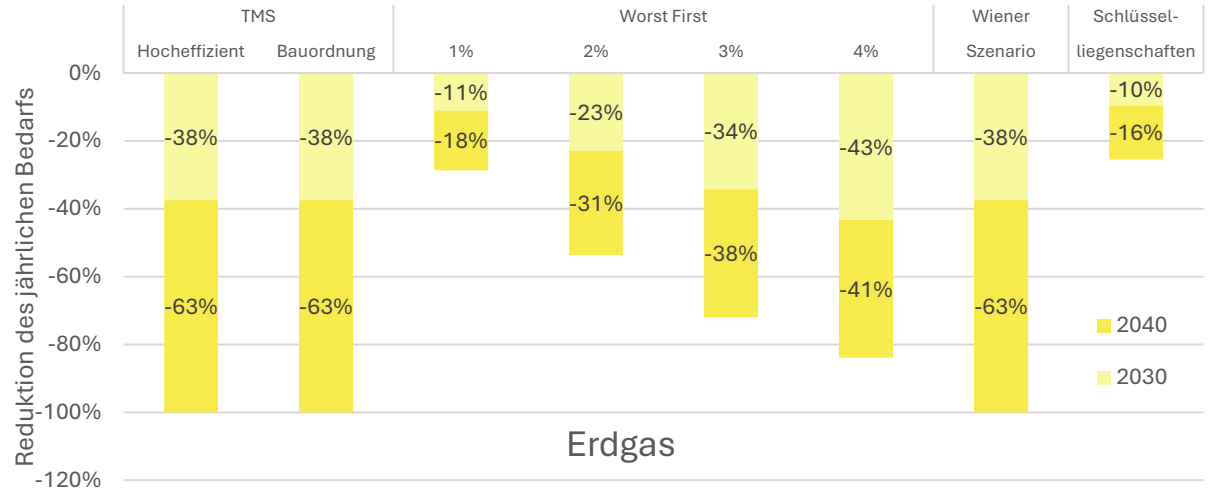


Abbildung 4-15 Reduktion des jährlichen Bedarfs an Erdgas für Heizen und Warmwasser je Szenario und je Zeithorizont

4.4 Zielwert-Abgleich klimaaktiv Plus-Energie-Quartier

Keines der betrachteten Szenarien erreicht den Plus-Energie-Standard auf der Ebene des Gebäudebetriebs. Das Szenario TMS Hocheffizient verfehlt den geforderten Grenzwert um 17 kWh/m²a, obwohl die Reduktion des Primärenergiebedarfs gegenüber der Ausgangssituation mit -168 kWh/m²a beachtlich ausfällt.

Die Analyse beschränkt sich auf den aktuellen Systemrahmen, in dem Mobilität und graue Energie nicht berücksichtigt wurden. Ebenso wurde das Solarpotential ausschließlich auf Basis des Solarpotentialkatasters als maximale PV-Nutzung bewertet, ohne zusätzliche Flächen wie Fassaden einzubeziehen. Maßnahmen zur Energieflexibilisierung, etwa durch Speicher oder Sektorkopplung, wurden ebenso wenig betrachtet wie mögliche Effizienzsteigerungen durch Wärmepumpen.

Eine erneute Bewertung unter Einbeziehung zusätzlicher Photovoltaikflächen, Energieflexibilisierung, Wärmepumpensystemen und weiteren Effizienzmaßnahmen – insbesondere im Bereich der Stromanwendungen – könnte neue Potentiale zur Annäherung an den Plus-Energie-Standard aufzeigen.

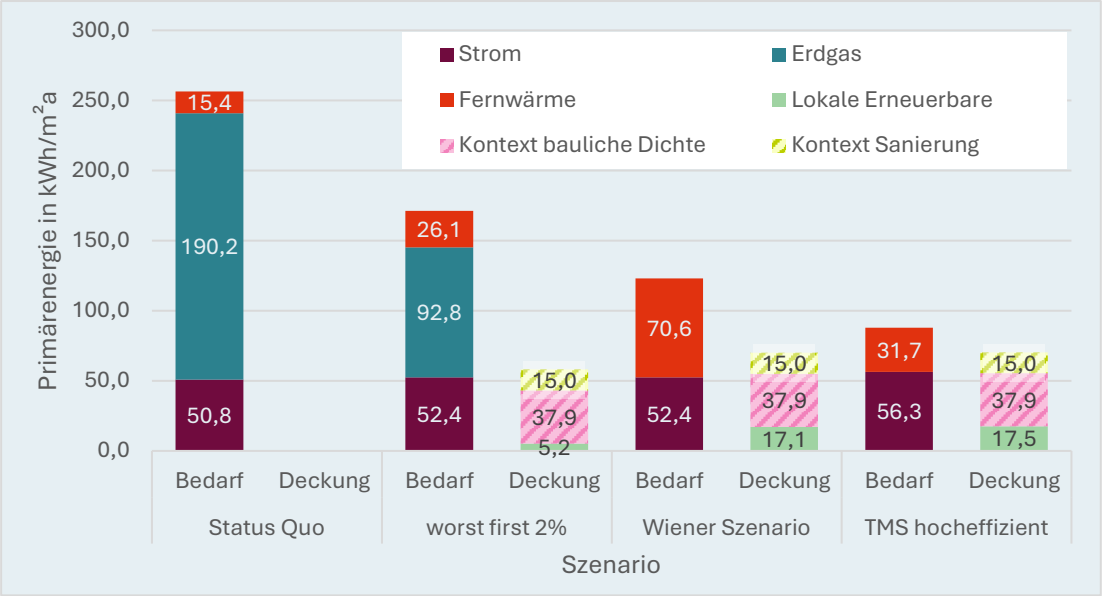


Abbildung 4-16 Bewertungsergebnis nach klimaaktiv Deklaration „Plus-Energie-Quartiere“. Der notwendige Primärenergie-Bilanzüberschuss von Deckung gegenüber Bedarf wird in allen betrachteten Szenarien verfehlt.

4.5 Vergleich mit den Wiener Dekarbonisierungstypen

Die Wiener Dekarbonisierungstypen (DKT) verfolgen wie die hier angewendeten Gebäudetypologie dasselbe Ziel, Gebäude mit ähnlichen Eigenschaften zu kategorisieren und clustern zu können – von Bestandsgebäuden bis hin zu dekarbonisierten Zuständen. Trotz des gemeinsamen Ziels unterscheiden sich die Ansätze in wesentlichen Aspekten des Detailgrads und der Systematisierung:

- Die Typologie dieser Studie berücksichtigen im Gegensatz zu den DKT keine Informationen zu **Kochsystemen** oder zur Differenzierung zwischen **zentralen und dezentralen** Versorgungssystemen.
- Während die DKT zudem die **Gebäudenutzung** und die Gebäudeart – insbesondere im Hinblick auf deren Größe – qualitativ thematisieren, wird dieser Aspekte hier außerhalb der Typologien in höherer räumlicher Auflösung im zugrunde liegenden geometrischen Modell direkt abgebildet, das zum einen extrudierte Baukörper und zum anderen mehr als fünf unterschiedlichen Nutzungen berücksichtigt.
- Hinsichtlich der **Energieversorgung** zeigen sich weitere Unterschiede. In den Typologien des Alliiertenviertels werden Erdgas, Fernwärme und Wärmepumpen jeweils als eigenständige Versorgungslösungen betrachtet, die alle Domänen versorgen. Ölvarianten oder kombinierte Gas-Fernwärme-Systeme werden hingegen nicht berücksichtigt.
- Der **Sanierungszustand** der Gebäude wird hier neben sieben Qualitätsstufen, die sich am Gebäudealter orientieren, zusätzlich durch drei Sanierungsqualitäten differenziert, was eine detaillierte Analyse von Sanierungsstrategien und deren Auswirkungen ermöglicht.

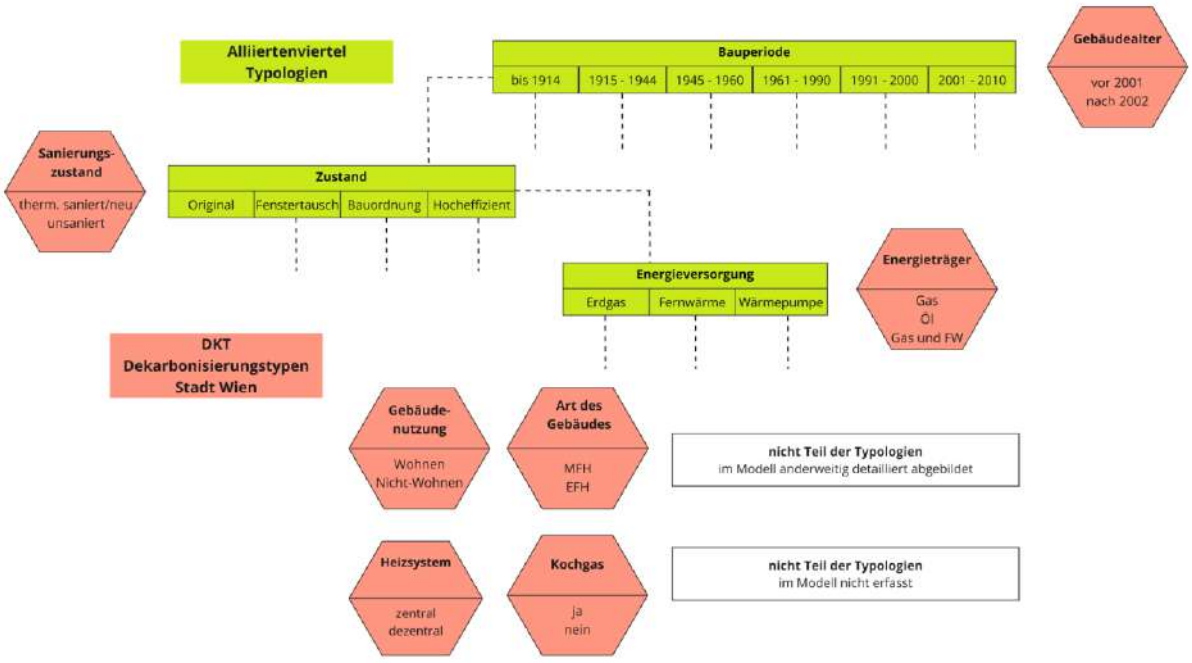


Abbildung 4-17 Gegenüberstellung und Einordnung Dekarbonisierungstypen Stadt Wien und Gebäudetypologien Alliiertenviertel

5 Fazit

Die vorliegende Studie zeigt, dass im Alliiertenviertel erhebliche Treibhausgas-Reduktionen im Gebäudebetrieb durch Umstieg von Gas auf Fernwärme und thermische Sanierungen möglich sind. Sie zeigt, dass die End-Energieeinsparungen bis 2030 und 2040 durch mittlere Sanierungsraten von 1,5 bis 2 Gebäuden pro Jahr – angefangen mit den ineffizientesten – erreichbar sind, aber dass die THG-Reduktion um **55 %** bis **2030** oder eine vollständige Dekarbonisierung bis **2040**, aufgrund der spezifischen Gegebenheiten des Alliiertenviertels herausfordernd sind:

- Bei einer vollständigen und ambitionierten Sanierung der unsanierten Gebäude im Quartier wäre eine Reduktion des mittleren Heizwärmebedarfs um mehr als **80 %** möglich.
- Eine **20-prozentige Endenergieeinsparung bis 2030** kann bereits bei einer thermischen Sanierungsrate von **2,5 %** pro Jahr erreicht werden, was etwa **2 Gebäudesanierungen pro Jahr** entspricht.
- Die Sanierung der ineffizientesten 17 % des Bestands ermöglicht eine Einsparung von 23 % im Bereich Heizen, Kühlen und Warmwasser.
- Bis **2040** kann eine Reduktion des Endenergiebedarfs um **30 %** mit einer Sanierungsrate von **2 % p.a. (1,6 Gebäude/a)** erreicht werden, wobei eine gezielte Sanierung der ineffizientesten 28 % der Gebäude Einsparungen von bis zu 35 % ermöglicht. Hierbei ist weniger die Sanierungsqualität als vielmehr die Anzahl der sanierten Gebäude ausschlaggebend.
- Die Umsetzung der **„Raus aus Gas“-Policy** erfordert bis **2040** eine Umstellung der Heiz- und Warmwassersysteme in **80 %** der Gebäude, was jährliche Kesseltauschraten von **5 %** impliziert.
- Ob alle vor 1980 oder alle vor 2000 errichteten Gebäude als sanierungswürdig eingestuft werden, hat nur einen geringen Einfluss auf die Gesamtbilanz, sodass die gezielte Priorisierung ineffizientester Gebäude zur Maximierung der Endenergieeinsparungen entscheidend ist.

Die Bewertung nach klima**aktiv** Deklaration für **Plus-Energie-Quartiere** zeigt, dass zwar

- noch kein betrachtetes Szenario den Zielwert vollständig erreicht, aber
- die ambitionierten Szenarien bereits in Reichweite der Zielwerte sind: Eine genauere Untersuchung zusätzlicher Maßnahmen wie
    - **energieeffiziente Nachverdichtung** mit
    - **erweiterter Photovoltaik-Nutzung**,
    - **Energieflexibilisierung** der Abgabesysteme sowie die
    - Berücksichtigung der **grauen Energie**

erscheint vielversprechend, um ein klimaneutrales Plus-Energie-Quartier zu erreichen.

Datenlage ist gut, aber ausbaubar

Die Analyse zeigt, dass zur Gebäudebeschreibung und Szenarienbetrachtung nach wie vor **Daten** zu Energie und Gebäuden nicht bekannt oder klar sind:

- Kritische Informationen zur Gebäudequalität und zum Sanierungszustand sind häufig unklar oder nicht definiert, etwa durch Inkonsistenzen in den verschiedenen öffentlichen und städtischen Datenmodellen (Gebäudehöhe, Geschossanzahl).

- Nutzungsänderungen in Sockelzonen sowie augenscheinliche bauliche Anpassungen, die der vorliegenden Datenlage widersprechen (z. B. Dachgeschoßausbauten, die als Neubauten erfasst werden), erschweren eine konsistente Modellierung.

Methodik leicht auf weitere Entwicklungsgebiete anwendbar

Dennoch ermöglicht die Entwicklung von 72 Gebäudetypen eine nachvollziehbare und skalierbare Analyse großer Teile Wiens, insbesondere weiterer *WieNeu*+-Areale, sofern sie eine ähnliche gründerzeitliche Blockrandbebauung aufweisen. Aktuell unberücksichtigt bleiben Einfamilienhaussiedlungen, prädestinierte öffentliche und Dienstleistungsgebäude sowie Sondergebäude. Eine Erweiterung der Typologien kann hier schnell eine detaillierte Analyse ermöglichen. Die eingesetzte Methode mit CEA erlaubt eine rasche Einbindung öffentlicher GIS-Daten und stellt eine Weiterverwendung der detaillierten Gebäudedaten in kompatiblen Formaten sicher. Damit kann das städtische Urban Building Energy Model (UBEM) künftig zur Planung, Monitoring und iterativer Verbesserung beitragen, insbesondere in Hinblick auf schnelle Potentialermittlung, Schwerpunktidentifikation und fortlaufendes Monitoring von Klimazielen.

Gebäudescharfe Ergebnisse in Reichweite

Aufgrund der teils vereinfachenden Annahmen in der Modellierung sind konkreten Einsparungspotentiale von Maßnahmen damit noch nicht quantitativ auf konkrete Einzelgebäude anwendbar. Für zukünftige Analysen erscheint eine Nachschärfung in Bezug auf den Kühlbedarf in oberen Geschossen sinnvoll. Die typologische Herangehensweise ermöglicht trotzdem eine fundierte Einschätzung der Sanierungspotentiale und Energieeinsparbandbreiten je Gebäudetyp, Gebäudezustand und Kompaktheit.

6 Anhang

6.1 Verfügbarkeit der Gebäudetypologien

Entwickelte Gebäudetypologien sind verfügbar unter github.com/simonschaluppe/ceautil. In technology\archetypes\CONSTRUCTION_STANDARD.xlsx sind die Typologien - jeweils universell nach Nomenklatur benannt (Abbildung 6-1) - mit den jeweiligen Eigenschaften (in Abbildung 6-2) verbunden.

Bauperiode_Gebäudezustand_Energieversorgung		
z.B. 1919-1944	orig	Erdgas

Abbildung 6-1 Nomenklatur der Typologien

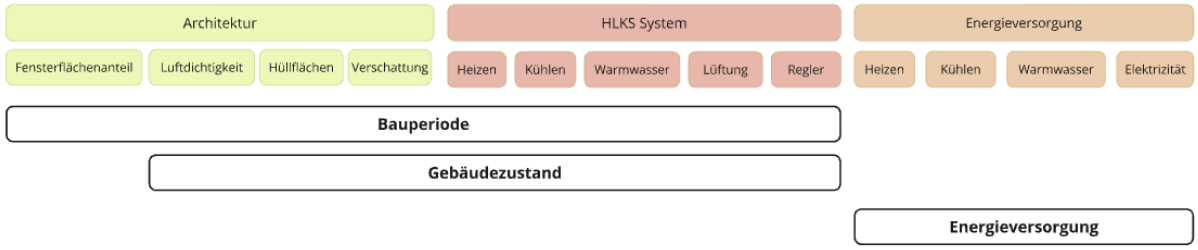


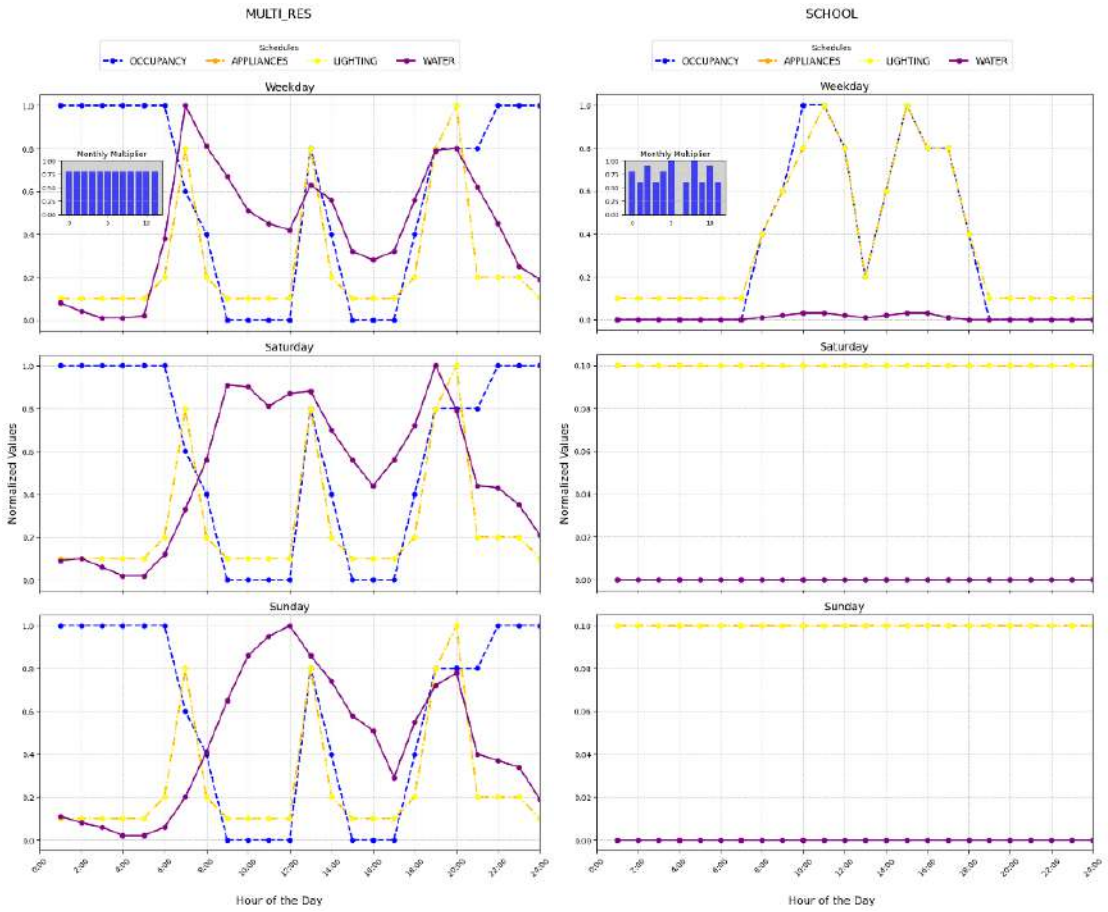
Abbildung 6-2 Abhängigkeiten der Parameterkategorien je definierender Eigenschaft

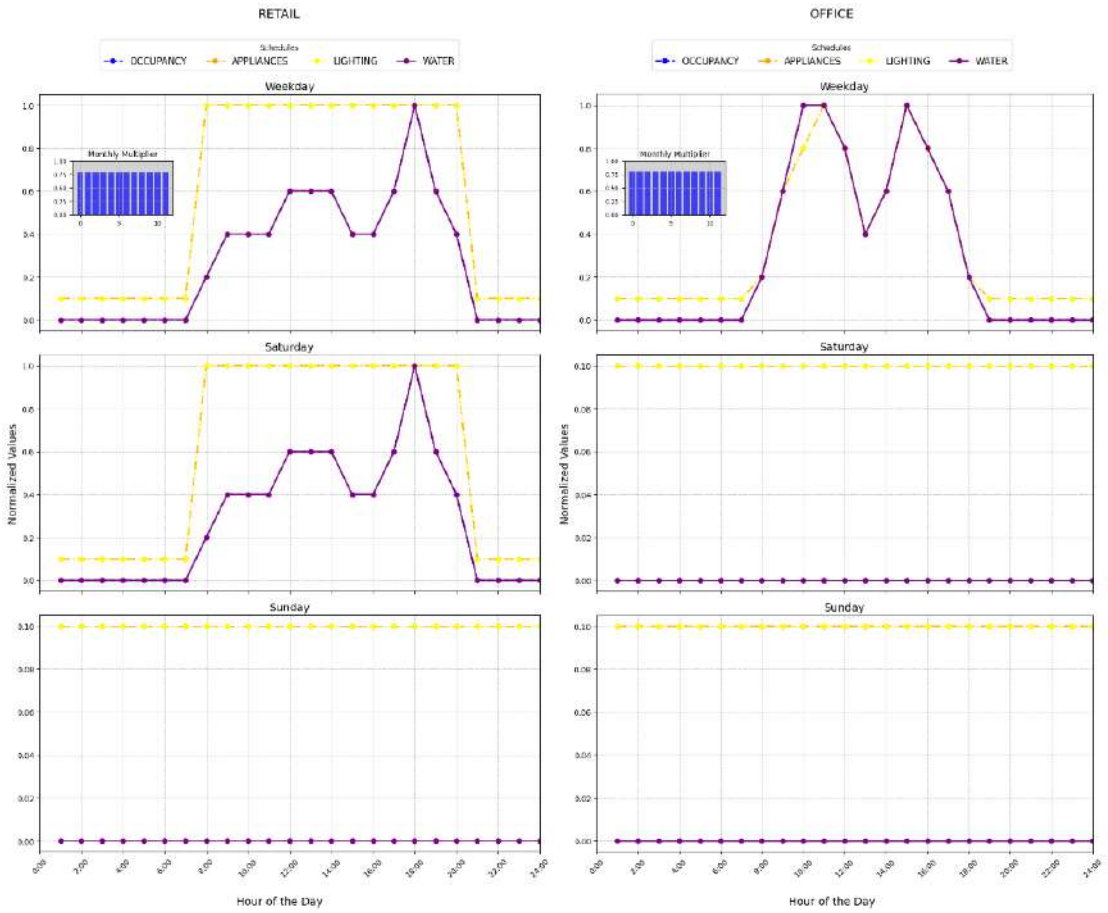
6.2 Nutzungsprofile

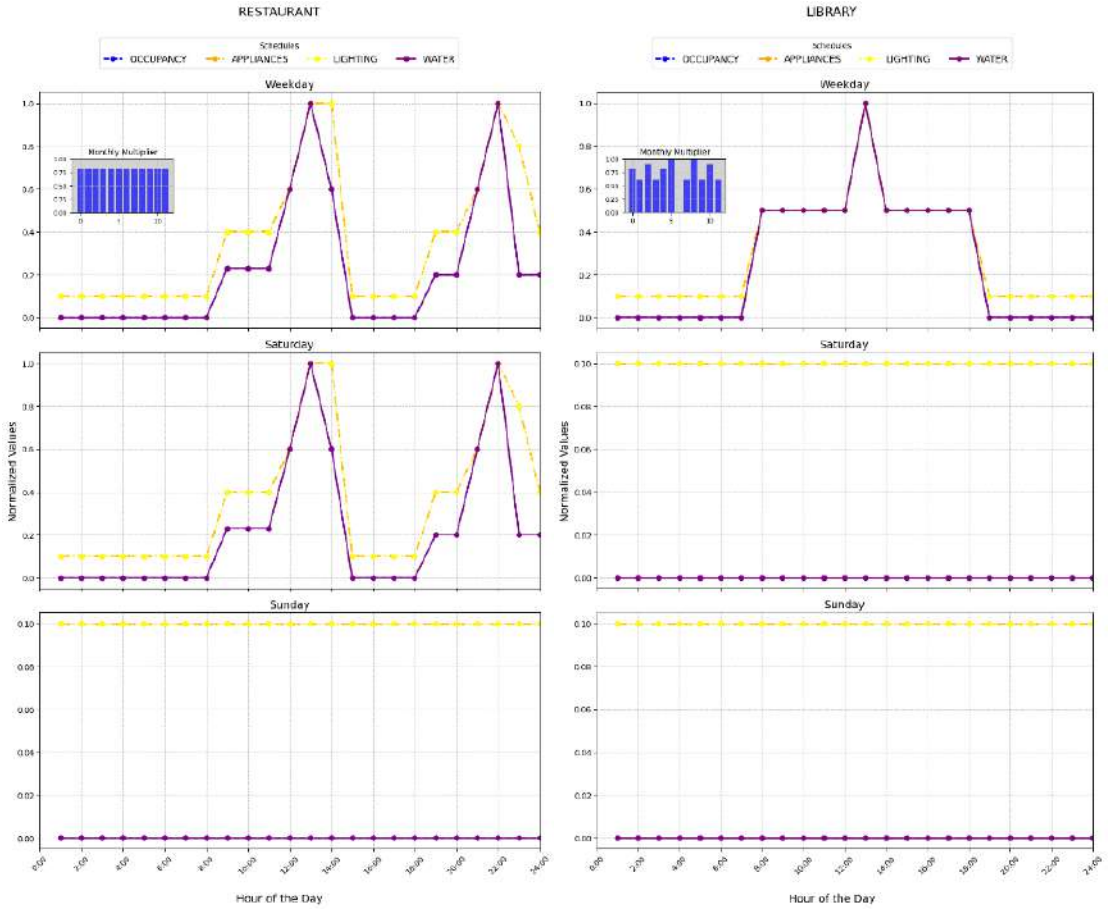
Bestehend aus prozentuellen 24-Stunden-Profilen je Werk-, Sams- und Sonntag, sowie den zugrunde liegenden Nominalwerten.

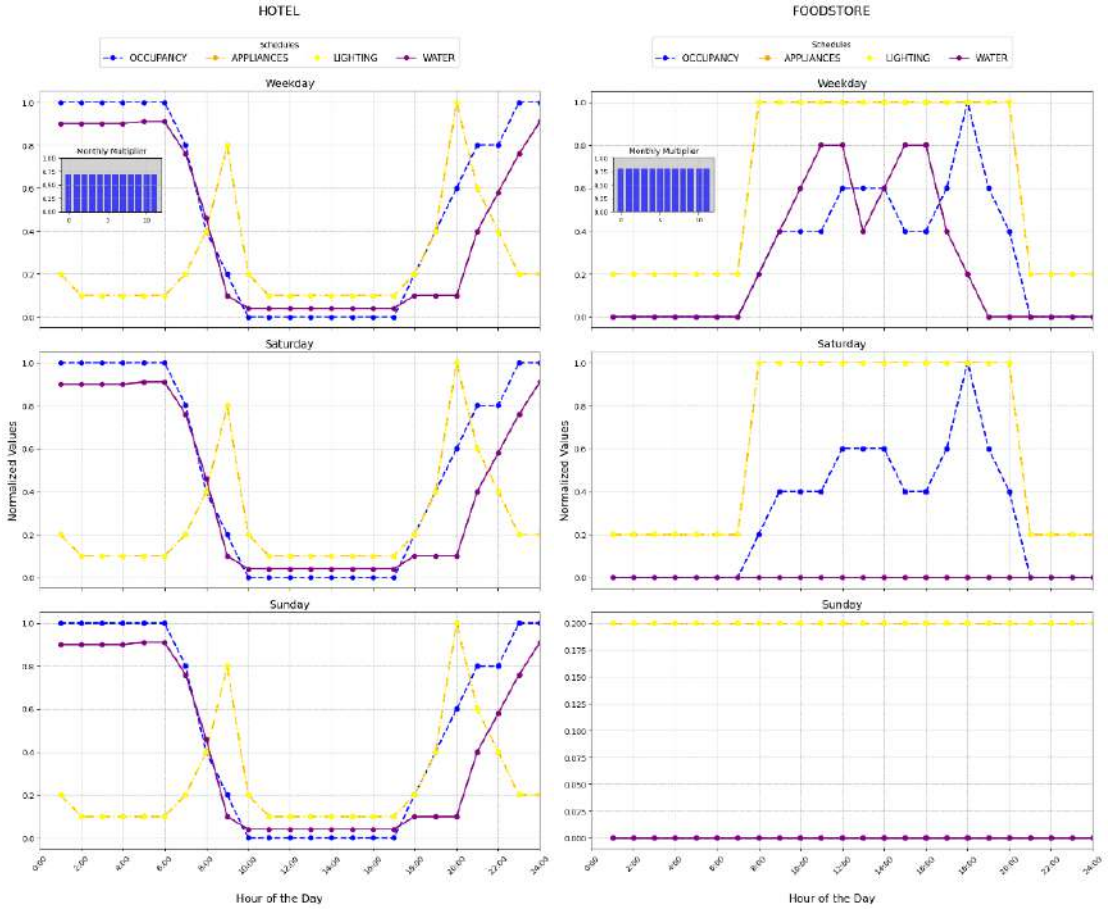
USAGE	Solltemperatur Kühlen °C	Solltemperatur Heizen °C	Luftvolumenstrom mindest Liter/Pers*Tag
MULTI_RES	26	21	8.3
HOTEL	26	21	10.0
OFFICE	26	21	10.0
RETAIL	26	20	8.3
FOODSTORE	26	20	8.3
RESTAURANT	26	21	10.0
SCHOOL	26	21	6.9
LIBRARY	26	21	10.0

USAGE	Belegungsdichte m²/Pers	Leistung Geräte W/m²	Leistung Beleuchtung W/m²	Warmwasserbedarf Liter/Pers*Tag
MULTI_RES	30.0	8.0	2.7	40.0
HOTEL	15.0	8.0	2.7	40.0
OFFICE	14.0	7.0	15.9	3.0
RETAIL	8.0	2.0	33.3	2.0
FOODSTORE	8.0	2.0	21.3	2.0
RESTAURANT	2.0	2.0	6.9	15.0
SCHOOL	3.0	4.0	14.0	2.0
LIBRARY	5.0	2.0	6.9	0.0


MULTI_RES
SCHOOL
Schedules
OCCUPANCY
APPLIANCES
LIGHTING
WATER
Weekday
Saturday
Sunday
Monthly Multiplier
Normalized Values
Hour of the Day


RETAIL
OFFICE
Schedules
OCCUPANCY
APPLIANCES
LIGHTING
WATER
Weekday
Saturday
Sunday
Monthly Multiplier
Normalized Values
Hour of the Day


RESTAURANT
LIBRARY
Schedules
OCCUPANCY
APPLIANCES
LIGHTING
WATER
Weekday
Saturday
Sunday
Monthly Multiplier
Normalized Values
Hour of the Day


HOTEL
FOODSTORE
Schedules
OCCUPANCY
APPLIANCES
LIGHTING
WATER
Weekday
Saturday
Sunday
Monthly Multiplier
Normalized Values
Hour of the Day

6.3 GIS Daten

Struktur der Modell- und Ergebnisdaten

6.3.1 Datei: Alliiertenviertel_Modelldaten.xlsx

Blatt: Glossar

Beschreibung der Spalten in den anderen Tabellenblättern

- Begriff
- Bedeutung
- Beschreibung
- Zugehörigkeit
- Kontext
- wo finde ich das?

Blatt: Alliiertenviertel

- Building_Code
- Type_CEA
- Bauperiode
- Gebäudezustand
- Energieversorgung
- Konditioniert
- Nutzung
- Schluesselliegenschaft
- Datenlage_unsicher
- Raus aus Gas
- therm. Sanierung 1%/a bis 2040
- therm. Sanierung 2%/a bis 20402
- therm. Sanierung 3%/a bis 2040
- therm. Sanierung 4%/a bis 20402
- 1ST_USE
- 2ND_USE
- 3RD_USE
- 1ST_USE_R
- 2ND_USE_R
- 3RD_USE_R
- RESIDENTIAL
- OFFICE
- LIBRARY
- RETAIL
- FOODSTORE
- RESTAURANT
- SCHOOL
- HOTEL
- type_cs
- type_hs
- type_dhw
- type_ctrl
- type_vent
- heat_starts
- heat_ends
- cool_starts
- cool_ends
- Hs_ag
- Hs_bg
- Ns
- Es
- void_deck
- wwr_north
- wwr_west
- wwr_east
- wwr_south
- type_cons
- type_leak
- type_floor
- type_part
- type_base
- type_roof
- type_wall
- type_win
- type_shade
- Tcs_set_C
- Ths_set_C
- Tcs_setb_C
- Ths_setb_C
- Ve_lsp
- RH_min_pc
- RH_max_pc
- Occ_m2p
- Qs_Wp
- X_ghp
- Ea_Wm2
- El_Wm2
- Ed_Wm2
- Ev_kWveh
- Qcre_Wm2
- Vww_ldp
- Vw_ldp
- Qhpro_Wm2
- Qcpro_Wm2
- Epro_Wm2
- type_cs_supply_systems
- type_hs_supply_systems
- type_dhw_supply_systems
- type_el
- height_ag
- floors_ag
- height_bg
- floors_bg
- Af_m2
- Aroof_m2
- GFA_m2
- Aocc_m2
- people0
- A
- V
- LC
- East Walls
- East Windows
- North Walls
- North Windows
- South Walls
- South Windows
- Top Roofs
- West Walls
- West Windows
- Adresse1
- Adresse2
- Adresse3
- Grundstücksnr.1
- Grundstücksnr.2

Blatt: standard_types

- STANDARD

- Description
- Bauperiode
- Gebäudezustand
- Energieversorgung
- YEAR_START
- YEAR_END
- type_cons
- type_leak
- type_win
- type_roof
- type_part
- type_wall
- type_floor
- type_base
- type_shade
- Es
- Hs_ag
- Hs_bg
- Ns
- void_deck
- wwr_north
- wwr_south
- wwr_east
- wwr_west
- type_hs
- type_cs
- type_dhw
- type_ctrl
- type_vent
- heat_starts
- heat_ends
- cool_starts
- cool_ends
- type_hs.1
- type_dhw.1
- type_cs.1
- type_el

Blatt: standard_values

- STANDARD
- Description
- Bauperiode
- Gebäudezustand
- Energieversorgung
- YEAR_START
- YEAR_END
- thermisch wirksame Speicherkapazität [Wh/m^{2}K]
- n50 -Wert
- Fenster U-Wert [W/m^{2}K]
- Dach/oberste Geschossdecke U-Wert [W/m^{2}K]
- Innenwand U-Wert
- Außenwand U-Wert [W/m^{2}K]
- Zwischendecken U-Wert
- Kellerdecke/Fundament U-Wert [W/m^{2}K]
- Verschattungsfaktor vereinfacht [-]
- Anteil BGF mit elektrischem Bedarf
- Anteil konditionierter BGF (oberirdisch)
- Anteil konditionierter BGF (unterirdisch)
- Anteil Nettogeschossfläche
- offenes Geschoss
- Fensterflächenanteil Nord
- Fensterflächenanteil Süd
- Fensterflächenanteil Ost
- Fensterflächenanteil West
- Abgabesystem Heizen
- Abgabesystem Kühlen
- Abgabe Warmwasser
- Regler für Heiz- und Kühlsysteme
- Lüftung
- Start Heizsaison
- Ende Heizsaison
- Start Kühlsaison
- Ende Kühlsaison
- Bereitstellung Heizen
- Bereitstellung WarmWasser
- Bereitstellung Kühlen
- Bereitstellung Strom

Blatt: CONSTRUCTION_env

- Description
- code
- Cm_Af

Blatt: TIGHTNESS_env

- Description
- code
- n50

Blatt: WINDOW_env

- Description
- code
- U_win
- G_win
- e_win
- F_F
- GHG_win_kgCO2m2
- Service_Life_win
- Reference Service life

Blatt: ROOF_env

- Description
- code
- U_roof
- a_roof
- e_roof
- r_roof
- GHG_roof_kgCO2m2
- Service_Life_roof
- Reference Service life

Blatt: WALL_env

- Description
- code
- U_wall
- a_wall
- e_wall
- r_wall
- GHG_wall_kgCO2m2
- Service_Life_wall
- Reference Service life

Blatt: FLOOR_env
- Description
- code
- U_base
- GHG_floor_kgCO2m2
- Service_Life_floor
- Reference Service life

Blatt: SHADING_env
- Description
- code
- rf_sh

Blatt: HOT_WATER_hvac
- Description
- code
- class_dhw
- Tsww0_C
- Qwwmax_Wm2

Blatt: HEATING_hvac
- Description
- code
- class_hs
- convection_hs
- Qhsmax_Wm2
- dThs_C
- Tshs0_ahu_C
- dThs0_ahu_C
- Th_sup_air_ahu_C
- Tshs0_aru_C
- dThs0_aru_C
- Th_sup_air_aru_C
- Tshs0_shu_C
- dThs0_shu_C
- Unnamed: 14

Blatt: COOLING_hvac
- Description
- code
- class_cs
- convection_cs
- Qcsmax_Wm2
- dTcs_C
- Tscs0_ahu_C
- dTcs0_ahu_C
- Tc_sup_air_ahu_C
- Tscs0_aru_C
- dTcs0_aru_C
- Tc_sup_air_aru_C
- Tscs0_scu_C
- dTcs0_scu_C
- Unnamed: 14

Blatt: VENTILATION_hvac
- Description
- code
- MECH_VENT
- WIN_VENT
- HEAT_REC
- NIGHT_FLSH
- ECONOMIZER
- Unnamed: 7

Blatt: CONTROLLER_hvac
- Description
- code
- dT_Qhs
- dT_Qcs
- Unnamed: 4

Blatt: HOT_WATER_supply
- Description
- code
- feedstock
- scale
- efficiency
- CAPEX_USD2015kW
- LT_yr
- O&M_%
- IR_%
- reference
- Unnamed: 10

Blatt: HEATING_supply
- Description
- code
- primary_components
- secondary_components
- tertiary_components
- feedstock
- scale
- efficiency
- CAPEX_USD2015kW
- LT_yr
- O&M_%
- IR_%
- reference

Blatt: COOLING_supply
- Description
- code
- primary_components
- secondary_components
- tertiary_components
- system
- feedstock
- scale
- efficiency
- CAPEX_USD2015kW
- LT_yr
- O&M_%
- IR_%
- reference

Blatt: ELECTRICITY_supply
- Description
- code
- feedstock
- scale
- efficiency
- CAPEX_USD2015kW
- LT_yr
- O&M_%
- IR_%
- reference
- Unnamed: 10

6.3.2 Datei: Alliiertenviertel_Ergebnisdaten.xlsx

Blatt: Ergebnisse_energetisch
- Building_Code
- Bezugsgröße_NGF_m^2
- Szenarien:
  - Bestand
  - TMS Bauordnung
  - TMS hocheffizient

- worst first 1%/a – 2040
- worst first 2%/a – 2040
- worst first 3%/a – 2040
- worst first 4%/a – 2040
- Schlüsselliegenschaften
- WIENER SZENARIO – hocheffizient
- WIENER SZENARIO – Bauordnung
- worst first 1%/a – 2030
- worst first 2%/a – 2030
- worst first 3%/a – 2030
- worst first 4%/a – 2030

- Ergebnisse je Szenario (Szenario Name wird dem Spaltennamen vorangestellt:
  - HWB_kWh/m²NGFa
  - KB_kWh/m²NGFa
  - WWWB_kWh/m²NGFa
  - Nutzerstrom
  - Beleuchtung
  - Hilfsstrom_kWh/m²NGFa
  - EEB_Netz_kWh/m²NGFa
  - EEB_Erdgas_kWh/m²NGFa
  - EEB_Fernwärme_kWh/m²NGFa
  - PEB_kWh/m²NGFa
  - THG_kgCO2eq/m²NGFa
  - Heizlast_W/m²NGF
  - Kühllast_W/m²NGF
  - Anschlussleistung FW_kW
  - Anschlussleistung Netz_kW
  - Volllaststunden FW_h

Blatt: PV

- Join_Count
- TARGET_FID
- STR
- ONR
- ADRESSE
- BEZ
- YR (kWh/kWp)
- Anteil Sehr gut
- Slope_MEAN
- Dachtyp
- Dachfläche
- Schlecht
- Mittel
- Gut
- SehrGut
- Gut+ Sehr Gut Flcähe
- Anlagenleisting [kWp]
- m2Dach/kWp
- kWp Gut
- kWp Sehr Gut
- kWh/kWp GUT
- kWh/kWp SEHR GUT
- kWh/kWp Insgesamt
- Delta zu SPK
- kWp GUT
- kWp Sehr Gut
- Belegunngsdichte Gut
  Belegungsdichte sehr gut
- Bezirk
- ORIG_FID
- Denkmalsch
- Shape_Leng
- Shape_Area
-

Blatt: PV Profile

- Ausrichtung
- Süd
- Ost
- West
- Ost.1
- West.1
- Horizontal Ost
- Horizontal West
- Gesamtquartier
- Unnamed: 9
- Unnamed: 10
- Unnamed: 11
- Unnamed: 12
- Unnamed: 13

6.3.3 Blatt: Zuordnung ID-Adr-Gstk

- PV ID
- Adresse
- BC
- Gst. Nr.
- Building_Code
- Adresse1
- Adresse2
- Adresse3
- Grundstücksnr.1
- Grundstücksnr.2
- TARGET_FID
- STR
- ONR
- ADRESSE

6.4 Attribute des Wärmeatlas

Name Ergebnisattribut	Attribut Bezeichnung	Attribut Beschreibung	Attributstyp	Attribut Einheit
Bauwerks-ID	ID der Gebäudepolygone	Eindeutiger Identifikator der Gebäudepolygone	-	[-]
BGF Wohnen beheizt	Bruttogeschoss-fläche	beheizte Bruttogeschossfläche, die auf Wohnnutzungen entfällt	stetig	[m^2]
BGF andere Nutzungen beheizt	Bruttogeschoss-fläche	beheizte Bruttogeschossfläche, die auf andere beheizte Nutzungen entfällt	stetig	[m^2]
Summe BGF beheizt	Bruttogeschoss-fläche	gesamte beheizte Bruttogeschossfläche	stetig	[m^2]
BGF unbeheizt + beheizt	Bruttogeschoss-fläche	gesamte Bruttogeschossfläche (Vollgeschoß und Teilgeschoß) ohne Differenzierung, ob beheizt oder nicht beheizt	stetig	[m^2]
Grundfläche	Außengrundfläche Hauptgebäude	untere Begrenzungsfläche des konditionieren Volumens, Shape Area des Gebäudepolygons	stetig	[m^2]
Gebäudehöhe	Firsthöhe	Maximale Gebäudehöhe	stetig	[m]
Gebäudevolumen	Bruttovolumen	konditioniertes Bruttovolumen	stetig	[m^3]
Bruttogrundfläche	Bruttogrundfläche	gesamte Bruttogrundfläche (Vollgeschoß und Teilgeschoß) ohne Differenzierung, ob konditioniert oder nicht	stetig	[m^2]
Außenwand-flächen	Außenwandfläche	alle Hüllfächen des konditionieren Volumens die nicht zu GF oder DF gezählt werden; reduziert um angrenzende konditionierte Flächen	stetig	[m^2]
Dachfläche	Dachfläche	obere Begrenzungsfläche des konditionieren Volumens	stetig	[m^2]
Hüllfläche	Hüllfläche konditioniert	ga_hf = ga_fl + ga_af + ga_df (gemäß Ö-Norm B1800)	stetig	[m^2]
Geschoss-anzahl	Anzahl der Geschosse		stetig	[-]
A/V Verhältnis	A/V Verhältnis	Verhältnis: Hüllfläche / Volumen	stetig	[1/m]
Kompaktheit	Kompaktheitsklasse (Code)	Klassifikation des A/V Verhältnisses	Klassifikation	[-]
Kompaktheit	Kompaktheitsklasse (Text)	Klassifikation des A/V Verhältnisses	Klassifikation	[-]
Bauperiode	Bauperiode des Gebäudes (Code)	Die Bauperiode ist jener Zeitraum, in dem das Errichtungsdatum des Gebäudes liegt.	Klassifikation	[-]

Bauperiode	Bauperiode des Gebäudes (Text)	Die Bauperiode ist jener Zeitraum, in dem das Errichtungsdatum des Gebäudes liegt.	Klassifikation	[-]
Baujahr	Baujahr des Gebäudes (wenn vorhanden)	Das Errichtungsdatum ist jener Zeitpunkt, zu dem das Gebäude seinem Bestimmungszweck entsprechend genutzt werden kann.	stetig	[jjjj]
Jahr der Sanierung	Saniuerungsjahr des Gebäudes (wenn vorhanden)	Datum oder Jahr der Sanierungsmaßnahme	stetig	[jjjj]
Schutzkategorie	Schutzkategorie des Gebäudes (Code)	Vorhandener Gebäudeschutz am Gebäude (Denkmalschutz, Altstadtschutz, Ortsbildschutz)	Klassifikation	[-]
Schutzkategorie	Schutzkategorie des Gebäudes (Text)	Vorhandener Gebäudeschutz am Gebäude (Denkmalschutz, Altstadtschutz, Ortsbildschutz)	Klassifikation	[-]
Sanierungsstatus	Sanierungsstatus des Gebäudes (Code)	Kategorien bzgl. Sanierungsstand	Klassifikation	[-]
Sanierungsstatus	Sanierungsstatus des Gebäudes (Code)	Kategorien bzgl. Sanierungsstand	Klassifikation	[-]
Energieträger Raumheizung 1	Energieträger 1 für die Raumheizung (Code)	Art des Energieträgers für die Raumheizung	Klassifikation	[-]
Energieträger Raumheizung 1	Energieträger 1 für die Raumheizung (Text)	Art des Energieträgers für die Raumheizung	Klassifikation	[-]
Energieträger Raumheizung 2	Energieträger 2 für die Raumheizung	Art des Energieträgers für die Raumheizung	Klassifikation	[-]
Energieträger Raumheizung 3	Energieträger 3 für die Raumheizung	Art des Energieträgers für die Raumheizung	Klassifikation	[-]
Gerät Wärmebereitstellung	Art Raumheizungssystem 1 / Gerät Wärmebereit-stellung (Code)	Beschreibt, welche Geräte zur Wärmebereitstellung im Einsatz sind	Klassifikation	[-]
Gerät Wärmebereitstellung	Art Raumheizungssystem 1 / Gerät Wärmebereit-stellung (Text)	Beschreibt, welche Geräte zur Wärmebereitstellung im Einsatz sind	Klassifikation	[-]
Gerät Wärmebereitstellung 2	Art Raumheizungssystem 2 / Gerät Wärmebereit-stellung	Beschreibt, welche Geräte zur Wärmebereitstellung im Einsatz sind	Klassifikation	[-]
Gerät Wärmebereitstellung 3	Art Raumheizungssystem 3 / Gerät Wärmebereit-stellung	Beschreibt, welche Geräte zur Wärmebereitstellung im Einsatz sind	Klassifikation	[-]
Hauptnutzung	Hauptnutzung des Gebäudes (Code)	Unterscheidung der Gebäude nach der vorwiegenden Nutzung. Bei mehreren vorliegenden Nutzungen wird die Hauptnutzung nach einer Nutzungshierachie zugewiesen	Klassifikation	[-]
Hauptnutzung	Hauptnutzung des Gebäudes (Text)	Unterscheidung der Gebäude nach der vorwiegenden Nutzung. Bei mehreren vorliegenden Nutzungen wird die Hauptnutzung nach einer Nutzungshierachie zugewiesen	Klassifikation	[-]

Detailnutzungen	Detailnutzungen des Gebäudes (Code)	Liste aller Nutzungen, die auf dem Gebäude vorkommen	Klassifikation	[-]
Detailnutzungen	Detailnutzungen des Gebäudes (Text)	Liste aller Nutzungen, die auf dem Gebäude vorkommen	Klassifikation	[-]
Anzahl aller Nutzungen	Anzahl der Nutzungen im Gebäude	Anzahl der Nutzungseinheiten je Gebäude (aus ZMR bzw. AGWR)	stetig	[-]
Anzahl Wohnnutzungen	Anzahl der Wohnnutzungen im Gebäude	Anzahl der Nutzungseinheiten mit Wohnnutzung je Gebäude (aus ZMR bzw. AGWR)	stetig	[-]
Spezifischer Wärmebedarf	Spezifischer Wärmebedarf	Spezifischer mit Verbrauchsdaten kalibrierter Wärmebedarf (heat delivered nicht gleich Endenergie) hd_loc / bgf_sum_be	stetig	[kWh/m^{2}a]
Raumwärme- und Warmwasserbedarf	Wärmebedarf Raumwärme und Warmwasser	Mit Verbrauchsdaten kalibrierter Wärmebedarf (heat delivered nicht gleich Endenergie)	stetig	[kWh/a]
Warmwasserbedarf	Wärmebedarf Warmwasser	Mit Verbrauchsdaten kalibrierter Wärmebedarf (heat delivered nicht gleich Endenergie)	stetig	[kWh/a]
Raumwärmebedarf	Wärmebedarf Raumwärme	Mit Verbrauchsdaten kalibrierter Wärmebedarf (heat delivered nicht gleich Endenergie)	stetig	[kWh/a]
Heizlast	Heizlast gesamtes Gebäude		stetig	[kW]
Kühllast	Kühllast gesamtes Gebäude		stetig	[kW]

B.10. From Fossil Fuel Dependency to Renewable Sufficiency: Predictive Energy Modelling for District Decarbonization Scenarios

Publication type: Conference paper

Authors: Simon Schneider, Raphael Drexel, and José Baptista.

Original publication:

Schneider, Simon, Raphael Drexel, and José Baptista. „From Fossil Fuel Dependency to Renewable Sufficiency: Predictive Energy Modelling for District Decarbonization Scenarios“. Conference paper presented at *CESB25 – Central Europe towards Sustainable Building 2025*. 17. September 2025. https://cesb.cz/cesb25-proceedings/.

Contribution to the thesis:

Comparative modelling approach combining building-level simulations using a secondary engine (City Energy Analyst, CEA) with district-scale assessment using PEExcel. Cross-Validation of aggregation logic and boundary assumptions across modelling scales, highlighting consistency limits between bottom-up building simulations and district energy balance frameworks.

From fossil fuel Dependency to renewable Sufficiency: Predictive Energy Modelling for District Decarbonization Scenarios

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Abstract. This study demonstrates the use of the City Energy Analyst (CEA) tool as a powerful method for modelling decarbonization scenarios in urban districts, particularly those with historic and varied building types. By combining accurate building geometry modelling from CEA with a robust, adaptable building typology from literature, this approach enables detailed analysis of energy performance and decarbonization potential across diverse urban areas. Key analyses focus on transitioning from gas to district heating, implementing heat pumps and thermal renovation based on specific building geometries, orientations and compactness.

1. Introduction

Urban Building Energy Models (UBEMs) can act as Decision Support Systems (DSS) by providing assessments on the effectiveness of retrofitting measures for the stakeholders involved. However, practitioners are often faced with the challenge of limited information on the necessary parameters for UBEM simulations, subsequently the required early integration of energy planning in the district development is lacking [9,23]: Even if sufficient data is available and modelling possible, using district simulation models can be too resource-intensive to deploy early, as the planning will inevitably change in the process, rendering detailed simulation results invalid [26]. Furthermore, assuming building data based on shared characteristics is error-prone, has potentially low accuracy, which can sometimes be increased by clustering-based approaches [11,14,17,24].

Modeling cities as 3D “digital twins” promises to provide highly detailed information for planning processes at the building level through a model encompassing urban objects linked to various datasets, including population trends, energy usage, maintenance management, sensor and real-time information, EPCs and material building passports [6]. In practice, it is rarely achieved due to complexity and resource intensity [18,22,25] and in particular semantic interoperability [25]. Special formats such as CityGML were put forward to address this, but still suffer from barriers in Tooling and complex data structures, that are further exacerbated by the necessity for application domain extensions for use cases such as energy modelling [1,7,20].

Another issue lies in the lack of dynamic models for climate neutrality assessment of energy flexible districts with suitable timesteps [5], which is necessary to assess its flexible operation. Studies have stated the importance of suitable timesteps in assessing energy flexibility [12]. Most

sources agree that stationary calculation of energy balances is insufficient due to its inability to resolve flexible control measures and interplay between district demand and onsite RES. High temporal resolutions on the other hand typically come with long compute times and resulting unwieldy models. The sweet spot for this application seems to be an hourly resolution on the scale of a full operational year [2,10].

City Energy Analyst (CEA) is an open source UBEM employing a bottom-up, physics-based RC model yielding hourly results [4], featuring open source databases on building typologies, which exist for Switzerland, Singapore, and recently Germany [3]. It uses building specific domain configurations as "types" to structure the input data required for irradiation, thermal and electrical energy balancing of heating, cooling, DHW, auxiliary and user electricity, as well as onsite renewable generation and operation properties of heating networks, which are well suited for re-use and re-arrangement of predefined types.

This study aims to **model decarbonization scenarios** for the case study of a dense urban vintage District in Vienna, Austria using the CEA software to investigate its potential as a DSS for municipal stakeholders. According to municipal decarbonization plans for 2040, a complete phase-out of gas heating is envisioned, not only for new buildings but also through comprehensive thermal and energy-efficient retrofitting of the existing building stock. The final energy consumption for heating, cooling, and hot water in buildings is to be reduced per capita by 20% by 2030 and by 30% by 2040, compared to the average value from 2005–2010. The associated CO_2 emissions are to decrease per capita by 55% by 2030 and reach net zero by 2040.

2. Methods and Data

The investigated district and its collected data on 76 existing buildings is shown Figure 1, indicating plausibility, energy supply, building condition and initial construction period as per municipal cadaster. Manual Data inspection based on site visits and satellite and Streetview

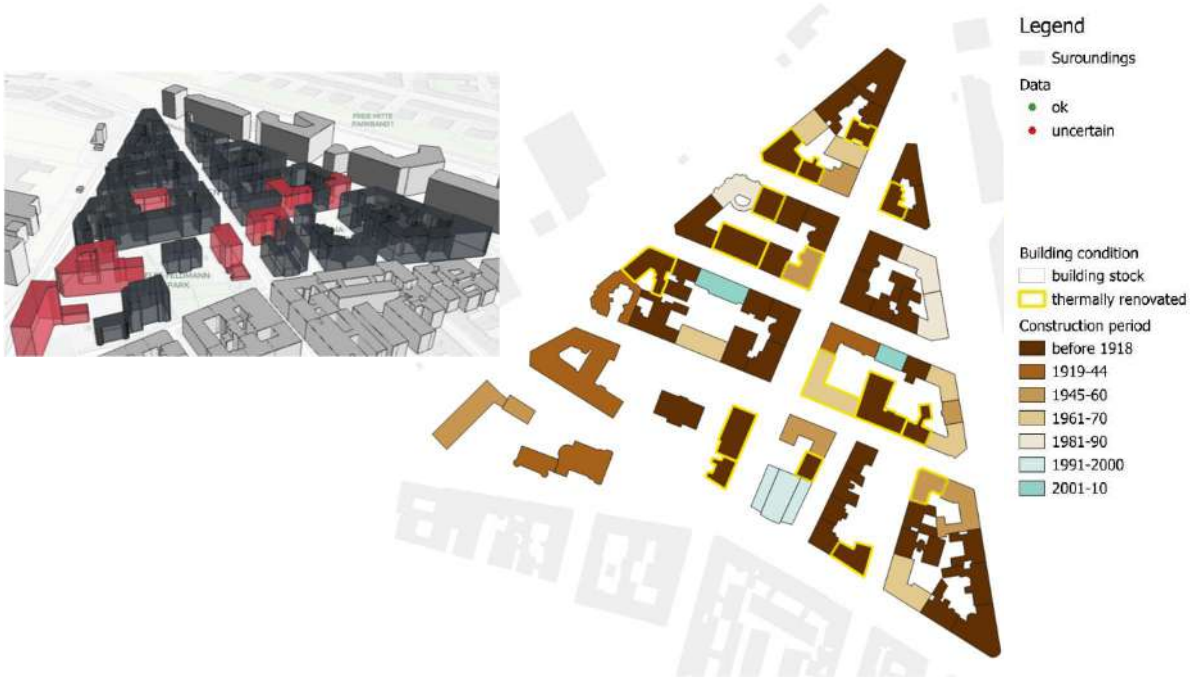


Figure 1. Left: 3D building geometry model in CEA, right: Collected district data for 76 buildings: Initial Construction Period, estimated Building Condition. Uncertainties in the initial dataset are shown as green/red dots

images indicated differences to the dataset such as unregistered window replacements and deviations in building heights, suggesting outdated or erroneous entries in the municipal cadaster, which were adjusted to the best of the authors' abilities. More than half of the 76 buildings in the district originated before 1918. About 30% of the buildings are mixed use with ground floor in non-residential use and the remaining district (69%) is residential. Currently, most of the district is fueled by natural gas, the remaining buildings are supplied by district heating and one lighthouse building renovation uses heat pumps sourced by ground water.

Four investigation scenarios were co-created with municipal stakeholders: **(0)** baseline conditions, **(1, BAU**) a phased, realistic retrofit approach representing business as usual with a 1.8%p.a. renovation rate, **(2, "Off-Gas")** targeting complete HVAC replacement of fossil Gas-Boilers to district heating and third **(3, "WAM")** a scenario representing the limit of feasibility with additional measures achieving nearly-zero-energy level Passive House standard in renovation for most buildings. Scenarios were simulated in CEA, and resulting heat demands validated by comparison with statistical top-down consumption data and secondary simulation results for individual buildings using an alternative single-zone RC-model [15,16]. The resulting energy end use for heating, cooling and DHW was extended with user plug loads from literature and separate hourly PV yield simulations and converted into primary energy and GHG emission equivalents for analysis and comparison with municipal goals.

2.1 Building energy modelling with CEA

Building modelling was carried out using open-source software City Energy Analyst (CEA), which requires definition of component and assembly types to be associated with each building geometry. The types were derived using three classifications consistent with existing structures in Austrian building code and literature: Construction period, building quality/renovation standard, and main energy supply, forming a consistent framework translated into CEA category types enabling compatibility and future scalability. The specifications and CEA type definition files can be found in the supplementary data.

Construction Period defines unchangeable building parameters such as window-to-wall ratios and building physics characteristics such as thermal mass. Based on the district data, seven types of Construction Periods are defined specifying baseline properties such as U-values, air

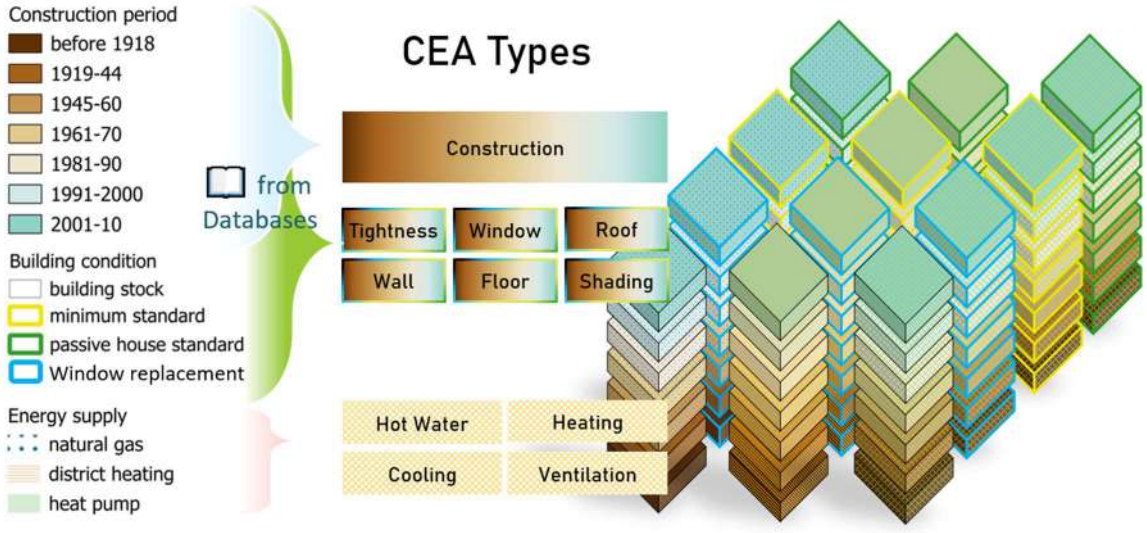


Figure 2. CEA Building Typology generation from available cadaster data (left), providing 84 unique building archetypes from 7 construction types, 4 building conditions/hull qualities and 3 energy supply systems. Specifications and CEA types can be found in the supplementary data.

tightness, and window-to-wall ratios according to Austrian building code classifications and other literature.

Building Condition determines the building physics properties and technical equipment features of the building, essentially characteristics subject to **potential change**. Five types were defined:

(1) **existing** condition of the building stock based on the EPISCOPE and Tabula [8,21]

(2) replacement of existing **windows** with SoA windows (thermal transmittance 1.1 W/m^{2}K)

(3) **minimum** thermal renovations based on Austrian building code [13]

(4) highly efficient passive house standard compliant retrofits.

Energy Supply defines the energy carrier(s) and the primary conversion process for delivering final energy within the building. Three Systems are defined as CEA types for Hot Water, Heating, Cooling and Ventilation and models focus on energy carriers and system efficiency, with options limited to district heating, natural gas and heat pumps.

By combining these three categories, each building geometry can be modelled with **84 possible configurations**, reflecting diverse districts and decarbonisation scenarios. The resulting 6132 single building simulations were grouped into four groups of compactness for sensitivity analysis.

Usage: Building usage types were adopted from the default CEA types based on Swiss SIA building norms [19], as they are also used in the Austrian assessment standard of Climate Neutral Positive Energy Districts [15].

Due to the high uncertainty of the true renovation status of individual buildings, simulated results of assumed data have low individual predictive value. To better accommodate inquiries and additional data from stakeholders and inhabitants and offer a catalogue of potential building performances to choose a given hull and HVAC system from, all building geometries were also simulated with all other defined typologies of construction period, renovation measures and HVAC systems. The 6384 results for 76 geometries with 84 archetypes were clustered into four groups of geometry compactness: “very high”: <0.6, “high”: 0.6-0.8, “medium”: 0.8-1 and “low”: >1], which were calculated with formula (1):

$$(1)\ Compactness = \frac{thermal\ \ Hull}{Gross\ floor\ area} = \frac{A_{roof} + A_{base} + A_{Facade}}{A_{GFA}}$$

3. Results

Figure 3 shows the resulting specific heat demand of 76 building geometries on the y-axis, plotted against the compactness on the x-axis, modelled for six initial construction periods in separate subplots, indicated by their background color, and for all four types of renovation measures, indicated by marker color: original building stock (black), window replacement (blue), minimum renovation according to Austrian building code (yellow) and Passive house renovation (green). The effect of renovation measures decreases for newer buildings as they already are of a higher standard. As expected, the results strongly correlate with the compactness. It indicates the potential spread for each combination of construction type, thermal hull and compactness and facilitates quick referencing for interested stakeholders.

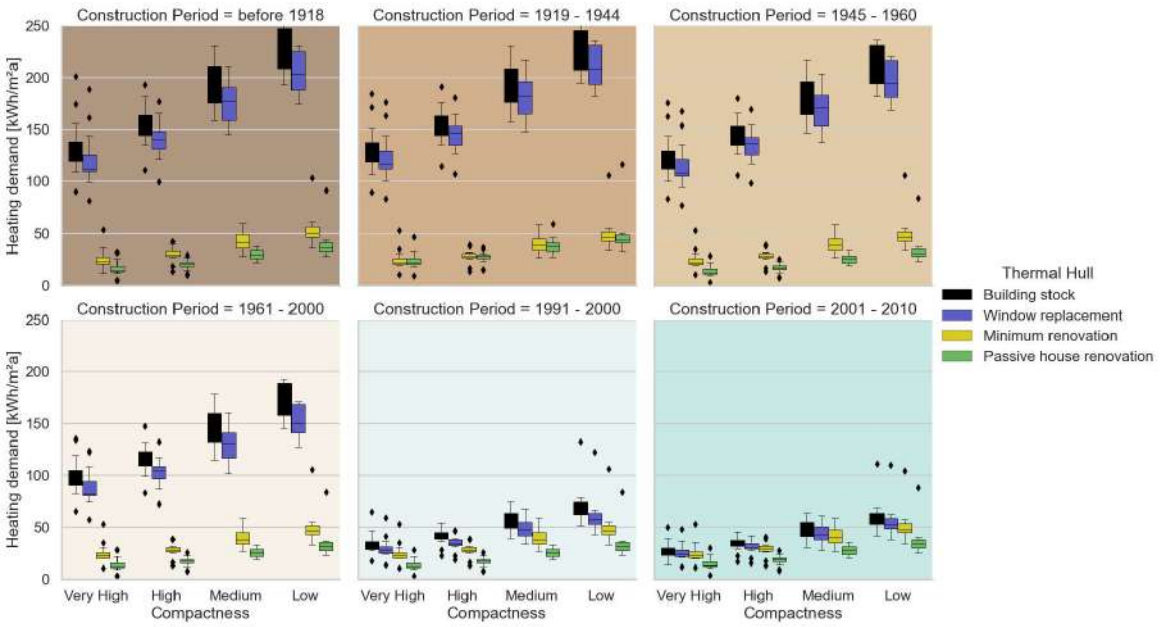


Figure 3. Heat demands for 6384 building configurations aggregated in four compactness categories as reference of potential savings for stakeholder and owners of individual buildings building archetypes from 7 constructions, 4 building conditions/hull qualities and 3 energy supply systems

Figure 4 shows the resulting heat demand, Primary Energy demand and GHG Emissions for building operation and use of the three investigated scenarios compared to the baseline. The aggregate specific heat demand, total primary energy demand and GHG emissions per gross floor area show that the municipal goal of reducing energy end use for heating by 30% until 2040 is achieved in all scenarios, including just business-as-usual (BAU) retrofitting rates, indicating lack of ambition. Note that the non-renewable primary energy and GHG emissions do not reach zero despite a complete switch to district heating in the rightmost two scenarios due to the use of current conversion factors as prescribed in [13]. The reduction in GHG emissions reaching net zero requires additional decarbonization of the surrounding electricity grid and district heating, which is not part of the building assessment in this framework.

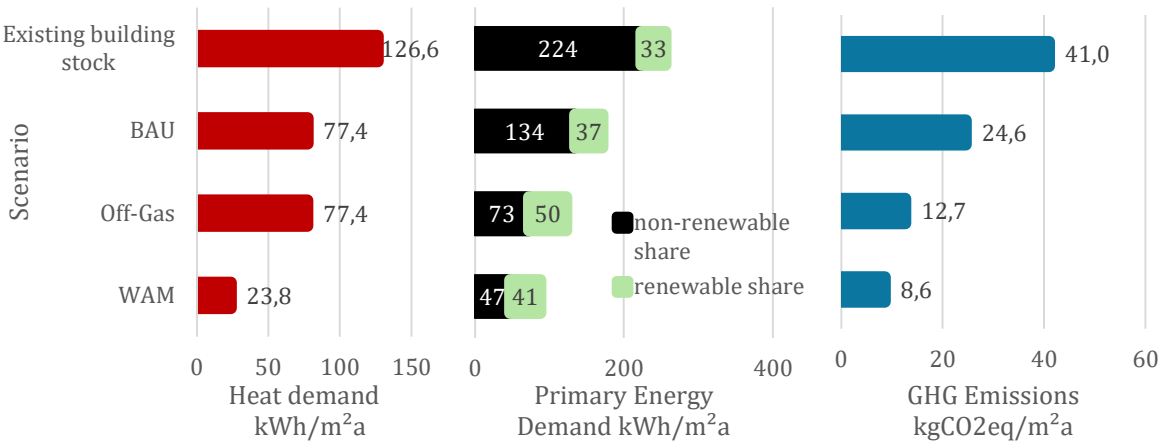


Figure 4. Heat demand, Primary Energy Demand and GHG Emissions of investigated Scenarios: Existing building stock, Business-as-usual (BAU), a complete HVAC switch to district heating ("Off-Gas") and a scenario with additional measures (WAM) and higher thermal retrofitting rates.

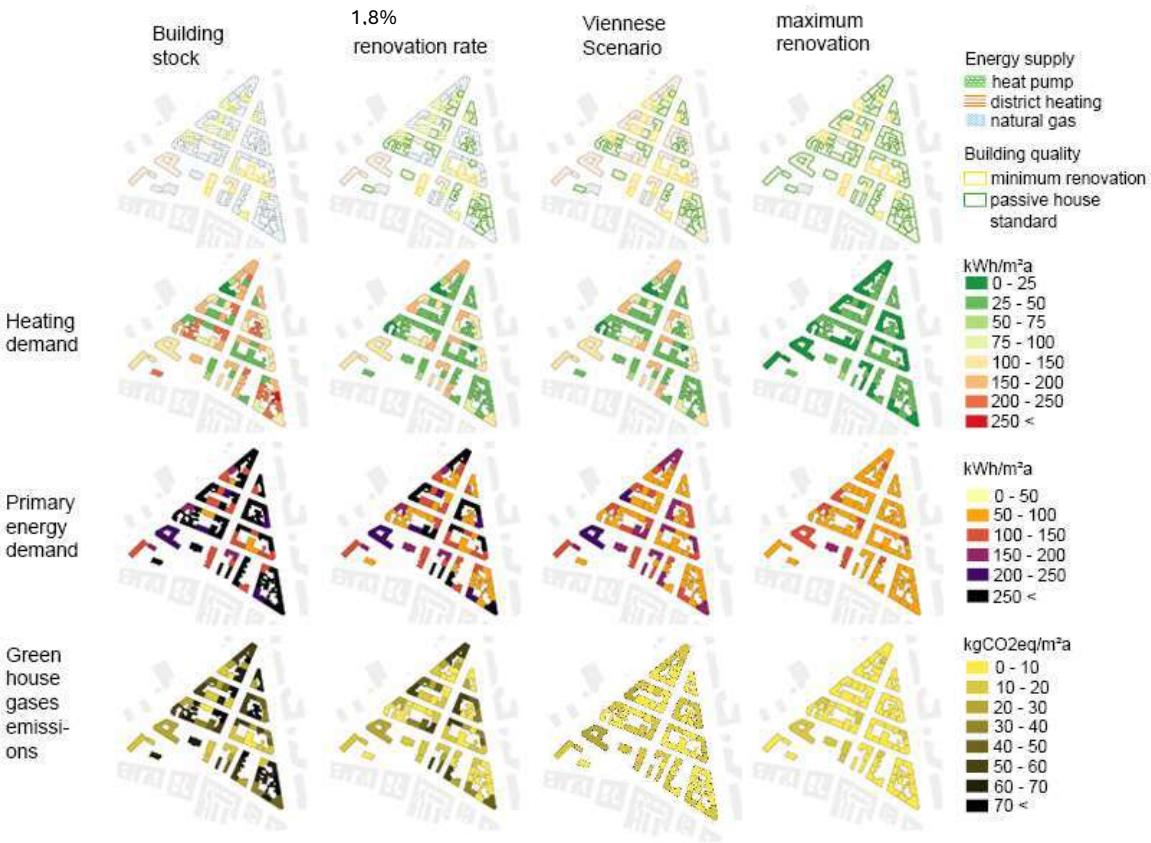


Figure 5. Scenario results per building: Heat demand, Primary Energy Demand and GHG Emissions (top to bottom) for scenarios: Existing building stock (left) , Business-as-usual (BAU, "2% renovation rate", middle-left), a complete HVAC switch to district heating ("Off-Gas", "Viennese scenario middle right) and a scenario with additional measures and higher thermal retrofitting rates ("WAM", "maximum renovation" right).

Figure 5 shows the assessment results in a GIS-based format per building and shows the potential of CEA as a DSS for municipal stakeholders. Since CEA inputs are GIS-based, the outputs too can be quickly analyzed and post-processed with common GIS-tools to navigate district scenarios and potential savings for individual buildings.

Discussion

District modelling and simulation with CEA

Using CEA, several uncertainties regarding the implementation of the model persist: First, changing ventilation type caused little change on the thermal energy balance, as indicated by the small difference between the minimal renovation and passive house renovation, where the former implements window ventilation and the latter a mechanical ventilation system with 75% heat recovery. Further analysis is needed.

Second, despite the principal possibility to simulate PV yields with CEA it was not employable in the study, as the system configurations cannot readily be changed to context-specific orientations considering real roof geometries.

Especially for highly efficient mixed-use buildings, CEA showed surprisingly low heat demands compared to similar simulation methods, potentially indicating overestimation of occupancy rates and subsequent higher internal gains and lower heat loss through ventilation or a combination of both. Further research and comparison with different district case studies is required. It should also be noted that the presented method makes use of building data compiled

and parametrized with the purpose of modelling Viennese building stock, rendering it not immediately replicable in other cities.

Nevertheless, CEA's ability to readily provide accurate thermal hulls for available 3D building geometry – given its accuracy – and combine it with pre-defined building stock typologies leads to quicker modelling and simulation cycles.

Scope limitations to Climate Neutrality assessment

The presented study is limited to investigating building operation and use and is omitting other aspects such as mobility and embodied emissions. It is a first step in integrating detailed building data into district scale assessments for climate neutrality and is well suited for future extension with the pursued framework of Austrian climate-neutral Positive Energy Districts featuring three expanding functional system boundaries: (1) Operation and Use, (2) Motorized individual Mobility and (3) Embodied Energy and Emissions over the life cycle.

Conclusion

The study showcases the use of City Energy Analyst to explore vintage, heterogeneous district decarbonization pathways that can be used for municipal stakeholders in communicating and implementing the district decarbonization plan. It adds re-usable components to future UBEM studies with the defined 84 typologies.

The study found that the municipal goals for reducing energy use intensity in the built environment do not go significantly beyond what is already happening by current trends in retrofitting to code.

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B.11. Declaration-Ready Climate-Neutral PEDs: Budget-Based, Hourly LCA Including Mobility and Flexibility

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Contribution to the thesis:

Declaration-ready assessment framework, Method of Emission balancing and budget derivation.

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Declaration-Ready Climate-Neutral PEDs: Budget-Based, Hourly LCA Including Mobility and Flexibility

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Abstract

In recent years, Positive Energy Districts (PEDs) have been interpreted in many—and often conflicting—ways. We recast PEDs as a vehicle for verifiable climate neutrality and present a declaration-ready assessment that integrates (i) a cumulative, science-based GHG budget per m^2 gross floor area (GFA), (ii) full life-cycle accounting, and (iii) time-resolved conversion factors that include everyday motorized individual mobility and quantify flexibility. Two KPIs anchor the framework: the cumulative GHG LCA balance (2025–2075) against a maximum compliant budget of 320 $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$ and the annual primary energy balance used to declare PED status with or without mobility. We follow EN 15978 and apply time-resolved emission factors that decline to zero by 2050. Its applicability is demonstrated on six Austrian districts spanning new builds and renovations, diverse energy systems, densities, and mobility contexts. The baseline scenarios show heterogeneous outcomes—only two out of six meet both the cumulative GHG budget and the positive primary energy balance—but design iterations indicate that all six districts can reach the targets with realistic, ambitious packages (e.g., high energy efficiency and flexibility, local renewables, ecological building materials, BESS/V2G, and mobility electrification). Hourly emission factors and flexibility signals can lower import-weighted emission intensity versus monthly or annual factors by up to 15% and reveal seasonal import–export asymmetries. Built on transparent, auditable rules and open tooling, this framework both diagnoses performance gaps and maps credible pathways to compliance—steering PED design away from project-specific targets toward verifiable climate neutrality. It now serves as the basis for the national labeling/declaration scheme klimaaktiv "Climate-Neutral Positive Energy Districts".

Keywords: Positive Energy Districts (PEDs); climate neutrality assessment; greenhouse gas (GHG) emissions; life-cycle assessment (LCA); embodied and operational emissions; building energy systems; mobility emissions

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1. Introduction

Achieving climate neutrality by mid-century, as stipulated under the Paris Agreement [1], requires rapid and deep decarbonization across all sectors. The IPCC Sixth Assessment Report confirms that limiting warming to 1.5 °C necessitates stringent carbon budgets, with global net-zero CO_2 emissions reached around 2050 [2–5]. The building sector is pivotal in this transition, contributing significantly to both operational greenhouse gas (GHG) emissions and embodied emissions from construction and refurbishment [6].

Over the past decade, ref. [7] have identified more than 25 different concept names for climate-friendly or energy-positive districts or neighborhoods. Most of these approaches emphasize improved efficiency and renewable integration at the energy-system level but vary widely in scope and lack consistent definitions—a critique common to many studies. As a result, many remain difficult to apply in practice, are hard to replicate, and have seen limited regulatory uptake. Among this broad landscape, four framings are of particular interest for this study: NZEB definitions, their extensions into renewable energy communities, the Zero-Emission Neighborhood (ZEN) framework, and Positive Energy Districts (PEDs). Each of these has shaped the discourse on carbon-neutral or energy-positive districts, but all remain closely tied to annual balances or ambition levels rather than finite, Paris-aligned carbon budgets:

- **NZEB and Net-Zero Definitions.** Classic NZEB frameworks focus on annual site/source primary energy balances, with variants on cost and emissions neutrality [8]. Their common core is a 12-month energy bookkeeping, sometimes coupled to EPB standards (EN/ISO 52000-1) for primary energy weighting [9].
- **NZEBs in Renewable Energy Communities (RECs) and energy sharing.** Recent work extends NZEB logic to building-to-building exchange, where prosumer surpluses buffer consumer deficits inside a REC [10]. While this can improve local self-consumption and grid friendliness, it remains annual-balance driven and typically does not enforce finite, Paris-compatible CO_2 budgets.
- **Zero-Emission Neighborhoods (ZEN).** The Norwegian ZEN program defines neighborhood-scale goals "towards zero life-cycle GHG emissions," explicitly emphasizing life-cycle accounting, high energy efficiency, high shares of local renewables, and smart/flexible operation. ZEN provides ambition levels and guidance but leaves budget allocation and declaration rules to localization [11]
- **Positive Energy Districts (PEDs).** IEA EBC Annex 83 positions PEDs as districts with net annual energy surplus and operational flexibility. Many PED implementations, however, still report annual net balances without translating them into cumulative, per-m^2 CO_2 budgets aligned with Paris [12].

Positive Energy Districts (PEDs) are increasingly promoted as a transformative model to support this transition, integrating high energy efficiency, local renewable generation, and energy flexibility to achieve an annual net-positive energy balance [13–16].

The PED Database compiled well over 50 projects on a self-reporting basis and showed that PED projects across Europe differ widely in energy performance, approach, and ambition and considered system boundaries [17]. This highlights the need for further clarification of the definition of PEDs, especially taking into account the specific context of the district (see, e.g., [18]). The increasing integration of volatile renewable energy (VRE) could significantly benefit from widespread demand-side response [19], but many actors in the field consider the assessment of energy flexibility in the context of a PED an important, yet unresolved question [20].

By contrast, climate neutrality requires aligning emissions with a finite, science-based GHG budget that limits global warming to 1.5 °C (or well below 2 °C), necessitating a lifecycle-based approach that captures embodied emissions, biogenic carbon flows, and time-resolved operational emissions [6]. Additionally, ref. [21] point out that low or zero carbon buildings require clear decarbonization targets as early as in the design phase.

Recent work has also examined the broader challenges of implementing energy-efficient and climate-neutral districts. For instance, ref. [22] identify governance structures, financing models, and integration of building- and district-scale technologies as critical enablers of high-performance districts. While their focus is on efficiency strategies, the findings underscore that achieving Paris-compatible Positive Energy Districts requires not

only carbon-budget compliance, as proposed here, but also systemic strategies to overcome institutional and financial barriers.

At the PED scale, three main challenges emerge:

1. Defining system boundaries that distinguish emissions under the direct control of developers from those managed at municipal or national scales [14,23–25].
2. Allocating GHG budgets consistently with global and national targets [4,26,27].
3. Accounting for temporal aspects, including the disproportionate climate impact of early emissions [6,28].

Life cycle assessment (LCA) standards, particularly EN 15978:2024 [29], provide a robust structure for defining system boundaries and modules for embodied and operational emissions in buildings and districts. Previous work [30–32] highlights the need to adapt these methods to the district scale while ensuring compatibility with sectoral carbon budgets. Dynamic emission factors—which reflect the projected decline in grid and district heating carbon intensities—are essential for accurately representing operational impacts over time [33]. International studies [34,35] demonstrate the influence of such temporal factors on lifecycle results, and highlight differences in the treatment of embodied emissions of energy infrastructure.

Novelty and Contribution of This Study

This study extends the established Positive Energy District (PED) assessment framework [24,36] by operationalizing climate neutrality through life-cycle carbon accounting and science-based carbon budgets. It (i) applies a life cycle GHG accounting method consistent with national decarbonization pathways, (ii) incorporates time-differentiated operational emission factors for multiple energy carriers, and (iii) integrates everyday motorized individual mobility as part of project-level design.

Building on EN 15978 system boundaries, it integrates dynamic, energy-carrier-specific emission factors and operationally includes everyday motorized individual mobility (EMIM) and embodied emissions from construction and maintenance from now until 2050. Within the broader assessment framework, this paper addresses the climate neutrality and life-cycle carbon accounting dimension, complementing the framework's energy system modeling and flexibility evaluation capabilities presented in earlier work [24,36]. The shared objective is to provide a methodologically rigorous yet practice-ready approach that captures both (i) the operational responsiveness of PEDs to system needs and (ii) their compliance with carbon budgets consistent with Austria's 2040 climate neutrality target.

The framework presented here is designed to be robust for research purposes—allowing sensitivity analyses with hourly emission factors—while remaining compatible with the monthly conversion factors of the Austrian building code for practical declarations of climate-neutral PEDs as part of the Austrian sustainability label family klimaaktiv [37]. Its core contribution is the derivation of a quantifiable carbon budget per square meter of gross floor area (GFA), integrating both operational and embodied emissions in a manner that is directly linkable to the performance outcomes of the broader PED assessment framework.

The novelty of this work consists of these three parts:

1. The integration of operational, embodied, and mobility energy and emissions into a combined assessment system with quantitative target budgets
2. The assembly of these factors into a nationally applicable declaration system.
3. The demonstration of achievability of this declaration for a diverse range of ambitious districts.

2. Materials and Methods

The methods build on previous work on primary energy-based PED definitions and extend them with a life-cycle, budget-oriented approach. To structure the framework, the development process and the subsequent application are separated. The development side describes the design choices, boundaries, and assumptions that define the assessment logic, while the application side illustrates how the framework is implemented using project data and simulation tools. This dual view ensures the transparency of our methodological decisions and shows how the framework translates into declaration-ready results. Figure 1 provides a schematic overview of the framework development and application.

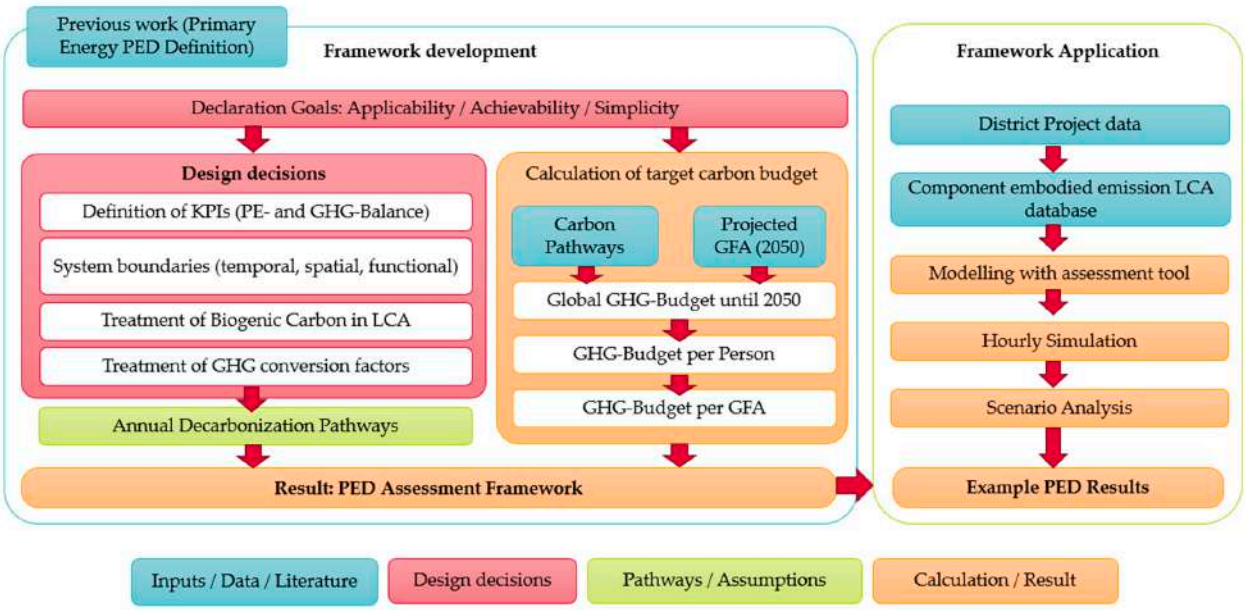


Figure 1. Overview of the climate neutrality assessment framework. The left-hand side illustrates the framework development process, including declaration goals, design decisions, carbon budget derivation, and integration of decarbonization pathways. The right-hand side shows the application to district projects, linking data inputs, modeling, and simulation to example results. Colors indicate categories: inputs/literature, design decisions, assumptions, and results.

2.1. Methodological Principles and Declaration Goals

The proposed climate neutrality assessment framework is grounded in a set of guiding principles that combine practical applicability with scientific rigor, ensuring relevance for both practitioners and researchers.

- Practical applicability and stakeholder relevance—The method is designed to be appliable for real estate developers, municipalities, and planning professionals, enabling its integration into certification schemes and project assessments [38,39]. Its scope is limited to emissions that can be directly influenced through planning and construction decisions—namely, operational energy use, EMIM, and embodied emissions—ensuring decision-making relevance.
- Integrated and adaptable system boundaries—Assessment boundaries extend beyond individual buildings to the full Positive Energy District (PED) system, covering on-site generation, storage, flexibility, and interconnections with surrounding grids. The framework retains adaptability in data sources, emission factors, and system configurations, allowing regional customization while ensuring methodological comparability.
- Temporal differentiation—Emissions are assessed over a 50-year timeframe (2025–2075), with the assumption that climate neutrality is achieved for all services by 2050. As a result, all conversion factors for both operational and embodied energy are assumed

to be zero by 2050, reflecting the expected decarbonization of energy carriers and building systems. This approach ensures alignment with long-term climate goals while considering the gradual reduction in emissions over the assessment period.

- Science-based, budget-oriented compliance thresholds—The method is anchored in downscaled CO_2 budgets consistent with Austria's 2040 climate neutrality target and the 1.5 °C global carbon budget [1]. It establishes a hierarchy of compliance thresholds: an ambitious target value representing a precautionary level for high-confidence Paris alignment, and a maximum compliance limit that must not be exceeded for a project to qualify as "climate-neutral PED."

These principles acknowledge that PED definitions inherently allow for multiple interpretations, particularly regarding system boundaries, included emission sources, and treatment of temporal dynamics. As noted in prior work [24,40], these degrees of freedom can lead to inconsistent assessments if not addressed explicitly. The framework therefore applies a design thinking approach to close such gaps: key methodological choices—such as which life cycle stages to include, how to allocate renewable generation, and how to integrate mobility—are made transparent, justified against policy and scientific references, and consistently applied. Figure 2 summarizes the methodological approach:

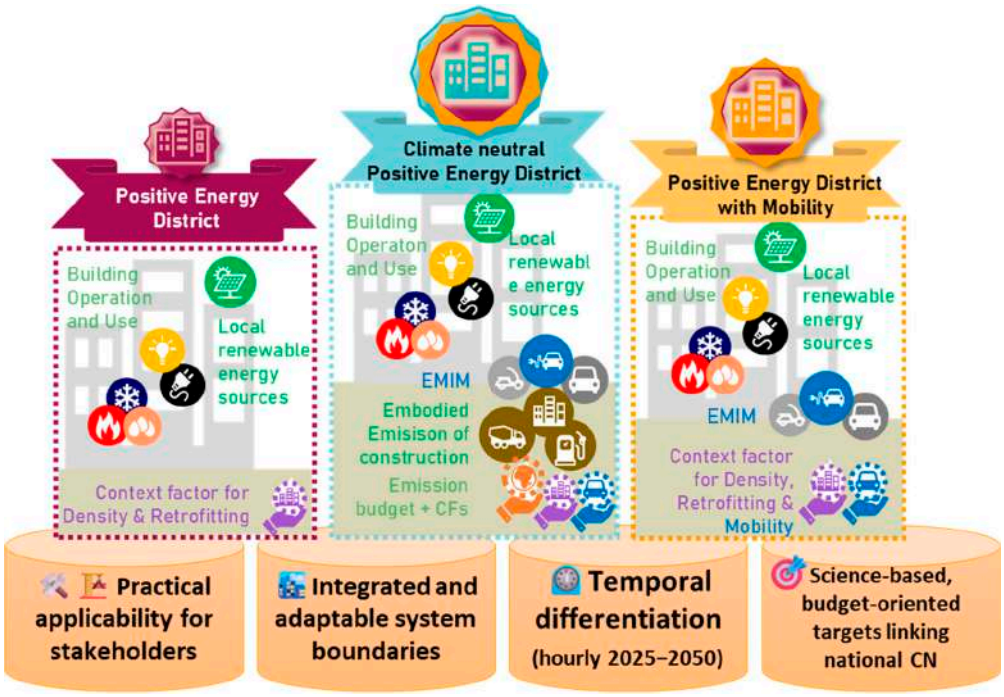


Figure 2. Methodological approach to design of climate neutrality definition for PEDs is based on four pillars, on which three practical PED definitions are based (boxes show conceptualized functional system boundaries).

The following section details how these methodological principles are operationalized within the proposed framework, from KPI definition and the derivation of target carbon budgets to the application of dynamic emission factors and the inclusion of mobility and embodied emissions.

2.2. Key Performance Indicators (KPIs) and PED Levels

We base our assessment framework on two main indicators as defined in [24]:

1. **Cumulative GHG balance** (2025–2075, $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$)—as defined in the framework above: cradle-to-grave embodied emissions (A1–A3; buildings, TES, PV, batteries, vehicles, C1–C4 before 2050), time-resolved operational emissions (B6; grid electricity

and thermal carriers), operational mobility (B8 EMIM; ICE/EV), and avoided grid emissions (D2; PV exports and EMIM substitution). Negative components reduce the balance. Compliance bands are L3 = 320, L2 = 196, L1 = 72 $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$.

2. **Primary energy balance** ($kWh_{pe} \cdot m^{-2}{}_{NFA} \cdot a^{-1}$)—computed per [24,36] as the difference between annual primary energy grid equivalence from local renewable supply and flexible grid use (PV self-use and feed-in, flexible TES and EMIM operation) and the district's primary energy demand (grid equivalence, HVAC and electricity; EMIM included or excluded depending on the level). A positive value denotes a "positive balance". Context factors for density, mobility and renovation may apply as virtual demands or supplies.

These two KPIs are reported consistently across case studies; figures annotate overshoot/undershoot against L-levels, and tables list the corresponding PE balances. Operational energy KPIs are expressed per conditioned Net Floor Area (NFA) because heating, cooling, and end-use electricity demands are specified for conditioned area in design practice and codes, so NFA best matches how these targets are set and tracked. Lifecycle GHG indicators are expressed per Gross Floor Area (GFA) to maintain comparability across typologies and density/fit-out choices; NFA↔GFA conversions are straightforward via use-type factors that are routinely applied (and refined from early planning estimates). Based on these indicators, we distinguish three PED levels as shown in Table 1:

Table 1. Types of PED as defined in [24].

Type of PED	Acronym	Requirements
Positive Energy District	PED	PE Balance > 0 excl. EMIM
Positive Energy District with Mobility	PED + M	PE Balance > 0 incl. EMIM
Climate-Neutral Positive Energy District (reported in this study)	CN PED	PE Balance > 0 incl. EMIM GHG LCA $\leq$ L3 over 2025–2075

Results are compiled with the scenario analysis tools provided by the framework (publicly available here [41]).

2.3. Carbon Budget for Climate-Neutral Districts

District-level carbon budgets are derived by downscaling a finite, Paris-compatible global CO_2 budget to Austria and further allocating a sectoral share to buildings, including construction, operation, and everyday motorized mobility (EMIM). This ensures consistency between global climate targets and district-scale declarations.

The calculation follows three steps. First, we adopt the +1.5 °C scenario with 66% probability of success, both with and without temporary temperature overshoot, and subtract materialized national emissions (2022–2024) from the 2022–2075 budget to obtain the available 2025–2075 budget. Second, we allocate a fixed share of the national budget (50%) to buildings and EMIM, expressed on a per capita basis. Finally, per capita values are translated into gross floor area (GFA)-specific thresholds using projected Austrian floor space statistics for 2050 (906.7 million m^2).

Natural carbon sequestration is included as a sensitivity, with additional sinks of 0.5 and 1.0 $t_{CO_2e}\ cap^{-1}\ a^{-1}$. These variants yield three levels of compliance:

- **L1** (Paris-compatible target): stringent limit without sequestration,
- **L2** (precautionary limit): moderate sequestration (0.5 $t_{CO_2e}\ cap^{-1}\ a^{-1}$),
- **L3** (absolute compliance limit): generous sequestration (1.0 $t_{CO_2e}\ cap^{-1}\ a^{-1}$).

Intermediate values and assumptions underlying this derivation are documented in Appendix A Table A1 for transparency. The resulting compliance thresholds, which serve as methodological reference points for district assessment, are summarized in Table 2.

Table 2. Target and compliance carbon budget limits for climate-neutral Positive Energy Districts (PEDs) in Austria for 2025–2075, expressed per gross floor area (GFA). These values represent the main methodological result of the budget downscaling process, derived from global Paris-compatible CO_2 pathways and Austrian building stock data.

Limit	Value	Unit	Description and Compliance Note
Paris-compatible target limit (GHG L1)	**72**	$kg_{CO_2e} \cdot m^{-2}{}_{GFA}$	Stringent, science-based carbon budget based on [2,27] ensuring a safe pathway to 1.5 °C. Assumes no contribution from biogenic carbon sequestration. Achieving this limit provides the highest certainty of Paris alignment but is challenging in practice.
Precautionary compliance limit (GHG L2)	**196**	$kg_{CO_2e} \cdot m^{-2}{}_{GFA}$	Upper bound for climate-neutral declaration, assuming 0.5 $t_{CO_2e} \cdot a^{-1} \cdot cap^{-1}$ sequestration. Offers flexibility while retaining a safety buffer; exceeding this value increases risk of overshooting Paris-compatible budgets.
Absolute compliance limit (GHG L3)	**320**	$kg_{CO_2e} \cdot m^{-2}{}_{GFA}$	Maximum permissible budget, assuming 1 $t_{CO_2e} \cdot a^{-1} \cdot cap^{-1}$ global sequestration. Exceeding this value disqualifies a PED from climate-neutral status under this framework.

2.4. LCA System Boundary and Scope of District

This section defines the temporal and functional boundaries for life-cycle assessment (LCA) within this proposed climate-neutrality assessment framework for PEDs. Unless stated otherwise, the assessment horizon is 2025–2075. All operational conversion factors decline to zero by 2050 for carriers on a national decarbonization pathway (e.g., electricity, district heating,); end-of-life and post-2050 processes are assumed carbon-neutral. As illustrated in Figure 3, the system boundaries include both operational and construction emissions, with a focus on the decarbonization of energy carriers and the treatment of biogenic carbon storage.

The assessment adapts life-cycle boundaries based on EN 15978 [29], with notable alterations: Unlike EN 15978, which references the EN 15804 [42] product standard that books stored biogenic CO_2 as an emission, this framework employs a time-dynamic LCA method for detailed declarations (Section 2.5.1). For practicability, a second, simplified method (Section 2.5.2) can also be used, which credits biogenic carbon in building materials in part as carbon storage (negative emissions, 55–100% depending on its origin and turnover period) over the horizon.

For mobility emissions, the framework includes parts of B8 that correspond to EMIM travels to the district and embodied emissions for vehicle construction (Section 2.4.2). Table 3 gives an overview of the concrete system boundary inclusions and exclusions.

To ensure consistent coverage of embodied and operational impacts, the framework uses building balance boundary BB6 (see Appendix A Table A2 for an overview of boundaries as defined by [43]) for A1–A3 (product stage), B2–B5 (use stage), and End-of-life prior to 2050 (C1–C4). BB6 covers the complete building: thermal envelope, interior partitions, basement structures, unheated buffer zones, technical systems, and exterior facilities (carports, bicycle shelters, auxiliary buildings).

Table 3. System boundaries of Life Cycle Stage Inclusion in the assessment framework.

Life Cycle Stage	Sub-Stage	Module	Considered	Notes/Comments
A—Pre-construction	Pre-construction	A0	No	Outside EN 15978 scope here
A—Product stage	Product stage	A1	Yes	Raw material supply (embodied)
		A2	Yes	Transport to manufacturing
		A3	Yes	Manufacturing emissions

Table 3. *Cont.*

Life Cycle Stage	Sub-Stage	Module	Considered	Notes/Comments
A—Construction process stage	Construction	A4	Yes	Assessed if practical/data available
		A5	Yes	Assessed if practical/data available
B—Use stage	Use (non-energy)	B1	No	Not considered
	Maintenance	B2	Yes	Included; post-2050 assumed carbon-neutral
	Repair	B3	Yes	Same as B2
	Replacement	B4	Yes	Same as B2
	Refurbishment	B5	Yes	Same as B2
	Operational energy use	B6	Yes	Dynamic/Monthly factors (to 0 by 2050)
	Operational water use	B7	No	Water-related GHG excluded
	Other building-related activities	B8	Partly	EMIM B8.1 and B8.2 included (Section 2.4.2) Operational mileage + one-time vehicle embodied (≤2050)
C—End-of-life stage	End-of-life	C1	Yes	Included; post-2050 carbon-neutral
		C2	Yes	Same as C1
		C3	Yes	Same as C1
		C4	Yes	Same as C1
D—Beyond system boundary	Reuse/recycling recovery	D1	Yes	
	Exported utilities	D2	Yes	PV grid feed in credited as substitution with negative hourly/monthly factors
Contextual	Off-plot public infrastructure	—	No	Optional reporting only (not for compliance)

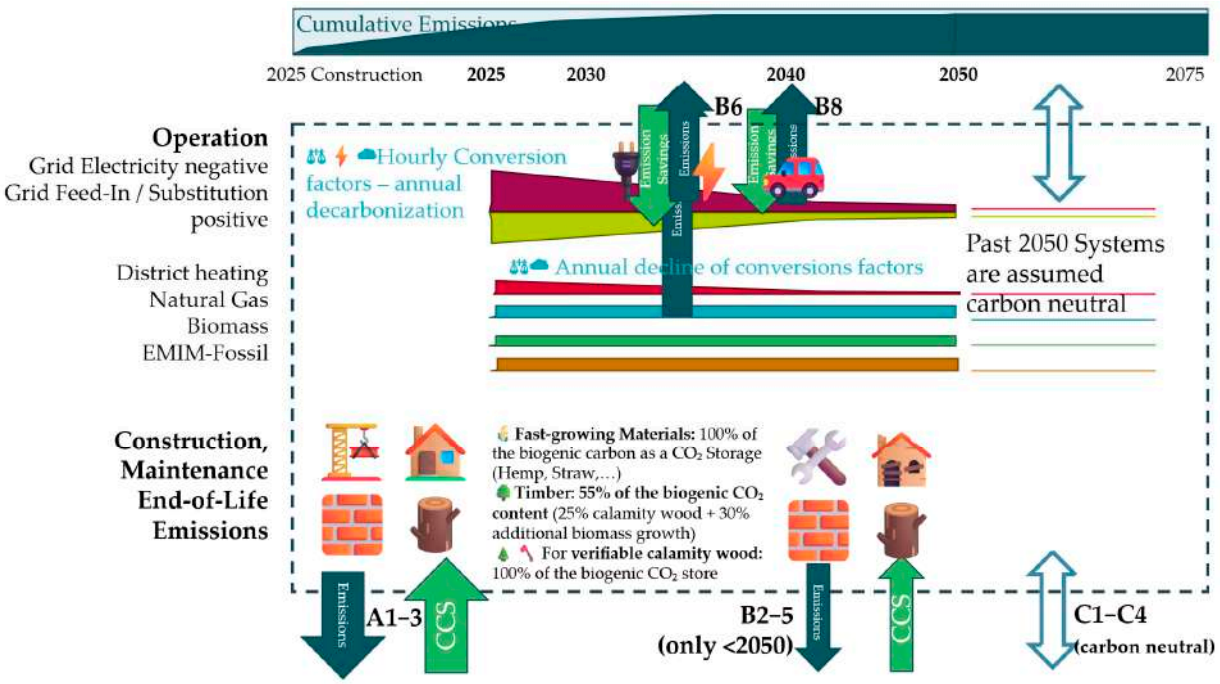


Figure 3. System boundaries and scope of the life-cycle assessment (LCA), covering 2025–2075 and including both operational and construction emissions. Operational emissions comprise grid electricity use and substitution as well as other carriers (natural gas, biomass, fossil, district heating and cooling), with annual decarbonization of conversion factors to zero by 2050 where applicable. Construction emissions (A1–A3, B2–B5) account for certain biogenic carbon storage in bio-based material.

2.4.1. Operational Energy Flows, Local PV, and Exported Electricity (Module B6/D2)

Local PV is handled operationally only: on-site generation replaces grid electricity using the framework's conversion factors (hourly for analysis and optimization; monthly building-code factors for declarations). Surplus feed-in is credited as grid substitution

potential (Module D2) using the same operational factors. This means that for declaration GHG and PE accounting purposes, feed in and self-consumption are equivalent. For research and future iterations of declarations, this can be detailed by setting the feed-in grid substitution factor to an appropriate/time-dependent fraction of the grid import factor.

Embodied impacts of PV systems are not counted at project level, as they are allocated to the national energy sector that underlies both the budget derivation and the operational factors used here (which exclude power-plant embodied impacts). This avoids double counting and maintains consistency with sectoral budgets. Crucially, it defuses the paradoxical tension of PEDs being penalized in the GHG balance for their role as energy providers: in our framework, EE from RES is already allocated and budgeted for in the energy sector, giving a clear energy and GHG incentive to include as much RES as is viable.

Waste heat imports are accounted for with zero direct emission intensity, since the associated emissions are allocated to the primary process generating the waste heat. District cooling supplied from a central city network is treated analogously to other imported carriers by applying the emission intensities provided by the utility or national statistics. Where district cooling is produced within the assessed district itself, the corresponding emissions are calculated based on the energy carriers used for its generation, most commonly electricity. See Figure 4 for a schematic overview of the energy flows, including the treatment of waste heat imports and district cooling emissions.



Figure 4. Schematic overview of emissions caused by operational energy flows in the climate-neutral PED framework, illustrating the treatment of local PV direct use and grid feed in, waste heat imports, district cooling emissions, and other energy carriers.

2.4.2. Mobility (Module B8)

Operational mobility emissions (Modules B8.1 for ICE and B8.2 for EV) are accounted for only for EMIM trips whose destination lies within the district. Public transport and active modes (walk/bike/other) are not directly modeled—either immaterial at district scale or outside district control and therefore allocated to the transport sector for GHG budgeting (as per [24]). However, district mobility measures increasing active or public modes of transport can decrease EMIM mileage, and thus, shifts in modal split are also considered under this framework. Trips whose destination lies outside the district are modeled, but considered the responsibility of the target location, omitting it in the district balance to avoid double counting. For EVs we cannot perfectly map charging emissions to discharge purpose; therefore we (i) compute emissions from all EV charging metered inside and outside the district using time-resolved grid factors, (ii) deduct a grid-equivalent credit for trips whose destination is outside the district (treated as an avoided-emission transfer to the destination system), and (iii) treat vehicle-to-building (V2B) as battery operation to the district with its charging emissions accounted for. This yields a net, destination-consistent EV operational footprint for the district, ensures fair effort towards sharing EMIM emissions between all buildings and districts, while retaining physical model accuracy and research

trajectories regarding charging infrastructure, where actual expected loads are critical. For ICE vehicles, B8.1 is computed from destination-based trip mileage and carrier-specific $g_{CO_2e} \cdot km^{-1}$ factors. Table 4 shows the resulting inclusion matrix.

Table 4. B8 Mobility emission inclusion matrix.

B8 Emission Cause/Use	Accounted	Modeled	Emission Balance Impact	Treatment/Rationale
Public Transport	✗	✗	None	Allocated to transport sector; outside district control.
Pedestrian, Bike, Other	✗	✗	None	Negligible at district scale. (Included if not)
EMIM **ICE** Trips to District	✓	✓	+	Included.
EMIM **ICE** Trips elsewhere	✗	✗	None	Not included here; attributed to destination system.
EMIM EV On-site Charging	(✓)	✓	+	Included using hourly source emission factors (grid, local PV = 0)
EMIM EV Off-site Charging	(✓)	✓	+	Included using hourly grid emission factors.
→ Trips to District	✓	✓	(+)	Implicitly included
→ Trips elsewhere	✗	✓	−	Deduct grid-equivalent credit to remove out-of-district travel from EV charging OE
→ V2B discharge	✓	✓	(+)	Like BESS: Charging included, but lowering building OE at discharge

Notes: "✓": Accounted/Modelled, "✗ ": Not accounted/modelled, "Emission balance impact": "+" adds emissions; "−" is a credit.

Embodied vehicle impacts are included once at purchase using static average factors (ICE $\approx$ 5 t_{CO_2e} per car; EV $\approx$ 9 t_{CO_2e} per car) without replacements or manufacturing decarbonization; this is a conservative snapshot and should be revisited in future work. Sensitivities can vary lifetimes and embodied factors, which should be further discussed; however, the mobility case study (Section 3.4) indicates that EMIM EE only plays a secondary role in district compliance.

2.4.3. Maintenance and End-of-Life: Post-2050 Assumptions (Modules B2–B5; C1–C4; Vehicles)

Building Maintenance (B2–B5) and End-of-life processes (C1–C4) are included. Since all processes taking place after 2050 are assumed to be carbon-neutral, in practice only those with a lifetime of <25 years need to be considered (or preferably avoided). The same convention applies to vehicles.

2.4.4. Public Infrastructure and Contextual Assets

For on-plot infrastructure within BB6 (e.g., internal access roads, on-plot sewers, bike shelters), impacts are included. Off-plot public infrastructure (municipal roads, trunk sewers, transit systems) is outside project control and not included in the core balance; where data are available, such items may be reported contextually but are not part of compliance checks. This keeps project-level accounting aligned with municipal/national responsibilities.

2.5. Treatment of Biogenic Carbon and Carbon Storage in Building Materials

The framework offers two admissible EE accounting options for declarations:

1. An optional but preferred dynamic LCA method, for time-explicit treatment where results materially depend on biogenic storage and timing effects (e.g., rotation periods, storage duration, end-of-life timing);
2. A default simplified LCA method provided for practicality and comparability, where biogenic carbon contained in bio-based building materials is partly credited as negative emissions (carbon storage) over the assessment horizon, based on carbon origin and use but independent of emission time.

The framework tooling accommodates both simplified and dynamic biogenic carbon methods, with the latter recommended where data availability allows.

2.5.1. Dynamic LCA Method

The dynamic method evaluates the climate impact of emissions and removals as a function of when they occur over the assessment horizon (here: 2025–2075). In contrast to static accounting, time-dependent characterization captures that (i) early emissions induce higher radiative forcing (RF) and contribute more to peak warming, whereas (ii) delayed emissions exert a smaller effect. Carbon storage in bio-based building materials is therefore credited according to its duration and timing [44].

Methodologically, dynamic LCA proceeds as follows. First, for each GHG, the atmospheric decay of a pulse emission is represented via an impulse response function (IRF): for CO_2 with the Bern carbon cycle model; for CH_4 and N_2O with first-order exponential decay. Second, instantaneous dynamic characterization factors (DCF_{inst}) are obtained by integrating the gas-specific radiative efficiency over the decay curve from the time of emission/removal to any time t. Third, cumulative warming impact is computed by summing the DCF_{inst} contributions over all time steps and flows, and results are expressed as dynamic CO_2 equivalents by normalizing to the absolute GWP of a 1 kg CO_2 pulse over the same horizon. This yields time-differentiated CO_2e values for both positive emissions and negative fluxes (e.g., biogenic uptake/storage) [45,46].

For biogenic carbon, two regrowth timing conventions must be declared transparently because they affect results: (a) "growth before harvest" (uptake occurs prior to material use) or (b) "regrowth after harvest" (uptake occurs during/after the building life). The latter typically yields smaller near-term credits due to delayed removals. Fast-rotation materials (e.g., straw, hemp) provide stronger near-term mitigation than slow-rotation timber; dynamic LCA makes these differences explicit. Compared with static "0/0" or "−1/+1" treatments, dynamic accounting reduces bias from ignoring timing and avoids net-negative artifacts when only early-stage credits are counted. The literature consistently finds the dynamic approach more robust and transparent for bio-based construction [44,47].

2.5.2. Simplified LCA Method

The simplified method accounts for biogenic carbon storage based on the origin of the material and its expected service life without differentiating the timing of emissions. All emissions are considered with equal weighting, regardless of whether they occur in the near or distant future. The following rules apply as described in [48], with mass-weighted averages for composite materials:

- **Fast-growing plant-based materials** (e.g., straw, hemp, flax; rotation period ~1 year): → 100% of the biogenic CO_2 content is credited.
- **Wood from managed forests** (typical rotation period ~100 years): Only the proportion representing additional sequestration beyond a no-harvest scenario is credited. This comprises: 25% of the harvested wood volume from calamity events (e.g., bark beetle infestation, storm damage), credited at 100% plus 30% of biomass growth from sustainable management practices. → This results in a total credit of 55% of the biogenic CO_2 content for standard harvested wood.
- **Calamity wood**: → 100% credit for the biogenic CO_2 content. Project declaration requires origin verification from pest or storm damage.

In all cases, the maximum biogenic CO_2 storage credit is limited to 100% of the Global Warming Potential (GWP-biogenic) associated with the material. This approach implicitly incorporates two climate-relevant effects:

1. Carbon uptake following harvest due to replanting in sustainably managed forests, compared to unmanaged forest growth.
2. Temporary sequestration of biogenic carbon, which reduces the peak temperature increase when paired with rapid and consistent GHG emission reductions.

2.6. Operational Conversion Factors/Dynamic Emission Factors

Operational conversion factors follow time-dependent trajectories per carrier. For electricity and renewable district heating, factors decline to zero by 2050, consistent with national pathways; fossil carriers (natural gas, oil, coal) remain relatively constant until phase-out in 2050; biomass follows the national profile defined in the scenario set; district cooling factors are included analogously and mapped to the respective supply mix. The framework uses hourly factors for analysis and model comparisons, and monthly building-code factors (linearly down-scaled to 2050) for declarations to maximize practical compatibility.

Data sources and projections:

- Hourly CO_2 intensity profiles for electricity are derived from Austrian grid mix data and adjusted to annual benchmarks from the Austrian National Energy and Climate Plan for 2030 and 2040.
- A linear trajectory is applied from current values to a CO_2 intensity of 0 $g \cdot kWh^{-1}$ by 2050, consistent with the 100% renewable scenario [49] adapted in [50], including seasonal balancing via methanation of summer electricity surpluses.
- For biomass and district heating from renewable sources, a separate decarbonization pathway is applied based on sectoral transition strategies, reflecting gradual replacement of fossil inputs with renewable heat sources.
- For natural gas and other fossil fuels, virtually no decarbonization is assumed; in line with their inherent combustion emissions, CO_2 intensities recede by 5% until 2050 and must thereafter be completely phased out or covered by CCS.

Use of hourly vs. monthly factors:

- For analysis and research purposes, the framework uses hourly conversion factors. Multiple factors sets from literature and modeling studies have been created and compared, enabling investigation of the timing effects of energy demand and generation. Section 3.3 presents a case study on these effects.
- For practical usability, regulatory consistency, and comparability with the Austrian building code, the official application of this proposed climate neutrality assessment framework for PEDs uses monthly CO_2 conversion factors to normalize hourly values to compliant annual levels (Appendix A Table A3). These are linearly downscaled from 2023 to 2050 according to the decarbonization path for each energy carrier (Section 2.5.1).

Application in operational assessment (Module B6/D2):

Electricity consumption: Hourly or monthly CO_2 intensities are multiplied with consumption values to determine operational emissions.

On-site PV generation: Electricity consumed on-site replaces grid electricity at the corresponding CO_2 intensity.

PV surplus exports: PV electricity exported to the grid is credited with the same CO_2 intensity, representing its grid substitution potential.

Other on-site renewable heat generation (e.g., solar thermal, biomass): Reductions are credited based on the carbon capture potential of the plant-based materials used, such as the production of biochar, rather than simply accounting for the avoided CO_2 emissions from displaced heat sources (e.g., district heating or fossil boilers). This approach emphasizes the importance of incorporating carbon sequestration into the lifecycle carbon

balance. By producing biochar, carbon from biomass is locked away in a stable form, thereby removing it from the atmosphere and contributing to long-term carbon storage. This method is vital for aligning the assessment with the broader goals of carbon neutrality and addressing the full carbon cycle, including both emission reductions and carbon sequestration. Additionally, this approach avoids the common pitfall of solely relying on emission avoidance through reduced fossil fuel use, which does not fully capture the potential of biomass as a tool for climate mitigation.

Treatment of embodied emissions of energy system: Dynamic conversion factors in this framework only include operational emissions from energy generation and supply. Embodied emissions from the construction and maintenance of generation assets (e.g., PV systems, district heating infrastructure) are excluded, as these are already covered within the national sectoral carbon budget for the energy system and deducted when deriving the building sector's carbon budget. In contrast, other methodologies (e.g., IEA lifecycle factors) include both operational and embodied emissions, which would cause double counting here.

Gradual Emission Intensity Reductions 2025 to 2050

The total operational emissions over a 50-year period are calculated by multiplying the simulated annual operational emissions by the integral of the decarbonization pathway, as shown in Figure 5. The annual interpolation of emission reductions is based on established literature sources: a transition to a 100% renewable electricity system by 2030 [51] and the achievement of a fully climate-neutral energy system by 2050 [52]. After this point, no operational emissions are considered (fossil fuels are assumed to be phased out).

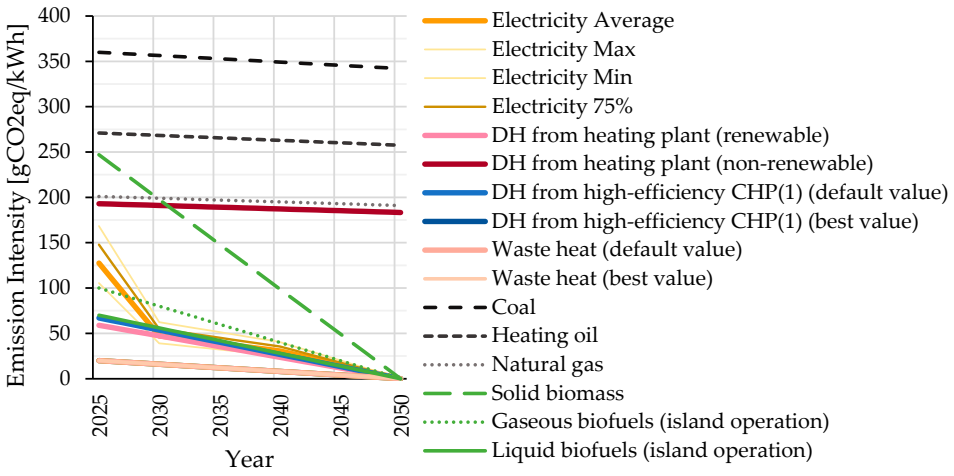


Figure 5. Projected emission intensity reduction paths for GHG balancing 2025–2050. Electricity corridors reflect differences in historic weather and production mix baselines (2016–2022).

2.7. Example Assessments

2.7.1. Selection of Use Cases

Six representative Positive Energy District (PED) projects were chosen to demonstrate the applicability of the proposed climate neutrality PED assessment framework. The selection was guided by three main criteria. First, all projects are located in Austria or in comparable climatic regions, ensuring consistent boundary conditions for energy and emission analyses. Second, sufficient data availability was a prerequisite, covering both life cycle and operational aspects. Third, the cases were selected to reflect a broad diversity of conditions relevant for PED development:

This diversity is expressed in several dimensions: building typologies range from renovation projects to new constructions, from low, rural to high, metropolitan density, and from small-scale interventions to large district developments. Different usage mixes are

represented, as well as various energy system configurations, including air and ground source heat pumps and district heating, with and without cooling, and with different ratios of mechanical ventilation and heat recovery. The projects also employ contrasting building materials, spanning from conventional reinforced concrete with EPS insulation to hybrid solutions, wood–concrete compounds, and timber structures with ecological insulation materials such as hemp or straw. Finally, mobility-related conditions differ significantly across the cases, reflecting settings from metropolitan to rural locations, with corresponding variations in annual mileage and electric vehicle penetration levels ranging from 10% to full adoption.

Each of the six investigated projects was characterized by a set of key input parameters relevant for energy and emission balances, as shown in Table 5. These include the gross floor area (GFA), use type (residential, office, education, retail), the district density in terms of floor space index (FSI), and the share of renovation versus new construction and prevalent construction type (conventional, ecological). For energy demand calculations, climate datasets (TRY datasets) were assigned to each location, and hourly outdoor air temperatures and irradiation were considered. In addition, parameters for user electricity, lighting, domestic hot water, heating, cooling, and ventilation demand were included. Table 3 provides an overview of the six projects and their key parameters. Note the specific PV yield's inverse proportionality to the district density, which complicates positive energy balances for higher densities. Operationalization for this effect in the form of target amending context factors is described in [24].

Table 5. Example district baseline parameters.

Project	D1	D2	D3	D4	D5	D6	Unit
Location	Innsbruck	Vienna	Salzburg	Vienna	Vienna	Vienna	
Gross Floor Area (GFA)	5886	13,069	31,799	9761	180,492	33,506	m^2
Net to Gross Floor Area	80	80	80	80	80	80	%
Construction Type	Conv	Eco	Eco	Eco	Conv	Conv	
Refurbishment Share	0	18	0	100	100	0	%
Buildings	5	2	16	2	71	3	
District Plot area	7341	14,095	28,000	4662	48,131	7323	m^2
Floor Space Index (FSI)	0.80	0.93	1.14	2.09	3.75	4.58	–
Context Factor Density [24]	−11.3	−3.7	5.6	26.2	37.9	40.7	$kWh_{PE} \cdot m^{-2}{}_{NFA} \cdot a^{-1}$
Context Factor Mobility [24]	14.1	12.9	14.9	14.1	16.7	25.6	$kWh_{PE} \cdot m^{-2}{}_{NFA} \cdot a^{-1}$
Context Factor Renovation [24]	0	2.2	0	15.0	15.0	0	$kWh_{PE} \cdot m^{-2}{}_{NFA} \cdot a^{-1}$
Residential Space Use	100		96	100	91	52	%
Office/Commercial Space Use			3		9	38	%
Education Space Use		100	1			2	%
Retail Space Use						8	%
Heating System	AS HP	GS HP	DH	GS HP	DH	GS HP	
Cooling System	None	GS HP	None	GS HP	None	GS HP *	
DHW System	AS HP	GS HP	DH	DH	DH	GS HP	
Window Ventilation	0100	0100	0100	0100	90	0100	%
Mechanical Ventilation					10		%
PV installed capacity	196	345	547	91.1	127	560	kW_p
PV specific yield	39.8	32.7	20.0	10.7	0.9	19.5	$kWh \cdot m^{-2}{}_{NFA} \cdot a^{-1}$
EV Share	70	70	70	70	70	70	%
EMIM Reduction	27%	20%			10%	10%	
Flexible TABS	+3K heat	+1K heat	None	+3K heat −2K cool	±2K	+3K heat −2K cool	

* In office and commercial spaces.

2.7.2. Data Collection and Assumptions

For each district, a consistent dataset from project declaration and assessment reports was compiled. To ensure transparency and reproducibility, a detailed catalog of all input fields of the framework (over 250 parameters, including units and default assumptions) is provided in the online data repository [41]. A summary of model categories and parameters can be found in Appendix A Table A4. The data covers building characteristics, energy

systems, mobility integration, life cycle inventories, and operational performance and were either taken from project documentation, energy performance certificates, and simulation models or derived from the Austrian LCA database *baubook.at*.

3. Results

3.1. Comparative Analysis

This section compares six baseline districts against the compliance framework using consistent project descriptors and harmonized KPIs. Lifecycle performance is summarized in Table 6 and Figures 6–8: cumulative GHG per m^2_{GFA} is decomposed into A1–A3 (construction incl. TES + EMIM embodied), B-ops (time-resolved electricity and thermal operation), EMIM operation, and PV substitution credits (D2); we also report the primary energy (PE) balance. Compliance bands are shown at L3 = 320, L2 = 196, L1 = 72 $kg_{CO_2e} \cdot m^{-2}_{GFA}$.

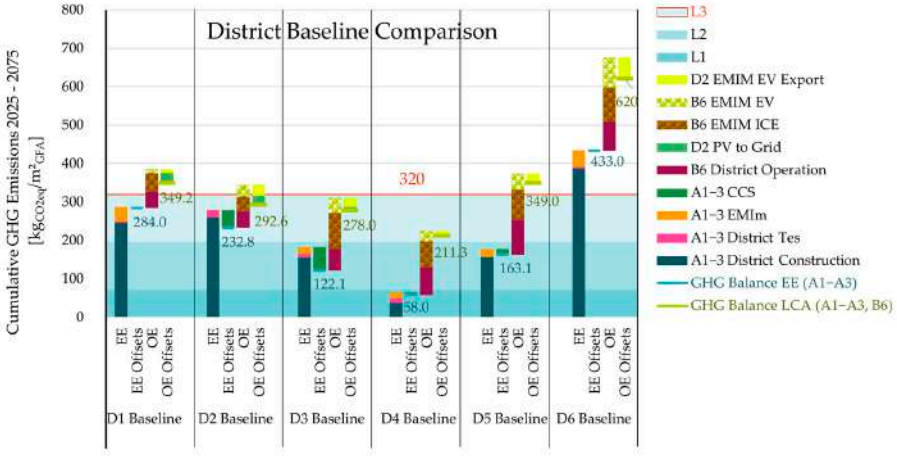


Figure 6. Baseline life cycle GHG balance by district. Cumulative emissions 2025–2075 per m^2_{GFA}, decomposed into A1–A3 (building, TES, and EMIM embodied), B-ops (building/TES with time-resolved grid factors), EMIM operation, and PV substitution credits (D2); CCS shown where applicable. Horizontal bands denote compliance thresholds (L3 = 320, L2 = 196, L1 = 72 $kg_{CO_2e} \cdot m^{-2}_{GFA}$).

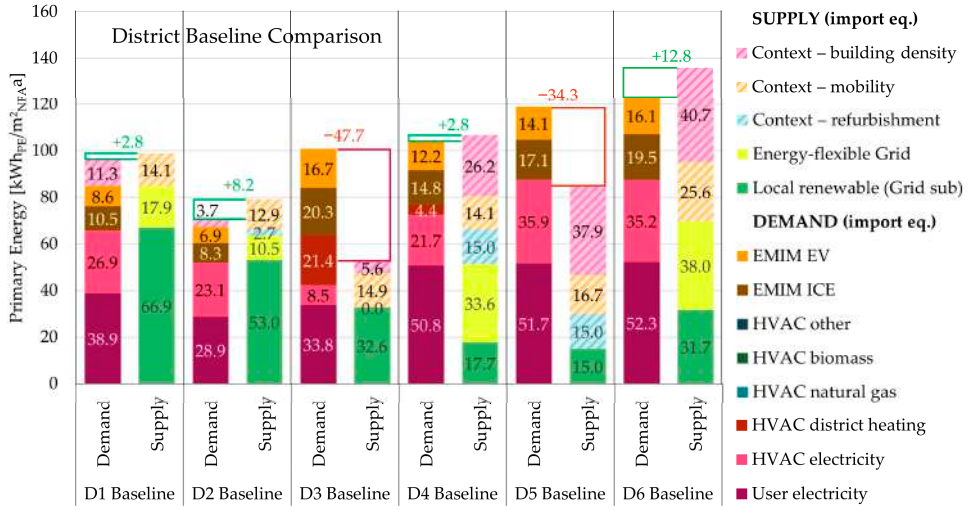


Figure 7. Annual primary energy balance by district. Positive energy criterium is met if supply (PV self-use and export as grid substitution, energy-flexible operation, contextual factors) is greater or equal (=green box) to the demand (import equivalent HVAC heat, electricity, mobility), otherwise (=red box) positive energy criterium is not met. Balancing definition and method is based on [24].

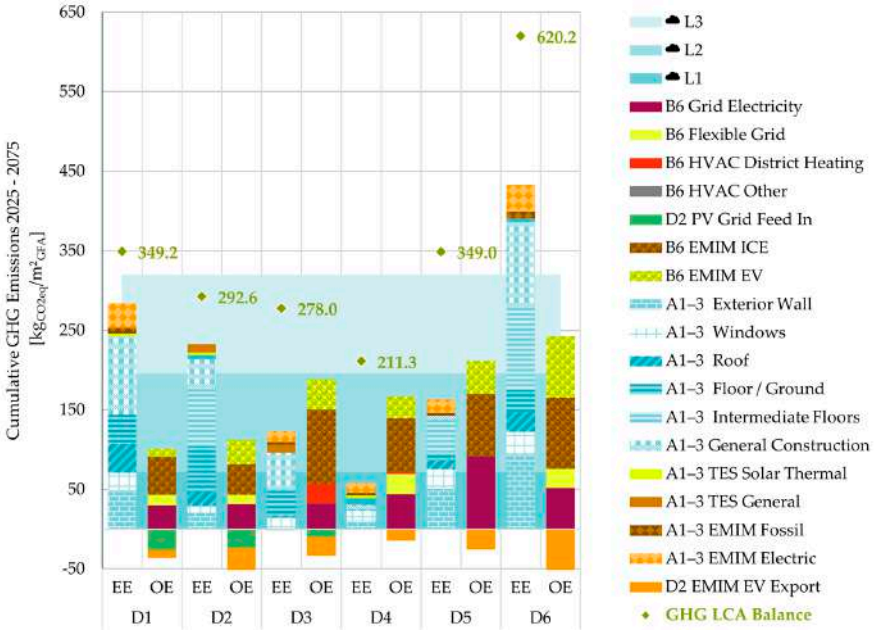


Figure 8. Cumulative GHG balance (2025–2075) with full LCA breakdown. For each district (D1–D6), the left bar (EE) shows embodied impacts (A1–A3, B2–B5) from construction and technical systems (TES, PV, batteries, ground probes) and embodied mobility. The right bar (OE) shows operational impacts (B6) from grid electricity and HVAC carriers (district heating, natural gas, biomass, other) plus EMIM (ICE/EV, B8.1-2) operation and flexible-grid effects; negative segments reflect avoided-grid D2 substitution (e.g., PV feed-in, EV other travel exports). Green diamonds indicate each district's net LCA total, while shaded bands mark according compliance thresholds L3 = 320, L2 = 196, L1 = 72 $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$. Positive stacks add emissions; negative stacks reduce the balance.

Table 6. Baseline district scenarios comparative compliance (✓= yes/✗ = no) summary.

District	Heat Demand $kWh \cdot m^{-2}{}_{NFA} \cdot a^{-1}$	GHG LCA $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$	Δ vs. L3 * $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$	<L3	PE Balance ** $kWh_{PE} \cdot m^{-2}{}_{NFA} \cdot a^{-1}$	>0	CN PED ***
D1	16.0	349.2	+29.2	✗	2.8	✓	✗
D2	21.3	292.6	−27.4	✓	8.2	✓	✓
D3	25.6	278.0	−42.0	✓	−47.7	✗	✗
D4	33.3	211.3	−108.7	✓	2.8	✓	✓
D5	37.9	349.0	+29.0	✗	−34.3	✗	✗
D6	13.8	620.2	+300.2	✗	12.8	✓	✗

* Δ vs. L3 = GHG LCA − 320; positive (red) = overshoot; negative (green) = undershoot/compliance. **, ***: ✓/green = compliance, ✗ /red = non-compliance.

Results are heterogeneous across use types and system choices. Relative to the L3 budget, the modeled GHG balances show: D1 +29.2, D2 −27.4, D3 −42.0, D4 −108.7, D5 +29.0, D6 +300.2 $kg_{CO_2e} \cdot m^{-2}$ (overshoot positive; undershoot negative). PE balance passes in D1, D2, D4, and D6 but not in D3 or D5. District-level drivers include the following:

D1 (greenfield, conventional): Non-compliant at L3 (+29.2). High A1–A3 intensity dominates; EMIM measures reduce operational mobility loads but cannot offset embodied peaks.

D2 (school campus): Benefits from the EMIM context (only commuting staff; high PV self-use + flexible charging), hybrid-eco materials achieving L3 compliance (−27.4) and a positive PE balance.

D3 (urban housing, DH): Meets L3 (−42.0) but fails PE (−47.7 $kWh_{PE} \cdot m^{-2} \cdot a$). Lack of supply side flexibility on DH limits net-positive operation despite acceptable GHG.

D4 (ecologic renovation + TABS/GSHP + mobility measures): Strongest performer across both metrics (−108.7; positive PE). Renovation and density factors reduce embodied intensity; mobility context halves residual EMIM; TABS/GSHP enable flexible, low-carbon operation.

D5 (conventional renovation, gas→DH): Switch to DH alone is insufficient for L3 (+29.0) and fails PE; Section 3.5 explores compliant variants (non-fossil TES + eco materials + flexibility).

D6 (low heat demand, high embodied/EMIM): Despite a positive PE balance, very high embodied and/or mobility-related loads lead to substantial L3 overshoot (+300.2).

Effects of PV TES Embodied Emission (Non-)Inclusion

As argued in Section 2.4.1, embodied emissions from PV systems are not part of this PED system boundary but instead are nationally budgeted in the energy sector. Table 7 shows the omitted PV EE in contrast to the remaining balance: especially for low-density districts that require a significantly positive PE balance and PV export to offset their negative density context factor, and for refurbishment projects with inherently low EE, PV EE can increase overall LCA emissions by more than 10% (assuming 1.16 $t_{CO_2e} \cdot kW_p^{-1}$ incl. inverters and cabling).

Table 7. Embodied emissions of example district PV TES EE in total and relative to the remaining EE and total LCA emissions.

Scenario	PV Yield $kWh \cdot m^{-2}a^{-1}$	EE A1–A3 w/o PV $kgCO_2e \cdot m^{-2}_{GFA}$	EE A1-3 TES PV $kgCO_2e \cdot m^{-2}_{GFA}$	ΔEE [%]	ΔLCA [%]
D1	39.8	284.0	38.6	+13.6	+11.0
D2	32.7	232.8	30.6	+13.1	+10.5
D3	20.0	122.1	19.9	+16.3	+7.1
D4	10.7	58.0	10.8	+18.7	+5.1
D5	0.9	163.1	8.1	+5.0	+2.3
D6	19.5	433.0	19.4	+4.3	+3.1

3.2. Case Study District 3: Redevelopment of Urban Medium Density Residential District: Interplay of Embodied and Operational Emission Reduction Measures

The case study evaluates multiple redevelopment scenarios for an urban medium-density residential district (District 3). Variants differ in construction materials (fossil-intensive vs. ecological), PV system sizing (1×, 2.6×, 2×, 1.7×), mobility assumptions (70% EV, 100% EV, reduced traffic), and heating/DHW supply (district heating vs. ground-source heat pump). The main assessment criteria are the cumulative GHG balance (2025–2075) and the primary energy balance for PED compliance.

The results in Table 8, Figures 9 and 10 show that all scenarios with ecological construction remain within the L3 compliance limit, while only the fossil construction variants exceed it. The combination of 2× PV oversizing with mobility measures (full electrification and reduced traffic) performs particularly well, even reaching the precautionary compliance limit for cumulative GHG emissions. This underlines the necessity of combining renewable supply expansion with systemic demand-side measures.

Table 8. Summary of case study results for District 3—urban medium-density residential redevelopment, reporting GHG LCA emissions (kg_{CO_2e}/m^2_{GFA}), L3 and L2 compliance, primary energy balance and zero PE-Balance and Climate Neutral PED compliance (✓/green = yes, ✗/red = no).

Scenario	Construction	PV Size	Mobility	TES	GHG LCA *	<L3	<L2	Primary Energy Balance $kWh_{PE} \cdot m^{-2}_{NFA} \cdot a^{-1}$		Climate Neutral PED
Fossil Constr	Fossil	1×	70% EV	DH	378.2	✗	✗	−47.4	✗	✗
Eco Constr	Ecological	1×	70% EV	DH	277.6	✓	✗	−47.4	✗	✗
EC + 2.6 × PV	Ecological	2.6×	70% EV	DH	225.6	✓	✗	+6.9	✓	✓
EC + 2 × PV + Mob	Ecological	2×	90% EV, 80% EMIM	DH	170.1	✓	✓	+5.7	✓	✓
EC + 1.7 × PV GSHP	Ecological	1.7×	70% EV	GS HP	254.7	✓	✗	+0.9	✓	✓

* Unit: $kg_{CO_2e} \cdot m^{-2}_{GFA}$.

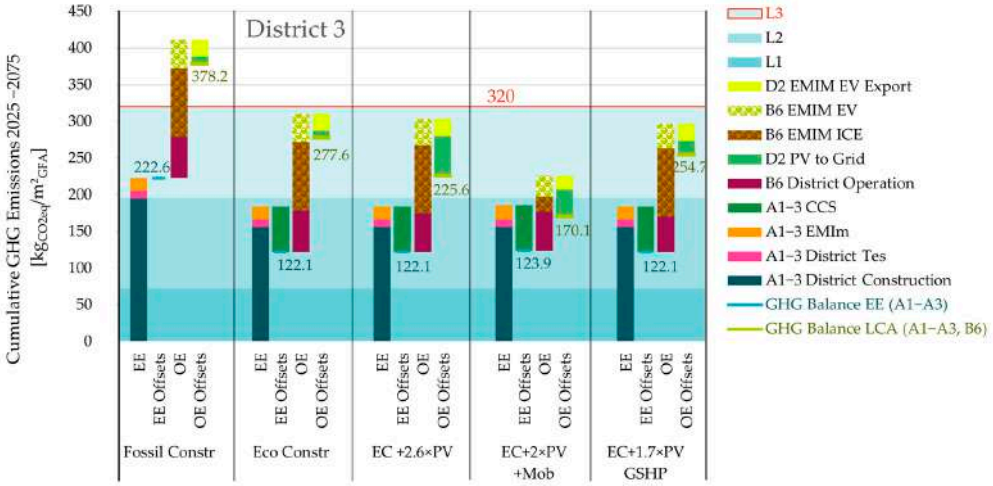


Figure 9. Cumulative GHG emissions (2025–2075) of district scenarios compared to compliance, precautionary, and Paris-compatible limits. Bars show embodied emissions (construction, technical systems, mobility), operational emissions, avoided emissions (grid feed-in), and carbon capture. Results highlight that fossil-based construction exceeds budget limits, whereas scenarios with extended PV and mobility measures approach or meet precautionary compliance thresholds.

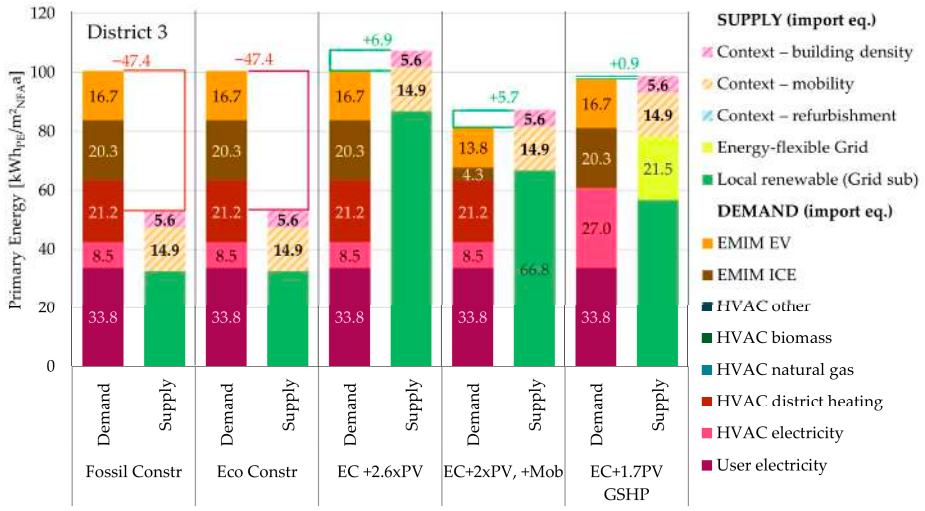


Figure 10. Primary energy balance of district 3 scenarios (demand and supply side contributions as grid equivalents). Stacked bars illustrate energy demand categories (HVAC, electricity, mobility) and supply contributions (PV own consumption and grid feed-in both visualized as grid substitution equivalent, energy-flexible grid demand equivalent, contextual measures). Scenarios compare fossil and eco construction, enhanced PV deployment, additional mobility measures, and renewable heat supply (GSHP, DH). "Positive Energy" criterium is met if supply is at least equal (=green box) to the demand (import equivalent HVAC heat, electricity, mobility), otherwise (=red box) positive energy criterium is not met.

At the same time, the assessment framework requires not only compliance with the finite GHG budget but also a positive primary energy balance. Here, the analysis reveals significant challenges: while PV oversizing improves operational balances, achieving a net-positive balance is difficult in configurations relying on district heating (DH) even when assuming a renewable CHP-based supply with favorable emission factors. In contrast, scenarios with on-site GSHP systems combined with PV come closest to fulfilling both compliance criteria simultaneously, emphasizing the importance of coupling low-carbon construction materials with renewable-based, highly efficient supply systems.

3.3. Case Study District 4: Effect of Dynamic vs. Static Grid Emission Factors and Flexibility Signals

For District 4 we quantify how the temporal resolution and level of grid-GHG signals affect (i) total operational emissions from B6 electricity and EMIM-EV charging, and (ii) the average effective grid-intensity seen by the project. We compare monthly, building-code style factors (OIB '19/'23, currently in effect) with hourly, live-tracking based factors for the same vintages. We first test flexibility via TABS GS HP heating (+3K) and cooling (−2K), then in combination with building battery (BESS) sized = PV installed capacity (kW_p) and EV bidirectional charging (V2B, down to 50% SoC_{RES}). Loads and controls held constant: Building demand, PV generation, HVAC control set-points (22 °C heating, 25 °C cooling), occupancy, and the EMIM travel demand are fixed across scenarios. Only the GHG emission factors and flexibility signals vary:

GHG Emission factors:

- Monthly and annual factors (2019 M, 2023 M, 2019 A) Twelve constant monthly values per vintage or a constant annual value (see Appendix A Figure A2).
- Hourly factors (2023 H). 8760 time steps based on historic live tracking; annual means match the corresponding OIB vintage 2023 (see Appendix A Figure A2)

Flexibility Signals:

- Wind Peaks (for 2020, above 50% nominal Power)
- GHG Emission rolling averages (RA): signal if current GHG EF(t) is y% below Rolling Average (x hour future window) (x ∈ {+24, +48, +72}, y ∈ {5%, 10%})

Detailed scenario definitions for case study 4 can be found in Appendix A Table A5. The hourly flexibility signals are visualized in Appendix A Figure A2.

This use case highlights why hourly emission accounting is essential. As shown in Table 9, with monthly factors the import-weighted average EI is higher than the annual grid average (≈241→231 $g_{CO_2e}\cdot kWh^{-1}$ for 2019 and 189→175 for 2023) because districts typically import in winter (carbon-intensive) and export PV in summer (cleaner). Annual bookkeeping would therefore "hide" the winter penalty; capturing it is critical—but without flexible dispatch comes at the cost of higher emissions—arguably as it should be.

Table 9. Operational life cycle Emissions (OE), PE Balance, Imported Electricity (ED), relative differences to the Baseline (and Emission Intensities (AVG = Annual Grid Average, Actual = Actual district import average).

Scenario	OE (B6-D2)	ΔBL %	PE Balance $kWh\cdot m^{-2}\cdot a^{-1}$	ED $kWh\cdot m^{-2}\cdot a^{-1}$	ΔBL %	EI_{Grid} $g_{CO_2e}\cdot kWh^{-1}$	$EI_{district}$ $g_{CO_2e}\cdot kWh^{-1}$	$\Delta EI/EI_{Grid}$ %
[%]19M	175.4	+14.4	+6.3	43.2	0.0	231	241.0	+4.2
19A	167.3	+9.1	+6.6	43.2	0.0	227	224.6	−1.1
23M (Baseline)	153.3	-	+2.8	43.2	-	175	188.9	+7.4
23H 24-5	141.3	−7.8	+22.2	44.2	+2.3	175	162.1	−8.0
23H 48-5	141.2	−7.9	+22.5	44.2	+2.3	175	161.9	−8.1
23H 48-10	139.4	−9.0	+14.4	43.6	+0.9	175	160.6	−9.0
23H 72-10	139.8	−8.8	+14.7	43.6	+1.0	175	161.1	−8.6
23H 48-10 BESS	137.3	−10.4	+19.3	44.2	+2.3	175	154.2	−13.5
23H 48-10 BESS V2G	137.9	−10.0	+25.6	45.4	+5.1	175	151.3	−15.7

Switching to hourly factors and applying rolling-average GHG signals enables flexible loads to target low-EI hours. In our runs the import-weighted EI drops from ~189 $g_{CO_2e}\cdot kWh^{-1}$ (2023 monthly) to ~160 $g_{CO_2e}\cdot kWh^{-1}$ with hourly signaling alone, and to ~150 $g_{CO_2e}\cdot kWh^{-1}$ when BESS and V2B are added. The grid EI for inflexible demand remains comparatively high (≈200 $g_{CO_2e}\cdot kWh^{-1}$) because it is drawn during scarcity events when storage is depleted—another effect only visible with hourly accounting.

The results presented in Table 9 and Figure 11 also show that flexibility generally slightly increases total electricity demand (round-trip and conversion losses; more charging activity), but because a larger share becomes flexible and is shifted to low-EI hours, net GHG emissions fall (our KPI), and the average effective EI of imports/charging (second KPI) improves materially. In short: hourly signals + storage/V2B make compliance harder to "paper over" yet provide a real, quantifiable pathway to lower life-cycle emissions.

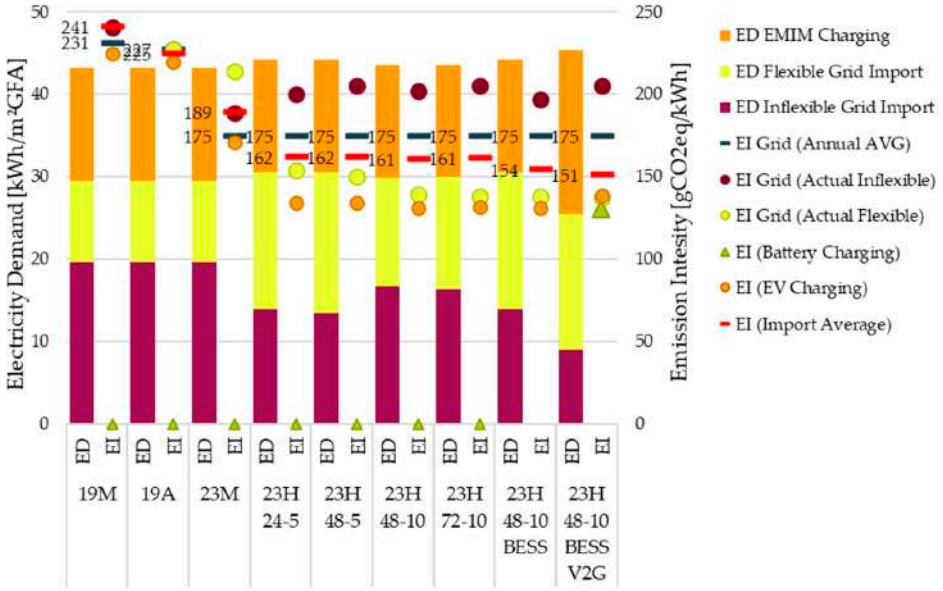


Figure 11. Temporal GHG signals and flexibility for District 4. Stacked bars (left axis) show annual electricity demand split into inflexible grid import, flexible grid import, and EMIM EV charging. Markers (right axis) report effective emission intensities (EI) for imports and charging. Scenarios progress from monthly factors (19M, 19A, 23M) to hourly factors with rolling-average signals (23H 24-5, 48-5, 72-10, 48-10) and three flexibility options (TABS—all, BESS = PV-sized battery; BESS V2G = battery plus EV V2B). The dashed blue line is the annual-average grid EI; filled markers show actual EI of inflexible imports, flexible imports, EV charging, and battery charging; the red bar give the realized import-weighted average EI.

Appendix A Figure A2 shows how energy demand, supplies and flexibility options are dispatched on an hourly basis for Scenario "23H 48-10 BESS V2G".

3.4. Case Study District 5: Mobility and Renovation Sensitivity

The case study illustrates the cumulative GHG emissions of an identical district configuration (70 multistory buildings) across different spatial contexts, with mobility-related emissions added. It shows that EMIM can dominate total emissions, even in not-decarbonized districts such as this gas-heated example. We investigate District 5 with a factorial scenario set spanning four drivers of mobility-related impacts and their interaction with energy supply and envelope choices as shown in Table 10. Figure 12 shows the mileage differentiation of [53] used in this framework as described in [24,54].

Table 10. Case Study District 5 Factor Variations and resulting Scenarios.

Factor	Levels (Examples)	Unit	How Operationalized
Location (EMIM mileage)	34 (Rural-2)/24 (Rural-1)/23/22/12 (Suburban)/11 (Urban)/91 (Metropolitan)	$km \cdot cap^{-1} \cdot a^{-1}$	Per capita annual motorized mileage profile
EV share	5/30/70/90	% of EMIM	Split km into ICE/EV; apply EI_c, $EF_c(t)$
TES	Gas/DH/GSHP + PV	—	Carrier paths; PV substitution and D2 for feed-in
Renovation	None/Conventional/Ecologic	—	A1–A3 inventories; B6 demand

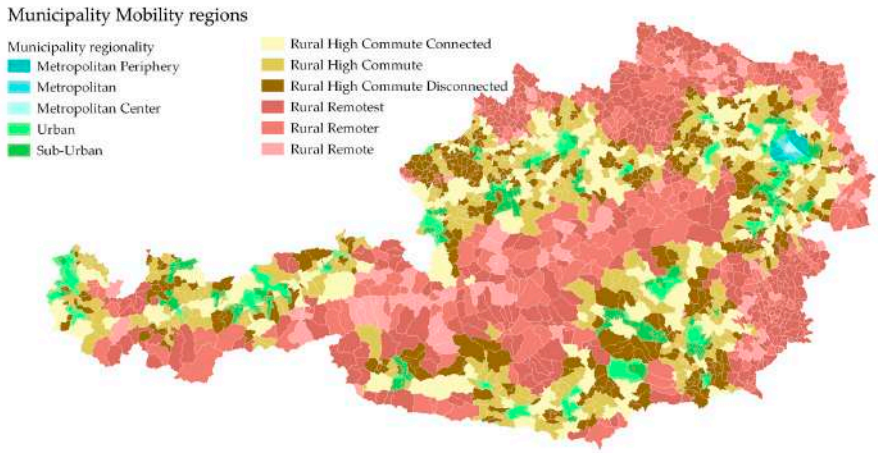


Figure 12. Mobility regions based on [53,55] determining annual EMIM mileage subject to public transport availability and average trip length.

District location and accessibility to public transport strongly influence travel demand, resulting in average annual personal motorized mileage ranging from about 4000 km in metropolitan contexts to 11,831 km in rural areas with high commute and low public transport availability (Region 24). Consequently, cumulative district emissions shown in Figure 13 vary widely—from about 740 $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$ in the fossil-fuel-based, rural commuter case down to 27 $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$ in the fully electrified, metropolitan case with high shares of alternative transport modes. Even at intermediate penetration levels (30–70% EV share), overall district mobility emissions remain dominated by fossil-fuel operation, accounting for roughly 90% of total EMIM-related emissions at 30% EV penetration and still around 60% at 70% EV penetration. These emissions can further be reduced by active mobility measures in the district such as bike and car sharing opportunities to reduce the total annual mileage by EMIM. This shows that L3 compliance (320 $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$ over 2025–2075) is possible in all locations, but complete electrification of vehicles until 2050 (averaging an EV share of 50%+ from 2025 to 2075) is essential. Measures to shift the modal split away from EMIM to public transport, pedestrian and bikes can help also but may not be sufficient by themselves.

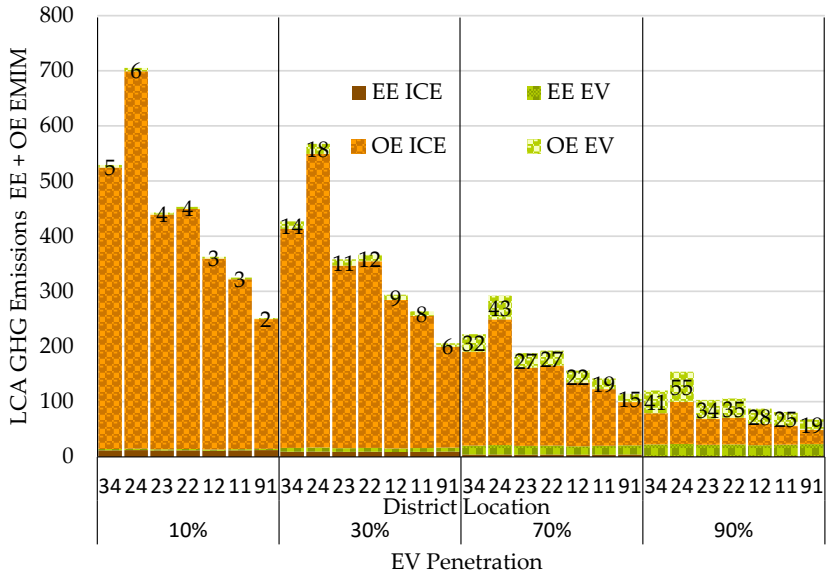


Figure 13. LCA GHG Emissions from EMIM (B8.1-B8.2) in $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$, bars showing embodied emissions (EE) and operational emissions over 50 years (OE) for vehicles with internal combustion engines (ICE) and electric vehicles (EV) in $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$ per district Location (region code in Figure 12) and EV Penetration rate (10–90%) on the x-axis.

3.4.1. Impacts of Refurbishment

Table 11 reports KPIs, and Figures 14 and 15 visualize composition for seven refurbishment options under the metropolitan mobility case with 70% EV: Unrenovated fossil gas (Gas SQ) violates all limits (~753 $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$). Switch to District heating without refurbishment (DH SQ) reaches GHG compliance (~196, L2) but without efficiency measures fails the PED/primary energy target. Conventional renovation is not reliably compliant (DH Conv-Reno ~334; GSHP Conv-Reno ~360), whereas ecologic renovations reduce A1–A3 and are robust (DH Eco-Reno ~246; GSHP Eco-Reno ~245). GSHP Conv-Flex remains within L3 (~293) thanks to higher avoided-grid credits and reduced operation, and—critically—enables a positive primary energy, illustrating that a climate-neutral PED is difficult but feasible when non-fossil TES is combined with flexibility even for an arguable worst case scenario of dense urban areas with little potential for energy efficiency and local renewable energy (Roof PV installations are required on all roofs up to conservative estimates).

Table 11. Sensitivity Analysis of LCA results to TES, renovation and flexibility for District 5. (✓/green = compliance, ✗ /red = non-compliance).

Scenario	TES	Renovation	GHG LCA $kg_{CO_2e} \cdot m^{-2}{}_{GFA}$	L3 Compliance	PE Balance $kWh_{PE} \cdot m^{-2}{}_{GFA} a^{-1}$	PED Compliance
Gas SQ	Gas	None	752.7	✗	−188.3	✗
DH SQ	DH	None	196.0	✓	−64.4	✗
DH Conv (Baseline)	DH	Conventional	349.0	✗	−28.5	✗
GS HP Conv	GS HP	Conventional	359.7	✗	−23.7	✗
DH Eco-Reno	DH	Ecological	245.7	✓	−28.5	✗
GS HP Eco-Reno	GS HP	Ecological	244.9	✓	−23.7	✗
GS HP Conv Flex	GS HP	Conventional	292.9	✓	+0.4	✓

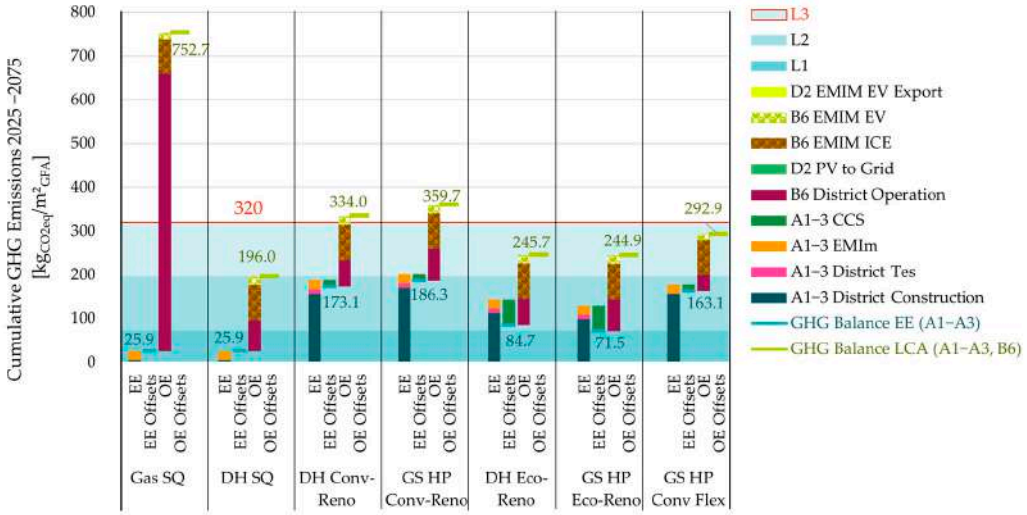


Figure 14. Cumulative GHG LCA emissions (2025–2075) per m^2 GFA for seven refurbishment options—status quo (Gas SQ, district heating DH SQ), renovated conventionally (DH Conv-Reno, GSHP Conv-Reno, GSHP Conv-Flex), and ecologically (DH Eco-Reno, GSHP Eco-Reno) under the metropolitan mobility case with 70% EV. Stacks decompose district-operation (B6), EMIM (ICE/EV), TES-related A1–A3, construction A1–A3, avoided grid emissions (D2), and CCS; shaded bands show compliance thresholds (L3 = 320, L2 = 196, L1 = 72 $kg_{CO_2e} \cdot m^{-2}$).

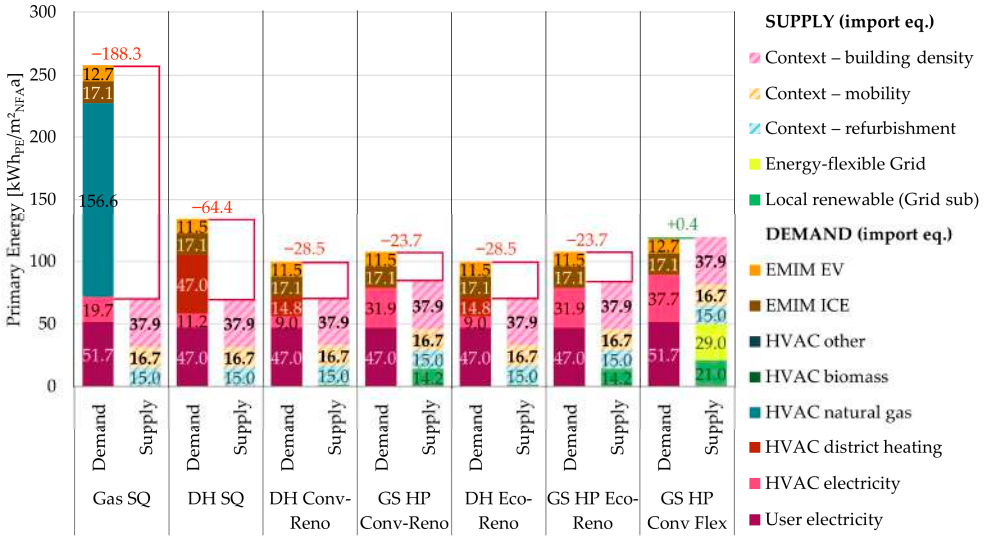


Figure 15. Annual primary energy balance (demand vs. supply) for D5. Demand stacks show user electricity, HVAC electricity/district heating/natural gas/biomass, other HVAC, and passenger-car fuels; supply stacks show local renewable substitution and energy-flexible supply (TABS load shifting) plus contextual credits. "Positive Energy" criterium is met if supply is at least equal (=green box) to the demand (import equivalent HVAC heat, electricity, mobility), otherwise (=red box) positive energy criterium is not met.

3.4.2. Sensitivity to Location and Renovation

Figure 16 plots embodied emissions on the x-axis and operational emissions (including EMIM) on the y-axis for five location types (rural-1, rural-2, suburban, urban, metropolitan). Blue diagonals mark the cumulative-budget limits L3 (320), L2 (196), and L1 (72) $kgCO_2e \cdot m^{-2}$GFA to 2050. Marker fill indicates EV share (5–100%), outlines the thermal energy system (blue = gas, red = district heating, yellow = GSHP), and shapes the construction choice (square = no renovation, triangle = code-compliant renovation, circle = ecological renovation).

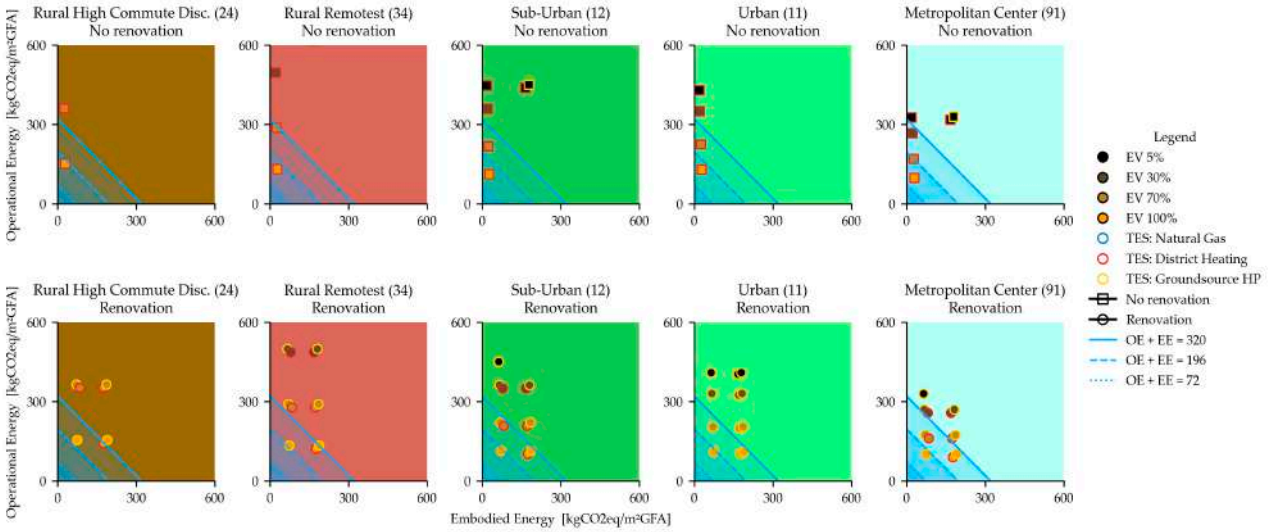


Figure 16. Embodied (x-axis) vs. Operational incl. EMIM (y-axis) emissions by location (background color matching Figure 12) and intervention. Blue lines show cumulative budget thresholds (solid L3, dashed L2, dotted L1). Fill = EV share (5–100%); outline = thermal system (blue gas, red DH, yellow GSHP); shape = construction choice (■ no renovation, ● ecological).

In the "no renovation" cases (top row), points cluster far left—reflecting very low embodied emissions—but sit high on the y-axis because the envelope is unchanged. Where district heating is available, several variants fall below L3 on GHG alone; gas heating generally breaches L3, particularly in rural settings where mobility dominates. Raising the EV share pushes points downward across all locations yet seldom achieves L2 or L1 without fabric measures.

With renovation (bottom row), points move right as embodied emissions rise, while y-values drop markedly—most strongly when GS HPs are combined with high EV penetration. In suburban, urban, and metropolitan contexts, many renovated variants approach or meet L2, and a few edge toward L1 when ecological materials are used. Rural districts remain the most challenging, because mobility keeps operational totals elevated.

Overall, if the sole objective were cumulative GHG-budget compliance, a "do-nothing fabric" strategy paired with low-carbon district heating could pass L3. However, it fails PED/primary energy goals: without efficiency upgrades, primary energy demand remains high, and the district cannot achieve a positive PE balance. Meeting both the climate-budget and PED criteria requires renovation (preferably but not necessarily ecological), electrified heat (GSHP or very low-carbon CHP DH), and high EV shares to reduce operational and mobility-related emissions simultaneously.

3.5. Case Study District 6: Differences in Construction Practice and Materials

This case study illustrates the application of the simplified biogenic carbon accounting method to the planned green-field development in District 6 and highlights that construction choices dominate D6's life-cycle GHG outcome (Table 12 and Figure 17):

The baseline structural design of the project is based on conventional building practice, with a reinforced concrete framework, thermal insulation using expanded polystyrene (EPS), and mineral wool in areas requiring enhanced fire resistance. The district benefits from high urban density and mixed-use zoning, supported by excellent public transport accessibility. As a result, the underground parking facility can be designed with a single basement level and a reduced parking ratio, further limiting construction-related GHG emissions. The high building compactness enables comparatively low EE during the construction phase.

Despite these advantages, the total CO_2-equivalent emissions from both construction and operation exceed the allocated climate neutrality budget. Even with relatively low operational EI for building use and EMIM, the project surpasses the allowable threshold by more than 75% over the 50-year assessment period.

Scenario "Eco Skeleton" introduces a more ecologically ambitious system, featuring a reduced structural frame with timber elements and mineral wool insulation. Although this scenario represents an improvement over the traditional concrete structure, the CO_2 emissions remain 30% above the maximum compliance limit due to the use of steel-reinforced concrete in the structural elements.

Table 12. Description of District 6 scenarios and GHG LCA results. (✓/green = compliance, d/yellow = near compliance, ✗ /red = non-compliance).

Scenario	EE A1–A3	Δ Baseline	EE Total	GHG LCA	Compliance
Conv-Build (Baseline)	386.5	-	433.0	620.2	✗
Eco-Skeleton	266.7	−31%	313.2	500.3	✗
Mixed-Use CLT	90.1	−77%	136.6	323.8	d (L3, <320)
Max-Eco	−91.0	−124%	−44.5	142.7	✓ (L2, <196)

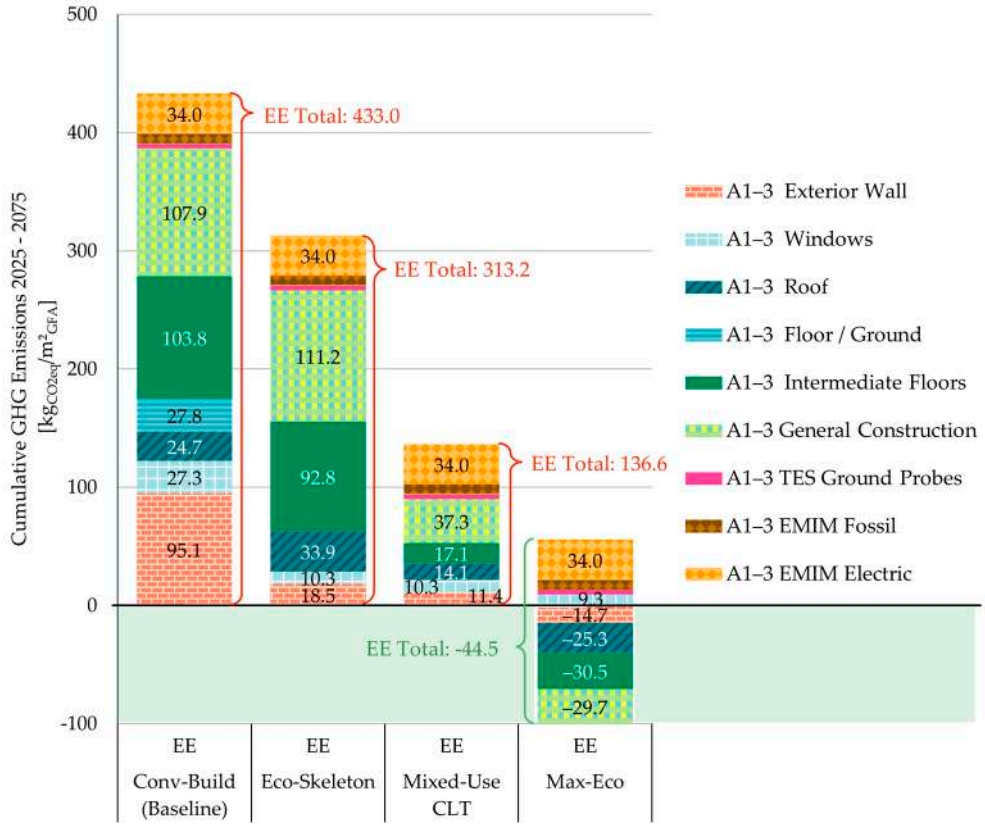


Figure 17. Cumulative A1–A3 GHG emissions by building component for four D6 concepts (EE stacks only). Bars decompose structure, envelope, interior, and TES-related items; negative segments indicate biogenic carbon storage (timber/straw).

Scenario "Mixed-Use CLT" incorporates a combination of concrete and cross-laminated timber (CLT). This mixed approach, including wood–concrete composite decks and CLT for upper floors, leads to a significant reduction in emissions compared to the first two scenarios, while still maintaining structural integrity and near L3 compliance.

Scenario "Max-Eco" represents the most ecologically advanced scenario, pushing the boundaries of climate-neutral construction. It utilizes CLT, straw insulation, wood-fiber insulation, and clay boards, minimizing the use of concrete to only necessary structural components. This results in the lowest emissions, making it the most sustainable option among the four, though it involves compromises, such as relaxing some fire protection regulations. An alternative pathway to compliance would involve further reducing operational and mobility-related emissions. For example:

- Increasing the installed PV Capacity,
- Raising the share of electric vehicles in the mobility mix,
- Implementing more ambitious ecological construction measures (e.g., low-carbon materials).

4. Discussion

The proposed climate neutrality assessment framework for PEDs aims to bridge the gap between research-grade, high-resolution analysis and practical, policy-aligned application. By defining clear system boundaries, integrating dynamic emission factors for multiple energy carriers, and embedding the assessment in a finite CO_2 budget, it offers a robust yet operationally feasible tool for evaluating district-level climate performance. The framework's treatment of local PV—crediting both self-consumption and grid substitution—along with its differentiated decarbonization pathways, supports both design optimization

and strategic policymaking. The following subsections discuss the methodological implications, applicability and limitations, and policy relevance of this approach.

4.1. Methodological Implications

Positive Energy Districts are framed here as vehicles for verifiable climate neutrality. The assessment boundary is refined to include operational emissions, EMIM, and selected embodied emissions, while transparently excluding categories allocated to other sectors in the national carbon budget (e.g., PV embodied impacts; end-of-life beyond 2050). This departs from conventional LCA allocations (e.g., attributing the embodied impacts of generation assets to the consuming district) and makes explicit how sectoral budgets are coupled. Many proposed PED assessment frameworks, such as those employed by POCITYF [56] or the SYNIKIA [57] project, use a wide range of KPIs covering many important aspects such as quality of life, comfort, planning processes, waste management [58], and the affordability and accessibility of PED measures. In our proposed declaration framework, in contrast, these aspects are deliberately not included in light of existing well-established national declaration systems covering these aspects in great depth: the klimaaktiv labels "Building" and Districts and Neighborhoods", as well as the TotalQualityBuilding labels all employ methodologically similar points-based approaches, with dozens of assessment aspects across different categories, such as (A) Management, (B) Communication, (C) Urban Development, (D) Building, (E) Supply, (F) Mobility, for the "District and Neighborhoods" declaration [37].

Temporal resolution is central. Hourly emission factors and flexibility (via BESS/V2B/TABS) are shown to significantly reduce operational emissions compared to traditional monthly or annual accounting methods, highlighting the importance of time-resolved emissions in accurate PED assessments. For declarations, monthly factors aligned with Austrian code are used to ensure comparability and administrative simplicity. Carrier-specific decarbonization pathways are differentiated: electricity and district heating follow a declining trajectory to (near) zero by 2050, while gas, biomass, and other fossil carriers retain constant intensities over the horizon. Using static, time-invariant factors would implicitly assume society fails to decarbonize by 2050; the framework instead embeds the policy trajectory, after which the budget is exhausted and historic LCA conventions become less informative.

Local PV is treated on a system basis. Onsite generation offsets grid consumption at the time-varying CO_2 intensity of the avoided imports; exports are credited for grid substitution (D2), recognizing their contribution to system decarbonization. This contrasts with net-zero approaches that ignore export credits, and it materially affects both design optimization and policy interpretation in PED contexts.

Separating measurement effects from control effects is important. Hourly factors expose seasonal import–export asymmetries that annual bookkeeping masks, while dynamic flexibility signals (wind-peak events, GHG rolling averages, price signals, etc.) guide storage, charging, and thermal shifting to lower import-weighted intensities. Declarations remain simple and comparable, while design and operation can be assessed with rigor and temporal fidelity. Crucially, it prevents projects from over-crediting flexibility measures and allows PED compliance to be evaluated transparently against both carbon budgets and energy balance criteria. This clarity improves interpretation but means results are sensitive to the chosen signal and available assets (BESS, TABS, V2B)—an interconnection which would benefit from further research so it can be better designed for. Overall, the approach is auditable and declaration-ready, but data localization and policy-trajectory uncertainty remain the main limits.

A finite-budget perspective tightens alignment with Paris-compatible trajectories. Front-loaded embodied emissions cannot be "repaid" later without crowding out the remaining budget, so low-carbon construction becomes a first-order design variable alongside operational efficiency and mobility measures. Annualized KPIs can conceal this timing problem by spreading construction impacts evenly over service life; the budget lens restores the correct climate signal by tracking accumulation and scarcity over time.

Sensitivity to decarbonization pace and policy timelines. Our baseline assumes electricity and renewable DH factors decline to ~0 $g_{CO_2e}\cdot kWh^{-1}$ by 2050. To reflect slower or partial decarbonization, we examined alternative factor sets: (i) a delayed trajectory achieving near-zero intensities by 2060; and (ii) residual post-2050 intensities (e.g., 30–60 $g_{CO_2e}\cdot kWh^{-1}$ electricity; non-zero DH), with end-of-life and maintenance after 2050 treated accordingly. Under these paths, the import-weighted emission intensity rises, and cumulative totals tighten the available headroom; projects that pass under the baseline may require stronger embodied-carbon reductions (materials, lean design), greater PV and flexibility (BESS/V2B/TABS), or stricter EMIM measures (higher EV shares, demand reduction) to remain below L2/L3. Because the framework is parameter-driven, declarations can be re-issued annually with updated factors and budgets as EU/national trajectories and EPBD/ZEB provisions evolve.

4.2. Applicability and Limitations

Applicability. In contrast to other research that found their PED concepts favored certain locations, conditions, and contexts, such as districts with low density [59] or favorable climate conditions [60,61], over others, our framework is applicable to a wide range of PED sizes, usages, and typologies, from small urban districts to large-scale developments, both for renovation and green-field developments in Austrian climates. Its applicability stems from the modular definition of system boundaries and context factors and the flexibility to select conversion factors based on data availability and assessment purpose. In practice, high-resolution operation and mobility inputs are not always available; calibrated defaults allow use at early stages but reduce fidelity.

In contrast to other definitions of PEDs that interpret the positive energy balance without contextual factors [25], the investigated projects show compliance with our assessment targets, even though their energy balance is negative or they have net emissions over the life cycle or both. This indicates their compliance with a net zero emission future, and the strength of the declaration system to map these differences.

Uncertainty and change in underlying assumptions. The calculation of the carbon budget relies on several assumptions about future developments that will need to be revisited as conditions evolve. Key drivers include the total gross floor area, which affects the per-m^2 allocation, and the decarbonization rates of other sectors such as industry. These parameters are outside the direct control of individual projects and require regular updates at national and international levels. A systematic treatment of uncertainties, for example, through Monte Carlo or scenario analysis, lies beyond the scope of this study but should be pursued in future work to provide clearer uncertainty ranges and to link results more closely with established scenario pathways.

In particular, the exclusion of certain embodied emissions, particularly those for PV systems, relies on the integrity of the national budget allocation methodology, which represents a methodological limitation. Future work should investigate and compare other allocation methods.

Regulatory alignment is strong within the Austrian building code, but deviations from international standards, particularly EN 15978, may affect acceptance in cross-border contexts. The focus on a fixed 2050 endpoint aligns with current climate targets but may

require revision in the event of accelerated or delayed policy timelines. In fact, current trends indicate that decarbonization might occur slower than mandated, which is why it is crucial for the assessment framework to annually update the remaining carbon budget based on national and international reports. The later sufficient action is taken, the smaller the remaining budget. This periodic update also includes other system developments such as gains in energy efficiency, or potential carbon budget undershoots of realized renovations, which could theoretically lead to a greater carbon budget for future projects if system decarbonization goes better than planned.

Transferability beyond Austria is methodologically possible but requires localizing decarbonization pathways, carrier factors, mobility contexts, and code constraints (e.g., tall-timber fire rules, façade-PV allowances).

The framework embeds a Paris-aligned trajectory to 2050 and assumes (near-)zero-carbon electricity/DH thereafter; if national timelines diverge, factors and budgets must be re-parametrized. Excluding PV embodied impacts at project level relies on coherent sectoral accounting; where this is not in place, a documented alternative (including PV embodied with a compensating sectoral adjustment) may be warranted.

4.3. Policy Implications

Adoption as declaration: The presented framework underpins a national labeling/declaration system in Austria ("klimaaktiv" [37,62]), providing consistent, comparable verification and a practical bridge between research-grade assessment and regulatory adoption. With localized factors, it can harmonize with emerging EU instruments (EPBD revisions, taxonomy), supporting broader uptake while maintaining the rigor needed for credible climate-neutral PED claims. To maintain the claim of "climate neutral districts", the remaining carbon budgets must be routinely monitored and adjusted, preferably on an annual basis.

Adoption in practice: The framework enables municipalities and developers to test whether districts are on Paris-compatible paths by combining two KPIs—a finite cumulative GHG budget and a positive primary energy balance—within transparent boundaries that include user-driven mobility. Moving from annual KPIs to a 50-year cumulative budget refocuses decisions on near-term embodied and operational peaks, closing a key blind spot in conventional PED assessments. Authorities can request framework compliance as part of development agreements and contracts, or set concrete, verifiable targets (e.g., max embodied GHG per m^2, minimum flexible PV/BESS/V2B capacity, DH decarbonization milestones), align procurement and permitting, and offer incentives accordingly. Separating measurement (hourly EF) from control (flexibility signals) also gives a policy lever to reward time-aware operation rather than nominal annual surpluses. This should further be aligned with national adoption plans for Smart Readiness Indicators (SRI).

5. Conclusions

We present a declaration-ready framework for assessing climate-neutral Positive Energy Districts (PEDs) that combines finite CO_2 budgets, explicit system boundaries, and dynamic emission factors. It reconciles research-grade accuracy (hourly CO_2 intensity and differentiated decarbonization paths) with policy usability. Core elements include the following:

- **Two KPIs with targets:** (1) cumulative GHG budget to 2050 and (2) annual primary energy balance for declaring PED status (with/without mobility).
- **LCA scope:** unified accounting of construction, operation, and EMIM (including embodied construction, maintenance, end-of-life) over the next 25 years within a single CO_2 budget.

- **Alignment with national sector targets:** PV embodied impacts remain in the energy sector and are excluded at project level; non-EMIM travel modes (PT, walking, cycling) remain in the transport sector. PV exports are credited with time matching.
- **Time-resolved emissions and pathways:** hourly (or monthly) grid factors; distinct decarbonization trajectories for electricity and renewables; constant intensities for gas/biomass/other fossil carriers (with DH inheriting its supply mix).

Design implications. Case study applications show that even highly efficient PED designs can exceed the available CO_2 budget unless embodied emissions are minimized (through low-carbon/biogenic options, lean structures), renewable energy and flexibility contributions are maximized (through flexible dispatch with BESS/V2B/TABS), and operational emissions from user-driven mobility are effectively reduced (by electrification and demand reduction). The main findings of the case studies indicate that:

- A diverse set of districts, ranging from low-density residential to high-density urban contexts, can achieve compliance with the proposed carbon budget (L3) by adopting low-carbon construction materials, maximizing renewable energy generation (PV oversizing), and electrifying mobility (e.g., EVs).
- Operational mobility emissions (EMIM) remain a significant factor for compliance, particularly in rural areas with lower EV penetration and less access to public transport.
- Hourly emission factors and flexibility (via BESS/V2B/TABS) can significantly reduce operational emissions compared to traditional monthly or annual accounting methods, highlighting the importance of time-resolved emissions in accurate PED assessments.

Limitations. The interpretational claim of "climate neutral" hinges on systemwide decarbonization trajectories and assumptions that remain a source of uncertainty and may require periodic re-adjustment moving forward. Transferability of results depends on localized data. Results obtained in early planning may be subject to change as planning progresses, potentially requiring separate declarations.

Next steps.

- Link to dynamic LCA databases to strengthen embodied-emission estimates.
- Adapt to diverse regulatory contexts and add uncertainty analysis for pathways.
- Automate compliance checks and embed the workflow in planning regulations and municipal decision processes.

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Abbreviations

The following abbreviations are used in this manuscript:

ACH	Air Changes per Hour
AS HP	Air Source Heat Pump (Air to Water)
BESS	Battery Energy Storage System
CLT	Cross-laminated Timber
COP	Coefficient of Performance
CO_2e	CO_2 Equivalents
DH	District Heating
DHW	Domestic Hot Water
EI	Emission Intensity
EMIM	Everyday motorized individual Mobility
EV	Electric Vehicle
FSI	Floor space index (Gross floor area over plot area)
GHG	Greenhouse Gas
GFA	Gross Floor Area
GS HP	Ground source Heat pump (boreholes)
HVAC	Heating, ventilation, air conditioning and DHW
ICE	Internal combustion engine (vehicle)
LCA	Life Cycle Assessment
NFA	Net Floor Area
PAX	Persons approximate
PEF	Primary Energy Factor
PE tot	Total Primary Energy
PE nre	Non-renewable Primary Energy
PE re	Renewable Primary Energy
PED	Positive Energy District
PV	Photovoltaics/Photovoltaic System
TES	Technical Building Energy Systems
TSD	Time Series Data
V2G	Vehicle to Grid
V2B	Vehicle to Building

Appendix A

Table A1. Input parameters and intermediate calculation values used to derive the floor-area-specific carbon budget limits: global budgets, national allocation, population, floor area statistics. The resulting thresholds are summarized in the main text (Table 2).

Variable	Unit	Timeframe	50%	66%	Source
Paris-compatible CO_2 budget (+1.5 °C, no overshoot)	Mt CO_2	2022–2075	510	280	[27]
Paris-compatible CO_2 budget (+1.5 °C, with overshoot)	Mt CO_2	2022–2075	610	340	[27]
Remaining budget after 2024 (no overshoot)	Mt CO_2	2025–2075	360	130	calc. from [27]
Remaining budget after 2024 (with overshoot)	Mt CO_2	2025–2075	460	190	calc. from [27]
Population (Austria)	million cap	2025	9.0	9.0	[27]
Per capita net budget (no overshoot)	t CO_2 cap^{-1}	2025–2075	40.0	14.5	calc. from [63]
Per capita net budget (with overshoot)	t CO_2 cap^{-1}	2025–2075	51.1	21.1	calc. from [63]
Natural carbon sinks (NCS)	t CO_2 $cap^{-1}a^{-1}$	—	1.0	1.0	[64]
Cumulative NCS added (50 years)	t CO_2 cap^{-1}	2025–2075	50.0	50.0	—
Per capita gross budget (no overshoot + NCS)	t CO_2 cap^{-1}	2025–2075	90.0	64.5	—
Per capita gross budget (with overshoot + NCS)	t CO_2 cap^{-1}	2025–2075	101.1	71.1	—
Buildings + EMIM sector share of per capita budget (A1–A3, B1–B6 + EMIM, C1–C4)	%	—	50	50	[49] based on [26]
Estimated national GFA (all uses)	million m^2	2040	906.7	906.7	[65]
Specific GFA per capita (all uses)	m^2 GFA cap^{-1}	—	101	101	calc from [65]

Table A2. Building LCA balancing boundaries defined in [43].

Boundary	Scope Description
BB1	Complete thermal envelope, including internal floor ceilings
BB3	BB1 plus all interior walls, basement structures, unheated buffer zones; excluding open circulation spaces (stairs, balconies, loggias, galleries)
BB5	BB3 plus shared circulation areas and building services
BB6 (used here)	BB5 plus exterior facilities (carports, bicycle shelters, etc.) and auxiliary buildings

Table A3. Total (PEtot), non renewable (PEnre) and renewable (PEre) primary energy conversion and GHG emission factors for 2024 of selected energy carriers based on [52,66].

Energy Carrier	PEtot	PEnre	PEre	GHG
		$kWh_{PE} \cdot kWh^{-1}{}_{EE}$		$g_{CO_2e} \cdot kWh^{-1}$
Grid Electricity	1.76	0.79	0.97	156
District heating—boiler plant (renewable)	1.72	0.4	1.32	59
District heating—boiler plant (non-renewable)	1.48	1.16	0.32	193
District heating—high-efficiency CHP (default)	0.59	0.41	0.18	67
District heating—high-efficiency CHP (best case)	0.33	–	0.33	20
Waste heat (default)	1.00	–	1.00	20
Waste heat (best case)	0.30	–	0.30	20
Coal	1.46	1.46	0.00	360
Heating oil	1.20	1.20	0.00	271
Natural gas	1.10	1.10	0.00	201
Biomass (solid)	1.13	0.10	1.03	247
Biofuels (gaseous, stand-alone operation)	1.40	0.40	1.00	100
Biofuels (liquid, stand-alone operation)	1.50	0.50	1.00	70
District cooling Vienna [67]	0.42	0.13	0.29	42

Table A4. Assessment model parameter overview.

Category	Description of Data	Parameters
District	Gross floor areas, net to gross ratios, renovation shares per usage	24
Weather	Hourly TSD of ambient temperature and N/E/S/W/H Irradiation	6 × 8760
Heights	Average room heights per usage	8
Building Hulls	Hull areas by type and orientation	8
Building Physics	Thermal transmittances, effective heat capacities, heat bridges, air tightness	10
Windows	Total transmittance, shading offsets summer, winter, mobile shading	32
Ventilation	Per usage: Share of mechanical/window ventilation, night ventilation, ACH, electricity demand, heat recovery rates	44
Usage	Concurrencies, DHW demands, auxiliary and plug loads per usage	35
Lighting	Daylight controls and coefficients, illuminations, light power loads per usage	12
Heating	Heating periods, Heating Powers and COPs of up to four heating systems (2 electric, 2 thermal), setpoint temperatures	33
Cooling	Cooling periods, Cooling Powers and COPs of up to 3 active and 1 passive systems (2 electric, 1 thermal, 1 free cooling), setpoint temperatures	37
DHW	Hot water demand profiles, storage volumes, generation efficiencies, distribution losses	17
PV Systems	Installed PV capacity, module and inverter efficiency, Hourly TSD of PV yield	5 + 8760
BESS	Battery capacity, charging/discharging efficiencies, c-rates, losses	11
Flexibility	Heating, Cooling, DHW, PV, BESS utilization ranges	11
Mobility	Usage intensity, concurrency, motorization, Annual mileage, EV penetration rates, charging infrastructure, minimum SOCs, utilization of bi-directional charging	32
Embodied carbon	Construction materials and components, GHG intensities, life cycles, reference areas	37
PE and Emission factors	Primary energy and emission conversion factors, annual weights, context factors	2 × 8760 13

Table A5. Scenario definition for case study District 4.

Scenario	Grid Emission [66]	Flexibility Signal	Flexibility Option (s)
19M	2019 M	Wind Peaks > 50% P_{nom}	TABS
19A	2019 A	Wind Peaks > 50% P_{nom}	TABS
23M (Baseline)	2023 M	Wind Peaks > 50% P_{nom}	TABS
23H 24-5	2023 H	RA x = 24, y = 5%	TABS
23H 48-5	2023 H	RA x = 48, y = 5%	TABS
23H 48-10	2023 H	RA x = 48, y = 10%	TABS

Table A5. *Cont.*

Scenario	Grid Emission [66]	Flexibility Signal	Flexibility Option (s)
23H 72-10	2023 H	RA x = 72, y = 10%	TABS
23H 48-10 BESS	2023 H	RA x = 48, y = 10%	TABS + BESS
23H 48-10 BESS V2B	2023 H	RA x = 48, y = 10%	TABS + BESS + V2B

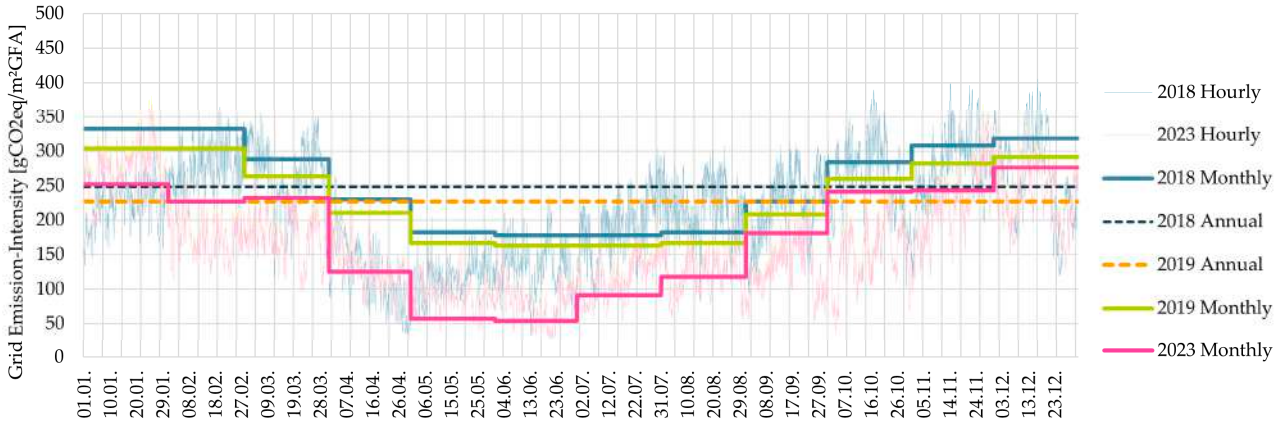


Figure A1. Grid emission intensities of different base years (A—Annual, M—Monthly, H—Hourly).

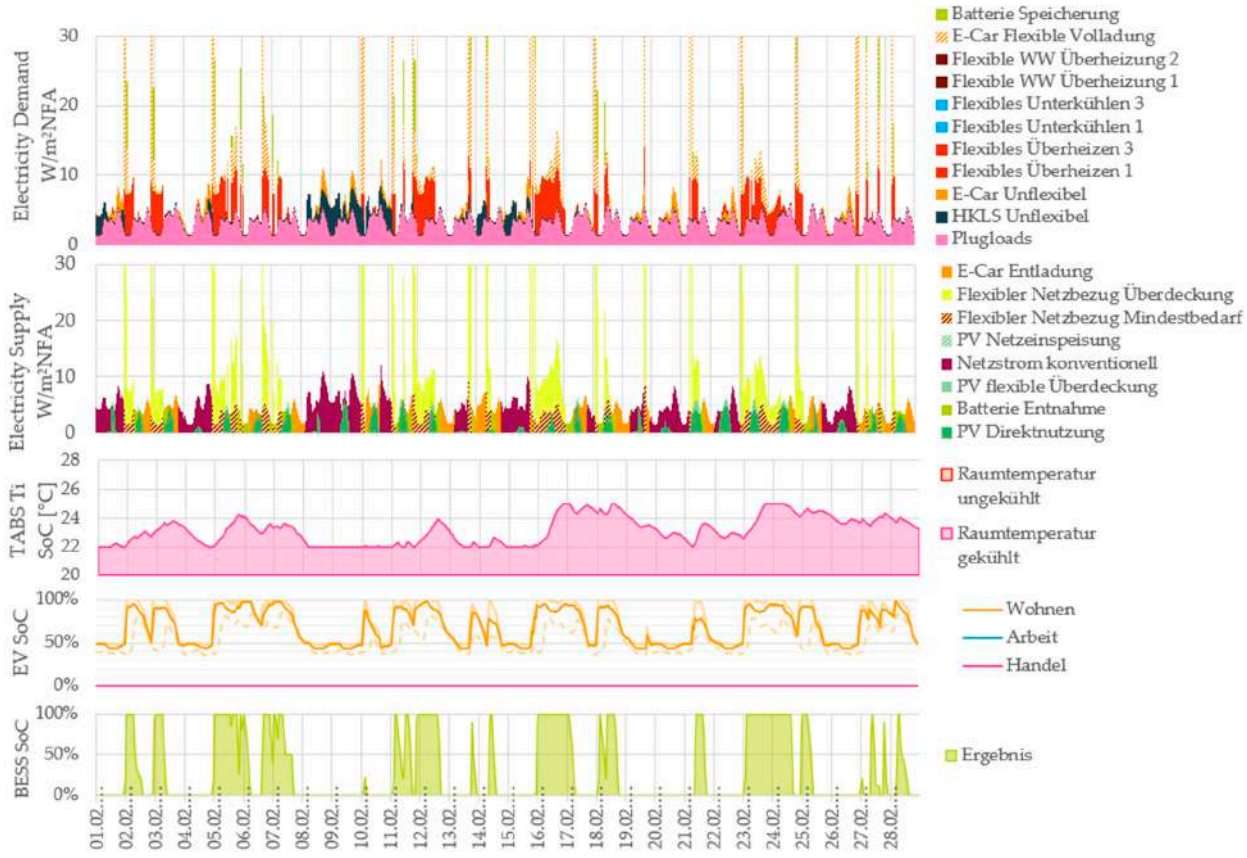


Figure A2. Hourly Flexibility Operation in February Scenario "23H 48-10 BESS V2G" showing electricity demands (Top), electricity supplies (below, note the flexible grid charging in lime and V2B in orange), thermal storage SoCs (TABS indoor Temperature, purple = zones with cooling, red = zones without cooling not present in scenario), EV Battery SoC (dashed = avg offsite SoC, solid = avg SoC, thin solid = avg SoC onsite, orange = residential, purple = retail is overlaying blue = office, both at 0%) and battery energy system SoC (lime green).

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